Volume 1

Multimedia as Meta-Art :

the processes and aesthetics of interactive digital art.

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This dissertation and folio of original multimedia art (in two volumes)
presented to the School of Contemporary Arts,
Southern Cross University, Lismore, N.S.W.,
in fulfillment of the requirements
for the degree of Doctor of Philosophy.

It is hereby certified that the content of this dissertation has not previously been submitted for any degree, nor is it currently being submitted for any other degree.

Signed : ……………………………………………………………………………………

Date :……………………………………………………………………………………

All sources used and any assistance in its preparation are acknowledged within.

Dedication :

This dissertation is dedicated to James and Midge Ferrier in grateful acknowledgement of their perennial support and encouragement.

Acknowledgements :

Thanks to the following for support during the preparation of this dissertation :

- Dr. Michael Hannan, S.C.U., for academic supervision.
- Australian artist and healer Duane Radford for her patience, encouragement, and the use of her unique paintings as the basis for the animations on the enclosed DVD "Take Your Space Now".
- Peter Cox for insight into the production of the "Real Wild Child" CD-ROM.
- Mark Hill for advice on cross-platform issues in producing "the Machine".

Abstract:

Concepts of meta-art, or a synthesis of the arts, have inspired many historically important artists, including Wagner, Scriabin, Kandinsky, Cage, and the Bauhaus and Futurist artists. Such thinkers dreamed of combining the diverse means of traditional artistic expression into a total, transcending art experience.

A meta-art *machine* was visualized by the German author Hermann Hesse in his Nobel prize-winning novel *The Glass Bead Game (Magister Ludi).* Hesse imagined a machine which has evolved to become the ultimate in art technology, a multidimensional machine/human interface, depicted (in a pre-computer time) as a fantastic creative and conceptual abacus, employing all the cultural and scientific knowledge of the ages. This once futuristic vision is becoming a reality, with the rapid evolution of the contemporary personal computer, which offers ever more sophisticated, multi-dimensional, multi-sensory art-making tools, within one machine. Wired to the Internet, this nascent meta-art machine can deliver multimedia art to the world.

The concept of a meta-art machine is used as a unifying theme in *Volume 1* of this dissertation, to help focus a broad exploration of the processes and aesthetics of interactive digital multimedia art. Arts associated with each of the primary sense modalities, it is argued herein, communicate on a number of discrete *channels*. The contributing arts associated with each of these sensory channels of communication are discussed, offering a broad overview of the processes and unique aesthetics of the newly emerged, digital multimedia (meta-) artform.

The author's personal exploration of multimedia art practice is represented in the accompanying creative works, and discussed in *Volume Two* of the dissertation. This creative folio comprises two works – a large scale video animation in four parts, with synchronized audio/music track, (on DVD), entitled *Take Your Space Now* ; and an experimental interactive art experience, on CD-ROM, entitled *the Machine*.

Contents (Vol. 1) :

2.2 *The Machine* - an interactive art experience, presented on CD-ROM.

The DVD and CD-ROM comprising the creative folio are included in Vol. 2.

Introduction

Author and Context: The background to the undertaking of this study

In 1980 I enrolled in a newly established “Diploma of Music Composition” course at what was then the Northern Rivers College of Advanced Education at Lismore, N.S.W. This course was the brainchild of composer Dr. James Penberthy. I had at the time just moved to the region after accepting an offer of a position as a member of a progressive funk/rock band based in nearby Byron Bay, called *The Feelers*.

Penberthy established this Diploma course with the vision of building a radical but rigorous alternative to the mainstream Conservatorium system, by offering a supportive musical environment for young composers from less traditional or academic backgrounds, such as the rock, popular music and jazz fields. I viewed the course as an opportunity to equip myself with some formal training after a decade of intense work as a professional musician and popular music and theatre composer, in Sydney and Melbourne. The Far North Coast Region of N.S.W. already had a reputation as a counter-culture destination, established from the time of the Aquarius Festival (1973) and my decision to move to Byron Bay had been made in the spirit of a working holiday. However I had ambition, and I hoped that a period of study would offer a modicum of intellectual discipline, a strategy to avoid simply marking time while experiencing the Byron Bay “alternative” lifestyle.

Many years have past and the NRCAE has evolved into Southern Cross University. I established a permanent home in Byron Shire, and, as one of the first graduates of James Penberthy’s noble experiment in academic optimism, I eventually found myself in the position of being the first S.C.U. post-graduate student to face the task of developing an approach to academic research into popular music, coming from a compositional rather than musicological background.

My admission into this candidature and the granting of a University Scholarship were based on an assessment of my activities as an experimental artist in a field where composition of original music is the recognized form of research. However, the newly established Graduate College had yet to confront the logical conclusions of the historical legacy of offering Contemporary Music, with a composition major, within the School of Contemporary Arts. Under the guidance of my supervisor Dr. Michael Hannan, a proposal was developed that I would undertake my doctoral work based upon the American model of the ‘creative’ degree. In other words, the major thrust of the work was envisioned to be original creative artistic experimentation, backed up by a documentation component to place it in context. This proposal was eventually approved by the Graduate College, and became a model for subsequent creative PhDs at SCU.

This doctoral dissertation, the end product of that original submission, thus consists of two parts:

i) A folio of creative work, (discussed in Part Two of the exegesis; its scope is outlined at the end of this introduction)

and

ii) supporting documentation, which places the creative work in context, gives an overview of relevant literature, and reveals key ideas emerging from or related to the field in which the creative work is placed.

Research in the performing arts is fieldwork, though a lot of the experimenting goes undocumented. This introduction is devoted to outlining the artistic career that provided the background to this present research, a career perhaps most notable for its sheer diversity of artistic activity or field work. Such diversity has been partly an imperative of artistic survival - organisms that grow in an arid zone, such as the Australian regional cultural climate, must be adaptive.

In an attempt to précis something essential about the diverse and decidedly non-linear nature of my artistic career, I've been led to the realization that if there is one constant, it has been a seeking out of situations which allows for immersion in the process of artistic creation. This word 'immersion' is also a buzzword associated with multimedia and as such has become an important concept at the core of this study. This is a concept closely akin to that frequently invoked in the performing arts area - suspension of disbelief.

Prior to beginning my career as a musician and composer I undertook undergraduate studies at University of Sydney's Faculty of Arts, with a double major, in Fine Arts (the academic experience which informs the art theory and aesthetics element to this dissertation) and Psychology (hence my interest in the psychology of art appreciation).

In the final semester of my Bachelor degree and armed with some self-taught music skills I walked into, virtually by chance, what became a successful audition for my first fully professional position, as a vocalist in Harry M. Miller's popular and spectacular theatrical production of *Jesus Christ Superstar* , directed by Jim Sharman - a stage-show famous for launching the careers of a whole generation of Australian popular music artists (most notably Jon English, Marcia Hines, Air Supply, the Ferretts and John Paul Young).

The decision to join *J.C. Superstar* proved to be a significant crossroads (though I returned to the University of Sydney years later to complete the Arts degree) and in retrospect I now consider this to be the beginning of my professional life as a multimedia artist. A recounting of this career is relevant here, as a background against which the present work has been formulated.

As a composer, the earliest significant professional experience was the opportunity to create the soundscape/score for the Lindsay Kemp Mime Company's interpretation of Oscar Wilde's *Salome* at the New Arts Cinema in Glebe (later performed to sell-out crowds at the Roundhouse in London). This involved composing, in collaboration with Andrew Wilson, a strongly spatial, quadraphonic electronic score, augmented by a range of exotic live instruments. Kemp's (for the times) avant-garde approach to theatre production was to challenge the audience's senses and sensibilities from the moment it stepped off the street. This was achieved, for example, by employing a one-armed trumpet player to provide an eerie musical ambience in the post-nuclear, graffiti-splashed foyer, the burning of frankincense, and the use of quadraphonic surround-sound in the theatre.

These first professional theatrical engagements were powerful experiences that nurtured a fascination for the processes of artistic immersion and left a strong impression of the power of music in a theatrical context.

Around the same time the then head of classical music for J. Albert & Son Publishing, Dr. Franz Holford, offered me the opportunity to produce an L.P. album of a musical I had composed with writer/lyricist (and later film director) Frank Howson, entitled *Magical Frank*. The CEO of Albert's at the time was the late Ted Albert and he took a personal interest in the project, coming on board as audio engineer. This work soon also enjoyed a season at the newly constructed *Total Theatre* in Melbourne. The recording of the work for Albert's featured Reg Livermore (at the peak of his stage career) and John Paul Young (then the reigning Countdown "King of Pop") among a talented cast, mainly drawn from *Jesus Christ Superstar* ranks. I found myself at the helm in a major recording studio, painting pictures and telling stories in sound, using theatrical voices and sound montage. The songs were pure melodic pop, influenced by the Beatles' *Sgt. Peppers Lonely Hearts Club Band* in the use of ambient sounds and sound effects to evoke imagery and help create a narrative structure.

This opportunity was soon followed by a second L.P. recording for Crest International in Melbourne of another 'musical fantasy' I had composed in collaboration with Frank Howson - *The Boy Who Dared to Dream,* featuring *Jesus Christ Superstar* principal Trevor White, and award winning Australian actor John Waters.

It became clear to me during the *Salome* experience that the possibilities offered by early multi-track home audio recording equipment (which had just begun to reach the market in the early seventies), were too creatively attractive to ignore, and also clearly the most immediate path available for creative experimentation with compositional ideas for unusual ensemble voicings.

In Australia, with its relatively small population, non-mainstream composers have proportionately limited access to funding or infrastructure to support performance opportunities. However, considering that research can be self-funded, I purchased a TEAC four-track open reel recorder, newly available Roland CR-78 analogue drum machine, and the Roland JP4 (a four voice polyphonic mono-timbral synthesizer), plus a microphone, and hooked the technology up to a home stereo amplification system for monitoring. I began working with these pre-digital tools, little realizing I was beginning a long journey into the world of technologically-based music experimentation.

Much of the compositional work done during this early career period found a performance outlet in an experimental cabaret format at such venues as Melbourne's Flying Trapeze Theatre Restaurant (*Kabaratz, The Silent Scream Show)*; and the Pram Factory (*The Astounding Optimissimos*[1]). The exciting and supportive artistic climate flourishing at the time in these venues offered me the opportunity to try my hand at writing, acting, lighting design and directing.

Between theatrical engagements I found work in a variety of rock bands (e.g. Jeff St. John Band, Phil Jones and the New Quintessence) and a vocal harmony group and continued to compose. I played with chart-topping pop/rock band *The Ferrets*, appeared in television commercials, acted in a graduation film for the AFTRS and

[1] the Australian debut of director Jean Pierre Mignon, later of *Anthill* fame

acted in another mainstream musical, Andrew Lloyd-Webber's *Joseph and the Amazing Technicolour Dreamcoat,* at the Seymour Centre in Sydney.

A further recording contract was secured, this time from the R.C.A. Corporation, in collaboration with talented vocalist and multi-instrumentalist Cammie Lindon, as Lindon Ferrier. The singles produced at United Sound Studios in Sydney by the eccentric Spencer Lewis received high-rotation radio airplay and led to several television appearances, but made little impact commercially because, at the time, multi-national recording label support for developing local artists was largely tokenism. On the positive side, I gained considerable performing experience, appearing on the ABC's *Countdown*, and as support artist for Ry Cooder (Palais Theatre, Melbourne) and a Norman Gunston Christmas Show tour. The recognition kept me hard at work composing, recording and experimenting.

As mentioned at the outset, by the early eighties when I was beginning to reassess my direction, and the opportunity arose to work in the regional centre of Lismore, northern N.S.W., while simultaneously gaining a more formal training in music composition under Dr. James Penberthy. Penberthy, though an inspirational teacher, soon found the administrative burden onerous. He was succeeded by percussionist and conductor/composer Richard Mills, who offered a completely different, more conservative, "craft oriented" musical focus and influence.

I was awarded the Diploma of Music (Composition). There followed a period of intensely practical experiences in the city of Brisbane. This began with the opportunity to stage a season of a rock musical I had composed in my spare time, entitled *Goodnight World*. The latter was a musical spoof of the doomsday theories and religious cults proliferating at the time. The show was set in a television station, where a late night current affairs programme called *Goodnight World* was being broadcast on the fateful night that nuclear Armageddon arrived. It played to full houses for 4 weeks at *La Boite* theatre. The musical accompaniments were recorded into an early MIDI sequencer, triggering a Yamaha R-8 drum machine and Korg Poly-800 8 voice synthesizer, with live saxophonist, doubling on flute & clarinet, five principal actor/singers and a chorus of 16 vocalist/actor/dancers[1].

Interest aroused by this production secured me a job in an epic stage production produced as part of the opening ceremonies for the Queensland Performing Arts

[1] a song from *Goodnight World*, entitled *Android,* is used as the soundtrack to the credits section of the accompanying interactive CD-ROM, *The Machine*, sung by the voices synthesis module.

Complex. This was Robyn Archer's *Three Legends of Kra,* directed by self-styled "king of visual theatre in Australia", Nigel Triffitt. This show had previously been a hit at the Adelaide Arts Festival and featured a life-size, balsa Viking Ship and six metre tall Japanese Samurai puppet. It afforded me the opportunity to write for and conduct a 150-piece choir and 37-piece orchestra (including 3 buffalo horn players). The success of my music direction and composition for this show led to 5 years experience as composer for a diverse range of productions at Q.P.A.C.

I also traveled regularly to the Northern Territory during the eighties, after being commissioned to supervise the equipping of the first aboriginal music recording studio for the Central Australian Aboriginal Media Association in Alice Springs. I carried out a host of diverse music-related activities for C.A.A.M.A. and the N.T. Arts Council, giving workshops in music technology, recording traditional and contemporary aboriginal music, and performing with aboriginal musicians. One of the most memorable experiences for me was the production, in collaboration with Bill Davis, of two series of 20 half-hour, original music-based radio programmes, aimed at fostering pride in local culture and awareness of health issues among aboriginal communities. The series, entitled *Bushfire Radio*, went on to win the international Golden Reel Award for community radio. Again music technology was central to the realization of this project.

In the mid-1980s the Queensland Performing Arts Trust (the body responsible for Q.P.A.C.) had purchased a Fairlight CMI IIx, an Australian-designed and manufactured, dedicated music computer, which at the time was state-of-the-art music technology. Though not a strictly logical purchase (over $50,000) for a theatre complex, at the time it clearly embodied the promise of the future of technologically-based music composition. My composition work at Q.P.A.C was especially interesting, in that it was unfettered by commercial restraints yet undertaken in atmosphere of total professionalism. Many productions were financed by the education arm of Q.P.A.T., under the auspices of Dr. Cathy Brown, a forward-thinking and highly intellectual administrator, who encouraged experimental work. One of the many theatre productions I worked on, as composer and live musician, included the innovative, environmentally-themed music and mime production for children, *Solomon and the Big Cat,* which won an *Australian Society for Excellence in the Arts* award and toured nationally.

This period of working for Q.P.A.T. climaxed with the Concert Hall performance, which I devised and directed, of a production with (for Brisbane in the nineteen-eighties) the impressive budget of over $36,000, entitled *Dreams and Machines.* This was an experiment in large scale performance art which featured the aforementioned C.M.I., a Fairlight Computer Video Instrument (C.V.I.), and the prototype of the Fairlight Voice-tracker (one of the earliest pitch-to-MIDI converters), as well as vocalists, rock band, classical wind ensemble, choreographer and modern dance group. The years at QPAT also offered the opportunities to begin experimenting with video technology, and several productions, installations and workshops were mounted over a five-year period, which emphasized the experimental combination of processed image and sound, with a high-tech focus. The work undertaken also included the mounting of experimental workshops for disabled people and terminally ill teenagers,utilizing MIDI-trigger pads, interactive composition and live video processing.

Parallel to this unique period of working as a composer for a government institution, I continued to work in the popular music field with a variety of idealistic rock bands, and began experimenting with the promising new multimedia art product, video clips.

It seemed a natural progression when offered the opportunity during Brisbane's Expo '88 of composing quadraphonic soundscapes for the 17 spectacularly large and zany animatronic floats, which made up the $3m *QANTAS Light Fantastic* parade (which toured the Expo site daily). A second equally challenging multimedia-slanted contract was simultaneously offered - to create a video to be shown on a giant screen behind the UNESCO Youth Orchestra's performance of *An American in Paris,* on the River Stage (which I worked on in collaboration with Paul Rainsford Towner).

During the nineties the career diversity continued - while employed as band leader / guitarist for Eartha Kitt's 1995 Australian tour, I was simultaneously appointed composer for the controversial $1 m joint production between the Aboriginal Development Corporation and private backers, at the Sanctuary Cove Theatre, on Queensland's Gold Coast. This production was supposed to be a permanent tourist attraction in the form of a multi-media theatrical production (featuring a cast of 9 aboriginal actors/dancers/musicians, video, laser lighting and multi-channel music). This was a naïvely idealistic concept, which confronted the political minefield of aboriginal-white reconciliation just a little before its time. Unfortunately (and ironically) the venture folded in acrimonious circumstances involving racial tension, after only a handful of performances (and sadly, outstandingly positive press revues). It was clear that this was a unique moment in Australian performance art and multimedia, and it subsequently became the topic of an honours thesis leading to the award of Bachelor of Letters (Honours) from Deakin University.

The early 1990s found me involved in a further progression through a period of composing and recording sound tracks for short films, which included *Green Tea* (winner of the Department of Ethnic Affairs' *Multi-cultural Award*) and the AFC funded *Moments of Cruelty* directed by the late Guy Morgan. Film scores were an obvious outlet for the pursuit of the digital musical experiments begun on the Fairlight C.M.I., but now usurped by an ATARI 1040ST and Mirage Sampling keyboard.

The digital revolution was in full swing, but a devastating house fire consumed my recording studio and all my arsenal of art-making technology, instruments and archives, including two years of Ph D. work. After a period of shock and adjustment, I refocused my energies. Rebuilding your life after a major catastrophe is inevitably a new beginning - one really has no other choice. I needed to take positive steps to re-invent myself, and I couldn't face the task of restructuring my first, lost attempt at building a Ph.D. dissertation based on the film-opera. After much introspection about where I wanted to go from here, I finally set out to consciously place myself at the cutting edge of multimedia. I set up a new studio, migrating to the Macintosh desktop computer platform, supplemented by a Roland S-770 sampler. These years saw the burgeoning of desktop digital art tools based on a PowerPC chip and it seemed inevitable that I expand my digital activities into visual and video art. Though, at the time, living in regional Australia I felt isolated in my artistic activities, I was one of countless people worldwide who was seduced by the challenge of the newly available art technology: PhotoShop, scanners, video cameras, analogue/digital video capture cards etc. It seemed within my grasp to make my own movies. It was no longer necessary to wait for film score commissions. I also started experimenting with web-design.

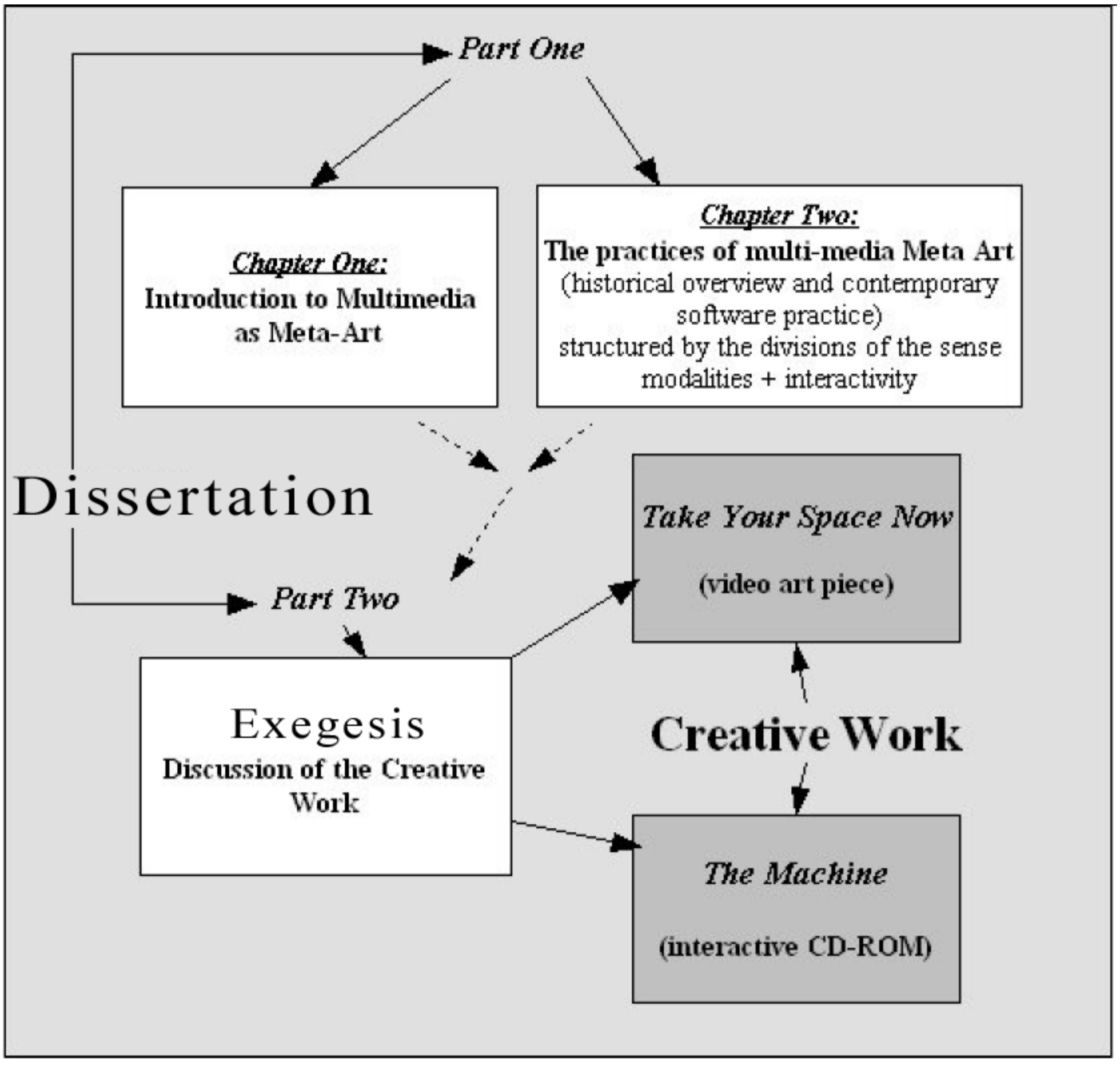
By this time I felt very confident with digital audio technology and music/soundscape composition, and during the latter half of the nineties my creative work began to increasingly focus on the visual - digital graphic art, 3D animation and experimental video art. I received commissions to produce a number of documentary videos for aboriginal communities in the Northern Territory, and completed a number of corporate and music video productions. I began to formulate this disertation and found that it became a constant fascination to me to attempt to process intellectually what all this burgeoning of digital technology might mean to the world of art making. I began the collaboration with painter Duane Radford, which was to become the major work central to this Ph. D. submission.

In 1998 I was appointed Assistant Professor, Multimedia at Bond University, Gold Coast, Queensland, and worked there for three years developing and teaching multimedia courses (graphic design, interactive media programming using Macromedia *Flash*, digital audio) in the School of Information Technology. During 1999-2001 I also taught video production, non-linear video editing and music video production for the School of Humanities at Bond. This was, to say the least, a challenging period in terms of maintaining impetus with my Ph.D. work, but gave me important practical experience and further broadened my knowledge base. The creative component of my Ph.D. component expanded dramatically during this period with my collaborator deciding to increase the scope of the work from one half hour videotape to two. I also did a substantial amount of experimentation in using low-band internet as a creative art space, building the *inya-face.com* web-site. This site came to the attention of Melbourne-based television producer Tony Skinner, at Channel 9, who encouraged me to develop part of my work into an animated television series. I produced a half-hour pilot animation, entitled *In Ya Face*, starring the fabulous Head Brothers, Boof and Dick, (the pilot of which appears in *Part Two* Part B of this submission, the interactive CD-ROM entitled *the Machine*).

During the post-September 11 travel slump, Bond School of I.T. experienced a dramatic drop in student demand and budgetary problems. My contract with the University came up for renewal at Christmas, 2001, and I was one of the unlucky ones retrenched. I was fortunate to immediately gain a position at Central Queensland University's Gold Coast International Campus, where I presently teach Multimedia Design and Multimedia Development using Macromedia *Director*. As I am employed by the School of Informatics and Communication, I was required to deliver a substantial 'communication theory' component to my courses. The readings associated with this preparation revitalized my thinking with regard to this research paper. It was since beginning work at C.Q.U. that I began to develop the accompanying interactive CD-ROM *the Machine*, as my teaching responsibilities centred around Macromedia *Director* and CD-ROM design/programming. In 2003 I taught a course in *Director* at Southern Cross University, Lismore and pushed the development of *the Machine* further, introducing the voice synthesis element. Parallel to these University teaching positions, I have also taught Digital Arts and Media part-time at the North Coast Institute of T.A.F.E, Kingscliff campus, for the past four years and I developed the Macintosh version of *the Machine* for this teaching programme.

This introduction has described the broad picture and significant details of a personal career path which has, from the point of view of this research, given me a wide-ranging practical experience in the diverse gamut of major skill areas associated with multi-media art production. It is a time when multimedia technology is maturing rapidly and much academic interest is focused on the infinite potential of the creative world accessible through these tools. The speed of change made completing this work an on-going challenge and frustration. In conclusion, this dissertation culminates thought processes associated with 25 years of music composition and art experimentation in theatre, popular music, multi-media and performance art contexts. It also offers an opportunity to take a very broad view of where this technological revolution could lead in the future.

2. The structure of the work



This PhD submission is divided into two major components – the dissertation, and the creative work . The written component is sub-divided into two parts, the first of which is an overview of multimedia as meta-art, divided into two chapters (Volume1), and the second is a discussion of the accompanying creative work (Volume 2). So essentially *Part One* of the dissertation is theory (an overview of multimedia art), and *Part Two* is practical - supporting documentation to aid in the understanding of the accompanying creative work. *Part Two* draws on ideas presented in *Part One*.

There are also two items in the creative component of this doctoral submission. The first, or Part A, is a large-scale experimental video art piece, with synchronized music score, entitled *Take Your Space Now*. This work was planned to be presented both in the form of a double-video cassette (two forty-five minute videotapes divided into four sections) and an interactive DVD version. Only the DVD version has been submitted within the PhD context.

This video art piece was commissioned by Queensland artist Duane Radford. Duane had painted a large body of oil paintings, and she planned a collaboration based on these, which would produce a video that was to be an animated journey through the creative “spaces” comprising this series of original oil paintings. This collection of paintings is her life’s work, and Duane’s personal style, at once expressionist and transcendental, exhibits a painterly quality and strongly intuitive sense of colour. She envisioned an ambient musical score as a major creative component of this video art piece.

The collaboration between natural media and digital art techniques, and the process of remediation of Duane’s ‘timeless’ and individualistic images into a time-based, digitally-manipulated, animated video, invoke theoretical issues which are central to the accompanying theoretical dissertation in *Part One,* Chapter One of this work. A discussion of issues related to, or arising from this interface between the natural media and digital media art processes has become a major link between the creative component and *Part One* of this PhD dissertation. The animations featured in *Take Your Space Now* were created using many of the digital art making technologies, both audio and visual, discussed in *Part One*. *Part Two* of the dissertation discusses the specific practical processes leading to the creation of this video art piece, including the music score.

Part One of the dissertation evolved into a broad-ranging discussion of multi-media art practice from the point of view of each of the senses, an analysis further unified by the idea that this technology is evolving in a direction foreseen by Herman Hesse in *Magister Ludi* (discussed below). But it is established in this dissertation that it is *interactivity* which sets contemporary multimedia art apart from other mixed-media forms such as cinema. As the creative work *Take Your Space Now, (*while comprising a large-scale experiment in digital art techniques) is essentially a linear, cinematic-style experience, it was decided towards the end of the development of this PhD submission to append a second creative element, which is non-linear and interactive, as a practical exposition of the discussion of non-linear interactivity in art and contemporary digital practice. This second item, or Part B, in the creative component is therefore a work of interactive art, entitled *The Machine.* This is an interactive CD-ROM created using Macromedia *Director* as the authoring tool. It too is experimental in nature, employing voice synthesis to weave in a literary strand. It is a non-linear experience which delves into the challenges of making art interactive, while also, in a tongue-in-cheek fashion, playing with the Art-Machine metaphor.

To aid the reader in orienting to this doctoral dissertation and linked creative work, a little more introductory explanation will be helpful. As stated, *Part One* of the dissertation (divided into two chapters) is an analysis/discussion of the digital revolution, its impact on art, with a treatment of various relevant background concepts. It provides a framework for discussing the various branches of digital practice which constitute the newly-emerged multimedia art form. As a unifying theme to help organize such a broad-ranging discussion, multimedia art is considered in the light of the futuristic idea of a comprehensive, meta-art machine visualized by the mystic German author Hermann Hesse in his Nobel Prize-winning novel *The Glass Bead Game (Magister Ludi*) – on which he imagined the meta-artist of the future extemporizing multidimensional, multi-sensory expressions of human culture and understanding. As will be seen in the body of Chapter One, there is a long history of experimental thought related to this idea of an epic synthesis of the arts, or meta-art, and this dissertation encourages the idea that contemporary digital computer technology is becoming a practical embodiment of this expansive theoretical tradition.

In *Part One* of the dissertation, the natural division of the sense modalities is used as a structural device for approaching the diverse range of activities comprising multimedia practice. In Chapter Two of *Part One,* the media forms targeting each sense modality are considered historically, and in terms of how digital techniques are currently being applied during the creative processes of digital art-making (i.e. what advantages and new techniques that computers offer to artists, and the ideas associated with these practices). The field of interactive design is included in this discussion as separate cognitive modality, based on the premise that it is the interactive element which separates multimedia from earlier mixed-media art forms. While this is an artificial construct, it helps to simplify the analysis and discussion. The sensory modalities are further sub-divided into channels of information (e.g. the visual modality is sub-divided into the image, moving picture and text channels). This concept will be discussed in depth in the body of the written work.

A basic premise driving the development of this dissertation is has become possible and creatively profitable to consider the technologically-driven phenomenon popularly known as multimedia to be a discrete art form, with its own unique aesthetic parameters, and as a separate field to the traditional contributing media forms (video or movie making, animation, graphic design, music composition and audio design, creative writing etc.) – a case of the whole is greater than the sum of the parts.

This study asks the question: *if multimedia is now a discrete art-form, then what are the processes, and aesthetic considerations which make it unique*. The dissertation attempts to draw together an overview of contemporary theories and practices relating to multimedia art. An in-depth discussion of contemporary theories and concepts of art is undertaken as a basis for understanding multimedia as art.

A summary of a wide selection of the different digital art-making processes that can contribute to multimedia is presented in Chapter 2 of *Part One* of the dissertation. It is acknowledged that the major limitation of the approach taken herein is that it is simply too large a field of study, and can never be exhaustive and complete. It is also possible to lose focus amidst all the detail and end up with a sort of huge "Compendium of World Facts". The very nature of the research field led to the study becoming much larger than anticipated. Nevertheless, it is equally worth the effort to gain an overview of multimedia art and bring the various fields of endeavour together under the umbrella of a unifying study. A study such as this may prove to be a useful starting point to more developed studies of each of these individual areas. Therefore, a comprehensive study of each of these sub-areas of digital art-making activity is not the objective, but rather a broad overview of the wide range of art skills that are possible using digital techniques, in each of the contributing channels of commun-ication. Each of the mini-studies undertaken offers an historical summary, to give a context, and attempts to give a sense of how the process of multimedia production comes together, what the issues in common within each media form are, and finally where all this activity could lead in the future. Each of these mini-studies culminates in an overview of contemporary desktop software applications being used in the digital art practice. Of central interest are contemporary ideas about any evolving rules and principles at work in multimedia production that are guiding practicing artists or theoreticians. Also included in the discussion are some current ideas about how the mind, emotions and senses relate to the artistic content of multimedia. Innovative,computer-specific art-making technologies are of special interest.

The use of internet referencing within this study.

James Lester in *Citing Cyberspace* [1997] states that:

> Currently, most Internet sources have no prescribed page numbers or numbered paragraphs. You cannot list a screen number because monitors differ. You cannot list the page numbers of a downloaded document because computer printers differ. Therefore, in most cases do not list a page number or a paragraph number… the marvelous feature of electronic text is that it's searchable, so your readers can find your quotation quickly with the FIND feature.

Hence, internet sources are not referenced with page numbers in this study. Some cited references appear in the footnotes, which include author and date, but no page number - these refer to internet sources. The rule followed is : if the author has not been credited in an internet research source (e.g. news reports) then this internet source is cited within the body of the text as a footnote, rather than in the bibliography.

Likewise, if the internet source is not dated, then the URL and access date is also cited as a footnote, for ease of access. Wherever possible Lester's guidelines for establishing the academic credibility of internet sources have been followed in choosing internet references.

Please note: This dissertation includes a wealth of illustrations because of the nature of the field discussed, necessitating the division of the printed document into two volumes.
The complete bibliography for this dissertation can be found at the end of Vol. 1 (Part One: Chapter Two). Vol. 2 includes a DVD (*Take Your Space Now*) and 2 CD-ROMS which are the *Macintosh* and *Windows* versions respectively of *the Machine,* discussed in Vol. 2 : Part Two.

PART ONE

CHAPTER 1

Multimedia in the context of the Western art tradition

The Digital Revolution

At the beginning of the third millennium, technological systems are being manufactured and marketed globally, which enable the creation and presentation of unified, time-based works of interactive multimedia art. If computer hardware and software sales figures can be used as a guide, the Digital Arts – electronic music, graphic design, interactive CD-ROM and web-design, digital video production, 2D and 3D animation – are being practiced in hundreds of thousands, if not millions of households daily, across broad sections of the developed world. A surge of creative energy has been released in affluent societies, encouraged by a proliferation of advertising and editorial hyperbole. Large numbers of people the world over have taken to dabbling in this new multi-dimensional information arts production and delivery medium, which has evolved in the relatively short period since the advent of commercially available, and affordable desktop computers. Australian universities and Institute of T.A.F.E. Colleges offer courses in Digital Arts and Media, or Multimedia as part of mainstream Degree, Diploma and Certificate courses.

While the sales figures of the major personal computer manufacturers and software vendors have (to some degree) fluctuated from year to year in accordance with economic cycles, nevertheless annual sales figures of desktop computers have for nearly a decade consistently numbered in the tens of millions of units. The following tables give snapshots of the global scale of the impact of the digital computer revolution in the home, in education and in business, as indicated by the numbers of people acquiring the basic hardware over this period. The first table shows sales for the top 5 PC vendors back in the second quarter of 1998 (over 8 million units).

Top 5 vendors, worldwide PC shipments, Q2 '98

Vendor	2Q 1998 Units	% 2Q 1998 Share	% 2Q 1997 Share	% Unit Growth
1. Compaq*	2,818,000	14.1	13	15
2. Dell	1,815,000	9.1	5.7	71.5
3. IBM	1,572,000	7.9	8.8	-4.1
4. HP	1,240,000	6.2	5.6	19.4
5. Packard Bell NEC	896,000	4.5	5.3	-8.7

* Compaq sales include all Digital Equipment Corp. PC sales for the quarter. Source: International Data Corporation

> According to preliminary estimates from Gartner-owned research firm Dataquest, worldwide PC shipments totaled 134.7 million units in 2000, an increase of 14.5 percent over the previous year, while domestic (U.S.) sales saw a modest 10.3 percent bump, representing 49.4 million units, in 2000. The year 2000 figures stand in marked contrast to 1999, which saw record sales growth of 21.7 percent. (Saliba, 2001).

The statistics related to global sales trends in personal computers indicate that the global digital revolution is continuing and consolidating. There was a contraction in PC sales in 2001, but nevertheless phenomenally high global sales figures were achieved, especially in Asia, which began to outpace global growth in the area of personal computer sales from 2002 (21.66 million PCs sold in the Asia Pacific region or an 8.6% rise, well above the 2.7% global rise for that year[1]). There is more recently an upward trend in PC sales, and the Gartner research company reports that Worldwide PC sales rose by 10.9% to 168.8 million units in 2003, with the fourth quarter showing a 12% increase from the same period in 2002, totalling 48.4 million units. [2]

Unique in the computer world in the way they sell hardware and OS software combined as a single off–the-shelf machine, the Apple Corporation is a smaller and gradually declining force in worldwide computer sales. Apple is placed 5th in terms US market share but services a specialist market, with in recent years, a changing emphasis to computing entertainment [3]. Historically Apple *Macintosh* has a larger percentage of its user base involved in multimedia activities (partly because of its early innovation and leadership in this area).

> In a nationwide survey of 22,000 creative firms, 77% of them planned to purchase *Macintosh* computers in 1999, outselling both *Windows* and *Windows NT* by an astounding margin of over 3 to 1. The research company stated, "Apple continues to dominate the creative markets."(TrendWatch 1999 Creative Atlas Guide). [4]

The next table below shows Apple *Macintosh* Quarterly Sales, 2000-2001 [5], indicating that consumption appeared to be on a downward trend, but sales for 12 months in 2000-2001 nevertheless *totalled* 3,158,000 Units:

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[1] http://it.asia1.com.sg/newsdaily/news002.20030212.html ; accessed 1-10-0.3
[2] http://www.itweb.co.za/sections/business/2004/0401161039.asp?A=COM&O=F ;accessed 16-02-04 .
[3] http://www.macobserver.com/article/2004/01/15.15.shtml ; accessed 16-02-04.
[4] http://www.13idol.com/mac/macfacts.html ; accessed 16-02-04.
[5] Miles, http://news.com.com/2100-1001-215316.html?legacy=cnet; accessed 15-05-02.

00-01 Quarterly Apple Mac Sales	Q1 '00 Actual	Q4 '00 Actual	Q1 '01 Actual	Sequential Change	Year/Year Change
Product Summary	**Units k**	**Units k**	**Units k**	**Units**	**Units**
iMac	702	571	308	-46%	-56%
iBook	236	89	100	12%	-58%
Cube	0	107	29	-73%	NM
G4	355	269	173	-36%	-51%
PowerBook	84	86	49	-36%	-42%
Total	1377	1,122	659	-41%	-52%

The slight downward trend for Mac has continued into 2003, yet they still managed to sell 1,675,000 total units[1].

Clearly computers have become a part of life for a great many people. Of course these huge unit sales figures do not represent a uniform worldwide pattern, and it is in affluent developed countries with strong economic growth, most notably the U.S.A., that the impact of the digital revolution has been greatest [2]. This trend in computer consumption and usage has been building for over a decade and 1994 marked the first year in which PCs out-sold TVs in the U.S.A. [Ebersole, 1995]. In a statement to the U.S. Senate Foreign Relations Committee, Colleen Poliot, General Counsel for Adobe Systems (leading multimedia and desktop publishing software vendor), puts the sales figures laid out above in the context of the overall economy:

> To illustrate our industry's phenomenal rate of growth, I would like to share a few statistics with you. Between 1980 and 1992, the U.S. domestic economy grew at an average rate of less than 3% a year. During that same period, the U. S. computing and software industry averaged 28% annual growth. From 1990 through 1996, the U.S. economy grew at an average of slightly over 3% a year; the software industry grew at an average annual rate of 12.5% during the same period. [3]

Of course, not all the computer hardware is used for multimedia art production, hence the need to look at the sales of multimedia-specific software for an indication of how significant the desktop multimedia art revolution really is, in terms of global impact. Multimedia software vending giant Macromedia, Inc. claims an installed base of 3 million developer/users[4]. Adobe Systems, Inc. is the second largest PC software company in the U.S. (also supplying software for *Macintosh*), with annual revenues

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[1] http://www.macobserver.com/article/2004/01/15.15.shtml ; accessed 16-02-04.
[2] Access translates into usage, e.g. *in the U.S. …whites are more likely than African Americans to use the Web because they are more likely to have access*.
(Hoffman, Novak & Schlosser, 2000, 1).
[3] http://www.techlawjournal.com/intelpro/19990429cp.htm ; accessed 17-05-02.
[4] http://www.macromedia.com/macromedia/ ; accessed 17-05-02.

exceeding U.S. $1 billion. Adobe and Macromedia both have a policy of not releasing sales figures of individual products, but more than half of Adobe's revenue comes from Web related products. [1]

It must be noted that, spectacularly large as they are, official sales figures for the major multimedia software vendors do not give the full story. It is difficult to determine the number of people carrying on digital art-making procedures using illegal “cracked” or pirated software. As Pouillot describes in her statement to the U.S. Senate Foreign Relations Committee enquiry into software piracy:

> Software theft comes in almost as many forms as software itself. In addition to end-user piracy, there is "hard disk loading," where manufacturers and retailers of computer hardware illegally load software onto computer hard drives in an effort to boost sales of their hardware products. Another form of software theft is the mass production of counterfeit product, packaging, manuals, and holograms. Most mass-produced counterfeit software is in the form of replicated CD-ROMs, and so we call this problem--which is also a major problem for the sound recording, motion picture, and video games industries--"optical disc piracy." (This form of software theft is particularly prevalent in Asia, Latin America, and Eastern Europe, although exported product also finds its way into the main distribution chain in Europe, the United States, and elsewhere.) And, last but certainly not least, there is a new frontier in software theft: Internet piracy. A tool of unparalleled economic promise, the Internet also threatens to exponentially increase the rate of software theft. The Internet has become an easy place to traffic in illegal software, with myriad "warez" sites offering thousands of illegal software products for download at no cost to the downloader. Adobe Photoshop, one of our flagship products, is particularly popular with the new breed of cyber-pirates.
>
> When one adds together all these methods of piracy, just how pervasive is software theft? Alarmingly so. International Planning and Research (IPR) estimates that, of all the software installed onto computers worldwide in 1997, fully 40% was illegal. The market value of this massive amount of stolen software was $11.4 billion--up $200 million from the year before. $11. 4 billion, to give this number some perspective, is almost the size of the entire economic output of some U.S. states. (Miles, 1998).

BBS/Internet Piracy, the fastest growing form of online counterfeiting, occurs through the electronic transfer of copyrighted software. Basically, pirate system operators upload copyrighted software and materials onto an electronic bulletin board on the internet for users to download without paying the purchase price of a proper per user license. Acquiring access to such files usually requires the participant to illegally upload other files onto the bulletin board database first, before being granted password access to retrieve sought-after pirated files from protected areas of this illegal database. Peer-to-peer file-transfer and search engine software, such as the notorious *Napster, Scour, Kazaa* and more recently *LimeWire,* facilitate the most popular form of this piracy, offering a highly professional and organized environment

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[1] http://www.adobe.com/aboutadobe/pressroom/companyprofile.html; accessed 17-05-02.

for illegal file transfer which gives new or naïve users a sense that the whole counter -feiting and piracy process is a normal part of the internet, and somehow legitimate.[1]

A press release by software vendor Symantec offers a graphic example of how unofficial distribution of pirated software in the Asian context has been well established for many years:

> **HONG KONG** -- *December 31, 1998* -- Two leading software companies, Adobe Systems and Symantec Software, have stated that total seizures of illicit copies of their software titles in Hong Kong in 1998 exceeded those of actual sales of legitimate products that they each made in the territory…Adobe *Photoshop* and Adobe *PageMaker* were by far and away the most popular Adobe[2] titles seized…Statistics recently released by the Customs and Excise Department show that the tally of seizures of all pirated disks in Hong Kong increased tenfold in 1998, to an estimated HK$1.4 billion from the corresponding period last year. [3]

This trend continues up to the present. As the Business Software Alliance reports:

> Average Global Piracy rate increases to 37 percent; Losses at $11.8 Billion; BSA, a watchdog group representing the worlds leading software manufacturers, today announced the results of its sixth annual benchmark survey on global software piracy. The independent study highlights the serious impact of copyright infringement with piracy losses nearing $11.8 billion worldwide in 2000.

If nothing else, these piracy figures signal a demand. It can be assumed that, if people are going to the trouble of acquiring pirated versions of this software, they are intent on using it.

The wave of cultural activity and naïve enthusiasm reflected in computer hardware and multimedia software sales figures (and also presumably the statistics reporting illegal software usage) has been inspired by a vision of creative freedom sold along with the technology. Is this just good marketing, or is it possible that there is a valid, discrete new art form emerging from these technological developments, which could take an important place in the evolution of human artistic activity?

This study adopts the latter view, and takes it a large step further. Herein the idea is pursued that contemporary interactive multimedia is the harbinger of a future hybrid meta-art. This idea has long been the dream of many famous visionary artists, including Wagner, Scriabin, Cage, the Bauhaus and Futurist artists, and most

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[1] Chew, Jr., Steven; COPYCAT: The Effects of Software Piracy on the Global Economy. http://www.stanford.edu/class/e297c/new/trade_environment/growing_pains/schew.htm ; accessed 12-12-0 .

[2] Adobe produce multimedia art / desktop publishing software tools, e.g. *PhotoShop* (graphic design and photo manipulation), *Illustrator* (vector art/illustration), *Premiere* (digital video editing), *After Effects* (digital video compositing) and *GoLive* (Web Page Design) etc.

[3] http://www.symantec.com/region/hk/press/hk_981231.html; accessed 17-05-02.

notably our inspiration for this study, Herman Hesse, whose novel *Magister Ludi* explores the concept a multimedia meta-art machine in poetic and mystic depth.

In the developed world, digital art activities have already reached a point of acceptance and maturity that makes it possible and relevant to attempt to describe this unique, fledgling multimedia art aesthetic and the range of practice associated with it. Other commentators (e.g. Holtzman, 1998, 17; Wilson, 2002) have recognized this need and have, for example, attempted to describe the World Wide Web, digital music forms (electronica), and interactive multimedia in these terms.

An emerging aesthetic of multimedia art

Origins of the term "multimedia"

An established aesthetic system, or at least an awareness of aesthetic values, is bound to emerge from any field in which substantial artistic energy is being invested. During the past decade there has been a rapid emergence of an arsenal of relatively low-cost digital art tools, and a proliferation of interest in the creative processes and activities involved in using these tools. It isn't difficult to suspect, however, that what we are witnessing is a very early stage of what will prove to be a long evolutionary process. It is a field which attracts futurists, so there is much speculation about where all this activity and development is leading (e.g. writers as diverse as Toffler, 1980; Rheingold, 1994; Negroponte, 1996; Kurzweil, 1999; Wilson, 2002).

As a beginning point to a study of the current understanding of the form, function and aesthetics of multimedia art, it is useful to consider origins of the term. As Burger (1993, p. x) tells us:

> The term "multimedia" has roots preceding the computer. The word has been used for decades to describe productions incorporating multiple slide projectors, video monitors, audio tape decks, synthesizers, and/or other stand-alone media devices. Upon the arrival of the microprocessor, the hardware tools used in various media disciplines became programmable.

Jordan (2002) argues that Vannevar Bush should be considered one of the important pioneering thinkers with regard to digital multimedia, when he proposed a visionary mechanical device that operated literally "as we may think." Jordan describes how, in an article published in 1945, Bush postulated an (analogue) machine, which he dubbed the *memex*, which would support the mind's process of free association in the act of creation. In these pre-digital times he was picturing a (real) desktop and storage space that gave access to microfilm, audio recordings, photographs, and movies - a kind of library, but with an important difference. Rather than arranging the information linearly,

> Bush was interested in rearranging information according to the idiosyncratic paths of personal association that each individual invents during the creative process. He wanted a machine that encouraged spontaneous, associative, stream-of-consciousness thinking, and then left a trail of that thought process behind so that it could be retrieved, not only by the individual who created it, but by others as well. In this way, the *memex* would allow people to share their private, unconsidered thoughts as they leap between ideas moment by moment.
>
> Bush was interested in identifying a central aspect of consciousness, and making a device that effectively expanded consciousness through mechanical means. If you look at the history of the personal computer from this perspective -- as an ongoing project to create a media machine that enhances the intuitive, associative tendencies of consciousness – it connects digital media inextricably to important currents that run through modernism. (Jordan, 2002). [1]

By making a connection between Bush's conceptualization of a technology that is designed to facilitate freely-associative thought and creativity on one hand, and contemporary multimedia on a computer on the other, Jordan is suggesting that multimedia technologies and multimedia products embody a new way of arranging and accessing information and of engaging with art, a significantly new human-technology interface.

Wise (2000, 1) states that it was in the 1980s that the term *multimedia* came to refer to a marketing niche for specialized products of the computer industry. Wise argues (ibid., p.2) that:

> The history of multimedia may be seen as a reciprocal relationship between three institutions: the state through its military and intelligence agencies, the computer and media industries, and various (counter-) cultural elements.

Wise describes how military interest in three areas, aircraft simulation, image analysis, and battlefield command and control, were strong motivating factors in the drive to develop digital technologies, which later adapted well to consumer marketing strategies developed by the computer and media industries. Wise [*ibid*., p.3] claims that the breakthrough for multimedia as a commercial product came with the development of the graphical user interface (GUI) for use with personal computers, combined with the development of the CD-ROM. Wise goes on to offer an argument which stresses the importance of the counter-culture movement of the 1960s in

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[1] *Art and consciousness* is discussed on page 46

shaping the way multimedia has been taken up and adapted by society. Wise [ibid., p.3] argues that:

> This movement has contributed two main themes to the discourse around multimedia technologies, neither of which excludes the other. One stresses the democratic and enabling potential of the computer while the other sees it primarily as an aid to spiritual evolution. Both of these perspectives have been adopted, to various degrees, by those in politics, commerce and education, who promote multimedia technologies as the herald of a new age. Most significantly it was out of the counter-culture movement that the personal computer emerged as a consumer product and commercialism supplanted radical idealism.

Burger [1993, p. xiii] describes how the existence of this new technology was embraced by consumers and quickly adapted for artistic and communicative purposes, so that:

> ...In only a few short years, multimedia has become powerful and accepted enough to change the way we communicate... Marshall McLuhan said “the medium is the message”, meaning, in part, that new media actually change the nature and aesthetics of the content and can impact (upon) society in ways unforeseen by its pioneers. Indeed, this is one of the most exciting aspects of multimedia’s future.

The broad area of activity enabled by these new technologies, and now known by the name *multimedia*, can be considered an emerging or perhaps even maturing art form. As such it, and the various sub-genres of digital art, have won an acceptance in the mainstream gallery/exhibition and academic worlds traditionally dominated by traditional media, and fine arts. However, many artists working in the digital domain primarily find an outlet and exposure through popular and commercial channels of distribution - a large proportion of the digital art industry is essentially commercial or popular art, overlapping into cinema and television (3D graphics, digital compositing of live action footage with 3D animation, special effects etc), radio (popular electronic or techno music), computer games, and advertising. It can be argued that Digital Arts and multimedia have driven much of the burgeoning popularity of the internet, by making web-surfing a richer sensory experience.

Screen-based media products offer curators and art entrepreneur’s challenges in terms of the mass-market consumer viewing experience. It is this writer’s contention that problems of physical presentation of multimedia art to large groups (to some degree) marginalizes multimedia and digital arts in the gallery context. Optimized, purpose-built facilities for exhibiting multimedia art are still to be designed.

However, the proliferation of glossy magazines dedicated to supplying the interests of the growing population of practitioners of digital art recognize that there is a substantial target market of users of digital art-making technologies who do in fact consider themselves to be producing art. Examples of such special interest journals widely distributed in Australia include:

Graphics Arts/Digital Video oriented:

Digital Publishing Design Graphics	Design Graphics Pty Ltd., Ferny Creek, Australia
Multimedia and Digital Video	VideoCamera Publications, Dee Why, Australia
Digital Media World	Digital Media World Pty. Ltd. Bondi Junction, Australia
Computer Artist	PennWell Publishing, Tulsa U.S.A)
Design, Typography & Graphics	http://www.graphic-design.com/DTG/
Computer Arts	Future Publishing, Bath U.K.

Computer Music oriented:

Computer Music	Future Publishing, Bath U.K.
Electronic Musician	http://emusician.com
Future Music	Future Publishing, Bath U.K.
Recording Magazine	MusicMaker Magazine ASIN: B00005N7UO
Mix Magazine	http://www.mixonline.com/
Keyboard Magazine	Music Player Network ASIN: B00005N7R2

Several of these magazines include, as a marketing attraction, a cover CD, which, apart from including demonstration versions of the latest multimedia software releases, also includes examples of readers work and gallery pages devoted to presenting digitally created visual, musical art or 'new media' works. Dedicated (on-line) e-zines devoted to digital arts also proliferate, e.g.:

limbozine	http://www.limbozine.com/
Churn	http://www.churnmag.com
ion	http://www.ion.sgi.com
restart	http://www.res.com
Artistry online	http://www.artistrymag.com/
Shadows In Light	http://www.shadowsinlight.com
recirca.com	http://www.recirca.com/
eclecticity	http://eclecticityezine.com/
designboom	http://www.designboom.com/eng/
Black Robin	http://www.blackrobin.co.nz/
ISEA Inter-Society for the Electronic Arts	http://www.isea-web.org/eng/index_eng.html

1.2 Towards a definition of digital multimedia

Demonstrably a phenomenon of late 20th century popular culture, it is not inappropriate to consider the commonly accepted definitions of the multimedia phenomenon found in popular dictionaries. *The Macquarie Dictionary* (2nd edition, 1991) offers a very straightforward approach:

> **Multimedia** - of, pertaining to or involving several types of media;
>
> **Media** -the materials, or techniques, or means which an artist uses;
>
> **Art** - the production or expression of what is beautiful (esp. visually), appealing, or of more than ordinary significance.

It helps to simplify an analysis of multimedia by adopting the natural division of media into the sense modalities to which each media form is addressed, which, at the present time means the visual and auditory. (In this dissertation the cognitive processes involved with engaging in interaction with multimedia works is treated as a separate modality).

Theoretically we could allow for future technological advances which would widen the list of multimedia sensory modalities to include the other sense modalities:

- tactile,
- gustatory, and
- olfactory

It is already possible to include a kinaesthetic modality, if we consider virtual reality simulators which manipulate the body's sense of balance and movement, e.g. arcade games such as skiing simulators.

Right: The *Karo* cross-country ski simulator [1] .

[1] http://www.karo.com/products/altero.asp ; accessed 17-05-02 .

Virtual reality theatres also usually include kinaesthetic manipulation, by incorporating computer controlled hydraulic motors, to tilt and shake participants, as a means of simulating motion sensations.

Above: Illustration from the *Warner Brother MovieWorld* web-site, depicting a family inside the hydraulically-powered, computer controlled *Batman Adventure* simulator. [1]

A popular example of the latter is the "*Batman Adventure Ride*" at Warner Brothers MovieWorld theme park on Australia's Gold Coast, which is essentially a virtual reality theatre, with video screens supplying visual cues, multi-channel sound systems delivering audio cues and music, and computer controlled hydraulic motors moving, tilting and shaking the module, to give participants motion cues.

As the *Warner Bros MovieWorld* promotional web-site describes:

> Every jolt, every turn and every lurch is experienced, as advanced computer technology makes the free flight sensation seem real and the battle frenzied. The multi-million dollar ride incorporates the latest in animation and computer generated technology.

This writer has had experience in producing video, animations and soundtracks for Hughes Engineering, a company which builds light aeroplanes, but also builds and markets a 9 seat and 27 seat virtual reality theatre, in Ballina, Northern N.S.W, Australia. However the kinaesthetic modality is not commonly addressed on desktop platforms, so it has largely been omitted from the present discussion.[2]

Sense modalities and channels of communication

A similar approach to discussing multimedia in terms of the sense modalities is taken by Elsom-Cook (2001, 3), who further sub-divides the modalities into *channels*, pointing out that:

> While the division of multimedia activities into modalities is clear, it is obviously not sufficient for us to be able to distinguish adequately between different sorts of multimedia artifact. If we consider the auditory modality, for example, there is something fundamentally different about receiving spoken communication, hearing noises, or listening to music. Yet they all operate within the same modality. We need to introduce something more sophisticated.

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[1] http://www.movieworld.com.au/rides_attractions/ride_batman.cfm ; accessed 17-05-02.

[2] **Note:** On page 129 other gestural- and movement-based digital interfaces are discussed.

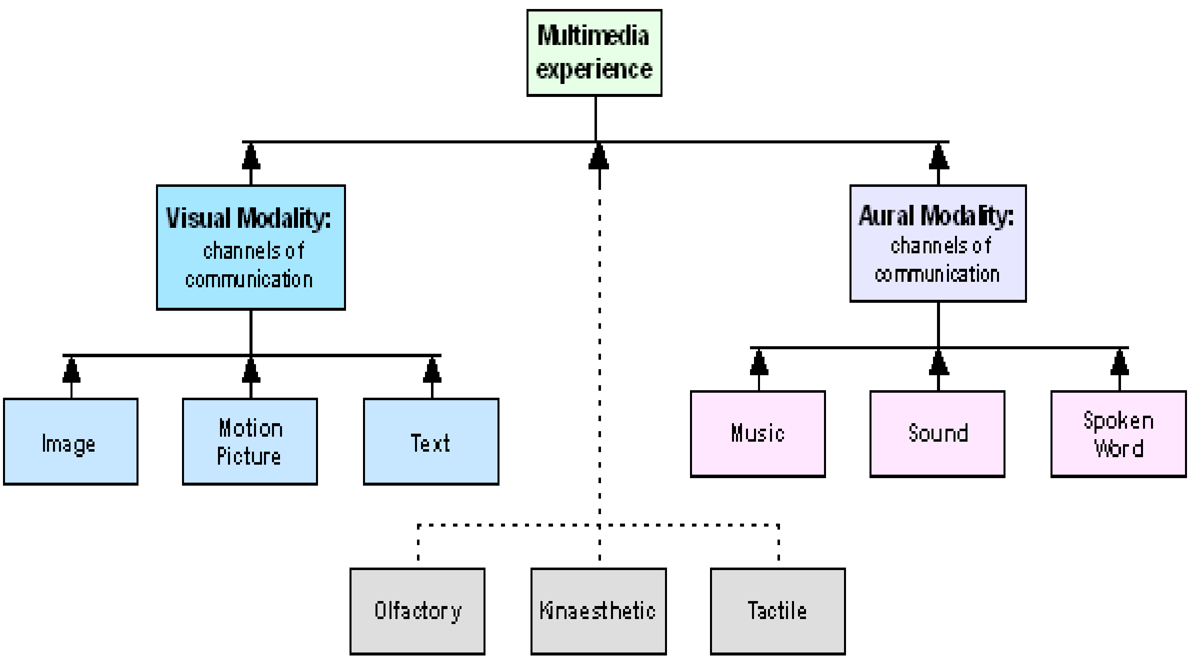
> The additional concept is the idea of a channel of communication. A channel of communication exists within a single modality, but one modality may contain many channels of communication. We can regard a channel as something like a form of encoding information within a particular modality.

Elsom-Cook’s useful analysis goes on (p. 3) to define a *medium* as:

> A set of co-ordinated channels spanning one or more modality, which have come by convention to be referred to as a unitary whole, and which possesses a cross-channel language of interpretation.

By *cross-channel language of interpretation* Elsom-Cook means that the information unique to each different channel is synchronized and co-ordinated, and builds a synthesis of communication in which the whole communicative experience is greater than the sum of the parts. For example, watching a documentary on television without sound, or listening to the same broadcast without the pictures can each convey a certain amount of information. However it is the learned cross-channel language of interpretation which enables us to build a complete understanding of the message that is being conveyed. The richness of the communicative experience is enhanced as the nuances unique to one sense channel resonate with and illuminate parallel sense experience.

Sense Modalities and Communicative Channels in the Multimedia Experience



© 2002 B.Ferrier (after Elsom-Cooke, 2001)

Elsom-Cook progresses from these concepts to a somewhat contentious definition of multimedia (p. 4) as:

> The combination of a variety of communication channels into a co-ordinated communicative experience for which an integrated cross-channel language of interpretation does not exist.

Elsom-Cook's definition here diverges from many other attempts to define multimedia, in that he is describing multimedia in terms of combining communicative channels in new, innovative and untried ways to produce new media. That is, for Elsom-Cooke, the defining characteristic of multimedia is novelty. From this point of view the term *multimedia* is restricted by definition to any new multi-sensory experience, in which the combination of media elements provides a genuinely new communication experience, without a previously developed cross-channel language of interpretation. This approach becomes unworkable however, because a novelty is only a novelty until it becomes popular or familiar, whereas the majority of existing works of multimedia use a standard and familiar off-line or on-line delivery format. By developing a cross-channel language of interpretation for new media we gain a greater understanding and perhaps become more immersed, but surely that does not then preclude the experience from being defined as a multimedia experience.

To further pursue a workable definition of multimedia, it is useful to consider a quote from Microsoft Corporation's *Encarta multimedia encylopaedia* (1995), which describes 'multimedia' as:

> the combination of sound, graphics, animation, and video. In the world of computers, multimedia is a subset of hypermedia, which combines the elements of multimedia with hypertext, which links the information.

Encarta's hypertext link for *hypermedia* offers the following description which resonates with the previously mentioned ideas of Vannover Bush and his *memex* (see p. 28):

> ...in computer science, the integration of graphics, sound, video, or any combination into a primarily associative system of information storage and retrieval. Hypermedia, especially in an interactive format where choices are controlled by the user, is structured around the idea of offering a working and learning environment that parallels human thinking - that is, an environment that allows the user to make associations between topics rather than move sequentially from one to the next, as in an alphabetic list. Hypermedia topics are thus linked in a manner that allows the user to jump from subject to related subject in searching for information. ... If the information is primarily in text form, the product is hypertext; if video, music, animation, or other elements are included, the product is hypermedia.

It is the introduction of interactivity and the resulting non-linear branching structures, which sets multimedia apart from traditional linear, mixed media products such as film. Multimedia art is herein defined as:

> *Time-based communication artifacts, created and delivered via digital computer technology, through processes which involve the organization and manipulation of digital media into a non-linear, interactive, multi-sensory experience which is designed to be perceived as a unified artistic expression.*

Such digital media would include digitally sampled and stored images, video motion pictures and animations, digitally recorded sound and music, including MIDI data, and /or text. However, this definition begs the perennially contentious question: What is art? When is expression artistic?

1.3 What is art?

The spectrum of art-making activities

If one were to attempt a short answer to this question, it would seem that art is whatever one defines it to be. For this is an epoch which sees self-styled artists setting up ‘art factories’, such as *Kostabi World*, where Peter Kostabi employs teams of young artists to produce and even design “art” in his style, and then signs the products. Kostabi’s method of producing paintings through committees (made up of idea people, painters, and occasional random visitors)

> … has posed a conundrum, through a kind of shape-shifting trickster persona, about the location of the artwork. Is the painting the artwork? Is the mode of production of the painting the artwork, with its elements of both conceptualism and performance art? Or is the constructed persona of the artist who controls that mode of production the artwork? Where does object art give way to conceptual and performance art? [1]

Indeed there is an endlessly fascinating, if somewhat bewildering variety of art products being released globally, along a spectrum from Kostabi World art products, to the conceptual extreme of Tehching Hsieh, who makes no permanent, saleable artistic product at all (Shaviro, p.1):

> Hsieh is an artist whose medium of expression is not words or sounds or paint, but his own life. His work consists of five "One Year Performances," done between the years of 1978 and 1986, and "Earth," a thirteen-year performance that stretched from the end of 1986 to the end

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[1] McEvilly in the foreword to *Conversations with Kostabi*, written by Kostabi himself. Excerpt found at http://www.chaplin.ee/english/naitused/past/kostabi/default.htm ; accessed 14-04-02.

> of 1999. Each of these Performances involves a particular vow, a particular constraint, and a particular mode of being. Each of them is meticulously documented in a manner appropriate to its content. And, although Hsieh never explicitly states his rationales for his pieces, each of them implicitly raises profound, difficult questions about life and art and being, and about what it means to live in the world we live in.

Maksymowicz, (1996) describes some of Hseih's obsessively disciplined performance art pieces, which require extremes of integrity and dedication, and which to many, it's fair to claim, would constitute acts of madness:

> (Hsieh) has made the boundaries between art and life disappear by undertaking year-long symbolic actions that bring attention to uncomfortable aspects of modern life. He once punched a time clock, every hour on the hour, for 365 days. He spent one year living on the street, documenting it all, and another year of forced interdependence, tied by a rope to another artist named Linda Montano.

Digital artworks in all their forms, must fall somewhere on this wide spectrum, and there is no logical reason to disqualify digital technology as a valid medium for human artistic expression. As a background to our understanding of digital art, consideration must be given to ideas associated with the role of artistic expression for the individual and in society.

Much has been written about this elusive human artistic spirit. One might argue that it is some kind of divine gift, unique to mankind. Hopwood, [1990, 7] is eloquent in his introductory section which is entitled *The Meaning of Art:*

> From the dawn of history art has been an essential ingredient in the life of mankind. Art reflects the spirit of man and the age in which he lives and is one channel through which man can reveal his ideas, longings and imaginings in a tangible way. It indicates his intellect and greatness, and it enables him to translate what he sees, or wants to see, of the physical world. Sometimes the artist is more concerned with human feelings and his thoughts than with seeing. Through art noble ideas and flights of imagination that well up in the human mind can be liberated and expressed. The artist is always seeking new and suitable ways to express his feelings, sometimes in an experimental manner. A work of art is man-made.

(used by permission of the Great Ape Alliance)

Wilson [2002, 17] supports Hopwood's last point, stating that there is some agreement on the following features: art is intentionally made or assembled by humans, and usually consists of intellectual, symbolic, and sensual components. Yet apes have been taught to paint on a canvas - is this art?

Perhaps the ape may be stimulated by the bright colours, and appears (much as a human child may become involved in finger painting) to be absorbed in the process[1]. The web-site maintained by the *Canadian Great Ape Alliance*, a group devoted to saving apes from "bush meat" exploitation (and possible extinction), which sells "ape art" as a means of raising funds for this cause, explains:

> Once the apes settled into the task of painting, they were extremely focused and determined in their work. Great Ape art is not utterly random, and is most often purposeful and representational.

Above: Ape art (used by permission of the Great Ape Alliance)

Kerry Bowman, a bioethicist at the University of Toronto's Joint Centre for Bioethics (and a representative of the Great Ape Alliance) responded in private correspondence to questions about the nature of this "representationalism" in ape art, by saying that:

> …apes that have been taught sign language will literally name their paintings (email).

Do possible arguments for the case that ape-art is valid art necessarily require the involvement of a human participant, at least as an aesthetically critical observer? This is an intriguing philosophical question, which illuminates the difficulties in establishing a definition of art predicated on humanist assumptions.

The mature human impulse to create art could be viewed as a learned response [Barry, et al., 1964, 8], or perhaps even, on some deeper level, a genetically based drive, arising out of a need for a conscious being to relate to, and make sense of the perceived outer and inner universe. This idea takes on poignancy when we consider the following item [Walt, 1999, 1] reporting the effects of war on the psyche of child victims:

[1] http://www.great-apes.com/apePaintings.html; accessed 14-05-02.

June 4, 1999; STANKOVIC 1 REFUGEE CAMP, Macedonia: In the refugee camps, there are children who draw burning houses, bombs exploding and dead people bleeding on the ground. But those are not the ones to worry about. The children who are truly troubled are those who paint sunflowers and daisies, cloudless blue skies and golden sun rays shining down on a cut lawn. On paper, everything is right with their world. Psychologists who deal with trauma have seen this before -- in Bosnia, Rwanda, Cambodia, Afghanistan -- in corners of the world where children's secure lives have been wrenched from their sockets. When a catastrophe like war occurs, children are tossed upon the emotional high seas and left adrift in the turmoil. As their reality begins to take on some grim new shape, they start drawing pictures of what they have seen -- and of what they wish they still could -- offering tentative glimpses into the chaos within them.

Or, then again: is art activity merely a sophisticated form of entertainment which has evolved socially, as a by-product of tool-making and ritual or religious observation, now become a cultural tradition? For example, Fiedler [1996, 1] points out:

> The Aborigines were hunter-gatherers with an oral tradition. This tradition includes music, song, dance, and graphic expression -- all of which can contain rich symbolic meaning. A combination of forms may, in fact, be required for a concept to be fully expressed. Graphic expression played an important role in Aboriginal temporal communication as well as in tribal ritual -- how important was not understood outside of the aboriginal world until the latter half of the 20th century.

Whatever its source, the fruits of this need to express are common to all societies, and they form a thread of culture which stretches back beyond the mists of antiquity, as demonstrated by prehistoric paintings on the walls of caves.

The value and meaning of art as a commodity remains subjective and context specific. Only the passage of time allows an emerging consensus of any greater or sustainable significance an individual work of art may have to society or humanity. While this was once not the case in relation to Tribal art, which embodied long-evolved codes of belief, the inexorable march of globalization (it seems to this writer) is changing the status even of this kind of art, as the pressure arises to feed tourist industries. There is a simultaneous movement of tribal cultures toward a developed economic form, connected to the global culture via telecommunications and the media, putting a different kind of pressure on the traditional role and relevance of tribal art.

The scientific measurement of art and aesthetic response

The complexity of levels on which an individual work of art may operate, and the complexity of the individual's response to art, create immense problems for an objective scientific study of the psychology of art. One of the earliest attempts at a psychometric approach to aesthetics was Fechner's *Vorshule der Asthetik* [1876]. Fechner gathered statistics on individual preference for various shapes of rectangle and other simple forms, working the curves of distribution objectively instead of philosophizing about what people 'ought' to like, or being concerned with why. Along with Galton in England, Fechner pioneered the experimental, quantitative approach to the psychology of art [Munro, in Hogg, 1969, 41].

Scientists might wish to remain optimistic and believe that given time it may become possible to verify the objective properties of works of art. However some investigators such as Moyles et al. [1965, 685-690] suggest that this search is, in fact, in vain.

> …it seems likely that for most subjects the determinants of preferences are item-specific interactions between the unique properties of each design and the subject's idiosyncratic aesthetics [Moyles et al., 1965].

This lack of behaviourist precision will not dampen enthusiasm for speculation and argument about the form, function and value of art, both socially and aesthetically. This applies equally to the art of the so-called digital revolution, which will take its place and be judged within the overall field of aesthetics, but it will have its own unique aesthetic characteristics.

Art can exist, firstly, as a simple entertainment or diversion. However, beyond its role as a novelty or curiosity, it is the present writer's premise that we as a society find value in art, it persists, primarily as a celebration of human faculties, and also for what it can contribute towards the quest for a wider understanding of life and the human condition.

Art, Ritual and Mysticism

Among the more generalized perceptions of art and art-making, there is a viable fascination with art that it is produced through some kind of transcendent, charged or inspired insight. There has been a long history to the belief that great geniuses of art can achieve an element of honesty and directness in relation with, or response to the experience of life, a certain intuitive immediacy or deeper understanding, which by-passes rational or language based analysis [Canaday, 1964, 163].

> Implicit in this religious explanation is a true psychological insight: that artists often feel and act as if their best creative work were directed by some outside power over which they have no control, not consciously willed or planned from within. [Monro, in Hogg, 1969, 34].

Such ideas were central to many modern art movements, especially Surrealism, [Pierre, 1970, 9-14], and can perhaps be traced back to the connection prehistoric art had with primitive religious beliefs, magic and ritual.

The ritual creation of artistic motifs associated with hunting that we know of from cave painting can be interpreted as a form of affirmation, of building up of a sense of power and confidence to help mankind face up to fear, and succeed in a mysterious and dangerous world [Barry, et al., 1964, 22-23]. Or is this interpretation merely a glamourization and dramatization of an ancient moment, which may have simply amounted to a rainy day 'doodle'? Similarly, why project your own hand on the wall as an image – is this not self-exploration, recognition of consciousness, affirmation of existence?

Or could it have been simply a child-like activity, an entertaining way of passing time? It is possible to speculate on a spectrum of possible scenarios which could have been associated with the genesis of these images, based on a knowledge of human life. And, given the vast gulf of time, it is easy to prefer to imagine an heroic creative act, significant and symbolic of a long lost time, and an ancient state of mind.

Such art, all art, must be intimately linked with the world view of the society which produced it. Thus, it is not uncommon to consider that art can function as a mirror to society, and, in an increasingly complex world, it can offer the promise of insights and clarity across a range of personal and social issues. If this social function of art is the historical legacy of the ritual and magical origins of art activity, it can be observed that the Western art tradition is an unbroken continuation of this ritual relation of mankind to art, largely within a Christian tradition.

Religion in its myriad forms seems resilient to the advances of rational science. The BBC web site devoted to Religion and Ethics claims 1000 million followers for contemporary Christianity [1]. Nevertheless, during the 19th century, the deep-seated and relatively unquestioned cultural dominance of institutionalized Christian thought in western society was beginning to shift. Science challenged basic tenets and beliefs such as creationism. The onset of these changed attitudes coincided with the shift in world-view brought on by the fruits of scientific investigation, and the ideas of the giants of modern thought, especially Darwin [Bergman, 2001, 1]. Further, despite the cultural arrogance of European imperialist nations during the age of colonialism, exposure to other world views, ideologies, religious and cultural practices, and art had an inevitable impact on Western society and culture. The phenomenon of cultural exchange through colonization was the beginning of the process of what is now recognized by post-colonial theorists as globalization [Myung Koo Kang, 1998; Fitzgerald, 1999; Afolayan, 2002].

Christian cultural absolutism may have weakened in the West, but nevertheless, during the 20th century and beyond, ritual and spirituality remain important elements in art-making. In Aniela Jaffe's chapter on *"Symbolism in the Visual Arts"* [Jung, 1964, 285-292], she states that it is:

> ...a psychological fact that the artist has at all times been the instrument and spokesperson of the spirit of his age. His work can be only partly understood in terms of his personal psychology, the artist gives form to the nature and values of his time, which in turn form him....
>
> Psychologically interpreted, this spirit is the unconscious. It always manifests itself when conscious or rational knowledge has reached its limits and mystery sets in, for man tends to fill the inexplicable and mysterious with the contents of his unconscious.

However, art not only grapples with meaning and the mystical, it also can, (and from a commercial perspective perhaps *must*) offer sensual rewards. Nevertheless, at its highest level (from a humanist point of view), art offers the hope of expressing the otherwise inexpressible. From the Jungian perspective [ibid., 297] Jaffe interprets

[1] http://www.bbc.co.uk/religion/religions/christianity/index.shtml ; accessed 15-05-02.

this as, in some way, a bringing into consciousness of unconscious knowledge, with consciousness playing the decisive part:

> Consciousness alone is competent to determine the meaning of (art) images and to recognize their significance for man here and now, in the concrete reality of the present. Only in an *interplay* of consciousness and the unconscious can the unconscious prove its value, and perhaps even show a way to overcome the melancholy of the void.

Art as commodity

The association with these sorts of intangibles has imbued the products of the artistic process with a sense of intrinsic importance and value, and, being valuable, art becomes a marketable commodity, an investment, and finally a status symbol. The socialist philosopher Karl Marx was largely responsible for a re-interpretation of the philosophy and psychology of art in socio-economic terms. The influence of Marx's thinking on the psychology of art is summed up by Munro:

> In general, the idea that socio-economic factors influence art and attitudes toward art is now generally accepted. Most historians of the arts pay some attention to the social background of each period and nation, and to the ways in which styles are related to socio-economic conditions. This bears upon the psychology in that the artist is said to acquire and express in his works, often without realizing it, the ideology and economic interests of his social class and the system of which it is part. It applies also to the phenomena of taste and preference and to aesthetic principles, which are likewise explained resulting mainly from social conditions. [Munro, in Hogg, 1969, 40].

Art in all its forms has the power to communicate experience and meaning. Thus art can be a powerful political tool, both as a form of propaganda, and a means of social criticism and subversion. Its power as a form of non-verbal communication has led to the appropriation by the advertising and marketing industry of the language and vocabulary, innovations and icons of art and the great artists.

Art remains a vital social phenomenon because the nature of art is to continually change and evolve, in response to changes in society and human life. Art is a unique form of human expression, and, by assuming an implicit audience, of communication.

Art as Experiment

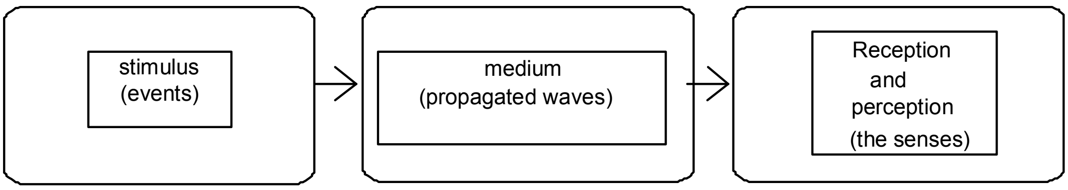
The twentieth century saw a burgeoning of experimentation in art activity and a radical broadening of horizons. Wilson [2002, 16] offers the following list of new experimental art activities that have taken art beyond the traditional, historically validated boundaries, into new media, new contexts and new purposes. He stresses this is at best a partial list:

- Extension beyond realistic representation (e.g. abstract pictures)
- Incorporation of found objects (e.g. Picasso's collage and Duchamp's urinal)
- Movement into non-art settings and intervention in everyday contexts (e.g. Schwitter's Merzbau apartment and Russian AgitProp)
- Presentation of live art (e.g. Dada and Futurist performance)
- Use of industrial materials, products and processes (e.g. Bauhaus, photography, kinetic art, electronic music and Warhol's Brillo boxes)
- Conceptual art (ideas as art, and de-emphasis of sensual form)
- Earth art (work with natural settings with resident materials)
- Interactive art (dissolution of the border between the audience and artists, for example, living theatre and interactive installations)
- Performance art and happenings (e.g. Allan Kaprow)
- Public art (work on site-specific materials, social processes and institutions, and community collaborators)
- Exploration of technological innovations (e.g. video, copiers, lasers, and holography)

Other new forms of experimental art not included on Wilson's list include kinetic and light sculpture, sound art, electronic music, and the subject of the creative component of this doctoral submission, animated video art, to name but a few.

Art and Technology

According to Burger [1993, 4] all communication requires stimulus, a medium, and reception / perception.



With the possible exception of some conceptual or performance art, art is universally created through some form of technology, whether it be ground ochre on a cave wall or marble blocks carved with hammer and chisel etc.   The history of art is delineated by landmark technological developments, which in effect have caused artistic revolutions, and precipitated outpourings of experimental art. Examples include:

1400s	oil paint
1452	the printing press
1722	the well-tempered clavier
1839	Photography
1877	sound recording
1895	the motion picture camera
1928	sound on film
1935	television/video
1937	the electric guitar
1940	plastic-based magnetic recording tape
1955	the sound synthesizer
1979	Fairlight C.M.I. – audio sampler/sequencer
1980	the programmable drum machine
1982	digital sound recording
1982	MIDI
…………	…………………………………..
1992	multimedia CD-ROM
1997	DVD

Ebersole, [1995] points out that technological determinism, and its subset, media determinism, are contentious philosophical positions, and the relationship between technological advances and society is complex.

> The issues raised by technological determinism question the role of technology in shaping our future. To what extent do the tools we make and use determine our behavior? … one of the tenets of hard determinism is that "the world is a mechanism”. Ergo if one believes in technological determinism, one could say that "mechanization is a mechanism", i.e., the creations of our hands determine our future. By creating technology, we create our future, which cannot be avoided. Those who fear the impact of technology are often the most ardent believers in technological determinism and are outspoken about our need to promote our humanity while at the same time subjecting technological progress to rigorous critique. Critics of technological determinism counter that the technology is not the sole determinant of change. Rather, it is the technology working within a complex social structure.
>
> Media determinism, a subset of technological determinism, is a philosophical and sociological position which posits the power of the media to impact society. As a theory of change, it is seen as a cause and effect relationship. New media technologies bring about change in society.

It may be simplistic to assume a simple cause and effect relationship between technological change and social evolution, but they are intimately bound together. Developed societies are presently experiencing a period of great technological change, which appears comparable in historical significance to the Industrial Revolution. Significant social and cultural ramifications accompany this digital revolution. The silicon chip has changed and is continuing to change society. If our discussion regarding the social relevance of art as a response to life has some merit, then art makers will inevitably and rapidly respond to this sweeping social re-organization.

Significantly the art being produced in response to the new technology is being recognized by some as embodying a paradigm shift in the concerns of art. For example Ascott [1993, 2] suggests that:

> It is the overarching concern with appearance and the representation which has hitherto characterized western art and which has made it the servant of ideologies, of both church and state. It is its concern with appearance which has kept it in line with classical science, looking no further into things than their outward forms allow, making of the world a clockwork machine of parts whose movements are regulated by rigid determinism, and seeing Man as little more than a material object. It is the art of appearance which is purveyed in boutiques, galleries, museums and on the pages of chic art magazines. It is International Art. And it is dying. It is dying because it is no longer relevant to a culture which is progressively concerned with the complexity of relationships and subtlety of systems, with the invisible and immaterial, the evolutive and the evanescent, in short, with apparition. Questions of representation no longer interest us. We find no value in representation, just as we find no value in political ideologies. We do not wish to keep up appearances.

Digital art has emerged against the back-drop of this new paradigm of art practice.

Art and consciousness

The empirical paradigm of mainstream science suggests that much of perceived “reality” is just that: a matter of perception. Colours have their existence as the human brain’s way of responding to light frequencies - we cannot see infrared, but of course that doesn’t mean it doesn’t exist as a physical phenomenon. Sounds, physically described, are merely variations in air pressure. It is only within the brain that sound becomes music, and noise can have a meaning. Art cannot exist without consciousness, and engagement with art changes our consciousness.

If our sense organs are indeed limited with regard to the range of stimuli they can respond to, then the logical conclusion (at least from an evolutionist’s point of view) is that adaptive processes brought about by the 'survival of the fittest' imperative have been responsible for us developing sensitivity to just those aspects of the physical universe which were/are of significance to our day-to-day survival. Such an argument leads to the conclusion that our brains and sensory systems are either not responsive to, or are filtering out unnecessary data.

In *The Doors of Perception*, Aldous Huxley [1954] proposes a similar argument when he suggests that psychedelic drugs such as mescaline and lysergic acid cause an overriding or bypassing of these sensory filter mechanisms, allowing us to perceive much more of the sense data available to us, giving us a transcendental experience of a heightened (more complete) perception of reality. Huxley's title quotes mystic English poet and mixed-media artist William Blake:

> If the doors of perception were cleansed everything would appear to man as it is, infinite.

It might be argued that such unnatural (drug-induced) sense experience only provides distorted data, mere hallucinations. Nevertheless, the point made is illustrative of the way in which notions of reality equate with individual consciousness. Increasingly western art-making is concerned with a questioning of consciousness itself, with playing with, and reflecting upon the construction of our notions of reality [Peat, 2002].

In his essay entitled *The Alchemy of Creativity: Art, Consciousness and Embodiment* [Peat, 2002], compares the traditional philosophical approaches (i.e. approaching consciousness by way of logical debate and didactic argument), with the way the technique of perspective is used to depict a street scene in a classical painting. In the latter, a wholistic view is created at the cost of a necessary distortion of constituent elements.

> Perspective provides an overarching point of view in which the various elements of the picture are forced to fit into a single geometrical scheme of projective geometry - single vanishing-point perspective. On the surface this may look like integration, but in another sense it is a false type of holism because although the streets and houses fit together within the same mathematical order they do so at the expense of circular openings being distorted into ellipses, parallel lines converging and right angles become acute or obtuse…
>
> Art, in its widest sense, is a form of play that lies at the origin of all making, of language, and of the mind's awareness of its place within the world. Art, in all its forms, makes manifest the spiritual dimension of the cosmos and expresses our relationship to the natural world. In this sense the act of making manifest, of bringing into the world, is the precursor of all human civilization.
>
> …all creative action - from painting and sculpture, to dance, music, poetry and science - involves acts of projection onto the material world. The power of this projection is that it gives us access to a series of codes, linguistic structures, archetypes, call them what you will, that we can internalize, only later to inject them into our perceptions and thinking about the world. These codes and structures are therefore the ways we make sense of the world, society and our relationships. They are the infrastructures that determine the way we act and have our being. Thus at times a single work of art enables us to see the world in a radically new light…

In this sense, art experiments of the twentieth century and beyond are experiments in consciousness.

Meta-narratives

Anderson, [1995, 4] points out the shifts in overarching social belief structures which he calls *meta-narratives*:

> A meta-narrative is a *story* of mythic proportions, a story big enough to pull together philosophy and research and politics and art, relate them to one another, and above all - give them a unifying sense of direction…examples (are) the Christian religious story of God's will being worked out on Earth, the Marxist political story of the class conflict and revolution, and the Enlightenment's intellectual story of rational progress…(we can) define the postmodern era as a time of "incredulity toward metanarratives" – all of them.

Our contemporary global society has developed a more questioning attitude towards the meta-narratives of traditional religious faith, but also has seen the collapse of Marxist-inspired atheistic communism. However, the needs which spiritual meta-narratives address still exist. There remains what could be described as an underlying human yearning for reassurance and guidance in the areas where once religion provided answers, beliefs structures and comfort. In this context, artists, by addressing their work to major human themes and life questions of all kinds, inherit some of the roles once taken by the witchdoctor or the parish priest.

Another way to view the transcendent nature of peak art is to consider the concept of what Peat [2002] calls *the alchemy of creativity* - an attitude to art which views artistic work by the recognized genius as the modern day equivalent to magic - conjuring from the depths of the artistic soul something valuable, even priceless. For it is a mystery indeed that mere charcoal squiggles on a piece of paper, smudges of oil-paint on canvas, or a certain combination of common strummed guitar chords and rhyme can become a seminal work of 20th century popular- and/or fine-art. This is indeed 'base metal into gold', and it remains the enticing mystery of art. Munro supports this idea [Hogg, 1969, 34], writing:

> The origin of psychological theorizing about the arts can be traced to the wide-spread primitive belief in the supernatural power and inspiration of the artist. It was thought that some artists, like shamans, priests and oracles, could be divinely inspired; the artifacts, chants and dances could have magical power for good or ill. The right kind of statue, temple or ritual dance could please the gods….

The acknowledged historical masters of art are seen as visionaries, even if sometimes in a modern world their visions are bleak.

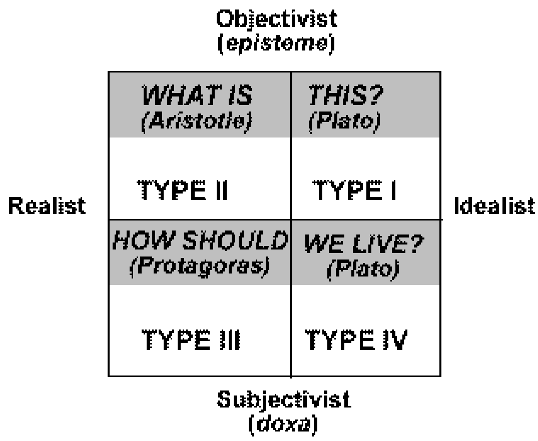
In a post-modern world, society has become incredulous with regard to the historical meta-narratives of science and religion. Global mass communication, and raised education standards have influenced the way these meta-narratives are understood. This understanding has been tempered by the idea that we construct our own local or cultural representations of reality over time. Such socially constructed realities exist within a perceptual framework prescribed by the orientations and limitation of language.

Four Meta-Paradigms of art

The astonishing diversity of art-making activity and the vast sweep of divergent thought regarding aesthetics and the philosophy of art make any serious attempt at systematizing and organizing an understanding of this field of human behaviour worthy of consideration. Pietersen (2000) has made such an attempt at elucidating the underlying meta-paradigms of art, offering a coherent view of art and aesthetics, Pieterson identifies the four meta-paradigms of art as: the search for universal form, representation (imitation), the expression of personal experience and feeling, and the expression of ideas concerned with social reform. Drawing on classical Greek philosophy (Plato and Aristotle), he identifies four basic modes of intellectual activity or fundamental orientations to knowledge which underpin all human intellectual behaviour, which he calls *the Meta-Paradigmatic Framework*. He presents this idea schematically as a matrix, divided on polarized axes designated as a Rationalist – Subjective axis and a Realist – Idealist axis.

Rationalist thought (types 1 and 2) essentially pursue the question: *what is this?*, whilst subjectivist thought (types 3 and 4) in varying degrees revolves around the humanistic question of: *how should we live?* At bottom, and despite the modern more flexible (fallibilist) view of human knowledge, the rationalist quest is for firm, once-and-for-all laws (regularities) of nature, including human nature and society. For Humanist thought, on the other hand, culturally embedded values, not Reason *per se*, are the central problem and purpose of human existence [Pietersen, 2000].

The Meta-Paradigmatic Framework



Having argued that all thought falls somewhere within a matrix of these four generalized polarities (though not necessarily exclusively confined to one domain) Pietersen offers more detail regarding each of the meta-paradigms:

Meta-orientations – the general framework

TYPE II: OBJECTIVIST – REALIST	TYPE I: OBJECTIVIST - IDEALIST
EXPLICATORY / SCIENTIFIC MODE	*VISIONARY / METAPHYSICAL MODE*
Empirical truths (Facts)	Essential truths (Ideas)
Impersonal / CONTROLLED INQUIRY	Impersonal / SPECULATIVE INQUIRY
Observation / measurement	Theoretical / mystical
Specialist / "Pebble-picking" / Differentiation	Generalist / "boulder-building" / Integration
Systematic analysis and prediction/ Description	Encompassing concepts ("patterns that connect")
Deterministic / foundational / immanent	Deterministic / foundational / transcendent
TYPE III: SUBJECTIVIST - REALIST	TYPE IV: SUBJECTIVIST - IDEALIST
EXPERIENTIAL – INTERPRETIVE MODE	*MISSIONARY / DEVELOPMENTAL MODE*
Existential truths (symbols, linguistic)	Ideological truths (concepts; principles)
Expressive – revelatory - poetical	Political – advocacy - action
Personal - engaged	Communal - engaged
Values (humanism) – empathic	Values (humanism) – developmental / reformist
Voluntaristic / contextual / immanent	Voluntaristic / contextual / transcendent
To praise, eulogize, tell inspiring STORIES; To unmask, debunk, critique and tell 'sad' stories	To influence and ENGINEER life/world/society according to valued ideals and principles

Pietersen argues that various approaches to art-making fall generally, though again not necessarily exclusively, within one of these quadrants. To follow this line of thought it could be argued for example that serial composition, with its emphasis on system over expression, could fall within Type II. Expressionist art could be said to fall within Type III. Works of a political nature would tend to belong to Type IV, etc. Pietersen then provides a schematic layout which formally applies his system of categories of intellectual activity to the philosophy of art.

Meta-orientations in the philosophy of art

REPRESENTATIONALISM – II	FORMALISM – I
Imitation – Aristotle/Hume Realist/Naturalist Descriptive	Universal forms – Plato/Kant Patterns of symmetry and measure Axiomatic/ Form of the Beautiful
EXPRESSIONISM – III	**FUNCTIONALISM – IV**
Expressionism Emotion/experience Impressionism; Surrealism; Dadaism; Nietzsche/Collingwood	Art and Society Propagandist/Promotional Normative/Ideological Plato/Marxism/Tolstoy

Type 1 art is concerned with form. Pietersen quotes Harrison-Barbet (1991)[1]:

[1] Harrison-Barbet, A (1991) *Mastering philosophy*, England: McMillan. No page number cited.

> What matters is the way the elements of the composition are arranged and interconnected. And by 'elements' is meant such features of the work as colors, lines, shapes, and the ways they are arranged and fused together in what is in fact a complex, organic' unity.

Type II art is concerned with representation. Pietersen elaborates on the examples in the table, discussing Realism in terms of its emphasis on 'telling or showing how it is' in nature and in the everyday lives of people; and Naturalism, with its preference for ‘depicting human conduct in its unadorned (even sordid) state of biological and or social determinism’.

Type III art is concerned with individual expression, and alludes to individualistic, Romantic approaches to art-making, often a protest in art against the establishment and rationality; e.g. Surrealism was a reaction to and exploration of dark, even absurd, images conjured up by the Freudian psychology of the unconscious. [Pietersen, 2000]

Type IV art is concerned with functional outcomes beyond the art work, for example political statement, the involvement of art in society, the reformist and promotional mode of subjectivism-idealism.

Art as apparition and performance

Ascott [1993, 5] argues that the action painting of Jackson Pollock should be seen as the turning point in contemporary western art history, away from a representative paradigm of static appearances, towards this new art he terms “apparition”. We can understand the use of this word in the sense of the act of appearing, the bringing into the physical world of images through a performance process. There is a connotation of magic or conjuring in the momentary aesthetic expressions unique to the artist. It is understandable that debate has arisen over the question: Expression of what?

Pollock’s paintings are abstract – they are not meant to be a representation of nature. In the act of rendering the paint to the canvas, the artistic process of action painting could be described as:

a) a direct expression of prevailing emotion,

b) a concerted attempt to get a flow of applied energy,

c) an aesthetic improvisation in the manner of jazz,

d) a new way of looking at and handling the picture plane

Beyond this, what was innovative was a simple proposition posed by this new approach to image-making: why shouldn't hurling paint at a canvas be considered for its potential to produce aesthetically pleasing results? Why shouldn't we just enjoy the accidental effects, the riot of colour and artistic energy? There is the implication, in the view of action painting proposed by Ascott, that the majority of viewers, though perhaps possessing minimal synaesthetic abilities[1], can nevertheless through exposure to action painting, come closer to an understanding of a sense of music in these visual rhythms. And why shouldn't such a use of paint produce a pattern just as interesting as representations of nature, by virtue of the fact that we can enjoy seeing something that never before existed? We discover how the play of layered colour excites or stimulates our visual sense in unexpected ways.

There may be a human propensity to try to see order in chaos, and perhaps we do employ some sort of survival-oriented data selection filters in the way we "see" the world. Many individual factors no doubt influence what we see in the patterns of paint. Because of our need for spatial orientation, we see depth in arbitrary cascades of paint, even though there is no classical perspective. In this way we tend to naturally perceive the field of colour as an environment or space, as though it *was,* after all, representational. We react intuitively, emotionally and individually to these images of energy frozen in time, in the sense that they could be large and complex Rorschach tests, open to personal interpretation and response.

Pollock's work was very much about an exploration of the outcomes of a range of non-traditional techniques developed in an attempt to realize this new aesthetic. By stressing the primacy of focused action, rather than pursuing a predetermined concept, or indeed being bound by an obligation to represent something beyond simply the effect of paint on a canvas, he opened up a new way of viewing art.

It is a paradox that Pollock's art is not about representing nature, yet this technique in theory is an embodiment of the flow of energy in nature. There is a parallel worth making here with Japanese calligraphy, which in its highest form is considered to be an embodiment of the spiritual state of the artist, a making visual of the "ki", the universal energy or life-force flowing through the artist (Stevens, 1995, 37). The strict Buddhist discipline aimed at the development of highly meditative, relaxed, even ultimately spiritually enlightened persona, is given expression in the flowing strokes of Japanese brush and ink calligraphy. The more spiritual attuned the practitioner, the more relaxed and free-flowing the personal energy - and thus the more sublime

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[1] see page 107 for an in-depth discussion of synaesthesia in relation to multimedia

and poetic the brush-strokes. Watts, [1957, 108] describes this calligraphic painting as:

> Closest to the feeling of Zen …usually a painting and a poem in one….it is a perfect instrument for the expression of unhesitating spontaneity, and a single stroke is enough to "give away" one's character to an experienced observer.

Pollock attempted to harness the intensity of extreme emotion (often it seems an anger fueled by alcohol) into this flow of physicalized human energy, as expressed in his art. There is a connection here with the concept of reading personality from handwriting analysis -— the state of mind influences the flow of energy and is self-realized, expressed in the signature. This is the organically human area Jackson Pollock was exploring as he attempted to harness intense emotional states and allow himself to "perform" them with paint on a canvas. It is a relationship to the art process that was new to the visual arts, though perhaps quite familiar to performing artists of other disciplines, especially jazz music. For example,

> During the 1940s Charlie Parker was pioneering a new musical vocabulary based on spontaneous improvisation -- one that went far beyond the method established by Louis Armstrong. Parker's radical approach to improvisation, the charts be damned, placed non-linear associative thinking above all else in jazz, and led to the free jazz of John Coltrane, Ornette Coleman, and others in the 1960s and 70s. [Jordan, 2002].

In Ascott's view [1993], Pollock's dynamic visual improvisations and experiments with the flow of paint in time showed the way forward from the stultifying tyranny of the gallery 'picture window', and the closely-guarded secrecy and artificial mystique of the art-process.

'Jackson Pollock, Blue Poles: Number II, 1952,
enamel and aluminum paint with glass on canvas, 82 7/8" x 15' 11 5/8"

By viewing the canvas as a performance arena, Pollock was the precursor of a fundamental shift in contemporary concepts of art, a taking up of a new attitude, which encourages interaction with the processes of art.

These ideas could be summarized by saying that the central concerns underpinning much of late twentieth century art, from the period of Abstract Expressionism on, are transformation, change, flux and flow. This embodied a new attitude, which values art as action and attitude, over product.

It is against this conceptual background that developed societies have moved into the era of digital art, of art in cyberspace, a time in which art activities are tempered by new processes of expressing the artistic impulse through technology. This has fueled a complementary philosophical shift towards an attitude, which interprets networks as art performance arenas.

Ascott [1993] makes this connection between late twentieth century ideas about art, and the possibilities offered by cyber art.

> Art in the cybersphere is emerging out of a fusion of communications and computers, virtual space and real space, nature and artificial life, which constitutes a new universe of space and time. This network environment is extending our sensorium and providing new metaphysical dimensions to human consciousness and culture. Along the way, new modalities of knowledge and the means of their distribution are being tested and extended. Cyberspace cannot remain innocent, it is a matrix of human values, it carries a psychic charge. In cyberculture, to construct art is to construct reality....

Digital Expressionism, technological mediation and natural media simulation

A distinction can be drawn between:

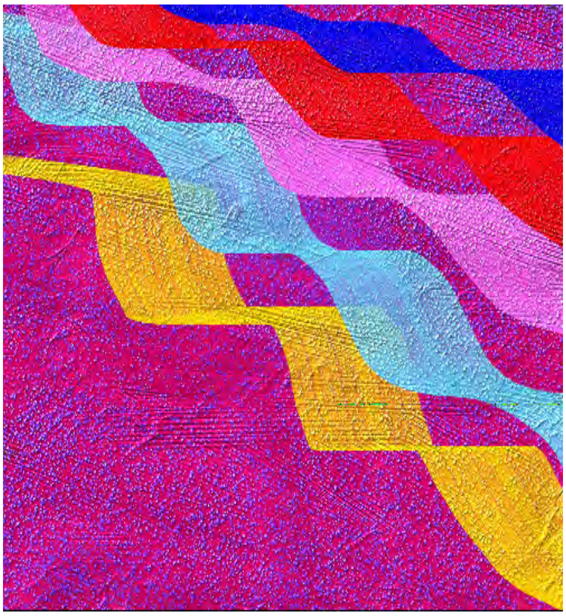
- approaches to using computers creatively which focus on reproducing traditional pre-computer techniques, and
- art activities or techniques that can only be achieved using computers.

Speaking of digital approaches to musical composition, Holtzman [1996, 240-241] writes:

> …almost all research to date on computer sound synthesis has used models drawn from existing disciplines…however comfortable or well accepted, these models are not new. They borrow from existing paradigms. They were conceived without computers.
>
> From a creative perspective, what is interesting is not how well computers can emulate traditional models for performing their tasks and solving problems, but, rather, the new territory that computers will reveal. What are the new possibilities opened by computers?

> What idea and means of expression will we discover that are only conceivable with computers? What new models will we develop for viewing the world in light of computers? *What means of expression are idiomatic to computers?…*
>
> Digital expression represents a radical new paradigm. It is a break with established tradition. Just as new computer tools in the creative process require a different way of thinking about expression – that is, in terms of explicit rules and abstract structure – the use of computers themselves will stimulate a shift in the very nature of expressive languages. Digital expression exploits the full expressive capability, the nuances and the subtleties, of a digital instrument. It is expression that could not have been conceived of without digital technology. Digital expression is of our time.

The concept of expression through technology is a challenge to the software and hardware engineers of the future. Certainly, we have the pressure sensitive art pad, which allows much the same free flow of energy that a pencil offers. Brush techniques can be digitally simulated. An embodiment of this idea is the so-called 'natural media' software application *Painter* [1]. *Painter* has a highly sophisticated set of tools and techniques, which have been developed in response to the desire to produce simulations of traditional painterly effects in a digital domain. Attempts to overcome the inherent limitation to expression in digital art, when contrasted with the organic techniques of natural media, have interestingly produced new techniques and concepts, such as the "image hose", which allows the digital artist to "spray" on a pre-determined, but randomly occurring set of sub-images.



Left: this image (produced by the author © 2003) using *Painter 7,* demonstrates a digitally simulated spatula (rendered with mouse strokes) being used to apply digitally-simulated oil paint, over a digitally-simulated grainy base-texture.

[1] now in its 7th version, a technology which was developed in the U.S. by *Fractal Design Corp*., later purchased by innovative software design firm *MetaCreations*, and then sold on to a company called *Procreate*.

The image to the **right** was created by the author in under three minutes, using the "image hose" tool in *Painter 5.0*. The pre-prepared “nozzles” (the terminology used within the software to designate a series of images that can be applied by clicking and dragging with the "image hose" tool) were, from top down: trees, houses, ivy, rocks, grass, poppies, ivy again, sand with a few “spurts” of rocks, and some sparingly applied poppies at the bottom.

Another innovative graphics application which combines traditional painting concepts with new technological approaches is called *Zbrush* (by Pixologic), a further example of how computers can offer art techniques only possible by digital means. Intuitive, and unique, *ZBrush* combines features of 2D and 3D software in one blended environment. The user can manipulate the mouse or art-pad to paint, sculpt and texture the screen-canvas effortlessly. *ZBrush* offers tremendous flexibility, allowing the user to change lighting, texture, materials and Z-Depth (3D dimensionality) simultaneously. All these characteristics are rendered in real-time while the user is at work. *Zscript*, is a proprietary technology which enables the user to record the steps in the creative process of generating an image, which can be recalled later or shared with others, and makes complex programmed features possible.

Above: This image shows five lateral mouse strokes (left to right) demonstrating the effect of five of the countless possible 3D “brush” techniques available in *ZBrush*. © 2003 Ferrier.

Interesting and creatively exciting these interface design and computer technologies may be, but we have a long way to go before reaching the level of plastic freedom implied by some digital equivalent of throwing and dribbling paint.

1.4 Multimedia, art and society

As a society we are only just beginning to develop a comfortable relationship with hi-tech art.

> Because of the accelerated pace of technological innovation, even newer technologies are rapidly passing into the stage of institutionalization. Fields such as computer graphics, computer animation, 3-D modeling, digital video, interactive multimedia and Web art, which were revolutionary a few years ago, have become part of the mainstream. [Wilson, 2002, 10].

Nevertheless, forces still exist in society that offer a conservative resistance to, or suspicion of the very idea of techno-based art. The fears expressed by the so-called Neo-Luddites are certainly not without basis. The predictions that digitization would usher in a Brave New World, made by Reinecke in *Micro Invaders* [1982, 15] have proven to have some validity:

> …the microchip is a threat to autonomy both at work and in our social activities. Whatever its claimed virtues, the explosion in the use of the microchip will mean that most people will have greater control exercised over their lives. Government and other agencies will know more about us than they ever have done. The frontiers of privacy will be pushed back further than they ever have been and for some of us the skills we struggled long to learn will be rendered useless. The scale on which the microelectronic invasion is being conducted is enormous.

Reinecke was voicing the doubts and unease that many feel about a technocracy out of control, and a life that is divorced from nature. This mistrust extends to a suspicion about the value of digital art. Sherman [1991, 164] observes that:

> The arts community in any nation with an advanced technological infrastructure seems to find comfort in the fact that the arts are a physical and spiritual link with the past. In the visual arts, painting and sculpture continue to satisfy most artists, patrons, curators, critics, dealers, publishers, bureaucrats and psychologists....

It must be said that this state of affairs can also be partly explained by the commercial imperatives of contemporary society in which art is a commodity needing a market. Marketing channels are well-established for traditional natural media art, profits are made and the survival needs of a relatively small number of professional artists are met. This marketing system is self-perpetuating, because a well-known name is easier to sell. Natural media works need no elaborate technological set-ups and expensive infrastructure to be displayed.

Meanwhile art has become a serious investment as well as an overt measure of affluence for social-economic elites. In a world of volatile share markets, rare art icons become secure conservative investments. But funds tend to be invested conservatively in recognized art commodities, offering the security of an accepted market value.

Marketing remains a fundamental challenge to digital artists. Nevertheless, there is the challenge and excitement of the new, and Sherman contends that:

> The choice an artist often must make is whether to work in well-developed or completely undeveloped territory. If one wants to make a painting the problems have been addressed over and over by countless numbers of very capable individuals. On the other hand, if one wants to work in video, computers or telecommunications networks, relatively few have looked at the territory from an art-making perspective.

Multimedia: art or craft?

Multimedia artists should be reassured by *Compton's Multimedia Encyclopaedia* [1993], which cuts through the complexity and esoterics of the art debate, and bluntly states:

> ... (Art) has been defined as a skill in making or doing, based on true and adequate reasoning ...fashioned by people, artificial, in contrast to everything that is natural — plants, animals, minerals. The Latin word ars (plural, artes) was applied to any skill or knowledge that was needed to produce something. From it the English word art is derived, as is artificial, which means something produced by a human being.

Ultimately, on a basis of simple definitions, the weight of press commentary and some serious academic dialogue, we may accept that there are artifacts currently being produced by people, using computers, which could be reasonably, legitimately labeled works of art.

Having said this, any consideration of the *aesthetics* of this new multimedia art is made complex by the blurred motivations occurring at the interface between some kind of notion of 'pure' or 'serious' art, and commercial art.

But of course this, too, is an age-old art debate. On this topic *Compton's Multimedia Encyclopaedia* [1993] invokes the art vs. craft controversy, tracing it back at least to the classical era, when the "liberal" arts[1], which required superior mental ability, were

[1] From the Latin liberalis, meaning "suitable for a freeman."

considered worthy of noble citizens, and craft was relegated to the “servile” arts, or products of the labours of the lower classes:

There is a wealth of discussion among art theory literature with regard to the status of crafted utilitarian items such as pottery within the pantheon of art. The art vs. craft debate is made somewhat more confusing when one considers the phenomenon of commercial art of the advertising world.

The revered status of the cultural icons exhibited in state-owned galleries and art museums is a reflection of the human desire/need to view human life as significant, important, and great art as one of the peak human achievements. The art thus preserved as a testament to this sense of self-importance becomes intimately linked with a framework of commercial forces, an all-pervading context in which the value of art is quantified by a market which seeks recognized art as a hedge against inflation and 'boom and bust' economic cycles. Art is an investment. It is improbable that more works attributable to Van Gogh will be discovered, and thus market forces push auction prices for the eccentric Dutch master’s works upwards into the realms of the incomprehensible, e.g. *Self-portrait without Beard* [1889] painted during his stay in the mental asylum at Sain-Remy, France, in order to reassure his mother he was alive and well on her 70th birthday, sold to an anonymous telephone bidder for $US71,502,500 [1].

In the world of the visual arts, digital art still largely tends to find its market as commercial art, and can, in the media/entertainment field, be considered a modern equivalent of the above-mentioned “servile arts” (crafts). Digital skills tend not to be discussed as craft and the craft/art continuum is an imprecise paradigm, but we see much commercial graphic art in print advertising, where fantasy and humour are pushed to highly sophisticated post-modern levels. 3D animation, while mainly a contributing craft of film making, is coming into its own with fully animated, team produced films such as *Toy Story* [1995], *Antz* [1998], *Shreck* [2001], *Final Fantasy* [2001] leading the way in recent years [2]. Once again the advertising world has adopted 3D animation as a powerful communicator and entertainer, the central tool of many contemporary television commercials. Digital music techniques have proliferated in popular music and the techno music genre is virtually *de facto* digital music, and while, by the nature of the culture which it informs it, it can be seen as

[1] source: www.vangoghgallery.com ; accessed 25-06-02.
[2] see page 242 for a discussion of 3-D animation

relatively disposable, at the same time it is regarded by practitioners and aficionados as “art” [1].

If the proposition that art is a mirror to society does indeed have some merit, and art history could therefore be viewed as a vertical stratification of cultural sediments that reflects human social evolution, then the status of digital art *will* inevitably change. Just as there is evidence to suggest that the dominance of the print paradigm is being eroded by the revolutionary proliferation of the Internet [Smart and Whiting, 1994; Rubinstein, 2002], so it seems likely that digital art will gradually gain in status, as society becomes more dependent on digital technology.

Post-modernism and multimedia

Post-modernism, as a movement in art and literature, developed as a reaction against modernism. It is a difficult but significant field of modern thought, and because it has been applied to a wide range of late-twentieth century culture, it is important to consider it in relation to multimedia. Klages, [1997] notes that post-modernism

> … is a complicated term, or set of ideas, one that has only emerged as an area of academic study since the mid-1980s. Post-modernism is hard to define, because it is a concept that appears in a wide variety of disciplines or areas of study, including art, architecture, music, film, literature, sociology, communications, fashion, and technology. It's hard to locate it temporally or historically, because it's not clear exactly when postmodernism begins.

The very term post-modernism is an awkward one, in that it is difficult to establish a completely satisfactory definition of the term *modernism* itself, and, as Barth [1980, 69] puts it, referencing an artistic era to its immediate past is

> …suggestive less of a vigorous or even interesting new direction…than of something anticlimactic, feebly following a very hard act to follow.

Modernism was a reaction to nineteenth century bourgeois realism, and was an art/literature movement characterized by a difficulty of access, due to the radical disruption of the linear flow of narrative or anti-linearity, the intentional frustration of conventional expectations concerning unity and coherence, the deployment of ironic and ambiguous juxtapositions, the celebration of private, subjective experience over public experience, and a general inclination towards the metaphoric [Barth, 1980, 71-72]. Post-modernism rejects modernism but is at the same time highly referential towards it. As Klages [1997] puts it:

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[1] See page 301 for a discussion of techno-music.

> Post-modernism, like modernism, …(rejects) boundaries between high and low forms of art, rejecting rigid genre distinctions, emphasizing pastiche, parody, bricolage, irony, and playfulness. Post-modern art (and thought) favors reflexivity and self-consciousness, fragmentation and discontinuity (especially in narrative structures), ambiguity, simultaneity, and an emphasis on the destructured, decentered, dehumanized subject.
>
> But--while post-modernism seems very much like modernism in these ways, it differs from modernism in its attitude toward a lot of these trends. Modernism, for example, tends to present a fragmented view of human subjectivity and history (think of *The Wasteland*, for instance, or of Woolf's *To the Lighthouse*), but presents that fragmentation as something tragic, something to be lamented and mourned as a loss. Many modernist works try to uphold the idea that works of art can provide the unity, coherence, and meaning which has been lost in most of modern life; art will do what other human institutions fail to do. Post-modernism, in contrast, doesn't lament the idea of fragmentation, provisionality, or incoherence, but rather celebrates that. The world is meaningless? Let's not pretend that art can make meaning then, let's just play with nonsense.

Post-modern techniques include a frequent use of historical material as quotation in a playful or critical way. The use of such quotes relies on a recognition of the relevance or meaning implied by the use of it as a quote. Barth [ibid., 73] illuminates the complexity of post-modern thought by setting up a series of dichotomies:

> If the modernists, carrying the torch of romanticism, taught us that linearity, rationality, consciousness, cause and effect, naïve illusionism, transparent language, innocent anecdote, and middle-class moral conventions are not the whole story, then from the perspective of the closing decades of our century we may appreciate that the contraries of these things are not the whole story either. Disjunction, simultaneity, irrationalism, anti-illusionism, self-reflexiveness, medium-as-message, political olympianism, and a moral pluralism approaching moral entropy – these are not the whole story either…

Contemporary multimedia technology offers the potential for an artform characterized by a blending of a multi-layered diversity of sensory and cognitive stimuli and a density of text. Because of their non-linearity, multimedia works can be labyrinthine and even fragmented, and the interactivity opens the possibility for the user to relate to and understand the work in his/her own unique way. If the ‘art is a mirror to society’ construct has any truth to it, then, in a post-modern society, the evolution of this new eclectic artform could be viewed as a projection of the social consciousness of the times that produced it, rather than just a matter of historical coincidence.

This point of view is supported by Soeborg [2000] in an essay discussing ideas presented in Lev Manovichass’ book *The Language of New Media*. Soeberg points out that, traditionally, the concept of an aesthetic is related to the concept of an *aesthetic object as an object*, i.e. as a “self-contained structure limited in space and/or time”. This is an assumption he claims is fundamental to all modern thinking about aesthetics. Soeberg refers to attempts by some artists from the 1960s onward to substitute a traditionally defined aesthetic *object* with other concepts such as *"process", "practice"* and *"concept".*

And indeed, does art necessarily involve a finite object? Considering that the work of conceptual artists such as Tehching Hsieh (see p.36) are considered by many to be valid art, why cannot telecommunication between computer users itself be the subject of an aesthetic? Soeberg goes further and asks whether a user’s search for information can be understood aesthetically? He argues that, if a user accessing information and a user telecommunicating with other(s) are as common in computer culture as a user interacting with a representation, can we expand our aesthetic theories to include these and other coming, new situations? The conclusion Soeberg draws from these speculations is that post-modernism is an evolutionary state that society has reached, simultaneously with the emergence of digital technologies, but again more importantly, this is no coincidence.

> The development of Graphic User Interface and software like *PhotoShop* … (which) legitimized cut-and-paste logic and media manipulation, took place at the same time as contemporary culture became "post-modern". Fredric Jameson characterizes post-modernism as a periodizing concept whose function is to correlate the emergence of new formal features in culture with the emergence of a new type of social life and new economic order.
>
> … by the early 1980s that culture no longer tried to "make it new". Rather, endless recycling and quoting of the past media content, artistic styles and forms became the new "international style" and the new cultural logic of modern society…rather than assembling more media recordings of reality, culture is now busy re-working, recombining and analyzing the already accumulated media material".

Following this logic, multimedia can be viewed as a “physicalization” of post-modernist consciousness.

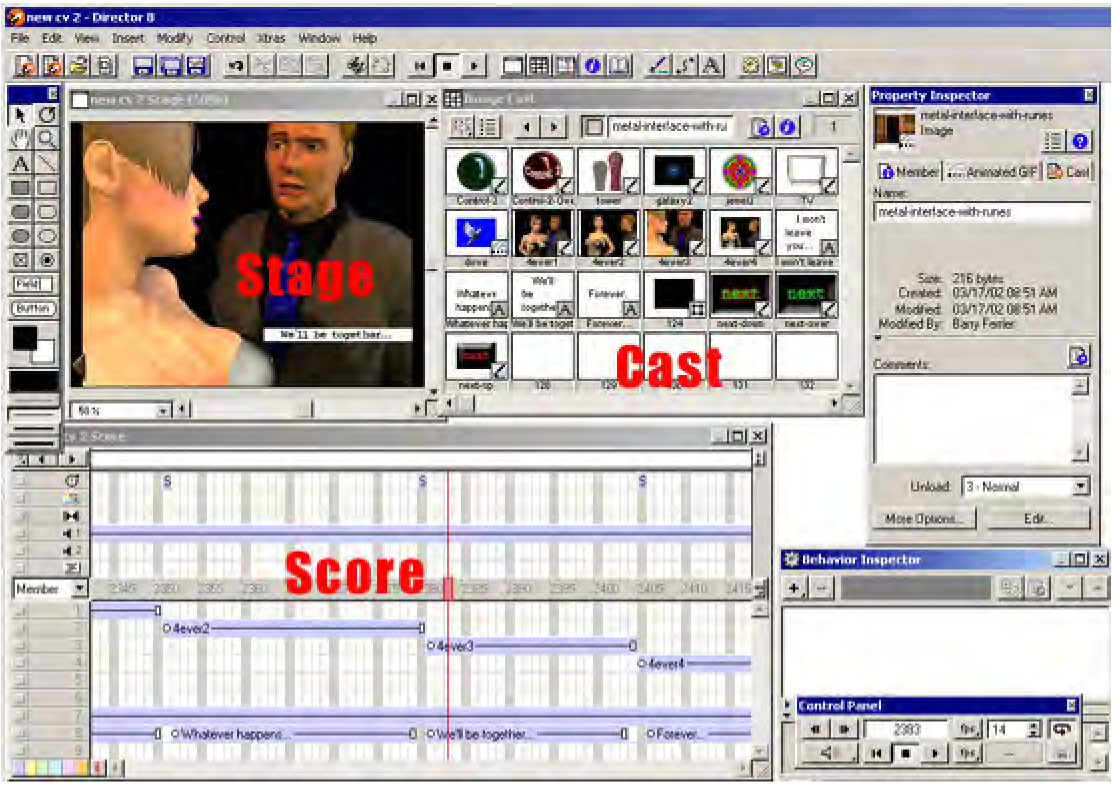
Technology as art-space, computer as theatre

With regard to the relative status of multimedia in the pantheon of art: while we may find naïve definitions of art in popular encyclopaedias, Dadaist ready-mades forced us to consider that it really is an open question as to “what is art?” [Hopwood, 1990, 118; Barry et al., 1964, 220; Pierre, 1967, 114]. For the purposes of this discussion this writer offers as a minimal working definition the suggestion that art is:

> *The process or outcome of placing an object or activity in an aesthetically considered “space”.*

The technological space offered by CD-ROM equipped computer workstations, the Internet, and associated or future electronic/digital media delivery technology, can become an art space. A multimedia artist is an auteur of this cyber-artspace.

This space is also a theatrical space, and our analysis can profit from viewing multimedia art activities within the theoretical multimedia artspace from a point of view of the history of performance art – with the multimedia artist as "director". This perspective is acknowledged by the interface design of the leading CD-ROM design software product entitled *Macromedia Director,* which uses a stage, cast and score metaphor in its interface.



Media elements are the "cast", there is a stage upon which the user "directs" the cast member/media element to "perform", and a "score" which the director uses to orchestrate changes over time and weave a soundtrack.

At an environmentally perilous point in human history, there is an elegance in an art that can be created and delivered in a virtual space (the computer workstation and the internet) which can be embraced, as part of the urgent shift away from toxic and resource consuming industrial processes such as the chemical manufacture of paints, wood-pulp paper products etc. Unfortunately we have developed a new reliance on the electronics and plastics industries which now contribute in dangerous new ways to the global ecological crisis.

Technological advances have now created a means of portraying and experiencing a real-but-illusory, multi-dimensional, multi-layered, non-linear, virtual space, and we are witnessing the first, no doubt clumsy attempts at colonizing it with our imaginations, using this technology as a mediator of human creativity, in the on-going process of responding to the human condition through art.

1.5 The psychology of multimedia art appreciation

Apart from considerations of how we relate perceptually to the content of this new art space, there is also a field of thought emerging which attempts to understand conceptually how this space works, its existential and ethical ramifications and how we relate to it cognitively as humans.

In his mystical book "*Music and the Power of Sound*", Alain Danielou [1995, 1-2] remarks that:

> The link between physical reality and metaphysical principles can be felt in music as nowhere else.... The strange phenomenon by which coordinated sounds have the power to evoke feelings or images is accepted as a fact..... the real problem is not to know how human beings may have acquired the knowledge of musical intervals, which always brings us back to a question of myth, ancient or modern, but to find out the real nature of the phenomenon by which some sounds can be combined to represent ideas, images or feelings..

Whether or not one agrees with Danielou's exotic ideas about the symbolism and numerological significance of musical intervals and scales, the field of study suggested here is essentially the psychology or phenomenology of music.

If the premise is accepted that multimedia should be viewed as a discrete new artform, then in order to understand the form, function and aesthetics of this phenomenon, it requires an approach focusing on the combined effect of its subsidiary arts. The present discussion, then, is especially concerned with a field that could be called the 'psychology of multimedia art', in the sense that there is a conceptual equivalent to the psychology of music, that takes in the aesthetic/semiotic experience of exposure to this new multi-dimensional art. Such art addresses the multiple human sensory channels, offering clusters of perceptual cues that remind us of the real world, as part of the process of engaging our minds and conveying meaning and novel experience.

This analogy with the psychology of music may seem a simple idea, however one can immediately recognize that this would be an extremely broad and complex field of study (which in fact must include, as sub-categories, consideration of psychoacoustics and the psychology of music within its ambit, especially with regard to how they influence the dynamics of 'immersion'). It would presumably find relevant a whole range of data derived from physiological studies of perception. On a technical level, devices, techniques and technologies that enhance immersion are of interest in such a study. Semiotics, or multimedia as text is another pertinent field of study. Beyond that, research into the psychology of multimedia art must delve into the relation of multimedia art to the society which produced it, including cross-cultural comparisons.

Ultimately, the 'psychology of multimedia art' is a subset of the broader field of the psychology of art. This is a field that is fascinating, historically diverse (with contributions from philosophy, general psychology, psychoanalysis, sociology or social psychology, art criticism, aesthetics and anthropology), and ultimately fraught with difficulties with regard to research methodology. Indeed students of 'the psychology of art' are far from being able to claim it to be a fully-fledged science or branch of science at the present time, though nevertheless there remains an enduring consensus that it is an important task to describe and explain the phenomena of human behaviour and experience, in relation to works of art. To contemplate the idea of a psychology of multimedia art, researchers must address multi-sensory input modalities, decision making or interactivity, and the potential for multi-layered conceptual content – our mind and senses are engaged with a virtual world coming ever closer to the complexity of the real world.

There is a range of significant philosophical and psychological concepts arising from the literature regarding the psychology of art, which are interesting to consider in the context of developing an overview of multimedia as a discrete artform.

Behaviourist or objective scientific studies reliant upon measurement seem unsatisfying in relation to a study of art, because such a large proportion of human thought is subjective and concealed from observation. Hogg begins his introduction to *Psychology and the visual arts* by writing:

> There is a very real ambivalence in our attitude towards the psychological study of art. On the one hand the experience of art is felt to be so personal and private, and so intimately related to our individual personality that no mere psychological theory, let alone one dependent on 'objective' techniques could possibly probe and elucidate its significance. On the other hand the very fundamental nature of, and importance we ascribe to such experience suggests that for any psychology to account successfully for the relations we form with the world, it must also say something about the function art serves. [Hogg, 1969, 9].

Neitzsche postulated a duality or polarity he designated as *Apollonian* and *Dionysian* in Greek art, and applied this theoretical dichotomy to the wider history of art[1]. His ideas become of contemporary relevance and new interest when applied to technology-based art. Apollo personified the calm and clear perfection of the dream-world, and is associated with the classical approach to art, signified by restraint, harmony, symmetry and order. This is contrasted with the romantic notion, personified by Dionysus, of the blissful, self-forgetful ecstacy of 'drunkenness', which Nietzsche used as a metaphor for inspiration and transcendence through art, as well as a merging of individuality in a sense of mystic oneness. [Munro, in Hogg, 1969, 42]. The modular and very rationally planned nature of the CD-ROM experience, of digital music-making, and technologically-mediated art processes in general, suggests a naturally Apollonian orientation in multimedia art. However, in approaching techno-based art, there is value in a healthy suspicion of the influence of the machine and technological systems over the creative processes of the artist, and of the regimenting nature of the man-machine relationship. With these thoughts in mind, the introduction of Dionysian elements can be seen as desirable, for what balance it may bring to this relationship, and for whatever organic element, or sense of natural flow it may introduce into advanced technological art processes.

Immersion

Using currently available digital mediators, the artist endeavours to address the sense perception mechanisms of the participant, enveloping them in the resulting virtual reality experience. There is no scientific benchmark for the creation of a successful multimedia experience. The commonly expressed gauge of whether multimedia works is the subjective term *immersion*. The Oxford Dictionary [2] defines immersion as:

1. Dipping or plunging into water or other liquid, and transf. into other things.
2. *transf.* and *fig*. Absorption in some condition, action, interest, etc.

Cognition involves an interplay between externally derived sensory phenomena and internally generated models and expectations [Smith, et al., 1998]. The success or completeness of the subjective state of immersion is proportional to the perceived fidelity and cohesion of the virtual experience, with reference to our inherent and conditioned sense of so-called reality, built from our memories of past experience, i.e. we view subjective reality as a standard of full immersion. The dreaming state and day-dreams are reference points to immersion in "non-real" experience. The

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[1] http://www.geocities.com/SoHo/Studios/3684/nietzsche.htm ; accessed 20-01-03.
[2] The Oxford English Dictionary, Second Edition, Vol. VII, Clarendon Press, Oxford, 1991.

long tradition of theatrical performance and the more recent success of cinema arts are a testament to our ability to allow a suspension of disbelief.

Immersion in a virtual reality environment is dependent on many factors, including the users physical senses and mental processes, the technology involved, the types of interaction involved and the tasks of the current application such as game-play. The subject’s interest in the presented material, local background noise and the subject’s current state of mind are all influencing elements [Smith, et al., 1998]. The synchronization and cohesiveness established between media forms, and the presence of a cross-channel language of interpretation, must also be factors in successfully engaging the user in an immersive interactive experience.

The immersive experience is subjective and difficult to measure or quantify. Perhaps there is a range of individual differences with regard to the ability to become immersed in multimedia. This has yet to be scientifically quantified. Acceptance in popular culture, as implied by purchasing behaviour, and the anecdotal evidence and specialist reviews found in consumer niche market magazines and the press, can give a relative sense of the immersive value of particular multimedia products, just as with cinema. But as technology evolves and multimedia experience proliferates, standards will change. Last year’s hot computer game is likely to have a short shelf-life in this rapidly evolving field.

Reading a good book can immerse a reader, and this is an acquired skill. Equally the ability to engage in multimedia, and the degree of immersion achieved by the user may develop with exposure and experience.

It is implied or assumed that the virtual experience must offer something beyond what ‘reality’ already offers to be of interest or value as experience and as research or experimentation. On a sense level, the aesthetic arrangement of sensory cues on each of the sense modality channels is overlaid with the meaning emerging from the interaction of these media elements and the embedded cultural texts. Each modality must work in its own space and simultaneously compliment and contribute to the multidimensional semiotic texture. Audio elements such as spatial cues (directionality and spatial dimension) and the temporal musical cues (a framework for organizing collapsed time) work towards maximizing immersion in the experience.

Should we fear an ultimate future in which reality is replaced by virtuality? Perhaps as the world becomes increasingly urbanized, and the natural environment increasingly impoverished and degraded, this is a necessary evolutionary step - to create an aesthetically pleasing, virtual world to replace the one we are destroying, so that we can survive the physical ugliness. What unforeseen psychological dangers lurk in a man-made virtual reality, in which the four walls ceiling and floor of our living rooms become giant video screens washed with artificial ambient sound and perfumes? This science fiction scenario, vividly envisaged by Ray Bradbury in *The Illustrated Man* [1951], could be the logical conclusion of a quest for total immersion in virtual reality.

Multimedia, Semiotics and Consciousness

Semiotics attempts to approach media texts from the point of view that media experiences such as cinema, radio and television have more in common with (what we treat as) ‘reality’ in the everyday world of our own experience, rather than a symbolic system such as writing. Semiotics is an attempt to avoid the pitfalls of trying to understand the form, function and aesthetics of media by forcing them into a linguistic framework. For example, using the analogy of photography, Burgin [1982, 505] says:

> There is no 'language' of photography, no single signifying system (as opposed to technical apparatus) upon which all photographs depend (in the sense in which all texts in English depend upon the English language); there is, rather, a heterogeneous complex of codes upon which photography may draw.

A multimedia text, then, consists of a diverse range of media forms that combine to produce a meaning. It would seem logical that, in order to be distinguishable, the codification of meaning within any one medium must be proportionately simplified or made bold, as layers of sensory information and complexity are added to the multimedia mix.

Artificial experience

Sensory cues that enhance the immersive experience are generally referenced to sensory experience in the real world. However, as members of an increasingly technologically oriented society, we have obviously become quite familiar with a wide variety of artificial experiences. Myron Krueger [1990, 31-34], in pointing this out, sees cyber-art as a natural progression:

> Vicarious experience through theatre, novels, movies and television represents a significant fraction of our lives. The addition of a radically new form of physically involving, interactive experience is a major cultural event which may shape our consciousness as much as what has come before.

That western consciousness is embracing the artificial as a commonplace part of the natural world can be seen reflected in the output of the traditional arts. In popular music, technology has transformed the way music can be made - the output of a machine playing simulations of tribal drumming, used as a sub-structure for a love song, is no novelty. Since the onset of the sampling revolution, experiments with hybridizing pop music with ethnic music styles has fuelled a burgeoning of “world music” and techno-music, in all its shades and sub-genres. Popular music culture of the past decade is a postmodern distillation of musical experience, using musical “quotes” to recycle and simultaneously re-interpret sound experiences. (Shuker, 1994, p.286)

Increasingly we are relating to the natural environment with a consciousness tinged by this knowledge of artificial realities. For example, Veronica Hollinger [1990, 1] quotes Gibson’s celebrated sci-fi novel, aptly published in the year 1984, entitled *The Neuromancer* (in which the term ‘cyberspace’ was first used) as an icon of postmodern cyberpunk art trends.

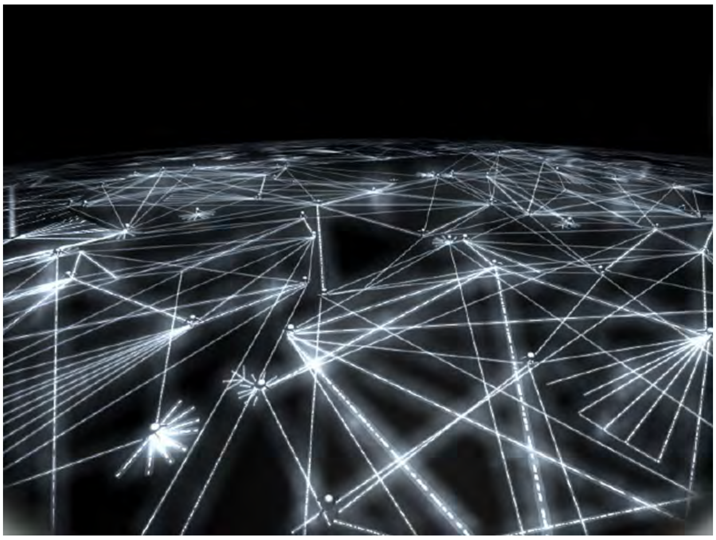
> Gibson’s first sentence - ‘the sky above the port was the colour of television, tuned to a dead channel’ - invokes a rhetoric of technology to express the natural world in a metaphor which blurs the distinctions between the organic and the artificial....

Cyberpunk capitalizes on our familiarity with artificial realities, invoking them as a backdrop to the new, hard-edged reality – a reality which is voguishly futuristic, yet tinged with an ambivalence that emanates from the associated angst of alienation from the bosom of Mother Nature.

1.6 Cyberspace

Coining the term *cyberspace*, American author William Gibson describes an eerie vision:

> Cyberspace. A consensual hallucination experienced daily by billions of legitimate operators, in every nation, by children being taught mathematical concepts...A graphical representation of data abstracted from the banks of every computer in the human system. Unthinkable complexity. Lines of light ranged in the nonspace of the mind, clusters and constellations of data. Like city lights, receding... [Gibson, 1984].



Above: *Warriors of the Net* is a short movie that shows how the Internet works from the viewpoint of individual data packets. It was created by Gunilla Elam, Tomas Stephansson, Niklas Hanberger and Monte Reid, at the Ericsson Medialab, in Sweden. The image above is the view of the global Internet from the movie, visualized as a complex mesh of glowing links floating in empty space.

Gibson's prescient concept of cyberspace is now a significant and celebrated day-to-day reality in the form of the Internet. The magnitude and complexity of the World Wide Web makes it difficult to construct an overview of the role it is playing in society. Sandbothe [1998] describes the internet as virtual reality:

> The Net opens up a new world to us. And it does this in a way differing from, say, a trip by car or aeroplane. When we fly from Berlin to San Francisco we also arrive in another world in which, partially, other laws dominate. But the basic coordinates of our understanding of reality - space, time, identity - remain unchanged. It is different when we leave 'real' life and proceed into the Net. The world becomes 'virtual'. The constitution of reality becomes a different one. 'Virtual reality' steps in taking the place of 'real life'. The terms 'real' and 'virtual' are reflexive terms similar to the opposition of natural and artificial. Things only ever appear 'real' or 'virtual' from a particular perspective. If one considers the observer's relativity then it's no surprise that the on-line world already seems more real to many professional Net-surfers than the 'real' world outside of the Net. I do not associate normative implications of any kind with the real-virtual opposition. I use this only to differentiate varying forms of construction of reality from one another on the descriptive level…The virtual world is independent of sunlight. There is no unitary, somehow natural time which partners in communication could presuppose as self-evident. Rather they must onerously inform one another about their respective local times and adjust for the differences if they want to meet on the Net. The horizons of time are in constant motion.

There is undisputed value in having such a direct, convenient source of information available to the household, school and office. However early excitement about the potential inherent in the internet for individual expression and liberation, and an anarchistic freedom of global communication beyond bureaucratic regulation (Post, 1996; Weil, 1998), has not been fulfilled and has given way to a different reality, shaped more by market-driven commercialization. The World Wide Web has in fact (to a large extent) taken on the appearance of a huge advertising directory.

Nevertheless it can still be imagined that the potential for delivery of significant new art may yet be fulfilled, with the global rollout of high-bandwidth optical-fibre connections. We can only wait and see whether the Net will eventually merge with high definition television broadcasting, to become the ubiquitous hybrid entertainment / information delivery medium.

Multimedia artists have the Internet as one possible artspace (and it's the most immediate channel for self-marketing and distribution). Multimedia artists who specialize in the web must find an audience, and to do this on-line they must understand the responses, needs and expectations of on-line of multimedia users, and understand web-culture.

Cyberspace is a shared 'mental landscape', a cultural terrain, mediated by technology. Information and ideas, which have an abstract existence in people's minds, are shared and evolved across an interactive computing network, which embodies a representation of this collective consciousness. The concept of a 'mindscape' is not unique to the digital age. It could be argued, for example, that intellectual disciplines such as mathematics and physics exist as an ongoing dialogue, based on a framework of shared mental constructs, with practitioners separated in space and time, building theorems upon an evolving framework of esoteric language. Pure mathematics exists on a level of platonic abstraction, but is also a sharing of consciousness between the community of mathematicians. The ideas that are shared are an evolving structure, which mediates an otherwise incoherent or unmanageably complex experience. Jones [1982, 51-52] makes a similar but illuminating argument with regard to distinguishing between the physical world and the conceptual metaphors of physics:

> Physics is a metaphor because it is a representation of a part of human experience and creates the meaning which that experience has for us. When we use a concept like space in physics, we are constructing an image, which despite its abstract character and quantitative description, evokes and summarizes what our being and experience feel like to us…
> What might otherwise be some kind of meaningless, inchoate experience of our own undifferentiated existence is transformed through the metaphor of space into an awareness of differentiation, diversity, separateness. Space organizes and gives meaning to our tangled, amalgamated experiences through its attributes of place and distance. Space has by now become so synonymous with all these experiences that we cannot see it as a representation of them, nor can we say clearly what is standing for what.

Abstract metaphors have, through experience, become an accepted, practical tool of orientation and communication. An analogous maturation process with regards to the concepts, jargon, and experience of cyberspace is proceeding within networked elements of society. Networked computer technology is a physical interface which can be viewed as playing the metaphorical role of the *TARDIS* [1] of television science fiction: a physical connection, which allows us to enter, interact with and travel through shared, digitally-encoded mindscapes.

What is especially fascinating is the very fact that the contents of the Internet, and the activities being played out there, are perceptually received or experienced, but these perceptions are of an apparition. This illusion is encoded into a physical reality made up of bytes and bits and electronic signals, which our senses are not equipped to perceive. So, in a sense, the coded apparition of cyberspace exists only as a form of stored mental energy, thoughts and imaginings, which we share - a shared mental landscape or 'mindscape'. Participants perceive its vague, shifting terrain, at first dimly, but with increasing confidence and facility as they gain cyber-experience. Benschop [2002] views cyberspace in sociological terms;

> Cyberspace is an illusion, it is a consensual hallucination that is not anywhere in our physical reality. It is a no-place that exists only within headspace. Cyberspace is something that cannot be demarcated in geographical terms at all. It is a reality that can be localized 'nowhere' and yet its presence is felt 'everywhere'. It is a new form of social reality that is a challenge for sociologists who don't recoil from analyzing such ostensible 'metaphysical' realities. One thing seems to be sure: more and more people define and experience cyberspace as real.

This metaphorical world is ultimately 'natural' in the sense that it is a social mechanism which has developed as a response to the human condition, and so it is valid to suspect it must have inherent 'natural laws' of its own which, inevitably, we attempt to discover. Researchers have already given much thought to a social science of cyberspace and internet communities[2] and a psychology of cyberspace. For example, Suler [1998] attempts to lay the foundations for such a psychology:

> The virtual world is quite different than the in-person world. Digitizing people, relationships, and groups has stretched the boundaries of how and when humans can interact… In different online environments we see different combinations of these features, thus resulting in a

[1] TARDIS is an acronym for "Time And Relative Dimensions In Space". It was the time-machine which looked like a telephone booth, featured in the TV series "*Doctor Who"* - the longest-running science-fiction series in the history of television. The series was created by the BBC in 1963.

[2] A search for the term *cybersociology* using the *Google* search engine produced 2860 (mainly academic) references in October, 2002. A search for the term *cyberecology* returned 46 references.

> distinct psychological quality to each environment which determines how people experience themselves and others. We may think of these features as the fundamental elements of a conceptual model for a psychology of cyberspace.

In attempting to analyze the internet experience Suler [1998] lists ten conditions, which differentiate psychological relations in cyberspace from real-world relationships. These can be paraphrased as:

1. **reduced sensations:** The sensory experience of encountering others in cyberspace - seeing, hearing, and COMBINING seeing and hearing - is still limited. For the most part people communicate through typed language.

2. **texting:** Drawing on different cognitive abilities than talking and listening, typing one's thoughts and reading those of another is a unique way to present one's identity, perceive the identity of one's online companion, and establish a relationship

3. **identity flexibility:** Communicating only with typed text, you have the option of being yourself, expressing only parts of your identity, assuming imaginative identities, or remaining completely anonymous. Anonymity has a dis-inhibiting effect that cuts two ways: it may be used to act out some unpleasant need or emotion, or it may allow the user to be honest and open about some personal issue that they could not discuss in a face-to-face encounter.

4. **altered perceptions:** Suler claims that:

    > While reading e-mail or text talk in chat rooms, some people experience a blending of their mind with that of the other person. In the imaginary multimedia worlds - where people shape-shift, speak via ESP, walk through walls, and spontaneously generate objects out of thin air - the experience becomes surrealistic. It mimics a state of consciousness that resembles dreams. These altered and dream-like states of consciousness in cyberspace may account for why it is so attractive for some people. It might help explain some forms of computer and cyberspace addiction.

5. **equalized status:** All users of the internet, - regardless of status, wealth, race, gender, have an equal opportunity to have their voice heard. Influence on other users is dependent on: skill in communicating (including writing skills), persistence, the quality of ideas, and sometimes technical know-how.

6. **transcended space:** The irrelevance of geography has important implications for people with unique interests or needs. In their outside life, they may not be able to find anyone near them who shares that unique interest or need. This can have both a positive and negative manifestation.

7. **temporal flexibility:** In both asynchronous (email) and synchronous (chat-room) communication there is a stretching of time. During chat. one has from several seconds to a minute or more to reply to the other person - a significantly longer delay than in face-to-face meetings. In e-mail or newsgroups, one has hours, days, or even weeks to respond. Cyberspace creates a unique temporal space where the ongoing, interactive time together stretches out. This provides a convenient "zone for reflection." Compared to face-to-face encounters, the user has significantly more time to mull things over and compose a reply.

8. **social multiplicity**: Users can contact people from all walks of life and communicate with hundreds of people. When multi-tasking, one can juggle many relationships in a short period of time - or even AT the same time (chat or instant messaging) without the other people necessarily being aware of one's juggling act.

9. **recordability:** Unlike real world interactions, the user in cyberspace can keep a permanent record of what was said, to whom, and when. One can re-experience and reevaluate any portion of the relationship.

10. **media disruption**: There will be moments when software and hardware don't work properly. There will be moments when telecommunication systems give us nothing, not even an error message. The frustration and anger experienced in reaction to these failures says something about our relationship to our machines and the internet - something about our dependency on them, our need to control them.

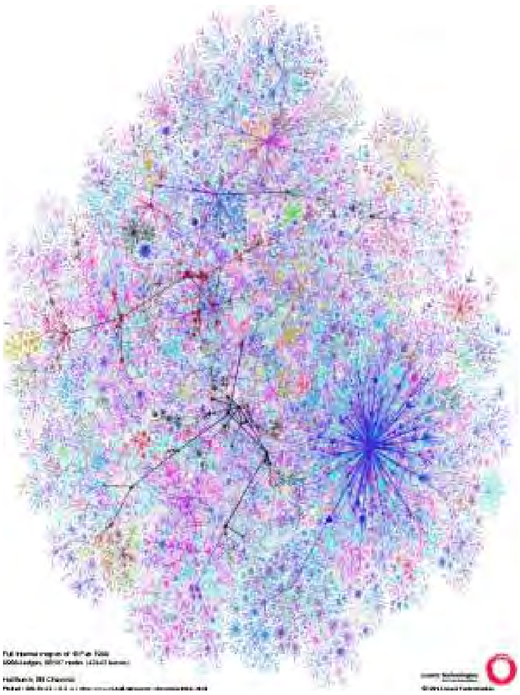
New kinds of personal relationships are available in the electronic village and they exist parallel to, but must to some degree supplant, traditional face-to-face relationships. Benschop, [1998] identifies three levels on which cyberspace has become integrated into human life in networked communities:

Levels of social integration	System Level	Action Level
Societal level	Comprehensive social systems	Societal action
Organizational level	Organizations	Organizational action
Interactional level	Direct social interaction systems: dyadic social relations and networks	Interactional or role action

> Modern communication technologies offer new opportunities for other kinds of social relations and communities. The development of electronic communication technologies hasn't destroyed time and space, to be sure (as McLuhan claimed in 1964), but time and space have been condensed in such a way that we actually live in an infinite 'global village'. Communication technology connects people in a new type of community, in virtual communities. These new technologies can unite people in coherent interest communities, but they may also atomize people, causing them to further withdraw in tribalism.

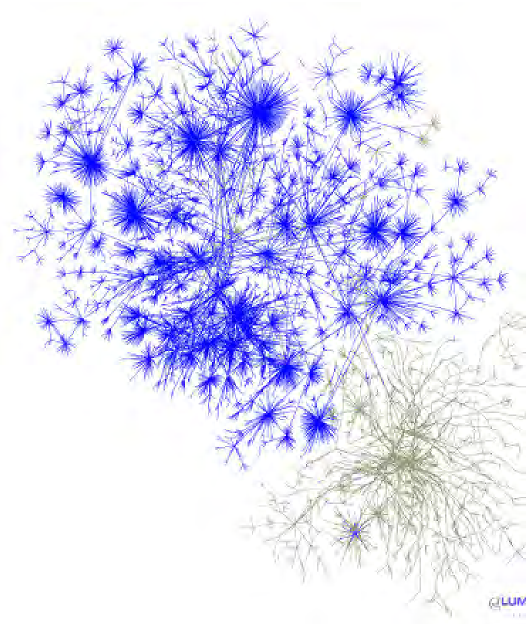
Examples of these macro-level internet structures influencing social and one-to-one/one-to-many relationships include the government level - attempts to regulate and censor the internet, and on organizational level - the paramount importance of email in day-to-day business and education.

Cyberspace represents a new frontier. It represents new challenges, new sorts of relationships, and new potential for creative experimentation. There is more going on here than is indicated by the much-vaunted term "information technology".



There is culture and commerce, there is art and entertainment, there is anarchy and there is absurdity. Jung's famous term "the collective unconscious" seems to have some relevance here, for no single person or institution is laying down the structure and shape which cyberspace is taking. It is an artificial construct, yet it seems to be growing organically. Topological maps of connectivity suggest the fractal structures of a Mandlebrot set.

Left: Visualisation of the organism-like growth and form of the internet: This graph shows "*the router level connectivity of the Internet as measured by Hal Burch and Bill Cheswick's Internet Mapping Project*". [Dodge, 2002].



Left: The unplanned, "organic" growth of the internet is suggested by the visualisations generated by *Lumeta*. *Lumeta* are developing network measurement and mapping tools for analysing large corporate intranets. This graph shows the portion of a corporate intranet that is 'leaking' with the Internet. [Dodge, 2002].

Like an amoeba or a virus in its expansion, the number of individual connections to the Internet is growing exponentially, with every indication that this growth will continue to accelerate. The following tables quantify this phenomenal, exponential expansion during the early nineties:

The raw internet growth data, 1993 -1996[1]; (after Gray, 1996):

Date	Domains	Replied to Ping
Jan 93	21000	NA
Apr 93	22000	0.4 million
Jul 93	26000	0.5 million
Oct 93	28000	NA
Jan 94	30000	0.6 million
Jul 94	46000	0.7 million
Oct 94	56000	1.0 million
Jan 95	71000	1.0 million
Jul 95	120000	1.1 million
Jan 96	240000	1.7 million

Growth in number of internet hosts (Coffman and Odlyzko, 1998):

Date	Hosts
August 1981	213
May 1982	235
August 1983	562
October1984	1,024
October 1985	1,961
November 1986	5,089
December 1987	28,174
October 1988	56,000
October 1989	159,000
October 1990	313,000
October 1991	617,000
October 1992	1,136,000
October 1993	2,056,000
October 1994	3,864,000
January 1996	9,472,000
January 1997	16,146,000
January 1998	29,670,000

Ollivier Dyens [1994, 327] takes this "organic" metaphor one step further. He begins by quoting the messianic words of the great philosopher of science, Jacob Bronowski:

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[1] *Domain*: Usually associated with some organization, e.g. university, company or other group.
Replied to Ping: Estimated number of machines actually connected to the net and responding.
Hosts: The number of machine addresses on the net, as reported by the nameservers.

> In every age there is a turning-point, a new way of seeing and asserting the coherence of the world.....

Dyens [*ibid*., 327] uses this as a basis to propose that:

> the new coherence of our age is that of cyber-ecology....the science of plastic interactions between artificial and natural organisms.....

Mapping Cyberspaces

Various attempts have been made to depict the internet in graphical form, the better to visualize its labyrinthine 'shape', magnitude and complexity. Martin Dodge's [2002] web-site entitled *An Atlas of Cyberspace* offers a comprehensive collection of such depictions, and it is a large and fascinating resource, a small representative selection from which is included below.

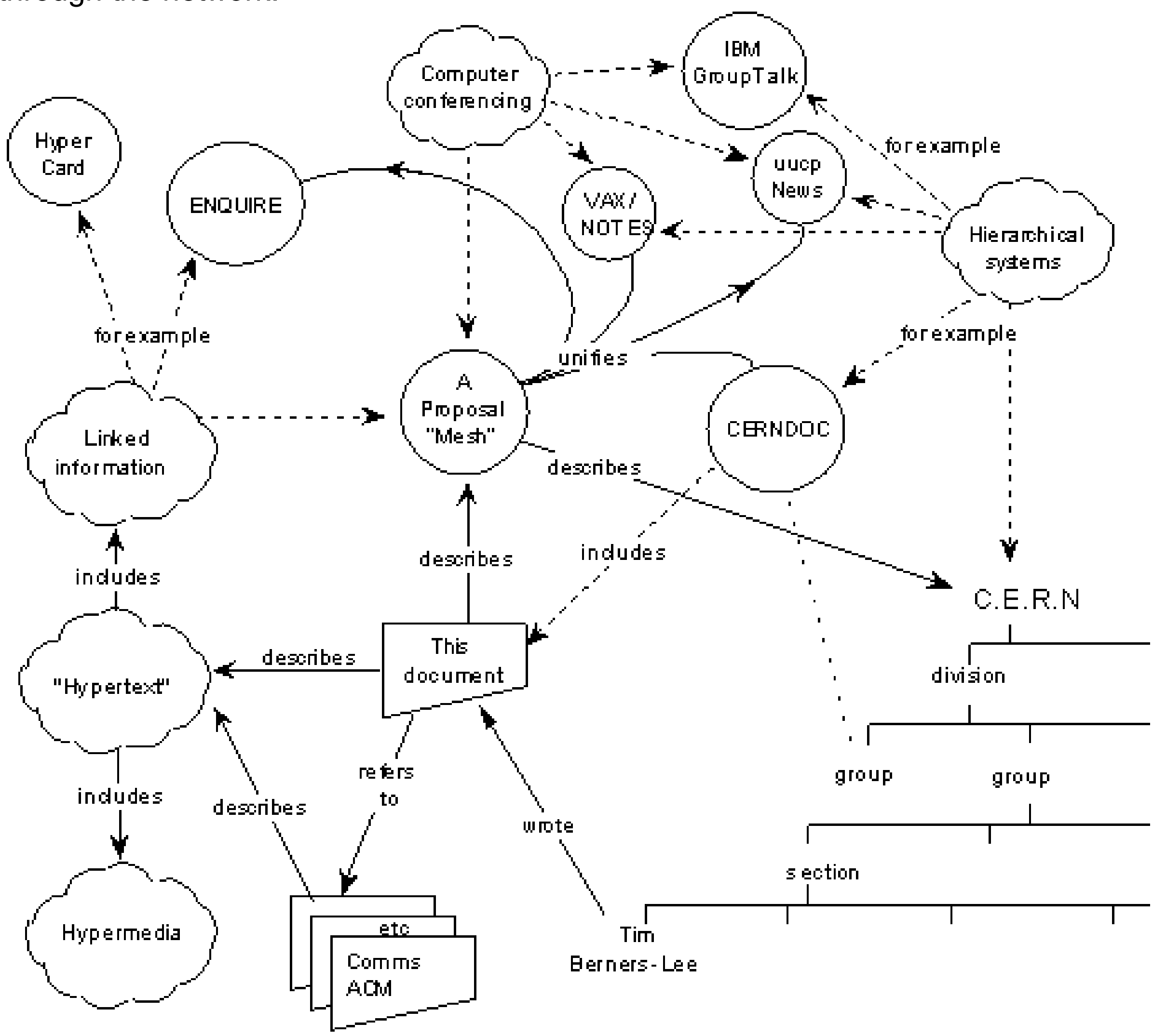
Gibson's term *cyberspace* has, in common usage, become almost synonymous with the internet. Most of the images that Dodge presents attempt to depict the internet in a global sense, though a few are devoted to depicting smaller or specific cyberspaces such as intranets and individual web-sites. As a collection, they cumulatively give some sort of impression of the magnitude and complexity of the internet phenomenon.

Such graphical depictions can be visually striking, though in many cases their individual efficacy in assisting with the visualization of the cyberspace phenomenon is questionable. Dodge classifies the many and various attempts to map or visually depict cyberspace into the following categories:

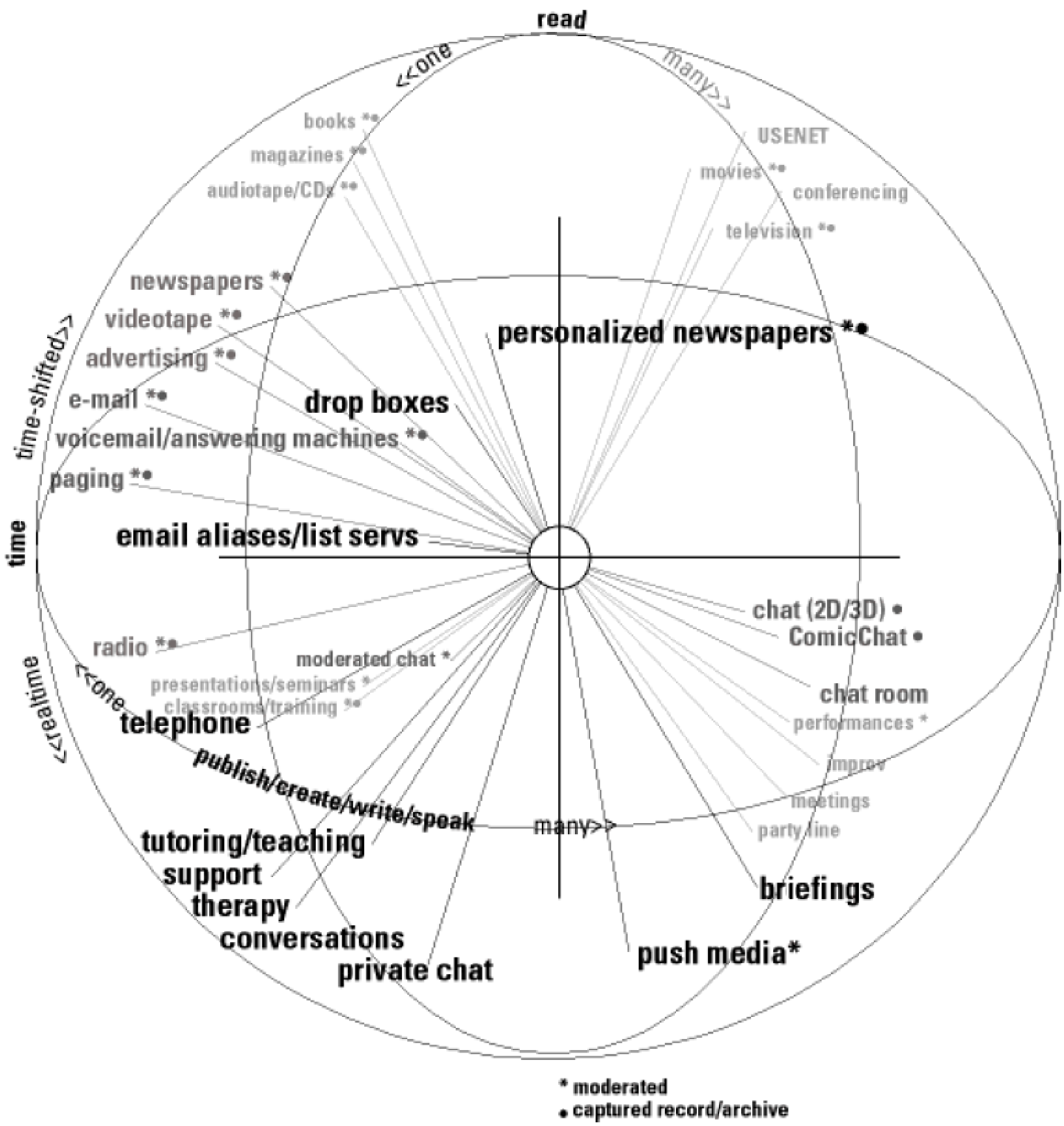
Conceptual Maps of Cyberspaces

Here Dodge uses the plural cyberspaces indicating the term can be used on both macro and micro levels, with regards to multimedia art practices over networks. A common, indispensable planning tool for developing complex abstract information hierarchies for non-linear interactive productions, during the multimedia design process is the *concept map* or *information map.*

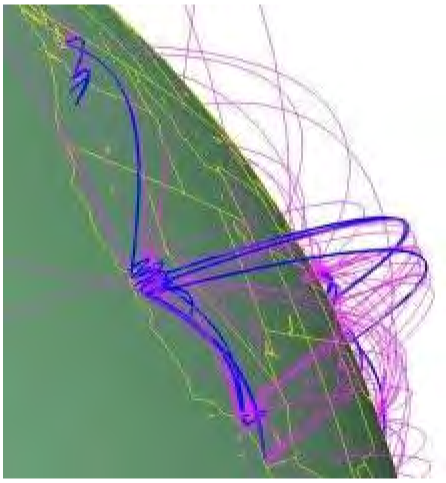
The next image is a classic concept map, used by Tim Berners-Lee in his original World-Wide Web proposal, a hypertext system called the "Mesh", presented in 1989 [Dodge, 2002]. It attempts to simplify a complex web of proposed inter-locking activities that could interrelate in new and previously unimagined ways. Berners-Lee had the insight to conceptualize this abstract web of potential relationships between information, media and forms of communication that would become possible through this proposed digital connectivity. With a touch of irony Berners-Lee uses the presentation of his internet idea as an example of the way information could flow through the network.



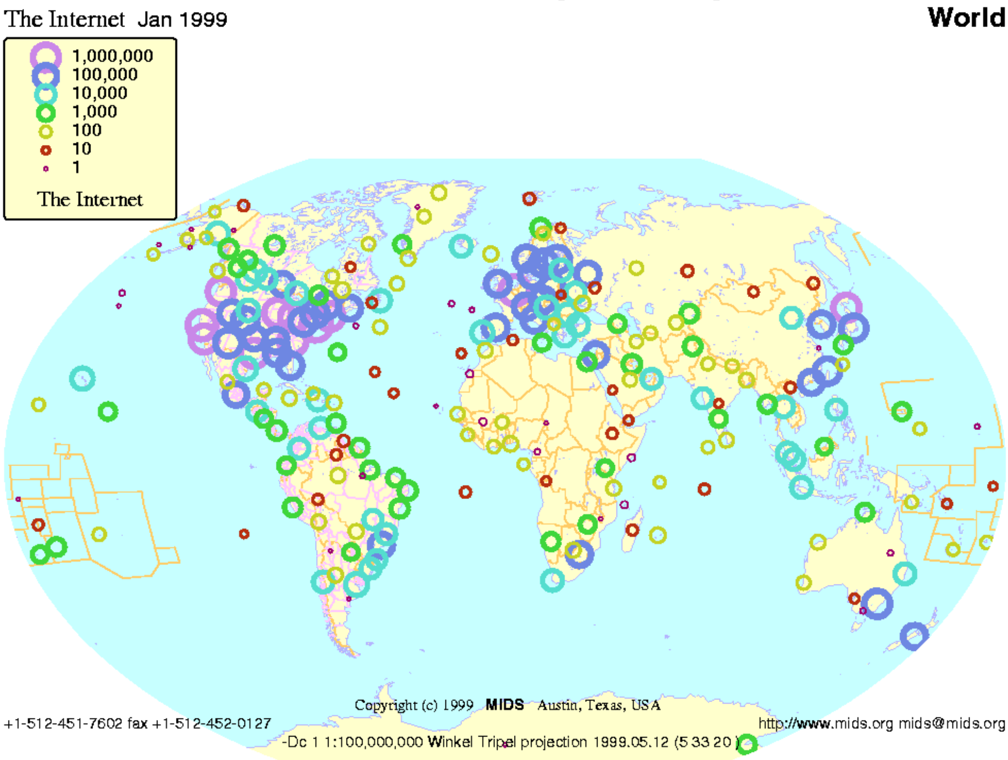
The following image is a complex but clearly constructed visual taxonomy of different modes of communication and interaction using the internet. It attempts to show a relationship between net media experience and traditional media. What distinguishes it from other approaches (covered below) is the way it attempts to depict the internet as an experience, or as an activity-based phenomenon, rather than a physical or abstracted geography. The active user is visualized as being at the nucleus of an orbiting network of media, information and communication. It was created by designer Nathan Shedroff as part of his research into “experience design” [Dodge, 2002]. The position of the individual communications nodes within the sphere indicates its particular characteristics. Proximity represents relationship between media forms or networked activities.



Mapping Cyberspace Using Geographic Metaphors



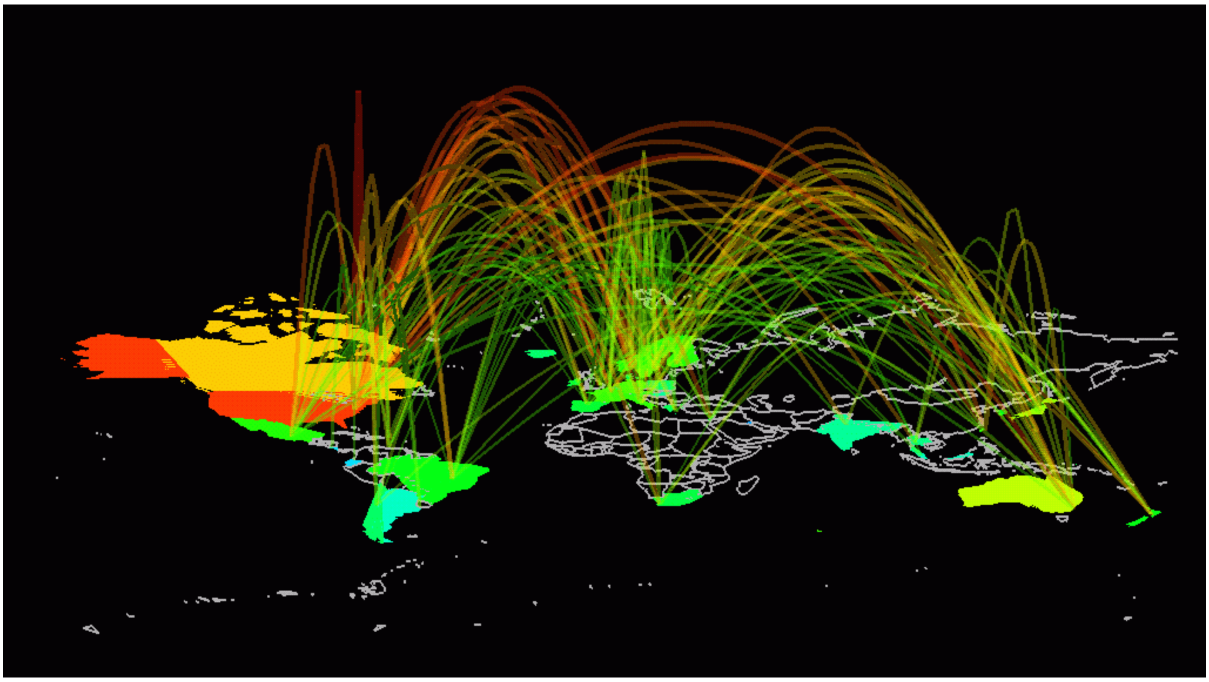
Because of our familiarity with the metaphor of a geographical map, maps of the internet which relate its reach to a depiction of the physical world are an accessible, comfortable, and perhaps most helpful way of visualizing the non-tangible nature of the internet. The following three maps use geographical metaphors combined with abstract depictions of connectivity, and try to relate to a physical scale the presence of hardware connected to WWW. [Leão, *ibid*.].



The above image is from *Matrix Information Directory Services* - MIDS - by John Quarterman, who, according to Dodge [2002], was one of the first scientists to face the challenge of mapping the Internet.

> Matrix, located in Austin, Texas, developed different softwares that measure, analyze and cartograph the Net. Among the maps of the web, the most recent, called **Internet Weather Report** - IWR: it is a dynamic representation with animated maps. The mapping method is based upon the locating of web knots and applies these data to a representation of classic geography, the map of the world [Dodge, 2002].

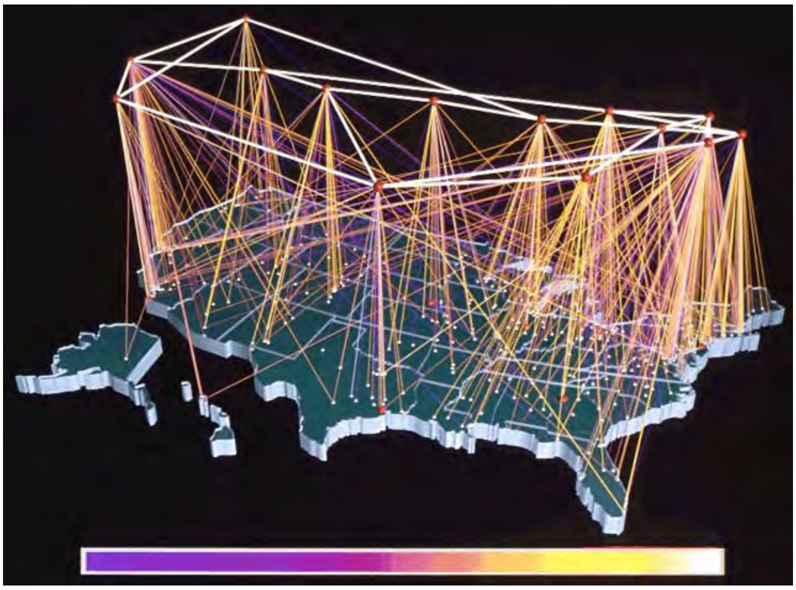
Quarterman's book, *The Matrix: Computer Networks and Conferencing Systems Worldwide* [1990], has become a classic in the discussion of problematical aspects in computer webs mapping. In the title of the book, *Matrix*, in the meaning used by Quarterman, is the set of all computers interconnected by webs [Leão, undated]. It shows clustered communication hubs, a new manifestation among the socio-economic structures of human society. This map's light colour scheme gives the impression of the internet being used day-by-day in the course of business, in a safe, ordered world.



Above: A vivid example of the research work of Stephen G. Eick (eick@acm.org) and colleagues (of Bell Laboratories) into "*the visualization and analysis of Internet traffic flows*".

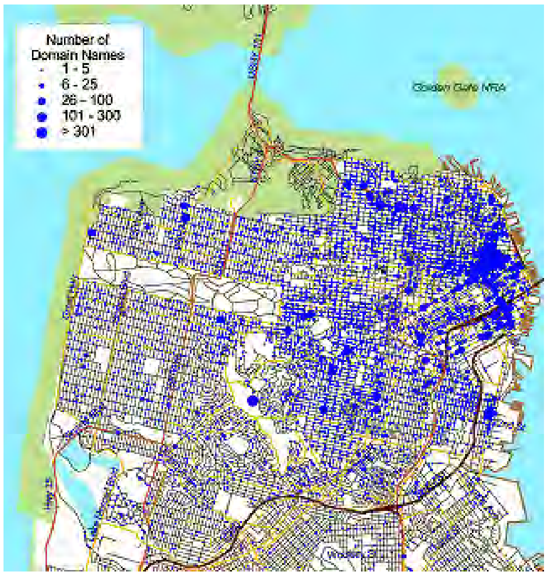
According to Leão, the map shown above, depicting three-dimensional arcs of energy, is perhaps the most famous image of what the Internet is. It is an easy to grasp visualization of internet traffic flows between 50 countries, measured by backbone NSFNET, in 1993 [1]. With darker tones it visualizes the ceaseless beaming out of information, against a night sky.

[1] Leão, undated; www.lucialeao.pro.br ; accessed 14-10-02.



Right: This image physicalizes the concept of “network”. It also gives a sense of the bombardment by invisible data streams and electronic energy, which is going on around us from moment to moment.

From *Visualization Study of the NSFNET,* undertaken by Donna Cox and Robert Patterson from the NCSA [1992].

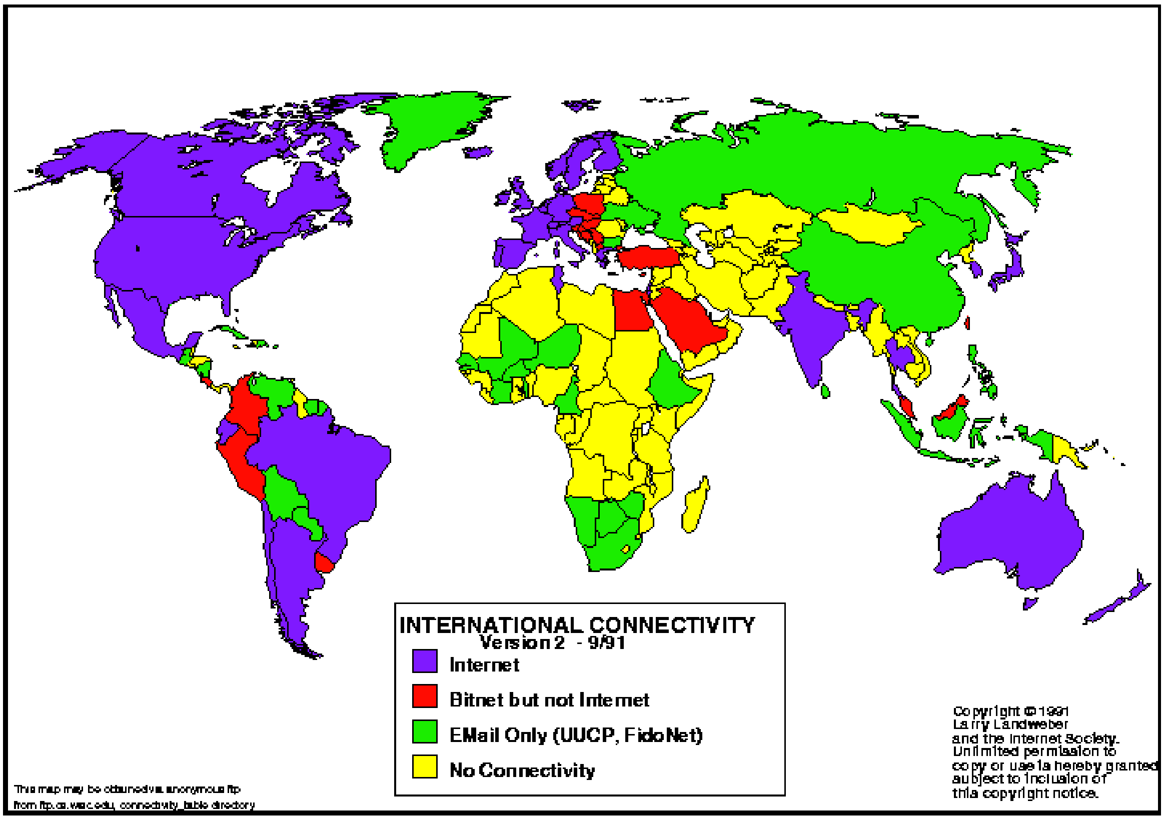


Right: “*A map of domain name ownership at street level for downtown San Francisco, circa summer 1998. The map was produced by Matthew Zook as part of his Internet Geography project analyzing the spatial patterns of the Internet industry*.” [Dodge, 2002].

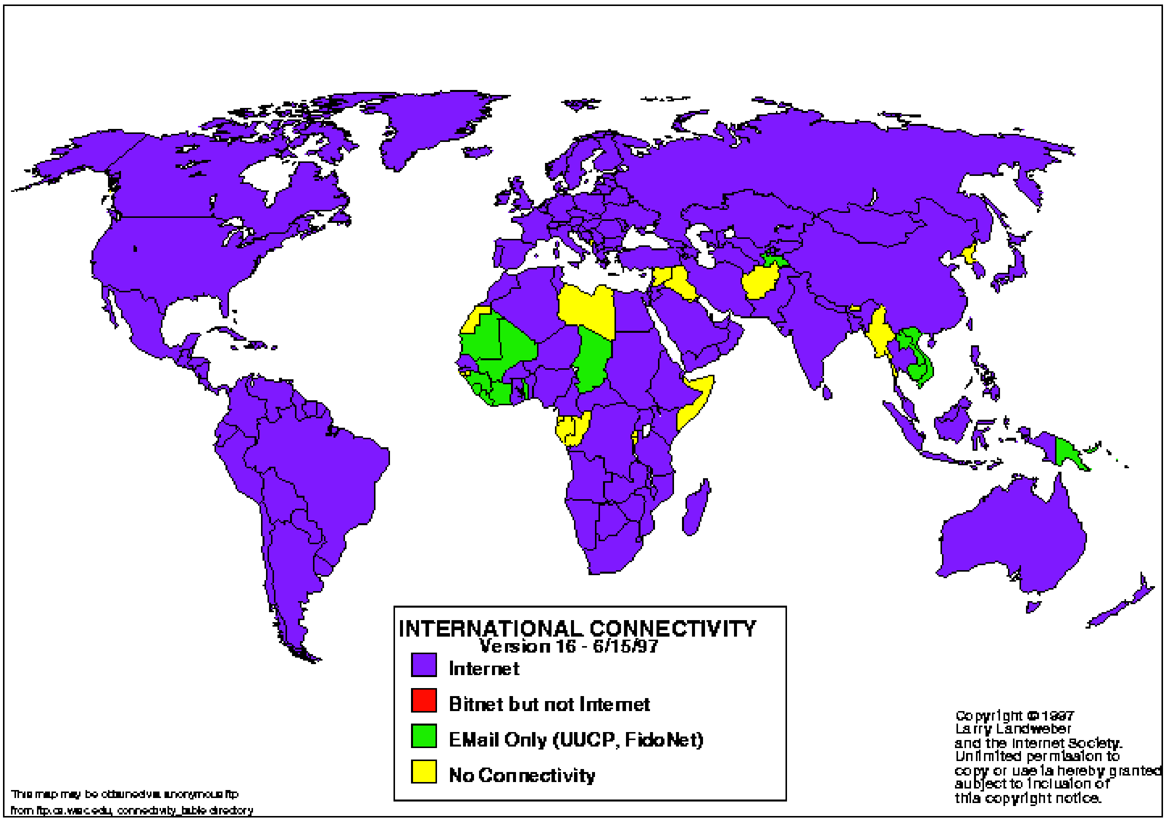
This map gives a vivid glimpse of the reality that the internet has truly arrived (at least in the affluent Western world). It is a graphic impression of just how densely connectivity has permeated contemporary lives, and how significant web-presence has become in the self-identification of suburban dwellers in developed societies. How many of these users only communicate by email or chat electronically with other San Franciscans, who are within walking or driving distance? This is the McLuhanesque electronic village, come of age.

Census and Statistical Maps of Cyberspace

e.g. A census of Internet connectivity by countries has been undertaken at regular intervals by Larry Landweber, of the Computer Science Department, University of Wisconsin, Madison, USA. They become valuable and interesting as a series because they assist in conveying the huge change and growth in cyberspace.

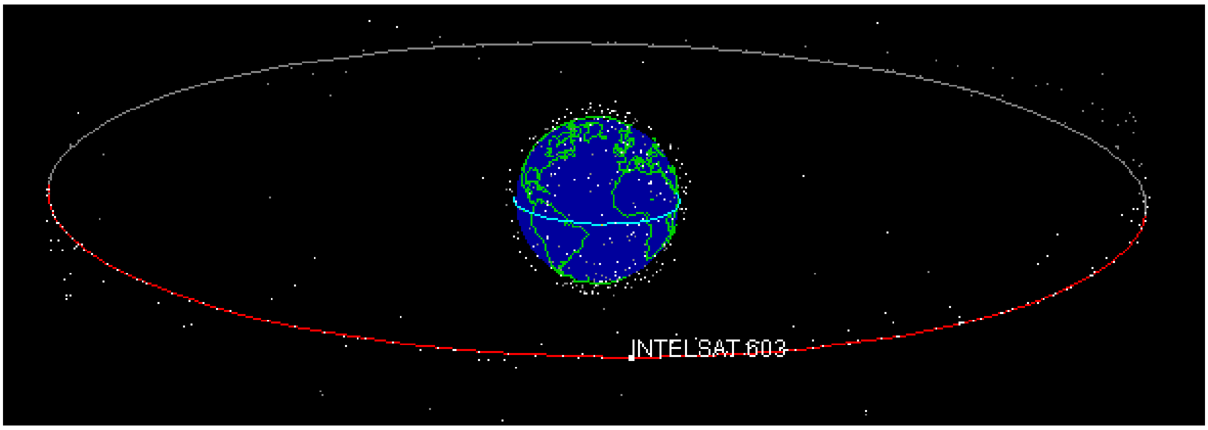


The map above shows the differential levels of network connectivity back in September 1991. The map below shows the comparative connectivity position in June 1997. The change in connectivity levels is clearly evident, assisting us to grasp the magnitude of the spread of the Internet [Dodge, 2002].

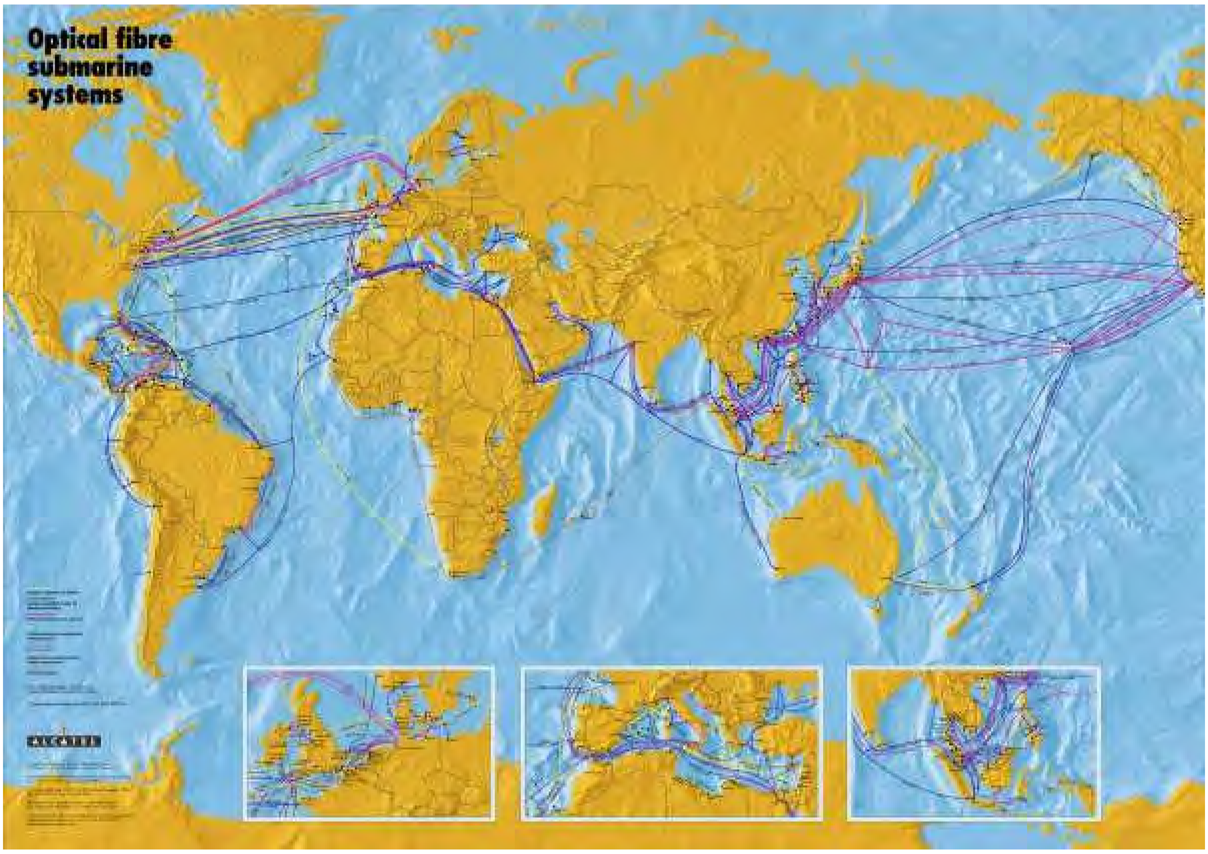


Mapping cyberspace in terms of hardware or infrastructure: *fibre-optic networks, submarine cables and telecommunications satellites*

These are also geographically-related maps, but they depict cyberspace in terms of real physical hardware or infrastructure, rather than an abstracted or conceptualized maps of connectivity, and emphasize the scale of this human achievement.



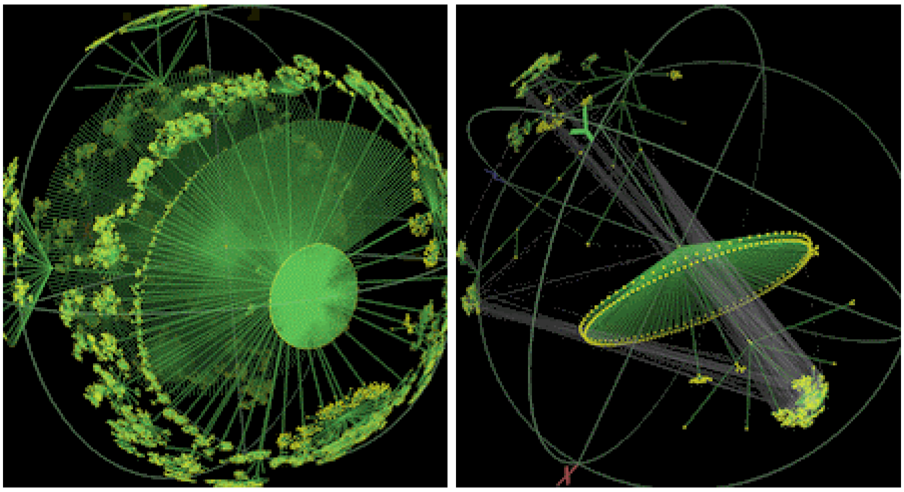
Above: This image shows "*J-Track 3D, an interactive visualization of satellite tracks created by the Mission Operations Laboratory at NASA's Marshall Space Flight Center*."



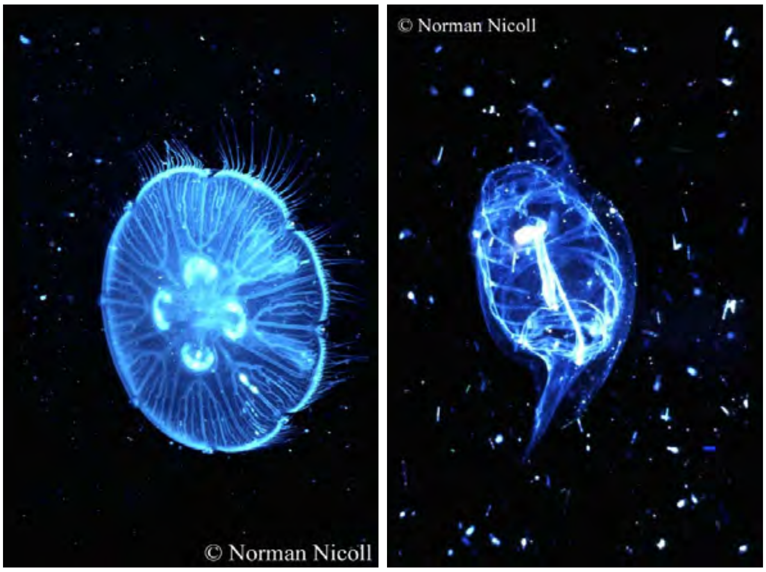
Above: A contemporary map of submarine communication cable systems across the globe.

Imagery depicting the hardware infrastructure of the internet gives a vivid sense of the magnitude of the physical investment of time, energy and resources into the establishment of these global systems.

Topology Maps of Elements of Cyberspace



While their meaning may be obscure to the layman, these strikingly beautiful images again vividly suggest a cyberspace that is a living, growing organism. Compare them with these images of plankton:



Above: Plankton: *Aurelia Aurita* (left); *Salpa fusiformis* (right) [1]

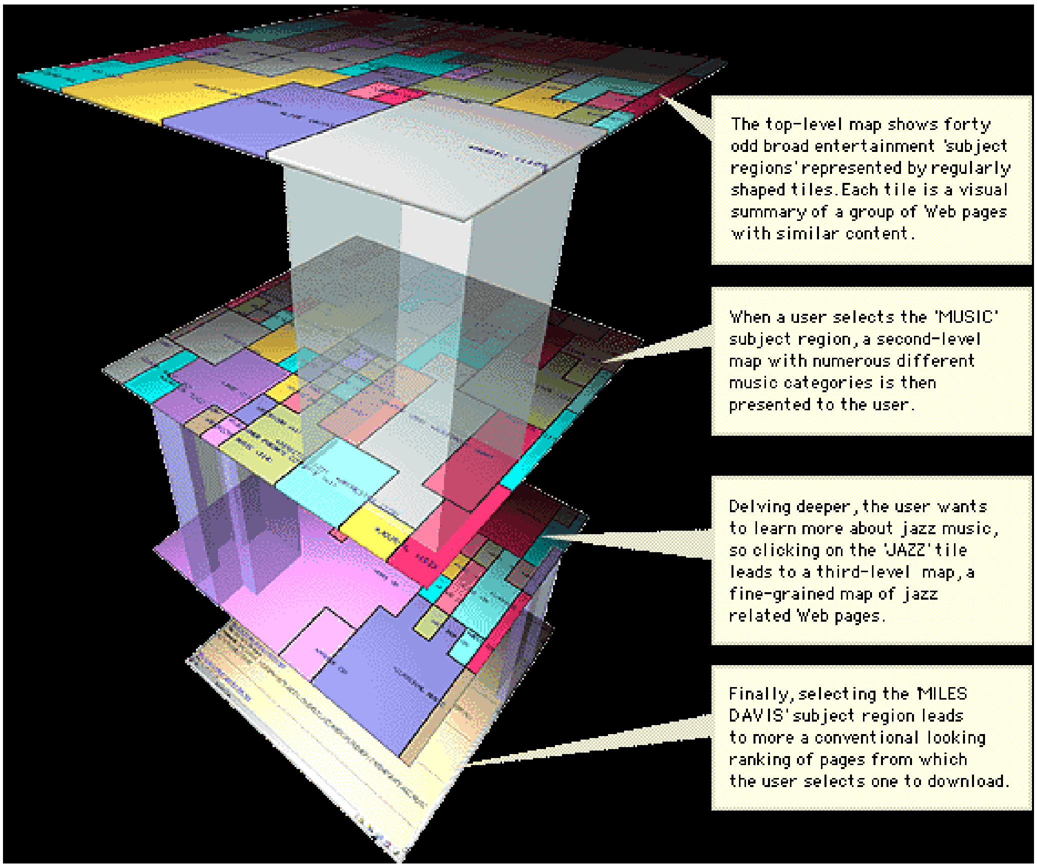
They are 3D hyperbolic graphs of Internet topology. They are created using the *Walrus* visualization tool developed by Young Hyun at the Cooperative Association for Internet Data Analysis (CAIDA). The underlying data on the topological structure

[1] Plankton Image Library http://192.171.163.165/image_lib.htm ; accessed 6-01-03. ©2002 Sir Alister Hardy Foundation For Ocean Science.

of the Internet is gathered by *skitter*, “a CAIDA tool for large-scale collection and analysis of Internet traffic path data”. [Dodge, 2002].

“Information Space” Maps -
Cyberspaces represented as 3-dimensional, hierarchical concept maps.

According to Dodge, these can be considered as analogous to conventional colour-coded land-use maps used in city planning. Often sophisticated information indexing/classification methods are employed to produce these maps. The aim of these maps is to provide a sense of the "lie of the land" of an information domain and thus to aid browsing and document retrieval. They should ideally help a user to grasp and orient to an abstract hierarchical relationships between categories of data more rapidly.



Above: "*ET-Map*" - a multi-level visualization or “category map” of the information space of over 100,000 entertainment related Web pages listed by Yahoo! *ET-Map* is one of a number of information visualization techniques developed by researchers, led by Hsinchun Chen, at the Artificial Intelligence Lab, University of Arizona, USA. [Dodge, 2002].

The “ET-Map” is a vivid 3-dimensional visualization and an excellent example of how mapping techniques can help clarify the complex hierarchical nature of cyberspaces.

Above: The metaphorical 3D "cityscape" view of the Web, generated by *Map.Net (*which has a parallel in the *GeoCities* concept). The user can experience a fly-through of a conceptualized internet world, with individual websites represented by different buildings. The larger skyscrapers represent the most popular and important sites on the Web.

Artistic Representations of Cyberspace:

The Internet is now a commonplace part of day-to-day existence in developed society, and has thus begun to play an appropriately important role in popular culture, such as cinema. Dodge, [2002] states that:

> The conceptions and representations of Cyberspace created by artists in literature, art, computer games, films and television have a powerful influence on how we perceive these new spaces.

Dodge offers a few "artistic" depictions of cyberspace. These stills from contemporary cinema are not literal and technical, i.e. they are obviously not grounded in real data or hardware, but are visually compelling images 'that have attempted to deal artistically and dramatically with issues arising from the cyberspace concept, and its ramifications for human life'. It is interesting to note that the examples below each depend on an illusion of deep 3-dimensionality, populated by imagery representing information in motion, and/or hierarchies of information, with a ghostly human element, and a brooding, neon, sci-fi mood. They are attempts to render a complex abstraction into an imaginative, evocative concrete representation.

Left: Image from the 1995 film *Hackers*, directed by Iain Softley (featuring Jonny Lee Miller and Angelina Jolie). The sequence was created by London-based *Artem Visual Effects*.

The key Cyberspace representation in the film *Hackers* was the *"City of Text"* dataspace, visualized as an eerie Gibsonian urban night scape, characterized by towers of energized data, and skyscrapers of pulsing information and computer circuitry, the motherboard as city, as shown in the still above.

Right: Image from the film *Johnny Mnemonic* [Tristar Pictures, 1995]; it was scripted by William Gibson himself, after one of his short stories. It was directed by Robert Longo and featured Keanu Reeves, Dina Meyer and Ice-T.

Futuristic and ominous, such depictions of cyberspace as the still from *Johnny Mnemonic* shown right, are an interesting blend of the abstract, metaphorical and representational.

Left: Image from *The Matrix* (1999) by Andy & Larry Wachowski (featuring Keanu Reeves, Laurence Fishburne, Carrie-Anne Moss). It was the creation of Australian sfx company, *Animal Logic*.

This labyrinthine representation of cyberspace from the landmark film *The Matrix* is thematically connected to one of the key, repeated images - the eerie green flowing computer code, and culminates in the final battle with the AI ‘agents’ in the 'corridor of code’.

The film impacted on the popular imagination to the degree that it has become something of a cultural reference point, and the green flowing code an iconic reference in advertising and on the web.

There are even screensavers available mimicking the flowing green computer text. (See p.269 for a computer-generated script analysis of *the Matrix*).

Cyberspace Paradigms

Guay [1995] attempts to explain the Internet phenomenon in terms of competing paradigms:

> The WEB is a new media, it is in its early evolution, still in paradigmatic flux. In this flux convergent paradigms compete with divergent paradigms, other paradigms converge fusing into new paradigms.
>
> By convergent paradigms I mean the repurposing of the paradigms of the older media; a WEB site using repurposed paradigms, converges with the older media. Whereas a divergent paradigm, pulls away from the older media, forging a distinct identity for the WEB. Communications history teaches us that if a paradigm is sufficiently divergent, it will profoundly impact the very structure of society itself. Truly we live in interesting times.

Guay [1995] sees *interactivity* as crucial to the special status of the Internet as a divergent paradigm, allowing it to break the intellectual stranglehold of the print medium. Guay views the non-linear structure made possible by interactive web-page design as a crucial historical step forward in intellectual and artistic evolution. He identifies three levels of interactivity at work in Internet design, which he places in an order of increasing sophistication. These can be paraphrased as:

1) Navigational Interactivity - the most basic form of interactivity, allowing the user control of what they access next

2) Functional Interactivity - where the user interacts with the system to accomplish a goal or set of goals. That goal may be winning, as in the case of a game, or ordering a product, as in the case of an online catalog. Throughout the interaction the user receives feedback on their progress, or lack thereof, towards the goal(s).
3) Adaptive Interactivity - This is the highest level of interactivity, while the boundary between functional and adaptive interactivity is blurred, there is a key difference. Adaptive interactivity offers a far higher level of creative control to the user, allowing the user to adapt the application or information space to fit their goals, or even their personality. At higher levels, adaptive Web sites allow the users to add or modify the site itself. At this level the distinction between author and reader becomes blurred.

Telaesthesia

The interactive element reinforces an active rather than passive role by the participant, whose primary senses are already all being stimulated. By offering branching structures instead of limiting the artist to a linear narrative, the multimedia artifact is placed in an ambiguous time frame and space, within and through which the cyber-traveller navigates, and can do so again with little immediate likelihood of choosing exactly the same path. Taking a different fork in the maze, the explorer discovers new meaning and experience. This is especially true of the internet experience, where interactive options are virtually infinite.

The Australian cyber-culture commentator McKenzie Wark has applied the term "telaesthesia" to cybernetic experience [Stanley, 1997]. This is a term originally used by Swedish 18th century mystic and scientist, Emanuel Swedenborg, to denote a kind of multi-sensory clairvoyance or supernatural sensation / perception, received across impossible distance [1], which he experienced in the form of visions of happenings in distant places (the most publicized being his simultaneous knowledge and description of a great fire in Stockholm while 250 miles away in Gothenburg).

Wark misappropriates the word *telaesthesia* to denote sensory and cognitive experiences in cyberspace. It could be said that "supernatural" is perhaps not so inappropriate, for we have transcended the natural with multimedia technology and play with experience which once we could only imagine.

> The art institution is finished. All it has left to do is record its own death throes and parade them around as art. Net.art that takes existing concepts of 'art' as a starting point won't amount to much, as a consequence. The real project is not art, its aesthetics. How can we

[1] Bourgeois, Robert [undated]. *Remote Viewing.* http://www.fpc.edu/pages/Academics/behave/psych/web93-2.htm ; accessed 13-01-03.

> think about the singular properties of the experience of the form of culture when it flows along the new lines of the net? That's the question. A lot of what art thinks of as its history will be of marginal interest to that, and a lot of stuff as yet unexplored, particularly in the aesthetics of media, is coming into focus.
>
> I think it belongs in a history of the experience of telesthesia, or 'perception at a distance'. This is a history that begins with the telegraph, the first moment at which the flow of information can really move at a different speed to the movement of people or things. Then there's the telephone, television — each of these is a more subtle and developed aspect of the vectors along which forms of experience might come to know themselves. Since the telegraph takes off in the 1840s, we can say that we have been living in cyberspace for 130 years, only we didn't really know it. Now that this whole terrain of telesthesia is becoming a self-aware experience, it is ready for its aesthetic moment, for its aesthetic revolution. [Wark, interview with Stanley, 1997].

Multimedia art transcends physical boundaries and is a medium for perceptions and sensations outside time, real space and physical reality. It is here an analogy with poetry becomes useful. The Macquarie Dictionary [2nd ed., 1991, p.1365] defines poetry as

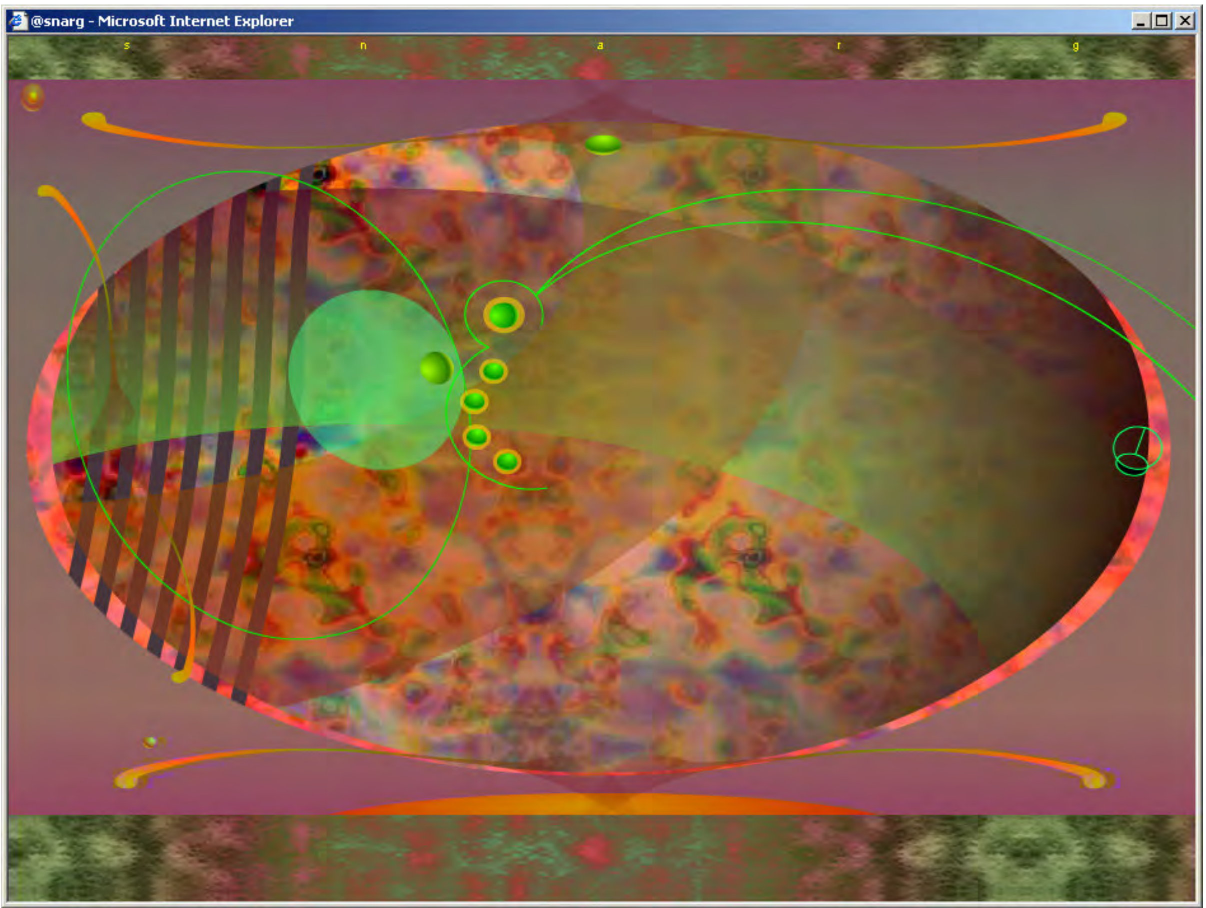
> Composition…characterized by artistic construction and imaginative or elevated thought.

Leonardo Da Vinci believed that *painting is poetry seen* [Barry, et al., 1964, 36].
If one extrapolates Leonardo's idea to a multimedia art, a user/participant can return to this work at different times in different moods and reconsider the artistic offering, gaining a cumulative, abstract, non-literal, and intuitive response to its inherent meaning. This involves at least a two-stage use of time by the participant. The first is most likely a "click-fest" in which the participant moves rapidly and randomly around the interactive branches of the work getting an overall feel for its concept, dimensions, quality, structure, framework, ambience etc. This is followed by a slowing down of time, where a more detailed, considered navigation is made, one which is more, one might suspect, in accord with the time-frame and manner of experiencing the work intended or imagined by the artist.

Cyberspace as artspace

If, as Marshall McLuhan famously advises, technologies are an extension of the human body/mind continuum [McLuhan and Zingrone, 1997], then cyberspace is a vivid example of this concept. Yet it is also a place of inter-connectivity between humans. What then are artists doing to utilize this extensibility as a creative art space, as a means of presenting something unique, in the form of an art experience to the interactively-connected potential audience?

The series of enigmatic web-sites associated with the name *snarg*[1] offers one such experimental art experience. Complete with eerie minimalist music, it exists mysteriously and may be explored by the curious, with no obvious utilitarian or commercial imperative. It is a compelling, if somewhat frustrating, deeply-layered and sensuous interactive visual and audio “trip”, a game that defies explanation, a confident step along the road to some yet to be fully-realized interactive artform. The very individual graphic style is colourful, almost psychedelic, with a strangely organic look and feel to the abstract designs and animations. There seems at first to be no discernible logic to the way the interactivity works, but with persistence deeper layers can be accessed and the user will unlock some beautiful animated sequences that defy description.



Above: Screenshot captured from one of the *snarg* web-sites.

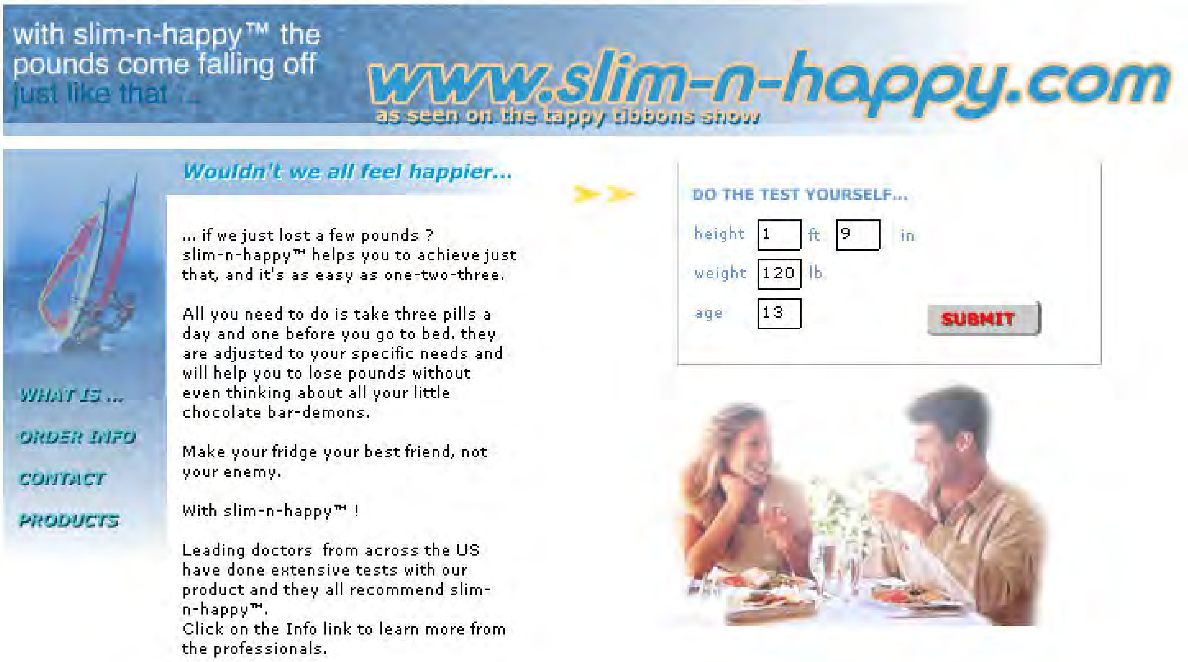
Among the interactive devices to experience, the user is confronted at one point with an empty text box. Only by keying in a text string can further progress be made, allowing the user to access the next stage/level. This forces the user to unwittingly contribute to an experiment in collaborative writing. The random snippets (often

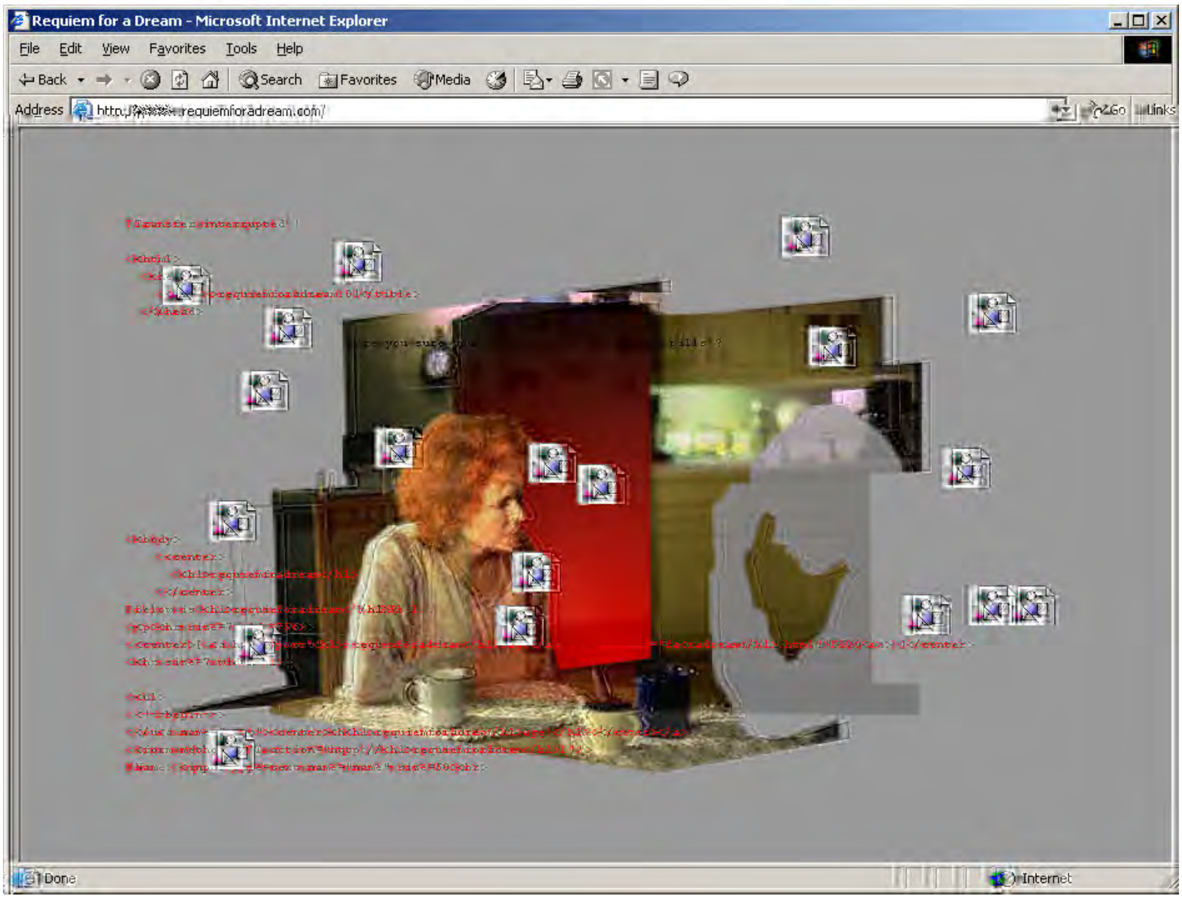
[1] http://snarg.net ; accessed 9-12-02.

tinged with the frustration or impatience of users of who are used to guided and obvious interactivity) accumulate, creating a fascinating, continuously evolving text. The sarcastic/poetic/humorous comments can be read back as a continuous flow of prose, a sort of communal stream-of-consciousness experiment.

Another brilliant use of the internet as an art-space can be found at www.requiemforadream.com. While this site does in fact have a commercial imperative, in that it is a promotional site for the dark movie of the same name, the creators have used the interactive web-authoring program Macromedia *Flash* to create a surreal and satirical, self-contained interactive movie, that is strong on mood and again has a powerful musical component.

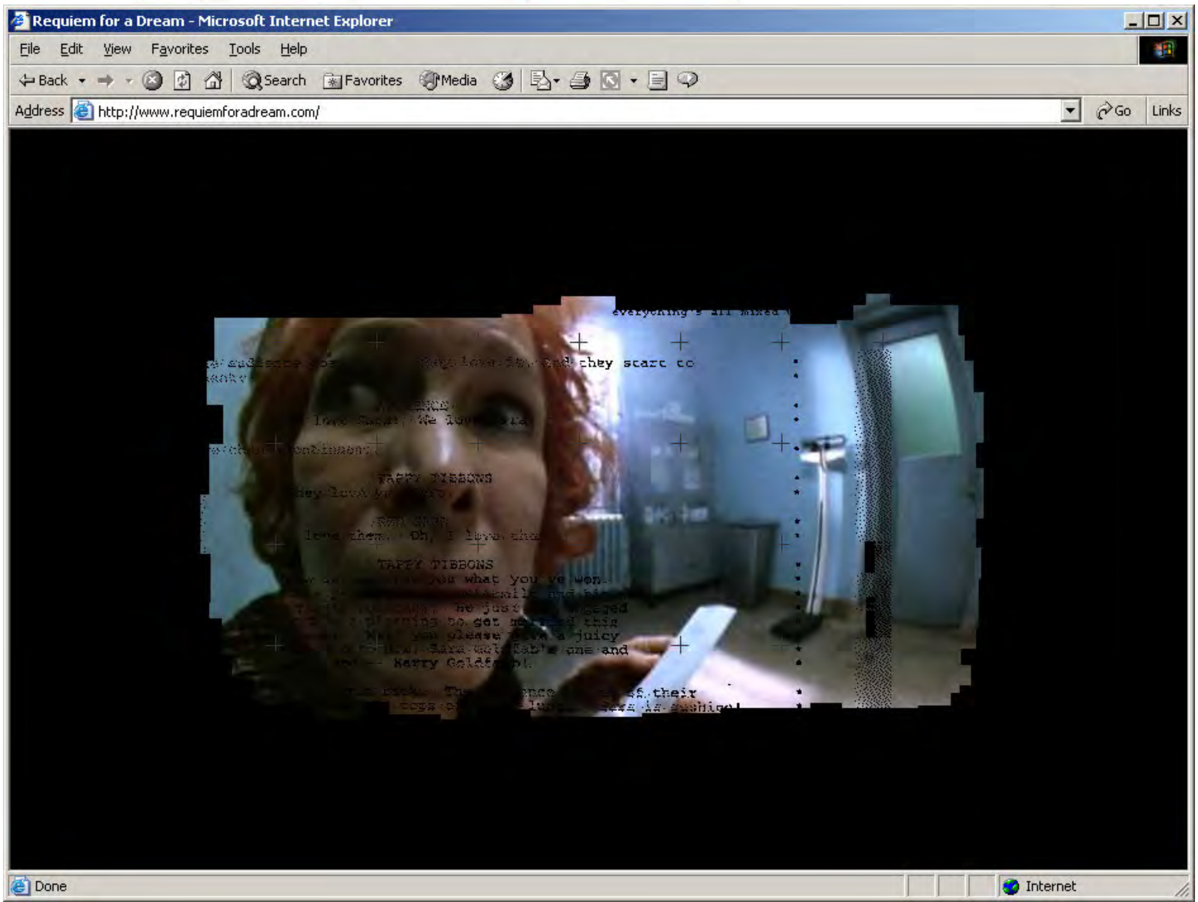
It includes in its interactive sequences some savagely humorous send-ups of the uglier side of the commercialized internet experience, with randomly occurring mock screens, such as gambling sites where the user "always" wins, interactive weight loss and self-improvement sites making ludicrous claims, and other apparently malfunctioning screens which break up as though the internet was experiencing melt-down. Familiar internet symbols and broken link icons begin to form patterns, swirling and stuttering around uncharacteristically, to finally fly off the screen, with a soundtrack of appropriately chaotic electronic interference and fragments of incoherent communication. These comments on the internet experience serve as transitions between the story nodes.





Above and next: Screenshots from www.requiemforadream.com:
strange snippets of pseudo-data, broken link symbols become "alive".

Passing beyond the transition sequences, the user discovers deeply-layered, animated, interactive graphics with poignant symbolism, and can enter a non-linear, interactive movie/story which encompasses four seasons and several characters. There is a sense that we are observing the lonely internet user from a camera placed within their computer monitor, with echos of Orwellian and Brave New World regulation and oppression.



While remaining enigmatic, there is a strong voyeuristic feeling of peeking into the desperation of ordinary suburban lives, in which the internet is just another commercial rip-off, and at best a kind of port-hole of distraction from the grimness and dark poetry of love and addiction.

1.7 Market forces and the emergence of a multimedia meta-language

Multimedia art has emerged as a result of a rapid, ‘organic’, or unplanned and spontaneous social process. As recently as 1947 leading American mathematician and computer engineer, Howard Aiken seriously predicted that only six electronic digital computers would satisfy all the United States’ computing needs. Now the digital revolution has placed the electronics industry as the third largest in the global economy after automobiles and oil. [Palfreman & Swade, 1991, 8]. Microcomputers became familiar items in the business world for purely economic reasons, through their ability to handle text manipulation, accounting and other data processing in new efficient (labour- and timesaving) ways.

Computers are unique machines in the way they can be software-adapted to a variety of uses – a general device for manipulating symbols. The application of fine motor control and visual co-ordination (mouse operation) to on-screen commands facilitated the desktop revolution of the 80s [Wise, 2000, 3]. Object-oriented interface design - the graphical approach to the screen interface originally popularized by Apple Corporation’s *Macintosh* operating systems and computers - has enabled a more natural orientation to digital information, engaging the visualizing abilities and offering icons for concepts, in a virtual space.

Digital multimedia has been made possible for an emerging consumer market through the rapid development of computer hardware and the visual interface and software, but has also helped drive this development. The tools are now available, the technology itself is no longer the limiting factor, and we are witnessing the gestation of multimedia as an artform. Still there is a distance to cover before we will see artists producing interactive multimedia productions with the same degree of certainty, streamlined process, and understanding of the medium and its sub-structures as in, say, oil-painting or a contemporary feature film.

The changes in the media production landscape have been so rapid that it is easy to see them as revolutionary. It is difficult to interpret a revolution when in the midst of one. The contemporary capitalist economic battlefield of desk-top computing technology is made confusing by this fast pace of technological change, competing operating systems and the proliferation of overlapping software applications. Paul

Brown, (1996) makes an analogy between the development and maturation of computing technology and the evolution of modern cinema technology. He offers a system for delineating the historical stages that the coded software environment has been through, along the road to the present situation, which sees the widespread availability of seamlessly integrated commercial suites of media production tools. He tries to give a sense of what these stages really meant in terms of being able to produce multimedia on the desktop by offering an analogous (fictitious) film production scenario. He also looks forward:

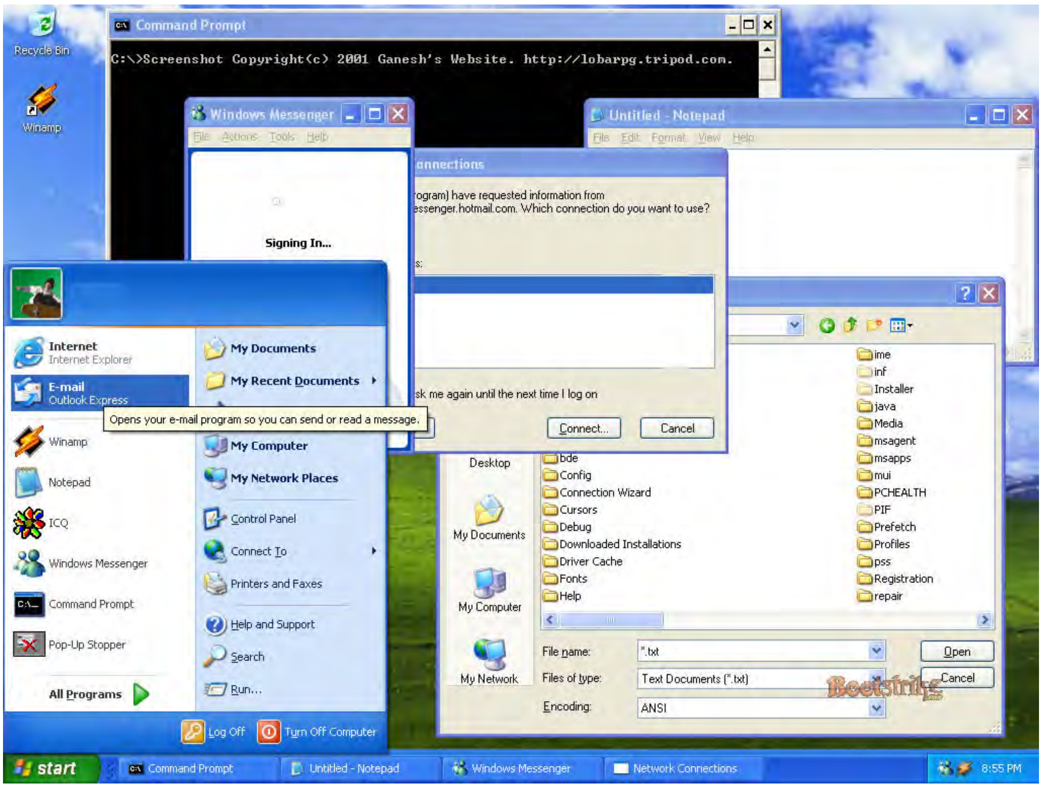
Era	**Computer evolution**	**Analogous period in cinema history**[1]
1940s	Machine Code	(filmmaker must go out and) buy metal ingots to build gears for camera
Early 50s	Assembly Code	(filmmaker must) buy pre-made gears to build camera
Late 50s:	High Level languages	(filmmaker can) buy pre-made camera, but (must rely on) total self-production (including processing film)
1980s	Object oriented OS	3rd party processing of film
1990s	software agents	Hire a professional film crew
Early 2000s	meta-agents	the first great silent movie directors (Eisenstein, Griffith etc.)
21st century	Meta-Language	The great modern directors: (Hitchcock, Bergman, Kubrick, Tarantino etc., etc.)

Brown's overview of the eras of computing evolution gives a sense of just how recently unfettered artistic experimentation in multimedia has become a practical possibility. His analysis supports the view that at the turn of the millennium we are witnessing the embryonic development of a mature multimedia art. That this evolutionary process will continue to unfold, is suggested by Brown's idea of a future meta-language. Brown is implying a future in which operating systems and software become standardized, optimised, and universalized – a ubiquitous, highly-evolved "universal interface". That the evolution of this meta-language is indeed underway can be given testimony by the observation that common graphical user interfaces, such as those offered by the Microsoft *Windows*, Apple *Macintosh,* and *Linux* operating systems, are moving closer in terms of screen interface conventions. The three examples below of the G.U.I.s offered by the major operating systems exhibit close similarity in the metaphors and organization of user operation, and data storage and access – including now widely understood icons in file menus and navigation bars (virtual trash cans and folders, torches and magnifying glasses for searching, scissors icons for cutting data etc.), and the convention of windows-

[1] Words in brackets were added for clarification.

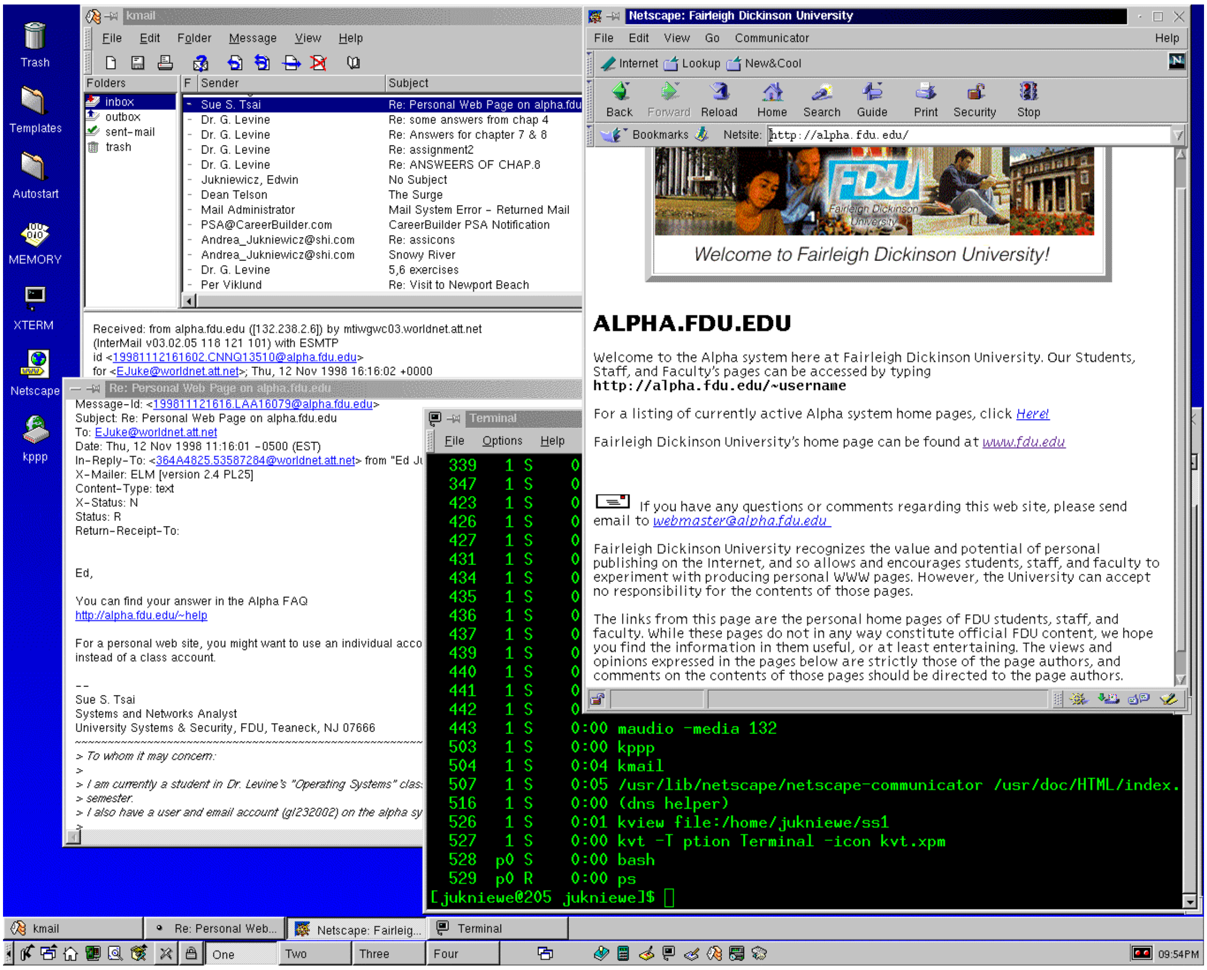
within-windows, a standardization supporting the idea of an emerging meta-language of computers.

Left: Apple *Macintosh OS X* G.U.I.



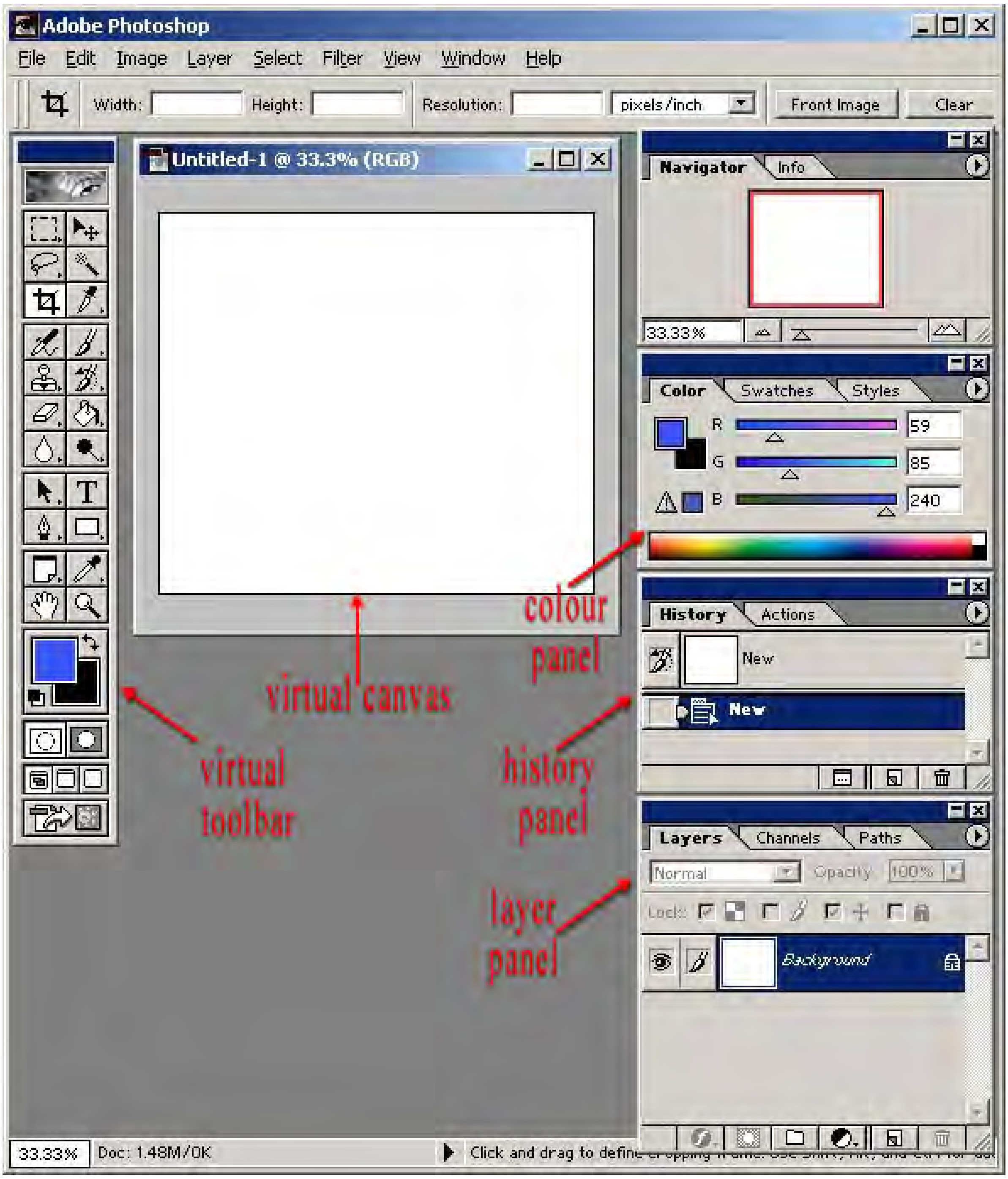
Above: Microsoft *Windows XP* G.U.I.

Even the use of blue as a dominant background/theme colour in each of these G.U.I.s is significant in the way that the designers of each OS seem to be symbolically expressing the concept of a space or virtual world offered within a personal computer.

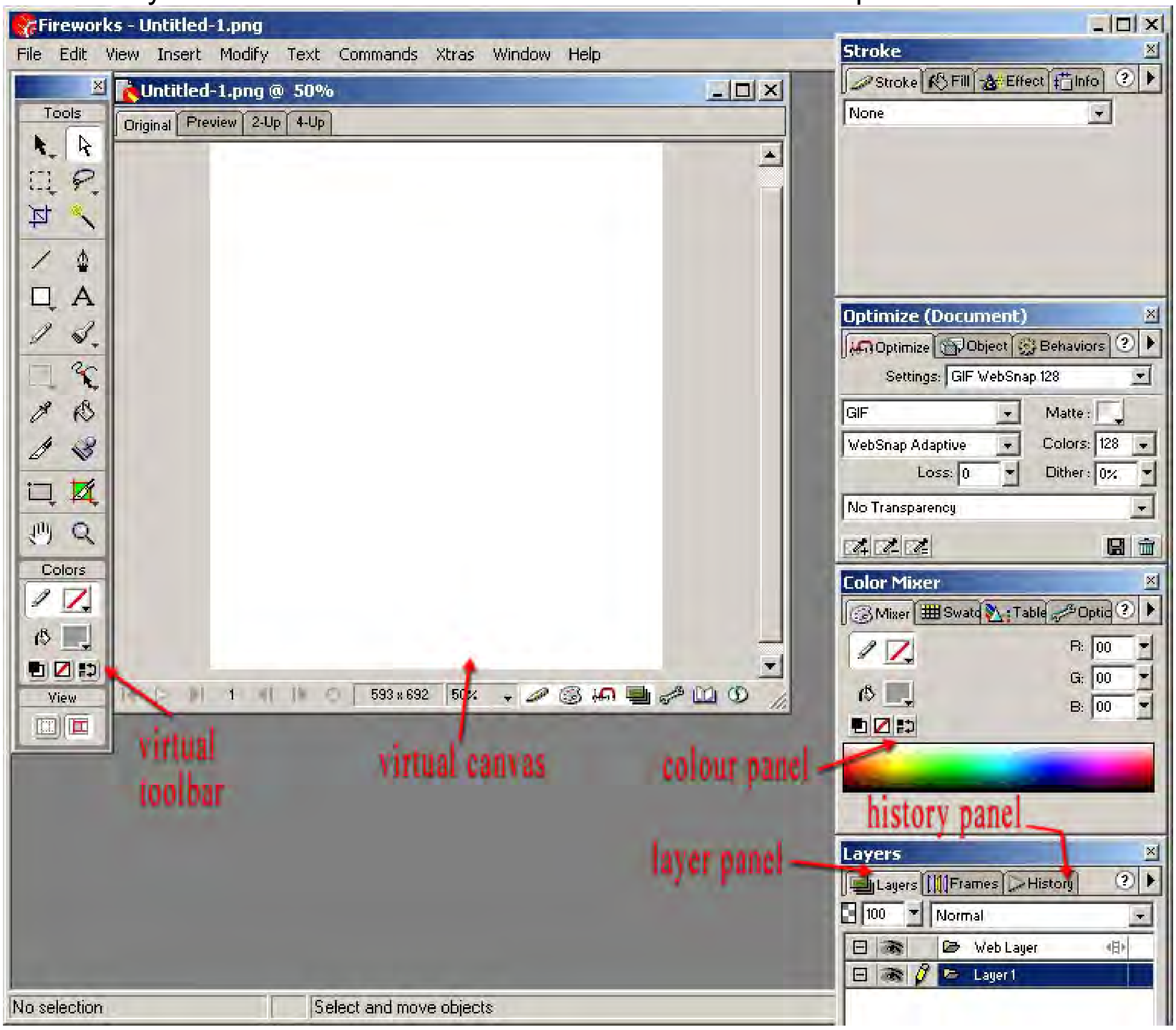


Above: *Linux Operating System* graphical user interface.

Similarly, the interfaces for the multimedia production environment offered by Macromedia and Adobe (two of the largest multinational vendors of multimedia software and market competitors) are also becoming increasingly standardized, with virtual toolbars and panel sets for manipulating variables looking increasingly similar with each new release. Below is a screenshot of the interface offered by the so-called industry-standard digital image manipulation software Adobe *PhotoShop*. The image which follows this shows the interface for a similar (though more specifically internet-oriented) image manipulation software package offered by Macromedia entitled *Fireworks.* They are very similar in terms of interface design. The virtual toolbar has become a universal feature of this kind of software, (and the toolsets almost identical), as are information panels (example images show colour, history, layers panels). Both applications offer a virtual canvas, and both have adopted the *layers* concept (virtual layers are visualized to be like plastic sheets laid over an image, allowing the user to independently manipulate elements of an image. These layers are of course just a way of organizing digital image data); both use the *history* concept, allowing the user a way of “undoing” steps in the design or image manipulation process, which encourages experimentation and creativity); both have very similar colour-picking options, etc. etc.



Research and development in popular software applications is an on-going process with a goal of broadening the customer base, but also of nurturing and retaining the loyalty of already committed customers. There are significant on-going revenue streams to be earned from the original investment in software design through an ongoing process of releasing upgraded versions of successful software offering additional facilities, tools, enhanced and streamlined processes, and promises of increased productivity. The competing software design teams inevitably keep a constant eye on the work of their rivals in order to remain competitive.



Above: The Macromedia *Fireworks* interface

Rather than trying to differentiate these products by designing a very different on-screen look, software engineers appear to be recognizing successful design implementations and responding to these by adopting the best ideas and incorporating them into upgraded versions of their products. Recent patent wars between Adobe and Macromedia are a testament to this analysis.

> Adobe filed (a) suit in August 2000, alleging that the user interface of Macromedia's *Flash* Web animation tool infringed on Adobe's patent for "tabbed palettes," a feature that allows users of design software to rearrange the work space on the PC screen.
>
> A jury in the U.S. District Court of Delaware agreed with Adobe and awarded the company $2.8 million in damages. Adobe said in a statement that it also expects a judicial injunction preventing Macromedia from selling the infringing software.
>
> A trial is scheduled to begin in the same court Monday regarding Macromedia's first countersuit, charging that Adobe's *Photoshop* image-editing software and its *GoLive* Web design software infringe on two patents that Macromedia holds for editing tools. A second Macromedia countersuit, filed last year in U.S. District Court for Northern California, is not yet scheduled for trial. [1]

According to Brown's cinematic analogy regarding the recent history and future of multimedia, the emergence of the first great masters of multimedia art will be tied to the emergence of a computer meta-language (see table, p. 97). Corporate competition seems to be driving a gradual evolution of a universally accepted approach to interface design for specific multimedia software applications, such as the image processing/graphic design examples shown above. It is argued here that this is tangible evidence supporting Brown's concept of an emerging meta-language, (which essentially reflects an ongoing development of 'industry best practice'). It matters not that this evolutionary process is being driven by market forces. It is easy to understand the commercial imperative to appeal to, and thus retain, an educated target audience which has already developed software skills based on earlier releases of a popular software application, and feels comfortable with certain familiar on-screen conventions. Developers no doubt recognize that users want to remain productive when they upgrade software, and that they have a natural resistance to acquiring new software which forces them to re-orient to unfamiliar interface designs, or re-educate themselves into new work habits. In rolling out the software upgrades, software developers have to balance this resistance against the need to ignite the popular imagination with something new. It is a fine balance that developers must strive to get right. This is the market mechanism driving research and development that will create the multimedia meta-machine of the future.

The emergence of the Multimedia Meta-Art Machine

Leisure activities designed to be enjoyed in a computer environment have had a rapid acceptance by a consumer market. Technical advances, economies of scale and competitive forces continue to bring prices of popular digital technology down, while at the same time there is upward pressure for more computational speed and

[1] Source: David Becker [May 2, 2002]. URL: http://news.com.com/2100-1040-898061.html ; accessed 2-12-02.

power. There is an ongoing development of processes of capturing, storing and processing still- and moving-images, and sound, digitally via the personal computer and of larger and faster storage media. Approaches to software design by corporations developing these technologies stress the primacy of ease of operation. Technological advances on all these fronts are cumulatively resulting in an unplanned creation: a new artistic delivery medium - the multimedia machine, the meta-art machine.

The graphic display unit (RGB monitor) is similar technology to the television – a technology which has evolved into the most powerful marketing tool yet invented - an entertainment and information vending machine with which even the less affluent sections of human society have developed a very comfortable, often pivotal relationship. In many parts of the world television is a central focus of the day-to-day domestic routine. It is possible to imagine a world where this focus is further entrenched if we consider the concept of a potential merging of the functions of the personal computer platform, and the on-line environment, with traditional television services.

Multimedia technology has had a remarkably rapid evolution. Enthusiasm by both artists, domestic consumers and marketing engineers have fueled a revolution which has led to the design of a host of sophisticated software, which can turn a single personal desktop computer into the equivalent of:

• a printing press	(hooked up to a colour printer)
• an image capture / manipulation machine and digital darkroom	(using scanners, digital stills cameras and image editors)
• an audio/music recording studio	(digital recording, sampling; MIDI composition; post production)
• music arranging / copying service	(notation/score print-out from MIDI)
• video editing and special effects suite	(digital video camera and firewire, video editing/compositing/processing applications)
• a 2-d graphic design studio	(pressure sensitive Wacom-style art tablets and vector/bitmap design applications)
• 3-d design and animation studio	(motion tracking, 3D model scanning, network rendering; 3D modeling / animation applications)
• secretarial or transcription service	(optical character recognition / voice recognition)
• a proof reading machine	(spell and grammar check)
• an interactive media authoring system	(interactive authoring applications)
• Web authoring and publication system	(Web design and ftp applications)
• Data/Audio CD and DVD mastering and duplication system	CD and DVD burners

These add-on peripherals such as printers, scanners, digital cameras, WACOM-style pressure sensitive art pads, 3-D model plotter/scanners, motion-trackers etc., enhance the potential crossover between media, and the potential multi-dimensionality of art manipulation. Marshall McLuhan popularized the view that technologies extend the human sensory systems and nervous system [Mitcham, 1994, 23], and echoing this concept, these peripheral devices extend the multimedia machine and open up channels to digitally sample the physical world, as though sensory organs were being added to the computer. CD-ROM/DVD burners offer a durable storage and delivery medium, and open up interactivity as a new cognitive entertainment modality. Finally using a modem or broadband optical-fibre connection to interface with networks and the Internet offers a totally new, low-cost distribution medium for the products of the above technologies, subverting the dominant publishing, arts funding, censorship and distribution powerbases and paradigms.

Of course, at the present embryonic stage in its evolution, one can't simply go out and buy an optimized multimedia art machine. It must be assembled from a range of available and competing items. An analogy might be offered of having to choose from a pile of cogs, wheels, gears etc., the ingredients for a horseless buggy, without a blueprint. While new generations of work-station/software packages usually stress multimedia capabilities in their advertising, one must still assemble the total multimedia machine somewhat in the fashion of the meccano set, after purchasing the particular configuration that an individual's research and budget has determined to be necessary. Through saturation marketing and widespread press interest, awareness of the possibilities offered by this technology has inexorably penetrated the demographic groups already accustomed to spending disposable income on consumer electronics and leisure entertainment. Mature markets have emerged and developed and entrepreneurs have responded to these new markets, to the point where large retail outlets (such as the Harvey Norman chain in Australia) have dedicated whole sections of their retail space to selling computer hardware and software. Specialist shops have made it increasingly easy to assemble the machine.

Semantic Compression

Guay, [1995], argues strongly for the potential importance of multimedia as a form of communication. In Guay's view, the inherent possibilities for "semantic compression" [*ibid*.] give multimedia an edge over the traditional forms of communication which have evolved under the print paradigm. This has particular relevance as it points the way to the future relevance of multimedia as a meta-art form.

> Due to the ever-increasing amount of information we need to take in to function in contemporary postindustrial society, the demand on our cognitive bandwidth has significantly grown. Properly blended multimedia reduces our cognitive bandwidth load by semantic compression. Semantic compression refers to the density of information transmitted by given

> media. The cliché, "a picture is worth a thousand words", describes the high semantic compression of graphic images over text.
>
> From low to high compression, media are ranked as follows: text, audio, still images, and video. The base semantic compression is increased if the message or effect triggers cultural archetypes or memories of individual human experience. Triggering these increases the overall information density by blending it in with previously stored information. For example, a simple phrase, such as 'first love' can trigger a very dense multisensory information stream, complete with reinvoked emotions.
>
> Semantic compression is also heightened by increasing the level of user interactivity with the information stream. This heightening is the result of the mind focusing its resources on the information.
>
> Thus the well thought out use of an interactive blend of high semantic compression media, with the message or effect designed to evoke cultural archetypes or individual memories, makes optimal use of the user's cognitive bandwidth. When the choice is between optimizing cognitive bandwidth or network bandwidth, choose cognitive bandwidth.

Guay's concept of semantic compression finds support in studies reported by the Fraunhofer Institute for Computer Graphics in *Acoustic Simulation for Visualization and Virtual Reality* [Astheimer, 1995]. A synchronized combination of communicative visual and audio material enables a receiver to perceive quantitatively, and analyze qualitatively, more data in the same amount of time. The concept of semantic compression resonates strongly with Elsom-Cooke's closely related concept (2001, p.3) of an emerging cross-channel language of interpretation, by which the cross-referencing of information gained on parallel sense channels results in an increased grasp of the communicated experience – the user receives a more comprehensive communicative experience through the inter-relationships perceived from co-ordinated parallel sensory stimulation (see p. 34).

McLuhan and multimedia

The Canadian media theorist Marshall McLuhan had a penchant for summarizing and representing complex ideas with succinct, though unconventional slogans. His influence on media theory is so significant that his ideas must be considered in an overview of multimedia. However some of these famous phrases and coinages are difficult to deconstruct without extensive argument, and it is only possible to make a few brief points within the present context.

Among his more resonant phrases is the prophecy that the electronic media would create a "global village" and lead us back to a sense of tribalisation from which we were divorced during the era of print media dominance. As early as 1967, in *The*

Medium is the Massage, McLuhan was able to articulate the enormous changes to the human condition wrought by electronic communications:

> Ours is a brand-new world of allatonceness. "Time" has ceased, "space" has vanished. We now live in a global village...a simultaneous happening.... Electric circuitry profoundly involves men with one another. Information pours upon us, instantaneously and continuously. As soon as information is acquired, it is very rapidly replaced by still newer information. Our electrically-configured world has forced us to move from the habit of data classification to the mode of pattern recognition. We can no longer build serially, block-by-block, step-by-step, because instant communication insures that all factors of the environment and of experience co-exist in a state of active interplay. (p. 63)

Satellite TV news, the Internet and chat-room cyber-cultures are an embodiment of this analysis. McLuhan's concept of "hot" and "cool" media is somewhat less illuminating and more difficult to clarify, but it does attempt to unravel the complex socio-psychological phenomena of media experience. Gow [1] makes some useful points with regard to this concept:

> A hot medium is one that extends one single sense in high definition. High definition is the state of being well-filled with data. A photograph is, visually, high definition. A cartoon is low definition simply because very little visual information is provided. Telephone is a cool medium, or one of low definition, because the ear is given a meager amount of information. And speech is a cool medium of low definition, because so little is given and so much has to be filled in by the listener. On the other hand, hot media do not leave so much to be filled in or completed by the audience. Hot media are, therefore, low in participation, and cool media are high in participation or completion by the audience… McLuhan was a poet. These are poetic probes and require a certain amount of figurative interplay. They are not tightly defined terms like "deoxyribonucleic acid" or "constitutional monarchy". Furthermore, it seems to me that hot and cool should be viewed as relative and not fixed labels. That is, hot and cool can only exist in relation to one another (within the larger spectrum of media). Print is neither hot nor cool on its own. Its temperature is gauged in relation to other forms of communication.

Multimedia can theoretically be a mix of hot and cool media and therefore does not sit easily within McLuhan's theoretical analysis. If multimedia is viewed as a discrete, wholistic new media form, and the concept of semantic compression is taken into consideration, then (following Gow's advice that hot and cool are relative labels) multimedia could be placed at the hot end of the McLuhanesque media spectrum – large clusters of cross-referenced information are presented to the users senses. However interactivity implies participation – a cool characteristic. Also, McLuhan describes film as a hot medium in contrast to cool TV, not only because of the higher definition of film but also because of the focused, low-interference viewing environment of the cinema experience. Multimedia, like TV is experienced on low

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[1] Gow, Gordon (undated) *Thawing out Media: Hot and Cool* http://www.peak.sfu.ca/cmass/issue2/july.html ; accessed 16-01-03.

resolution screens, usually in distracting home or office environments. Gow clarifies this contradiction succinctly, coming down on the cool side:

> The more that meaning is derived from a single point of intense focus, the hotter the medium. The more that meaning is derived from a spectrum of sources (especially non-linear)--a sensory buffet if you will--the cooler the medium. A single point of focus requires heavy filtering of (other) sensory input. A wide spectrum of focus on dynamic forms involves less filtering and greater all-round sensory engagement.

McLuhan’s most famous phrase, "the medium is the message", conveys the idea that technologies provide an environment for types of expression that would otherwise not arise, and suggests a moulding influence of technology on culture. In a multi-layered, culturally blended, information-rich society, McLuhan’s slogan can also be seen as supporting the idea that multimedia technology is a “physicalization” of the post-modern consciousness (see p.60-62).

Wallace [1996, 5] seems to naively misunderstand McLuhan’s most famous of phrases, but succinctly presents the commonsense ‘content over form’ argument:

> The medium and the method of distribution is just so many atoms, arranged in a moderately interesting pattern. Regardless of which technology you are discussing, film, TV, print, CD-ROM or the Net, it isn’t what you deliver on, or even how you present it - it’s what you have to say.....
>
> Most of the industrialized world have the power to see nearly anywhere on the planet, anytime of the day or night. And most of us can reach out and contact almost anywhere via phone, fax or the Internet....
>
> Now, more than ever before, we must remember that even though how we say it is important, it is what we say that matters most in the exchange.....When you pick up a great work of literature, you are not affected by the wonderful quill or parchment technology that went into creating the original...it is the art, the message, that matters - not the canvas or how they mixed their paints....

Wallace carries his concerns for digital art further, making an interesting point with regard to future histories of digital art when he worries that:

> .....digital creations...are in danger of being as fleeting as chalk drawings on the sidewalk. We have no paper trail to refer to. Historians will not be able to peruse papers and documents, looking for those notes in the margins that often provide so much insight.” (p.4)

Synaesthesia

In this dissertation a simple modular strategy has been adopted in exploring the various media streams which, in combination, constitute a multimedia artwork. This modular analysis is based on what is accepted as the natural division of human perceptual mechanisms – the senses. Motluk, [1995] explains that this modular theory of sensory perception is thought to be an adaptive strategy.

> Modularity is held to be the answer of evolution to the development of maximally efficient information processing problems.

While it is convenient to think of sense experience in this way, human experience could in fact just as easily be viewed as wholistic - the sensory information combine to give a complex, merged perceptual experience. Sense perceptions interact to give the whole experience, hence it is important in the present context to also consider the *relationship* or *interaction between* the various sensory modalities.

Ultimately any analysis of perceptual responses to multimedia must be underpinned by an understanding of the neurophysiology of sensation itself. Motluk, [1995] points out that:

> Neuroscience has provided a number of models of how the brain might operate, though currently there is no consensus regarding which model is likely to prove the most fruitful. Current perspectives appear to owe much to the notion of a hierarchy of science and so do not as much compete as models of brain function as provide explanations couched in vocabulary appropriate to the discipline... In order for us to 'know' that a percept is visual, auditory, olfactory, etc. we must have developed a method of identifying information as being of one sensory kind or another. There is therefore presumably a modular structure to sensation which allows for discrete identification of information as being specific to a sensory system…

It must however remain beyond the scope of this study to exhaustively summarize the scientific literature with regard to this complex field. Nevertheless, concepts introduced herein, such as Guay's "semantic compression" raise intriguing possibilities in relation to the form, function and aesthetics of multimedia art, and suggest the need for an understanding of the cognitive interface between sensory modalities (i.e. the biochemical interaction which occurs in the brain when the two main sensory input channels, sight and audition, are manipulated simultaneously).

A fascinating branch of neurophysiology, which has resonance with regard to the multimedia experience, has developed in response to the phenomenon known as *synaesthesia*. Derived from the Greek *syn* - union and *aesthesis* - sensation, synaesthesia means the involuntary physical experience of a cross-modal association. In this condition a sensory experience normally associated with one sense modality occurs when another modality is stimulated. Frequently the merging of the sensory input and the secondary sensation seem to occupy the same sensory space. A common example would be the idea that hearing a certain sound induces the visualization of a certain colour.

Gerstley [1996] describes it thus:

> Synaesthesia is a literal crossing of the senses. Those with the condition (may) "smell" colours, "see" sounds, "feel" tastes. The quotes around these associated senses are appropriate only for individuals without synaesthesia; for those who enjoy the condition, the secondary sense is as real as the primary sense.

Gerstley states that estimates on the incidence of synaesthesia vary considerably. He mentions one (undocumented) estimate of incidence at 1:300,000, but points out that this can vary with the definition of what types of experiences constitute synaesthesia. Motluk [1995] puts the incidence of some forms of synaesthesia much higher:

> New surveys show that around one in every 2000 people hear words, letters and numbers as distinct colours. The results also show that the condition may be inherited… possibly X-linked and dominant.

Sceptics see no neurological basis for synaesthesia, believing it to be simply an extreme form of association. For example, the not uncommon colour-to-letter association may be 'imprinted' upon a child by using lettered building blocks of a certain colour, or by reading and re-reading an ABC book (A = apple = red/green etc). However there is evidence that the phenomenon of synaesthesia has a physical basis.

> When neurologists carried out brain scans on colour synaesthetes, they discovered increased blood flow in those sections of the brain dealing with the perception of colour - a phenomenon not present in non-colour synaesthetes. This implies that synaesthesia has a genuine neurological basis and that perhaps there are additional links between perception areas in the brains of those with synaesthesia. Another implication that synaesthesia is neurological in origin can be seen in that many non-synaesthetes will experience synaesthetic effect as a result of neurological change, due to migraine, epilepsy, brain injury or the effects of certain (especially hallucinogenic) drugs [1].

Costa [1996] warns that much discussion in the literature of the phenomenon of synaesthesia is based on subjective reports, and consideration of these must be tempered by the observation that they are only as scientifically valid as the thousands of UFO sightings. Nevertheless, Costa goes on to report that:

[1] http://www.bbc.co.uk/dna/h2g2/alabaster/A558371 ; accessed 29-12-02.

> There is new neurophysiological evidence strongly supporting the fact that cross-modal associations do take place in the mammalian brain. In one experiment described in Rauschecker (1995), portions of the anterior ectosylvian cortex typically dedicated to vision (AEV) became responsive to auditive and somatosensory input as a consequence of cortical plasticity responses to visual deprivation in cats. Such a phenomenon implies not only improved accuracy for analysis of auditory and somatosensory information, but also raises the hypothesis that part of the auditory (or somatosensory) stimuli can be "seen" by some individuals with specific lesions or developmental abnormalities. Similar results have also been reported respectively to monkeys and humans (PET and ERP were used in the latter case). Based on such findings, it seems that cross-talk between different modalities can indeed be produced by rewired afferent connections. It is also possible that cross-modal interferences are present even in the normal cortex, though their diminished scale prevents them of being clearly perceived….

The English language is rich with metaphor that has evolved with a synaesthetic connotation.

> Commonplace examples of synaesthetic metaphors in English include phrases such as "loud colors", "dark sounds", and "sweet smells". Tabulations of the frequency of types of synesthesia and synesthetic metaphors in English reveals that for physiological synesthesia, colored sounds are most common; in English literature, synesthetic metaphors employed for descriptions of tactile sound predominate. Of the various senses, hearing is most frequently expanded and elaborated upon by both synesthetic sensory perceptions and synesthetic metaphors. Synesthetic "visual hearing", which antedates language, may have influenced language development. [Day, 1996, 1].

The widespread acceptance of these kinds of metaphor in literature and everyday speech is suggestive of the possibility that human sensory mechanisms are not the fixed, closed systems we generally assume them to be. Perhaps we all fall somewhere along a variable scale of synaesthetic sensitivity, however slight. This view is offered some support by Watkins [1997], who argues that:

> Synaesthesia can sometimes be induced in non-synaesthetic people through the use of drugs such as L.S.D. Such people show brain activity which correlates with emotional change, thus suggesting that the emotions may be related to synaesthetic perceptions. This, however, is an empirical question yet to be tested. The fact that synaesthesia can be induced in non-synaesthetes implies that the ability to experience it lies dormant in everybody, and will be brought to consciousness when conditions are optimal.

According to Day [1996], theories pertaining to the understanding of synaesthesia fall into two streams. The first is a linguistic approach, which follows from theories of metaphor, connotation, and association. This stream of thought he traces back to early Greek rhetoric, and gives the example of Aristotle's discussion of "common sensibles" (c.330 B.C.); the implication of the linguistic approach to synaesthesia is that an understanding of the phenomenon must ultimately be tempered by considerations of cultural relativity – [p.8], i.e.:

> …other cultures have other concepts as to what comprises a mode of perception and have other counts and divisions of those perceptions

The second major strand of thinking with regards to synaesthesia has only emerged since the age of empirical science, attempting to employ measurable scientific approaches and to follow on from theories of physical, neurological disorders and/or psychosis.

The 17th Century French philosopher, Rene Descartes, proposed the notion of Dualism, which posits the existence of two domains, *res extensor* (the physical world) and *res cogitans* (the mental world). Motluk [1995] rejects Cartesian Duality, with a claim that most scientists work from the assumption that the relationship between 'mind' and 'brain' is that they simply constitute different vocabularies for discussing the same thing. In this sense mind is a functional concept and brain the structural equivalent, i.e. mind is the activity of the brain.

Motluk [1995], pursuing the logical consequences of an acceptance of modularity, points out that there are no differentiated physical structures specific to sense modalities – i.e. the same cellular structures process both visual and auditory perception:

> …a break-down in this modularity may yield a percept which whilst auditory in origin, has a visual quality to it ... Our explanation (of the physiology of sensation) suggested that in the normal case, speech and colour perception rely on entirely independent modular systems… In cognitive terms, this would be true if speech and colour perception were 'informationally encapsulated' (Fodor, 1983). In neural terms, this would be true if they had their own class of cells and structures.

The neurophysiology of sensation is a broad field of enquiry, both extremely complex and far from mature, and appears to be a long way short of reaching an accepted scientific model. Therefore science cannot yet claim to understand synaesthesia. Motluk [1995] offers one possible explanation:

> … that neonatal pathways between auditory and visual areas of the brain are perpetuated beyond the neonatal period in which they are found in normal individuals. Consequently, auditory information finds its way to those parts of the brain that normally deal exclusively with visual information.

There appears to be no universally accepted criteria for even diagnosing synaesthesia. One of the leading commentators, Cytowic, [1995] has proposed five elements as a practical basis for such a diagnosis:

- The secondary (and possibly tertiary and more) sensations are elicited involuntarily.
- The secondary sense is projected
- The sensations are durable, discrete, and generic
- The sensations are memorable
- Finally, the elicited sensation are emotional and noetic.

To put it simply, genuine synaesthesia is spontaneous, specific, consistent and durable. Synaesthetic form is usually referred to by its secondary sense first. Thus, a coloured-hearing synaesthete is one who experiences coloured photisms in response to hearing stimuli.

Synaesthesia is said to be additive - that is, another sensation is added to the initial or primary sensory perception, rather than replacing one perceptual mode by another. For example, with synaesthetically coloured perception of musical keys, an ability attributed, for example, to American pianist and composer Amy Beach[1], the synaesthete can both hear and "see" the sounds; the visual images do not replace the auditory sensations. Both sensory perceptions may thus become affected and altered in the ways they function and integrate with other senses [Day, 2001].

Coloured-hearing is usually referred to as the most common form of synaesthesia. Although there are theoretically a potential 20 types of two-sense synaesthesia, not all of these forms have been recorded. Some of the pairing that are most commonly recorded [Cytowic, 1995] are:

- Coloured-Hearing (also known as chromaesthesia)
- Coloured-Olfaction
- Coloured-Gustation
- Tactile-Gustation

Synaesthesic experience is generally a one-way interaction; that is, for example, for a given synaesthete, though tastes may produce synaesthetic “sounds”, but sounds will not produce synaesthetic responses. However, there have been a few rare cases of bi-directional synaesthesia, in which, for example, music induces (synaesthetic) colors and seeing colors induces (synaesthetic) sounds. To further complicate an understanding of the phenomenon, the correspondences are not in fact the same in both directions [Day, 2001].

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[1] 1867-1944 (her career flourished 1900 - 1920's).

Gerstley [1996] claims there are individuals in whom three or more senses are crossed, and also subjective reports of those that have bi-directional synaesthesia. He refers to the most famous case of synaesthesia:

> ... described in great detail in Luria's wonderful book, *The Mind of a Mnemonist.* In the book, Luria recounts a patient known as S., who was noted for having an almost unlimited memory.[1] S. was synaesthetic through *four senses,* and his synaesthetic perceptions were not only vital to his memory strategy, but also interfered with his day to day activities. For example, the images evoked by a person's voice could interfere so much with the content of the person's speech that S. could not understand what was being said. Another curious problem suffered by S. was his inability to understand metaphoric speech. In a sense, his normal perceptions were concrete metaphors, and artificial ones imposed upon him conflicted in a disagreeable and incomprehensible fashion. In addition to S., a woman in Britain is reported to have bi-directional coloured-hearing. In her case, it has been reported that a traffic light changing colour may evoke a bell-like tone, as well as sounds producing "photisms".

There are forms of synaesthesia that do not seem strictly to belong in a category with the above-mentioned forms. These are synaesthetic associations that pair a secondary sense (usually colour or shape) with language, or particular elements of a category. These forms are referred to by a variety of titles, including pseudo-chromaesthesia. Gerstley [1996] states that this latter term was in vogue in the early part of the century, and then seemed to lose favour and vanish from use. Gerstley [1996] offers the following examples of these pseudo-chromaesthetic pairings:

- Coloured-Numbers
- Coloured-Letters
- Coloured-Graphemes
- Coloured-Phonemes
- Shaped-Numbers (also referred to as number form)

Pythagoras considered synaesthesia to be the greatest philosophical gift and spiritual achievement. John Locke also raised the subject in *An essay concerning human understanding* (1690). [Lyons, 2002].

Of documented chromaesthetes, one of the most distinguished was the Russian Romantic composer Alexander Nikolayevich Scriabin (1872-1915). However, it remains uncertain as to whether colour-music associations were personally experienced by Scriabin, or whether he simply used what was, at the time, a

[1] One of the crucial side-effects claimed for synaesthesia is that it improves memory and recall. *"The synaesthetic experience gives additional associations for names, numbers and sounds, which can provide a vivid link to the information, and the more forms of synaesthesia a person experiences, the better their memory is likely to be".* (http://www.bbc.co.uk/dna/h2g2/alabaster/A558371 ;accessed 30-12-02.) It is interesting to speculate what this effect has had on the careers of the extraordinary number of notable artists who claim to experience the phenomenon of synaesthesia.

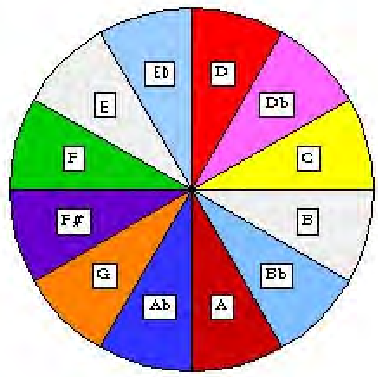
'fashionable' concept as a basis for an artistic exploration of music[1]. Either way, Scriabin was an important pioneer of multimedia art, visualizing his musical compositions to be multi-sensory experiences, involving smell, colour, feeling, etc. In his 1911 composition *Prometheus*, one of his most controversial works, Scriabin scored a separate instrumental part for clavier luminiere or colour organ, also called *Fiestra per Luce*, or keyboard of light [Milicevic, 2000].

> The score of *Prometheus* contains a line designated *"Luce";* this was for a light organ, playing two lines: one to correspond to Scriabin's concepts of the "correct" colors for each musical key, as he modulated from key to key; the other, to counter the first lines colors. Scriabin and others were unable to realize a light-music performance of *Prometheus* until its premier performance in New York, in 1915, where, rather than using a color organ, colored light was projected onto a screen set above the orchestra performers' heads, using a system designed by Preston S. Millar, W.F. Little, and William McKay. [Day, 2001].

Scriabin wanted everyone in the audience to wear white clothes so that the projected colours would be reflected on their bodies and thus possess the whole room [Moritz, 1997].

This table below shows Scriabin's key-colour equivalences:

C#	-- Purple
F#	-- Bright Blue/Violet
B	-- Blue
E	-- Sky Blue
A	-- Green
D	-- Yellow
G	-- Orange
C	-- Red
F	-- Deep Red
Bb	-- Rose/Steel
Eb	-- Flesh
Ab	-- Violet
Db	-- Purple (same as C#)
Gb	-- Bright Blue/Violet (same as F#)



Above: This colour wheel, purporting to depict Scriabin's key-colour correspondences, by Dozier-Stefanuk [1998], shows inconsistencies with Moritz's table on the left, but also internal inconsistencies.

Dozier-Stefanuk [1998] accepts Scriabin as a bona fide synaesthete and makes some interesting observations regarding his synaesthesia. However her colour wheel visualization of his key-colour relationships seems flawed, reminding us that while this area of study continues to fascinate, research material can often be unreliable. Nevertheless her dedicated web-site is mentioned because it makes one valuable observation:

[1] While there is no hard evidence of Scriabin being a poly-sensory synaesthete, according to Lyons, [2002], a subjective account of his experience of synesthesia was published in a 1915 article in the *British Journal of Psychology*.

> It is interesting to try to imagine what Scriabin saw in his mind's eye when he heard more than one pitch at a time. Was it a mixture of the colours, or a layering of colour?

While the literature attributes Scriabin's correspondences as being between *key* and colour, it remains an interesting point to consider in relation to this field of research.

Scriabin spent the last 12 years of his life, from 1903 to 1915, planning what he was to call *Mysterium*, a composition of inconceivable proportions, that would combine elements of all the senses, including odours[1]. This monolithic composition was never realized however. [Dozier-Stefanuk, 1998].

Another major musical figure associated with synaesthesia is the composer Olivier Messaien, who:

> …gives a particularly florid description of his colour sound associations in his 1947 account of "the gentle cascade of blue-orange chords" in the piano part of the 2nd movement of *Quator pour la fin du temps*. He is later more specific about the nature of his synaesthesia when he states he sees "colours which move with the music, and I sense these colours in an extremely vivid manner". [Baron-Cohen, 1996].

Synaesthesia has had an on-going influence on the arts. Day [2001] lists composers Franz Liszt, György Ligeti, and Jean Sibelius as "true" synaesthetes. Synaesthetic tendencies have been claimed for British visual artist David Hockney, Russian film-maker Sergei Eisenstein and French poet Charles Baudelaire. Russian novelist Vladimir Nabokov was a self-confessed synaesthete, who wrote vividly of his childhood explanation of the phenomenon is his autobiography[2].

Day, who claims to personally experience synaesthesia, discusses the use of synaesthetic concepts in *Fantasia* [1940, Disney]:

> One of the main themes of this animation film was to put pictures to pieces of orchestral music – in a sense, an early version of MTV. The results, however, did not produce much in the way of synaesthetically motivated art – with the noticeable exceptions of the opening piece, the abstract colors accompanying J.S. Bach's Toccata and Fugue in d-minor. More direct to synaesthesia, but constantly overlooked in the movie Fantasia, is the short divertissement section featuring an animated oscilloscope-like "sound track" which presented shapes and colors for various instruments. These colors and shapes bear some similarities to the types of actual perceptions colored-music synaesthetes experience.

In the same article Day reports that contemporary composer Michael Torke (who Day considers is "definitely" a synaesthete) experiences an unusual form of

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[1] http://www.psychiatry.cam.ac.uk/isa/arts.html ; accessed 28-12-02.
[2] http://www.bbc.co.uk/dna/h2g2/alabaster/A558371 ; accessed 30-12-02.

synaesthesia - coloured time units (days of the week, years, and such). Torke composed *Color Music* in 1991, and (according to Day) is currently on contract with Walt Disney studios to write music for films.

Reference has been made in an earlier section (p. 46-47) to suggestions that the focus of contemporary art has shifted during the twentieth century from a concern with the appearances of things towards an exploration of consciousness itself. If, as Watkins [1997] suggests, all humanity falls somewhere on a spectrum of synaesthetic ability, then a multimedia has tantalizing potential in relation to artistic explorations of consciousness.

> The phenomenon, synaesthesia, has been shown to be real in some people, and it has been suggested that it actually occurs in most people, but at a level beneath which we can access consciously. In light of this, suggestions have been made which would apply synaesthesia to popular studies of consciousness in order to throw light on the previous results...synaesthesia may be the key which will help unlock the secrets of the mind. [Watkins, 1997, 10].

In his multimedia performance piece, David Cubby [1995] eloquently places the concept of synaesthesia in relation to multimedia art:

> The notion of synaesthesia culled from the traditional psychology of perception has yet to be adequately invoked around an accredited dialogue across art and design. It appears now as an invigorated concept, a term on the rise, interestingly appropriate to multimedia, referring to similarities between cognitive and digital confusion, (and to) fusion, manipulation and morphing of sensory data. Multimedia provides a synergetic state similar to the deep neurologic interplay of sense data that supplies the notion “synaesthesia “- the dynamic point where vision/sound (static and mobile), smell, taste, touch enhance existing forms or draw on the possibility of new form.

Hugo [1995] takes this ‘invigorated concept’ one step further and hypothesizes what he calls *tele-synaesthesia*.

> The new media and the Internet enable us to experience different kinds of information which are of a specifically telematic nature and for this reason effectively differ from the usual forms of communication. By linking the concepts tele and synaesthesia to each other, we deal with the fact that the transmission of data creates a synaesthetic effect: tele-synaesthesia. At the end of the 20th century, the practicable units of time have become digitalised, magnified, and incredibly accelerated. The modalities of our sensorial perception become interactive by means of electronic mechanisms of control and selection. A tele-culture is emerging, subjecting both the perceptual and the conceptual to —strictly speaking— continuous revision.

Technological simulations of synaesthetic phenomena

There is a history of experimental technology inspired by, or aimed at reproducing, the synaesthetic experience, which constitutes a sort of pre-history of multimedia.

> Ancient Greek philosophers, like Aristotle and Pythagoras, speculated that there must be a correlation between the musical scale and the rainbow spectrum of hues. That idea fascinated several Renaissance artists including Leonardo da Vinci (who produced elaborate spectacles for court festivals), Athanasius Kircher (the popularizer of the "Laterna Magica" projection apparatus) and Archimboldo who (in addition to his eerie optical-illusion portraits composed of hundreds of small symbolic objects) produced entertainments for the Holy Roman Emperors in Prague. [Moritz, 1997].

Athanasius Kircher's system of correspondences between musical intervals colours in his *Musurgia Universalis* of 1662, was based upon complex traditional symbolisms rather than synaesthetic perception. These symbolic associations are presented in the table on the right [Day, 2001]:

octave	green
seventh	blue-violet
major sixth	fire red
minor sixth	red-violet
augmented fifth	dark brown
fifth	gold
diminished fifth	blue
fourth	brown-yellow
major third	bright red
minor third	gold
major wholetone	black
minor second	white
minor wholetone	grey

Marin Cureau de la Chambre [1650], proposed a different scheme of coloured musical intervals, derived from Aristotle:

double-octave	black
twelfth	purple
eleventh	blue
octave	green
fifth	red
fourth	yellow
base	white

In the 18th century Isaac Newton compared the spectral colours that were produced by splitting the suns light with a prism with the musical notes of the tempered scale.

> Newton's premise was that color and sound were, at the very least, analogous. He observed that both light waves and musical tones involved vibrations that could be measured and thus arranged in a scale-like pattern. Since there were seven pitches in the diatonic scale, he came up with a spectrum of seven colors: red, orange, yellow, green, blue, indigo and violet, corresponding to the pitches of a B-flat major scale beginning on the second note. Later scientists argued that Newton's division of the spectrum was arbitrary, that there was an

> infinite number of shades in between these colors. But for Newton that was an insignificant detail. Like Rimsky-Korsakov, he would have seen the color green and heard, in his mind's ear, the pitch F. [Wierzbicki, 1990].

According to Day [2001], Newton later discarded notions that there was any true connection between colours and the musical scale.

Castel, an 18th century mathematician and physicist, possibly influenced by Newton, was a firm advocate of there being a direct solid relationship between the seven colours and the seven units of the scale. Milicevic [2000] recounts that:

> Following Newton, the Jesuit, Bernard Castel constructed his *clavcien oculair* (ocular harpsichord) around 1725, which consisted of two colored discs (in 12 parts corresponding to the 12 semi-notes of the tempered octave) connected to a harpsichord. He believed that a base note, (which we call C) rendered a firm, tonic color serving as a foundation for all colors, and that that was blue.

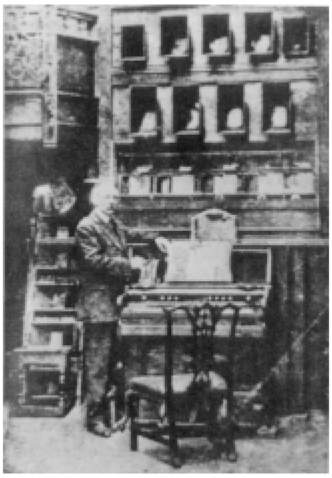
B	=(dark) violet
Bb	=agate
A	=violet
Ab	=crimson
G	=red
F#	=orange
F	=golden yellow
E	=yellow
Eb	=olive green
D	=green
C#	=pale green
C	=blue

The relationship between sounds and colours, according to Castel, is presented in the table on the right (Day, 2001).

Castel's working model of the Ocular Harpsichord had a 6-foot square frame built above a normal harpsichord; the frame contained 60 small windows each with a different colored-glass pane and a small curtain attached by pullies to one specific key,so that each time that key would be struck, that curtain would lift briefly to show a flash of corresponding color. High society was dazzled and fascinated by this invention, and flocked to his Paris studio for demonstrations. A second, improved model built in 1754 used some 500 candles with reflecting mirrors to provide enough light for a larger audience, but the clumsy technology generated fumes and heat, with considerable chance of noise and malfunction between the pullies, curtains and candles. No physical relic of it survives. [Moritz,1997].

That the synaesthetic experience has fascinated generations of enquiring minds is given testament by Wierzbicki [1990] who states that:

> Erasmus Darwin, in 1789, produced a similar contraption that used oil lamps and colored glass instead of candles and tapes. In 1844 there was another one, by someone named D.D. Jameson. And in 1877 there was still another, a light-emitting organ built by the American composer Bainbridge Bishop.


Rimington at his colour organ.

Electricity opened new possibilities for projected light. The work of Scriabin, mentioned above, which fell within this experimental techno-synaesthetic tradition, led to a collaboration with the British painter A. Wallace Rimington, whose electric Colour Organ facilitated the moving lights that accompanied the 1915 New York premiere of Scriabin's synaesthetic symphony *Prometheus: A Poem of Fire* (which, as we have seen, had indications of precise colours in the score). Rimington's book describing the colour organ and the colour theories on which it was based was published in 1911 as *Colour music: The Art Of Mobile Colour* [Lyons, 2002].

Apart from Scriabin, a second important Russian composer, Nikolai Rimsky-Korsakov [1] is another artist of historical significance said to have synaesthetically associated musical keys and colours.

His set of colour-key relationships is presented in the table on the right. That this is a rather murky area of research is evidenced by the fact that discrepancies arise between the above table and another list of Rimsky-Korsakoff's colour-key correspondences, quoted by Wierzbicki [1990], who cites:

B major	gloomy, dark blue with steel shine
Bb major	darkish
A major	clear, pink
Ab major	greyish-violet
G major	brownish-gold, light
F# major	greyish-green
F major	green, clear (colour of greenery)
E major	blue, sapphire, bright
Eb major	dark, gloomy, grey-bluish
D major	daylight, yellowish, royal
Db major	darkish, warm
C major	white

> …the article on Color and Music in the wonderfully musty 1938 edition of the Oxford Companion to Music.

Wierzbicki also lists Scriabin's colour-key correspondences, and here again there are discrepancies with Day's list. Both researchers seem to slip between referring to a key-colour relationship, and a pitch–colour relationship, as though it has no bearing on the interpretation of the phenomenon, which seems sloppy. If it is actually key-

[1] Day [2001] attributes this claim to an article in the Russian press (Yastrebtsev 1908), cited by Galeyev & Vanechkina (2000). Wierzbicki [1990] states that Rachmaninoff confirmed through personal contact that Rimsky-Korsakoff made claim to synaesthetic ability.

colour correspondences that these eminent composers were referring to, it is difficult to understand how the major – minor key dichotomy is not mentioned as a factor influencing perceptual responses. If it is key related, then it seems to this writer that synaesthetically produced colour associations would produce a slowly changing internalised colour experience that reflected the architectural/structural form of the music, rather than an rapidly changing subjective colour experience, of the kind associated with contemporary stage lighting technology effects, i.e. a kaleidoscopic lightshow with its emphasis on rapid rhythmic changes, mimicking musical events.

Wierzbicki sees the few instances of agreement between the two Russian composers established by this comparison of their stated key-colour associations as significant:

> Why Rimsky-Korsakov was unable to "see" the pitch A-sharp remains a mystery. But he did claim to "see" all the other pitches. Like Scriabin's, his musical scale was chromatic in more ways than one. And different though they are in their details, the scales' occasional correspondences seem more remarkable than their contradictions (Wierzbicki, 1990).

Key	**Scriabin**	**Rimsky-Korsakov**
C	red	white
C#	violet	dusky
D	bright yellow	yellow
D#	steel gray	bluish gray
E	bluish white	sapphire blue
F	red	green
F#	bright blue	grayish green
G	orange-rose	brownish gold
G#	purple-violet	grayish violet
A	green	rosy
A#	steel gray	----
B	bluish white	dark blue

The following table compares the pitch or key-colour relationships proposed by Castel, Rimsky-Korsakov and Scriabin, (using Day's correspondences for Rimsky-Korsakoff and Scriabin, so that the reader may inspect the discrepancies between the two sets of listings). It shows Castel's system has no correlation with those of Scriabin and Rimsky-Korsakov, but of course there is no evidence to suggest Castel was a true synaesthete:

Pitch or key	**Castel** (pitch)	**Rimsky-Korsakov** (key)	**Scriabin** (key)
B	=(dark) violet	gloomy, dark blue with steel shine	Blue
Bb	=agate	darkish	Rose/Steel
A	=violet	clear, pink	Green
Ab	=crimson	greyish-violet	Violet
G	=red	brownish-gold, light	Orange

F#	=orange	greyish-green	Bright Blue/Violet
F	=golden yellow	green, clear (color of greenery)	Deep Red
E	=yellow	blue, sapphire, bright	Sky Blue
Eb	=olive green	dark, gloomy, grey-bluish	Flesh
D	=green	daylight, yellowish, royal	Yellow
C#	=pale green	darkish, warm	Purple
C	=blue	white	Red

However Rimsky-Korsakov and Scriabin agree with regards to the key of D, and there is a reasonable overlap between their respective synaesthetc perceptions for the keys of B, Ab, G and E. For Wierzbicki [1990],

> …all it proved was that Scriabin and Rimsky-Korsakov "saw" musical sounds, and "heard" colors, in their own ways. That stimuli of one sense organ triggered responses in another was something in which they adamantly believed. For them, the so-called synesthetic reaction was both constant and consistent.

The next interesting experimentation in the area technologically-based synaesthetic experience was carried out by members of the Italian *Futurist* movement in the early years of the 20th century[1]. One of the co-authors of *the Futurist Manifesto*, Bruno Corra, constructed his own version of the colour organ using a series of 28 coloured electric light bulbs, corresponding to twenty-eight keys, and composed colour sonatas for the new instrument. He also published an article entitled *Abstract Film – Chromatic Music* in 1912, in which he described experiments carried out with his brother Arnolda Ginna, hand painting some nine abstract films directly on film-stock in 1911. Unfortunately these films have been lost [Milicevic, 2000; Moritz 1997].

Bruno Corra's overlap of interests between a colour organ approach, and the use of film technology to merge audio and visual perception in an abstract, non-narrative space indicates the beginning of a new direction that this form of experimentation began to take, with the emergence of film technology. This present discussion will deal primarily with the colour organ and essentially theatrical approach to simulating synaesthetic experience. In a later section the history of experimental abstract film will be discussed in detail (see p. 220).

Wassily Kandinsky, one of the co-founders of non-objective painting, claimed a synaesthethic ability, enabling him to experience certain colour tones in relation to musical notes. This ability inevitably permeated his artistic sensitivity.

> He describes an experience of Wagner's Lohengrin during the early 1890s: "*All my colours were conjured up before my eyes. Wild, almost mad lines drew themselves before me. I did not dare to tell myself in so many words that Wagner had painted `my hour' in music. But it*

[1] The *Futurists* ideas about a wholistic meta-artform are discussed in the next section, see p. 136.

> *was quite clear to me that painting was capable of developing powers of exactly the same order as those music possessed."* (Lyons, 2002).

Apart from producing an historically important body of paintings, Kandinsky attempted to connect music and visual images through his ideas on abstract theatre, published in 1912 in an essay entitled *On Stage Compositions* in the journal *The Blue Rider* [Milicevic, 2000].

> His central idea, which he geared to his entire stage works, was that of a "totally synthetic work of art". Kandinsky connected with the term "synthesis" the idea of fundamental "translatability" of the various arts: namely that color impressions triggered sounds in him and vice versa, that hearing music transformed itself into artistic impressions. In view of this, Mussorgsky's *Pictures at an Exhibition* presented itself almost ideally as a "stage synthesis" for a scenic realization on the stage.[1]

Kandinsky was granted only a single opportunity of realizing his concept of a “totally synthetic work of art”. Playing the role of director and set-designer, he staged a visual realization of Modest Mussorgsky’s composition, *Pictures at an Exhibition,* at the Friedrich Theater in Dessau in 1928 [2]. The production of the 16 musical episodes or “pictures” resulted in 16 choreographed sculptural stage events, created from mechanically moving geometric shapes, colours, lines and lighting effects. The audience experienced the “visit to the exhibition” as a succession of abstract, moving pictures on the stage. The choreographed scenic designs followed Mussorgsky's music “to the letter” [3].

> The effect of individual acts was large and magical: the floating down of the circle in front of mysterious dark, its lighting up and its cooling down - or the movement of a white square over the stage. The floating down of the arch in the catacombs was unforgettable. You could feel the drama, the effectiveness of shapes and colors. The floating, gliding and standing of the shapes, the changing of the colors depending on the type and intensity appeared as a dramatic sequence full of excitement. [Ludwig Grote: Bühnenkompositionen von Kandinsky, 1928] [4] .

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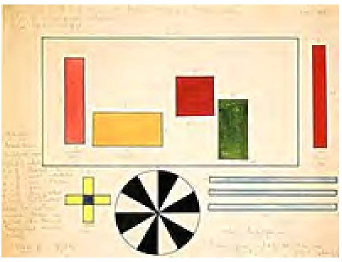
[1] www.bildklang.de/Englisch/Initiation_2.htm ; accessed 26-12-02.
[2] http://www.bildklang.de/Englisch/Bildklang_E.htm ; accessed 29-12-02.
[3] www.bildklang.de/Englisch/Performance_2.htm ; accessed 26-12-02.
[4] http://www.bildklang.de/Englisch/Bildklang_E.htm ; accessed 29-12-02.

Right: *The Great Gate of Kiev,* 1928.

Kandinsky's closing set design for *Pictures at an Exhibition*, musical score by Modest Mussorgsky.

(*Picture XVI*, Tempera and watercolour).



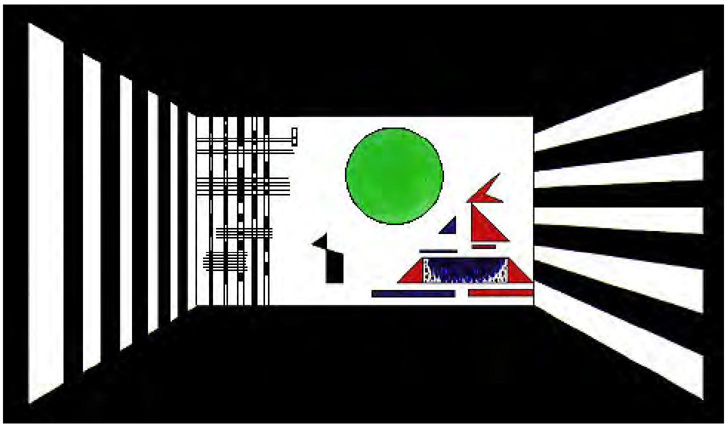
Picture VII, Bydlo, 1928.

Picture XIII, Catacombae, 1928.

This work has recently been recreated in the form of computer animation. This was a collaboration between post-graduate researcher and animator Karoline Hofman, pianist Stefan Danhof, with technical implementation by computer scientist Martin Hofman. They attempted a precise implementation of the director's instructions. Kandinsky's scene designs are shown on a screen in a darkened room in the form of moving images. These visual elements are stored in a computer as multimedia sequences and are projected on a large scale with the aid of a connected video projector. The synchronization of music and video projection is achieved through the interaction between computer animation and the live musician. As well as the live performance, it is also possible to set up the presentation as a permanent installation, e.g. in museums, using a recording of the music [1].

[1] http://www.bildklang.de/Englisch/Bildklang_E.htm ; accessed 29-12-02.

Right: Still from the Bildklang *Shockwave* visualization (computer simulation) of Kandinsky's abstract stage realization of Mussogorsky's *Pictures at an Exhibition*.



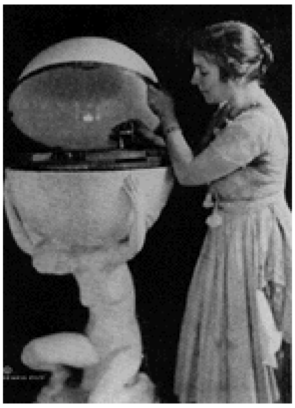
During the 1920s two prominent colour-organ artists entertained American and international audiences. One was Danish-born Thomas Wilfred, who originally arrived in America as a vocalist, specializing in early music. He came into contact with a group of Theosophists who were interested in building a colour organ, with the idea of using it to express spiritual principles. Wilfred developed his own colour organ, which he named the *Clavilux.* He also conceptualized an artform of colour-music projections, which he called *Lumia*. He described it in terms of polymorphous, fluid streams of slowly metamorphosing colour. Wilfred went on to establish an *Art Institute of Light* in New York, and gave *Lumia* concerts in the U.S. and in Europe at the Art Déco exhibition in Paris. He also built "lumia boxes," which were self-contained television-like units, which could play for days or months without repeating the same imagery. (A parallel between Wilfred's *Lumia* and the video animation of Duane Radfords paintings which comprise one section of the accompanying creative component is drawn in Part Two of this dissertation.)

The second significant inventor and performer of a colour-organ type of technology was Mary Hallock Greenewalt. A renowned concert pianist and an ex-student of the eminent Theodore Leschetizky, Greenewalt built a career based on skillful interpretations of Chopin, including recordings for Columbia Records [1]. Wishing to control the ambience in a concert hall for sensitive music like Chopin's, she began to experiment with light modulation [Moritz, 1997] devising a keyboard instrument that interpreted music into light and colour.

[1] www.wichitaartmuseum.org/character.html ; accessed 26-12-02.

This remarkable woman was a skilled electrician, a member of the Society of Electrical Engineers, who eventually held 11 patents to her name. These included the rheostat, enabling smooth fade-ups and fade-outs of light, and the liquid-mercury switch, both of which have become standard electrical devices, but were fundamental to her colour organ [1]. She published a book of instructions on the playing of her instrument. She continued to perform on her colour-organ, which she called the *Sarabet*, creating a special notation that recorded the intensity and deployment of various colours in a piece.

In her *Notation For Indicating Lighting Effects* (US Patent 1,385,944, July 26, 1921), the notation was written left to right on a row of dots, one per beat of the light score, allowing it to be interlaced with a traditional music score and synchronized with sound events [2].



Right: Mary Hallock Greenewalt with another synaesthetic invention, her Visual-Music Phonograph (1919.) [Moritz, 1997].

Synaesthesia and computers: the digital interface between sound and image

The Fraunhofer Institute study [Astheimer, 1995] concludes that the visual sense is the most important means of perceiving information in day-to-day human experience.

Sound is the next most important sensory information input modality. Humans discriminate between sounds, filtering for cues, which aid us in navigating through our local environment. Auditory cues offer us directional information and also offer us warning information about sources of danger (e.g. a hungry tiger, an approaching car, etc.) including spatial, and proximity information. We have the ability to prioritize sound, for example learning to ignore repetitive traffic sounds, the sound of a flowing creek etc. Sounds are accents to visual information. We develop a subconscious sensitivity to sounds and silences.

[1] When competitors (including Thomas Wilfred) began infringing on her patents by using adaptations of the rheostat and mercury switch, a judge ruled against her application for damages, denying her case on the grounds that these electric mechanisms were too complex to have been invented by a woman [Moritz, 1997].
[2] W. Shawn Gray, http://www.auzgnosis.com/akns/akns1-1_a1.pdf. Accessed 26-12-02.

The combination of auditory and visual stimulation increases our knowledge and awareness of our immediate environment exponentially. Visual and sound cues interact.

Acoustic events tell us where to look, focus our attention. In nature the sense inputs are “natural”, i.e. parallel and synchronous. Dislocating the synchronization of parallel or “appropriate” stimuli results in disorientation, “unnatural” experience, or what could be described as a kind of altered state of consciousness. (See the table on page 135.)

The tantalizing enigma of synaesthesia offers suggestions of art possibilities which have been taken up by software engineers. An interesting attempt at playing with the possibilities occurring at the interface between the visual and the aural is embodied in the software sound synthesis technology or “graphic sound design and composition environment” marketed under the name *MetaSynth* (created by programmer and artist Eric Wenger, the inventor of equally innovative graphic design software *KPT Bryce*) The promotional web-site for this product describes it thus:

> MetaSynth combines sampling, wavetable, additive, subtractive, granular and frequency modulation techniques with the most intuitive means for visualizing audio information ever created. MetaSynth's Image Synth palette literally lets one paint with sound. Draw in, harmonics and undertones, or paint a vivid tone picture in broad strokes. Users can even import any PICT file and use that picture as a starting point for a new sound or composition.
>
> MetaSynth works with audio just like a graphics program; grab with the hand tool to flip or resize the image, apply brushes, filters or gain/contrast tools. Even non-musicians can instantly interpret and use the Image Synth's picture window ... it's light-years away from confusing conventional notation and grid displays. Audio professionals will love the deep effects toolset, powerful real-time wavetable and FM windows, elegant envelope generators and the sheer flexibility of MetaSynth's design.
>
> Metasynth maps picture coordinates into time and frequency domains. The gray scale of each pixel controls the volume of a specific harmonic at a specific moment in time. Stereo position is determined by pixel color. [1]

[1] . URL: http://www.harmony-central.com ; accessed 6-12-02.

Above: Interface for MetaSynth: white jagged lines are the “painted” grayscale image. The sound generated algorithmically by the software is depicted by a waveform envelope rendered in green.

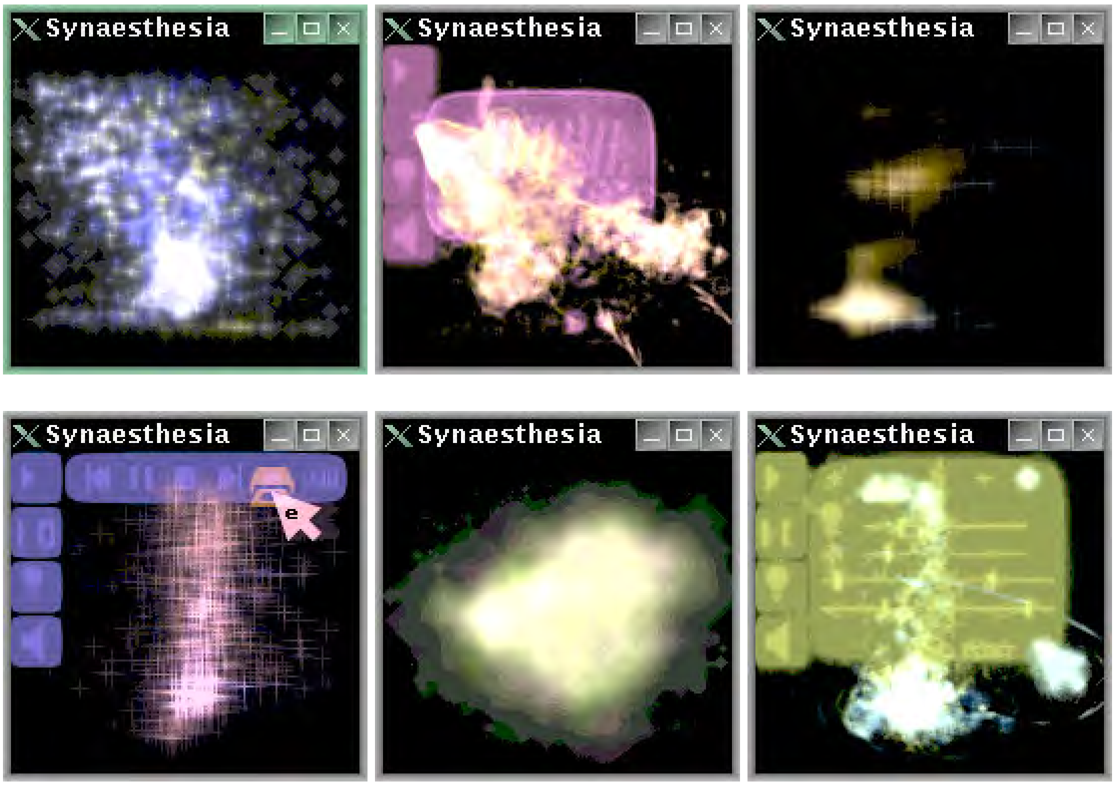
How *MetaSynth* “plays” (or sounds) a picture:

- vertical axis represents pitch
- horizontal axis represents duration
- brightness intensity represents volume (Black being silent)
- colour represents stereo placement (Red is left, yellow is center, and green is right)

Another attempt at grappling with this concept via software (this time with music being used to generate images) is the Australian freeware programme *Synaesthesia,* written by Paul Harrison, while a computer science student at Monash University. *Synaesthesia,* according to its author

> … is a program that represents music graphically in real time as coruscating field of fog and glowing lines. It is intended as a visual accompaniment to music. [1]

[1] URL: http://yoyo.cc.monash.edu.au/~pfh/synaesthesia.html ; accessed 12-06-02.



Above: screenshots of visual output from the *Synaesthesia* programme, interpreting musical sounds in a visual form.

Synaesthesia is intended to function as a visual accompaniment to music. Harrison's web-page notes offer these minimal further details:

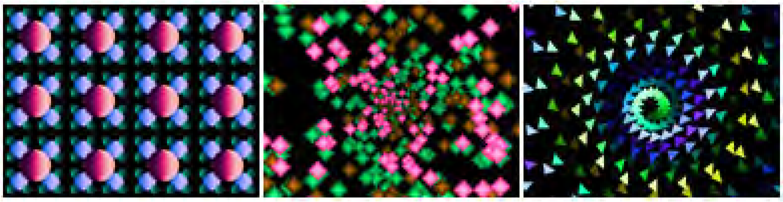
> "Synaesthesia:" seeks to provide not just a visual representation of sound, but a representation of how sound is perceived. Its display combines information about the frequency, location and diffuseness of sound.
>
> The display is sufficiently detailed to make it possible to distinguish several individual instruments, singers, or special effects on screen by their location, shape and color, and sufficiently fast to distinguish individual drum beats and notes.[1]

Bliss Paint is a commercially available programme, which also includes the facility to link visual output to sonic input, or in other words generate imagery which is algorithmically derived from parameters inherent in sound, over time. It is marketed by Imaja [2] who claim that:

> MIDI instruments may be used to control the color synthesizer and shape selection in Bliss Paint. Sound input through a microphone or line input may be used to control color synthesis. With sound and MIDI input, Bliss Paint animations respond rhythmically to live music or music

[1] "*Synaesthesia*" runs under Linux or DOS and is available from pfh@yoyo.cc.monash.edu.au.
[2] http://www.imaja.com/BlissPaint.html ; accessed 7-12-02 .

> from your stereo. Many of Bliss Paint's features are designed for live performance animation for music.[1]



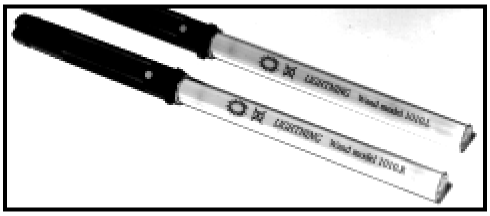
Above: Screenshots showing *Blisspaint* output**.**

Human Movement Tracking Technology

A different perspective on creating an interface between visual and auditory signals to produce artistic results is utilizing the MIDI (Musical Instrument Digital Interface) protocol to control lighting effects. This conceptual approach to experimental music performance can take into account human gestures or movement as a controlling variable [Mulder, 1994].

> With current state-of-the-art human movement tracking technology it is possible to represent in real-time most of the degrees of freedom of a (part of the) human body. This allows for the design of a virtual musical instrument (VMI), analogous to a physical musical instrument, as a gestural interface, that will however provide for much greater freedom in the mapping of movement to sound. A musical performer may control therefore parameters of sound synthesis systems that in real-time performance situations are currently not controlled to their full potential or simply not controlled at all. In order to decrease the learning and adaptation needed and avoid injuries, the design must address the musculo-skeletal, neuro-motor and symbolic levels that are involved in the programming and control of human movement. The use of virtual musical instruments will likely result in new ways of making music and new musical styles.

LIGHTNING II is a specialized MIDI controller developed by legendary electonic instrument maker Don Buchla. By tracking tiny infrared transmitters that are built into baton-like handheld wands, it senses their position and movement and transforms this information to MIDI signals for expressive control of electronic musical instrumentation. In addition to functioning as a MIDI controller, LIGHTNING II, with its self-contained 32 voice synthesizer, comprises a complete, ready to play instrument.



[1] URL: http://www.imaja.com/blisspaint/ ; accessed 12-06-02.

Lightning's promotional web-site describes some of the interesting creative possibilities this technology opens up:

> Basically, LIGHTNING II senses the horizontal and vertical position of each hand, for a total of four independent coordinates. From this information, LIGHTNING's digital signal processor computes instantaneous velocity and acceleration, and performs detailed analysis of gesture. An easily mastered, musically oriented interface language allows the user to define relationships between various gestures and potential musical responses.
>
> In one sort of implementation, LIGHTNING's coordinates might be mapped to various MIDI controllers on multiple channels. Spatial pitch wheels, pan pots, level sliders and modulation wheels are easily defined and great fun to play. Performance gestures can be analyzed for direction and velocity and can be used to generate a variety of notes as well as other musical events. Multi-dimensional zoning capability can be used to create different musical responses in different regions. Everything you need to create the conceptual ensemble (an invisible, acoustic virtual reality).
>
> User definable scale and tuning tables allow one to determine the range and selection of notes occurring along a horizontal or vertical axis. Pitches can be in any order, and the boundaries can be set where-ever desired, facilitating the creation of spatial instruments and imaginary orchestras.
>
> LIGHTNING II features a conducting facility that can analyze a conductor's gestures, display deviations from a preset tempo, and signal errors such as missed beats. Simultaneously, LIGHTNING can transmit a synchronous MIDI clock for controlling external sequencers and output programmed note data to accompany specific beats within a measure. [1]

Mulder [1994] attempts a comprehensive list of other experimental approaches to gestural- and movement-based musical controllers, in a series of tables, which are included below. Hesse's concept of the Grand Multimedia Art Machine (discussed on pp. 140-141) was left largely to the reader's imagination in terms of functional details, and it is interesting to consider that this possible future merging of technologies into an ever-evolving meta-art form, could include some yet to be invented interface devices and expressive controllers. Mulder's tables give a hint of what lines research is pursuing in this regard:

[1] URL: http://www.buchla.com/lightning/descript.html ; accessed 5-12-02.

Table 1a: *An overview of recent experiments with new musical instrument designs*	
Author	**Musical instrument**
Rubine & McAvinney (1990)	Videoharp MIDI controller (optical sensing of fingertips)
Mathews & Schloss (1989)	Radiodrum MIDI controller (short range EM sensing)
Palmtree Instruments Inc., La Jolla CA, USA (no published date)	Air-drum MIDI controller (acceleration sensitive)
Machover & Chung (1989), Neil Gershenfeld	Hyperinstruments: acoustic instruments extended with extra control abilities

Table 1b: *An overview of recent experiments with devices for conducting*	
Author	**Conducting device**
Keane & Gross (1989)	MIDI baton (AM EM sensing)
Bertini & Carosi (1992)	Light baton (LED sensed by CCD camera)
Don Buchla (1992-94)	Lightning MIDI controller (Infrared LED sensing)
Machover & Chung (1989)	Exos DHM glove for conducting MIDI devices

Table 1c: *An overview of recent experiments with musical instrument designs involving gloves*	
Author	**Glove application**
Pausch & Williams (1992)	Handmotions control speech synthesizer
Fels & Hinton (1993)	GloveTalk: CyberGlove controls a speech synthesizer
CNMAT, Berkeley; Scott Gresham-Lancaster, Berkeley; Mark Trayle, San Francisco; James McCartney, University of Texas; Thomas Dougherty, Stanford University; William J. Sequeira, AT&T; Tom Meyer, Brown University; Rob Baaima, Amsterdam; Lance Norskog; The Hub and probably many others (no date cited).	Powerglove as a MIDI controller
Gustav's Party (no date cited)	Virtual reality rock band using a.o. Datagloves as MIDIcontrollers

Table 1d: *An overview of experiments with new musical instrument designs involving whole body movements*	
Author	**Motion to sound design**
Lev Termin (see Vail, 1993)	Capacitively coupled motion detector controls electronic oscillator
Chabot (1990)	Ultrasound ranging to detect whole body movements and to control MIDI devices
Bauer & Foss (1992)	GAMS: Ultrasound ranging to detect whole body movements and to control MIDI devices
Camurri (1987)	Costel opto-electronic human movement tracking system controls a knowledge based computer music system

Table 1e. *An overview of experiments with new musical instrument designs involving bioelectric signals*	
Author	**Bioelectric design**
Knapp & Lusted (1990)	Biomuse EMG signals control a DSP
Chris van Raalte / Ed Severinghaus, San Francisco CA, USA (no date cited)	BodySynth MIDI controller

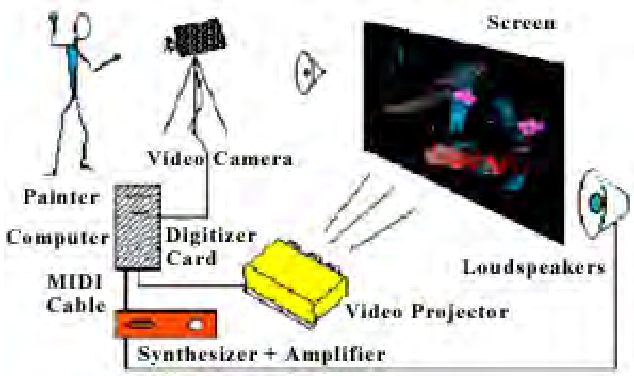
Rosenboom (1990), Richard Teitelbaum, Germany, Pjotr van Moock, the Netherlands and probably others	EEG/EMG interface to synthesizer setup

The **Imaginary Piano** consists of a real-time video-captured image-analysis system: here, the interaction "tools" are the mere bare hands of a pianist. The pianist is sitting as usual on a piano chair and has in front nothing but the camera (a few metres away pointed on his hands). The system establishes an imaginary line at the height where usually the keyboard lays:

when a finger, or a hand, crosses that line downward, a specific message (actually a *NoteOn* MIDI message) is issued; the position at which the line is crossed states the key number, how fast the line is crossed, states the velocity. The promotional web-site states:

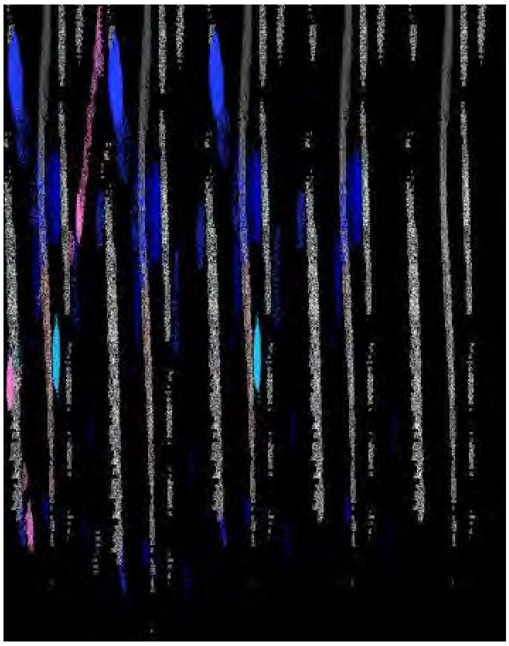
> This application should be considered an original performance rather than an original instrument due to the inaccuracy of gesture when striking the right key.[1]

PAGe systems (Painting by Aerial Gesture) In this application, positions and movements of a performer's hands are recognized in an wide vertical x-y area: the performer acts as a "painter" who uses his hands for selecting colours and nuances of colour and for actually drawing a picture.



[1] http://spcons.cnuce.cnr.it/music/Gesture.html ; accessed 1-01-03.

Movements are performed in the air and the resulting picture is projected on a large video-screen. An UV lamp is placed very close to the performer and whatever white-coloured item he uses (such as white gloves) is interpreted by the computer as a drawing brush; the left hand is used as a real-time controller for nuances. By freezing real-time produced images, “paintings” can be printed using wide ink-jet printers [1].



Right: Laser Harp used as MIDI trigger mechanism.

The effect of audio design on the perception of image.

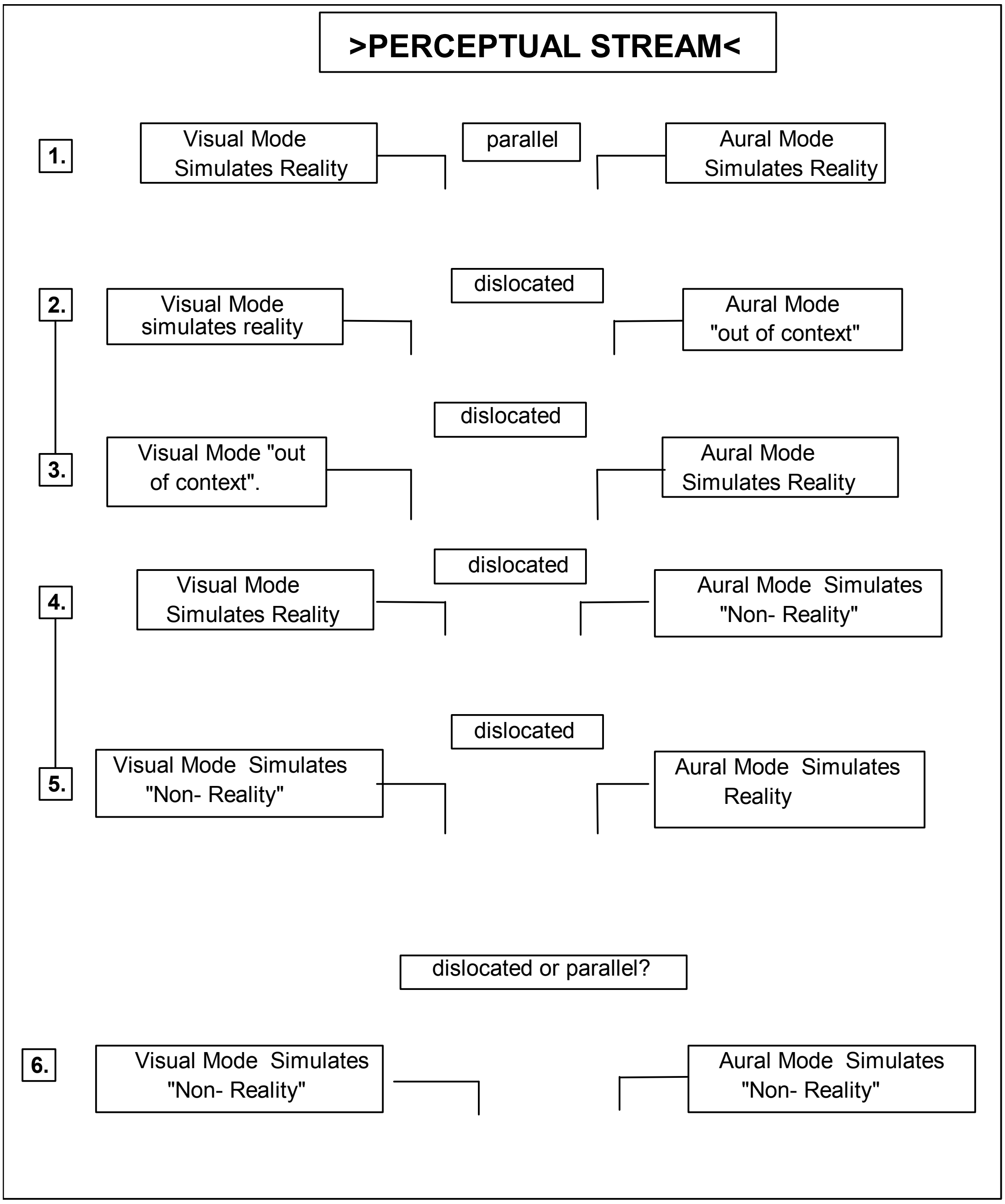
The artist’s entrance point to the field of multimedia art is a significant variable in the development of an individual aesthetic, style, mode of working etc. For instance it is feasible that a multimedia artist coming into the field from a music technology background will have a different way of working, a different sense of psychological timing, and therefore produce a different genre of artistic product to, say, a graphic artist, animator, or music video *director*.

[1] http://spcons.cnuce.cnr.it/music/Gesture.html ; accessed 01-01-03.

The effect of sound design on the perception of image is a central interest in the understanding of how multimedia art works. The convenience of referring to musical terminology and analogies in the analysis of multimedia art aesthetics is significant, for indeed the history of music is in one sense the history of an evolution of a means of shaping time through a sonic architecture. The experience of the passage of time in life teaches us about rhythms. Multimedia artists must, whatever their background, develop skill at shaping the experience of time.

In multimedia art, as in cinema before it, careful attention to the sound element is crucial in building up a perception of a believable, transparent reality in the mind of the beholder. The effect of sound on the perception of image has long been a subject of fascination in film studies (see pp. 347-9 for further reference to this issue). But, with the exception of experimental abstract films, it is generally true that in the vast majority of cinema art, film is used as a narrative, storytelling medium, constructing an experience of compressed and interpreted reality.

Multimedia considered as an artform offers the opportunity of presenting non-narrative visual material synchronously with audio designs that can include real-world sound recordings, synthesized elements and traditional music structures, allowing the artist to produce simulations of experiences and realities not available as a part of everyday reality. The table below attempts to summarize the possible combinations of representational and non-representational sound/image streams.



This figure above is not attempting to be a portrayal of set or specific choices facing multimedia artists, but rather a range of possible outcomes arising out of different artistic approaches. A distinction is made here between naturally recorded sound or image that is used “out of context”, and the use of any visual or sound element which is synthesized or depicts an imaginary space, alternative reality, fantasy, abstraction etc. In the aural modality this can of course mean simply adding music (in the sense of underscoring, as opposed to a diegetic portrayal of a specific live musical performance).

Model 1 obviously refers to traditional narrative filmmaking, of which there are countless examples.

Model 2 is a technique used in film for a variety of purposes, the most obvious being to convey a hidden emotion or an underlying motivation, or to establish a link with an important element in the story (e.g. a significant childhood experience that might be the motivation for a characters behaviour [1]). We are offered meaning that would not be available if 'natural' sound was used.

For an example of *Model 3* we might turn to Music Video, where we accept the convention of a band performing a song as the referent reality, but also the convention that all manner of visual image may be offered as an accompaniment. This out of context visual material may offer another layer of meaning to the song, or it may be presented intuitively for its non-quantifiable, transcendent or surreal value - in that the juxtaposition of the "out of context image" causes an unexpected, unpredicted experience considered valuable or interesting in the artistic judgment of the creator.

In the case of *Model 4*, any piece of film-making which uses underscored music could legitimately be considered an example, but also other creative use of sound to evoke a heightened sense of reality, create a mood or dramatic tension, humour etc.

Examples of *Model 6* can be found in many abstract and experimental films as well as music videos, in which artificially produced sound and image are combined arbitrarily and randomly, or with the purpose of creating tension.

Any multimedia art which deals with indeterminacy and chance outcomes may also deal with one or some of the dislocated models.

1.9 Multimedia as a Meta-Art

German opera composer Richard Wagner believed that the future of all the arts lay in a *Gesamtkunstwerk,* or total artwork, a fusion of the arts. Wagner wrote an essay in 1849, *The Art-work of the Future*, defining a synthesis of the arts in which he outlined his vision that opera could serve as a vehicle for the unification of all the arts into a single medium of artistic expression. Packer and Jordan [2000] summarize

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[1] e.g. as pioneered by Bernard Herman with the "Rosebud theme" in Orson Welle's *Citizen Kane.* (Carlsson, Mikael (undated) *Citizen Kane: Bernard Hermann.* http://www.musicfromthemovies.com/pages/reviews/citizen_kane.html ; accessed 26-01-03).

Wagner's ideas, which have resonance in the context of a study of multimedia as meta-art:

> Wagner's description of the Gesamtkunstwerk is one of the first attempts in modern art to establish a practical, theoretical system for the comprehensive integration of the arts. Wagner sought the idealized union of all the arts through the "totalizing," or synthesizing, effect of music drama : the unification of music, song, dance, poetry, visual arts, and stagecraft. His drive to embrace the full range of human experience, and to reflect it in his operas, led him to give equal attention to every aspect of the final production. He was convinced that only through this integration could he attain the expressive powers he desired to transform music drama into a vehicle capable of affecting German culture.

In 1876, twenty-seven years after *the Artwork of the Future,* Wagner took this approach even further when he opened the famous Festpielhaus Theater in Bayreuth, Germany, where his theatrical innovations included: darkening the house, surround-sound reverberance, the orchestra pit, and the revitalization of the Greek amphitheatrical seating to focus audience attention on stage. [Jordan, 2002]. There is incisive foresight in Wagner's approach to the integration of the arts, which resonates with contemporary notions of interdisciplinary multimedia art.

The 20th century nurtured a variety of other attempts to heighten the experience of art by integrating traditionally separate disciplines into single works, beginning with the ideas of the *Futurist* art movement. The Futurists celebrated the speed, noise and technology of "modern life, continually and riotously transformed by all conquering science", and were excited by where this technological evolution could lead [Pierre,1969, 13].

In the 1916 manifesto *The Futurist Cinema*, Marinetti and a number of other collaborators declared the imminent disappearance of the book. Film was seen by this group to be the art of the times and indeed the supreme art, embracing all other art forms through the use of (then) new media technology.

> One must free the cinema as an expressive medium in order to make it the ideal instrument of a new art, immensely vaster and lighter than all the existing arts. We are convinced that only in this way can one reach that *poly-expressiveness* towards which all the most modern artistic researches are moving. Today the *Futurist cinema* creates precisely the poly-expressive symphony that just a year ago we announced in our manifesto "Weights, Measures, and Prices of Artistic Genius". The most varied elements will enter into the Futurist film as expressive means: from the slice of life to the streak of color, from the conventional line to words-in-freedom, from chromatic and plastic music to the music of objects. In other words it will be painting, architecture, sculpture, words-in-freedom, music of colors, lines, and forms, a jumble of objects and reality thrown together at random. We shall offer new inspirations for the researches of painters, which will tend to break out of the limits of the frame. We shall set in motion the words-in-freedom that smash the boundaries of literature as they march towards painting, music, noise-art, and throw a marvelous bridge between the word and the real object. [Marinetti et al., 1916].

Concepts of meta-art or a synthesis of the arts continued to fascinate in ensuing decades and many other leading artists and schools of thought examined ideas associated with combining media to create a total art experience. For example, while *the Bauhaus* is best known for its contribution to industrial design practice, leading artists, who included László Moholy-Nagy, Paul Klee, and Wassily Kandinsky, used the school as a laboratory to further examine principles of abstraction in such media as painting, photography and sculpture. These adventures led Moholy-Nagy, along with the sculptor Oskar Schlemmer, to postulate a theatre of abstraction for *the Bauhaus*, based on a reductive approach to the synthesis of its primary elements, declaring that only the synthesis of the theatre's essential formal components: "space, composition, motion, sound, movement, and light" - into an organic whole, could give expression to the full range of human experience.

Right: Light Play with Projections from Bauhaus Theatre[1].

In a 1924 essay describing this Bauhaus theatre, entitled "*Theater, Circus, Variety,"* László Moholy-Nagy called for a shift in emphasis away from the actor and the written text, and bring to the fore every other aspect of the theatrical experience. Moholy-Nagy referred to this idea as "the theatre of totality", a reinterpretation of Wagner's concept of total theatre. The Bauhaus associates believed they were creating an abstract theatre of movement, colour, light, form in which *Soundlanguage* would be added later, once the stage had been purged of its literary encumbrance. They believed that humanity's essential nature--freed from history, tradition, class, and nationality--would find expression in theatrical works that incorporated pantomime, masks, dance, and acrobatics. [Packer and Jordan, 2000].

During the 1940's John Cage undertook some collaborative theatrical experiments combining the use of mixed media with ideas of artistic indeterminacy, which became a significant catalyst for an explosion in this kind of artistic experiment after World War II. He developed his provocative "theatre of mixed-means" in

[1] URL: http://www.artmuseum.net/w2vr/timeline/Moholy.html# ; accessed 23-11-02.

collaboration with the artists Robert Rauschenberg and Jasper Johns, and the choreographer Merce Cunningham, during a residency at Black Mountain College, introducing all types of actions, artifacts, noises, and images [Miller, 2002].

> The dancers' movements were sensed by multiple, five-foot long theremin antennae, and their interruptions of light beams were sensed by photoelectric cells. The output from these devices fed into a complex sound system which included multiple tape machines and short-wave radios, affecting in turn the output and distribution in space of these sound sources. Altered television and film imagery were also shown, and while these images were not modified in real time by movement, at least some of the original film footage was drawn from dance rehearsals. It was a remarkably complicated piece for the time. Mumma has written, "With virtually no precedent, this work established at once a coexistence of technological independence and artistic nondependence ... Every complicated production of *Variations V* was logistically precarious ... I loved the work and absolutely dreaded the exhausting preparation of every performance." [Miller, 2002].

Left: scene from the influential electronic theatre work, *Variations V* (1965) showing Cage and Tudor in the foreground. [Packer & Jordan, 2000].

Cage's work proved to be extremely influential on the generation of artists that came of age in the late 1950s. Allan Kaprow, Dick Higgins and Nam June Paik[1] were among the most prominent of the artists who, inspired by Cage, developed non-traditional performance techniques that challenged accepted notions of form, categorization, and composition, leading to the emergence of genres such as *the Happenings*, electronic theatre, performance art, and interactive installations [Packer and Jordan, 2000].

Twenty years ago (1984), pioneering software designer and computer theorist Alan Kay (p.41) wrote:

> It [the computer] is a medium that can dynamically simulate the details of any other medium, including media that cannot exist physically. It is not a tool, although it can act like many tools. It is the first metamedium, and as such it has degrees of freedom for representation and expression never before encountered and as yet barely investigated.

[1] See p. 202 for a discussion of Nam June Paik's work in the context of the history video art.

Michael Hamman [1], discussing the HCI design for computer music composition, emphasizes the creative possibilities which will open up for multimedia artists as HCI design evolves, advocating that:

> … the computer be understood not merely as a tool for building tools, but as a tool for constructing conceptual frames by which we come to understand and extend problem domains. Such an understanding of the computer allows for evaluation of its use in the creation of progressive art. Use of the computer in music composition and sound synthesis has a long history (more than 40 years) and provides a wealth of theoretical and experimental data on creative idea generation and human/computer interaction. Recent theories in cognitive psychology indicate that creative thinking involves a process of iterative activation of "cues" which help direct the generation of ideas and the search for problem solutions. Along the way, however, thinking can become *fixated* on particular cues, thus leading to mental blocking and failure to creatively solve a given problem or formulate a new idea. Such a deadlock can be broken by shifting the conceptual framework through which cognitive patterns are engendered. The computer can be understood as a tool for designing interfaces and, as such, as a tool for constructing conceptual frameworks. In this capacity, computers assist not only problem solving, but in designing the very conceptual means by which problems might be formulated. It is precisely this capacity of the computer—the capacity for constructing representations of problem domains and spaces—that compels its use in creative activity like music composition.

While multimedia in its present CD-ROM delivery format is a means of conveying information in a format which is clearly structured (hierarchical), entertaining (sense stimulating, immersive), and rational (short-term and long-term outcome-oriented), it also offers possibilities for experimentation in communication in a range of non-linear, non-rational, expressionist, abstract, intuitive and surrealistic modes. If we use the commonly accepted analogy of navigation through cyber-space, then the artistic and academic ideal of successful multimedia art could be described as an experimental or knowledge-gaining journey through artificial constructed experience. This view seems to be built on an underlying assumption that novel experience is an enrichment or extension of life – i.e. the positive worldview that by extending knowledge we enrich life experience.

It could be imagined that the CD-ROM (or DVD-R) drive-equipped computer workstation with colour (HDTV?) screen, stereo (multi-channel?) speakers and interactive (multi-dexterous?) interface is but the primitive forerunner of some fantastic future input/output-module, the ultimate multimedia art machine. The meta-art machine was visualized by the German author and mystic Hermann Hesse [1978, 18], in his esoteric and visionary, Nobel prize-winning novel *The Glass Bead Game (Magister Ludi*). Hesse draws the reader into a futuristic, spiritually advanced, yet

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[1] *Priming Computer-Assisted Music Composition through Design of Human/Computer Interaction* (undated); URL: http://www.shout.net/~mhamman ; accessed 24-02-04.

somehow almost medieval and hierarchical society, in which the realm of culture is set apart to pursue its goals in splendid isolation. Mankind has developed the ultimate in art technology, a multidimensional machine/human interface which is translated as *the Glass Bead Game*, depicted obliquely (in a pre-computer time) as a fantastic creative and conceptual abacus (computer?), that employs all the cultural and scientific knowledge of the ages. It is set in the fictional land of Castalia where academic study, the arts and nobility are the factors that determine the governing of the country. Of all the noble arts, the Glass Bead Game commands the highest respect, and Castalia's greatest minds have sought to unravel the mysteries of the game for centuries. In this story we follow central character Joseph Knecht's life from childhood through to manhood, until he is appointed to the loftiest position in the land, the Master of the Glass Bead Game. In this strongly atmospheric novel, Hesse carefully avoids actually describing this central conceit. His main interest lies in exploring the moral discipline and personal sacrifice required to master such a demanding machine. He restricts himself to one direct description of the concept. In the following ambitious passage, Hesse's poetic vision touches on one of the core ideas of this dissertation:

> The Glass Bead Game is thus a mode of playing with the total contents and values of our culture; it plays with them as, say, in the great age of the arts a painter might have played with the colours on his palette. All the insights, noble thoughts, and works of art that the human race has produced in its creative eras ...on all this immense body of intellectual values the Glass Bead Game player plays like the organist on an organ.

Immense dedication and a lifetime of meditation and study are required to become a virtuoso of this grand instrument of human aspiration and inspiration, on which can be improvised a symphony of knowledge and ideas [1].

The desktop multimedia workstation, mounted with the already standard, off-the-shelf battery of multimedia software tools, such as word processor/DTP software, 2/D digital image capture hardware and image manipulation applications, 3D modeling, animation and rendering capabilities, digital sound recording, signal processing and audio editing tools, video digitizer and non-linear video editing suite, with compositing, motion graphics and special effects facilities, internet content creation tools and browsing interface, the facility to access on-line databases, this whole conglomeration of digital technologies could be seen as, conceptually, the first manifestation of Hesse's prescience. It's possible that it stands in relation to the central idea of *the Glass Bead Game* in some sort of equivalence to the historical

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[1] There are a number of web-sites set up by people who want to actually *play* the Glass Bead Game. e.g. http://www36.pair.com/waldzell/GBG/

stage that the Wright Brothers flying machines represented in relation to a contemporary Boeing jet or even NASA space craft. Its input modules are QWERTY and MIDI keyboards, WACOM-type pressure sensitive art pads, audio & video digitizers connected to transducers (microphones, scanners, and cameras).

While Hesse’s metaphor for the Grand Art Machine is an organ or keyboard – we don’t get much insight into who is listening, or witnessing the multimedia meta-art performance. Hesse did not perhaps foresee a networked world, in which an individual artist could broadcast multimedia art across space and time.

The foregoing short history of the intellectual pursuit, during the 19th and 20th centuries, of a concept of a merging of the arts into a meta-artform demonstrates that it has had an on-going impetus as an idea or vision. An evolutionary progress fueled by technological change has finally led to this vision having the possibility of a physical reality via the mediation of the networked multimedia workstation.

Digital improvisations

It is interesting to consider Hesse's vision that the grand multimedia machine allowed the artist a facility to improvise in a multimedia fashion. In early experiments leading to the formulation of this dissertation, I carried out experiments using a video camera, based on the previously mentioned 'art as performance' paradigm discussed in relation to the work of Jackson Pollock (see pp. 51-54). These were attempts to improvise a work of video art in response to nature, and later, to urban landscapes, in real time. The idea was to employ the video camera in a controlled but extemporaneous manner, conceptually similar to that with which Pollock approached action painting, and jazz artists approach sonic expressionism. In the latter examples, each expressive action (brush-stroke or musical phrase) is built on previous expressive actions within the performance, shaping or building an unplanned work intuitively, but based on the artist's experience and mastery of the medium. To do this in the video medium requires a disciplined sense of in-camera editing. The video camera also acts as a digital audio field recorder, and these experiments have simultaneously utilized the camera as a means to selectively edit ‘found sound’ in real time, with a composer’s and musician’s aesthetic. A simple example is driving around in a city, responding to and sampling found visual environments and action, with random sound from the car radio providing a found (unplanned) soundtrack and musical pulse, which the video camera operator can use as a rhythmic/atmospheric motivator. These experiments attempted to focus on the artist’s response to the aural environment, cross-referenced to visual meaning.

Making sense of this approach to technological expressionism may not be easy, but the idea of it is especially stimulating. It is the brain's facility to process complex clusters of perceptual input intuitively, unconsciously, and almost instantaneously, that makes art-making such a mysterious yet fascinating process.

In digital art the ease of editing and the lossless re-ordering of data allows the artist to then move from the spontaneity of sampling with input transducers, to the cerebral re-ordering of perceptual stimuli. At this signal processing/digital editing stage of the creation of a digital artwork, new levels or layers of meaning are recognized and/or introduced.

Technological developments during the past decade have included trends towards the integration of different software, the establishment of universal file formats, the embracing of plug-in schemes, and a gradual standardization of interface design. The overall thrust and impact of this progression is to make it increasingly possible for an artist to produce multimedia works as an independent operator and retain sole creative or artistic control.

There are, of course, opposing forces or dynamics with which the practicing multimedia artist must contend. Firstly, there is the sheer immensity of the task of mastering the endless detail of the necessary software tools, in order to address a diverse range of media forms and artistic processes. Then there is the considerable scale of work involved in bringing a solo project to completion. There are also a range of lifestyle negatives: physical effects of radiation, lack of physical activity, eyestrain and fatigue, and possibly psychological syndromes, such as stimulation dependency. The artist must spend long periods of time relating to a machine and avoiding human interaction, with resulting relationship outcomes (e.g. computer-widow syndrome). Further, there is the stress resulting from computer crashes, lost data, technical breakdowns and faults.

But of course artists are often driven to pursue their creative ideals, despite such obstacles. Michelangelo was willing to lie on his back for four years and permanently weaken his health in his heroic drive to single-handedly complete the vast painting on the ceiling of the Sistine Chapel [Barry et al. 1964 , 26].

It is intriguing to ask: what form will the digital equivalent of the ceiling of the Sistine Chapel take?

The democratization of art through the availability of multimedia technology

From the standpoint of a computer-literate generation which has experienced interactivity in art, works delivered in traditional art media formats offer an experience which is passive, one way or linear. They are also, it can be argued, elitist. For centuries before the age of mechanical reproduction, proximity to great visual art, literature and music was the by comparison restricted to members of the privileged minority i.e. wealthy patrons, and the Church hierarchy [Barry et al., 1964, 68].

Printing, photography and audio recording made the images and music that represented the highest cultural achievements of society gradually more available to the common (Western, but increasingly, global) man. The great works of art, previously distant and unattainable to the point of gaining an almost mythic status, were subject in the process of widespread dissemination, to a dilution of what German socialist philosopher Walter Benjamin called their "aura" [1]. For Benjamin the aura of a work of art emanated from what he called its *authenticity*.

> Even the most perfect reproduction of a work of art is lacking in one element: its presence in time and space, its unique existence at the place where it happens to be. The unique existence of a work of art determined the history to which it was subject throughout the time of its existence. This includes the changes which it may have suffered in physical condition over years as well as the various changes in ownership…..The presence of the original is the prerequisite to the concept of authenticity.

Benjamin's concept of a work of art having an aura and a unique authenticity cast an interesting light on the debate about the validity of multimedia as art. This process, the withering of the aura of art, which Benjamin describes in relation to mechanical reproduction technologies, is now going through an exponential acceleration. Multimedia technologies, with all their power to clone art, have become available at shopping malls and by mail order, and are increasingly becoming household items. Hertz [1995, 164] comments that:

> ...the contradiction between culture as commodity and culture as communication is determined not by differences in the intelligence or creative capacity of persons, but by the ownership of the technologies of production and distribution. Electronic reproduction and networked communications now promise to push the contradictions of culture and society to new extremes. Digital electronic media don't merely reproduce, they clone. Every digital artwork is an original.

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[1] In his prophetic work *"the Work of Art in The Age of Mechanical Reproduction"*. [Benjamin, reproduced in *Illuminations* 1968, p 218].

Whether there is an ultimate devaluation of the status of sound art resulting from the proliferation of music technology that allows a person possessing minimal traditional music technique to sample, quantize and sequence sound, and algorithmically generate recognizable, professional-sounding music recordings, compositions and performances is an issue worthy of debate. The result may have been an outpouring of low-budget music releases, of commercially and artistically aspiring attempts to create a relevant and popular musical product. Of course each example of this multitude of art products may succeed to some degree in achieving preconceived artistic goals, across a wide spectrum of intention from naive to ultra-sophisticated. Success in the commercial music industry, including implicitly artistic success, is ultimately gauged by unit sales, as reflected in the industry award system (e.g. the APRA awards). Whatever the ultimate meaning and value of the phenomenon of a burgeoning wave of home recording of original music, there is clearly no doubt that a lot of creative activity has ensued, fueling a march towards an ever richer global culture, and a broader acceptance of art as a civilizing force.

It is indeed a testament to Cage's philosophical foresight to see the idea that all sound is potentially musical sound[1] becoming such a practical day-to-day reality. Music technology has made the process of transducing real world sound into musical raw material a more immediate and seamless one, and parallels can be drawn in the arena of the visual. The availability of low cost flatbed scanners and digital still cameras in combination with the relatively effortless process of digital image manipulation via the Adobe *PhotoShop* software application and its competitors, has shifted the whole playing field in the art of photography and associated fields of advertising and graphic design, print layout and television graphics.

The possibility of high quality full-motion video capture and playback on the 'desk-top', in combination with the development of "firewire" and the availability of low-cost non-linear video editing, has had an equal impact on how things are done in the industrial video, music video and semi-professional video fields, and what is possible on a grass-roots level. Video art is one area where the entry-cost differential between traditional analogue and the new digital technology is immense. Artists can now realistically conceive of accessing video techniques previously the exclusive province of high-end broadcast professionals and corporate media production houses. Non-linear video editing techniques have changed the creative process of video making to a degree analogous with the way the introduction of PC-based

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[1] Source: "American Masters " URL: http://www.pbs.org/wnet/americanmasters/database/cage_j.html accessed 15-11-02.

word-processing technology has influenced processes of creative writing, facilitating creative experimentation, continuous reworking and encouraging a non-linear flexibility.

Whether the result of this democratization of access to technology is an ocean of "bad" art doesn't matter, for there will inevitably arise the equivalent of digital Da Vincis. It is also feasible to suggest a positive educational outcome of a widespread exposure to digital art techniques. As computers invade the workplace, school and home, and become increasingly integrated into human life, community expectations and standards of aesthetic awareness could be raised by constant exposure to the products of multimedia design. If we could speak of some kind of lowest common denominator of art appreciation and indeed art skills in a developed society – then perhaps there is also some kind of upward artistic mobility, through the widespread acquisition of previously inaccessible and esoteric specialist skills such as afforded by the availability of complex but user-friendly graphic design software such as Adobe *PhotoShop* or video editing software such as Apple's *Final Cut Pro*, or music composition software such as Steinberg *Cubase*.

This aspect of democratization, it is reasonable to suggest, may work to *enhance* the role of digital art in society - for it will surely be the resonance of ideas, concepts, and ultimately the power of what is communicated that will stratify art in terms of its impact on and value to society.

Another danger or potential negative outcome of the democratization of art through the wide availability of easy-to-use software tools, is a *homogenization* of digitally produced art - the trend for everything to look and sound the same. Rather than necessarily setting the artist free, the technology may have the effect of overly influencing the work process of creating art and pushing the artist towards certain stylistic approaches, through a "path of least resistance" syndrome. Innovative interactive film artist Chris Hales [1] [1996] has indicated that:

> The recent growth and investment by corporate developers and publishers with huge budgets to throw at multimedia, with the idea that something good will drop out of it at the end, has already produced a stereotypical CD-ROM or game-type product. Individuals with modest or even non-existent budgets, trying to produce personal and innovative work are few and far between, and are disadvantaged by the 'industry'. In addition development tools such as *Macromedia Director* tend to produce work that looks the same. It seems that in this case advances in technology seem to produce items that do the same old thing but quicker, or better, or more conveniently, rather than differently.

[1] See pp. 408-411 for a discussion of Hales' experiments with interactive films

Nevertheless, the relative ease with which reasonably professional art can be produced using digital technology, and the resulting proliferation of art activity, may ultimately be a factor in promoting widespread interest in interactivity. Aspiring artists are commonly motivated to emulate resonant ideas, and while there is elegance in simplicity (and the introduction of branching structures inevitably increases the complexity of the production process), the goal is an indeterminacy of outcome akin to life itself.

Multimedia art is hailed as offering a promise of new opportunities in the relationship between the work of art and its consumer. Whether interactivity will prove to be a positive addition to the artist's palette or, in fact, will dilute the power of art (by creating an expectation of distracting interactive rewards for the participant) is yet to be discovered. There is little doubt these areas of interactivity and non-linear process will be increasingly explored.

A fascinating field of study, once the province of science fiction, is presented by any attempt to understand the unique psychological forces at work as the human spirit seeks to express itself via the output of machines. Research and Development decision-makers and marketers of multimedia technology are increasingly attuned to an innate resistance in the market to machine-imposed limitations on freedom of creative expression. Breakthroughs in user-interface technologies have an immediate and dramatic effect on the market. Market forces fuel the advance towards perfect transparency (by which it is meant the efficiency and elegance of interface design. Transparency in interface design equates with intuitive, flowing ease of use. A good interface becomes invisible, as the user becomes immersed in the experience). The ultimate transparency will come, we can imagine, with some kind of direct brain-wave transmitter/receiver, or remote brain-to-brain data transfers, telepathy via technology, already long foreseen by science fiction writers. This concept fires the imagination with images of a future cyber-ecology where intelligent machines merge with modified living organisms, and technological applications are derived biologically. It is possible to extrapolate these ideas toward a future in which carbon based life-forms prove to be a mere stepping stone towards silicon- or some other technologically-based life-forms, a future in which humanity fulfils its destiny by creating an intelligent, conscious, self-perpetuating machine. Futurist/inventor Ray Kurzweil [1999, 280] describes a time (which he argues will be reached by 2099 A.D.) when:

> Machine-based intelligences derived from extended models of human intelligence claim to be human, although their brains are not based on carbon-based cellular equivalents. Most of

> these intelligences are not tied to a specific computational processing unit. The number of software-based humans vastly exceeds those still using native born neuron-cell-based computation.
>
> Even among those human intelligences still using carbon-based neurons, there is ubiquitous use of neural-implant technology, which provides enormous augmentation of human perceptual and cognitive abilities. Humans who do not utilize such implants are unable to meaningfully participate in dialogues with those who do.

Will these post-human entities express themselves through art?

It must be said that, in a global context, the democratization of art through the increasing availability of multimedia technology is a relevant concept in developed societies only, and in the present historical epoch, multimedia art remains and activity for affluent elites. In less developed countries computers remain rare, as is the infrastructure to support modern computer systems – lack of reliable power supplies, ubiquitous telecommunications and surplus income excludes large section of the world's population from participation in the digital revolution.

But for technocratic societies the sci-fi angst with regard to technological dependence remains poignantly relevant, perhaps symptomatic, and an inevitable by-product of the central concerns of such societies, which have sought through technology to fulfill dreams, to remove physical danger and minimize physical work, and to fill the resulting leisure time with safe but stimulating entertainment, personal empowerment and quasi-companionship.

Some writers warn of the dangers inherent in the new obsession with, and proliferation of digital technology in terms of creating cultural elites (e.g. Malina, 1994, 2) and destroying or distorting human relationships. But equally, it may be argued, the cultural changes and the challenging issues arising from the digitization of human society, are a fertile field for artists to explore.

The multimedia experience

In approaching an aesthetic of Multimedia Art, we must consider it to be a case of the whole is greater than the sum of its parts. As cited on p. 104, Guay's [1995] concept of "semantic compression" finds support in studies reported by the Fraunhofer Institute for Computer Graphics in *Acoustic Simulation for Visualization and Virtual Reality* [Astheimer, 1995]. These studies have shown that during the process of information transfer via a computer-based audio/visual workstation, the combination of visual and sonic material enables a receiver to perceive quantitatively, and analyze qualitatively, more data in the same amount of time. The information transfer is exponentially greater than the result obtainable by offering the same visual and audio material non-synchronously. This effect may indeed be based on experience and will possibly be greater as society develops a new-media literacy.

The experience offered to the end-user of multimedia artifacts is a synthesis of the constituent artistic elements from the following cognitive/perceptual modalities:

1. *Idea:* Theme, script or underlying structural/conceptual framework

2. *Visuals:* 2D and pseudo-3D graphics or stills, 2D and 3D video animation; motion-picture video; text as graphics (literary content considered purely for its aesthetic role as a visual element)

3. *Audio:* music, ambient sound, spoken word (including literary content considered purely for its aesthetic role as an audio element)

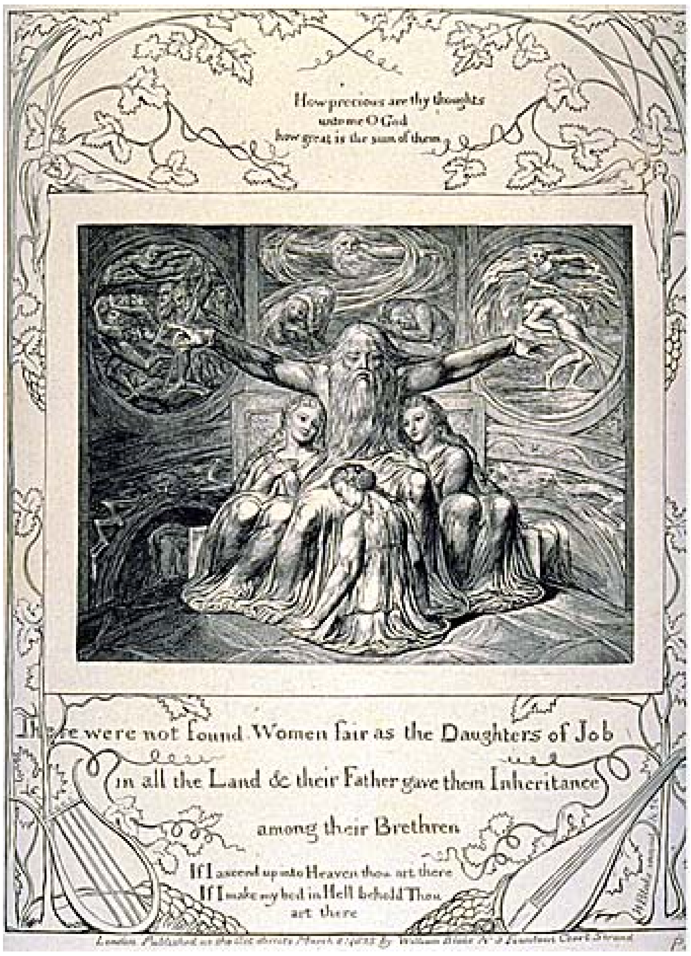
4. *Literary Content*: text; graphics and audio considered as text

5. *Programmed interactivity:* user-interface metaphor which allows access to hierarchical or branching design / structure, introducing real or perceived possibilities of user input to / control of artistic or experiential outcomes

6. *Kinaesthetic simulation:* The use of computer controlled motion (e.g. hydraulic motors rocking and tilting the users body) to simulate motion.

Aesthetic approaches to multimedia design

In Multimedia art, the melding of these constituent elements results in an artistic experience for which, it is argued [Hales, 1996, 2], a useful analogy is the kind of (often) non-narrative, non-literal, subjective aesthetic experience offered by poetry. It is not necessarily goal oriented (i.e. a narrative leading to a climax or conclusion) and may in fact have no sense of resolution. Reading poetry can be a process of interpreting and reinterpreting, reacting and reassessing, of absorbing meaning and mood.



The Daughters of Job, 1825 by William Blake.

Medieval illuminated manuscripts and the work of early mixed-media artists such as William Blake incorporated design elements, which were not necessarily literal illustrations, but whose decorative embellishment offered rather a sense of style and overall design and unity– a case of blending the print and graphic art medium into a new aesthetically cohesive work, whose imagery enhances the experience of reading the text (Onuf, 1995)[1]. From this point of view they can be considered the very faintest early glimmer in a tradition that has led down through centuries of evolution of technologies and design, to contemporary multi-media art forms.

Allen, [1994, 347] has explained the multi-dimensionality of multimedia:

> One of the projects I (have) worked on was called “Books Without Pages”. Having been a traditionally trained graphic designer, I suddenly found myself thinking about a book that included the dimensions of time, space, motion, sound, rhythm, and interactivity. As an electronic book designer I had to interweave layers of multimedia information and ask such questions as: Where does text belong in the electronic medium? What are the design criteria for timing and pacing? When should something move and when should it be static? And how does sound integrate with imagery?

When making comparisons within the ambit of traditional artforms such as oil-painting, classical music composition etc., apart from interactive possibilities, the most obvious difference in approach offered to artists using digital technology is the concept of *the sample*. There is precedent in the Western classical repertoire for quotation from other pieces, and in the visual arts for ‘found art’. However, the artistic

[1] Interestingly, Blake’s illuminated manuscripts are being converted into on-line hypermedia texts, archivist having realised that this is the only cost effective means of distribution in a form that is true to the original. See http://www.blakearchive.org.uk/cgi-bin/nph-dweb/blake ; accessed 30-01-03.

method of digital art often involves, as a central process, the capture of small slices of sound, image and video into a digital storage medium, for future manipulation and deconstruction-reconstruction. It is then often a process of re-ordering, layering and editing these samples, building up collages and sound-mixes that refer, in most cases (by virtue of the raw material), to the "real" world. The work to succeed must be driven by a concept, a process, a feeling, an idea, an agenda. There are potentially many ways open to the imagination to reach the goal of a fully realized multimedia art piece, whether logical, or intuitive, poetic or anarchistic, inspired, market-researched, or simply amusing.

Shaping Time

Time is a concept that eludes precise definitions, yet is basic to the human condition. We intuitively understand time, even if we cannot explain it. Newton [1687, 6] refers to time as

> ..of itself and from its own nature...(which) flows equably without relation to anything external

The operative word for the artist is *flow*. Art succeeds in simulating an abstract or imaginary sense of time to the degree that it is perceived to flow. The visual arts (painting, sculpture, graphic arts etc.) can be said to be timeless, in the sense of asynchronous. Film and music are temporal (synchronous) arts. Film, (abstract or experimental film aside) uses narrative techniques, and in this sense is more concerned with realistic representations of the passage of time.Yet this simulated "real time" experience has been selectively filmed, edited, condensed, compressed, in order to entertain and bring into focus dramatic events and emotional insights.

Music manipulates time in a more abstract sense. Real time passes, but also there is a symbolic or representational sense of time unfolding. In this author's view, musical events and textures build towards climaxes, and are appreciated at a deep cognitive / emotional level, psychologically referenced to our innate feel for, or inherent sense of natural rhythms and dramatic pace, derived over long periods of learning from the experience of life, as well as based on a genetically-based sense of inner circadian rhythms. One might validly argue that the rhythms and episodic structures of music are created against the backdrop of our experience of the rhythms and dramatic shape of life, the shape within shape, and rhythm within rhythm of a fractal universe.

The introduction of interactive branching structures allows multimedia art to simulate the multiplicity of possible outcomes in an indeterminate world, and also a freedom of movement back and forth in virtual time, and thus the generation of some other sense of imaginary time we are now discovering.

Pure music is a non-utilitarian aesthetic of sound organization. Because of its time element, its use of repetition, and the need for composers to address the process of shaping time, traditional music theory offers many concepts relevant to the analysis of a potential multimedia art aesthetic, from the perspective of time, e.g.

1) Rhythm:
   i. pulse
   ii. motive
   iii. phrase
   iv. movements - larger segments

2) tempo:
   i. rubato - accelerando/ritard
   ii. metric modulation, tempo changes

3) Metre
   i. beat
   ii. bar or measure (interesting because it is a relative measure dependent on stated properties of musical time such as time signature and tempo)
   iii. Hypermeasure or larger structural units such as verse, chorus or segment (segmental units of a sonata for example)

4) dynamics

5) The tension between order and anarchy

6) Serial processes vs. Chance and expressivity

7) texture
   i) harmony
   ii) counterpoint

Many of these concepts at some time derived from traditional music analysis have been applied to the analysis or description of the visual arts - e.g. rhythm, dynamics, texture, harmony. They also to some degree are of use in critical analysis of literature, film and dance. Critics and theorists turn to the language of music because of the difficulty in grappling linguistically with the subtle structure of visual art. For example, Ballo [1969, 198] (discussing the visual arts, primarily painting) writes:

> Rhythm in art is an internal equilibrium of spaces, achieved by means of beats and pulses. It implies modulations and consequently motion, even if everything may appear to be stationary. It means the movement created by spaces, by similarities which may correspond to beats or pauses. That is why proportion is a variant of rhythm, since it involves a concordance of harmonies. Proportion is born of the need for restraint (whereas rhythm, …, knows no limits). Proportion presupposes some kind of scheme, but one consisting of multiples and sub-multiples, from the relationship between which the rules governing composition can be deduced.

The Royal Academy web-site has this to say about Van Gogh's *Sunny Lawn in a Public Park* (Arles)', (oil on canvas, 1888), pictured above: "*The variety of colour in the trees and shrubs in the background, and in the newly mown grass in the foreground, is rendered in long, fat, juicy, marks of thick paint. These directionally applied brushstrokes create an alternating, visual rhythm across the picture surface. Everything in the painting can be identified and the composition is artfully constructed to remind us that it is an arrangement of brushmarks on a flat surface as well as a representation of nature. The picture is dominated by what we might expect to be the least important aspect of the landscape — a stretch of grass. There is no real foreground since the viewer looks down at the grass and shrubs from a high viewpoint. This elevated position makes it difficult to judge the distances within the painting. Everything seems more or less flat and equally close to us and our attention is repeatedly brought back to the surface, and its play of brushstrokes and colour*". [1]

While different to the rhythms, pulses and motion of time-based art, visual rhythms are structural/aesthetic mechanisms which use repetition to enhance flow and cohesion. Epstein [1995] argues that there are other important aspects of time that bear upon a discussion of musical time (which in turn are relevant to an understanding of multimedia as a temporal artform). The first of these is *Duality***.** Epstein contrasts the measurement of time into mechanically or electronically defined units - e.g., Greenwich Mean Time, hours, minutes seconds etc., - with experiential or subjective time, which we experience as a flexible phenomenon, depending upon the emotional content or relative personal significance or drama inherent in a particular experience:

> Experiential time...is the stuff of life, endlessly interesting, rarely the same filled with the episodes of living by which it is framed...unpredictable, its particular structures drawn from and framed by those very elements that make a given experience unique. [p.7].

[1] source: http://www.royalacademy.org.uk/?lid=647 ; accessed 23-11-02.

The second aspect is *Hierarchy***.** In his analysis of musical time, Epstein considers it worth mentioning that the basic hierarchical nature of time as we perceive it, as represented by the natural units of time which occur according to the cycles of nature - the periods defined by the cycles of the heavenly bodies e.g. the day, the lunar month, the solar year, and also as referenced by our so-called "biological clocks" or genetically given sense of time - are so fundamental to our life experience that they cause us to be predisposed to view musical time in a hierarchical way.

The third aspect is *Motion***.** Epstein argues that because motion can only be measured with regard to units of time, there is a sense in which time is connected with motion. Thus in a psychological sense we tend to view time in a goal oriented fashion - something must happen for us to get a sense of period or passing of time, and this predisposition to view time as a motion towards some climactic event or demarcation point is a fundamental to the understanding of musical time. [Epstein, 1995, 6-12] Further, there is a sense of motion implicit in harmonic structures, polyphony, melodic line, and resolution or closure.

The Rock/Pop Music Industry and Multimedia

An analysis of the aesthetics of multimedia art would be incomplete without consideration of the influence of an important socio-economic/cultural tributary - the popular music industry/culture. The marketing arm of the music industry has seized upon the CD-ROM format as a natural extension of its marketing cycle, specifically in its relation to the music CD (artwork / consumable product) and the music video clip (quasi-artwork / marketing tool).

As is discussed in detail in a later chapter (see p.195), music video has worked primarily as a marketing tool, a unique form of advertising, and is linked in a unique relationship with the musical product produced by the artist. This is because it is both an extension of the music artists image, an artistic artifact in its own right and thus enjoys a different relationship with its target consumer than other forms of mainstream media advertising.

> The distinctive look that a culture gives to things it makes is called *style….*Any object that "speaks" in the idiom of the society that produces it has *style*. It has a definite visual relationship to when and where it has been made, to the materials available to the maker-designer, and to the ideas that have dominated the life of that society. [Barry, et al., eds., 1964, 134-5].

It is a spirit, and sense of style, which has gone on to inform the multimedia product of artists from a music technology and music video background.

The popular music industry's approach to CD-ROM design has been focused on the opportunity it offers to make a further consumable out of ingredients already produced in the process of music distribution, while at the same time offering the necessarily image-conscious popular musician the chance to be perceived to be moving with the cutting edge of techno-culture.

But the music industry is fuelled by innovations and a thirst for the new, and those involved in marketing popular music are constantly looking for marketing advantages. Hence there has been a rapid recognition of and response to at least some of the creative possibilities offered by multimedia art as an adjunct to music industry practices. However, despite the widespread awareness of this apparent potential, this is yet to translate into revolutionary or remarkably innovative results, and the CD-ROMs produced by the music industry can largely be seen as simply a minor evolutionary step into a multimedia format, of the album cover with liner notes, artwork, lyrics, biographical material. It remains in essence an advertisement for the creative talent of the music artist, though, firmly in the popular music tradition of the rock era, it has a strong sense of style.

Among the pioneering music industry based multimedia presentations, there has been a range of (so far relatively clumsy) attempts to incorporate interface metaphors to engage the participant. While it is beyond the scope of this work to offer a comprehensive survey of the pop music industry CD-ROM output to date, nevertheless some random examples of the way the music industry has approached incorporating multimedia and interactivity into the marketing of music artists and their work are offered here, to put this aspect of multimedia evolution in a context.

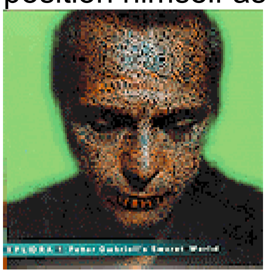
The Rolling Stones have released a CD-ROM entitled *Voodoo Lounge* (1996) based upon the fantasy that when the group is not touring or recording, they relax in the virtual reality of a place known as 'Voodoo Lounge' - a mansion on a grand old Louisiana plantation that has been converted into:

> …a wicked house of pleasure by its mysterious owner. (From the enclosed manual, p.1).

The user is offered an interface which allows an exploration of an exotic and outlandish virtual house. The house contains hot-spots, which the user must discover, each of which trigger *QuickTime* video footage of jam sessions and concerts. This is the extent of interactivity, with no deeper unifying concept or game-play, and *Voodoo Lounge* must be regarded as a token attempt at harnessing interactive multimedia technology to the album liner notes tradition.

Tommy: The Interactive Adventure (1996), based on the famous rock opera by Pete Townsend and *the Who*, takes a more multi-layered approach, but is still essentially an interactive coffee-table book - and hardly qualifies as an interactive adventure as claimed in the advertising descriptions.
It contains a behind-the-scenes perspective of the creative evolution of the work, using video and audio content and text that follow the development of the opera from The Who's original demo recording through to the various incarnations, including Ken Russell's film and the Broadway musical version. It includes documentary style interviews with those involved in the original productions, clips from the film version, concerts and the stage show, as well as some photos, original notes and drawings from Pete Townsend's personal archives. The CD-ROM includes a so-called "new multimedia version" of *Tommy*. The user can search for hidden video clips, as well as jump to a different recording or to another song (though the latter transitions are not seamless, but result in an awkward pause or momentarily jerky playback because it plays directly from the CD and is subject to the slow data transfer rate). The work is divided into three main areas. *The Experiential Tommy* is the main menu area. Here the user can interact with the music from Tommy; listen to the songs, see the lyrics, view clips from the movie or play. The second area is the *Documentary*. This timeline allows the user to see how *The Who* and *Tommy* have evolved through the years. The third area, *Pete's Archive*, is a collection of notes, profiles, and trivia, which have been part of the *Tommy* legend.

Peter Gabriel has been one celebrity musician who has seen it as important to position himself as an active multimedia producer. His high profile CD-ROM



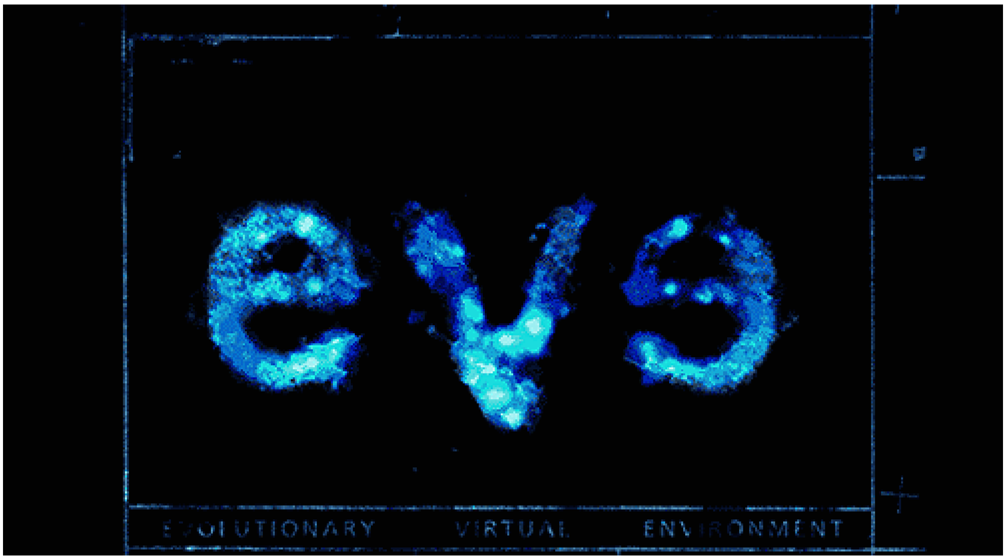
Xplora1 – Peter Gabriel's Secret World (1993), featuring a strong visual style with bold colour combinations, was developed with *Macromedia Director*. Screen backgrounds are made up from still images of grass and flowers, sky and clouds, gravel, fire, water, and (incongruously) of electrical cables.
The ever-present colour-bar at the side of the screen interface allows easy-to-understand navigation to a series of short interviews, as does a taskbar along the top

of the screen that offers icons for each the main sections of the CD-ROM which the user can enter: *US, Real World, Behind the Scenes*, and *Personal File*. The *US* section explores four videos from the artist's *US* album. The *Real World* section allows the user to explore ‘world music’ instruments and sounds, offering the opportunity to hear the instruments individually.

Above: Peter Gabriel’s *Xplora*. Colour-coded interactive bar at left and interactive icons at top.

The *Behind the Scenes* section offers backstage video footage, in *QuickTime* form, from various events involving Gabriel. The *Personal File* section includes photos, videos, and information on Gabriel's personal life. The work was designed for the Apple *Macintosh* platform. An attempt by the producers to port this work to *Windows* 3.1 is compromised by major technical difficulties, and a planned *Windows 95* version was never released.

The virtual experiences are aimed at uncovering Gabriel’s creative processes, and it also attempts some educational insights regarding World Music, with just a touch of the political, through references to Amnesty International. Three of these experiences are interactive puzzles. On start-up, the user must complete a simple ‘jigsaw puzzle’ of Peter Gabriel's face in order to proceed to the rest of the program. Once this problem is solved, the user is confronted with a second on-going, rather naïve puzzle, which is a search for items to put in a mysterious briefcase, offered by an on-screen Peter Gabriel at the onset of the journey through the CD-ROM. Filling the briefcase results in being awarded a prize. Four of the twelve items in the case are “passes” which allow the player access into certain sections of the CD-ROM program, such as a “backstage pass” to see the Drummers of Burundi, or to see a clip of Gabriel's performance at the Grammy Awards. The third puzzle is a musical jam session inside Gabriel’s *Real World* Studios. The object of this puzzle is to try to find the 33 singles/combinations of musicians to play together and “please” the producer, Brian Eno, who is an on-screen assistant. Another rather naïve attempt at interactivity is the music mixing facility, allowing Gabriel’s song *Digging in the Dirt* to be remixed by the user. Unlike the production values for most of the CD, this facility is quite primitive, limited to four tracks with sluggish controls, controlling alternative sound files with very poor sound quality.



Gabriel followed up the now dated *Xplora* with a second foray into a multimedia delivery format, entitled *Eve.* Much contemporary popular music focuses on the theme of relationship – through love songs in all their variety– and *Eve* explores the theme extensively, using multimedia techniques. It is based on a puzzle-solving metaphor and once again great emphasis is placed on delivering a visually attractive interface. We enter a world in which Paradise has been lost, and it is up to the user to find hidden elements, as part of a quest for Paradise regained. This requires careful exploration of each environment. There are three separate worlds to explore, *The World of Mud, The Garden,* and *Profit*. Each world features one Gabriel song, and the artwork from one of three acclaimed visual artists, Helen Chadwick, Yayoi Kusama, Cathy de Monchaux. Each of the chosen artists have created a specific piece for this CD, and they are interviewed with a focus on their art and philosophy. The user cannot access the interviews until certain items in the “world” have been found. This part of the interactive conceit becomes frustrating, in that each screen world has a limit of 2 or 3 wrong answers before the user *must* change screens. The user cannot avoid, for instance, testing every insect in a garden, and therefore must “leave” the garden every few minutes. As the user explores the virtual world, the cursor changes from blue to yellow when detecting a roll-over hot-spot (an icon or image etc. pre-programmed to allow an interactive response) through mouse movements. For instance, while exploring a *QuickTimeVR* rendered 3D panorama[1], which allows the user to scroll continuously through a 360 degree first- person view of an enchanted forest, we see a silver light hovering over a pool of water. One can 'catch' this light with the mouse, and use it as a magical force, which opens otherwise closed 3d objects, such as a suitcase or a door, revealing new spaces or possibilities for interaction. One room that is revealed features a tiny television screen (81x56 pixels), on which appears a series of short video clips of people from many walks of life speaking about their relationships or their view of an ideal relationship. (These clips have been compressed and scaled down to a file size of only a few hundred kilobytes on average. There are nearly 400 short video segments contained on the CD-ROM). At other 'locations' in the navigation Adam and Eve figures appear, and there is a sensual, sometimes erotic atmosphere and visual style. Gradually, as the user navigates through the work, there emerges a sense that a philosophy of relationship is being explored, (though this never comes clearly into focus). Apart from the typical looped sounds, and accenting sounds synchronized to events, there are musical rewards for puzzle solving which us allow us to hear Gabriel's songs. Valazza [1997, 1-2] a programmer on the project, describes the authoring process:

> EVE was created with a proprietary multimedia authoring tool developed at Starwave. Known internally as *Dr. Seuss*, the tool was developed as the foundation of Starwave's CD-ROM efforts and was used to create the critically acclaimed yet financially unsuccessful titles: *Eastwood, Sting*, and *Muppets Inside*. For the creation of EVE, the *Dr. Seuss* authoring environment was extended to support lightning-fast alpha-channeled sprites and a *Basic*-like programming language called *Possum* that opened up new levels of interactivity not previously possible in the original Dr. Seuss. In addition, the EVE player was ported to the *Macintosh* making it possible to ship a dual platform CD-ROM with one set of audio, video, and art assets….
>
> Many programmers leave some personal mark on their creations. Known informally in the industry as *easter eggs*, these undocumented features often manifest themselves as secret messages, hidden animations, or images of wives and cats. When faced with extra time between design revisions, I contemplated what mark I wanted to leave hidden in the depths of EVE. With the help of Martha, the project's media manager, we decided to draw on the relationship themes of EVE and based a number of secrets on the number four, symbolic of the group of four of us who became close friends from across continents.

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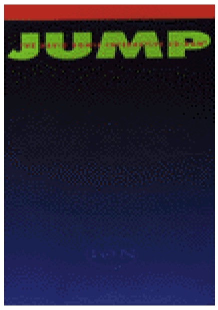
[1] see p. 178

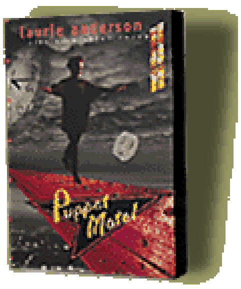
Left: Screenshot from "*The Mud Theme Riddle*" (Peter Gabriel's *Eve*). Numbers indicate hidden "easter eggs" accessible by a particular combination of mouse movements, inserted by the programmer and not documented in the original release.

Another style-conscious rock musician who was quick to position himself as a multimedia artist was David Bowie with *Jump: The interactive CD-ROM.*



The CD was produced with minimal input from the artist however.[1] It allows the user to edit their own "*Jump They Say"* video, mix their own version of the music, and, as advertising material describes, once again "*explore a virtual world of hidden animations, sounds, pictures and other surprises*," none of which is very surprising or interesting. The whole product is seriously flawed in that it "crashes" repeatedly. It contains four videos from the album of the same name (*Jump They Say*, *You've Been Around*, *Black Tie White Noise* and *Miracle Goodnight*) and some interview clips.

Laurie Anderson, while not attaining the celebrity status of the popular artists so far discussed in this section, has created a niche market as an innovative and multi-talented musician/performer with a strong sense of style and an on-going interest in the nexus between art and technology.



Anderson made her interactive CD-ROM debut with *Puppet Motel* (1998), one of the more interesting, aware and medium-specific forays into the hybrid world of music industry-driven CD-ROM design.

Anderson and designer Hsin Chien Huang collaborated to create an interactive environment in which the user becomes both audience and performer. The CD-ROM contains 33 virtual rooms for exploration, made of vivid 3D graphics and animation, including video documentary, spoken-word performances, and more than an hour of avant-garde instrumental electronic music – which includes performances

[1] Source: *David Bowie Live Chat Transcription - 1/7/94*; URL: http://www.bowiewonderworld.com/chats/dbchat0794.htm ; accessed 1-07-02.

from Anderson's albums *Bright Red* and *The Ugly One With the Jewels.* The interactivity is extensive—the user can record a message on an answering machine, create a constellation, reshape the human ear, play the viophonograph (or the tape bow violin), ask questions of an *Ouija* board, and watch videos. *Puppet Motel* allows the user to connect to the World Wide Web and download video clips from Anderson's Web site to view as one plays the CD-ROM. The work avoids the obvious gaming conventions of many interactive CD-ROMs, but instead offers the user a hypnotic flow of repeated images, mysterious voices, and rhythms. Navigating through *Puppet Motel* is challenging in itself; the only “map” is one’s memories of what has already been seen, with objects functioning as icons/signposts. In addition to the repeated objects, a virtual Anderson appears from time to time, offering explanations of, for instance, a particular set, or the metaphysics of dance. In place of a coherent linear narrative, electrical sockets, ventriloquists' dummies, chairs, clocks, telephones, musical instruments, and a range of shadowy icons drift in and out of view, underscored by Anderson's distinctive techno-rhythms. With overtones of science fiction, digital voices chant mundane phrases in a monotone - "*You are out of memory. Save. Save now", "This is a recording”,* etc. Huang's rich, photorealistic textures and minutely detailed virtual spaces are important in retaining user immersion in Anderson's work, which demands a high level of concentration. The user can interactively supplement Anderson's rhythms. In one section, for instance, surreal, hybridized musical instruments float on the screen. The user can drag across the virtual strings to produce speech instead of music (violin is Laurie Anderson’s first and ‘trademark’ instrument). In another narrative section the user can interactively insert a virtual needle into an image of an ear-lobe, at various pressure points, which activates a virtual Anderson telling a story about an acupuncture session. By jabbing the needle at various intervals, one can produce rhythms, one medium merging with another.

While obviously it is impossible to give more than a very small sample of this kind of multimedia product, the examples reviewed herein are quite representative. For example “The artist formerly known as Prince” has produced a *Prince Interactive* (1996) which is very similar in concept and execution to the works in this genre cited above – a simplistic puzzle-solving game and already hackneyed idea of a virtual mansion to explore.

The symbol that TAFKAP uses as his personal identification was used as the centerpiece for his interactive experience. The object of the user's efforts were to be put towards solving a puzzle; a game if you will. The symbol was broken in pieces and scattered throughout a mansion devoted to TAFKAP. When all pieces are recovered, the user is treated to a never before seen video of TAFKAP's single "Endorphinmachine". The mansion was a reflection of TAFKAP himself. Mirrored white marble floors, lavish gold Egyptian styled trim and a lavender carpeted, sky-domed master bedroom. Of the seven 3D rooms, CyberPipe's founder was the digital artist and texture mapper of three. All throughout, maintaining a very magical feel, most of the objects in the mansion were "hot areas" which linked to interviews, games, video clips, photographs, or music videos, including the song "Interactive" written especially for the CD-ROM, by TAFKAP. [1]

Though not strictly speaking a music industry product, *Real Wild Child!* is a popular music history resource on CD-ROM that was inspired by the exhibition of the same name held at Sydney's Powerhouse Museum.

It won 7 major awards including the *Gold Award* for best multi-media product of the year, 1997. This irreverent and quirky CD-ROM was compiled with the co-operation of many of Australia's major, and some independent music labels, plus the resources of the ABC (including its national youth radio network *Triple J* FM), the Powerhouse Museum, and a number of music publishing houses. Technical production was by *Pacific Advanced Media*, and it was funded through the federal government's '*Australia on CD'* program[2].

Real Wild Child! offers an overview of the key performers, musical trends and significant events in forty years of Australian popular music. It attempts to be a thorough, accurate and balanced account of the subject. Although the CD-ROM provides detailed information about hundreds of bands and performers, it is not constructed as an encyclopedia and does not pretend to be a complete database of Australian rock, but is lighter, concentrating on using multimedia techniques to providing an entertaining insight and to capture the "look and feel" of each era of popular culture. On accessing the CD-ROM, the user is presented with a surreal

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[1] http://hometown.aol.com/freefaller79/cdroms.html ; accessed 12-01-03.
[2] Department of Communications, Information Technology and the Arts. URL: http://www.dcita.gov.au/Article/0,,0_1-2_1-3_461-4_11768,00.html

opening cartoon sequence, which unfolds into an interactive suburban street scene. This cartoon street is the main interface for the CD, and includes some clever interactive devices. The bold and colourful cartoon artwork for the piece was created by Reg Mombassa (of novelty rock group *Mental as Anything* and *Mambo Art* fame) and is eccentrically stylized and strong in visual appeal.

Above the buildings in the main street flies an angel figure, which is actually an interactive icon that can be clicked on for "help", or to adjust the volume of your speakers, and also revealing a menu of the relative icons and their functions. Clicking on a chicken returns the user to the street scene interface. Here the user must simply drag the mouse cursor left or right to move along the street. A skateboarding figure, representing the user, is cruising along the street/interface. This figure frequently passes over the top of a manhole cover, which the designers have included as a humorous way of exiting the program. Clicking on this manhole cover moves the cover away from the hole, which causes the skateboarder proxy to fall into the hole, in turn prompting an option pop-up asking whether the user would like to "drop out" or "keep looking".

There are seven different musical 'eras' for the user to enter and explore. Each is represented by a stylized building on the main street, and clicking on each of these buildings allows access to animated imagery and information on an era of Australian popular music history. Each of these eras are introduced by a short introductory sequence, consisting of a cleverly animated collage of period images, such as publicity photos and posters from the era, with a concise, documentary-style voice - over explaining the major themes, the various bands and events of major importance pertaining to that particular popular music era. The various eras are represented by:

- 1956 – 1963: A 50's Milkbar
- 1964 – 1968: A Teenage Girl's Bedroom
- 1969 – 1973: A "Hippie" House
- 1974 – 1977: The ABC TV Studio
- 1978 – 1983: The Star Hotel
- 1984 – 1989: A Nightclub
- 1990s: A Garage

Once the user interactively enters an era's representative building, (having viewed its introductory sequence), the user experiences an interactive “room”, featuring humorous and cleverly put together references to that eras musical history and its cultural icons. Each of these satirical spaces in turn conceal links to other interactively accessed collages pertaining to key artists or styles of the decade in question. The user discovers humorous animations of various famous or notorious Australian personalities including Prime Ministers and media stars (Robert Menzies as a milk bar operator, *Countdown* television presenter Ian “Molly” Meldrum mumbling incoherently, fugitive corporate business pariah Christopher Skase in oxygen mask etc.). Roll the cursor over them and they speak a suitably quoted and invariably funny line, using the technique made famous by Terry Gilliam of Monty Python, of moving a cut-out of the jaw.

The *Real Wild Child!* CD-ROM contains a large body of media and information constructed as entertainment, including 2200 photos, 300 audio excerpts, 50 video *QuickTime* segments, and approximately 170,000 words in the text. Over 150 artists or bands are profiled, and it includes over 100 discrete stories, covering musical trends, youth culture and current events – from teenage rebellion in the 1950s to dance music in the '90s. Additional content includes a quiz, original Top 40 charts of each era, select discographies, an Australian rock bibliography, information on the cost of living in each era, and a defence of the teenage music fan. The CD-ROM’s historical material is fully indexed, allowing the user to search for any of the 900 bands or performers that are mentioned, from anywhere in the text. In selecting which bands and performers to include, the creators looked beyond mere chart entries and record sales, and gave consideration to the lasting influences some bands have exerted on Australian music. Space limitations prevented the inclusion of every Aussie band ever to have a hit record. Clips and music from some performers were omitted for copyright reasons [Cox, 1997, email]. It was produced using the *M-tropolis*, multimedia authoring application, a short-lived competitor to Macromedia *Director*. (See next section for more information on the production of *Real Wild Child!*). The choice of subject matter is fortunate for the chosen medium, in that there is a serious data base of information underpinning, enriching and structuring the work, which is given expression by a rich and vivid visual and auditory history. This allows for a tightly structured, detailed yet entertaining and sensorily stimulating experience, overlaid with the irreverent energy and exuberance of youth culture.

As a concluding note to this section, it is not the intention here to give an exhaustive analysis of interactive CD-ROMs produced by the music industry, but rather, by offering a few high-profile examples, to offer an understanding of the general thrust of work produced for this market.

Multimedia and the marketplace

Economies of scale are driving down the costs of capital investment in multimedia. But as a self-conscious artform it began as a high cost exercise. To reach a marketable professional level, including promotion and wide distribution, remains a big budget exercise.

For instance, according to Peter Cox, curator of the *Real Wild Child!* exhibition at the Power House Museum, Sydney, which was used as the concept for a CD-ROM of the same name (discussed in the last section, and also under the co-ordination of Cox):

> The ‘Australia on CD’ funding made the *Real Wild Child!* CD-ROM possible. It was a once in a lifetime opportunity…I think the exact amount of the grant from DOCA was $645,000. A lot of the money went to copyright owners, including record companies, music publishers and owners of photographs and film clips…
>
> Over and above this figure, you must take into account the enormous in-kind contributions from the four consortium organizations. The ABC provided many other resources, facilities, personnel (including legal advice and copyright administration!!) etc at no charge or at subsidized rates. To record the voiceovers Triple J provided announcers and a studio with a producer. The ABC charged a small amount from the budget for the services of the project manager Ben Cardillo. The Powerhouse provided Julie Donaldson as a member of the management committee without taking a fee. Similarly Mushroom Pictures provided Martin Fabinyi on the management committee without taking a fee. If you cost the in-kind contributions, I reckon the budget would add up to close to $1 million! [Cox, 1997, email].

This particular project was burdened with a disproportionate royalty cost, but even if we allow 50% of extraordinary costs were due to this factor, it is indicative of the scale such projects can reach in terms of budget. It follows that the freedom to experiment on a release-quality level has been largely limited to those industry market leaders who have already developed a following, and a market. It has generally been artists who have already achieved a celebrity profile and widespread social awareness of artistic stature and authority etc., that have been able to migrate into multimedia delivery formats - e.g. Peter Gabriel, David Bowie, The Rolling

Stones, The Who, Bob Dylan, Prince. Such artists had access to the concomitant resources to underwrite or generate the requisite budgets.

Until the explosion of the Internet, publishers and distributors were the vital conduit between the artist and the consumer / audience. Without distribution and marketing, art cannot find its all-important audience.

But, as leading digital commentator Nicholas Negroponte [1995, 11-13] claims, multimedia art as a product in the marketplace (presently delivered in the CD-ROM format) has an inherent digital propensity to circumvent traditional market structures. Instead of being delivered as “atoms” (packaged product), digital information can arrive as bytes (electronic data transmission).

The large-scale utilization of telephonic systems to interconnect institutional and personal computer systems has led to a situation where the costs of, and the infrastructure involved in packaging, warehousing, freighting and retailing of consumer entertainment and art product can theoretically be eliminated from the chain between artist and consumer. This inherent propensity of digital data to be transmitted directly to consumers - to be “beamed” into their loungerooms - has helped feed the mutation and rampant growth of a new system of social communication and information exchange we know as the World Wide Web. At the same time accepted perceptions of what is possible have changed dramatically. Simultaneously it is shaking up the publishing and entertainment industries, and also exerting an influence on the cinema and the music industries.

A direct, virtually instantaneous means of delivery between supply and demand, artist and art lover or student, the curious and the knowledgeable, the innovators and the audience - is now potentially in the hands of the two respective parties.

Practitioners constantly recognize that the telephone system is an inefficient way of facilitating this exchange. Low-band internet is presently suitable for text and limited graphics delivery. Sound is used, but to a limited degree, and the use of video is still relatively marginal. There are major research and development programmes being undertaken by a wide variety of competing software companies aimed at producing technologies which will facilitate data transfer in more continuous, speedy, user-friendly or efficient ways. An increasingly sophisticated variety of audio and video streaming plug-ins are appearing which attempt to overcome the bandwidth bottleneck and allow Internet browsers to offer an immediate multimedia experience including real-time audio and video.

And meanwhile the practical potency and economic potential of the Internet in a large and complex society will continue to drive a gradual move towards technological solutions to the bandwidth bottleneck. Once these systems are in place we will begin to fully realize the potential of this medium. We may witness the promised revolution in entertainment, and (one would hope) art, once we can receive transmissions of an infinitely variable virtual reality on a high definition interactive television receiver, with multi-channel high-quality sound from the comfort of the home.

Conclusion to *Chapter One*

Statistics relating to the sale of computer hardware, computer software, and also software piracy, support the widely-held view that developed society is in the midst of a digital revolution. This revolution is impacting on global communications, leisure and work practices. It has also opened up new avenues for art-making and artistic expression, by making available affordable technologies that have brought sweeping changes to the way people work in graphic arts, popular music and audio recording, video production and animation. The desktop computer workstation has become a ubiquitous multi-purpose technology. It has democratized art-making by making sophisticated technological art processes (accessible through user-friendly interfaces) a reality for broad sections of society, in affluent / developed nations. High tech art has migrated into the private domain and is being integrated into the mainstream social fabric.

The introduction of programming for interactivity to this mix has produced a new art form – interactive digital multimedia. The rampant growth of the internet represents a new global interconnectivity which has opened up a direct channel for the dissemination and exchange of products of this digital art revolution.

This dissertation attempts to draw together an overview of contemporary theories and practices related to new media / multimedia art. During the production of multimedia art the broad gamut of art skills that are possible using digital techniques must be integrated, and a synergy between them developed. This dissertation proposes a unifying perspective on the contributing digital multimedia arts.
As a foundation to this overview, the origins of the new media form are considered, and an attempt is made to place multimedia within the context of contemporary western art practice and the broader cultural context.

This introductory section progresses to a discussion of multimedia as meta-art. To simplify the discussion of multimedia as an emerging meta-art, a modular approach was taken to the primary sensory modalities engaged by contemporary multimedia, sight and hearing. In order to differentiate the various forms of information flow occurring within the multimedia experience, a further conceptual device was introduced - that of *channels of communication.* Each communicative channel is a discrete form of encoding and communicating information.

Of the primary sense modalities, vision can thus access information asynchronously via image, and text, and synchronously via motion picture. By ‘asynchronous’ it is meant that information can be accessed outside the constraints of time. Motion picture is a time-based medium and the experience requires time to pass in a linear fashion in order to access the visual experience, and it is thus designated synchronous. For the auditory mode the channels of communication are: natural sound (sound effects), music, and spoken-word, all of which a synchronous or time-based phenomena. We learn a cross-channel language of interpretation as we gain experience as users of multimedia artifacts. This cross-channel language of interpretation allows us to interpret the cumulative experience offered via the respective channels of communication by cross-referencing between the information streams – a case of the whole experience being greater than the sum of the parts.

An analysis of the origins of the term *multimedia* shows that it had roots preceding the computer and was used to describe productions and co-ordinated art installations incorporating multiple slide projectors, video monitors, audio tape decks, synthesizers, and/or other stand-alone media devices, to produce a coherent mixed media experience. The media forms represented by these hardware devices continue as contributing communicative channels in digital multimedia – the means of handling these media forms has been changed by the merging of digital media production technologies within one meta art-making machine.

What sets contemporary digital multimedia apart from its analogue predecessors is the availability of, and increasingly realized potential for, programmable user interactivity, and non-linear structure. Hyper-linked interactivity supports the processes of free association and choice in the act of creative navigation through inter-related information, and is an essential and defining ingredient in the multimedia experience.

A working definition of multimedia was established:

> *Time-based communication artifacts, created and delivered via digital computer technology, through processes which involve the organization and manipulation of digital media into a non-linear, interactive, multi-sensory experience, which is designed to be perceived as a unified artistic expression*.

In pursuit of an understanding of multimedia as an autonomous new art-form with a unique aesthetic, it was necessary to consider definitions and conceptions of what constitutes art in the contemporary, post-modern world. Considering the range of art-making activities that have been undertaken and recognized during the 20^{th} century, any valid discussion of the state of contemporary art practice requires the acceptance of the broadest possible definition of what art can be.

Consideration was given to the origins of artistic activity. The discussion was extended to include a range of issues related to the role of art in human society. This led to a consideration of the evolutionary connotations of art-making, and of the human need to express through art responses to life that are beyond the scope of everyday language. This included reference to the ritual and mystic origins of art, and led on to an analysis of art as a commercial commodity within the gallery system, which imbues art with value as an asset, speculative investment and a symbol of socio-economic status. The revered status of the cultural icons exhibited in state-owned galleries and art museums can be seen as a reflection of the human desire/need to view human life as significant, important, and great art as one of the peak human achievements.

However much digital art is produced in a commercial context and this fact in turn brings up issues related to the "art vs. craft" debate. While much sophisticated and well-crafted multimedia is produced that falls within a category we could designate as commercial art – there is every potential for interactive artwork to be created that belongs in the realm of fine art. Public exhibitions of interactive electronic art have special technical needs which make it more challenging for curators of exhibition spaces however.

Post-modernism, as a movement in art and literature, which developed as a reaction against modernism, is a difficult but significant field of modern thought, and because it has been applied to a wide range of late-twentieth century culture, it is relevant to consider it in relation to multimedia. Because of their non-linearity, multimedia works can be labyrinthine and fragmented. The introduction of interactivity opens up possibilities for the user to relate to and understand the work in his/her own unique way. Sampling technologies facilitate and encourage the quoting from and remediation of cultural and media artefacts. In a post-modern, globalize society, the evolution of this new eclectic, interactive digital artform can be viewed as a projection of the social consciousness of the times that produced it, rather than just a matter of historical coincidence. The suggestion was put forward that it is indeed no co-incidence that digital interactive multimedia, and the technology to sample, clone and collage sounds and images, emerged simultaneously with the post-modernist movement. With the idea that “art is a mirror to society” as a premise, the argument was put forward that interactive digital art can be can be viewed as a “physicalization” of the contemporary consciousness identified by post-modernist thought.

In taking the discussion of the nature of art in contemporary society forward, art was also considered as a form of self-referential intellectual experimentation. Technological developments have opened broad new frontiers for art-makers to explore, mirroring the important issues confronting society. The sweeping changes brought about in society by the digital revolution were bound to resonate in the artistic output of the generations experiencing these changes, and the new media technologies present a fruitful field for a wide range of experimentation at the interface between art and technology.

Approaches to contemporary art which go beyond traditional concerns about the appearances of things and which attempt to interpret art in terms of raising or exploring human consciousness were also considered. Art (in the latter sense) is seen by Jungian psychologists as an activity that serves as an interface between the unconscious and the conscious, a way of bringing into the conscious realm a range of unconscious responses to the archetypical forces and issues central to human life. In these terms, the transcendental nature of peak art can be seen in the light of the concept of the *alchemy of creativity*. The role of art in society was then considered in relation to the changing historical social meta-narratives, ranging from a traditional Christian world view, in which the Church valued art as a tool of spiritual propaganda, through to the Marxist perspective of art as a socio-economic, class-based phenomenon.

A summary was then presented of the ideas of Pietersen [2000], who identifies four underlying meta-paradigms of art, reflecting the four basic modes of intellectual activity. These are: *Formalism* (concerned with universal form and the structural relationships within an art-work, devoid of political or personal meaning), *Representationalism* (Realism and Naturalism, concerned with imitating and celebrating nature and reality), *Expressionism* (concerned with the expression of personal or subjective experience and feeling), and finally *Functionalism* (concerned with social reform and political statement).

The discussion of the nature of art was extended to encompass those contemporary approaches of to art, introduced by Jackson Pollock and the Abstract Expressionist movement, which place emphasis on art as a process or expressive performance, rather than on the simple consideration of art as object or outcome. This revealed the difficulties associated with the migration of natural media techniques into digital forms of expression, in an art context that celebrates dynamic improvisation, change, flux and flow, and which values art as action and attitude over product.

The continuing importance of the role of technology in driving forward the historical evolution of art-making practices was discussed, and in relation to digital art, examples were given of how new artistic techniques are emerging which are specific and unique to the digital domain.

The concept of a broad-ranging psychology of multimedia art was also considered, which included reference to behaviourist theories of art and the difficulties associated with approaches to the scientific measurement of subjective aesthetic response.

Contemporary social life is inextricably dependent on communication technologies and now includes considerable exposure to artificial experience, through cinema and television. Reference was made to how this has contributed to the current acceptance and understanding of virtual experience in cyberspace and multimedia. Cyberspace is an important concept with relation to digital art and in real terms a significant new factor in contemporary life. Any discussion of interactive multimedia must deal with the impact the internet has had, with regard to opening up new channels of artist-audience relationships, subverting the dominant economic paradigms in terms of distribution of art products, and the way in which the internet can become an art-space in its own right. Concept maps are a visual aid used in the pre-production and conceptualization of complex, multi-layered, non-linear multimedia designs, and in an attempt to grasp the scale and complexity of the internet, contemporary approaches to mapping or visually representing this illusory reality were surveyed. Visualization approaches to understanding the scope and nature of cyberspace include concept maps, geographic metaphors, census and statistical maps, maps of hardware and infrastructure, topological maps, 3d visualizations of information spaces as well as artistic representations of cyberspace found in popular entertainment.

It is postulated herein that current multimedia art practices on a desktop computer workstation represent the early signs of an emergence of a multimedia meta-artform. An important complementary trend will be the development of a computer meta-language or universal interface, and early signs that operating systems and art-making applications are moving in this direction were presented. For example, Microsoft *Windows XP*, Apple *Macintosh OS X* and the *Linux* Graphical User Interfaces have adopted almost indistinguishable visual iconography and metaphors, interactive conventions, and even blue sky colour themes. Similarly, major competing graphic design applications are so close in presentation and design as to have initiated copyright infringement litigation between the major multimedia software vendors.

The multimedia art experience was discussed in terms of the concept of *semantic compression* – that a synchronized combination of communicative visual and audio material becomes immersive and engaging because of the exponentially greater communication of information involved in this multi-channel experience. The multi-channel experience is made richer by an emergent cross-channel language of interpretation, by which the cross-referencing of information gained on parallel sense channels results in an increased grasp of the communicated experience – the user receives a more comprehensive communicative experience through the inter-relationships perceived from co-ordinated parallel sensory stimulation.

The dissertation adopts a modular approach to sensory input, in its analysis of the experience of multimedia. The neurophysiology of receiving sensory stimulus, while a far from mature field of scientific study, remains an important consideration in terms of optimizing the effectiveness of multimedia. A stimulating side-issue related to the neurophysiology of sensation is the phenomenon of *synaesthesia* - the involuntary physical experience of a cross-modal stimulation and association, a condition reported by many highly regarded art-makers. Genuine synaesthesia is spontaneous, specific, consistent and durable. Comparisons were offered between differing sets of colour and sound correspondences used by historically significant synaesthetically endowed artists, including Scriabin and Rimsky-Korsakov, with the suggestion emerging that, different though they are in their details, the correspondences seem to support the objective existence of this phenomenon. Studies were cited which suggest all humanity falls somewhere on a spectrum of synaesthetic ability, a possibility intriguingly relevant to multimedia art experimentation.

The phenomenon of synaesthesia has long fascinated artists and thinkers. A short history of experimental technology inspired by, or aimed at reproducing, the synaesthetic experience was presented. This history reviewed many different attempts at constructing a musical instrument which played with colour (the "colour organ"). Then the abstract approach to a synthesis of sound and image in the experimental stage designs of Wassily Kandinsky was discussed. Next the extensive history of thought and experiment associated with technology-based, quasi-synaesthetic experience was presented as a evolutionary strand important to the development of contemporary multimedia. This section concluded with a description of a range of contemporary digital software approaches to exploring the synaesthetic experience (including the *MetaSynth*, *Bliss Paint* and *Synaesthesia* desktop applications), which concluded with an outline of current human movement tracking technologies being used experimentally at the interface between sonic, aural and performance art.

Much of the theory and experiment in the field of technology-based simulations of the synaesthetic experience have been guided by a vision that is also central to this dissertation: the concept of multimedia as a nascent meta-art. A survey of major artists who have pursued this as an ideal was presented, including Wagner's *Gesamtkunstwerk* or total art-work; the dynamic and influential ideas of the Italian *Futurist* movement which saw cinema as the cornerstone technology to a future technological synthesis of the arts; the Bauhaus dream of a *theatre of totality*, an abstract theatre of movement, colour, light, and form, purged of its literary encumbrance; John Cage's *theatre of mixed means,* which was in turn so influential on the emergence of later mixed-media genres, such as *the Happenings*, video art, electronic theatre, performance art, and interactive installations; and finally the futuristic, comprehensive, meta-art machine visualized by the German author and mystic Hermann Hesse in his Nobel Prize-winning novel *The Glass Bead Game (Magister Ludi*) – on which he visualizes the meta-artist of the future extemporizing multidimensional, multisensory expressions of human culture and understanding.

The effects on art-making practices and products resulting from the wide availability of affordable and highly sophisticated computer-based technologies was then discussed, beginning with the seminal ideas of German socialist philosopher Walter Benjamin regarding the degradation of the 'authenticity' and 'aura' of a work of art brought about by technologies of reproduction, and leading on to ideas associated with the democratization of art-making resulting from the digital revolution.

A small representative sample was then presented of early attempts by leading music industry celebrities to incorporate multimedia into their artistic output, which included works by Peter Gabriel, the Who, the Rolling Stones, David Bowie and Prince, which concluded that, generally speaking, early attempts at remediation or crossing over from popular audio art to multimedia by celebrity musicians were superficial and relatively clumsy. A more integrated attempt at bringing a multimedia approach to the pop music world was outlined in an analysis of the work of Laurie Anderson's *Pet Motel*, and in the award-winning *Real Wild Child!* interactive CD-ROM – a history of four decades of Australian popular music successfully remediated into a cohesive interactive form.

Commercial pressures shaping the form and development process of multimedia art products were discussed, in the light of multimedia production being an exercise requiring relatively large budgets.

Consideration was given to the elements of the multimedia experience and related concepts, including the difficult to quantify, but frequently invoked goal of user-*immersion*, and the concept of *shaping time*. With regards to the latter, it was seen how the language of musicology (for example terms such as rhythm and harmony), which have evolved out of a need to conceptualize temporal art structures, have been appropriated to serve in an analysis of other non-temporal and visual art forms. As cross-media terminology they become important tools within multimedia analysis.

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Please Note: For the bibliographical references cited in this dissertation including *Part One*: Chapter One, please refer to the complete bibliography at the end of *Part One:* Chapter Two .

Multimedia as Meta-Art:

the processes and aesthetics of interactive digital art

Ph.D. project

Part One: Chapter Two

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Contents : *Chapter* 2 - The Multimedia Modalities of Communication

Chapter 2: The Modalities of Communication in Multimedia

This chapter will look at the two primary sense modalities and consider the history or evolution of the relevant art activities which operate within these modalities, with the view to understanding how these traditions play a part in, or contribute to contemporary multimedia art practice. Each of these sections will conclude with an overview of popular desktop software applications that are being used within each of these fields of artistic endeavour. The final section of Chapter Two considers the field of interactive design, which is explored using the same approach.

Media forms contained within multimedia art address the sense modalities with communicative content. Blinkley [1993] points out some of the key differences between traditional analogue mixed media productions and contemporary digital multimedia:

> Analogue media imbue objects with resilient marks perceivable either directly through the senses, or indirectly through a display process that carries out an additional transcription. Photography is of the former type, video the latter. By transferring reflected light onto reflective pigment, a camera generates visible film whose recorded images are simply looked at. But when light patterns are embedded in an electrical signal by a video camera, the analogous forms imparted to the current are not visible until they are transcribed back into light as glowing phosphor on a monitor. When a video signal is recorded on tape, yet another transcription imprints electrical impulses onto analogous magnetic ones, and the playback process involves two transcriptions that first conform electric current to the recorded magnetic fields and then match the flicker of light on the screen to the varying flow of electrons. Transcriptions can be joined together in a cascading sequence of media that vastly extend the potential impact of a work of art or a noteworthy event…. Digital media by contrast convert, rather than transcribe, the information they preserve. Whereas analogue media store cultural information in the material disposition of concrete objects, digital ones store it as formal relationships in abstract structures. A phonograph record and a compact disc both archive music, but the methods they use are fundamentally dissimilar. Analogue media maintain a concrete homogeneity with what they represent, while their digital counterparts transform originating impulses into heterogeneous quantifications of their sources. An analogue-to-digital conversion process transfigures physical quantities into numbers [Blinkley, 1993].

This chapter looks at the processes and aesthetics involved with artistic communication, once this digitization process has been carried out, and ways in which digital software applications subsequently enable the artist to manipulate digital information for artistic ends, all from within a tradition that precedes and predates the digital age. This chapter culminates with a discussion of interactivity in art and in contemporary multimedia.

2.1 The Visual Modality

Elsom-Cook [2001, p 3], sub-divides the ways of receiving information via the sensory modalities into *channels* of communication (see p.33). The visual modality uses three main channels of communication – the image channel, the motion picture channel (including animation techniques), and the text channel.

2.1.1 The Image Channel

The visual image is one of the fundamental building blocks of the communication process in multimedia art. It is difficult to imagine a multimedia art which did not in some way address the creative process of presenting imagery.

> Prehistoric artisans wrestled their entreaties to immortality into recalcitrant physical substances long before Plato spurned images in favour of ideas as the best route to eternal verities. Ever since, supplicants striving for an audience with future generations have struggled to chip stone into beautiful shapes or plaster walls of dark rooms with arduously prepared pigment. The earliest creative labours left legacies enshrined in material objects with more staying power than mortal flesh. They have since evolved through a long history of increasing automation so that now anyone can capture an image for posterity with pushbutton immediacy [Blinkley, 1993].

Multimedia artists must become skilled at creating and processing imagery and integrating images into their interactive art pieces, and software available for these purposes has become increasingly sophisticated as demand has climbed.

A case study in Universal Principles and The Meta-Art Machine:
Divine Proportion and the Golden Section

In the process of seeking a conceptualization of a multimedia meta-art, universal design principles which apply across the arts are of special interest. An example is the idea of ‘divine proportion’. Symmetry may seem harmonious and balanced, but the golden section is thought by many people to offer the most aesthetically pleasing division of a line, shape or other structure[1]. The golden section is a precise way of establishing such a pleasing, proportional but non-symmetrical relationship using mathematics. It wasn't until the 1900's that American mathematician Mark Barr used the Greek letter *phi* or Ø to designate this ubiquitous proportion[2], which has been

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[1] Freitag, Mark (undated). Phi: *That Golden Number.* http://jwilson.coe.uga.edu/emt669/Student.Folders/Frietag.Mark/Homepage/Goldenratio/goldenratio.html ; accessed 10-01-03.

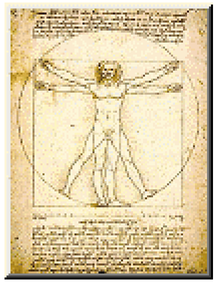
[2] *Phi* is the first letter of name Phidias, who used this *golden ratio* in his sculptures, as well as being the Greek equivalent to the letter "F," in turn the first letter of the mathematician Fibonacci's name.

called the golden mean, golden section and golden ratio, as well as the divine proportion.

The 12th century mathematician Fibonacci discovered a simple numerical series that is the foundation for the fascinating mathematical relationship behind *phi* (but it's not actually certain that he realized its connection to phi and the Golden Mean). Starting with 0 and 1, each new number in the Fibonacci series is simply the sum of the two before it. 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, . . .

It is significant that the *ratio* of each successive pair of numbers in the series approximates *phi* (1.618…), i.e. 5 divided by 3 is 1.666..., and 8 divided by 5 is 1.60 etc.[1]

The Golden mean or *Phi* was described by Johannes Kepler as one of the *two great treasures of geometry* [2]. Luca Pacioli (1445-1517) published his *Divina proportione (On Divine Proportion),* with illustrations by Leonardo da Vinci (right), in which he described the golden section, (but for the first time referred to it as *the divine proportion*):



A M B
| **1-x** | **x** |

The line AB above is divided at point M so that there is a ratio between the two parts, the smaller to the larger (or AM and MB),which is the same as the ratio of the larger part (MB) to the whole AB.

If AB is of length 1 unit, and we let MB have length x, then the definition (in italics) above becomes: *the ratio of 1-x to x is the same as the ratio of x to 1* or, in symbols:

$$\frac{1-x}{x} = \frac{x}{1} \text{ which simplifies to } 1-x = x^2$$

This gives two values for x, (-1-√5)/2 and (√5-1)/2.

The first is negative, so does not apply here. The second is just *phi* (which

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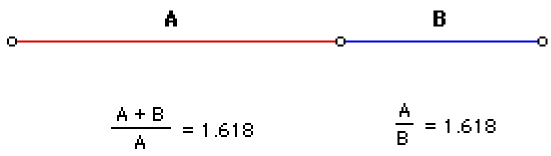
mathematician who first unlocked the numerical series fundamental to phi. http://goldennumber.net/history.htm ; accessed 9-01-03.

[1] http://goldennumber.net/fibonser.htm ; accessed 9-01-03 .

[2] The other is the Theorem of Pythagoras. http://goldennumber.net/geometry.htm ; accessed 9-01-03.

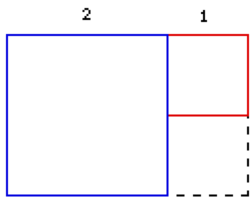
has the same value as 1/Phi and as Phi-1). [Knott, 2002]. Another way to express this is: A:B=B:A+B

In other words, if a proportional relation (ratio) is obtained by dividing a line so that the length of "A" in relation to "B" is the same as the relation of "B" is to the length of "A" and "B" put together, then a Golden Section has been established [1]

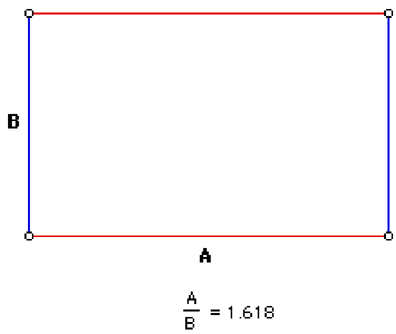


Another geometrical figure that is commonly associated with *Phi* is the Golden Rectangle.

Suppose we take a rectangle of side 1 unit and a rectangle of side 2 units and we put them side to side in the following fashion and draw the appropriate segments to form a rectangle whose sides are related by *phi.*



This particular rectangle has sides A and B that are in proportion to the Golden Ratio. It has been said that the Golden Rectangle is the most pleasing rectangle to the eye.[2]



Stakhov and Sluchenkova, (2002) state that Egyptologists date references to *the Golden Section* as far back as the 27th century BC, on panels found in the crypt of Khesi-Ra. Euclid described it in his major work, the *Elements* (approximately 300 B.C.) referring to dividing a line at the 0.6180399... point as dividing a line in the extreme and mean ratio[3]. This later gave rise to the

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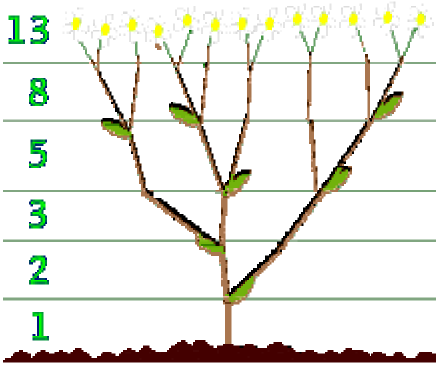
[1] http://www.spyrock.com/nadafarm/html/gm.html ; accessed 10-01-03.
[2] Freitag, Mark (undated). Phi: *That Golden Number.* http://jwilson.coe.uga.edu/emt669/Student.Folders/Frietag.Mark/Homepage/Goldenratio/goldenratio.html ; accessed 10-01-03.
[3] http://goldennumber.net/history.htm ; accessed 9-01-03.

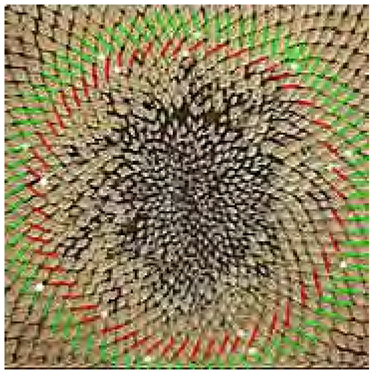
use of the term “mean” in *the golden mean*. Moreover, the Pythagoreans apparently knew about the golden section around 500 b.c.

The oldest examples of this principle, however, appear in nature's proportions. This is the reason it is called Divine - it appears to be a fundamental principle inherent in the structure of creation. Below are a number of examples from the plant and animal kingdoms [1]:

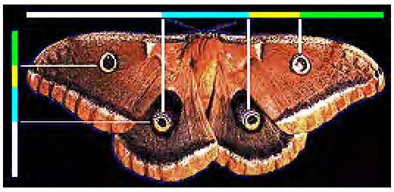
Above: Many flowers have petals that total a number in, or very close to, the Fibonacci series:



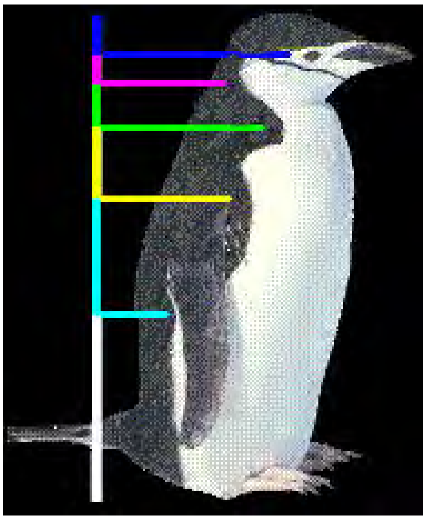
Above: a plant structure which illustrates how each successive level of branches is often based on a progression through the Fibonacci series.

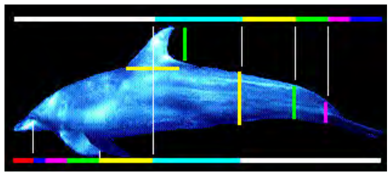


Above: a sunflower seed illustrates this principal as the number of clockwise spirals is 55 (marked in red, with every tenth one in white) and the number of counterclockwise spirals is 89 (marked in green, with every tenth one in white.)



Above: The eye-like markings of this moth fall at golden sections of the lines that mark its width and length.





Above: The eye, fins and tail all fall at golden sections of the length of a dolphin's body.

The dimensions of the dorsal fin are golden sections (yellow and green). The thickness of the dolphin's tail section corresponds to same golden section of the line from head to tail.

Both the ancient Greeks and the ancient Egyptians used the Golden Mean when designing their buildings and monuments, with many *phi* relationships in classical orders. The builders of Paestum used the Golden Mean in their temples[2].

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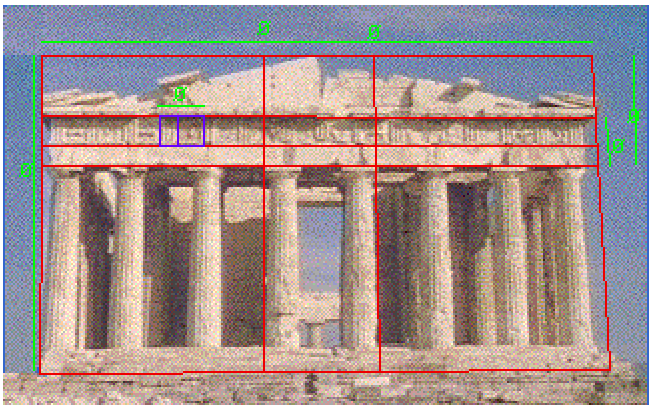
[1] http://goldennumber.net/plants.htm ; accessed 9-01-03.
[2] http://galaxy.cau.edu/tsmith/kw/golden.html ; accessed 10-01-03.



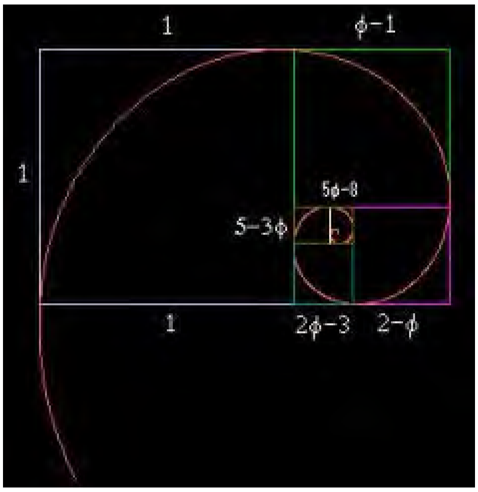
Left: The *Parthenon*, Athens, superimposed with a diagram demonstrating the underlying use of the Golden Mean in many visual relationships within the Parthenon's design[1].

Historically, significant artists as diverse as Leonardo da Vinci, Edgar Degas [Barry et al., 1964, 47], and George Seurat used the ratio when constructing their paintings.

For instance, Leonardo used it to define all the fundamental proportions of his masterpiece *The Last Supper*, from the dimensions of the table at which Christ and the disciples sat, to the proportions of the walls and windows in the background[2].

Above: Leonardo's *The Last Supper* [3].

As we have seen, any rectangle whose sides are related by *phi* (say, for instance, 13 x 8) is said to be a *Golden Rectangle*. It has the interesting property that, if one creates a new rectangle by 'swinging' the long side around one of its ends to create a new long side, then that new rectangle is also Golden. In the case of our 13 x 8 rectangle, the new rectangle will be (13 + 8 =) 21 x 13.



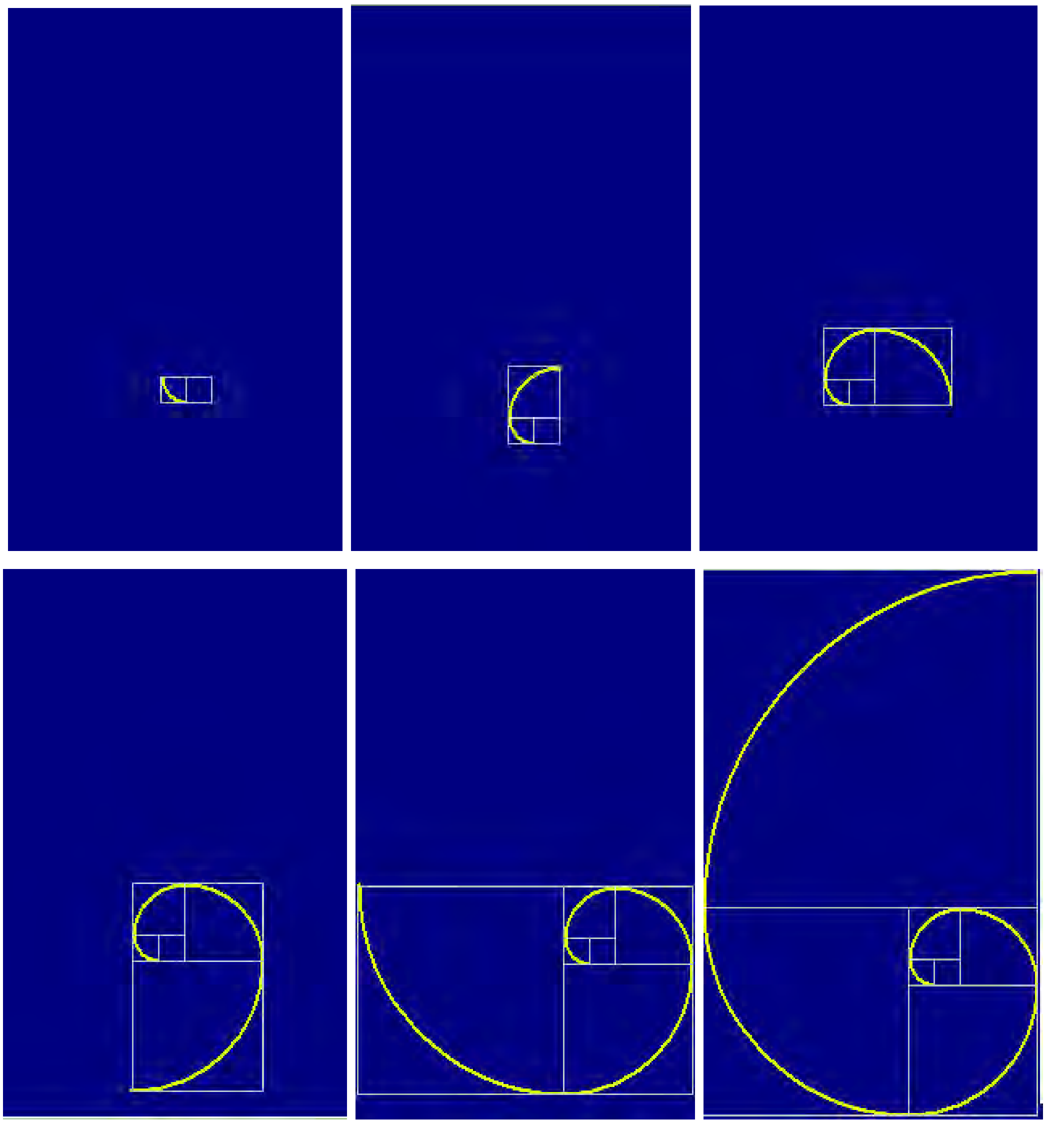
Using the same relationship as found in the Fibonacci series, beginning with a square (1 x 1), "swinging" the sides to make rectangles, produces a fractal series of Golden rectangles:

1 x 1   2 x 1   3 x 2   5 x 3   8 x 5
13 x 8   21 x 13   34 x 21 etc.

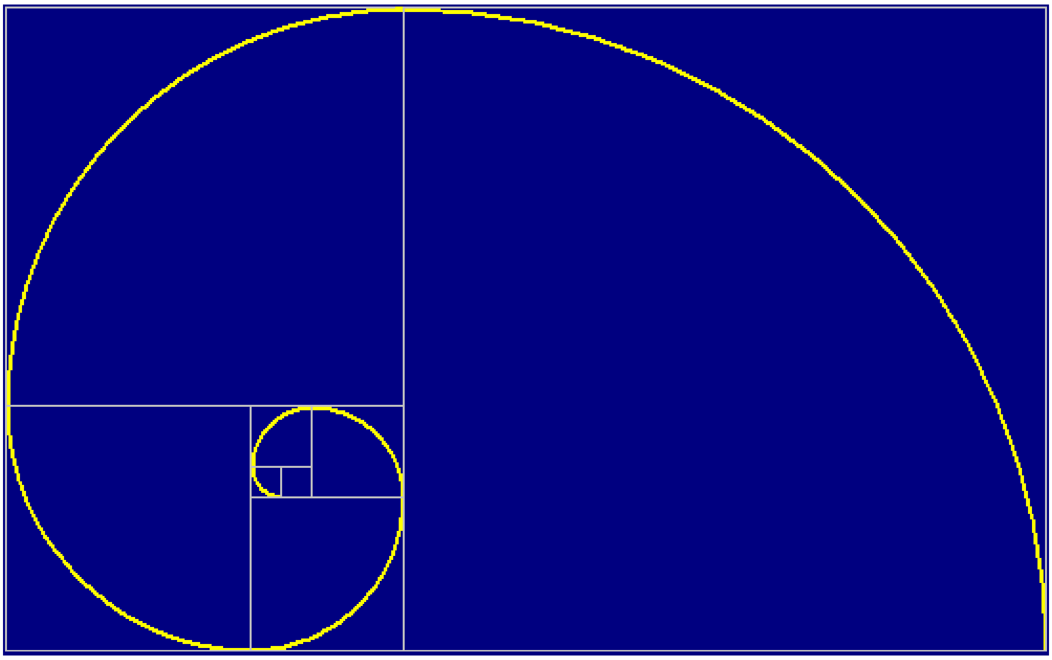
with, again, each addition coming ever closer to multiplying by *phi* [4]. Joining the proportional division points with an arc

[1] http://www.artlex.com/ArtLex/g/goldenmean.html ; accessed 10-01-03.
[2] http://goldennumber.net/history.htm ; accessed 9-01-03.
[3] http://goldennumber.net/goldsect.htm ; accessed 9-01-03.
[4] http://www.vashti.net/mceinc/Golden.htm ; accessed 9-01-03.

produces a perfectly balanced spiral. We can see from the following series of diagrams how this extrapolation from the concept of divine proportion, "the unfolding of the Golden Section", produces such a spiral [1].



[1] http://www.vashti.net/mceinc/unfold7.htm ; accessed 10-01-03.



Again, there are intriguing structural parallels to be found in the microcosm and macrocosm of natural phenomena.

Satellite photo of a hurricane

The Whirlpool Galaxy

Divine Proportion is sometimes loosely applied as the *rule of thirds*, especially as a framing concept in film and television camera work [e.g. Burger, 1993, 441-442].

This approximately one third - two thirds relationship has also been discerned in other art structures, including the temporal divisions of music. For instance, in the October 1995 issue of *Mathematics Magazine* [275-282], Putz described his investigation of whether the golden ratio appears in Mozart's piano sonatas. He concluded that:

> Mozart did create divine divisions in his piano sonatas - making the interplay of sections shine like sunlight. Yet he apparently timed those divisions with his mind--not with math, or at least not with the (precise) golden section.

Tempered Western Music itself can also be said to have a foundation in the Fibonacci series, as:

13 notes separate each octave of

8 notes in a scale, of which the

5th and 3rd notes create the basic foundation of all chords, and are based on whole tone which is

2 steps from the root tone, that is the

1st note of the scale.

Fibonacci Ratio	Calculated Frequency	Tempered Frequency	Note in Scale	Musical Relationship	When A=432 *	Octave below	Octave above
1/1	**440**	**440.00**	**A**	**Root**	**432**	**216**	**864**
2/1	**880**	**880.00**	**A**	**Octave**	**864**	**432**	**1728**
2/3	**293.33**	**293.66**	**D**	**Fourth**	**288**	**144**	**576**
2/5	**176**	**174.62**	**F**	**Aug Fifth**	**172.8**	**86.4**	**345.6**
3/2	**660**	**659.26**	**E**	**Fifth**	**648**	**324**	**1296**
3/5	**264**	**261.63**	**C**	**Minor Third**	**259.2**	**129.6**	**518.4**
3/8	**165**	**164.82**	**E**	**Fifth**	**162 (Ø)**	**81**	**324**
5/2	**1,100.00**	**1,108.72**	**C#**	**Third**	**1080**	**540**	**2160**
5/3	**733.33**	**740.00**	**F#**	**Sixth**	**720**	**360**	**1440**
5/8	**275**	**277.18**	**C#**	**Third**	**270**	**135**	**540**
8/3	**1,173.33**	**1,174.64**	**D**	**Fourth**	**1152**	**576**	**2304**
8/5	**704**	**698.46**	**F**	**Aug. Fifth**	**691.2**	**345.6**	**1382.4**

Knott sounds a warning against elevating Divine Proportion to the level of some kind of mystical truth however:

> There are many books and articles that say that the golden rectangle is the most pleasing shape and point out how it was used in the shapes of famous buildings, in the structure of some music and in the design of some famous works of art. Indeed, people such as Corbusier and Bartók have deliberately and consciously used the golden section in their designs. However, the "most pleasing shape" idea is open to criticism…

At best, the golden section is a practical tool and should be regarded as just one of several possible subjectively evaluated design methods, which can help artists structure what they are creating. At worst, some people have tried to credit the golden section with magical attributes beyond what can be scientifically verified [Knott, 2002].

Colour Theory

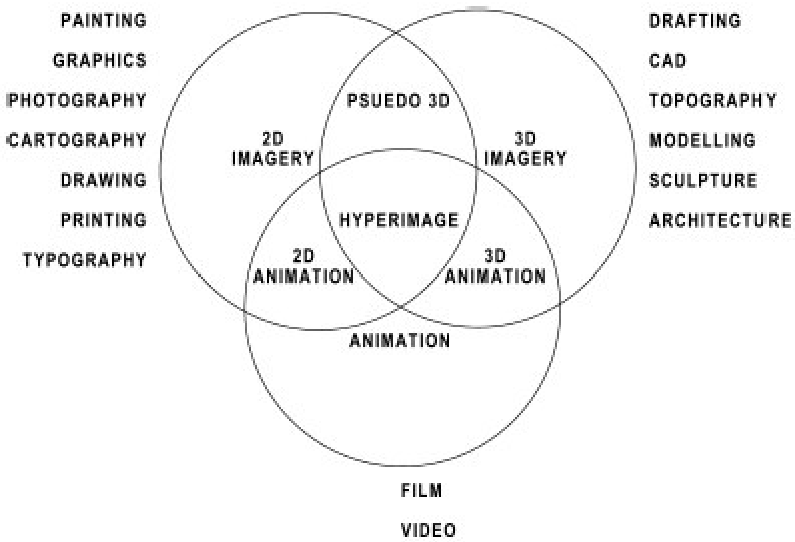
There is a large body of knowledge, theory and tradition to do with understanding the use of colour in art. As in the above section on divine proportion, some of this theory of colour and practical tradition related to the use of colour in art should ideally underpin the universality of the conceptual meta-art machine. Certainly present day graphics manipulation software such as Adobe *PhotoShop* and natural media simulation software such as *Painter* boast highly-developed and congruent interface access to colour manipulation, based on systemisation of colour theory. The accompanying interactive CD-ROM *The Machine* has a section (accessed from the sub-menu called *the Recreation Centre*) which is an interactive exploration of a range of ideas associated with colour theory. This should be viewed both for its interest as an example of an educational of instructional design genre of multimedia, (with the associated difficulties in presenting traditional linear arguments in a non-linear form) and for the content itself – a presentation of the broadly accepted fundamentals of colour theory in design.

Hyperimage and Hypermedia

An understanding of the role of Image in multimedia has become more complex by virtue of the use of hyperlinked image as navigational device, or metaphor for the nodal point of linkage between data available over networks. Holmes, [2000, 7] offers the following definition of a hyperimage:

> *a computer mediated image that is coded to perform the function of a node in a data network.*

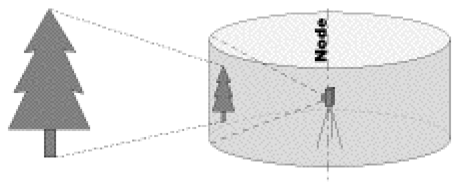
To add further complexity to a discussion of image in multimedia, new technologies (e.g. QuickTimeVR, IPIX, animated gifs, Shockwave Flash files, etc.) have blurred the boundaries of what an image can be. Holmes offers the following Venn diagram as an overview of the antecedents to this new media phenomenon, and overlapping formats that could be subsumed within the concept of hyperimage.



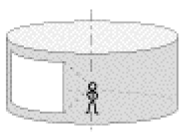
Above: *Hyperimage and its antecedents* (used by permission of the copyright holder).

The diagram above shows overlapping spheres representing the possible contributing fields of image generation activity that combine to make up hyperimage (2D image formats, 3D image formats, and motion graphics, which by implication include attached sound), but disregarding interactive coding.

New media formats include Pseudo 3D (also known as 2.5D), which refers to “cylindrical” and “spherical” panorama formats such as *QTVR*™ and *IPIX*™ respectively. While related to photography, they are distinguished from photography in that they present an experience to the computer user that is impossible to experience in print. *QTVR* panoramas are essentially a simulation of the view from a single point in space out to a surrounding environment.

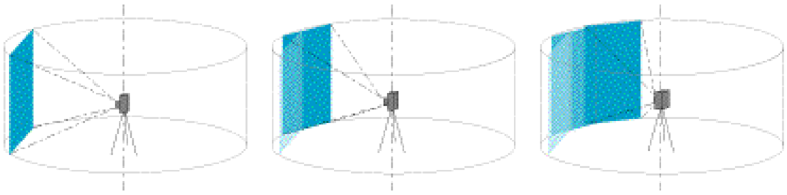


Panorama view from a single point in space



From the central observation point, called a *node*, a viewer can look in any direction and may ‘zoom’ into or out from a particular view by changing the *zoom angle* of their view.

The most common method of creating panoramas is to capture a series of source images around a single point of rotation, then digitize them into source bitmap files. As Apple developed this technology, the bitmap file format commonly used is the Apple PICT file.



Individual bitmapped images captured around a point of rotation .[1]

These source images are then digitally “stitched” together using the *QTVR* software to create a single panoramic image file that represents a conceptually “cylindrical” view, from the point of rotation. The following is such an image laid out flat.

Alternatively, panoramic cameras can create a single panoramic image directly on film, and these may also be converted into a *QTVR* panorama.

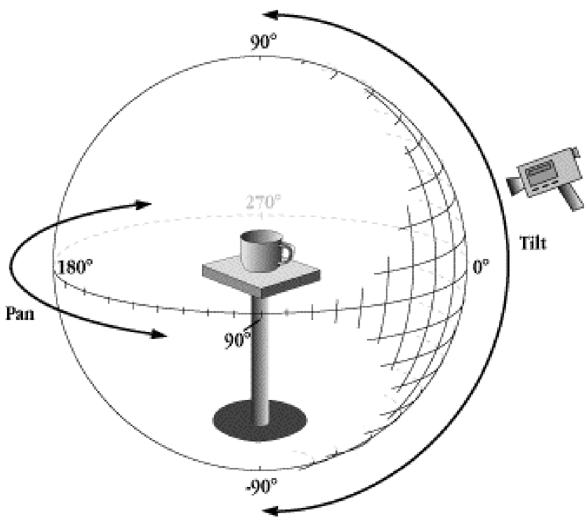
IPIX is a competing proprietary system which achieves a similar result, but requires a plug-in in order to be viewed within a browser, which does not enjoy the large installed user-base that *QuickTime* has achieved. Two overlapping 185 degree images are captured using a camera with a fish eye lens, mounted on a rotator. The patented technology corrects the distortion in the resulting fisheye images and “seams” the images together into a conceptually spherical format. Below is an example of two complementary 185-degree images captured by the *IPIX* camera.

Right: IPIX camera

[1]URL: http://www.letmedoit.com/qtvr/qtvr_online/Mod1/m1_panos.html ; accessed 17-09-02.

When viewed on screen, it is implied that the QTVR or IPIX (pseudo-3D) image exists beyond the viewing window. Mouse/cursor movements allow the user to reveal obscured portions of the projected image by accessing appropriate stored pixel data on demand. In this way, user interaction simulates rotating the digital projection. The technology allows the user a sense of being at the center of a 3D scene, despite the 2D nature of screen technology. The rotation of the image may also be scripted, and 'hot spots' for linking to other stored images and media may be embedded in the picture plane, making it a very useful mapping device, used for instance in, web-sites for art galleries allowing a virtual tour[1]. The bubble-like interface of the spherical panorama is limited, in that it is an inherently outward looking view. The inverse effect, that of looking in at the subject, is achieved using incrementally rotated views.

"Pseudo-3D" images of objects are therefore not 3D models but single frames, animated and scripted for interactivity. In other words, pseudo-3D images are essentially 2D images masquerading as 3D, by virtue of simulated spatial projection and/or frame animation techniques.

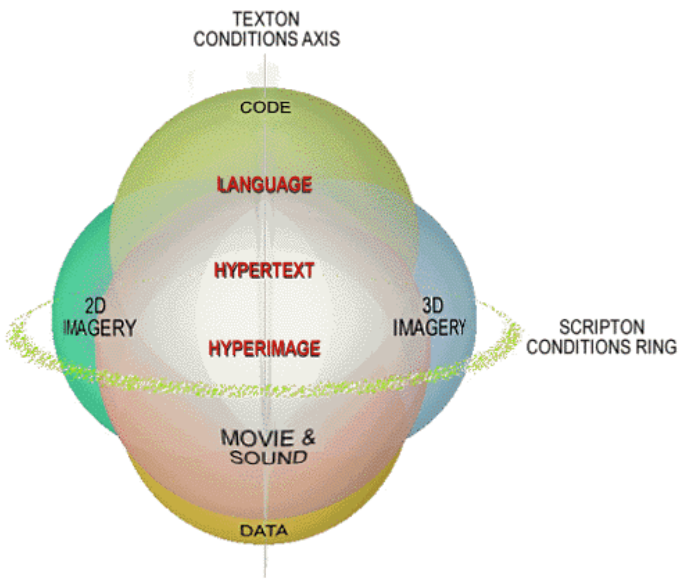


Left: This diagram illustrates the procedure for making a QTVR "pseudo-3D" image of an object, by exposing successive images as the subject is rotated in equal increments in space. These are then digitally "stitched" together into what can be conceived as a single "cylindrical" view, using the QTVR software.

To incorporate the coded linkages and embedded text elements available in hyper-image, Holmes [2002, 10] offers a more advanced model, in the form of a second Venn diagram (below right).

[1] e.g. Sydney Opera House web-site: http://www.soh.nsw.gov.au/

Right: a heuristic model of hyperimage; (used by permission of the copyright holder).



Holmes [2002, 11] modifies his definition of hyperimage accordingly, to include *scriptonic* and *textonic* conditions:

> *a computer mediated image that is coded to perform the function of a node in a data network, is represented screenically and, may dynamically operate in either or both of textonic and scriptonic conditions.*

As Holmes explains [2002, 10], the new definition allows for embedded code and other non-traditional new media phenomena that make such images qualify as "hyper":

> The *code* sphere accounts for the software systems and applications responsible for utility, interactivity, and digital communication of all media channels. The *data* sphere is actually more like a well or a nebula when knowledge is shared via distributed networks. Code and data are aligned on an axis that is representative of text, alpha-numeric and binary codes, called in the figure, the *texton conditions axis*. This axis represents the so-called 'texton' condition- and the ring of spheres qualified by graphic, musical, spatial and kinesthetic attributes, as represented by the spheres: *2D imagery*, *3D imagery* and *movie & sound*, comprise the 'scripton' condition; represented in the figure as the *scripton conditions ring*. These conditions were first described by Arseth and elucidated by Hayles. The model (in the figure above) shows that in hypermedia, hypertext and hyperimage share a semiotic that is modified by texton conditions and scripton conditions.

2.1.2 The 'Motion Picture' Channel - digital video and animation

The Multimedia artist creates a virtual reality, stimulating the senses and infiltrating the mind, by offering an experience which is intended to be referenced to the perceptual background of an individual's received knowledge of life and the physical world. Immersion in the space occupied by this artificially created, charged reality is a qualitative experience made more vivid by particular sensory cues, which echo our experiences in the natural environment. This becomes a feedback loop as virtual experience changes and heightens our perception of reality.

One of the important steps in the emergence of personal computer platforms as a delivery medium for a new multimedia art-form was the development of faster hard-drives which enabled the possibility of storing and replaying digitized motion video. Advances in the speed of storage media were complemented by parallel developments in the effectiveness of data compression schemes in lowering the sheer amount of digital data required to store and reproduce digital video images.

A third process of technological evolution which has influenced the development of multimedia art has been the development of relatively cheap but technically sophisticated domestic video “camcorders”. A quantum leap occurred with the release in 1996 in Australia of hand-held *digital* video cameras, which approach the traditional industry-standard BetaCam for image quality. The commercial availability of such devices has put in the hands of the independent multimedia artist a tool capable of delivering relatively high quality, potentially beautiful images and digital stereo sound recording.

35mm film is the standard of the motion picture industry, against which video must be measured in terms of its artistic potential as a motion picture medium. Video is vastly inferior, both technically and subjectively. In the real world occupied by the independent multimedia artist, however, the one supreme advantage of video is clearly the relatively low cost of videotape. This encourages the potential for *spontaneity* in the visual sampling process - it is (relative to professional film formats) not a financial daydream or nightmare for the independent artist to shoot literally hours of video footage without scripting, pre-production, even a plan. This way of generating visual material as a response to life, the passing of time, motion and the interplay of light in the world, has its own unique artistic parameters. The portability and sometimes the concealability of the camcorder supports this process. Motion-picture art can take on board some of the conceptual processes of junk sculpture and found art. With the gradual move towards HDTV the positive attributes of video will be combined with increased resolutions, screen size and image quality.

The cinema/television tradition as an antecedent to multimedia

The medium of film — motion pictures — is arguably one of the most influential and important of the arts of this age. This can be supported by observing that, though a mere 100 years since motion pictures were first demonstrated publicly, they have since become a common experience in the lives of people the world over.

Multimedia art can be viewed as the artistic offspring of a new technology, just as cinema was before it. The art of feature motion picture production is exceedingly complex, requiring contributions from nearly all of the other arts as well as multifarious technical skills. Much of the language of the video element of multimedia art has grown out of cinema language. In a study of the interface between sound and image in multimedia it is necessary first to look at current theories of the emerging role of music in cinema, and how the use of music and sound in classical narrative film was challenged by avant-garde/modernist film techniques

The other important antecedent to be considered is *the music video clip*, a popular culture phenomenon whose existence can be obviously traced to the era when television/video began to be offer a serious competitive medium to film.

The music video has become a unique genre of electronic communication distinguished by the introduction of non-narrative, evocative, stream-of-consciousness approaches to the visual/audio interface [Klieler & Moses, 1997, 28-29].

For the purposes of this discussion these historical antecedents to multimedia art are here differentiated by way of a broad but convenient dichotomy:

i) within the cinema or motion picture tradition the sound/visual interface is characterized by the visual element being the dominant sensory and semiotic input mode. Sound/music plays a powerful but usually supporting, accentuating role. *(Sound supports image)*

ii) This is contrasted with the music video clip, an audio-visual art medium, in which the sound (music) track is the dominant sensory and semiotic input mode. Video clips are referenced to a tradition, which asks us to assume that images play a supporting, accentuating role to the experience of listening to music. *(Image supports sound)*

What is of interest is how these two evolving streams of artistic endeavour feed into, and stimulate a multimedia artform, an artform in which sound is no longer a subservient element within a signifying system bound to a narrative mode, nor imagery just a novel way of getting the audience to listen to music.

Rather, an ideal is pursued involving a synthesis of these two traditions, which aims to offer new virtual artistic experiences, and ultimately even suggests the possibility of contributing to the social evolution of human consciousness, by offering opportunities to experience richly textured semiotic texts, multidimensional poetic and artistic insight, and simulations of new and challenging realities, all against a background of sense environments which could not otherwise exist. This is the world awaiting the intrepid multimedia artist – a creative exploration of the technological and artistic frontiers, where multidimensional mindscapes may be carved from or by the unfettered imagination.

Music Video as a stylistic and technical influence on multimedia

The contemporary era can be styled as an age of visual literacy as opposed to the more predominantly literature-based culture of previous generations. The computer revolution has contributed to this trend, with video games, television and cinema playing important roles. For example, the Christian Science Monitor web-site makes the claim that American children *average* three hours television per day throughout their childhood[1].

Multimedia commonly includes images, animations and video sequences within the visual modality. The multimedia artist can access all these media forms in the process of constructing a multimedia artwork, and thus must develop both the skills to handle, and a sophisticated understanding of, each of the contributing media forms. The way these visual elements are incorporated into the multimedia work, and the subsequent meaning that is derived from them by the viewer, are both inevitably influenced by the experience that creators and users of multimedia have had of other cultural contexts in which imagery has been presented. Cinema and television are the two primary sources of this cultural experience. We must turn to the literature devoted to an understanding of these media forms in our quest to understand the form, function and aesthetics of multimedia.

The interactive, non-linear form of multimedia tends to fragment narrative development. The very nature of multimedia – a melding of a variety of media forms into a new interactive media – presents challenges to the creator in terms of developing a cohesive, flowing, immersive unity.

Multimedia works most commonly include quite short video segments, if at all (constrained by the limited data storage capacity of the CD-ROM). These short video pieces can be a montage of shots and images illustrating an accompanying voice-over and music. Such video sequences frequently serve to create an

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[1] URL: http://www.csmonitor.com/2002/0329/p01s05-ussc.html ; accessed 15-11-02.

impression and add visual impact rather than stand as a coherent narrative in their own right. Because of their brevity and subsidiary role within the interactive multimedia piece they can be strongly influenced in their construction by the aesthetics of music video, itself a miniature form usually serving as a visual accompaniment to the three minute pop song [Kleiler and Moses, 1997, 40].

A maverick descendant of experimental film-making, video art and television commercial advertising, the music video clip is firmly established as a thriving and vital genre in its own right, though not without some paradoxical aspects. With its own unique visual language and sense of style, (which can place many examples more in the category of graphic art than narrative art), music video warrants consideration as an artistic phenomenon in this study because of the influence it continues to exert over the contemporary generation of consumers and artists in terms of visual sensibilities. An understanding of the style-conscious visual language of music video and its relationship to music is also an important and informative departure point for an understanding of the temporal aesthetics of multimedia art.

It may be observed that television content, whether naturalistic, period or stylized, has tended towards a narrative approach in its structure - television has become the electronic fireplace in our global version of tribal life, replacing the oral story-telling traditions of traditional tribal society, and the reading habits of post-Gutenberg, pre-digital, literate generations. In fact, the great media commentator Marshall McLuhan based his idea that in the *global village* we have returned to an oral-culture, on observations of the effects of television on Western lifestyles and culture [Andrews, 1995, 5]. As an habitual television/movie audience, we become ever more sophisticated in our understanding of the language of film and television.

In recent years a significant contribution to this contemporary visual language has been made by the new music video genre, whose format requires very concise telegraphing of fragmented narrative elements, primarily because of the relatively short format.

From a technical point of view music video has in turn become a significant influence on the language of film and television in a variety of ways. Music video camera work and (especially) editing have become virtuoso crafts, in which technique often overshadows meaning or content to a degree not possible within structures tied to narrative continuity. It could be said that the form tends to explore an emphasis on visual-auditory spectacle. Closely linked to this emphasis on visual spectacle is the most obvious innovation implicit in the concept of music video - the shift away from the dialogue-directed continuity and linear narrative of traditional cinema and television.

By virtue of a sometimes subtle, sometimes blunt but inherently anti-establishment stance, and this emphasis on attention-grabbing images and constant technical innovation, music video has become a rich field of media experimentation. The historical value of the video clip, in terms of popular art and culture, may prove to reside in the way that it has introduced new techniques and standards of visual taste (e.g. multiple fast jump-cuts, the hand-held camera look, unmotivated lighting, emphasis on style and feeling over content, graininess, blur and grunge, mixing colour and B & W etc.).

Joseph Ebben [1992, 88] explains that the music video form allowed a freedom to experiment which produced radical innovations. By offering a new forum for this experimentation, a whole generation of viewers were conditioned to new visual conventions:

> Rules were tested and broken. Risks too high for the world of movies and commercials were taken, and lessons were learned. As the medium became more and more popular, molding the visual tastes of a new generation, feature and television filmmakers - as well as Madison Avenue - sat up and took notice.....videos, by conditioning viewers to a looser, less reality based image, have ...allowed cinematographers and directors a freer hand in using that image to draw a particular emotional response.

Music videos offer a chance to send visual as well as musical messages to a wide youth audience of just what the difference is, with regards to particular music artists attempting to set new trends.

Shuker's [1994, 166] discussion of music video as an audio-visual text takes a semiotic approach, but also it can be taken as implicit recognition that music video can be regarded as a work of art, an artistic statement, with a unity and cohesion in its own right, rather than merely as a visual interpretation of and/or advertisement for a music artist's song. Who is the artist here?

Putting aside the cultural status and social perception of music video activity - there is one significant question with regards to music video's place in art history which seems useful to ask at this point: What precedents are there in the history of art for the recently-established music video convention, in which communication artifacts are produced that can (usually) be described as a visual artist's interpretation of another artist's work in a different medium (i.e. music), but which are then viewed on the assumption that they represent an artistic whole, an art offering in their own right?

Some parallel can be drawn with choreographers producing ballets based on the musical works of composers - such as Nijinsky's *Rite of Spring* to the music of Stravinsky. We could also consider here Walt Disney's *Fantasia* - an animated cartoon interpreting (imposing?) a programme upon non-programmatic music, and the film version of Gershwin's *An American in Paris*, starring Gene Kelly.

Music video has been an innovative cultural development, in that imagery was developed to accompany pre-existing music recordings of short popular songs. The result of this merging of lyrics, musical genre and experimental film techniques produces a complex, many layered cultural experience, which is difficult to analyze. The literature which has been produced as a response to this new media form is also relevant to an understanding of the use of imagery in multimedia.

Shuker [1994, 184] offers a good starting point for an understanding of music video within a popular culture context:

> The primary focus in the study of music video has clearly been on the MV as an audio-visual text. Various attempts to read music videos as texts have necessarily adopted the insights and concepts of film (and television) studies. There is also some recognition, at times rather belated, of the point that MVs are not self-contained texts, but reflective of their nature as industrial and commercial products....
>
> Two general points frequently made about MVs as individual texts are their preoccupation with visual style, and, associated with this, their status as key exemplars of 'postmodern' texts. Music videos are clearly pioneers in video expression, but their visual emphasis raises problems for their musical dimensions. As some threequarters of sensory information comes in through the eye, the video viewer concentrates on the images, arguably at the expense of the sound track...
>
> cultures interpenetrate….We are living in a new world, a world that does not know how to define itself by what it is, but only by what it has just-now ceased to be.

What complicates the aesthetics of music video is the paradox that, from the music video director's point of view, the clips do stand alone as artistic offerings, yet ultimately they must work as an advertisement for another artistic product - the music artist's musical offerings (compositions and recordings) and also live performances. Music videos are thus equally a promotional vehicle, and as such can also be analyzed as advertising art.

The Rolling Stone Book of Rock Video claims a general consensus that the video for Queen's epic, original and formula-defying song *Bohemian Rhapsody* [1975, directed by Bruce Gowers], was the first example of a video being "primarily responsible for making a song a hit" [Shore, 1984, 19-21].

This advertising element can be simply a subtle emphasis on the artist's sense of style and fashion, but is very significant in establishing product differentiation, by making clear appeals to particular niche markets. The hit song *Video Killed the Radio Star* by British studio group *the Buggles* is a clever and surreal parody of the emerging pre-eminence of the music video in pop culture.

No statistics are available to indicate to what degree the accompanying video promoting this song influenced its popularity, but with a suitable touch of irony it did receive extensive radio airplay. And with an extra twist to this irony, the video made to promote this song was the first ever played when MTV began broadcasting, heralding the re-ordering of the industry, and subsequently posing a challenge to the dominance of radio [1].

Below: Still from the video clip for the Buggles' song *Video Killed the Radio Star.*

The following news item identifies the significance of the introduction of MTV to the music industry:

> … the birth of the dedicated music channel did mark a watershed in the history of pop and a change in the relationship between popstars, audiences and record companies.
>
> "The creation of MTV and the emphasis on videos removed the need to see the artist performing," says Stephen Armstrong, a music writer for *The Face* and *Arena*. Armstrong says the creation of MTV allowed record companies to market and package bands to a much greater degree than before. "The Stock Aitken and Waterman acts, such as Kylie and Jason Donovan, had a huge advantage in the MTV age because they came from a telly background and were not about live performance." Mr Armstrong believes the rise in popularity of boy and girl bands over the last 20 years is directly attributable to MTV. Record companies were able to push their bands through videos and without relying on live performance, which honed the skills of bands in the past[2].

There are echoes of the unfolding drama and crisis that popular cinema actors of the silent era faced once Al Jolson had spoken his famous *"Come on, Ma! Listen to this…"* in *The Jazz Singer* [Oct. 6, 1927, New York]. Some matinee idols were not able to make the transition to sound – they didn't have the voice to go with the look. Similarly, from this point in popular music, telegenic glamour, physical appearance,

[1] http://www.gabrielmedia.org/news/mtv17.html ; accessed12-01-03.
[2] http://news.bbc.co.uk/1/hi/entertainment/tv_and_radio/1465495.stm ; accessed 12-01-03.

and youthfulness (of course already important in show business) became suddenly an even higher priority to A & R managers – musicians had to have “the look” to go with the sound. Dancing skills also have become highly valued through the influence of music video, and, because of MTV, of great significance to the careers of such music stars as Michael Jackson and Madonna. MTV may have had a similar impact on the success of other forms of theatricalism in pop music, most notably the quasi-gothic, highly theatrical posturing of Heavy Metal bands.

Shuker [1994, 185] points to the extremities of this trend in the marketing of popular sonic art, often bitterly resented by lovers of rock music culture, where the music begins to lose its central role in the overall process of selling the artist and becomes

> …merely part of an overall style package offered to consumers.

This tendency reached some kind of peak with the media scandal surrounding award winning vocal group *Milli Vanilli*, after it was revealed that they did not actually sing on their hit recordings. [1]

In terms of post-modernism, Shuker [ibid., 185] points to the work of cultural historian and theorist Jameson [1984, 185], summing up his ideas about the paradoxical intertwining of these complexly related artistic and commercial elements by offering the observation that Music Videos function as:

> “....’meta entertainments’ that embody the postmodern condition. It is certainly clear that MVs do indeed merge commercial and artistic image and its real life referent.... making them a part of a blatantly consumerist culture.”

The use of the word “noise” in video production designates randomly occurring pixel excitation which we perceive as a dirty or blurry image, unacceptable by professional or so-called broadcast standards, yet, as has been mentioned, often a conscious aesthetic choice in nihilistic style, grunge music video clips e.g. successful heavy metal band *Motley Crue’s* video to the track *Afraid* (from the album *Generation Swine*,1997, Elektra, Beyond Music, BMG). The use of “grunge video” can most fruitfully be read as a political statement against slick Hollywood or Madison Avenue influenced star-vehicle music videos (see p.197 for a background to this idea).

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[1] They became the first group to be stripped of a Grammy Award.
http://www.allmusic.com/cg/amg.dll?p=amg&sql=Bfudjyl3jxpsb ; accessed 23-11-02.

Music video and genre

Attempts to classify and deconstruct music videos into genre are difficult because of the complex overlapping between genres of semiotic text, technical experimentation and the conflict between artistic and promotional goals. Shuker [1994, 185-186] attempts to simplify the analysis by differentiating three focal areas with which an analysis can begin, which can be summarized as:

> *1. Cinematic aspects:*

Shuker lists *camera techniques, lighting, use of colour, editing.* He appears to mean purely technical, rather than formal, structural or narrative issues. Use of video "effects" i.e. digital signal processing techniques could be included.

> *2. Different styles of music utilize different video conventions e.g. Heavy Metal videos make frequent use of wide-angle lens and zoom shots in keeping with their emphasis on a large arena 'live concert' format*.

Genre specific visual conventions can often be considered separately to macro-issues, such as narrative versus graphic arts approaches etc.

> *3. Point Of View …who is looking at whom, from what point of view, and what is conveyed by the use of these conventions in terms of power relationships, gender stereotypes, and the social construction or definition of self?*

A concept central to the understanding of music video is the nature of "the gaze" in music video [Chandler, 2000]. Schroeder simplifies the concept as it has been presented in relation to film:

> Film has been called an instrument of the male gaze, producing representations of women, the good life, and sexual fantasy from a male point of view [Schroeder 1998, 208].

The concept of "the gaze" originally derived from feminist film studies, (specifically a seminal article called '*Visual Pleasure and Narrative Cinema'* by Laura Mulvey).

> … psychoanalytically-inspired studies of 'spectatorship' focus on how 'subject positions' are constructed by media texts ... Mulvey notes that Freud had referred to (infantile) *scopophilia* - the pleasure involved in looking at other people's bodies as (particularly, erotic) objects. In the darkness of the cinema auditorium it is notable that one may look without being seen either by those on screen by other members of the audience. Mulvey argues that various features of

> cinema viewing conditions facilitate for the viewer both the voyeuristic process of *objectification* of female characters and also the narcissistic process of *identification* with an 'ideal ego' seen on the screen. She declares that in patriarchal society 'pleasure in looking has been split between active/male and passive/female' [Chandler, 2000].

Gender is not the only important factor in determining the nature of the gaze in the cinema experience - *race* and *class* are also key factors [Lutz & Collins 1994, 365; Gaines 1988; de Lauretis 1987; Tagg 1988; Traube 1992]. The concept of gaze is valuable in music video analysis because of the frequently overtly sexual or even sexist nature of music videos and the traditional male dominance of the music industry, but it also has other relevance in terms, for instance, of videos produced for black American niche markets etc.

Shuker goes on to list the following analytical 'subdivisions' as an aid for understanding works of music video, which are presented here (quotes in italics), with comment:

1. *The mood of the video: the way in which music, the words, and the visuals combine to produce a general feeling.*

Possibilities mentioned can include:

i) *nostalgia* - looking back with longing on the past. Thematically this can include the representation of simpler, happier times (political theme); nostalgia for the innocence of childhood; the nostalgic celebration of retro music styles.

ii) *romanticism*. Here Shuker appears to be intending the word in the sense used by art historians, and the Macquarie Dictionary definition could well be a description of music video:

> Art characterized by freedom of treatment, subordination of form to matter, imagination, experimentation with form, picturesqueness, ...(as opposed to classicism)…fictitious, fabulous[1].

iii) *nihilism*[2]. Popular music has, since the period of early rock 'n' roll, been associated with youth rebellion, which can take the form of cynicism, alienation and an anti-establishment stance that is not directed towards

[1] The Macquarie Dictionary , 2nd ed., 1991, p.1524.
[2] *a skepticism towards or total disbelief in "the establishment", i.e. accepted mainstream social values, religious or moral principles and obligations*. [The Macquarie Dictionary, 2nd ed., 1991, p.1204]

political solutions, just a simple rejection of contemporary society as a cold, dark ugly reality.

2. *The narrative structure: the extent to which the video tells a clear, time-sequenced story, as opposed to a non-linear pastiche of images, flash-backs etc.*

This reveals a dichotomy evident in music videos between narrative, cinematic approaches to the form and a non-narrative, more abstract, surreal or graphic arts approach. The narrative cinematic approach tells coherent stories using standard film language and editing conventions, perhaps narrating song lyrics; but also sometimes a parallel narrative is woven into the text, upon which the song lyrics then seem to comment; both narrative approaches are frequently interspersed with footage of the artist(s) performing the song. The graphic arts approach is characterized by virtuoso video editing and a flow of not necessarily connected images [Kleiler and Moses, 1979, 26]

3. The degree of realism or fantasy of the settings, environments and events in the video.

Within the narrative or cinematic genres there are a range of approaches, from politically-conscious, hard-edged representation of socio-economic realities and inequalities, through to total fantasy influenced by popular cinema genres.

4. The standard themes evident

Music video is primarily a youth culture phenomenon with adolescents and young adults the target audience. Thus the predominant themes evident in music video tend to be to with identity, sexuality, authority and rebellion. Shuker offers the following examples:

the treatment of authority
love and sex
growing up
the loss of childhood innocence
masculinity and femininity

Of course many other non-sexual themes occur. These however may often be related to youth rebellion stance and include:

political and social conscience
the environmental crisis
celebration of youth culture
fashion

5. The importance of live performance (or simulated live Performance).

This is bound up with the need to gain promotional advantage from video clips – thus the presentation of the artist in a carefully constructed simulation of an exciting live performance is the most common visual element appearing video clips. It is a cross-marketing exercise which sells live concert tickets. It is also the simplest way to portray the music visually.

6. Different modes of sexuality

Partly because of the youth culture context, but also through the influence of recent important political movements such as gay liberation and feminism, depictions of sexual identity are a constant theme of music videos. Some of the most celebrated pop/rock stars have built an image by manipulating sexuality and sexual identity (e.g. David Bowie, Boy George, Michael Jackson, Madonna, Annie Lennox).

David Bowie

Annie Lennox of *the Eurythmics*

Michael Jackson

Boy George

Madonna

Shuker lists sexual themes which are explored as:

- *the female as mother/whore*
- *androgyny and the blurring of dress codes*
- *homoeroticism*

7. *The nature of music videos as "star texts", centred on the role played by the central performer(s) in the video and the interrelationship of this to their star persona in "rock culture" more generally.*

This can be read as part of the Image building / reinforcing, publicity and public relations process.

8. *The Music - what we hear and how it relates to what we see*

Shuker warns that many visually-oriented analyses of music videos overlook the sound-vision relation in their analysis. This can influence both:

i. *formal organization, and*
ii. *questions of aesthetic pleasure,*

which need to be considered in relation to the musical text.

Despite himself offering a detailed categorical analysis, Shuker states [p.187 –189]:

> Perhaps the eclectic nature of music video makes impossible anything other than a basic distinction between performance and fictional narrative MVs?...(it is possible to demonstrate) the intertextuality of most contemporary MVs, and the difficulty of neatly slotting them into a classificatory system.
>
> ...While there is no denying the creativity evident in many MVs, analysis must not be simply preoccupied with their textual codes and conventions; it is also necessary to relate these to their role as industry adverts, and the varying manners in which they are read by viewers.”

The difficulty of neatly categorizing a music video clip is demonstrated by Shuker’s analysis of *The Escape Club’s* 1991 music video, *Call It Poison* (promoting their 1991 album *Dollars & Sex*, on Atlantic), in which it is described as exhibiting:

> ...elements of social consciousness...with a form of struggle for autonomy. There are also elements of Kaplan’s nihilist category of video, though these are muted by the celebratory manner in which the flouting of social conventions is presented. The MVs treatment of women is reminiscent of the classic mode of video, constructed for the male gaze, but it does not display any other feature of this category. Despite the elements of pastiche and intertextual references present, it is difficult to classify the video as postmodern, since there is a clear narrative and strong elements of realism throughout.

Music videos indeed perform socially influential functions as what Shuker calls “polysemic narratives” presenting images of viewer fantasy and desire. This is difficult semiotic terrain open to cultural interpretation and academic struggles over meaning.

Bechdolf et al. [1995] makes an interesting contribution to semiotic arguments by stressing sub-cultural fragmentation in the perception of meaning in music video:

> ...watching pop is - along with other social categories as class, age, ethnicity - always defined as a gender-specific activity. Therefore, the social functions and personal meaning constructions differ along gender lines, especially because music videos address issues that are of crucial interest to young women and men, and girls and boys who are searching for significance, and designing their own identities and identity conglomerates in a world of gender dichotomies and hierarchies. They are exploring prescribed gender roles and trying out alternative possibilities, living through the exciting and painful period of the first love relationships and sexual encounters.

Bechdolf [*ibid*.] makes another observation concerning generation gap issues and the objectivity of the researcher:

> In watching pop rather than just listening to it, the younger generations distinguish themselves from their parents' preoccupations with the purity of sound and infect living rooms now with visual noise. In conjunction with the music, the images address and express in a more overt fashion their fears and desires, hopes and utopian fantasies.

Jameson [1979, 130-148] also draws attention to the possibility of a sort of sub-conscious reading of music videos. Music videos, like all products of contemporary culture

> …have as their underlying impulse - albeit in what is often a distorted and repressed, unconscious form - our deepest fantasies about the nature of social life, both as we live it now, and as we feel it in our bones ought rather to be lived.

As a counter-argument amid the prolific discussion of the industrial and commercial attributes of music video, Jamieson argues that audiences of popular music videos cannot be regarded as passive consumers, being manipulated by the capitalist music industry into buying the music recordings that the videos advertise. Music videos offer not only "imaginary solutions to real problems", but also genuine content, and a great deal of utopian fantasy, in order to fulfill their ideological function. And finally, music video is ideological (in Jameson's terms), because it

> …remains implicitly, and no matter how faintly, negative and critical of the social order from which, as a product and a commodity form, it springs [Jameson (1979, 130-148].

Music Video and the future of interactive art

This writer has for some years held the hope that music videos status might change from that of an ambiguous hybrid commercial appendage of pop culture and (at the worst) watered-down and dubious works of art, if a market for the videos themselves as consumable products emerged. De Lancie [1996, 96] quotes *Recording Industry Association of America* video sales figures for 1993:

> 11 million, up 44.7% over 1992. That number is a drop in the bucket compared to the 495 million CDs that shipped in the same period.

The current emergence of the new DVD technology holds the promise of the beginning of such a new era in the pop music industry, when music video matures as an independent artform. The technology and viewing habits are rapidly being acquired, as represented by this breakdown of the US market:

> …the first quarter of 2002 show continued transition to digital video (reflected in) manufacturer-to-dealer sales of digital video products.
>
> Factory-to-dealer sales of DVD players rose a total of 11 percent during the first quarter. Unit sales gained more than 32 percent in February, compared to the same period in 2001.
>
> First quarter sales also saw DVD sales edge ahead of VHS sales by approximately 100,000 units. The gap is expected to increase to almost 2 million by the end of the year.
>
> According to CEA president and CEO Gary Shapiro, digital TV product sales amounted to 148,369 units in March, an 86 percent increase from that of 2001[1].

While this market activity is in progress, however, the effect has been to divert available spending money to these technologies at the expense of music recording sales (which are already in crisis because of the Internet file-sharing and CD-burning revolution), so for the time being DVD has not been established as significant part of the industry [Bricklin, 2002]. Nevertheless the potential seems certain to be realized.

At present interactive CD-ROMs have made minor impact on the music industry in market terms and must be considered still little more than a novelty, notwithstanding that it is a novelty which may foreshadow the future. The video content is severely restricted by the limited data storage capacity (650 to 700 megabytes) of a CD. An artist must be sparing in the linear time allocated to half size (at best), half-speed (15 frames per second or less) video element in the composition, when this represents a data flow of about 1.5 megabytes and upwards per second. DVD offers the capacity for approx. 110 minutes of full frame PAL or NTSC video, optimum digital sound quality, as well as interactive elements, depending on the format. Whether this brings music videos into a new relationship with consumers remains to be seen, although there is evidence that DVDs are indeed finally beginning to make a significant impact on the music industry. The Australian Record Industry sales figures for 2003 showed a rise of 8% in CD album sales to 50 million units from 47 million in 2003, but music DVD sales *doubled* to nearly 5 million units over the same period.[2]

Such a commercial vision of an interactive future for music industry products has been taken up by Marc Canter, founder of major multimedia software developer Macromedia, with the project *MediaBand.* The first CD-ROM release is called "*Meet MediaBand* ". It is an attempt to publish what Philip De Lancie; [1995, 96-101] describes as

> "new interactive music videos that defy the term song, movie or game. They are, in fact, all three combined into one."

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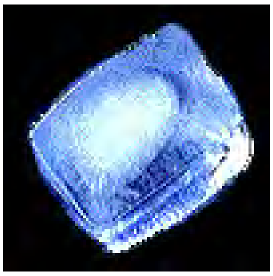
[1] http://www.globalsources.com/MAGAZINE/AV/0207/N050701.HTM ; accessed 11-01-03.
[2] Zuel, Bernhard. Sydney Morning Herald, March 18, 2004, p.3.

One track from *Meet MediaBand*, entitled *Undo Me*, allows the user to create a story line out of several parallel tracks of video, choosing from four prospective romantic partners for the featured singer Kelley Gabriel. The song and story progress through a series of fourteen decision points (represented by film-cel icons along the bottom of the screen).

At each decision point the user can inject passion into (click flame), or cool off (click ice-cube), the developing fictional relationship narrative.



Click to "cool off"

What makes it especially innovative is that these decisions also influence the music, with different musical arrangements, varying appropriately in mood, acting as accompaniment to the video material chosen at the branching points.

Click to "inject passion"

Thus the standard (approximately) three minute pop song video in this case offers sixteen possible endings, utilizing twenty-six minutes of video and parallel musical arrangements.

This approach to interactive music video is an indication of the way the multimedia strands are weaving together into a new art form. Compare *Meet MediaBand* with the winner of the 1996 ATOM Australian MultiMedia Awards event (including Gold ATOM and awards for best game and best dramatic production) - the CD- ROM entitled *The Dame was Loaded* featuring fictional hero Scott Anger. Here game conventions, cinematic conventions and interactivity meet. Cited for setting "new standards for production, technical qualities and richness of play", its most innovative feature was the use of live action video in a computer games context, drawing on the 1940s cinema and pulp fiction tradition of the gumshoe private investigator and film noir style. The game is delivered as a double CD-ROM featuring one hour of live action video. It took the one hundred contributors to the production thirteen months to create the product, the three week video shoot being the most technically demanding aspect, with multiple versions of each scene, correct to the second being required:

> ...players' actions (including the objects they possess) determine how the narrative unfolds...for one scene (lead actor) Berger had to deliver 26 variations on one line. Actors positions and postures at the end of each scene had to be exactly the same in each variation to ensure smooth transitions between segments.... Special lighting (was) needed to cater for the limited range of colours available on the computer screen. (Nelson, 1996, 31)

Randall Berger, one of the leading actors in *The Dame Was Loaded*, considers that performing in such multimedia productions as

> ...one of the most difficult projects for an actor. The first-person perspective means actors worked directly to the camera, something that runs counter to their training and experience. (Ibid.,)

Interactive cinema is an emerging strand of multimedia art experimentation with roots in music video and the games industry. Both have a relationship to cinema history. For the emerging multimedia art form there is an ambiguous cultural relationship between the artist and the cyber-audience. As in the case of music video, ambiguity arises out of the blurred boundaries between the commercial and the artistic elements.

Narrative in film as an influence on multimedia

The complexities and mysteries of life and the human condition are most readily understood in terms of one's own life history. We are trained as young children to enjoy stories, and are soon familiar with the shape of stories, beginning middle and end, the concept of cause and effect, as well as the conventions of a story having a climax and an underlying message. Film-making has evolved into an extremely sophisticated means of story-telling. Branigan [1992, 2] argues that narrative should be understood as a feature of general perception, a special form of assembling and understanding data. Narrative theorists characterize the building blocks of plot in terms of agents (called actants), events and the causal and temporal relationships between them[1].

Narrative is a way of organizing data into a special pattern which represents and explains experience. Classical narrative is thus shown to function as a recursive structure of representation, which organizes

[1] *Creating Interactive Narrative Structures: The Potential for AI Approaches*
R. Michael Young (undated). URL http://liquidnarrative.csc.ncsu.edu/pubs/potential.pdf. Accessed 26-02-04.

> …spatial and temporal data into a cause-effect chain of events with a beginning, middle, and end that embodies a judgment about the nature of the events as well as demonstrates how it is possible to know, and hence to narrate, the events. [p.3].

Knowing, and thus being able to tell, is a fundamental property of narration. In Branigan's view, narrative, cognition and knowledge pertain equally to film and life. According to this account of narrative, instead of recalling in a linear fashion, i.e. detail by detail, we tend to understand narrative by

> …constructing large-scale hierarchical patterns which represent a particular story as an abstract grouping of knowledge based on an underlying schema [p.16].

Film and other specific forms of describing the world through events, can only exist based on an ability to use a narrative schema to model a version of the world [*ibid*., p.17]. Branigan [p.18] clarifies this with a breakdown of such schematic constructions, typically occurring in film, listing 8 functions in narrative development:

1. abstract (title) or prologue;
2. orientation or exposition;
3. initiating event;
4. goal (statement of intent);
5. Complicating action;
6. climax and resolution;
7. epilogue; and
8. narration

All stories conform to the basic conventions of narrative structure, and will include some or many of the narrative elements listed above. In simplest terms there is generally an enigma (a puzzle of some sort: who is the murderer; a quest--to find the treasure, recover the grail, to win the girl, etc) and a resolution (finding the murderer/the treasure/winning the girl etc). Stories begin with a *disruption* that upsets the stable world-within-the-fiction (the body on the carpet, a stranger arrives, etc). The narrative will then unravel in such a way that the enigma is resolved and the initial equilibrium--the balanced, tranquil world--is restored, or a better order is put in place (the good King takes the throne). Narrative is regulated through what is called 'the chain of cause-and-effect'; this is to say that every event has a cause, every cause results in an effect. In classic narrative, there are no 'effects' without a 'cause'. Verisimilitude derives from both the setting (location) of the action (some knowable place, never 'nowhere') and the order in which the action unfolds--not necessarily in chronological sequence, but in a manner which conforms to our notion of time (temporality). Typically, the main characters will be individuals with knowable motives, psychological drives, character traits, etc. The chain of events will be driven by the traits and actions of these individuals. It is unusual to find a narrative which begins with one set of characters and ends with another[1].

In film, *editing* moves the story along and maintains the causal logic, that is, maintains the chain of cause and effect. Editing also controls time and delineates space, which maintains verisimilitude. Conventions have evolved which represent a tacit agreement between genre film-makers and their target audiences about the accepted meanings of certain techniques, and visual and sound/music cues.

The above brief breakdown of the possible elements of narrative has relevance to an understanding of the processes and aesthetics of multimedia, in that most complex experiential art contains, at the heart, some form of story telling. Engendering immersion in a good story/scenario must be a fundamental engagement strategy in multimedia design. However, multimedia is by definition interactive, and the introduction of branching structures brings new levels of complexity to concepts of narrative. On page 362 the issues related to narrative within a non-linear structure are taken up, including a discussion of interactive story shapes.

Video Art

Since the introduction of domestic video technology some artists have chosen the video medium to explore art concepts, sensory perception and experience. Video art is a significant and dynamic communication palette in its own right among the various dimensions of multimedia meta-art, quite separate to the other video delivery formats, music video and mainstream television. The 1960s marked the infancy of this ‘video art’. Video art can be clearly distinguished from the cinematic arts based on film technology:

> Video art does not necessarily use actors, may not contain dialogue, may have no discernible narrative or plot, or adhere to any of the other comfortable conventions that construct cinema as entertainment. This distinction is important because it delineates video art not only from cinema but also from the sub-categories where those definitions may become muddy (as in the case of avant garde or short films). Perhaps the simplest, most straightforward defining distinction in this respect would then be to say that cinema's ultimate goal is to entertain (i.e., to get someone to watch the film) whereas video art's intentions are more varied -- be they to

[1] http://www.ejmd.mcmail.com/filmntiv.htm ; accessed 12-01-03.

simply explore the boundaries of the medium itself (e.g., Peter Campus, "Double Vision") or to rigorously attack the viewer's expectations of video as shaped by conventional cinema (e.g., Joan Jonas, "Organic Honey's Vertical Roll"). [1]

Video art has an unusual history in that it has developed along a distinct path oriented towards self-conscious, often self-referencing public installations.

Korean-born Nam June Paik, widely acknowledged as “the father of video art”, was the first video artist to gain significant recognition, creating a robust personal aesthetic [Nguyen, 1998]. In his first solo exhibition of 1963, Paik already had a vision of a how he wanted to present his ideas, scattering televisions around a gallery, on their sides and upside down, in an installation which displayed distorted screen images of electronic waves. He pursued this modus operandi with ingenuity and has sustained an interesting creative career marked by constant intellectual exploration of this new technology-based art-form. For the premiere of the *TV Bra* (1969), viewers witnessed Charlotte Mormon playing cello topless, save for three-inch television monitors mounted on her breasts [Zigerlig and Ebersman, 2000].

Left and Above: Nam June Paik, *The TV Bra* (1969), [2]

Such comic irony, intellectual puzzles and oblique post-modern references are a feature of much of Paik's work, e.g. *TV Buddha* (1974, pictured right), in which a junk-sculpture Buddha sits cross-legged and childlike, transfixed by his own image on a TV screen[3]. Paik addressed the power of video with a very lateral approach, adapting video and television technology in his electronic junk sculptures, which are comprised mainly of TV sets. An extreme example was the

[1] http://www.wikipedia.org/wiki/Video_art ; accessed 12-12-02.
[2] www.heise.de/tp/deutsch/ inhalt/sa/3476/1.html **;** accessed 10-12-02.
[3] http://www.artandculture.com/cgi-bin/WebObjects/ACLive.woa/wa/artist?id=23 ; accessed 12-12-02.

traveling exhibition called *Electronic Superhighway,* composed of 313 TVs, arranged approximately in the shape of the United States, on which were projected a constant variety of clips of old game shows, the flag, landscapes, political images, electronic disturbances, news broadcasts, cartoons, and other random images. The entire exhibit totals more than 30 individual works and depicts, through video montage, counterpoint, cross-referencing, and accidental or random synchronicity, his futuristic view of life in cyberspace [Nguyen, 1998].

In 1965, Sony introduced the *Portapak*, a portable analogue camera with instant playback and sound. Prior to the introduction of the Sony Portapak, motion picture technology was only available to the consumer (or the artist for that matter) by way of 8mm film -- which was not only more expensive, but did not provide the instant playback that video tape technologies offered [1]. With this new portable video technology, people could make their own motion picture documents, see them immediately (a revolution), change them, craft them in a primitive fashion, and even distribute the output on cable television. With such affordable, versatile equipment and an intimate familiarity with the culture of television and film, early video artists branched off in two directions—those who used video to create narrative, documentary or dramatic forms—and those who addressed video culture through sculptures / installations, and abstract/conceptual processes [Zigerlig & Ebersman, 2000], inevitably informed by the counter-cultural imperatives of the late 1960s.

Inspired by the bizarre but incisive inventiveness of Nam June Paik's 60s installations, many artists focused on the aesthetics of video display over content, in what was essentially installation art. They used televisions, video monitors, and creative projection surfaces in innovative juxtapositions. In the early 70s, Vito Acconci, Keith Sonnier, Nancy Holt, Bruce Nauman, Shugeko Kubota, Chris Burden, and Les Levine, and others, began to investigate the role of artist as “art creator and interpreter” through the video medium, by allowing their bodies, movements, actions and words to be conscripted into comment on the art-making process [Zigerlig and Ebersman, 2000]. The pioneers of video art mostly hailed from backgrounds in performance and conceptual art and saw video art as an extension of these activities, a decisive factor in shaping the look and feel of the work[2].

[1] http://www.wikipedia.org/wiki/Video_art ; accessed 12-12-02.
[2] http://www.artspace.org.nz/shows/97_2.html ; accessed 11-12-02.

Above: Eleanor Antin – from *The Loves of a Ballerina*

During this period, several women artists such as the prolific Lynda Benglis, and feminist Eleanor Antin began to explore the possibilities of video art in order to comment on political issues and challenge the portrayals of women in media [Zigerlig and Ebersman, 2000].

> Antin played a formative role in the development of performance and feminist art, pioneering the use of autobiography, nontraditional narrative forms and fictive personae (the nurse, the ballerina, the king etc.) Yet, whether installation, performance, video or feature-length film, Antin's diverse projects are linked by their subversive humor, searching intelligence and common concern with issues of identity [1].

In 1975 Cuban artist Ana Mendieta, took a very different quasi-ritual approach and began exploring experimental video images of transient impressions of her body in the sand or burned into the earth, the so-called *Silueta* (silhouette) works. In these works she achieved a unique and mystical synthesis of imagery.

Linking avant-garde forms of expression with the spiritual mysticism of SanterÌa, a syncretistic religious movement, but also with issues of gender and questions of migration, her work is still relevant today[2]. Mendieta's *Silueta* images:

> … related her actions to the form (silhouette) of her body, to land art and to the typical esoterically symbols of religions and primitive rituals. [Nederend, 1999].

Joan Jonas was another pioneer of video/performance art. Her experiments and productions in the late 1960s and early 1970s, which featured, among other things, the use of mirrors and electric currents, to play with perceptions of the self, were essential to the formulation of the genre [3]. Jonas is still active today, using unlikely combinations of objects in performances concerned with the image as metaphor:

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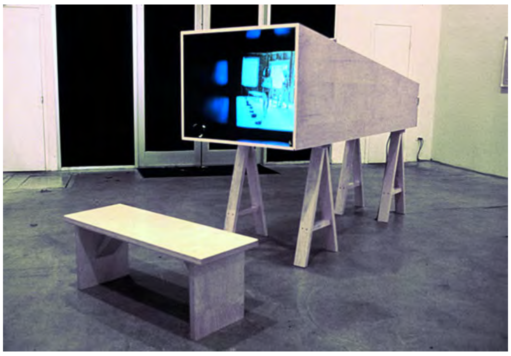
[1] http://galleryofart.wustl.edu/archives/eleanorantin/pressreleases/retrospective.html ; accessed 12-12-02,

[2] http://www.absolutearts.com/artsnews/2002/12/25/30602.html ; accessed 25-02-04.

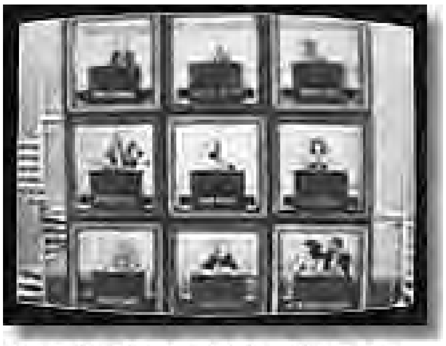
[3] http://architecture.mit.edu/people/profiles/prjonas.html ; accessed 10-12-02.



Above: *My New Theater III: In the Shadows of a Shadow*, 1999, Video projector, speakers, wood structure.[1]

Dara Birnbaum made her name with video art installations which commented on and satirized the presentation of female figures in 1970s television programs.

Technology / Transformation (1978 / 79)



Kiss The Girls: Make Them Cry (1979)

> Dara Birnbaum's video works are among the most influential and innovative contributions to contemporary discourse on art and television. In her videotapes and multimedia installations, Birnbaum applies both low-end and high-end video technology to subvert, critique, or deconstruct the power of mass media images and gestures in order to define mythologies of culture, history, and memory. Analyzing TV's idiomatic grammar (reverse shot, crosscuts, inserts) and genres (game shows, sitcoms, crime dramas), she recontextualized pop cultural icons — *Kojak, General Hospital* — through fragmentation and repetition. [2]

Meanwhile, there were artists exploring documentaries and narratives and these were gradually branching out into innovative avant-garde approaches. Groups like the *Viennese Actionists,* the Japanese Gutai[3] movement and the multifaceted Japanese performance art ensemble *Dumb Type* used video art to challenge the traditional dominance of gallery painting and to make powerful comments on war. Artists have increasingly turned to video in order to dramatize their ideas and concerns. By the 90s, video art had evolved from its early experimental installation

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[1] http://www.eloupe.com/Pages/felsen/jonas/jonas07.html ; accessed 12-12-02 .
[2] http://www.ithaca.edu/handwerker/global/birnbaum/ ; accessed 12-12-02.
[3] Japanese avant-garde art movement founded in Osaka in 1954, which specialized in elaborate "happenings" and public events.

approach to a form which featured complex video installations, enriched by sculpture and theatrical elements [Zigerlig and Ebersman, 2000].

Recent video art has drawn upon the tradition of experimental installation of the preceding four decades, since the technology became available, but with an expanding aesthetic and an involvement with current issues. Video artists take the viewers into a wide variety of plot, situation, or philosophy. Shirin Neshat's films,

Above: still from Shirin Neshat's *Rapture*

such as the trilogy *Rapture,* which must be viewed on two screens, focus on male/female relations in countries of Muslim faith,[1].

In the early nineties, Matthew Barney challenged the New York art scene with all the force of his controversially physical videos, and soon earned acclaim. By 1994, Barney was embarking on his defining work, the five part *Cremaster* series of hour-long videos [Zigerlig and Ebersman, 2000], one of a cycle of five films named after the "thermometer" muscles that retract the testicles in cold weather[2].

They feature the artist as a number of protagonists in fantastic, surreal stories, among them Gary Gilmore, a satyr and a Hungarian prince [Zigerlig and Ebersman, 2000]. The first of the *Cremaster* series (which was actually entitled "*Cremaster 4*") was an ambitious hybridization of video and film projections, each one-hour piece including fantastic sculptures, original photographs, and drawings.

Left: Image from the *Cremaster* series[3] .

[1] http://www.haberarts.com/neshat.htm ; accessed 12-12-02.
[2] http://www.artandculture.com/cgi-bin/WebObjects/ACLive.woa/wa/artist?id=814 ; accessed 12-12-02.
[3] http://www.c3.hu/events/96/barney/ images/cremaster.gif ; accessed 12-12-02.

In the mid-nineties, celebrated American video artist Bill Viola began to create videos which continue in the tradition of exploring the creative possibilities of the medium both conceptually and sensually [1]. He has never been concerned with using video as a narrative document or film substitute, but rather, through dramatic use of space, his installations function with a poetic and sometimes spiritual feel [Zigerlig and Ebersman, 2000]. They can be approached

> … less as concrete works and more like enveloping, temporal environments aimed at creating visceral experiences for the viewer -- sort of like Happenings with a rewind button.[2]

Their presentation, as well as visual content, is sparse yet direct, often grounded in explorations of archetypal natural forces such as earth, wind and fire. *The Crossing* (1996), for example, depicts a man (Viola) emerging from a black void, growing larger as he walks towards the camera. When he finally stops, he suddenly bursts into, and is slowly consumed by, flames, while an adjacent screen depicts the image of a man gradually inundated by water[3]. It requires a 14 by 10 foot two-sided video screen which stands like a monolith in the center of the room, plus two video projectors, and a very powerful sound system [4].

Above: Scenes from Bill Viola's: *The Crossing* (1996). Video/sound installation.

Specifications for installation: Two-channel colour video and stereo-sound installation, continuous loop, ideal room dimensions: 16 feet x 27 feet 6 inches x 57 feet. Edition 1/3.

Viola's motivating idea is to blur the boundaries between the mind and body, microcosm and macrocosm, and interior and exterior space. Using and manipulating the raw materials of sound (tone) and time (duration), Viola's images reach all the senses, heightening awareness. He creates an experience, not a view of an experience. In essence, the viewer

[1] http://www.artandculture.com/cgi-bin/WebObjects/ACLive.woa/wa/artist?id=42 ; accessed 12-12-02.
[2] http://www.artandculture.com/cgi-bin/WebObjects/ACLive.woa/wa/artist?id=42 ; accessed 11-12-02.
[3] http://www.artandculture.com/cgi-bin/WebObjects/ACLive.woa/wa/artist?id=42 ; accessed 12-12-02.
[4] http://www.flash.net/~curtis3/crossing.htm ; accessed 11-12-02.

> becomes both witness and world. A mediation on the phenomenon of sensory perception, Viola's work illuminates how the integration of all senses and the connection between mind and body are avenues to self-knowledge. [1]

More recently, Tony Oursler's installation sculptures, which juxtapose "live" animated talking faces with inert puppet bodies, projecting the videotaped faces onto fiberglass shapes and dummies in order to delve into psychological issues. His more recent video projections examine the effects on human psychology of our mass media-oriented society, delving into the negative effects of contemporary society's obsession with television, and suggesting we have reached a point where media consumption and psychosis seem to converge. [2]

Right: Tony Oursler *Eye in the Sky,* (1997).

Specifications for Ousler's *Eye in the Sky* (pictured right) are *:* Sony CPJ 200 projector, videotape, VCR; white acrylic paint on fiberglass sphere 18 inches diameter, 20 mins.; performance by Mary N.

Above: Still from *Selfless in the Lava Bath*, (1994), by Pipilotti RIST (1962 -) [3].

Pipilotti Rist is another creator of elaborate installations, but with an environmental art slant. His videos are projected, for instance, onto the contents of an apartment, creating "a world of simulacra amidst domestic stability" [Zigerlig and Ebersman, 2000]. Water is a recurring theme in the art of Pipilotti Rist, with evocative references to be found in titles like *Selfless in the Lava Bath*, *Sea of Coffee* and *Sip My Ocean*.

Rotating movement is a recurrent feature of her art and generates a flow.

The main thrust of video art during its relatively short history has been towards an abstract, conceptualized approach, drawing focus on the viewing medium itself and media in general. It has been largely exhibited in an installation context, utilizing projected imagery. A computer with RGB monitor offers a now commonplace

[1] http://www.dm-art.org/Collections/Contemporary/coll_crossing.htm ; accessed 10-12-02.
[2] http://www.ackland.org/art/exhibitions/illuminations/image5.htm ; accessed 12-12-02.
[3] http://www.mbam.qc.ca/expopassees/a-pipilotti_2.html ; accessed 12-12-02.

technology in the home and workplace that can be utilized as a mini-installation/exhibition space by multimedia artists, allowing a new, more accessible forum for art which draws on the tradition discussed above, outside the rarified atmosphere of specialised gallery environments. The Video art tradition can inform multi-media through its sophisticated, abstracted handling of political, humanistic and art concepts through montage and juxtaposition of communicative imagery.

The evolution of video art has produced interesting, intense conceptual approaches to harnessing the technology for artistic expression over its brief history, a strong and serious tradition of avant garde experimentation. With the advent of digital video, fire-wire connectivity, and desk-top video editing and compositing, motion picture media has seamlessly merged with a variety of other digital media to take its place in the service of meta-art.

Experimental / abstract film and animated kinetic art

Visual art has fragmented from being a largely representational or narrative medium into a myriad of new forms. These new concepts of visual art represent different approaches to what can or should happen within the image plane. A progression towards a pure abstraction has been one significant strand of this visual art history during the twentieth century. This century has also seen the rise of an important new technology based art medium – film. There is a very large body of work that falls within the domain of experimental film-making. Inevitably some film artists have been become interested in experimentation with abstract film-making. In this section an emphasis has been placed on kinetic animation, and especially any work which attempts to explore an synthesis of the visual and auditory arts.

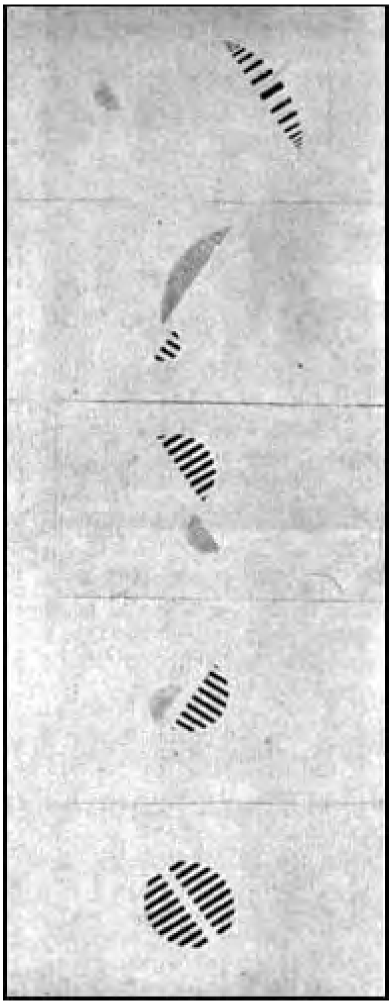
As has been mentioned in the earlier section which discusses the history of technological approaches to reproducing the synaesthetic experience (see p.117), in 1912 Italian *Futurist* artist Bruno Corra published an article entitled *Abstract Film – Chromatic Music*, in which he described experiments carried out with his brother Arnolda Ginna, creating some nine abstract films by hand painting directly on film-stock (in 1911). Unfortunately these early experimental abstract films have been lost [Milicevic, 2000; Moritz 1997].

In Germany about this same time, Hans Stoltenberg also experimented with hand drawn abstractions on film, none of which have survived [Moritz, 1997]. Meanwhile in Paris Leopold Survage (of Finnish/Danish/Russian descent), who was in contact with a circle which included Picasso, Braque, Leger, Brancusi and Modigliani, was preparing literally hundreds of sequential paintings which he planned to photograph for an abstract film of "symphonies in movement" he called *Rythme Coloré* (Lyons, 2002). He had hoped to make the film using one of the new multicolor processes that had recently emerged, but was prevented by the outbreak of World War I. He eventually sold a number of the paintings, breaking up the series, and as a result of this dispersal his vision still has not been filmed [Moritz, 1997]. Survage said of his work in the area that,

> Coloured music is in no way an illustration or an interpretation of a musical work. It is an autonomous art, although based on the same psychological principles as music. [Quoted in Lyons, 2002].

The ideas of the Bauhaus art movement of the "roaring" 20s, influenced some of the earliest abstract film-makers. Pioneer German abstract film-makers Viking Eggeling, Hans Richter and Walther Ruttmann developed the idea of an "optical music", described in terms of concepts from music composition: orchestra, orchestration, symphony, instrument, fugue, counterpoint, and especially the term *score* - the exact notation of events in time depicting the flow of movement. They found a medium suitable for the realization of these ideas in the hand-drawn animated film. Werner Graeff's film *Filmscore 1/22*, consisted of shapes painted onto the picture frames in colour. It was an attempt to produce a film equivalent of the play of light, and linking it with music [Milicevic, 2000].

Viking Eggeling also believed that music, because of its abstract qualities, could be "transposed" into the visual, time-based medium of film.

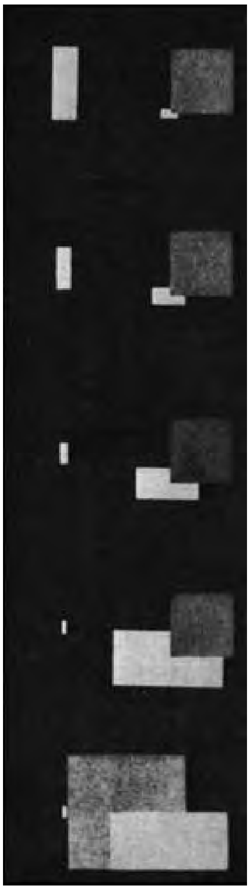


Right: Frames from the film *Diagonal Symphonie*, by Viking Eggeling, 1921-23 [1]. *Diagonal Symphony* exists today only in fragments [Middleton, 2000].

[1] http://www.mtc-arts.org.uk/arts/visual/surreal/rhythmns/abstract2.htm ; accessed 30-12-02.

Diagonal Symphony, the first purely abstract film, was a continuous sequence of abstract images that evolved and transformed, based on this musical analogy [Collopy, 2000]. *Horizontal-Vertical Orchestra.*

Hans Richter, who collaborated closely with Eggeling, expressed an interest in musical organization of film time in his films *Rhythm 21* and *Rhythm 23. Rhythm 21* is a study in geometric composition in which variously-shaded rectangles grow, shrink, and alter their proportions [1]. *Rhythm 23* (1923, 3 mins., silent) features criss-cross patterns, negative reversals, intercut stringed forms and further variations of the rectangle and square[2].



Frames from the film *Rythmus 21*, by Hans Richter, 1921.[3]

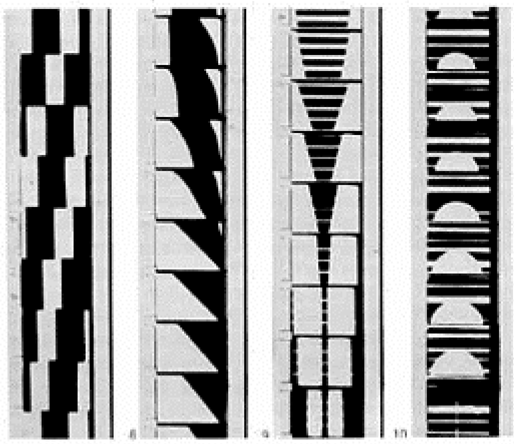
During the 1920's cinematic forms of animated kinetic art were beginning to appear in the visual entertainment arena. Walter Ruttman screened his 10-minute film *Lichtspiel Opus 1* in the spring of 1921, in Frankfurt, Germany - the first abstract film to be actually shown in public, and the first in his *Opus* film serial, I to IV, (1920-1923). The film was reviewed by Bernhard Diebold in the Frankfurter Zeitung under the heading *A New Art, The Vision-Music of Films* [Lyons, 2002]. Ruttmann, like Eggeling and Richter, was interested in the idea of "optical music". *Lichtspiel Opus 1* was hand tinted in striking and subtle colours, with an especially composed, live, synchronous musical score [Lyons, 2002].

In 1928 Ruttmann gained access to the best sound recording system available at that time in Germany. With this technology he produced a challenging "film", entitled *Weekend,* which actually contained no pictures, just a sound montage on the optical sound track, recording the noises of a working day and Sunday in the country-side [Milicevic, 2000].

[1] http://www.ubu.com/aspen/ ; accessed 30-12-02.
[2] http://www.iotacenter.org/tape.descriptions.html ; accessed 30-12-02.
[3] http://www.mtc-arts.org.uk/arts/visual/surreal/rhythmns/abstract2.htm ; accessed 29-12-02.

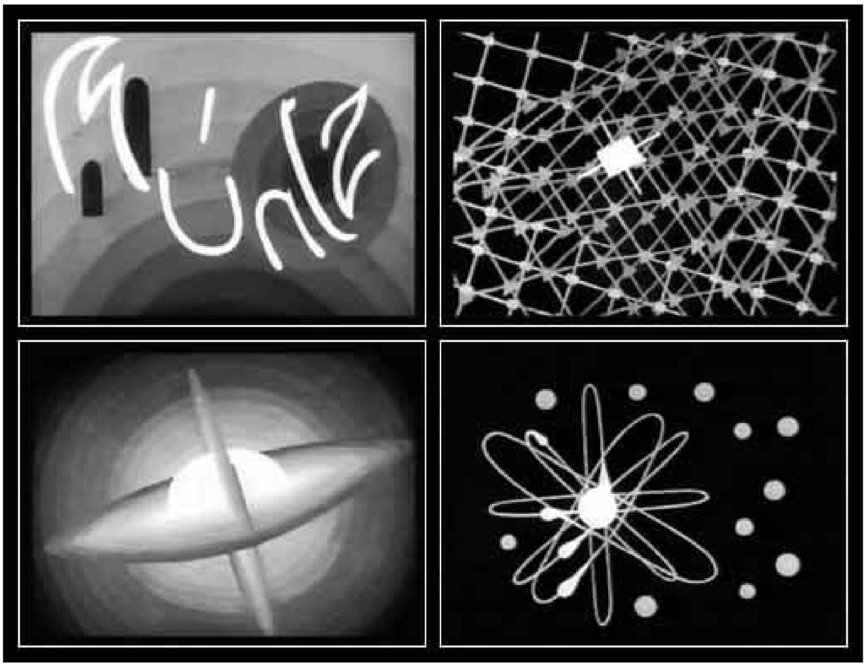
Along with Russian film-maker Vertov (see p.225), Ruttman must be regarded as pre-empting *Musique Concrète*. Ruttman's short but brilliant career was cut tragically short when he was killed while making a newsreel on the frontlines of World War II[1].

Right: Frames from Walther Ruttman's *In the Night (*1931) – Schumann's composition recreated in abstract picture and motion.



Parallel to Ruttmann in the 1920s, the youngest of the German Bauhaus abstract film-makers, Oskar Fischinger was pioneering 'visual music' films, using tinted animation to live musical accompaniment [Moritz, 1997].

Fischinger, was extremely impressed by the premiere of Ruttmann's *Opus 1,* which stimulated him to begin practical work as a film maker.



By 1922 he had begun to produce abstract films, and in a few years was synchronizing abstract imagery to popular records with a series he called *Studies (NB: see above image)*. These films were shown in theaters as advertisements for the recordings. Sixty years before MTV, they were the first music videos. Each of these studies ran three minutes in length and included approximately 5000 drawings coordinated to the music. [2]

Fischinger, regarded as one of the original wellsprings of experimental animation [3], devoted his career to this "visualized music", making many films which "illustrated" the flow of music in time. An analysis of Fischinger's films, even when viewed

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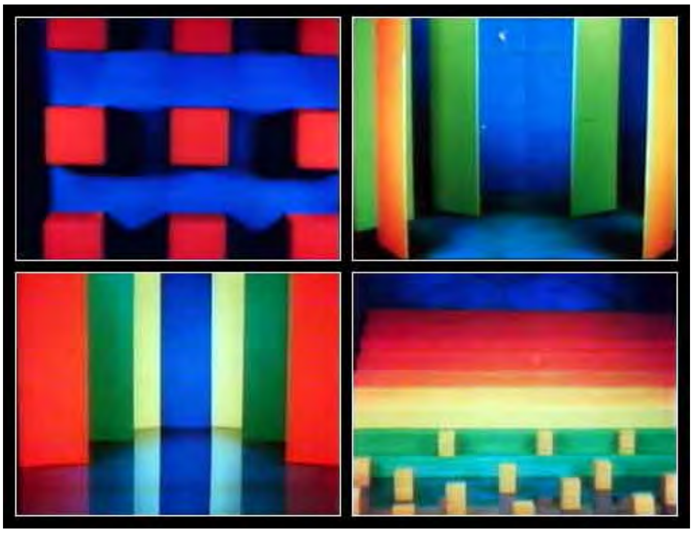
[1] http://www.artandculture.com/cgi-bin/WebObjects/ACLive.woa/wa/artist?id=1345 ; accessed 30-12-02
[2] Zone, Ray (undated). *Oskar Fischinger.* http://artscenecal.com/ArticlesFile/Archive/Articles1998/Articles0498/OFischingerA.html ; accessed 30-12-02.
[3] http://www.miaf.net/siaf/Fischinger.htm ; accessed 30-12-02.

without the soundtrack, reveals a clear musical structure, yet he himself only reluctantly accepted the burden of musical accompaniment [Milicevic, 2000].



Right: Stills from Fischinger's abstract four-minute film *Composition in Blue* (1935).

Under pressure from the Nazis, who regarded his abstract cinema as degenerate [1], Fischinger signed a contract with Paramount Studios and emigrated to America in 1936, where he produced the two and a half minute 35 mm abstract film *Allegretto*. In 1937 Fischinger made a 5 minute film for MGM, entitled *An Optical Poem* which was widely distributed to theatres in the USA. In 1938, Fischinger was hired by the Disney studios to design and animate portions of *Fantasia*. After a year, Fischinger resigned from the project because of the conflicts between his goals as an artist, and his encounters with the Hollywood committee system, which over-ruled many of his artistic decisions, exacerbated by his inability to speak English and dependence on studio interpreters [Rosenbaum, 2001]. Zone [undated] claims his influence is obvious in the initial Bach episode of the film[2], though Fischinger himself disowned it [Rosenbaum, 2001]. He also had difficulty with the contrast between Disney Studio's use of music to illustrate, set the mood, and provide temporal structure (sometimes disparagingly called "Mickey Mousing") and his own use of music as rhythmic counterpoint [3].

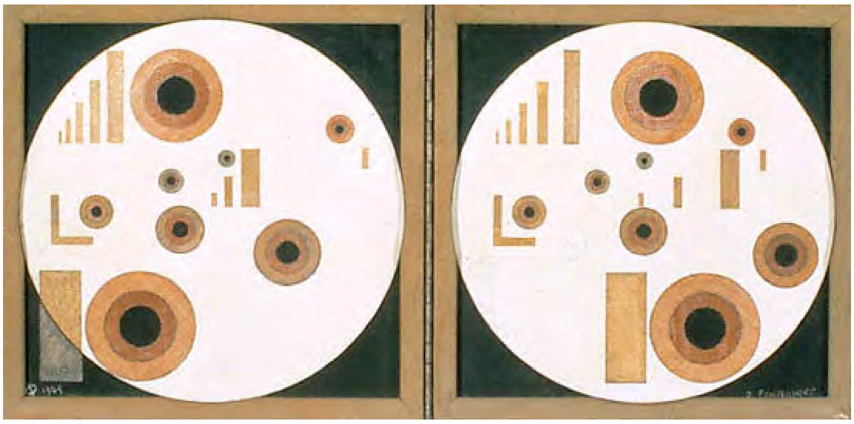
[1] http://www.roberthaller.com/firstlight/fischinger.html ; accessed 30-12-02.
[2] Zone, Ray [undated]. *Oskar Fischinger.* http://artscenecal.com/ArticlesFile/Archive/Articles1998/Articles0498/OFischingerA.html ; accessed 30-12-02 .
[3] Grush, Byron [undated]. *Space, Oskar Fischinger, and Desktop Computer Animation* . URL: http://www.swcp.com/~asifa/grush2.htm ; accessed 30-12-02.

Fischinger produced two self-financed film masterpieces in the late 1940's. *Radio Dynamics: a colour music composition* (1943, 3'45") is "an intentionally silent meditation, a visual invocation of a prayerful interior language" [1].

Motion Painting No. 1 documents the growth of one specific painting, uniquely combining the arts of film and painting as the work evolves before the viewer's eyes into the final intricate array of colourful motive lines. From 1952

Above: still from *Motion painting No 1*, (1947).

Fischinger also produced stereo paintings, in addition to a stereo film, in his drive towards a perceptual synthesis of visual space.



Above: *Circles in Circle*, oil on masonite, stereo painting, 12 x 12" each, 1949

Another interesting early pioneer of experimental, avant-garde and radical non-narrative approaches to filmmaking was the Russian Dziga Vertov. He spent time as a young man studying music at the Bialystok Conservatory of Music, but was forced to flee to Moscow in 1915 when Germany invaded Poland. After a year he moved to Petrograd, where he became interested in the ideas of the Russian avant-garde. During this period his interest was aroused by the possibilities of early sound recording technology, and he built his own primitive *Laboratory of Sound* in 1916. Vertov used a *Pathephone* disc recorder to collect sounds, developing an interest in experimental sound-montage, through the experience of working as editor on the weekly newsreel *Kino Nedelia* (Movieweek). He attempted to use sound editing

[1] Zone, Ray [undated]). *Oskar Fischinger. URL:* http://artscenecal.com/ArticlesFile/Archive/Articles1998/Articles0498/OFischingerA.html ; accessed 30-12-02.

processes to create montage “symphonies”. These were in effect precursors of Musique Concrète, but Vertov was hampered by the limitations of the rudimentary technology. This experience was an important formative influence on his later work in film however [Milicevic, 2000]. Many of his sound-montage ideas found their way into his later experiments with film, notably the *Man with a Movie Camera* (1929), an avant-garde, documentary meta-narrative, which celebrates Soviet workers and filmmaking. The film uses radical, tightly structured montage, and innovative cinematic techniques, such as variable camera speeds, dissolves, the use of prismatic lenses, superimpositions and split-screen forms. The film is a portrayal of a typical day in urban Moscow, from dawn to dusk, with a self-conscious use of the camera as the eye to replace the human eye.

> Vertov was a working-class artist who desired to link workers with machines. His film opens with a manifesto, a series of intertitles telling us that this film is an "experiment," a search for an "absolute language of cinema" that is "based on its total separation from the language of literature and theater…. The *Man With a Movie Camera* is divided into nine orchestral-type movements, and several of them use rapid-fire editing, wild juxtapositions (e.g. blinking eyes with shutter blinds) and multiple exposures to mesh workers with machines in a simultaneity of reverence and celebration. " [1]

The *kino-eye* in Dziga Vertov's *Man With a Movie Camera.*

Vertov's camera finds mechanical imagery in the beauty salon.

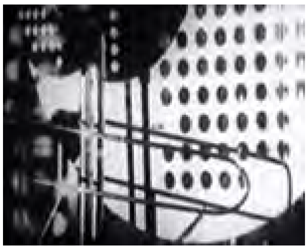
Vertov developed strong ideas about film, which he linked to his belief in the new Soviet system and the ideas of Marx.

> Attacking Russia's "film-dramas," Vertov criticized the industry's movement toward sophisticated narratives that relied on the same fantastical techniques as literature and theater. In his manifesto, he equated contemporary Russian cinema with religion, replacing the "religion" in Marx's famous maxim, "Religion is the opium of the masses" with "film-drama." [2]

[1] Tracey, Grant (undated). *Man with a Movie Camera: the Soviet Avant Garde.* Images Journal, Issue 5. URL : http://www.imagesjournal.com/issue05/reviews/vertov.htm ; accessed 29-12-02.
[2] http://www.artandculture.com/cgi-bin/WebObjects/ACLive.woa/wa/artist?id=1284 accessed 29-12-02.

In 1930 Vertov shot his first sound film *Symphony of Donbas*, or *Enthusiasm* in which he used a variety of innovative sound manipulation techniques [1] [Milicevic, 2000].

Hungarian Laszlo Moholy-Nagy, who was also involved in the Bauhaus, is primarily remembered for his work as a photographer. However he also had innovative ideas about film and film theory [Milicevic, 2000]. Many of his ideas for film projects e.g., his *Dynamic of the Metropolis* of 1921, (which he conceptualized as a purely visual experience that would "bring the viewer actively into the dynamic of the city") were never realized however. [2]



He did complete two documentary films in the late 1920s, and in 1930 shot the kinetic animation *Lichtspiel, schwartz-weiss-grau*, ("Light-play, black-white-grey"). This film is an intensely visual exploration of a purpose-built kinetic sculpture *Light Modulator*, focusing on the play of shade, light, and rhythm.

It is an extraordinary cinematic exemplification of Moholy-Nagy's concept of 'building an art-object of different pieces through a preconceived mathematical pattern'. The original film consists of only 49 shots (many of them with multiple exposure), mostly close-ups of the rotating modulator, which, under strong light and continuous motion, produces intricate optical effects on the screen [1].

In making a film, Moholy-Nagy was more interested in light, than in motion (Forgács, 1995). Like Fishinger, he was also interested in experiments with direct (hand-drawn) manipulation of the optical sound track. In his film *ABC of Sound (*1933*),* which he described as a lighthearted experiment, he re-photographed the optical soundtrack for simultaneous projection onto the screen [Milicevic, 2000].

Milicevic claims that Moholy-Nagys *ABC of Sound* was shown in London in 1934 at a Film Society screening, where Scottish-born abstract film artist Norman McLaren was able to see it. However McLaren himself describes his inspiration for becoming

[1] It wasn't until eighteen years later that broadcast engineer Pierre Schaeffer introduced those same techniques in Paris as his own invention, naming this music *musique concrete* (Milicevic, 2000).
[2] http://www.artandculture.com/cgi-bin/WebObjects/ACLive.woa/wa/artist?id=340 accessed 30-12-02.

an abstract film-maker as a prior screening he attended while studying interior design at the Glascow School of Art. He states:

> The films which excited me most were [Oscar] Fischinger's *Hungarian Dance No. 5* (1932), an abstract film which realized in its fusion of visuals and music what I had only dreamed of at that time; Emile Cohl's *Drame chez les Fantoches* (1909), a witty linear cartoon of metamorphosing people and objects; and [Alexandre] Alexeieff's *Night on Bare Mountain* (1933), one of the masterpieces of poetic animation made on his new invention, the pin-screen. These three films influenced me greatly in my early approach to film-making. [2]

McLaren's first "direct" film, *Love on a Wing* (1938) used continually metamorphosing symbols, drawn directly onto the film stock against photographed multi-plane backgrounds. Later, after McLaren had moved to the USA in 1939, he made a film called *Allegro* one of his first experiments in drawing both the image and sound track on directly onto the film. The film was purchased by the Guggenheim Museum but was never copied and, tragically, was eventually worn out by repeated projection [3].

In 1939 the Swiss "Musicalist" painter, Charles Blanc-Gatti, made a five-minute film called *Chromophonie,* based on the principle that each tone of music ought to have a single, consistent corresponding colour in the spectrum, continuing experimentation with the concept of correlation between auditory and visual tones, which had been pursued by so many artists [Milicevic, 2000].

> In his book *Concerning Sounds and Colors,* Blanc-Gatti says that Walt Disney came to an exhibition of his paintings in Paris during the early 1930s, and that he spoke to Disney about his ambition to make a feature-length musical animation film. After the war, when *Fantasia* was finally released in Europe, Blanc-Gatti became outraged and attempted to sue Disney for stealing his idea [Moritz, 1997].

In 1940 James and John Whitney developed a new approach for combining sound with film. John Whitney was a composer and his brother was a painter. John had been introduced to Schoenberg's twelve-tone row serial composition technique when taking lessons from one of Schoenberg's pupils, René Leibowitz. He was also acquainted with European expressionist music. The Whitney brothers became interested in the work of the pioneers of abstract film, and especially in Fischinger's

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[1] Petric, Vlada (undated). *Light Play: A Tribute to Moholy-Nagy* . URL: http://www.roberthaller.com/firstlight/petric.html ;accessed 30-12-02.
[2] Russell ,Steve (undated) *The Films of Norman McLaren* http://www.mic.org.nz/public_html/13art3.html ; accessed 30-12-02.
[3] Russell ,Steve (undated) *The Films of Norman McLaren* http://www.mic.org.nz/public_html/13art3.html ; accessed 30-12-02.

experiments with ‘visual music’, but were disappointed by the use of pre-existing symphonic music[1].

In the years which followed, John and James Whitney collaborated on a series of five small films, *Five Abstract Film Exercises*, (1942-1945) which radicalize the concept of technologically-mediated synaesthesia, thanks to an apparatus invented by John Whitney, which combines an optical printer with the movement of a gauged pendulum. The great innovation of this apparatus lay in its capacity to simultaneously create images and sounds directly on the film stock. Their approach was to use a series of manually activated pendulums whose movement controlled the amount of light passing through an aperture to which a pendulum was connected. An optical sound track was guided past the opening so that the changing light patterns were registered on the film. Since the rate of the pendulum’s movement was slow, the optical sound track had to be played back at a faster speed in order to avoid sub-audible sounds, (frequencies below 16 Hz).

Influenced by his brother’s interest in Schoenberg, these first films by James Whitney explore the formal development of elementary geometrical figures, elaborating on a serial model (transposition, inversion, retrogradation etc.). Because they had been forced to work at such slow speeds, the Whitneys were able to precisely synchronize the temporal relationship between sounds and their visual/ graphic imagery. In addition, they became aware of and utilized the musical timbre change when they increased the playback speed. Whitney’s soundtracks sounded like early electronic music, and their technique of speeding-up was employed by German composer Karheinz Stockhausen twenty years later in his mixed-electronic composition, *Kontakte.* In the film *Variation* (1940) which, as its name indicates, is analogous to a musical development of a visual idea, James Whitney continued to seek a kind of audio-visual synaesthesia (ibid.).

Viennese film-maker, Peter Kubelka, was strongly influenced by the music of Anton Webern. Webern reduced music to single tones and silent intervals between them, and similarly Kubelka developed what he called “the metric film”, by reducing film to the film-frame and the interval between two frames. Pioneering Russian film-maker

[1] www.olats.org/pionniers/pp/whitney/whitney.shtml ;accessed 30-12-02

Eisenstein established the use of shots as the basic unit and edited them together in a pattern to make meanings. Kubelka wished to go back to the individual still frame as the essence of cinema. Kubelka is a kind of perfectionist of the film medium - every part of the film is precisely measured and set in a relation to the film as a whole, and every part of the film communicates with all the other parts [1]. In his film *Adebar* (1957) Kubelka constructed a sound-track consisting of African Pygmy music in four phrases, each exactly 26 frames long. The film clearly demonstrated his interest in the alternation of rest and motion, making an analogy with the way sound and silence are equally important in music [Milicevic, 2000].

In 1957, Peter Kubelka was hired to make a short commercial for *Scwechater* beer. The beer company no doubt hoped they were commissioning a film that would help them sell more beers; Kubelka had other, more radical ideas. He worked with black & white and red frames and pure electronic sounds composed in the order 1 red frame, 1 image frame, 2 red frames, 2 image frames, 4 red frames, 4 image frames and so on. Whenever a red frame appears the two sine-tones (pips), are heard, higher introducing lower [Milicevic, 2000].

> He shot his film with a camera that did not even have a viewer, simply pointing it in the general direction of the action. He then took many months to edit his footage, while the company fumed and demanded a finished product. Finally he submitted a film, 90 seconds long, that featured extremely rapid cutting (cutting at the limits of most viewers' perception) between images washed out almost to the point of abstraction — in black-and-white positive and negative and with red tint — of dimly visible people drinking beer and of the froth of beer seen in a fully abstract pattern. [2]

Kubelka achieved extreme experimental minimalist reduction with the film *Arnuf Reiner* (1960). The film was made entirely without a camera, and consists of black and white frames which correspond to sound, in one-to-one relationship. Black alternates with white for durations as long as 24 seconds, and as short as a single frame [3]. Black represents no-light, translated in the music as a silence, white is simply light (consisting of the whole spectral range), translated into music as a burst of white noise, (consisting of the whole range of audio frequencies). Visual and auditory rhythms are synchronized [Milicevic, 2000].

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[1] http://www.kortfilmfestivalen.no/arkiv/english/articles/99_PeterKub.html ; accessed 30-12-02.
[2] http://www.kortfilmfestivalen.no/arkiv/english/articles/99_PeterKub.html ; accessed 30-12-02.
[3] *ibid*.

In the *Unsere Afrikareise* (1960-1966) Kubelka recorded approximately ten hours of sound, including random conversations, animal sounds, industrial and other noises, random samples of radio music, and shot several hours of images in Africa.

> In the six years between shooting and finishing *Unsere Afrikareise* Kubelka focused his work on the speed of perception on the instant-to-instant relationship between sound and image [Milicevic, 2000].

Soundtrack in this film is again a kind of Musique Concrète*,* continuing the same path Vertov had begun in his *Enthusiasm.*

In the U.K. Anthony Scott used indeterminancy, in John Cage's sense (see p. 312), in the making his film *The Longest Meaningless Movie*. It consists of adverts, complete and incomplete sequences from feature films, out-takes, sound-only film, and home-movie material. Sometimes whole lengths of film appear upside down and running backwards, and in the 35 mm version it flips left to right to reveal the soundtrack Milicevic [2000].

"Structural" film-maker, and unclassifiably versatile artist Michael Snow has been experimenting since the 1960s in almost every field of artistic endeavor, including photography, cinema, music, sculpture, painting and holography. In his *Wavelength,* Snow conceptually links a long camera zoom with a pure sine wave. The whole film consists of a single un-edited shot, which takes 45 minutes to reach its maximum zoom extension, i.e. to move from its widest field to the narrowest, the finally framing a black and white photograph of a wave, pinned to the far wall of his loft. The soundtrack parallels this movement with an ascending sine wave, starting at 50 Hz and ending around 12 kHz. Despite its apparent simplicity, *Wavelength* won first prize at the prestigious experimental film festival at Knokke-le-Zoute, and had a tremendous impact on the evolution of film as a medium, and became an instantaneous avant-garde classic, setting a new standard for originality and rigour [1].

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[1] http://www.frif.com/new2002/snow.html ; accessed 30-12-02.

Above: Still from *Wavelength* showing optical soundtrack.

Animation

The evolution of 2D animation

Multi-media meta-artists of the future will inevitably draw on the rich cinema and video art traditions discussed so far in this dissertation. Motion graphics -- 2D and 3D animation -- will make an equally important contribution to multimedia as meta-art, as the creative artist attempts to address the user's mind via the senses and weave meaningful visual experience into engaging cultural atefacts. There has been a long and interesting evolution of best practice in the field of pre-produced animation techniques which will be outlined in this section. In 3D animation software environments, among the most sophisticated and complex of human/computer interface designs, the creative artist can literally "play god" constructing and populating an imaginary world down to the smallest detail. Effective story-telling is fundamental to this artform as to other forms of cinematic engagement, though the histories of abstraction in art are not without relevance.

Winsor McCay, is considered to be the father of the animated cartoon. From 1911-21 McCay nursed animation from the unrealized potential of a simple camera "trick", to full-blown character animation, that would take two decades to be surpassed in terms of technical animation skills[1]. McCay introduced the world to the true potential of the animated cartoon in his landmark animated film *Gertie the Dinosaur* (1914) [Crandol, 1999].

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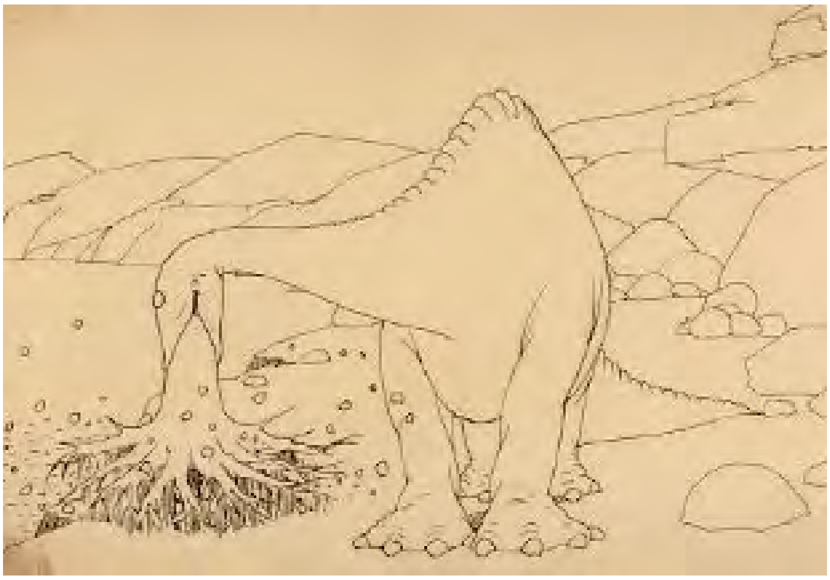
[1] http://www.toonopedia.com/gertie.htm ; accessed 7-12-02.



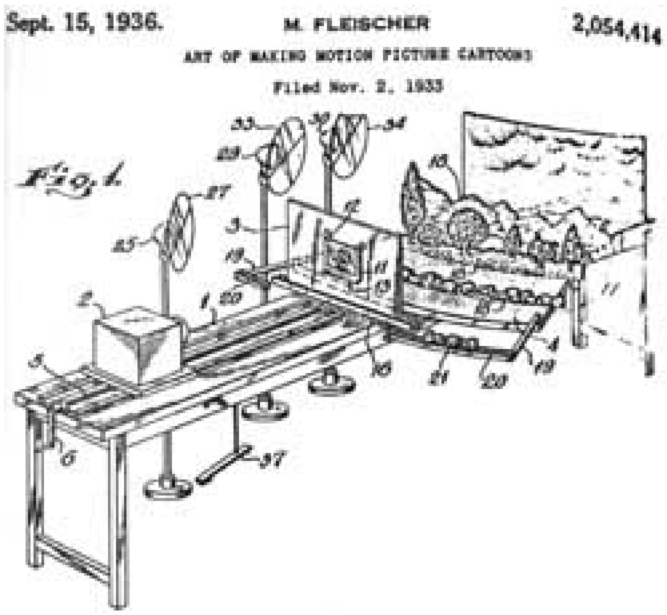
McCay animated his films almost single-handed; from inception to execution each cartoon was his and his alone. He took the time to make his films unique artistic visions, sometimes spending more than a year to make a single five-minute cartoon. [Crandol, 1999].

But commercial pressures within the rapidly evolving cinema industry created a need for a faster, more efficient way to produce works of a greater length and so the modern animation studio came into being. At the same time that sound and colour film technologies were being popularized, studios also found ways to streamline the animation process by using storyboards (small drawings of frames that represented different key moments in the cartoon action) to plan the cartoon and modularize the steps of the process [Butler, undated [1]]. The art of hand-drawn film animation was no longer the work of one man, it was an efficient, industrial assembly-line process in the tradition established by Henry Ford. Crandol [1999] argues that the most successful and influential of these early studios was the John Bray Studio, creator of the first successful cartoon series, *Colonel Heeza Liar*, in 1914. The Colonel was based on Teddy Roosevelt and his exploits emphasized exotic locales and predicaments (Johnson, 2000). The Bray studio's most important contribution to the medium was the introduction of cels. The cel is a sheet of transparent celluloid that is placed on top of a background drawing. By using cels, the animator need only re-draw the portions of the image that move, thus saving considerable time and expense. The process of inking the animator's drawings onto clear pieces of celluloid and then photographing them in succession on a single painted background was invented by Bray employee Earl Hurd in late 1914 [Butler, undated [2]].

[1] Jeremy G. Butler (undated) *"Cartoons",* at http://www.museum.tv/archives/etv/C/htmlC/cartoons/cartoons.htm ; accessed 7-12-02.
[2] Jeremy G. Butler (undated) *"Cartoons",* at http://www.museum.tv/archives/etv/C/htmlC/cartoons/cartoons.htm ; accessed 7-12-02.

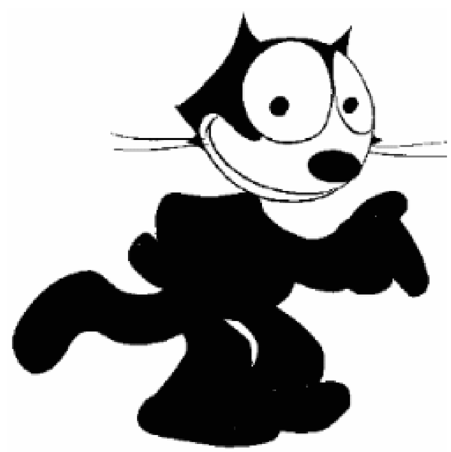
Rotoscoping and Fleisher’s set-back machine

Another technological breakthrough which pushed the evolution of animation forward was rotoscoping, the process of filming human action and using it as a guide to creating drawn/painted film animation which exhibits naturalistic fluidity of movement, a process explored successfully around 1915 by Max Fleischer.



Another Fleisher innovation was the "setback machine" which allowed for a realistically moving background in the two dimensional cartoon. The machine consisted of a turntable twelve feet in diameter on which was built miniature sets of various background scenes. The cels were hung vertically as far as six feet away from the set and the camera shot horizontally past the cel to the set. It revolved a fraction of an inch with each change of a cel [1].

Character animation

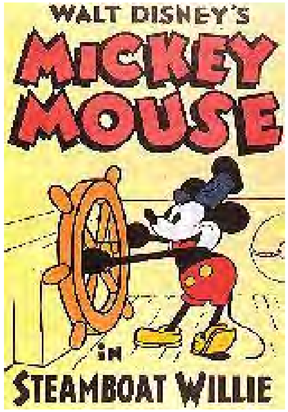


The next milestone in the development of animation as an artform occurred in 1916, when Otto Messmer, employee of the Pat Sullivan Studio, created *Felix the Cat*, a cartoon character exhibiting a strong and appealing personality [Johnson, 2000]. As a studio character, audiences could get to know him over time, while affording Messmer and his co-workers the opportunity to explore the possibilities of ongoing character development in animation.

Disney and synchronized sound

[1] http://www.popeye.dk/ ; accessed 7-12-02.

By far the most influential studio (both artistically and commercially) in the history of animation is the Walt Disney Studio. Disney's epoch-making entry into the industry was in 1928 with Mickey Mouse in the seven minute *Steamboat Willie* (the first significant cartoon to feature synchronized sound), and the corporate version of the company has continued to dominate the field to the present era [Butler, undated[1]].



> "With the debut of Mickey Mouse, something extraordinary happened. Suddenly an ungainly rodent with a falsetto voice squeaked his way not only into his own newly-created niche but soon had an entire world waiting to cheer on each new exploit. This unprecedented phenomenon, whose impact and popularity continues to this day, defies any simple solution" [Johnson, 2000].

Johnson cites both the advent of sound as well as the Great Depression as possible explanations for the phenomenal popularity of the cheerful and essentially good Mickey Mouse character. For the first time, an audience could hear as well as see a cartoon character. However the simplicity of emotions made the characterizations universally recognized and accepted.

Even foreign language audiences such as the Japanese became entranced, indicating the clarity of characterization and the strong physical humour [Johnson, 2000].

Walt Disney encouraged his studio artists to develop a realistic, naturalist style of animation in the early 1930s. Later he was the moving force behind such groundbreaking films as *Snow White and the Seven Dwarfs* (1937), the first full-length animated feature, and *Pinocchio* (1940). [Crandol, 1999].

Left: Production cel from Snow White

[1] Jeremy G. Butler [undated] *"Cartoons",* at http://www.museum.tv/archives/etv/C/htmlC/cartoons/cartoons.htm ; accessed 7-12-02 .

The multiplane camera

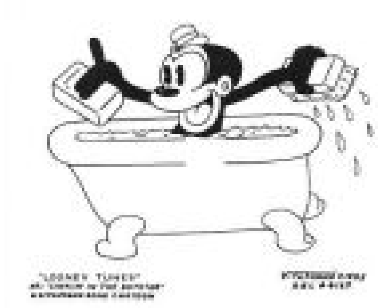
Both of these films reached new levels of technical brilliance, partly due to the use of Disney's patented invention, the multiplane camera, which dramatically enhanced the illusion of three-dimensionality and depth [1].

The system was to put a camera at the top of a multi-level structure containing glass panels. On each glass plaque was painted one element of the background or of the foreground. The technique involved taking one picture at a time, slowly moving each of the several glass plaques. There are only two multiplane cameras in the world, at Disney's Californian and Paris studios.[2]

Disney was also deeply involved in story development and after five months studio work on the *Pinocchio*, it was begun anew, with the decision to give Pinocchio a

[1] http://www.uspto.gov/web/offices/com/speeches/02-42.htm ; accessed 7-12-02 .
[2] Sebastien Barth (undated); *The Art of Disney Animation at Walt Disney Studios.* URL: http://perso.wanadoo.fr/sebastien.barthe/animationwds.htm ; accessed 7-12-02.

childs face (and not a puppet face) and *give Jiminy Cricket* the role of narrator of the story [1]

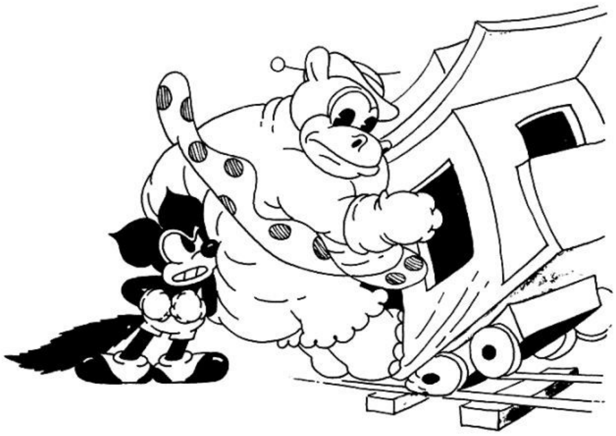


Promotional drawing for *Sinkin' in the Bathtub*

Looney Tunes began in 1930 when Disney animation studio veterans Hugh Harman and Rudy Ising teamed with producer Leon Schlesinger to make cartoons, to be distributed by Warner Bros. The first *Looney Tunes* cartoon released was an eight-minute musical adventure called *Sinkin' in the Bathtub*, starring two new characters named Bosko and Honey [2].

Though the Looney Tunes name is an echo of Disney's *Silly Symphonies*, this team is credited with breaking from the Disney tradition that most other studios had begun to mimic, and filled their animated films with highly exaggerated slapstick comedy, which ultimately proved to be the theatrical genre animation was best suited for (Crandol, 1999). After producing one *Looney Tune* a month for a year, a second series of cartoons called *Merrie Melodie*s was created. The *Merrie Melodies* cartoons were designed to showcase songs from Warner Bros' vast music library.

During this period the major film studios also owned the preeminent theatres and the standard way of exhibiting films at the time was two feature-length films separated by a newsreel and a cartoon. Because of this arrangement, animation



[1] http://www.ricochet-jeunes.org/eng/biblio/films/pinocIIcinema.html ; accessed 7-12-02.
[2] http://www.toonzone.net/early-years/ ; accessed 7-12-02.

studios and departments had a steady, constant demand for their product. The late 1930s to 1950s were a golden era for the cartoon and it is from this era that most theatrical cartoons on television are drawn [Butler, undated].

Cartoons made exclusively for television were introduced with Jay Ward's *Crusader Rabbit* in 1949, which was syndicated to, rather than produced by, a network. Production of TV animation didn't really become a major factor in programming until the1960s, by which time most of the cinematic cartoon studios had gone out of business [Crandol, 1999]. The networks' first cartoon series was *The Ruff and Reddy Show*, which was developed by a team that became historically the most successful producers of television cartoons, Bill Hanna and Joe Barbera, former MGM directors and creators of *Tom and Jerry*. The *Ruff and Ready Show* was the first made-for-TV cartoon show to be broadcast nationally on Saturday mornings and created impetus for the children's TV market. Hanna-Barbera was also responsible for bringing cartoons to the prime-time network schedule--though the prime-time success of *The Flintstones* (1960) did not result in an on-going trend, and no other series has enjoyed prime-time success until Matt Groening's *the Simpsons* (1987).

Above: *Ruff and Reddy*

Limited Animation

Hanna – Barbera dominated the TV market almost from its inception and continued to do so through the 1970s. Commercial forces fostered the introduction of "limited animation" introduced by Hanna-Barbera and their imitators, firmly establishing this limited animation style for television cartoons. The pressure to produce for weekly episodes, the budget restraints and hurried deadlines of the television industry prohibited artists from crafting the kind of art their cinematic predecessors achieved [Crandol, 1999].

In limited animation, the amount of movement within the frame was substantially reduced. Rather than have a character move the entire head in a shot, a cartoon might have just the eyes blinking. Secondly, in limited animation, figure movements are often repeated. A character waving good-bye, for instance, might contain only two distinct movements, which are then repeated without change. Full traditional cinematic animation, in contrast, included many unique movements. Thirdly, limited animation uses fewer individual frames to represent a movement. Full film animation uses 24 discrete frames to represent movement. Limited animation, in contrast, might cut that number in half. The result is a slightly jerkier movement (Butler, undated[1]). Other cost-cutting techniques used to mass-produce cartoons on a low budget included [2]:

- cels and sequences of cells were reused over and over again -- animators only had to draw a character walking one time.
- only portions of a character, such as the mouth or an arm, would be animated on top of a static cel.
- the visual elements were made subsidiary to audio elements, so that verbal humour and voice talent became more important factors for success.

However, a second more cultural force was also at work that gave an artistic direction to the changed approaches to animation being introduced in the name of commercial viability. Up until the 1950s animation cartoonists, especially those working for Disney, strove for a naturalistic aesthetic, attempting to make their drawings look as much like real world objects as was possible in this medium. The pinnacle of this style was Disney's *Snow White*, in which Disney artists roto-scoped (traced on film) the movements of dancer Marge Champion and transformed her into Snow White. Butler [undated[3]] describes the work of United Productions of America, who pioneered a new cartoon aesthetic:

> …post-World War II art movements such as abstract expressionism rejected this naturalistic approach and these avant-garde principles eventually filtered down to the popular cartoon. In particular, United Productions of America (UPA), a studio which contained renegade animators who had left Disney during the 1941 strike, nurtured an aesthetic that emphasized abstract line, shape, and pattern over naturalistic figures. UPA's initial success came in 1949 with the *Mr. Magoo* series, but its later, Academy Award-winning *Gerald McBoing Boing* (1951) is what truly established this new style.
>
> The UPA style was characterized by flattened perspective, abstract backgrounds, strong primary colors, and "limited" animation. Instead of using perspective to create the illusion of depth in a drawing, UPA's cartoon objects looked flat, like the blobs of color that they were. Instead of filling in backgrounds with lifelike detail as in, say, a forest scene in *Bambi*, UPA presented backgrounds that were broad fields of color, with small squiggles to suggest clouds and trees. Instead of varying the shades and hues of colors to imply the colors of the natural world, UPA's cartoons contained bold, bright, saturated colors.

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[1] Jeremy G. Butler [undated] *"Cartoons",* at http://www.museum.tv/archives/etv/C/htmlC/cartoons/cartoons.htm ;accessed 7-12-02.
[2] URL http://www.wikipedia.org/wiki/Limited_animation ; accessed 8-12-02.
[3] Jeremy G. Butler [undated] *"Cartoons",* at http://www.museum.tv/archives/etv/C/htmlC/cartoons/cartoons.htm ; accessed 7-12-02.

The quality of television animation improved in the second half of the 1980s when the two largest traditional studios, Disney and Warner Bros., entered the market. Disney's *DuckTales* (1986) and Warners' *Tiny Toon Adventures* (1989) were considerably better than anything their competitors were producing [Crandol, 1999].

Above: Mr. Magoo an example of the UPA limited animation cartooning style.

Finally in the 1990s the artists in the TV cartoon industry began to establish ways to effectively work within the commercial (and thus technical) limitations of the field.

1992 saw the debut of Warner Bros. *Batman: The Animated Series*, which, according to Crandol [1999] combined deep characterizations and strong stories.

The Simpsons (1989) has been the longest-running cartoon on American network television. Critically acclaimed, culturally cynical and economically highly successful, the series has helped to define the satirical edge of prime-time television in the early 1990s. Despite its ‘family sitcom’ format, two of the most striking characteristics *of The Simpsons* are its strong and consistent element of social criticism and its references to other cultural forms. Cartoon technique allows free movement and in *The Simpsons*, the manipulation of visual qualities, often mimicking comic strip perspectives and cinematic manipulation of space, creates a strong sense of time, place, and movement. On occasion *The Simpsons* has reproduced the actual camera movements of the films it models or lampoons. At other times the cartoonist's freedom and ability to visualize internal psychological states such as memory and dream have produced some of the program's most hilarious moments [1]. Another challenging technique is to use celebrity artist’s voice-overs and vocal performances in send-ups of themselves.

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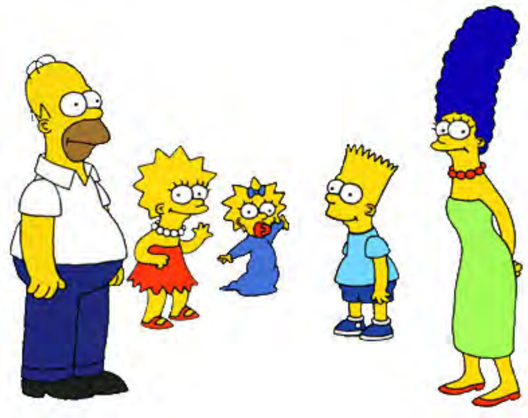
[1] http://www.museum.tv/archives/etv/S/htmlS/simpsonsthe/simpsonsthe.htm ; accessed 7-12-02.

Right: Characters from *The Simpsons* - only the second TV cartoon series to be successfully scheduled during prime-time in the USA and Australia.



Compositing live action with animation

Returning to animation in the cinema, the influence of television on audience's attendance at cinemas meant that, by the mid-50s, while there was the occasional non-Disney animated feature, no other studio was producing them on a regular basis. By the 1980s no studio was producing "shorts" on a full-time basis. However, a new generation of Disney artists revitalized cinematic animation with films such as *Who Framed Roger Rabbit?* (1988) and *The Little Mermaid* (1989), well-crafted cartoons which were celebrations of animation's glory days [Crandol, 1999]. *The Little Mermaid* was the last Disney animated feature film to exclusively use traditional ink and paint production techniques.

Who Framed Roger Rabbit? was a milestone in animation history, one of the top-grossing films of its year, receiving 4 Academy Awards, one of which was a Special Achievement Award for Animation Direction for its blending of ink-and-paint cartoon characters and flesh-and-blood live actors. Earlier efforts to combine humans and ink-and-paint cartoon characters side-by-side in a film e.g. Disney's *Song of the South*, 1946, and *Mary Poppins*,1964, (see right, promotional still) are considered primitive next to this film[1].

[1] http://www.filmsite.org/whof.html ; accessed 8-12-02.

Left: Promotional still from '*Who Framed Roger Rabbit?'* showing pen and paint cartooning composited with film of live actors[1].

Chroma-key Techniques

Blue-screen or chroma-key techniques are used for film and music video, most commonly to composite live actors over computer-generated or other non-real backgrounds. Increasingly, it is also being used by interactive CD-ROM and video game production companies [Gosh, 2002]. The process of removing the background is called Chroma Keying, or bluescreen extraction.

Creating a blue screen composite image starts with a subject that has been photographed in front of an evenly lit, bright, pure blue background. The compositing process, whether photographic or electronic, replaces all the blue in the picture with another image, known as *the background plate* [2]. This process is commonly used in TV news broadcasting.

The way the process works is that while the foreground newscaster is shot against a bluescreen background with a video camera, a technician running a piece of hardware known as *a keyer* replaces the blue background with an image, such as a weather map, in real time.

The technician adjusts the tolerance level of the keying process to compensate for uneven lighting conditions or bluescreen spill (such as unwanted splashes of reflected blue light on the weather person's face). Of course, the newscaster can't wear a blue tie on a bluescreen set because the viewer would see a hole in the middle of the newscaster, revealing the background map.

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1 www.pyramid.net/natesmovies/ inrogerrabbit.htm ; accessed 8-12-02.
2 http://www.seanet.com/%7ebradford/bluscrn.html ; accessed 14-01-03.

Right: a Chroma-key scene from the movie *Forest Gump*. To enhance the illusion that the Tom Hanks character and interview footage of the late John Lennon were present in the same scene, a special method was used that allows the background to move with the cameras automated movements, adding more realism to the blue-screened scene.

An example of chroma-keying can be found on accompanying CD-ROM *the Machine*, in the *Pay TV* section.

3D Animation

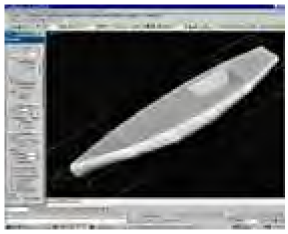
The oldest form of performance animation is probably found in the art of puppetry. 3-D animation is virtual puppetry. 3D animation is a significant new digital art-form and also a subsidiary art of multimedia. An overview of the developments in entertainment resulting from the emergence of 3D animation and also a brief review of major commercial software applications facilitating this art on personal computer workstations is presented below. As a basis for this discussion a simple, brief description of the major issues involved in 3D modeling, animation and rendering is presented first. Many important and complex algorithmic process used in modeling, texturing/shading, lighting, animation, and rendering cannot be introduced because of space.

Descartes' Cartesian System allows us to identify a point on a flat surface, using two coordinates, (x, y) where x is the coordinate on the horizontal axis and y on the vertical one. To describe a 3rd dimension, we add an axis z, and usually it is assumed to represent depth. So to represent a point in 3D space requires three numbers: (x, y, z). With the advance of computer hardware and software it has become possible to simulate or describe complex imaginary worlds, in terms of this virtual 3-dimensional geometric system. 3D Modeling is the process of defining the shape and other characteristics of objects by reference to specific co-ordinates or

points in this 3 dimensional matrix, producing a “wireframe” model by linking these points.

Curves are usually supported in the form of *splines,* which are essentially curved lines with a mathematical relationship to two or more points.

Solid modeling builds complex objects from simple 3D “primitives” such as cubes, cylinders and cones. Various wireframe elements can be joined at their vertices to form a more complex object. In many 3D applications these vertices may be moved individually in order to deform or reshape the object. Alternative approaches include *skinning* – a series of linked geometries serve as a skeleton over which the computer drapes a theoretical skin. *Extrusion* is adding a third dimension to a two dimensional shape, such as when a letter outline is extruded into the z dimension to create the illusion of 3D text. An extruded circle becomes a cylinder. *Lofting,* in its simplest form, is an extrusion for some distance, relative to some shape. Complex lofting involves multiple shapes, directions or paths (see right).



Polygonal-surface modeling employs 2-D polygons (usually a triangle) that are linked together and grouped to construct objects. This technique is less useful for curved surfaces. [Burger, 1993, 203-209].

The introduction of a key-framed timeline introduces the possibility of varying parameters of 3 dimensional geometry over time. Geometric transformations can be used to modify the shape, proportions, size, position, and orientation of three-dimensional objects over time using interpolation. Interpolation describes in-between states, based on information defined at start and end points (or key frames), i.e. the computer calculates the transition states between two defined sets of parameters, relative to a given frame rate, and generates images accordingly, producing the illusion of movement or change over time.

Complex surface characteristics for 3D-modeled objects can be defined in the form of surface shading or “textures”, beginning with simple colours defined by RGB values (relative proportions of red, green and blue light). Other surface properties which can be specified include specular reflection, diffusion coefficient, opacity, index of refraction (all of which interact with simulated variable ambient light and external point light sources) and artificial “glows”. *Texture maps* are 2-D rectangular images that can be “applied” to polygonal or curved surfaces. The 3D application maps the pixel data from the image to the defined 3D geometry. 3-D wireframe models (consisting of points mapped to 3-D co-ordinates) can in this way be surfaced with 2-dimensional textural imagery. The illusion is completed by computing the visual changes caused by introducing simulated light sources and the resulting shadows.

> Shadows provide a visual cue to the spatial relationships among objects in a given scene. Simulating hard shadows is possible using clever approximations. More robust algorithms that simulate shadows cast by moving complex objects onto multiple planar surfaces have also been developed. Soft shadows caused by extended light sources, however, require greater computation; those areas with an umbra (fully shadowed regions) and penumbra (partially shadowed regions). At true interactive rates, simulating soft shadows in a dynamic and complex scene require high-end, parallel hardware.[1]

A virtual camera lens defines how the objects in a three-dimensional scene are projected onto the projection plane. A virtual camera can be placed interactively by clicking on its on-screen icon and dragging it to a new location. The 3-D application computes what the user should see, once all of these variables have been defined.

The three dimensional illusion is finally “rendered”, i.e. pixel data for an image or image sequence is computed from all the above data, as it has been input by the user, and saved to an appropriate file format (often including a huge variety of other possible variables and algorithmic techniques, visual effects simulations, dynamic techniques etc). The introduction of a key-framed timeline and interpolation, allows for the illusion of movement and other changes over time to be rendered out as a series of images that can be played back at a designated frame-rate per second to simulate action, either as a video sequence, or ouput to film. Much of this animation process is synchronized to pre-recorded voice tracks for character animation, and then sound effects and music are added to complete the illusion.

NURBS

NURBS is an abbreviation of Non-Uniform Rational B-Splines, which is one of the 3D industry standard tools for the representation and design of geometry. NURBS enables the users to design free-form shapes in general mathematical terms, using control vertices (CVs) which control the shape of curves (see diagram below). It offers users a simple process for the design of complex shapes that could not be

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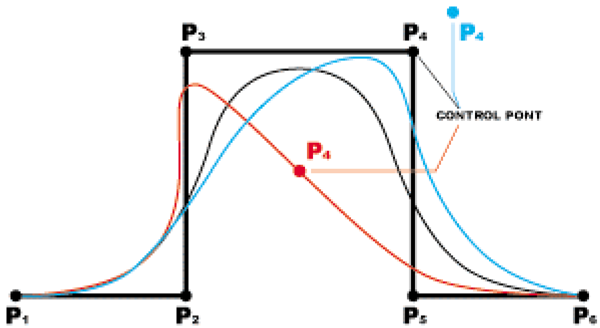
[1] http://xarch.tu-graz.ac.at/autocad/wiki/ShadowRenderingAlgorithms attributed to Hin Jabg (2001). Accessed 8-03-04.

achieved using 3-D mesh polygons. Some reasons for the use of NURBS are, that they:

- offer one common mathematical form for both, standard analytical shapes (e.g. cones) and free form shapes;
- provide the flexibility to design a large variety of shapes;
- can be evaluated reasonably fast by numerically stable and accurate algorithms;
- have powerful texturing capabilities

The relatively complex mathematics of NURBS are handled by the 3D software application, but NURBS are defined by control points, which have vector values. They are comprised of control point, weight and variable values. In simple terms, the curve is transformed by weight through the control point.

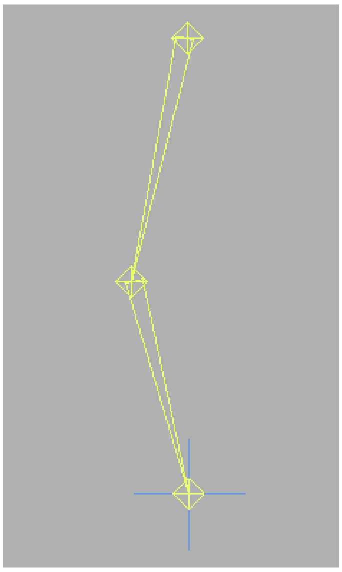


NURBS offers high flexibility, as it not only has control points, but also other variables. However, NURBS has a drawback because of this flexibility. NURBS requires relatively high storage space, even if users want to do just simple and conventional objects such as a circle [1].

Inverse Kinematics

The introduction of Inverse Kinematics (IK) algorithms to 3D animation software is an innovation that has increased the efficiency of 3D animation artists.

[1] Altman, Markus (undated) http://www.cs.wpi.edu/~matt/courses/cs563/talks/nurbs.html ; accessed 5-01-03.



The aim of kinematics is to create smoother, more realistic motion/mechanics in the 3D world, while speeding up the work-flow of 3D animation artists. *Forward Kinematics* is the process of calculating the position in space of the end of a linked structure, given the angles of all the joints. *Inverse Kinematics* does the reverse. Given the end-point of the structure, what angles do the joints need to be in to achieve that end-point? It provides a method for rotating a chain of objects (typically bones or joints) by manipulating a control object at the end of the chain. This control object is typically called a goal (or ``handle'' in some programs), because it's as if the object is the "goal" that the chain is reaching for [1]. For visually skilled animators, the inverse kinematics solution is of great importance, in that it is far simpler to express *spatial appearance* rather than the mathematics of joint angles. Applications of inverse kinematic algorithms include the facilitation of interactive manipulation during development, designing customized animation control, and the simulation of collision avoidance[1].

Inverse kinematics animation techniques require that the three-dimensional models to be animated are built as hierarchical structures (parent/child relationship to 3D elements, e.g. hand>forearm>shoulder etc.). In a practical sense, if the software knows where the hand object is (i.e. its co-ordinates in 3D space) in relation to, for example, a linked shoulder joint, it simply needs to calculate the angles of the joints of the IK chain required to reach it. This effectively allows an entire arm or leg to be posed by moving one object. As the Goal is moved around, the extremities follow it, and the joints above it in the hierarchy readjust accordingly. In order for the joints to readjust appropriately, constraints or ``degrees of freedom'' have to be assigned to the joints, so they don't bend beyond a realistic range, and for instance elbows don't bend backwards [Vaughan, 1998].

Rendering is the process of turning descriptions of 3-dimensional objects into standard 2-dimensional graphic files that can be displayed on a computer monitor, or sequences of images that can be viewed as film or video footage. The mathematics of the behaviour of light has been introduced to further enhance realism, all adding to the illusion of a virtual reality. *Ray tracing* technology simulates what the viewer

[1] http://webreference.com/3d/lesson97/ ; accessed 5-01-03.

would see when the light waves from all light sources in the 3D scene reach the viewers eye, including reflections and refractions - a global illumination-based rendering method. It renders shadows, multiple specular reflections, and texture mapping in a seemingly straight-forward manner - it “traces” rays of light from the eye back through the image plane into the scene. Then the rays are tested against all objects in the scene to determine if they intersect any objects. If the ray misses all objects, then that pixel is shaded the background colour. In other words, we start from the eye or camera and trace the imagined ray of light through a pixel in the image plane into the scene and determine what it would hit. The pixel is then set to the colour values returned by the ray, depending on other variable data. Even at its simplest, ray tracing must take into account objects and surfaces obscured by other objects in the scene, using a *hidden surface algorithm.* [Burger, 1993, 206-207].

3-dimensional scanners are used to gather precise modeling data about the outside and inside shapes of real models to further enhance realism and facilitate an efficient workflow [2].

The motion of live actors can be captured from sensors applied to the body at control points (right) and the gathered data used to specify the motion of animated 3D characters, accurately capturing the subtle and complex variations in human movement patterns, and the details of unique human movement ‘signatures’. The system pictured right works by radio transmission with a wireless range of up to a ½ mile (.8 km) outdoors. This area is larger than 450 football fields [3].

[1] http://xarch.tu-graz.ac.at/autocad/wiki/InverseKinematics attributed to Hin Jang, 2001. Accessed 8-03-04.
[2] http://www.3dscanners.com/ ; accessed 5-03-01.
[3] http://www.metamotion.com/gypsy/gypsy-motion-capture-system.htm ; accessed 5-01-02.

Real-time trackers for facial performance animation generally consist of a very lightweight video camera that is trained on the performer's face via straps on top and back of the head, or a lightweight helmet (right). The video camera watches the 2 dimensional motion of a set of markers or dots that are attached to the face. Various configurations and marker sets are possible, but the following basic configuration is generally the starting point: Sides and tops of lips, cheeks, eyebrows, and perhaps top and bottom of eyelids. The recorded motion can be used one of two ways: Swapping of previously created morph targets, to represent the positions of markers as accurately as possible, or to move groups of controllers that move

Above: Facial Tracker

Above: 14-sensor data glove

clumps of vertices in the 3D model mesh.[1] There are a variety of gloves which can be used for capturing hand motions (see image above right). Common uses range from capture hand motions to be used on a virtual character to puppeteering of a character's facial expressions, or humanoid or non-humanoid appendages[2].

Toy Story (1995, Pixar) was the first feature length animated film to be entirely created with three-dimensional computer animation techniques.

Left: Promotional image from Pixar's *Toy Story.*

The 1993 film *Jurassic Park* (see image top of next page) is a successful example of inverse kinematics and compositing of live action with computer-generated images.

[1] http://www.metamotion.com/hardware/face-trackers.htm ; accessed 5-01-03.
[2] http://www.metamotion.com/hardware/motion-capture-hardware-gloves-Datagloves.htm ; accessed 5-01-03 .

The 1997 film *Titanic* included sequences with dozens of virtual actors composited next to human actors, and 3D animation played a major part in the camera fly-overs of the ship, in which all of the “people” walking on deck were entirely animated [1].

Right: Screenshot of computer simulation of the breaking in two of the Titanic.

Animated series like *Beast Wars*, *Reboot* [2]and *Rednecks* redefined the use of computer animation in TV programming during the late 1990s, with a garish, cartoon-like style, influenced by print comic super-heroes and Japanese *anime*, that celebrated the freedom of 3D rather than striving for realism.

Left : Characters from *Beast Wars.*

[1] http://www.digitalanimators.com/2001/09_sep/editorials/virtualactors.htm ; accessed 12-12-02.
[2] 'Reboot' was created by Gavin Blair and Ian Pearson of Mainframe Entertainment, who originally teamed up to make the computer-generated music video 'Money for Nothing' for pop group Dire Straits.

Above: Characters from *Reboot.* A computer game of the same name has been remediated from the series.

The sci-fi feature film *Final Fantasy: The Spirits Within* (Square, 2001) [1], with a futuristic ecological theme, set new standards for realism in a totally computer-generated movie, depending heavily on motion capture for fluid and naturalistic simulated human motion. It was a case of remediation from a successful computer game (usually the flow is reversed – i.e. famous film is remediated into a game. See p. 399 for a discussion of remediation in multimedia games). It was directed by Hironobu Sakaguchi, the director of the original *Final Fantasy* series of games [2].

[1] www.finalfantasy.com
[2] http://www.arstechnica.com/wankerdesk/01q3/ff-interview/ff-interview-1.html ; accessed 5-01-03.

Production Systems Supervisor for the film,Troy Brooks, describes the production process:

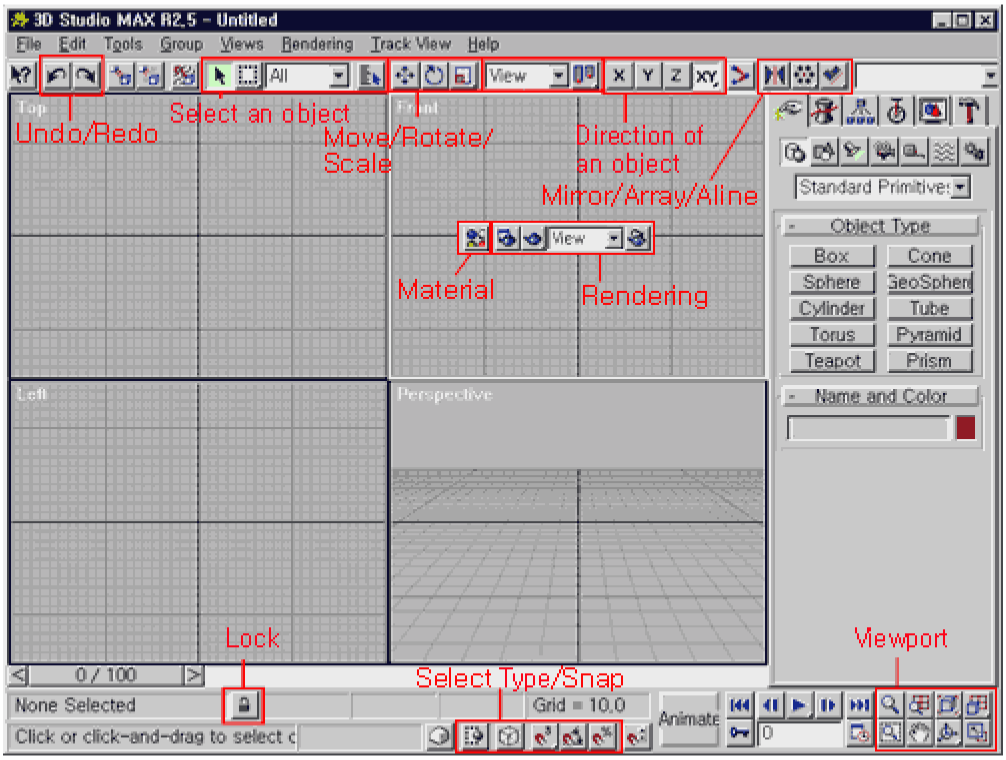
> The production process is very iterative. From the script/treatment, we get conceptual drawings and sketches, and as the script matures, that becomes storyboards - just like a "regular" movie. The layout section uses low-res sets, to define the camera movement and staging, and that's when things become "3D". That layout info is used by the animation section, who work on lower-res versions of the characters that are easier to manipulate in real time. They do frequent renders to check the camera, the animation and deformation of the characters. That animation is ultimately applied to the highest-res models for the final rendering. Sets and Props (and Character) deliver increasingly refined models, and the Lighting section lights the set, with all the characters, breaking the scene down into layers (foregrounds, backgrounds, reflections, shadows), which are carefully combined by the compositing group. All the way through, we do test renders, checking the camera, the pacing, the animation, the lighting, the texturing, and refining them all as we go along.[1]

Examples of popular 3D animation /modeling / rendering applications

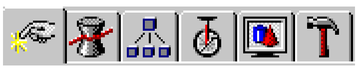
An overview of the major commercial desktop 3-D modeling and animation applications is offered here. Where possible a screenshot of the interface is included, and if they are compared, support is given to the idea that applications are evolving towards a universal interface (see p. 96).

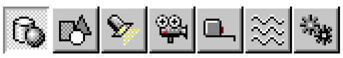
3D Studio MAX / Version: 3 by Autodesk [2]
One of the most popular professional-level 3D modeling and animation solution for PC. It was first introduced as the 3D Studio in a DOS version and later released as the 3D Studio MAX R1 for Windows NT users. *3D Studio MAX* is fully integrated with other Discreet software products, including paint, edit, and effect. The ancillary *Character Studio* package is perhaps the most commonly used character generator in the film, video and game industries. It combines motion capture, editing, and blending technology with traditional keyframe animation, and skin deformation tools. The 3D Studio MAX program is less commonly used in movies, though it was used in *Lost in Space* (1998) for some of the special effects. Example output:

[1] http://www.arstechnica.com/wankerdesk/01q3/ff-interview/ff-interview-1.html ; accessed 5-01-03.
[2] http://www.discreet.com



Above: *3Dstudio Max 2.5* interface (now up to version 6), in 4 screen view mode, with attribute editor panel on right, presently showing options for choosing geometry primitives.





Typical of complex 3D interfaces, icons are used to access and activate different sub-sections of the program.

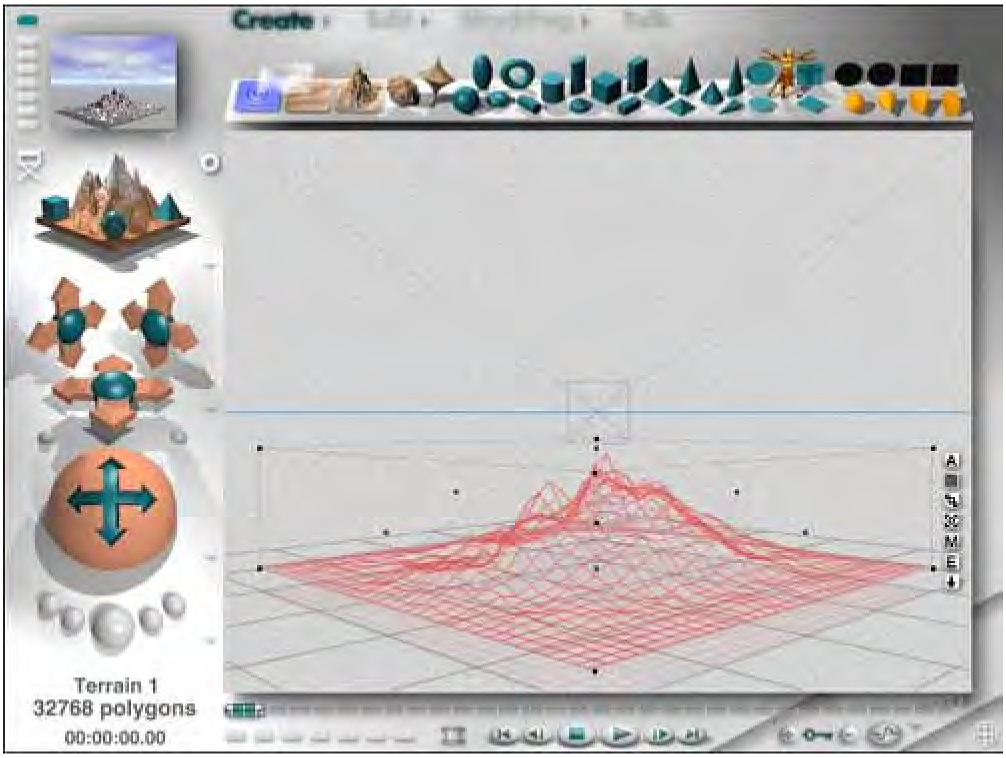
Above: Icons for selecting variable (key-framable) parameters: Geometry, Shapes, Lights, Cameras, Helpers, Space Warps, System. Appropriate attribute editor appears.

The Animation Stand / Version: 4.3 by Linker Systems [1]
One of the earliest professional tools on the Macintosh platform (now cross-platform), *Animation Stand* is a 2D, cel-animation oriented, computer-assisted animation and special effects system. Its functions include scanning, ink & paint, 3D shading, auto-painting, animation direction with exposure sheet control, special and optical effects. On the output side it can lay-out to tape, disk, or film, and HDTV, video, or QuickTime and files and management reporting. Other key features of *Animation Stand* include, multiplane camera control, automatic cel painting, audio editing, production cost reporting. It is available in professional, multimedia and personal editions. Example output:

Bryce 3D / Version: 4.1. Corel Corporation [2]
Bryce is a standalone Macintosh/Windows application that designs and renders natural and supernatural 3-D landscapes and terrains. *Bryce* enables users to create skies with natural cloud formations, humidity and light refraction, as well as landscapes with plateaus, rivers, snowy peaks, desert floors and an array of other special effects -- all with complex surface textures, transparencies and reflective surfaces. The program was originally designed by the MetaCreations Company, well-known for innovative software interface design. The interface is very different to the standardized interface design adopted by most other 3-D software vendors. It includes an animation timeline, but is extremely slow to render and frequently produces undesirable aliasing effects in the resulting animations. However it is

[1] http://www.animationstand.com/
[2] http://www3.corel.com

popular for producing 2D images and backdrops for use in other animation and video compositing applications.



Left: *Bryce* 4.0 interface showing wireframe view. Greater than usual emphasis is placed in visual icons for navigation in this interface design. Interactive camera controls are available on the left. Small preview panel in top left corner. Terrain models, planes, geometry primitves and lights are available from icons on the top bar.

Right: *Bryce* interface showing rendered view

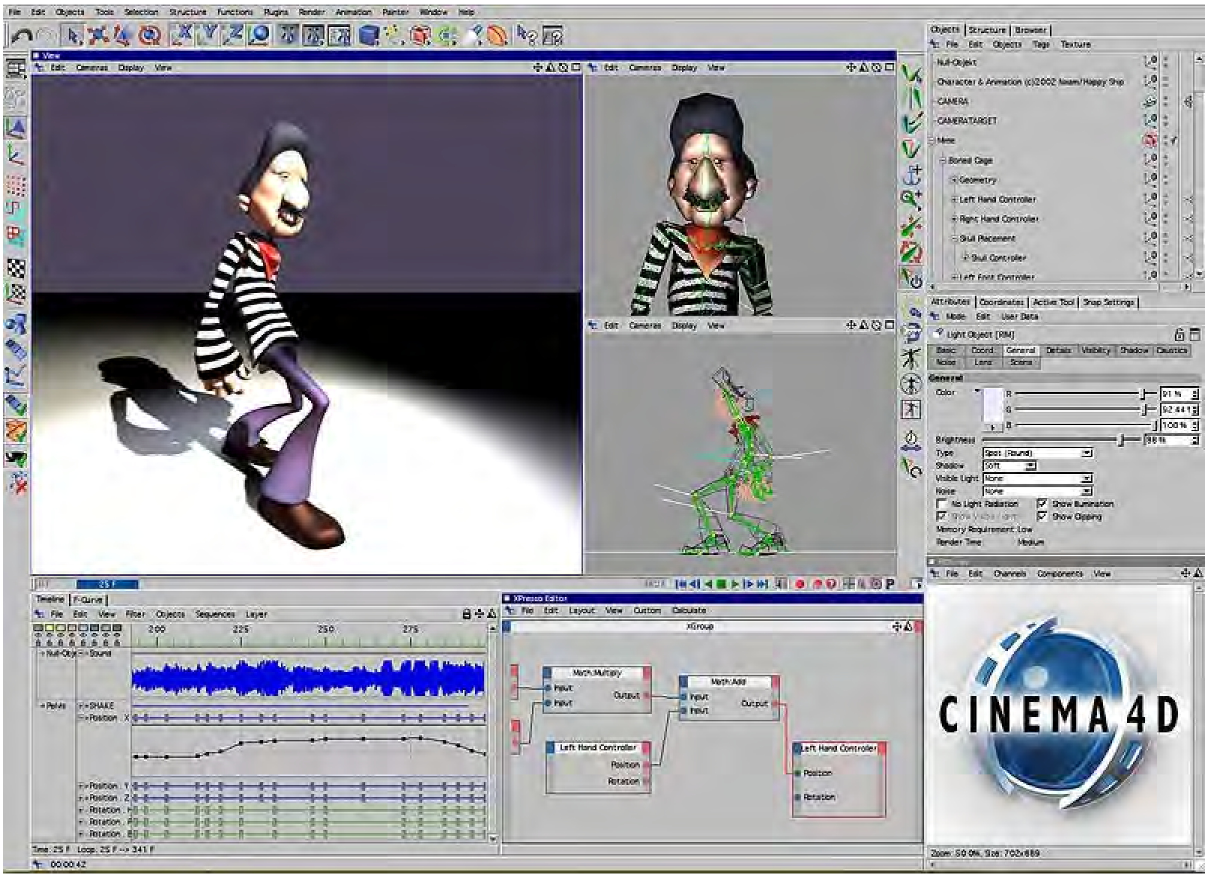
Bryce Skies can be custom created, or selected from a library (left). Skies can change over time. Similar libraries are available for complex shaders and texture mapping, and models can be stored for re-use.

Cinema 4D XL / Version: 6. MAXON Computer GmbH[1]
CINEMA 4D is a relatively low-cost modeling, animation and rendering (raytracing) application, with the power, speed and image quality to compete well with more expensive workstation-based software. CINEMA 4D offers optimized anti-aliasing with over-sampling up to 64 times and several render qualities, from scanline with transparency and reflectivity, up to real ray tracing. CINEMA 4D's adaptive technology is claimed to allow the user to render complex scenes in a matter of seconds or minutes, rather than hours. The manufacturer Maxon claims that CINEMA 4D is claimed to be 6 to 10 times faster than other currently available programs on the Macintosh. Seven additional modules are available with Release 8: Advanced Render, Pyrocluster, MOCCA, Thinking Particles, NET Render, Dynamics and BodyPaint 3D (realtime 3D painting – similar to Painter 3D).

Above: *Cinema 4D XL* output

The example interface below gives a good indication of the challenge facing designers of 3D modeling and animation software in building a copherent and user friendly interface, because of overall complexity of the programs, the huge number of sub-sections and the need to devote a large portion of the screen to visual feedback of the work in progress.

[1] http://www.maxon.net/

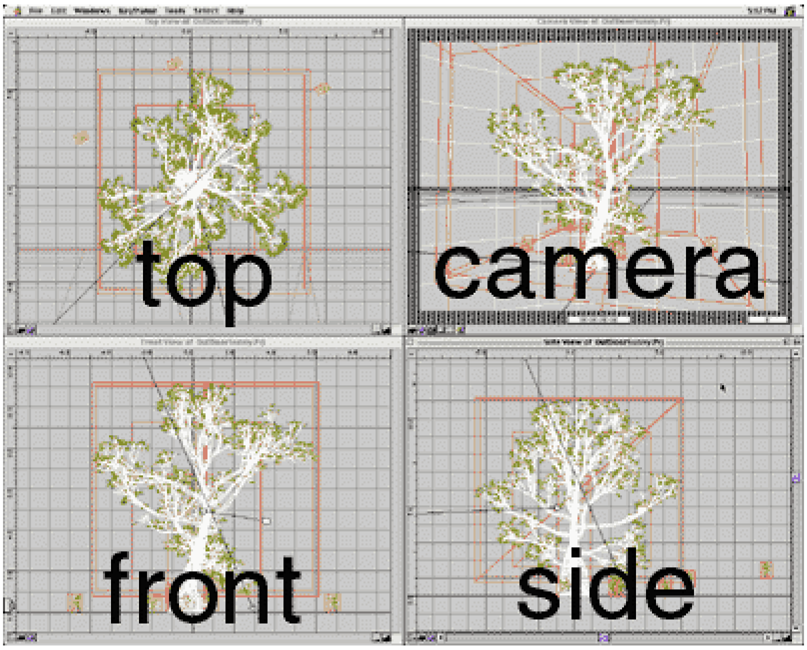


Above: *Cinema 4D XL* interface, featuring audio displacement graph to assist lip-sync

ElectricImage / UNIVERSE Version: 4 . Electric Image, Inc. [1]

For many years, ElectricImage Animation System was the fastest software-only animation system on the Macintosh platform. *Electric Image* was used in *Terminator II, Mission Impossible, Jurassic Park*, and *Star Wars: Episode I*. The program actually consists of three separate applications : Electricimage (modeling), Camera (Animation) and Projector (rendering).

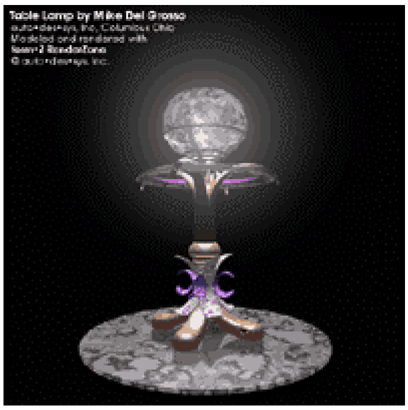
[1] http://www.electricimage.com/main.html



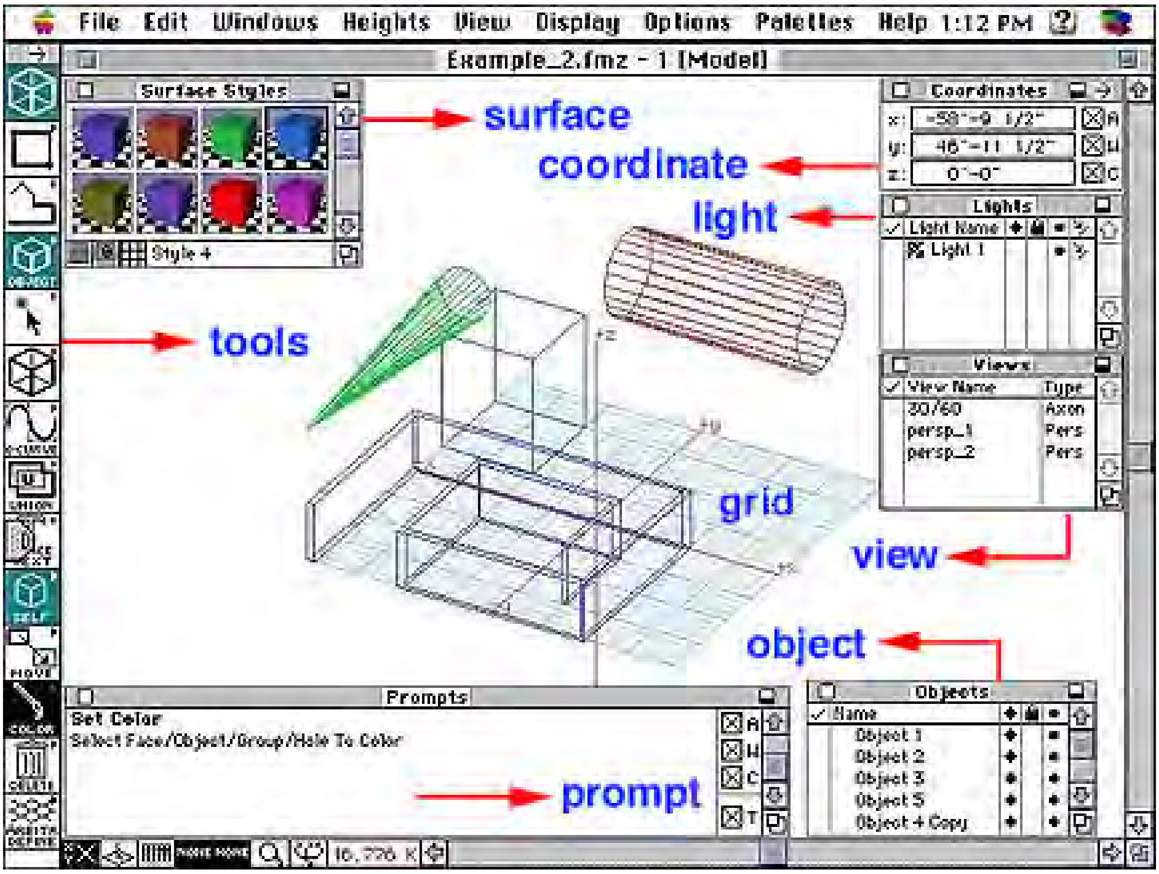
Above: The *ElectricImage* interface in typical 4-view mode – the top right panel showing a perspective view.

Form Z / Version: 3.6.5 by Autodessys Inc. [1]

Form-Z is a general purpose 3D solid and surface modeler with drafting and rendering and featuring a highly interactive interface. It performs boolean operations, dynamic 3D form editing and sculpting, terrain modeling, curved splines and meshes including NURBS, 3D text, object rounding, symbol instances and libraries, and has a number of direct file format translators, including DXF, PICT EPS, RIB, TIFF, STL, FACT, Illustrator, IGES, 3DGF. Although the learning curve is quite steep, it can be used to model just about any imaginable 3D form. *Form-Z* is also widely used for construction/architecture drawing.



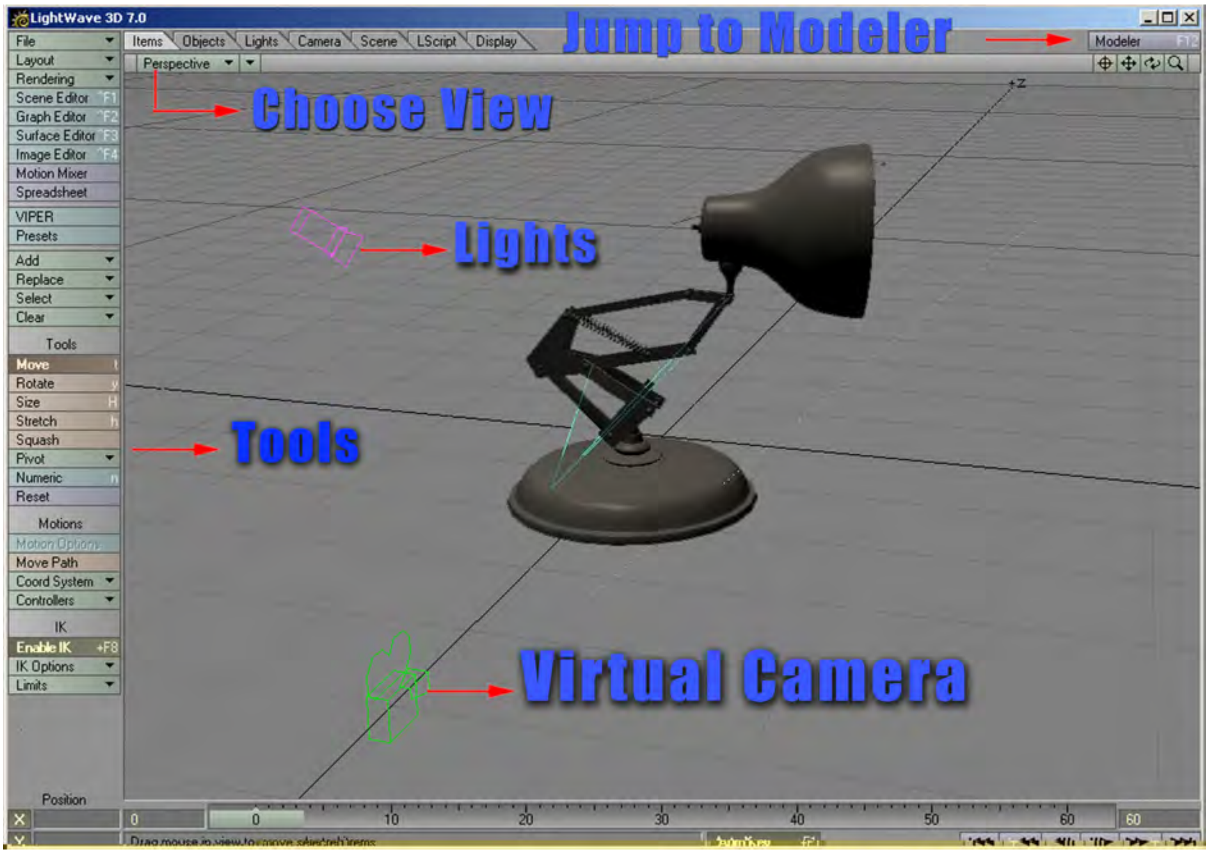
[1] http://www.formz.com/web_site_2000/content_pages/home/home_index.htm



Above: The *FormZ* interface showing shader panel in top left corner, visual icons for transformations on left.

LightWave 3D / Version: 6 by NewTek, Inc.

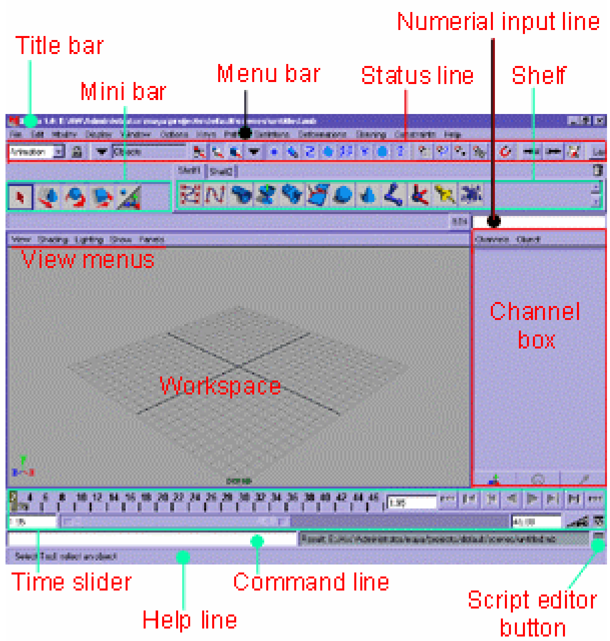
With a long history, *LightWave* was available on the AMIGA in the 1980s to the 1990s. With Version 4.0 *LightWave 3-D* was introduced for PC users. Newtek produced *LightWave* 3D V5.0 for PC's, Workstations, and Mac's. LightWave 3D V5.5 introduced "plug-in software". Now it is one of the most popular cross-platform 3D modeling package currently on the market. It features; ray tracing, depth of field, variable lens settings, a very functional lighting package (via a 3rd party plug-in) and convincing motion blur. 'MetaNURBS handles' facilitate realtime transformations between polygons and splines, while MetaBalls facilitate fast approximation of complex shapes. The specular optics and negative light values are one of the reasons that *LightWave* 3D creates convincingly realistic inanimate objects.



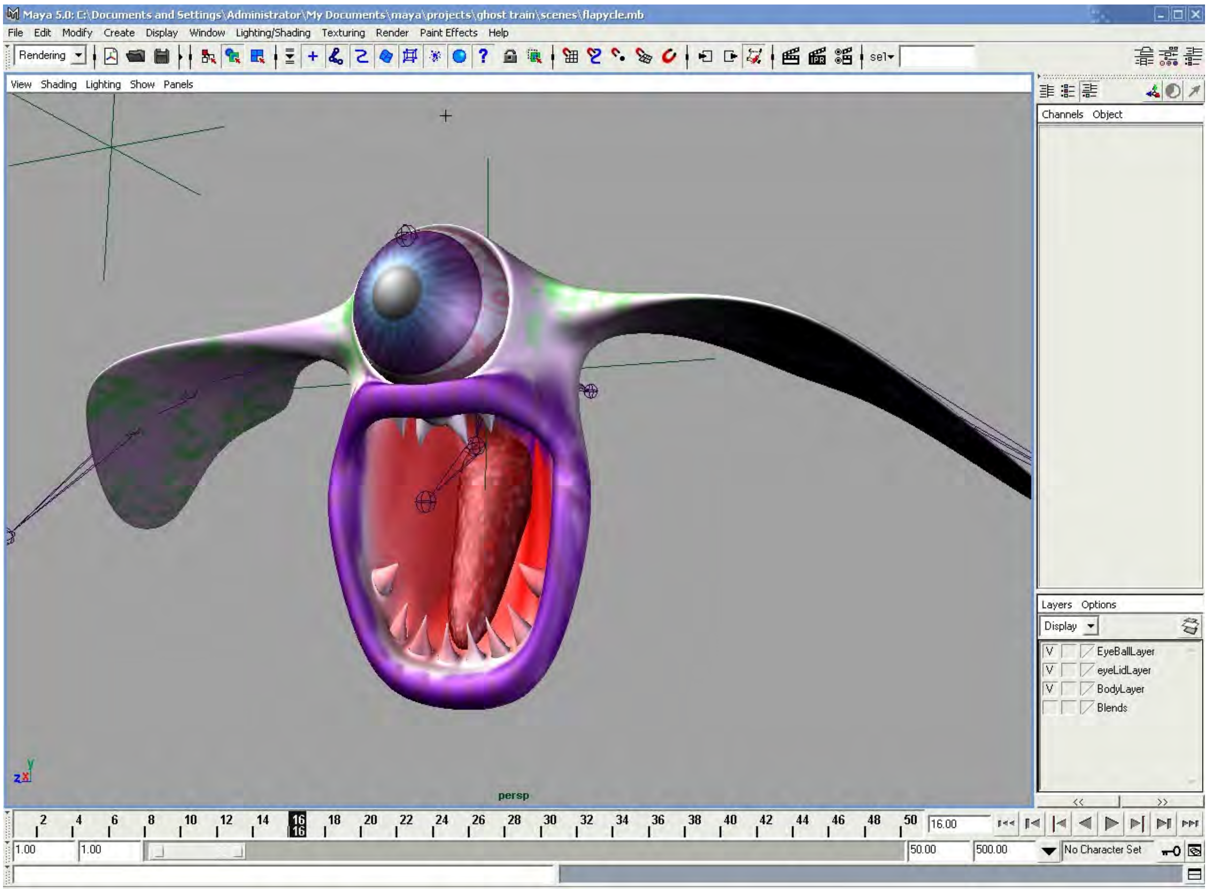
Above: *LightWave's* animation module, with its notoriously unorthodox interface.

Maya Alias/Wavefront

An award-winning 3D animation and visual effects software which has been used extensively for film, broadcast television, video, game development and location based entertainment markets. With an integrated, single environment that is completely customizable, *Maya* claims to be optimized for maximum productivity on the Windows platform and the SGI UNIX environment, and is now also available for Macintosh. *Maya* offers modeling, animation, dynamics, and rendering all in one setting (right, interface showing a single-view perspective viewmode). *Maya* has powerful NURBS modeling and animation. It was developed by the integration between *Alias/Wavefront* software with the *Explore Profession* and *Advanced Visualizer*. The proprietary scripting language MEL can be used to control modeling, animation,



and dynamics effects. *Maya's* modules consist of *Maya* base, *Maya* F/X, *Maya* Power Modeler, *Maya* Artisan, and *Maya* Cloth. *Maya's* strength is character animation.

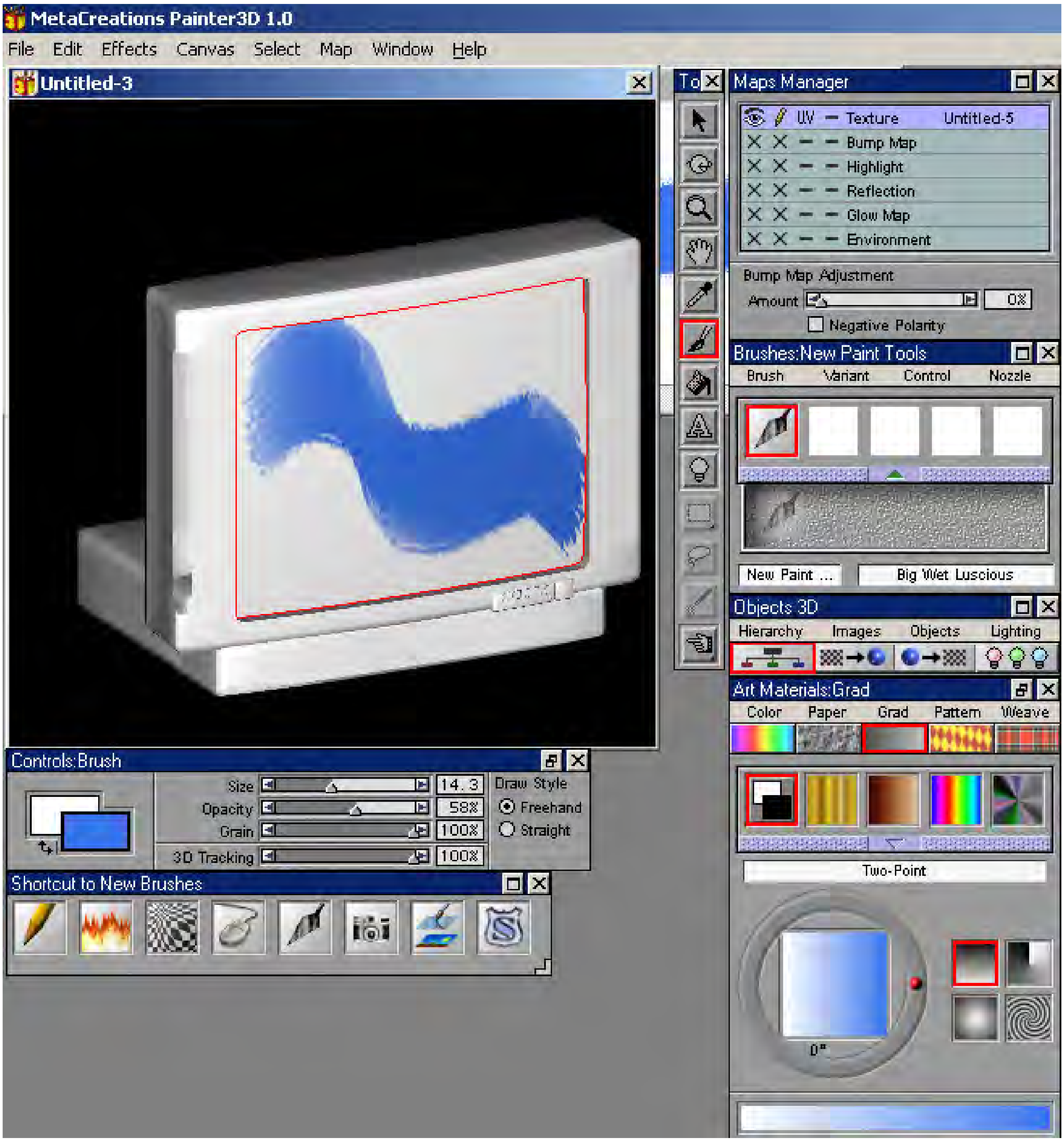


Above: Maya 5.0 showing a 3D character under construction (©2004 Barry Ferrier)

Painter 3D / Version: 1.0 Corel Corporation[1]

Painter 3D is a computer "paint" product that had a revolutionary impact on 2D and 3D design. Exhibiting the same interface as the popular 2D version of *Painter* (natural media simulation tools), it allows 2D designers to use their image-editing skills to create compelling textured, rendered 3D objects for use in 2D designs, with real-time feed-back in a 3D environment. With integrated support for *Painter* floaters and *Photoshop* layers, users can map photos onto imported 3D models in real time, arrange and modify maps and bring the final rendered objects into their 2D compositions. It was formerly known as *Detailer*.

[1] http://newgraphics.corel.com/order/painter3d.html

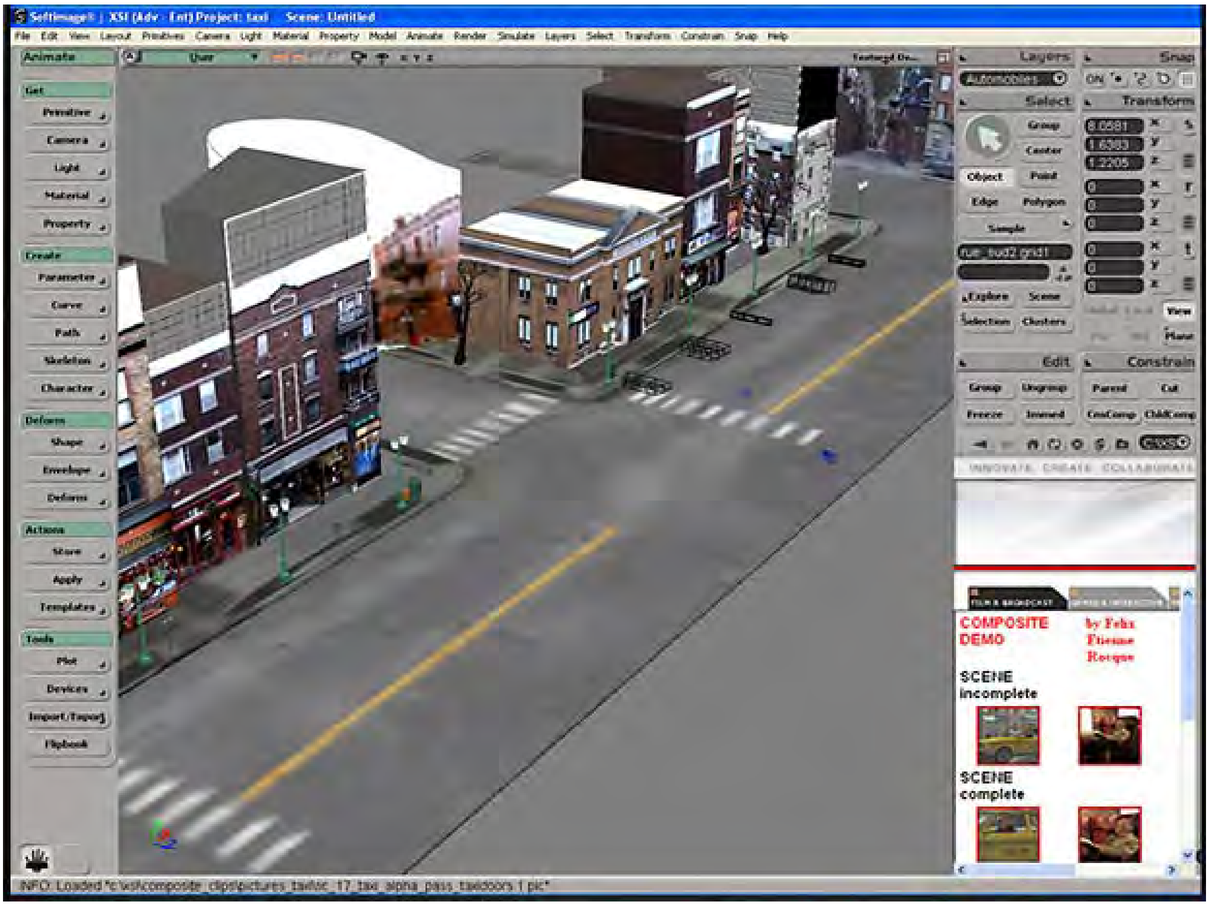


Above: *Painter 3D* interface showing "brush" (mouse) stroke applied to a 3D model of a television.

SoftImage / xsi Softimage

Softimage has been developing 3D animation software for over 14 years and has a history of producing award-winning commercials, games, TV-series (*Reboot, Walking with Dinosaurs*) and contributing to movies such as *Jurassic Park*, *The Matrix*, *Gladiator* and *The Cell*. Originally created by an independent Canadian company, it was acquired by Microsoft which established it strongly in the PC market. It has now become an Avid product, a company famous for digital video editing software, and changes to the program have been strong in the compositing section. The highly sophisticated GUI sports a clean, customizable look.

SoftImage displayed using a Matrox card, which allows the many menus and 3D views to be spread over 3 monitors for ease of use.

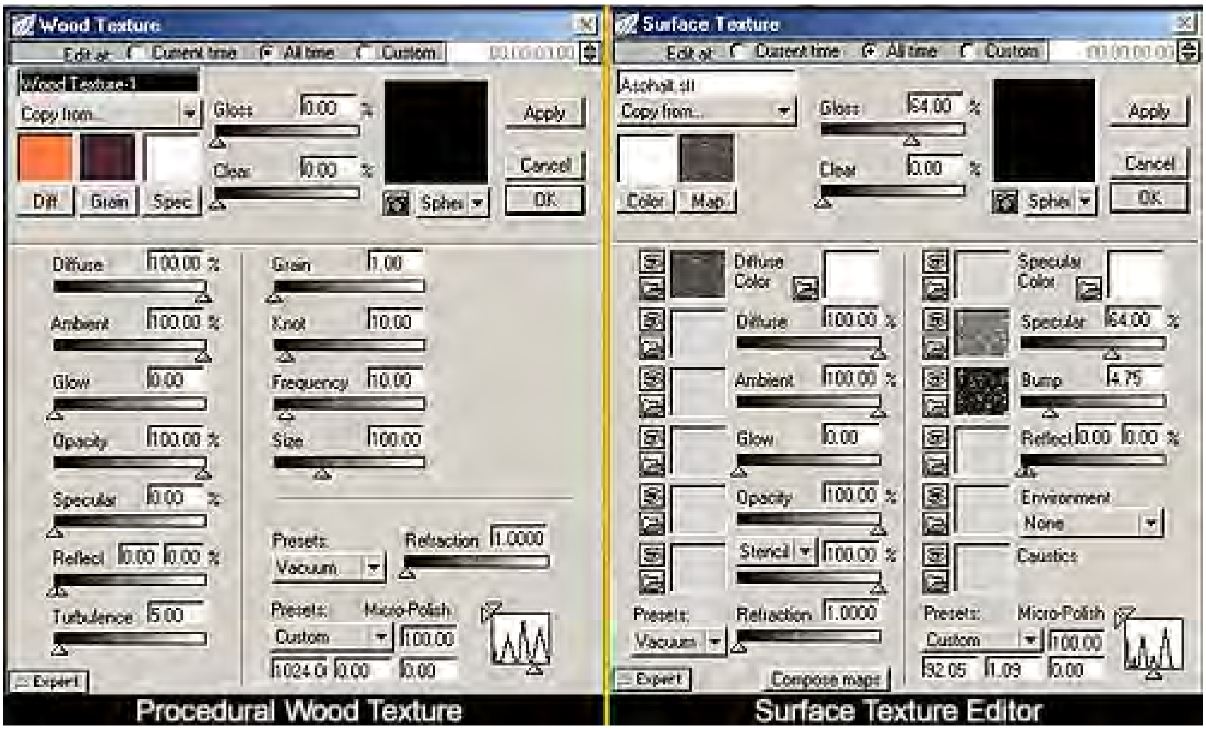


Strata 3D StudioPro / Version: 3.7 by Strata Inc.[1]

Another relatively low-cost (Macintosh only) 3D modeling, animation and rendering

[1] http://shop.strata.com/Product.cfm?Product=83

application, which also includes a powerful compositing module. *StudioPro* is a comprehensive 3-D modeling, animation and special effects package that supports a vast selection of pre-defined procedural textures, solid textures, surface maps, and volumetric effects, as well as multiple texture layering. Below is the texturing interface:



Strata *StudioPro* gives users the capability to use Ambient, Specular Colour, Bump, Reflection, Environment, Stencils, Caustic maps to define rich surface textures. Textures can be animated, either by changing their parameters over time, or by using an animated texture map such as AVI, or QuickTime. *StudioPro* provides controls for many realistic 3-D special effects, including atmospheric and environment effects, lens flares, auras and visible light rays. It can render in all the standard OpenGL Modes: Raytrace, Renderosity, or Scanline. Strata *3D StudioPro 's* animations can be exported to Macromedia *Flash* for use on the web, using Electric Rain's *RAViX2* technology.[1]

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[1] http://www.creative-3d.net/strata2.cfm ; accessed 5-01-03.

TREE / Version: 4.0 by Onyx Computing, Inc.[1]

- a 3D and 2D vegetation modeling program. It allows the user to create biologically photo-realistic trees for both broadcast and motion picture work. It comes with a very extensive library of models of over 350 broadleaf and conifer trees, bushes and palms, grasses and bamboos that are easy to modify. The user can also build customized trees, shrubs and bushes from scratch. There is now a plug-in for *ElectricImage* (see p. 256) that enables the user to model trees into the scene with an unprecedented speed - 20,000 polygons in a mere two seconds.

True SPace / v4.3 Caligari Corp.[2]
TrueSpace4 includes a photorealistic renderer based on *Lightworks Pro* rendering engine from Lightwork Design. Its features include Hybrid Radiosity, Volumetric Rendering, adaptive Anti-aliasing, Area and Projector Lights, Advanced Shaders, Anisotropic Reflectance Shaders, and Layered Materials.

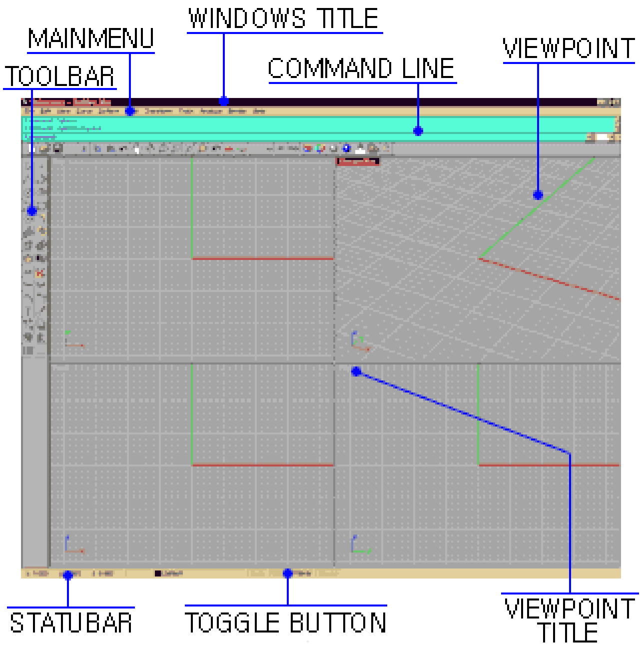
Left: example output from *TrueSpace4.*

Houdini / Vers. 2.0 Side Effects Software (SESI) of Canada[3]

Based on a previous product called *Prisms*, this 3D Animation tool was first introduced in August 1997 at the Sigraph93 held in Los Angeles. It is capable of handling modeling, compositing, rendering, motion, lighting, and performance simultaneously. It also can express not only nurbs, polygon, metaball, and particle, but also growth and behavior. *Houdini* includes Side Effects' proprietary distributed raytracer, *Mantra.*

[1] http://www.onyxtree.com/
[2] www.caligari.com
[3] www.sidefx.com

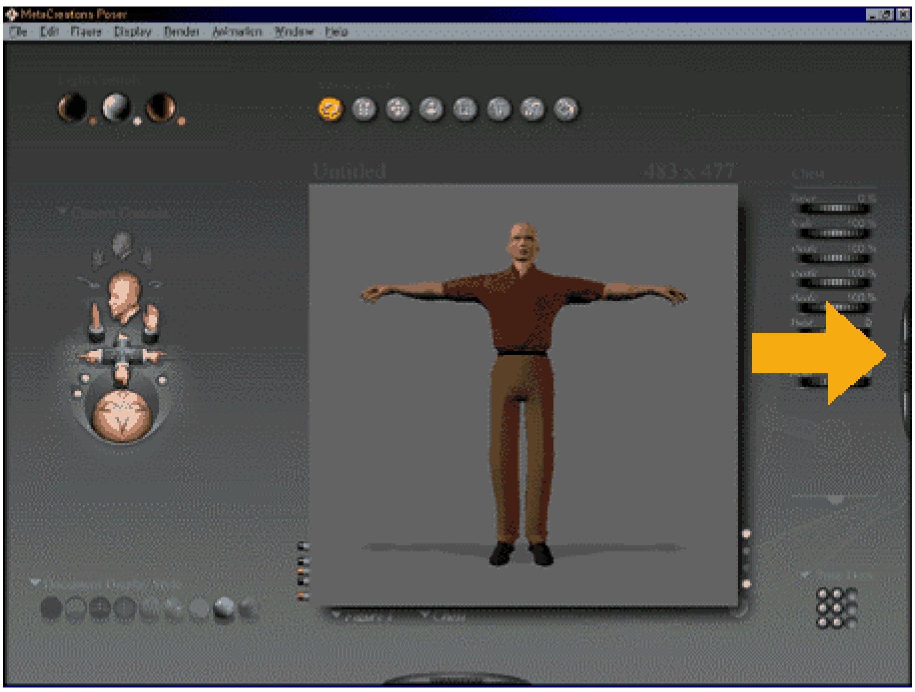
Some of the effects created by *Houdini* can be seen in the scene where the Red Sea splits in the animated movie *Prince of Egypt*. Also used in *Jurassic Park, Independence Day, Star Treck,* and *Contact*



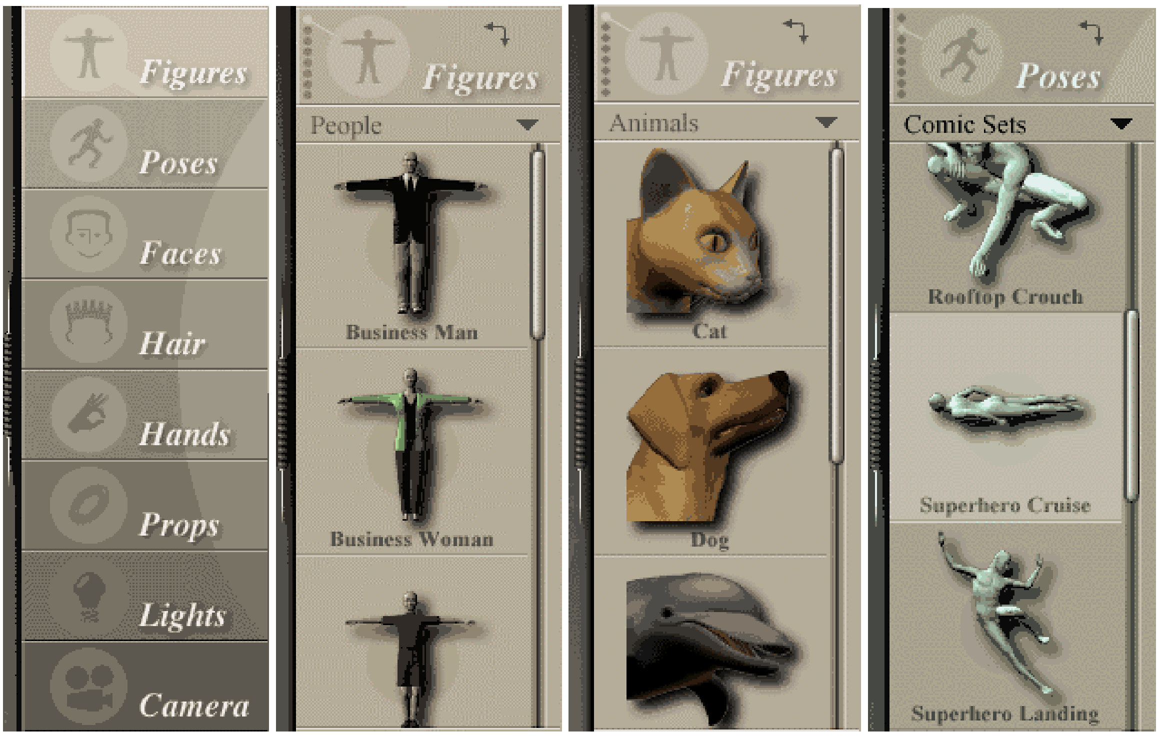
Above: *Rhinoceros* interface

Rhinoceros 3D / Robert McNeel Ass's A NURBS Modeling application, introduced in October, 1998 after undergoing a long process of Beta testing. This software has a high reputation as a power modeling program, (which is an additional module of the *Alias* application of SGI), with results that are clean and sharp. It takes up only a little hard-drive space and achieves high performance speed. It lacks rendering and animation functions however.

Curious Labs *Poser* / Vers. 5



Poser is a timeline-based character animation application offering pre-modeled libraries of human and animal figures, which may be customized and textured. Facial expressions, lighting and camera movements may also be animated. Poser does not offer modeling, ray-tracing or volumetric lighting and so does not produce the level of photorealism offered by higher end 3D programs.



Poser libraries offer customizable human, animal and clothing models and can be used to store keyframed poses or animations. Its timeline-based animations can be output to video file formats. Props, hair and clothing objects may be attached to the figures.

Right: Image output from Poser 5.0
©2004 B.Ferrier

The program *Poser* was used extensively in the production of the accompanying interactive CD-ROM *The Machine* (see *Part Two,* p.547 for a description of its use in this context).

Photorealistic Renderman / Pixar [1]

Pixar's Renderman (abbreviated PRMAN) is the most widely used rendering solution for feature films, responsible for rendering most of the 3D scenes in productions made by Pixar, Industrial Light and Magic, and other leading commercial animation companies.

The latest version (11) adds support for raytracing and global illumination, which previously had to be faked or computed by other renderers. Renderman is widely used in film because it excels in rendering high-geometry scenes, including motion blur and detailed displacement maps, at high resolutions, very quickly. Unlike many other renderers, Renderman does not tesselate all of the smooth NURBS or subdivision surfaces into polygons prior to rendering, but instead smoothly renders each region. PRMAN shaders are programmed in a scripting language that is designed to be powerful and flexible, but also the easy to write.

3D modeling/animation/compositing programs are among the most complex and powerful applications available in the contemporary multimedia world, drawing on a range of traditions and capable of producing animated and visually powerful virtual worlds – this is virtual cinema where the artists literally “play god" designing and building and controlling every element of the virtual world. Apart from the new 3D animated cinema, 3D animation is central to the immersive power of contemporary interactive games, although this is a highly specialized field requiring design work to be integrated with high-level computer programming skills. The industry standard multimedia authoring software Macromedia *Director* has recently been up-graded to include a 3D module, allowing multimedia authors without specific 3D programming skills to incorporate 3D worlds in interactive multimedia art. While this interface improves accessibility to non-programmers, it is still a highly complex process. However it opens up future possibilities for more immersive interactive non-linear art.

2.1.1 The Text Channel

Computers can assist artists with text and creative writing tasks in a number of ways. The written word is fundamental to script-writing and structured story-telling within a multimedia context. (For convenience, non-linear narrative within a multimedia context is considered in the later section devoted to interactivity in art - see p. 362) The written word can also perform the role of a graphic element within a design.

[1] https://renderman.pixar.com/main.html

Computer assisted storytelling

Despite the potentially bewildering variety of options, the technical sophistication, and the potential for high orders of complexity in multimedia art, the artist must in most cases ultimately have an (engaging) story to tell.

Software support is available for story development and script writing, which can aid plot and character development. (e.g. *Dramatica Pro* by Screenplay Systems). Such applications tend to fall into a role of facilitating the organization of ideas, checklisting the "through-lines" implied by script decisions, also aiding in the cross-referencing and linking of ideas.

The table set out below is the *story engine settings report* (from *Dramatica Pro*) in textual form, for the popular science fiction film *the Matrix* (Warner Brothers, 1999). According to *Dramatica Pro's* promotional web-site, each report is a fairly complete list of the "story points" found in that story's underlying storyform, which is established by entering responses to a set of questions about the story's plot and characters. To put *Dramatica Pro* story analysis output into meaningful context, here is a précis of the movie's complex plot:

> On an Earth in the future all appears well and normal but to some, things are not right. To a handful it becomes apparent that the life that humanity knows is that of a computer simulation. The human mind is utilized as a generator of energy for a grand machine superstructure. Man is plugged in, sensations are projected in the mind in such a way as to make everything appear real. Every sensation, every lusting, every taste, every touch; it's all made to feel real but in reality it isn't.
>
> … one man (Neo) wakes from this dream world. He is unplugged from his resting place where he was plugged into machine-bred reality. But can anyone else wake from this false world of beauty rather than reality? Some, the so-called "hackers" of humanity, can break into this machine dream world at will. They can interact with men who are a part of this dream world. But lurking within the dream world are computer controlled "agents" that constantly monitor and eradicate the interventions of those that enter this world with the truth. Agents can do the impossible almost in this computer world. Their actions and abilities are only limited by the Matrix or the computer program that runs this dream state. Anyone absolutely at anytime can be converted into an agent as needed… If someone dies in the dream world of the matrix then they die in reality as well. The body cannot exist when the mind believes that it is dead. … Neo, (known to the machine agents as Mr. Anderson) in one dramatic scene arrives with a female who has hacked into the Matrix with him. They are in search of their former leader Morpheus who is in the Matrix, as agents attempt to extract information from him.

> In the movie's end, agents are finally able to be killed and Keanu Reeves (Neo) realizes that he is the chosen one meant to awaken man from his dream state.[1]

The table below is the "storyform report" for the popular sci-fi film "*The Matrix*", generated by the *Dramatica Pro* application, based on entering responses to leading questions[2]. It is based on the idea that character transformation is the central process undertaken during the unfolding of a human story and that this is also what audiences find interesting about storytelling. It seems, from a superficial view, to simply reduce a complex plot and character details down to key focus words. But by stressing character and plot dynamics, the intention is to draw out and clarify the most important threads of the story themes. The following example helps one grasp this process and is therefore a logical marketing tool to explain the function of the software, however this sort of application is not really meant to create a sophisticated analysis of a finished product. One must therefore beware of underestimating the power of such a conceptual approach to script development, which seems to supply simplistic results when applied in retrospect. During the writing process important decisions must be made if the story is to move forward. *Dramatica Pro* encourages the writer to reduce plot and character details down to key words and summaries, assisting the would-be author to stand back from vaguely expressed characters and sub-plots that emerge during the early development stage, and follow the seeds of these ideas to logical and clearly defined outcomes. This can give clearer definition to the story themes and story shape, by emphasizing central story elements over distracting detail.

CHARACTER DYNAMICS:

- MC RESOLVE: Change
- MC GROWTH: Stop
- MC APPROACH: Do-er
- MC PROBLEM-SOLVING STYLE: Intuitive
- IC RESOLVE: Steadfast

PLOT DYNAMICS:

- DRIVER: Decision
- LIMIT: Optionlock
- OUTCOME: Success
- JUDGMENT: Good

---

[1] http://www.movieprop.com/tvandmovie/reviews/thematrix.htm ; accessed 30-11-02.
[2] Source: http://www.dramatica.com/story/storyforms/storyforms/matrix_storyform.html ; accessed 21-06-02.

MAIN CHARACTER

(Neo)

THROUGHLINE: Situation

CONCERN: The Future

ISSUE: Openness vs. Preconception

PROBLEM: Disbelief

SOLUTION: Faith

SYMPTOM: Reconsider

RESPONSE: Consider

UNIQUE ABILITY: Openness

CRITICAL FLAW: Denial

BENCHMARK: The Present

SIGNPOST 1: The Present

SIGNPOST 2: How Things are Changing

SIGNPOST 3: The Future

SIGNPOST 4: The Past

MAIN VS. IMPACT STORY

(Teacher/Student)

THROUGHLINE: Manipulation

CONCERN: Changing One's Nature

ISSUE: Responsibility vs. Commitment

PROBLEM: Uncontrolled

SOLUTION: Control

SYMPTOM: Temptation

RESPONSE: Conscience

CATALYST: Rationalization

INHIBITOR: Attitude

BENCHMARK: Conceiving an Idea

SIGNPOST 1: Developing a Plan

SIGNPOST 2: Playing a Role

SIGNPOST 3: Changing One' Nature

SIGNPOST 4: Conceiving an Idea

OVERALL STORY

(Finding "The One" who'll save humanity from the Machines)

THROUGHLINE: Activity

CONCERN: Obtaining

ISSUE: Morality vs. Self Interest

PROBLEM: Disbelief

SOLUTION: Faith

SYMPTOM: Temptation

RESPONSE: Conscience

CATALYST: Approach

INHIBITOR: Obligation

BENCHMARK: Gathering Information

SIGNPOST 1: Doing

SIGNPOST 2: Obtaining

SIGNPOST 3: Gathering Information

SIGNPOST 4: Understanding

IMPACT CHARACTER

(Morpheus / Trinity)

THROUGHLINE: Fixed Attitude

CONCERN: Innermost Desires

ISSUE: Closure vs. Denial

PROBLEM: Hinder

SOLUTION: Help

SYMPTOM: Temptation

RESPONSE: Conscience

UNIQUE ABILITY: Closure

CRITICAL FLAW: Preconception

BENCHMARK: Contemplation

SIGNPOST 1: Memories

SIGNPOST 2:Impulsive Responses

SIGNPOST 3: Innermost Desires

SIGNPOST 4: Contemplation

Significant terms such as “Signpost”, “Benchmark” “Catalyst” etc. point to specific key concepts implicit to this stylized approach to story development. It is probably possible to create software that will generate stories for the writer, however this is not the aim here, but rather to offer a potentially powerful labour and time-saving “brainstorming” device in the creative process of narrative development, especially for less experienced writers.

Type in Motion

The written word obviously plays a significant role in multimedia production and represents a primary channel of communication within the visual modality. But it is not only in terms of the literary or literal content, meaning and the form embedded in patterns of words that the written word should be considered. Type as graphic element is also worthy of discussion.

Graphic Symbolism: the Work of Saul Bass

Saul Bass (1920-96) was a pioneer in the field of what could be called “kinetic typography”. He was an originator of vibrant new approaches to title sequences, transforming the function of film titles from staid, formal, and merely pragmatic communication to complete mini-narratives, which used a metaphorical approach to establishing the mood and visual character of the film [Bellatoni and Woolman 1999, p.14]. He single-handedly revolutionized expectations with regard to the design of film title sequences, beginning with Otto Preminger's *Carmen Jones* (1954).

He went on to create a rich body of title designs for a diversity of motion pictures: *North By Northwest, Vertigo, Cape Fear, Casino,* and many more. (Bass also worked at different stages as an actor, producer, and director). Bass, recalling his early work, writes:

> Design is thinking made visual...I always had a notion that film title credits were a portion of the film that was under-utilized, a time when people went to the bathroom, chatted or ate popcorn.

An inspiration for Bass was Paul Rand's[1] use of shape and asymmetric balance (1940).

The signature design methodology of Saul Bass embraced modern art tenets of minimalist reduction and introduced an acute attention to pace, rhythm and detail.

Bass went a step further and reduced the graphic design to a single dominant image, usually centred in deep space. Bass saw the titles of a film as an opportunity to establish the attitude and feel of the film through the use of a reductive symbol, *a universalizing metaphor*, so that by the time the film began, the audience was already placed in the intended emotional area. Along with his wife, Elaine, he collaborated on titles for movies, such as "*The Age of Innocence*."

Bass, commenting on his design approach, states:

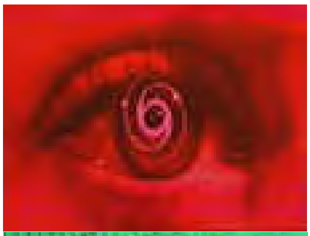
> When we undertake a new film, we begin to try to absorb everything we can about it. We read the script and the novel, if the film is based upon one, talk with the director about his point-of-view and his approach, see pieces of the film if they are available, and then agree with the director on an attitude.

For *The Age of Innocence*, (the original, not the remake) the title was envisioned as ambiguous and metaphorical, projecting a romantic aura and expressing suppressed sexuality.

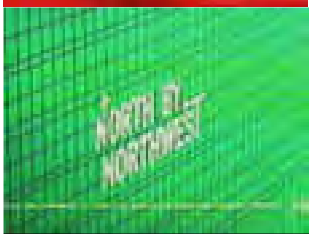
Saul Bass animated 2-dimensional shapes and cut-out forms in various ways that would demand both an emotional and intellectual response from the viewer. For instance, for the opening sequence of *Anatomy of a Murder* simple hand-cut shapes of body parts shift in scale, move and juxtapose one another to create abstract relationships, before they assemble into a complete human body, sprawled as if found at a murder scene.

Through his unique ability to identify the nucleus of a design problem, a movie title was released from its static origins. Bass saw to it that the beginning of a motion picture would never be the same.

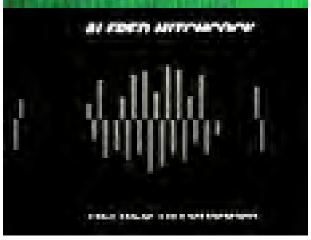
There is a robust energy about his forms. He introduced the idea of using a simple pictographic image to dynamically symbolize a movies theme, now so familiar to audiences everywhere. He pioneered the use of animation techniques to achieve a range of psychological and emotional effects unobtainable with conventional straight type. His collaborations with Otto Preminger and Alfred Hitchcock were outstanding, particularly *Vertigo* (1958), *Psycho* (1960) and *Bunny Lake Is Missing* (1965). Saul Bass designed the best opening credits among Hitchcock's many films [2]:



Vertigo (1958)
"Spinning vortex" credit design.



North by Northwest (1959)
"Lines going up (north) and left (west)" credit design.



Psycho (1960)
"Lines symbolizing stabbing (knife)" credit design, and input to the famous shower murder scene.[3]

The *"Vertigo"* title sequence combines live action, typography and advanced optical effects that anticipate and even now compete with the capabilities of computer technology almost 40 years later. The camera pans up to a closely cropped face of a woman, and then zooms into her eye, which becomes the focal point in the viewing frame. As a red hue dramatically envelopes the image, the main title treatment emerges from the centre of the pupil and zooms forward out of the frame. Following

---

[1] *Paul Rand was a seminal figure in American graphic design who explored the formal vocabulary of European avant-garde movements including cubism, constructivism and de stijl and developed a approach to problem solving - one of the "fathers" of modern design (Bass et al.,1997, 187-88).*

[2] http://nextdch.mty.itesm.mx/~plopezg/Kaplan/people/bass.html ; accessed 12-01-03.

[3] *As an interesting aside, Bass is usually credited with the design and execution of the shower scene in Psycho, something which has sparked a great deal of controversy. Hitchcock supporters like Janet Leigh have repeatedly stated that Bass is lying and Hitchcock directed the scene, but most of the experts agree, that the shower murder has Hitchcock's thematic drive and Bass's technical expertise.)* http://nextdch.mty.itesm.mx/~plopezg/Kaplan/people/bass.html ; accessed 12-01-03.

the title, a spinning, spiral-shaped web of lines zooms forward to suggest the dizzying sensation felt by those afflicted with vertigo. The final credit, that of director Alfred Hitchcock, emerges from the pupil. The main title and credits are set in a bold, outlined slab serif typeface. This highly functional font allows the letters to be tightly spaced while maintaining readability and enhancing the visual impact of the sequence [Bellatoni and Woolman 1999, 17].

Bass exhibits a Bauhaus influence in his most notable work. Middleton [2000] claims that the opening sequences for *Vertigo*, *North by Northwest,* and *Psycho* all bear striking resemblances to films from early Bauhaus abstract film-makers, especially Walther Ruttmann (see p. 222). Middleton also points out that the swirls of *Vertigo* recall the silent-era “visual music” studies of Oscar Fischinger (see p. 223), and the linear work was reminiscent of the opening for Ruttman’s *Berlin*.

2.2 The Aural Modality

This dissertation accepts an assumption of the modularity of sensory input to the brain. Multimedia art appeals to the ear as well as the eye. Adopting a cyberpunk analogy, in which consciousness is a television receiver, this section is a switch to the auditory channel.

However, since writing this opening statement, research has led to the adoption of Elsom-Cook’s [2001, 3] useful suggestion that the sensory modalities are further sub-divided into “channels”. So to be precise and consistent within this classification of media forms, this chapter is a switch to the auditory modality, within which the music channel, the language or spoken word channel, and the natural sound or “sound effects” channel, will be considered.

2.2.1 The Music “Channel” - histrorical antecedents and unique issues

Music is "an art of organizing sound in significant forms to express ideas and emotions through the elements of rhythm, melody, harmony and colour." [1] It is therefore an important channel for communicating meaning in multimedia art. A discussion of its functional role in the multimedia experience must begin with an historical overview of the evolution of relevant music technology.

[1] The Macquarie Dictionary, third edition

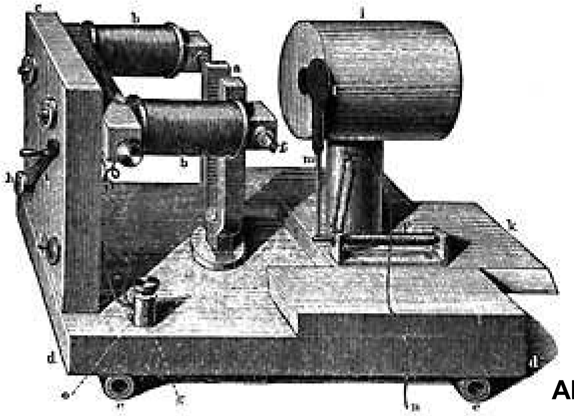
> Computer music is nearly forty years old. Electronic music is twice that old, dating back to the invention of the Theremin Vox. In that time, computer music has brought together many diverse disciplines, creating hybrids such as psycho-acoustics and algorithmic composition, aswell as spawning its own diverse branches. These range from performance instruments to music printing, from MIDI sequencing to automatic transcription. Such diversification is an indication of the success of the field. [Jaffe, 1995, p.1].

Digital music and audio techniques are an important contributing sub-set of multimedia as meta-art. As in other sections of this dissertation, a context for the discussion of contemporary approaches to this field is offered in the form of a brief history of electronic music

For the purposes of this discussion, electronic musical instruments are defined as:

Instruments that synthesize sounds from an electronic source.

This definition leaves out a whole section of hybrid electronic instruments developed during the last century that used electronics and tape recorders to manipulate or amplify sounds in the style of *Musique Concrète* (see below).



Above Left: The Helmoltz Resonator.

The origins of electronic music can be traced back to the audio analytical work of Helmholtz (1821-1894), the German physicist, mathematician and author of the seminal work *Sensations Of Tone: Psychological Basis for Theory of Music*, c.1860, [Trenn, 1998].

Helmholtz built an electronically controlled instrument to analyze combinations of tones, the *"Helmholtz Resonator"*. It used electromagnetically vibrating metal tines and glass or metal resonating spheres. The machine could be used for analyzing the constituent tones that create complex natural sounds (Helmholtz was concerned solely with the scientific analysis of sound and had no interest in direct musical applications). The theoretical musical ideas were provided by Ferruccio Busoni, the Italian composer and pianist, whose influential essay *"Sketch of a New Aesthetic of Music*" was in turn inspired by accounts of Thaddeus Cahill's[1] *Telharmonium* [2]. The latter instrument, also known as the *Dynamophone*, weighed over 200 tons and was designed to transmit sound over telephone wires. The generators produced pure tones of various frequencies and intensity; volume control supplied dynamics[3].

The first electronic instruments built from 1870 to 1915 used a variety of techniques to generate sound:

- the tone wheel (used in the Telharmonium and the Chorelcello) - a rotating metal disk in a magnetic field causing variations in an electrical signal.

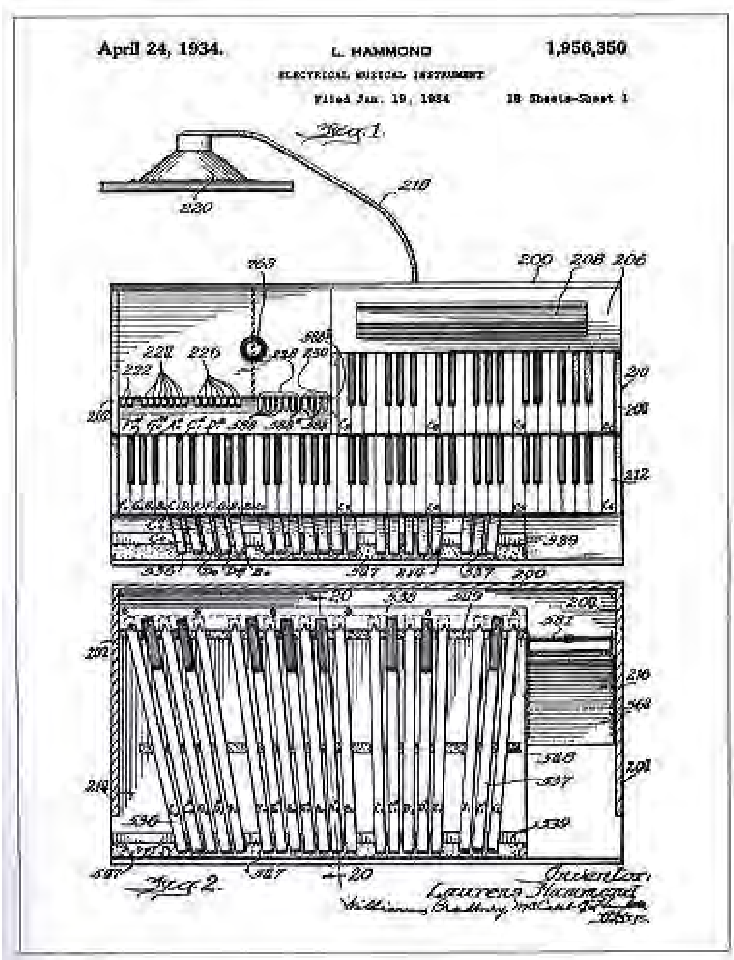
Left: Two players operating the 200 ton Telharmonium.[4]

- an electronic spark causing direct fluctuations in the air (used uniquely in William Duddell's "Singing Arc' in 1899), and

- Elisha Grey's self-vibrating electromagnetic circuit, used in the 'Musical Telegraph', a spin-off from telephone technology [Trenn, 1998].

[1] inventor of the electric typewriter
[2] http://www.obsolete.com/120_years/intro.html ; accessed 7-01-03.
[3] http://www.telecom.at/umm/ToneWheel/the_hammond_organ.html ; accessed 7-01-03.
[4] http://fr.audiofanzine.com/apprendre/dossiers/?idossier=12 ; accessed 6-01-03.

Right: Elisha Grey's *Musical Telegraph,* (1876). It could play notes over a two octave range and transmitted oscillations made by steel reeds over a telephone line using electromagnets [Trenn, 1998].



• *Page 1 of the 18-page patent filed on January 19, 1934, an overhead illustration of the Hammond Model A organ.*

The tone wheel was to survive until the 1950's in the *Hammond* Organ, pictured above, (patented by the American inventor and clockmaker Laurens Hammond in 1934)[1]. The Hammond name and sound has since become legendary (made

[1] http://www.hammond-organ.com/ ; accessed 7-01-03.

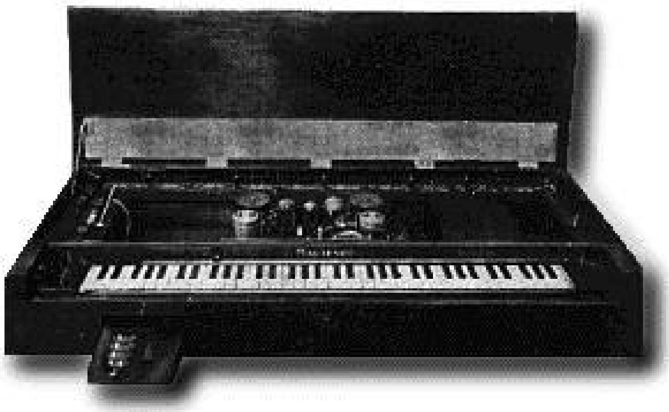
famous in the classic 60s song "*A Whiter Shade of Pale*" by British band *Procul Harum,* among many others), but the experiments with self-oscillating circuits and electric arcs were discontinued with the development of vacuum tube technology.

The engineer and prolific US inventor Lee De Forest patented the first Vacuum tube or *triode* in 1906, a refinement of John A. Fleming's electronic valve[1]. The Vacuum tube's main use was in radio technology, but De Forest discovered that it was possible to produce audible sounds from the tubes by a process known as *heterodyning*. The Heterodyning effect is created by two high radio frequency sound waves of similar but varying frequency combining and creating a lower audible frequency, equal to the difference between the two radio frequencies (approximately 20 Hz to 20,000 Hz)[2]. De Forest was one amongst several engineers to realize the musical potential of the heterodyning effect and in 1915 created a musical instrument, the "Audion Piano"[3]. Other instruments to exploit the vacuum tube were:

- the *Theremin* (1917)

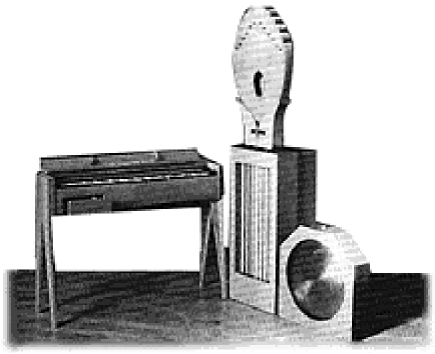
- Ondes Martenot (1928);

"Martenot's Waves" consisted of two table-mounted units controlled by manipulating a string attached to a finger ring (using the body's capacitance to control the sound characteristics in a manner very similar to the Theremin - discussed below).



[1] http://www.pbs.org/transistor/album1/addlbios/deforest.html ; accessed 4-01-07.
[2] http://www.obsolete.com/120_years/intro.html ; accessed 28-12-02.
[3] http://music.dartmouth.edu/~wowem/electronmedia/music/eamhistory.html ; accessed 3-01-03.

This device was later incorporated as a fingerboard strip above the keyboard. Later versions used a standard keyboard. Hundreds of symphonic works, operas, ballets, film scores, and other works were composed for the *Martenot* by composers such as Olivier Messiaen, Milhaud, Varese, Boulez, Honneger and more [1].

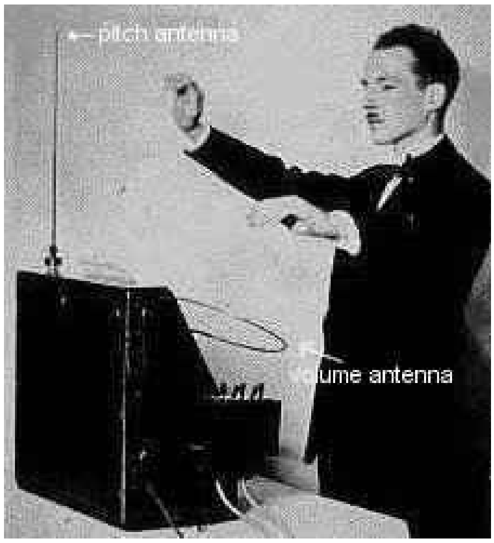


- the 'Sphäraphon' (1921)
- the 'Pianorad' (1926).

Technology based on the Vacuum tube was to remain the primary type of audio synthesis until the invention of the integrated circuit in the 1960s.

The Theremin

The world's first *practical* electronic instrument was invented by Leon Termin (often anglicized toTheremin) in 1920. The *aetherphone* as he first called it (sound from the 'ether'), later became known as the *theremin*.



Above: Lev Sergeivitch Termin (or Theremin) playing his invention the *Theremin*.

As we have seen, the principles of beat frequency or heterodyning oscillators were discovered by chance during the first decades of the twentieth century by radio engineers experimenting with radio vacuum tubes.

A major problem with utilizing the heterodyning effect for musical purposes was that, as the body came near the vacuum tubes, the capacitance of the body caused variations in frequency. Leon Termin[1] realized that, rather than being a problem, body capacitance could be used as a control mechanism for an instrument, in one stroke of insight finally freeing the performer from the keyboard and fixed intonation.

[1] http://csunix1.lvc.edu/~snyder/em/mart.html ; accessed 4-01-03.

It is thus one of the few electronic instruments of the pre-MIDI era that does not require physical contact to play.

Termin's first machine, built in the USSR in 1917, was thus the first instrument to exploit the heterodyning principle.

The original *Theremin* used a foot pedal to control the volume and a switch mechanism to alter the pitch. This prototype evolved into a production model *Theremin* in 1920, this was a unique design, resembling a gramophone cabinet on 4 legs with a protruding metal antennae and a metal loop[2].

Above: The 1920 production model .

This *Theremin* model was first shown to the public at the Moscow Industrial Fair in 1920 and was witnessed by Bolshevik Revolutionary leader Lenin, who requested lessons on the instrument. Lenin later commissioned 600 models of the *Theremin* to be built and toured around the Soviet Union[3].

Two antennas protrude from the theremin - one controlling pitch, and the other controlling volume. As a hand approaches the vertical antenna, the pitch gets higher. Approaching the horizontal antenna makes the volume softer. Because there is no physical contact with the instrument, playing the theremin requires the user to develop a precise skill which must be based on an accurate sense of relative pitch [1].

The output was a monophonic continuous tone modulated by the performer. The timbre of the instrument was fixed and, when played with vibrato (microtonal variations in pitch), can resemble a violin string sound. The sound was produced directly by the heterodyning combination of two radio-frequency oscillators: one operating at a fixed frequency of 170,000 Hz, the other with a variable frequency between 168,000 and 170,000 Hz. the frequency of the second oscillator being determined by the proximity of the musician's hand to the pitch antenna. The difference of the fixed and variable radio frequencies results in an audible beat frequency between 0 and 2,000 Hz. The audible sound came from the oscillators, later models adding an amplifier and large triangular loudspeaker.

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[1] To separate man from machine, Termin is used for the inventor’s name, and *Theremin* for the instrument he invented, in this text.
[2] http://www.obsolete.com/120_years/intro.html ;accessed 28-12-02 .
[3] http://www.obsolete.com/120_years/intro.html ; accessed 28-12-02.

Clara Rockmore established that it was possible to achieve the level of artistry required to gain the instrument serious concert hall acceptance. Composers such as Varese and Cage (*Variations V*), rejoice in the aural 'novelty' aspect of the instrument, using its unique means of expression and timbre to create new modes of music, radically different to those which had gone before [2]. It has been used in countless musical contexts e.g. the soundtrack to Alfred Hitchcock's *Spellbound* (1945), the cult-status science fiction film "*Forbidden Planet*" (MGM, 1956), and the classic hit pop song "*Good Vibrations*" by the American vocal harmony group *The Beach Boys*, and is still used today. However it never overcame its novelty appeal and was mainly used for a special effect, and rather more rarely as a 'serious instrument'. Most recordings employ the *Theremin* as a substitute solo string instrument rather than exploiting the microtonal and pitch characteristics of the instrument.

Termin left the Soviet Union in 1927 for the United States where he was granted a patent for the *Theremin* in 1928. The *Theremin* was marketed and distributed in the USA by *RCA* during the 1930's and continues, in a transistorized form, to be manufactured by Robert Moog's *Big Briar* company[3].

Termin went on to develop variations on the original *Theremin* which included:

- the "Terpistone",
- The "Rhythmicon" - the first electronic rhythm generator**,**
- the "keyboard Theremin" and
- the "Electronic Cello" - A fingerboard operated, Cello-like variation of the *Theremin,* that enabled continuous glissando and tone control.

The *Terpistone* was an adaptation by Termin of the *Theremin*, for use by dancers, the control antenna being made in the form of a huge metal sheet hidden under the floor. Movements of the dancers in the area above the sheet caused variations in pitch of the *Terpistone's* oscillators due to the capacitance of the dancer's bodies.

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[1] http://www.thereminworld.com/learn.asp ; accessed 7-01-03.
[2] http://www.theremin.info/modules.php?op=modload&name=News&file=article&sid=38 accessed 3-1-03.
[3] http://www.obsolete.com/120_years/intro.html ; accessed 28-12-02.

This instrument was used for several 'exotic' dance, music and light shows throughout the 1930s [1].

Robert Moog developed his ideas for an electronic instrument by starting out at age nineteen (1954) building and selling *Theremin* kits[2] and absorbing ideas about transistorized modular synthesizers from the German designer Harald Bode[3]. After publishing an article for the January 1961 issue of the magazine '*Electronics World*', Moog sold around a 1000 *Theremin* kits (The Vanguard and the Professional models) from 1960 to 63 out of a three-room apartment.

> The four tone colors of the Professional Model *Theremin* are named PRINCIPAL, HORN, WOODWIND, and STRING... The Principal tone is mellow and ethereal, like a flute, and is the timbre traditionally associated with the *Theremin*. The Horn tone is sharp and nasal, like that of an oboe. The Woodwind tone is hollow and woody, like a clarinet. The String tone is rich in overtones, like that of any stringed instrument...[4]

Musique concrète

It was around the 1950s that electronic instruments began to be treated more seriously, and simultaneously the musique concrète style established itself. The concepts and techniques evolved in the era of musique concrète remain relevant today. Many of the concepts fundamental to contemporary digital music production techniques originated in this period of experimentation with a linear, tape-based sound recording medium, and a thorough understanding of contemporary digital music technology is enhanced by knowing this history.

Musique concrète was an experimental music composition style which focused on taking sounds out of the real world, mixing them, and processing them to construct an experimental music composition or soundscape. This style depended primarily on the (then) newly-invented audio recording medium: magnetic tape. It excited creative sound designers because the sounds produced by manipulating tape had never been heard before, opening new vistas of creative sound production.

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[1] ibid.
[2] http://moogarchives.com/ ; accessed 07-01-03.
[3] http://www.obsolete.com/120_years/machines/moog/index.html ; accessed 3-1-03.
[4] http://moogarchives.com/ ; accessed 07-01-03.

Musique Concrète was recorded directly to tape with real (concrete) sounds, retaining the living characteristics of the original sound[1], while *musique abstraite* was the traditional way of composing by writing down the score to be played later[2]. History credits Pierre Schaeffer (1910-1995) with pioneering this form of musical experimentation. Schaeffer was not a musician but an engineer and broadcaster, who worked for the RTF Corporation during the Nazi occupation of France. He had at his disposal the resources of RTF such as phonograph turntables, disc-recording devices, a direct disc-cutting lathe, mixers, and a large library of sound effect records owned by the studio, and spent months experimenting with the technology available to him. He discovered that he could lock-groove records - instead of spiraling toward the center of the record, the needle could be made to stay in one groove creating a loop. He was drawn to the possibility of isolating naturally produced sounds[3]. Sounds could be recorded, then played backwards or at fast or slow speeds (i.e. have their pitch and tempo raised or lowered), processed through a variety of methods, spliced, looped, and ultimately an experimental musical piece could be structured which could not be produced in the real world. His first official composition*, Étude aux chemans de fer* (Concert for Locomotives), was a montage of sounds recorded at the train depot in Paris. Sounds included six steam locomotives whistling, trains accelerating, and wagons passing over the joints in the tracks[4].

Musique Concrète was thus largely based on manipulation of tape. It also concentrated on 'found sounds' or natural recordings rather than electronically produced sounds such as created by synthesizers. Pieces that would last only a few minutes could take months of recording, cutting and splicing to create.

In 1948, Schaeffer studied the effect of striking percussive instruments in different ways. He observed that a single sound event could be characterized not only by timbre, but by attack and decay as well. Cutting the tape at different angles was used to experiment with different attacks and decays.

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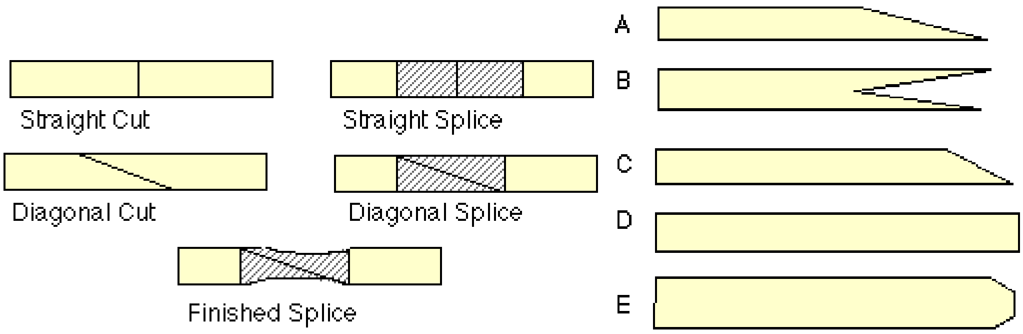
[1] http://www.music.columbia.edu/~cathy/BEaesthetics/musconcret.htm ; accessed 3-01-03.
[2] http://csunix1.lvc.edu/~snyder/em/mc.html ; accessed 3-01-03.
[3] http://personal-pages.lvc.edu/~snyder/em/schaef.html
[4] ibid.



A. soft attack or decay

B. combined attack and decay of two sounds

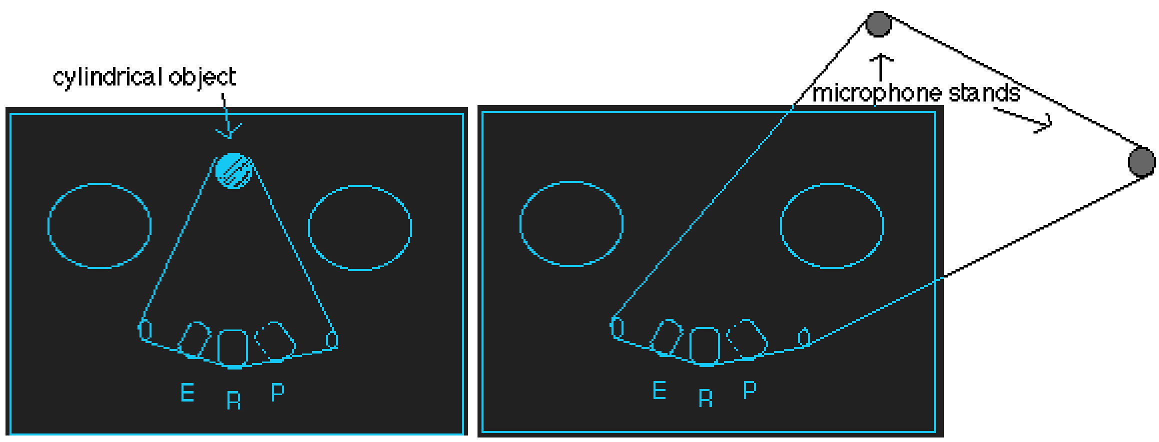
C. medium attack or decay

D. hard attack or abrupt finish

E. softer and less abrupt than D [1]

Not only did musique concrète composers use the cuts shown above, but they would go so far as to take a long horizontal cut, cut it into smaller calculated sizes, and splice the cuts together vertically or at different angles[2].

Creating a loop consisted of taking tape with recorded material and splicing the ends of the tape to make a loop.



Tape Echo Effects

Straight line delay:

The simplest echo effect was created by using a 2 channel recorder. The signal was recorded, monitored by the playback head, and sent back to the lower track of the record head. At 15 ips, the delay is around 100 milliseconds.

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[1] http://www.music.columbia.edu/~cathy/BEaesthetics/musconcret.htm ; accessed 3-01-03.
[2] http://csunix1.lvc.edu/~snyder/em/mc.html ; accessed 3-01-03.

Feedback delay:

Created by feeding the signal back into the record head of the channel originally recorded on. It sometimes involved multiple channels and the creation of a number of repetitions.

Pitch Changes

Obviously changing the speed of playback will affect the pitch: lower pitch if slower, higher if sped up. Using 2 decks, pitch changes can be recorded to tape. Early flanging effects were created by lightly touching the reel as the track is being bounced to another track.

Backwards Masking

Reversing the reels of recorded material and playing the tape backwards can be used to create reverse reverbs, attacks, etc. Recording the reversed track to another tape deck is the easiest way, although bouncing the reversed track to an empty track on a multi track can be done. [1]

Early Tape Decks

Pierre Schaeffer created the *Phonogenes* - two decks designed to play prerecorded loops at different speeds. With it he was able to transpose a loop in 12 distinct steps, using a keyboard (this later evolved into the mellotron keyboard). The keyboard selected one of 12 capstans of different diameters, like changing gears on a bike. A 2-speed motor allowed for octave transposition. He also used *The Morphophone* in the Paris studio. It was a specialized loop deck. It had an erase head, record head, and ten playback heads, which allowed for tape echo and a pseudo reverb effect, with an adjustable filter for each to create special timbre effects. One of the recorders had 5-track capability, giving Schaeffer the means of playing up to 5 separate tracks with 5 separate speakers[2]. This allowed for spatial experimentation of sounds. Four speakers were used for playback. Two speakers were located in front of the stage on the left and right, one was placed directly at the back in the

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[1] http://csunix1.lvc.edu/~snyder/em/mc.html ; accessed 3-01-03.
[2] Contemporary DVD MPEG-2 technology allows for 5 distinct audio outputs for spatially specific sound environment simulation, but here we see the idea in affect almost 50 years ago.

centre, and one was suspended from the ceiling. The ceiling speaker allowed for experimenting with vertical sound placement as well as the usual horizontal placement. The fifth track contained an additional channel spread between the four speakers that represented a performer using a hand-held coil, which could be positioned near one of four wire receiving loops that sent the signal to each speaker[1].

Electronic Music and Late 20th Century Popular Music

Many popular artists, especially of the 60s and 70s, incorporated Musique Concrète techniques into their recordings. *The Beatles*, with George Martin, experimented with tape cuts and loops, e.g. on the ground-breaking experimental album *Revolver,* (Parlophone, 1966) the Lennon-McCartney song *Tomorrow Never Knows* includes tape loops of the playing of each of member of *the Beatles* [2]. It was

> …a pure nightmare where John sang portions of the *Tibetan Book of the Dead* into a suspended microphone over Ringo's thundering, menacing drumbeats and layers of overdubbed, phased guitars and tape loops [3].

Revolver also featured backward-recorded guitar on *I'm Only Sleeping.* John Lennon describes creating *Revolution #9* on *the White Album* thus:

> It has the basic rhythm of the original 'Revolution' going on with some twenty loops we put on, things from the archives of EMI. We were cutting up classical music and making different size loops, and then I got an engineer tape on which some test engineer was saying, 'Number nine, number nine, number nine.' All those different bits of sound and noises are all compiled. There were about ten machines with people holding pencils on the loops - some only inches long and some a yard long. I fed them all in and mixed them live. [Miles, 1997, 484.] [4]

John Lennon and Yoko Ono later ventured further into this avant garde style of recording as heard on *Two Virgins* and other albums. *Steely Dan* used a long tape loop to try to find a perfect drum loop on *Gaucho,* long before drum-machines were used. *Pink Floyd* made the use of tape loops the basis of the definitive *Dark Side of the Moon,* most notably the ominous heart-beat[5] (like the Beatles albums, recorded at Abbey Road)[6].

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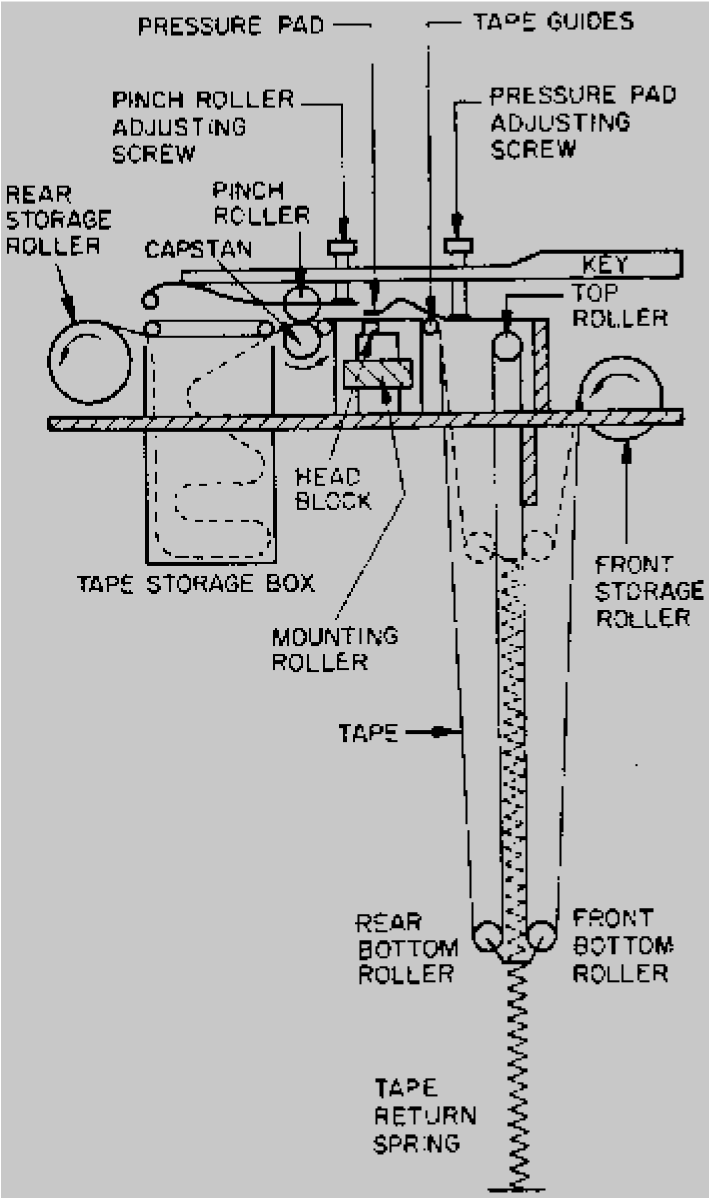
[1] http://personal-pages.lvc.edu/~snyder/em/schaef.html ; accessed 3-01-01.
[2] http://www.geocities.com/remark70/lp-revolver.html ; accessed 15-01-03.
[3] http://www.xs4all.nl/~fsgroen/Albums-B/BeatlesRevolver.htm ; accessed 15-01-03.
[4] McCartney hated it, and did not want it on the album, but finally agreed with profound apprehensions, and George Martin hated it more. http://www.iamthebeatles.com/article1246.html
[5] http://utopia.knoware.nl/users/ptr/pfloyd/interview/dark4.html ; accessed 3-01-03.
[6] http://csunix1.lvc.edu/~snyder/em/mc.html ; accessed 3-01-03.

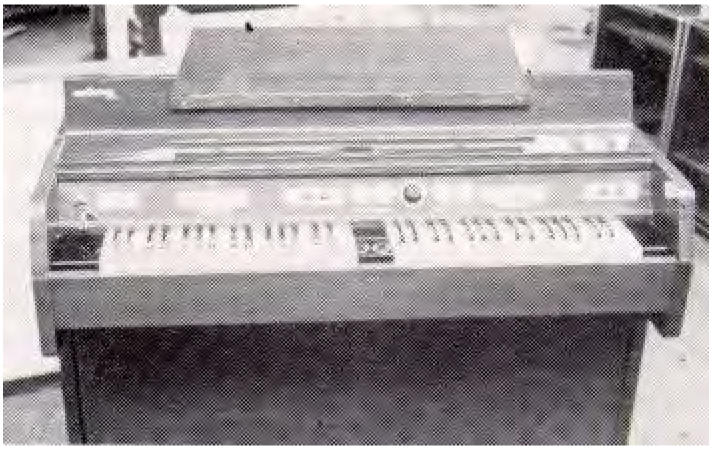


Tape manipulation reached a different level of control with the *Mellotron.* This device is based on a magnetic tape replay unit, triggered from a keyboard - it could be called a primitive analog sample playback unit. Under every key is a length of magnetic tape that is moved past a playback head whenever that key is pressed. Each piece of tape has a sound (for example a sustained flute note) whose pitch corresponds with the keyboard key by which it is triggered. After a key has been pressed the tape plays the note, and when the key is released, the tape is pulled quickly back to its starting position by a spring. Since each sound is produced by a linear piece of tape rather than an endless loop, the sound can faithfully reproduce the attack phase and then (a pre-recorded simulation) of the decay phase of a percussive instrument such as a piano[1].

While the designer, Harry Chamberlin's goal was probably to make it sound exactly like the original instrument that was recorded, the quality of a *Mellotron's* sound is, perhaps accidentally, distinctly its own - moody and full, giving a haunting, dream-like quality.

The *Mellotron* can be heard on almost every song on seven *Moody Blues* albums, *Strawberry Fields Forever* by *The Beatles,* and on albums by "progressive" British rock groups of the period *Genesis, King Crimson*, and *Yes*[2]. Bob Seger's live recording "*Turn the Page*" is another vivid example. Most of this "classic" music that is associated with the *Mellotron*, was produced on the Mark II (see below). *The Moody Blues'* early work was done on the Mark II also, but later work was done on other models, such as the 300 and 400, and the *Chamberlin* M1. The Mark II had two 35-note keyboards. The tape under each key had three tracks on 3/8 inch wide tape. Six different stations on the tape with three tracks each, means that each key can play 18 different sounds.

The left hand keyboard is further broken down into two sections. The left side is loaded with rhythms and the right side contains accompaniment and fills. The right hand keyboard has 18 different lead instrument sounds [3].



The pre-history of mass-produced Synthesizers

Synthesizers had existed prior to the late 60s, in various forms – e.g. the RCA Mark I and Mark II synthesizers (pictured right). In 1955 Harry Olson and Belar, both working for RCA, invented the Electronic Music Synthesizer (aka the *Olson-Belar Sound Synthesizer*)[4]. This device used sawtooth waves that were filtered for other types of timbres.

RCA Mark II

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[1] http://www.till.com/articles/Mellotron/ ; accessed 3-01-03.
[2] http://www.till.com/articles/Mellotron/
[3] http://www.geocities.com/SunsetStrip/Studio/7846/mtron_stats.html#Mark1 ; accessed 7-01-03.
[4] http://music.dartmouth.edu/~wowem/electronmedia/music/eamhistory.html ; accessed 15-01-03.

This synthesizer evolved into the *RCA Electronic Music Synthesizer Mark I.* The *RCA Mark I* has 12 fixed tuning-fork-based oscillators in equal temperament, whose frequency could be divided down to produce different octaves. Originally, tracks were recorded onto disc (up to six at once, replayed by six styli). The whole machine took up seven tall 19" racks[1].

Until the sixties, synthesizers were prohibitively expensive, and generally restricted to experimental composition in an academic environment. In the late 60s, however, popular musicians were searching for new sounds. *The Beatles,* as mentioned*,* were experimenting with tape loops and backwards recording ideas and exotic intrumentation in several songs, apart from their use of *the Mellotron*.

Film composers were also looking for ways to create unique sounds. In response to this demand, commercial synthesizers were eventually developed.

Integrated Circuits came into widespread use in the early 1960s. Inspired by the writings of the German instrument designer Harald Bode, Robert Moog, Donald Buchla and others pioneered a new generation of easy to use, reliable and popular electronic instruments.

Moog, after his initial minor success with his commercial version of the *Theremin*, decided to begin producing instruments of his own design. After toying with the idea of a portable guitar amplifier, Moog turned to the synthesizer. Whilst attending a convention in the winter of '63, Moog was introduced to the idea of building new circuits that would be capable of producing sound. In September 1964 he was invited to exhibit his circuits at the Audio Engineering Society Convention. Shortly afterwards Moog begin to manufacture electronic music synthesizers. Moog's synthesizers were designed in collaboration with the composers Herbert A. Deutsch, and Walter (later Wendy) Carlos. Walter/Wendy Carlos produced an album entitled *Switched on Bach***,** the first commercially successful recording to be entirely recorded using Moog synthesizers, an intriguing and charming adaption true to Bach's counterpoint and harmony, but with a catchy, futuristic effect which went on to win three Grammy Awards [2].

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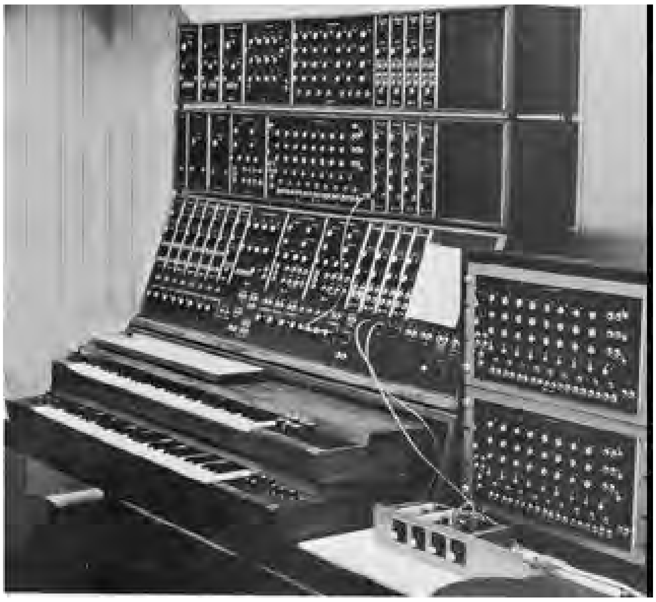
[1] http://www.synthmuseum.com/rca/ ; accessed 7-01-03.
[2] http://www.wendycarlos.com/biog.html ; accessed 3-01-03.

Modular synthesizers

The earliest synthesizers consisted of a series of modules that both created a signal (VCOs, noise generators) and processed a signal (VCF, VCA, etc.). These modules were mounted in large racks and connected to each other with 1/4" cables[1].

These modular synthesizers were usually very big, sometimes taking up entire rooms. Compared to the contemporary portable keyboard, a modular synthesizer was complex and cumbersome.



Above: a large modular Moog system, circa 1970.

However with a modular synthesizer, the user is not restricted to one particular signal path. Modular synthesizers have the greatest freedom in terms of composing sound "patches", and for this reason they are still used in the studios of film composers and studios that require the ability to make new, original sounds - rather than using preset (factory-programmed and thus potentially clichéd) patches.

Moog's instruments were the first to make the leap from the electronic avant garde, into commercial popular music. *The Beatles* purchased one, as did Mick Jagger of the *Rolling Stones*, who bought a hugely expensive modular Moog in 1967 (this instrument was only used once, as a prop on a film set and was later sold to the influential German experimental rock group, *Tangerine Dream*)[2].

[1] Because one patched together connections with these cables, a particular sound was called a *patch...* and the name has stuck, even though modern patch cords are virtual.

[2] http://www.obsolete.com/120_years/machines/moog/index.html ; accessed 3-1-03.

The *Buchla* (pictured right, showing patch cords) was an early competitor to *Moog* modular synthesizers. Although the *Buchla* was technically the first enter the commercial marketplace, the *Moog* modular synthesizer defined the standard which all analogs use to this very day: All functions of the synthesizer are controlled by a control voltage (CV) of one volt per octave.
Though setting a future standard for analogue synthesizer, the *Moog* Synthesizer Company did not survive the decade, larger companies such as *Arp* and *Roland* developed Moog's prototypes into more sophisticated and cost effective instruments. Moog returned to his roots and set up Big Briar, a company specialising in a transistorised version of the Theremin. Modular synthesizers are still being produced by small companies such as Doepfer and Serge.

Modular synthesizers are quite impractical for the performing musician - they are physically large, and not very reliable as musical instruments - their tuning is quite unstable, they need a long warm-up period and they can also quickly drift out of tune. Of course, this didn't deter several big-name artists from using a modular on stage, because they made great visual props. But it was for such practical reasons the *pre-wired synthesizer* was developed in the early 70s.

The pre-wired synthesizer took the most common modules of a synthesizer, each routed through the most common path, with the addition of various control options (for different patches) to the front control panel, through a variety of methods. The resulting device was a synthesizer that was much more practical (and affordable) for stage musicians.

An analogue synthesizer is constructed with analog-based (pre-digital) circuitry. These synthesizers, in general, use analog-style units (oscillators, filters etc.) that are controlled by voltage (like the early modular synthesizers, only pre-wired - no patch bay or patch cords).

Name: SH-1
Manufacturer: Roland
Years manufactured: 78-81
Type: Analogue **solo** synthesizer ;

(monophonic, monotimbral).

An example of an early *Roland* anologue synthesizer

Early analogs, such as the *Minimoog,* have what is called *discrete circuitry* (transistors, resistors, etc. - no integrated circuits). Later synthesizers, however, used other, more compact analog technologies such as op-amps and linear integrated circuits. Analog synthesizers are considered to have a warm, more pleasing sound, compared to most digital synthesizers [1]. Analog synthesizers are usually equipped with an array of hardware buttons, dials, switches and sliders on the front panel of the instrument. These were mainly designed for programming the synthesizer, but the player could also adjust and modify sounds in real time while playing, thus creating dynamic special effects and a more versatile, expressive and dynamic sound. They thus had the advantage of offering a more tactile, intuitive human-machine interface. Parameter changes on digital synthesizers are usually accessed through sub-menus on a small LED screen, requiring a studied knowledge of the hierarchical design or architecture. Acclaimed producer, recording artist and synthesist Thomas Dolby explains:

> If you benchmarked the numbers side-by-side, today's synth is so advanced it's like alien technology compared to what we had early on. Ergonomically, though, the design of classic synths was ultimately more satisfying to work with than the sort of 'modular' approach of modern synths. A couple of years ago, I went back to England for three weeks to do a session. I still have a lot of my old gear in London, so I pulled out my Roland JP-8, which I hadn't used in maybe twelve years. I started playing it, and when I wanted to alter the sound, my hand instantly went to the right slider to do the right thing. After twelve years, I was still able to very quickly program sounds and get the machine to do what I wanted. After three

1 http://archive.keyboardonline.com/features/thomasdolby/index.shtml ; accessed 15-01-03 .

weeks, I came back here, and I had already forgotten how to use my Roland JV-1080, which has millions of sub-pages to toggle through [1].

Today, synthesizers are used in a wide spectrum of music making. The synthesizer has undergone a radical evolution, but the basic concept remains the same: To start with an electronically generated signal, process the signal through electronic methods, and emerge with a unique sound that often cannot be duplicated by any other instrument.

The relevance if this brief history of sound synthesis is brought into focus when considering the new type of modular synthesizer which has emerged as an important potential tool of multimedia artists: *A virtual modular synthesizer*. These are totally software based and allow the user to create customized sound on a computer, using virtual patch cords and modules. These applications can synthesize a very similar sound to a commercial “modular synth + keyboard” unit, without all of the bulk and cables, inconsistency and expense. In the future they will appear in some form as a significant sub-structure of the multimedia meta-art machine.

Above: Interface for *The Tempest* software modular synthesizer. It provides a rich environment for sound synthesis which uses a model of a traditional analogue modular synthesizer to support a number of synthesis paradigms. More than an emulation of an analog modular synth, it will support sampling, additive synthesis, physical modeling synthesis, granular synthesis, as well as a full complement of effects modules [2].

[1] http://archive.keyboardonline.com/features/thomasdolby/index.shtml ; accessed 15-01-03.
[2] http://www.vlabs-online.com/index.html ; accessed 15-01-03.

Computers and Music

There are several areas where computers have facilitated the artistic manipulation of sound:

1) The capture or sampling of digital representations of real world sound (digital sound recording);

2) The processing and manipulation of the resulting digital waveforms, in order to in some way enhance or filter the aural experience;

3) The synthesis of sounds, either to simulate real world sounds, such as instrumental sounds, or to create new expressive sonic experiences as elements in experimental music composition and the creation of new soundscapes.

The area of sound synthesis can be further broken down into:

1) fm synthesis
2) additive synthesis
3) granular synthesis
4) physical modeling

Compositional activities using computers to generate musical material, and/or to trigger sound generation devices have become a commonplace part of popular music.

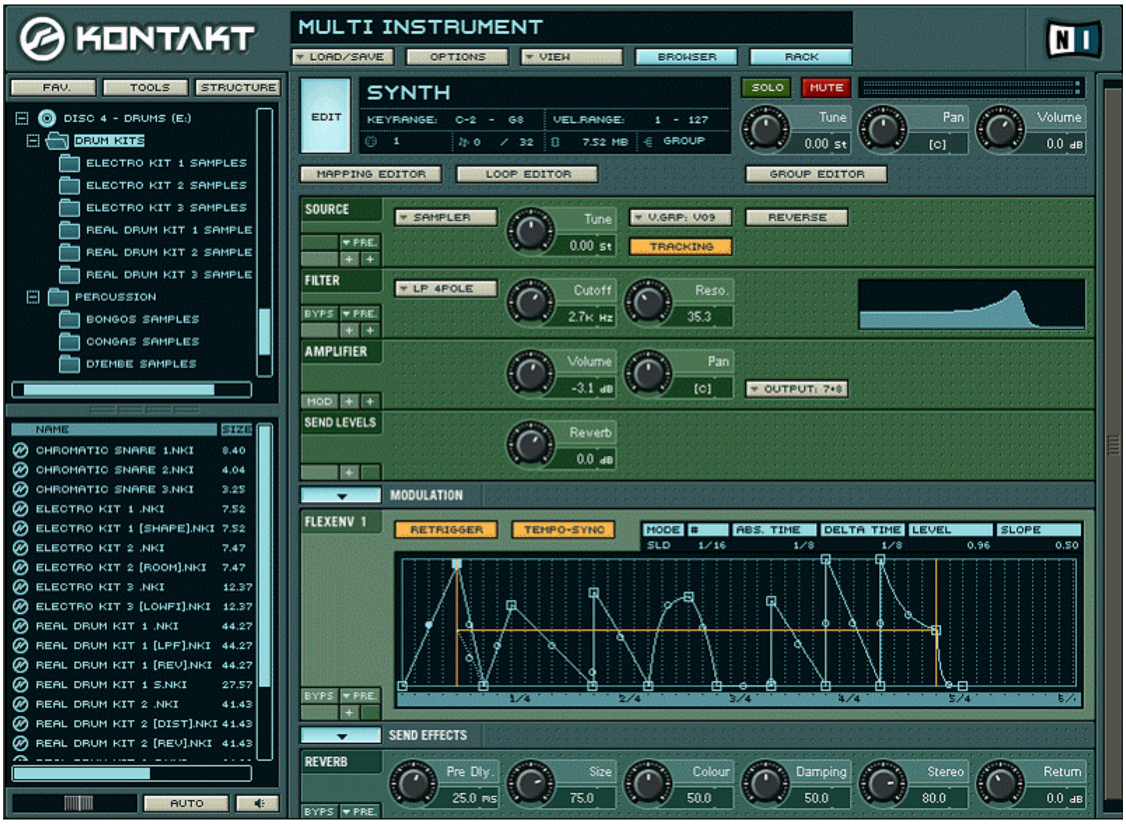
As outlined in the intoduction to this dissertation, one of the original but naïve motivations for embarking on an investigation of digital music, synthesis of sound, and computer-aided music composition was, for this writer, a desire to hear exotic or unavailable musical performances in a simulated arrangement. It was simply beyond available resources to arrange a real, skilled performance, and thus gain the aural feedback in an experimental composition context.

In the early days of commercially available synthesizers, the ability to approximate instrumental resources was highly valued. This made the concept of the sampler extremely exciting to composers in the years when the Fairlight C.M.I. (pictured right) first became commercially available. Of course, composers with a classical electronic music background might have also seen as equally exciting the inherent possibilities for the streamlining of classical musique concrète techniques that becomes possible with non-tape based, random access recording systems such as this.

An historical schism between recording-based and synthesis-based electronic music was reflected in hardware approaches to digital technology-based music making, which Jaffe [1995, 1] describes:

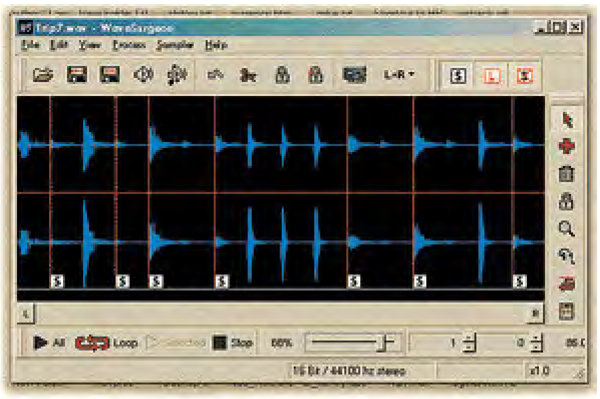
> One of the earliest dichotomies in electronic music was between recording-based music (i.e. Musique Concrète) and pure synthesis. In today's terms, this expresses itself as sampling versus synthesis. The strength of the recording-based approach is that the source material is itself of great richness. However, this complexity is achieved at the expense of flexibility. While the recording accurately depicts a snapshot of an instrument, as soon as it is placed in a musical context requiring variety of expression, it falls flat, no more than a cardboard cut-out with nothing behind it. In contrast, synthesis allows for great malleability but is hard-pressed to create realistic sounds. If we examine the current crop of samplers and synthesizers, we find manufacturers sensing the limitations of the extreme positions and moving toward the center, creating hybrid instruments. So-called "samplers" are enhanced by synthesis-like techniques such as pitch-shifting, looping, dynamic filtering, amplitude enveloping and other modifications that move far away from the pure recording-based paradigm. In fact, a looped sample is nothing more than a synthesizer's wave table oscillator with a wave table much longer than the period of the fundamental frequency. Meanwhile so-called "synthesizers" now incorporate sampled wave tables, which may then be used as (for example) FM modulation signals. The distinction between sampling and synthesis is then reduced to a matter of degree: how long is the wave table and what other techniques are applied?

Hardware-based synthesizers and samplers are giving way to a new generation of software synths and AV desktop computers with ‘off-the-shelf’ digital recording capabilities. An example of a software-based approach is *Kontakt.*



KONTAKT by Native Instruments fuses an innovative design with an advanced sampling engine. In addition to supporting all the standard sample playback and manipulation abilities of its hardware and software predecessors; KONTAKT includes realtime time-stretching and resynthesis, graphical breakpoint envelopes, an integrated loop editor, analog-modelled filters, visually displayed modulation. A sample library containing more than three gigabytes of sounds in various styles and categories is also included. [1]

Even low-cost audio recording applications such as the PC based *WaveSurgeon* have evolved capabilities previously confined to expensive hardware samplers:



The addition of *GigaSampler* support allows users of the software-based audio sampler to integrate *WaveSurgeon's* advanced sample editing features into their music production.

The pairing of *WaveSurgeon* and *GigaSampler* allows musicians to quickly create and edit loops, as well as work with large segments of sampled audio for post-production and Foley effects work.[2]

[1] URL: http://www.nativeinstruments.de/index.php?kontakt_us ; accessed 23-11-02.
[2] URL: http://www.prorec.com/prorec/pressrel.nsf/articles/A5F66D96C7F594798625683900 01D053 accessed 7-01-03 .

Physical Modeling

A commentary has been offered (see p. 54) regarding similarities and differences between natural media techniques (such as oil painting) and digital art practices, in the visual art context. There is an analogous field in digital music, worthy of discussion, embodied in the concept of "physical modeling". Currently the *de facto* standard in instrumental synthesis is a MIDI keyboard controlling a sampler. Performances are recorded with the MIDI keyboard controller, and on play-back these MIDI recordings cause the reproduction (via a number of MIDI channels) of the individual parts of the arrangement, by sounding stored samples loaded into RAM, or by triggering a synthesizer.

In a popular music industry context, arrangements must be built in the process of developing songs for a consumer market. Arrangers and composers (consciously or unconsciously) attempt to engage the widest possible audience, in order to succeed in this highly competitive marketplace. Proven successful production and arrangement formulae will be emulated, perpetuating musical genres and styles, etc. Meanwhile, there is a downward pressure to reduce labour costs by substituting a single keyboard player with a range of sound modules or MIDI keyboards with on-board sounds during recording sessions, rather than have brass and string sections on call. Hence there is an industry-driven market for commercial MIDI keyboards and sound modules which can output suitable sounds to reproduce familiar arrangements and arranging styles, voicings etc. What follows from this is a pressure on research and development engineers and scientists to come up with ways of more accurately reproducing the instrumental resources this technology is supposed to replace. Physical modeling is at the forefront of research into this challenge.

Nevertheless, there are clearly limitations with this approach in terms of reproducing the expressive flexibility of natural instruments (e.g. the particular slurs, ornamentation, vibrato, sustained or doubled notes and harmonisations natural to a violinist because of this instrument's historical tuning and the average size of the human hand; or the manner in which wind instrument players in effect play a continuous sound based on the breath, despite pitch changes). The challenge engineers have tried to meet is to design sound technology capable of reproducing all the subtle nuances of natural instrumental playing using a MIDI keyboard as a trigger device.

Sampling is relatively inflexible when it comes to simulating the continuous performance variables of expression and the nuances of individual instrumental techniques. Taking the violin as an example, traditionally a composer needs an intimate knowledge of real-world technique to effectively write for this subtle, mature and complex instrument, in order to produce a score that can be practically realized in performance by a skilled player. This background knowledge must include a pragmatic understanding of tuning, finger and hand positions, string crossing, slurring, slides, as well as the various bowing techniques, both traditional and avant-garde, limitations of interval leaps and double stopping, pizzicato etc., Many or most of these techniques would require a discrete sample or multiple samples in order to build a simulation of virtuoso playing using sampling technology.

Another obvious example is any attempt to simulate sophisticated electric guitar solos from a series of sampled guitar sounds – the countless subtle slurs, pull-offs and hammering techniques, string bends and finger vibrato techniques which are a part of the electric guitarist's repertoire are difficult to convincingly reproduce using a keyboard trigger device with pitch-bend and modulation wheels.

Physical modeling attempts to get around the inherent problems of MIDI-keyboard triggered sampling by beginning, not with an acoustic waveform, but with an analysis of the physics of the sound-producing mechanism itself. Physical modeling parameters correspond, as nearly as science allows, to real-world performance parameters, and make it possible for a much wider range of expressive variety in the context of a single perceived instrumental source. Jaffe [1995] has stated that:

> Marvelous realism can now be combined with the malleability that is necessary to synthesize expression. Furthermore, this malleability is of exactly the right kind--the parameters of the synthesis technique are precisely those that go into the construction and performance of the real-world sound-production system.

Stringed instruments, while so central to the creation of mainstream Western orchestral music, are especially difficult to simulate in a MIDI performance context. In the mid-80s Smith developed a violin model based on “non-linear bow dynamics” [Smith, 1986, quoted in Jaffe & Smith, 1995].

According to Jaffe & Smith (1995) this model, while “extremely effective”, was of limited practicality, in that it was prohibitively expensive due to the requirement of a high-order filter to model the body of the violin. The authors report that it also suffered from a certain unpredictable sensitivity to parameter values. Smith developed an alternative model by 1993. Jaffe and Smith describe how, with the new physical model:

> …the body filter is permuted with the string and combined with the excitation function, greatly reducing the computational expense of the model and removing the problem of the parameter sensitivity …The new version is capable not only of representing a specific type of ... string instrument (e.g. a viola), but even a particular instrument (e.g. an Amati with a 19th century top and an open crack in the back.)

The authors claim the new physical model allows for the creation of smooth transitions between notes, and enables a variety of bowing styles to be synthesized, such as legato, marcato and martele. The new model also offers a method for supporting such left-hand techniques as vibrato and glissando, and the efficient simulation of pitch-correlated bow noise. The authors claim that examples of string music from various periods of music history have been convincingly synthesized in real time. The digitally realized bowed string model Jaffe & Smith describe consists of three computational blocks which can be summarized as:

1) the "bow",

based on a periodically-triggered ‘excitation’ table (PET), where the excitation table is derived (off line) from measurements of real instruments.

2) the "string"

The string portion of the model is a digital waveguide string, in which a delay line holds approximately one period of sound at all times. The output of the delay line is lowpass filtered, summed with the bow output, and fed to the delay line input.

3) a pitch/vibrato module.

The pitch/vibrato module, controls both the bow and the string. The bow module feeds its output to the string module, which in turn produces the sound. Jaffe and Smith (1995) describe how the illusion of bowing is enhanced by complex computation:

> The effect of bow position illusion is further strengthened by simulating the burst of noise that is produced when a string slips under a bow … This noise occurs naturally when the string is slipping under the bow, and its duration per period is obtained by multiplying the period by the

> distance from the bow to the bridge divided by the string length. In principle, the string-slip noise must be convolved with the impulse response of the instrument body.

Making an expressive bowed-string "score" using physical modeling synthesis must inevitably still draw upon the traditional body of experience needed to coach a human violinist. Thus a sophisticated knowledge of instrumental technique and style remains invaluable to securing a satisfactory simulation, despite the new found expressive power brought to the world of digital music making by research into physical models of traditional real-world instruments.

But if one ventures beyond the issues involved with synthetic simulation of real world instruments, using digital techniques based on theoretical physical models, it is important to realize we are witnessing the dawn of a new and exciting world of creative possibilities previously confined to the realm of fantasy. Our imaginary future multimedia art machine is evolving, and the field of research touched on here has immense implications for how we will perform our multimedia art in the new millenium. Who can now imagine what form it will eventually take, when we consider that:

> Such duplication of existing instrumental sounds is only the starting point. By extending the physical parameters far beyond their usual values, it is possible to distort and abstract the simulated physical mechanism, creating powerfully-expressive new instruments. The physical modeling approach assures that these imaginary constructs behave in ways that make intuitive sense, drawing as they do on our real-world experience. [Jaffe, 1995].

One of the most popular applications of physical modeling has been in the recreation of early analog synthesizers. One of first companies to offer this technology was Clavia with the *Nord Lead* keyboard [1]. Physical modeling synthesizers like the *Nord* use computer technology to model the old analog devices, and even emulate the many knobs and sliders on the front panel that control the different components. They can reproduce the warmth and quirky sound characteristics, but are stable, tunable and MIDI compatible.

Techno-music

The huge impact low-cost digital music technology has had on popular music is of great importance to an overview of multimedia. The changes brought to the production of contemporary popular music by the introduction of what have become

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[1] http://www.macmidimusic.com/nordrack2.html ;accessed 15-01-03 .

the two primary tools of digital music - the digital audio ‘sampler’ and the MIDI system - have had a profound effect on popular music, contributing to the emergence of a significant cultural phenomenon known broadly as *dance music,* also referred to as *techno-music* or *electronica*.

Techno-music had its stylistic roots in what became known as *Disco* music, and in fact it can be argued [Lopez, 1996] that contemporary electronic “dance music” and “disco” are a contiguous phenomenon – the name change is arbitrary when one considers that:

> Things that we now take for granted came about from the pioneering days of Disco. Such as the extended single, the concept of multiple mixes or remixes (which itself dates back to the 60s Dub scene in Jamaica), the Break, mixing between songs (as opposed to a simple segue) and sampling, all came about or were popularized by Disco. It also brought about improvements in sound with the introduction of the higher fidelity twelve-inch single which is still with us today. More importantly, Disco is the foundation of all modern dance music and a force that shook up the establishment and tore down many socio-economic barriers. People of all incomes, races and sexual preference were able to enjoy Disco and have fun…. Dance music, Techno and House are just the logical evolution of the Disco-period. [Lopez, 1996].

Disco is an up-tempo style of dance music originating in the early 1970s. It was in turn strongly derivative of *funk* (a vigorous, hard-edged, rhythmic, electro-musical style, popular with Afro-American audiences in large U.S. cities[1]). The name *Disco* is a derivation of the French word *discotheque* and disco had no pretensions other than its primary goal - to facilitate dancing, in specialist venues set up specifically for this form of entertainment.

Stylistic elements of disco appear on earlier records (notably the 1971 theme from the movie *Shaft* by Isaac Hayes), but in general it can be said that first “true” disco songs were released in 1973. One of the earliest was *The Love I Lost* by Harold Melvin and the Blue Notes from Philadelphia, featuring the vocals of Theodore Pendergrass. Initially, disco songs catered to a nightclub/dancing audience, rather than general audiences such as radio listeners. By 1975 disco began to reach mainstream popular recognition, with hit songs like *The Hustle* (popularizing a disco dance style) and *Love To Love You Baby* achieving wide commercial success. Disco's popularity peaked in 1978 and the first half of 1979, driven in part by the success of the late-1977 film *Saturday Night Fever*, featuring actor John Travolta and the music of the *BeeGees*.

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[1] Source: *Wikipedia*. URL: http://www.wikipedia.org/wiki/Disco ; accessed 20-11-02.

Techno music has evolved out of the beats of the "disco" era, but was also heavily influenced by the industrial synthesizer music of German avant-garde electro-pop bands, most notably *Kraftwerk*. *Kraftwerk* have been especially influential in introducing a technological approach to rock music, combined with sci-fi robotic imagery. *Kraftwerk's* creative masterminds, Ralf Hutter and Florian Schneider, pioneered the use of Moog synthesizers and drum machines in popular music in the 1970s, an innovation which has heavily influenced the later evolution of techno-music.[1]

Above: Kraftwerk Playing *the New Song* at the *Tribal Gathering* 1997.

The evolution of Techno-music has been further driven forward by emerging music technologies. Early drum machines, such as the Roland TR-808 Rhythm Composer, designed as rehearsal devices, but capable of producing powerful cyclic rhythms, were especially important to the evolution of this popular new electro-rhythmic genre.

As Blashill [2002] describes:

[1] RollingStone Artist Biography http://www.rollingstone.com/artists/bio.asp?oid=910&cf=910 ; accessed 6-12-02.

> In the early '80s, Detroit techno pioneers like Juan Atkins and Derrick May - inspired in part by club DJs who'd begun using turntables as instruments, and limited by a profound lack of expendable dough - began picking up secondhand analog drum machines and keyboards that had been dumped by musicians who thought the equipment sounded too mechanical. Atkins & Co. were able to create techno music with these seemingly clunky tools precisely because they didn't *want* machines that sounded like human drummers. "For some, I guess, 'synthesize' means 'duplicate,'" Atkins says. "But for me, 'synthesize' is synonymous with 'create.'"

Blashill [2002], in his article entitled *Six Machines Which Changed the Music World* lists the Roland TR-808 Rhythm Composer, Roland TR-303 bassline synthesizer, the Technics SL-1200 turntable (on which "scratching" was first introduced), the Nord Lead 1 keyboard (by Swedish Company Clavia, 1995), and Amek System 9098 Equalizer (a device that producers typically use to attenuate, emphasize, or otherwise "pump up" selected frequencies in the audio spectrum) as landmark technologies which enabled, defined and drove forward the new electronic dance music.

Above: the Nord *Lead 1* keyboard

For example, Blashill recounts how TB-303 bassline synthesizer[1] (below)

[1] A dedicated digital music generating device produced by the Roland Corporation in 1982 to be used by musicians as a rehearsal device and for recording "demos", or rough sketches of songs, which could be later re-recorded with "real" instruments. The 303, however, sounded nothing like an actual bass guitar for which it was meant to substitute.

was influential in the evolution of the techno-music sub-genre *acid house*, but only because new creative possibilities, not intended or anticipated by its designers, were discovered through experimentation.

Blashill goes on to describe how Earl "Spanky" Smith purchased a second-hand TB-303 bassline synthesizer and, by tinkering with its controls, discovered a new sound:

> Jones, who was beginning to DJ under the name Pierre, played with a row of knobs - Resonance, Decay, and Cut Off Freq - for adjusting the bassline; the controls were meant to be set, then left alone during recording or rehearsal. But Pierre programmed a bassline, hit the Run button, then cranked each of the knobs to its upper limit as the bassline was playing back. The 303 reacted with a piercing, almost obscene screech…
>
> Though they didn't know it at the time, by adding the warped, druggy 303 sound to then-standard club beats, Pierre and Spanky had invented a new genre of dance music: *acid house.* They dubbed themselves *Phuture*, then released their 303 experiment as "*Acid Trax."*

Bull [1996] makes an interesting connection between *acid-house*, and the work of Italian Futurist L. Russulo. While Bull (p.1) seems to put the birth of house music in the '70s, which seems contentious, his argument is nevertheless thought-provoking and relevant:

> the development of electronic musics and proliferation of technologies through the 1970s to the 90s has seen the amalgamation of a multitude of cultural music 'systems'- the formation of a universal and hybridised(ing) 'pop aesthetic'...
>
> The Roland TB-303a, was a small and unassuming 'synth box', released by the Roland company in the early 80s - a technology so paradoxically convoluted that it ushered in an entirely new era of technological appropriation. Along with a number of cheap drum machines (the '808', '909' '606' and '303'), it became the tonal basis for the minimalist electronic dance movement known as 'acid house', indicative of a revival in a pseudo Russulolian mode of experimentation. ...Repetition, tonal graduation, ambiguous structure and the sound of the 'machine' were all crucial, if initially accidental, elements that helped form the basis of the new 'techno' aesthetic. '

Russulo's *The Art of Noises*[1] suggests a theory of culturally derived 'sound education', whereby, through an historical process, mankind's aural perception skills have been gradually attuned and continuously re-attuned to differing planes of 'cultural frequency' [Bull, 1996]. Bull quotes Russulo:

> 'the ear of an eighteenth century man could never have endured the discordant intensity of certain chords produced by our orchestras'.

[1] Direzione del Movimento Futurista, Milan, 1913.

Building on this hypothesis, Bull argues that:

> It is (the) very ‘exposure’ and ‘control’ of certain tonal elements afforded through electronic media, that has perhaps caused us to ‘re-hear’ sounds in terms of new psycho-acoustic frameworks, and to begin to deconstruct the largely representational metaphors of traditional orchestrated and acoustic forms.... the ‘pioneers’ of techno were borne on an enigmatic wave only now reaching the main-stream ear...Russulo and the futurists sought to move deliberately away from the purely representational tastes of traditional orchestral music forms, stating defiantly in their manifesto that ‘the art of noise must not limit itself to imitative reproduction’, and harping upon the fact that in order to experience a poignant ‘freshness’ - a ‘new musical reality’, the ‘limited circle of pure sounds must be broken’, and staunch representationalism would have to yield to Russolo’s ideal of abstract tonal impressionism.

The desk-top computer revolution has allowed for the absorption of the techniques and concepts at the core of this recently evolved, technologically-driven music genre into the design of dedicated computer music programs.

One of the most influential programs in this evolution from hardware to software-based dance music composition was Swedish software company Propellerheads’ *Rebirth*.

Right: *Rebirth* interface. Patterns or musical fragments can be stored and accessed from the numbered tabs on the left, and these can be chained into songs. There is a synthesizer (top centre), and drum machine, (bottom centre), and mixing / audio sweetening (on the right).

Propellerheads have developed the concept of *Rebirth* into a meta-program they have named *Reason*, the interface for which depicts a virtual electronic dance music studio on screen (see next image).


The Big Picture.rns*
MIDI IN DEVICE
AUDIO OUT
DDL-1
DIGITAL DELAY LINE
DELAY 1
RV-7
DIGITAL REVERB
REVERB 1
SUBTRACTOR
POLYPHONIC SYNTHESIZER
NN-19
DIGITAL SAMPLER
RHODESMKII
CONTROLLER
MOD WHEEL
VELOCITY
LFO
FILTER
ENVELOPE
AMP
OSC
PITCH
PEQ-2
TWO BAND PARAMETRIC EQ
EQ 1
COMP-01
COMPRESSOR 1

Above: Interface from Propellorheads "*Reason*" electronic music composition application which allows the user to build a virtual on-screen "rack" of standard electronic music generating devices, including a virtual mixer, digital signal processing devices such as simulated reverberation, digital delay and distortion (several examples occur in the illustration above), drum machine (labeled "Redrum" above) modeled on the early Roland drum machines, MIDI sequencer (the "Matrix"), synthesizers (labeled "Subtractor") and samplers (NN-19). The metaphor is fully developed, including depictions of knobs, sliders and buttons (controlled by the mouse), even going to the detail of screws, "hand-written" labels, and "device" serial numbers.

Reason is a comprehensive music production software design (which only omits actual digital recording from its available facilities[1]) and is a development which can be viewed as an interesting and importantt step in the evolution of the multimedia meta-machine, in which all art technologies merge within a universal interface.

Above: The reverse panel of the virtual rack available in the electronic music generating program *Reason* (accessed by hitting the Tab key) which allows users to connect the virtual sound generating and processing devices, using virtual patch cords. This program celebrates the evolution of techno-music by enshrining generic graphical interpretations of the seminal devices found in an electronic dance music studio.

In terms of practical multimedia music and sound production, the data limitations of the present CD-ROM delivery format, and similarly, bandwidth restrictions on the Internet, put a strict “minimize the megabyte” pressure on the audio element of multimedia art. Certain broad attributes of Techno-music tend to make it eminently suitable for inclusion in multimedia art pieces, not the least of which are:

[1] Presumably to avoid a commercial confrontation with the major established vendors of digital music software such as Steinberg (*Cubase*) and E-magic (*Logic)* - the latter recently purchased by Apple Corp.)

- The fact that reasonably professional sounding music can be created and edited by “non-musicians” or operators with only limited (perhaps even no) traditional music skills, but with a command of the digital technology, a new media literacy and, most importantly, a sense of style, imagination etc.

- The use of ostinati: bass-lines, drum loops and repetition of samples - a stylistic attribute of techno-music, lends itself well to a work process which is, in the age of the CD-ROM, ruthlessly trying to shed megabytes from the art project. A common technique is to use repeating sound loops with variations provided by accents of (often non-pitched) musical events triggered by interactive events (e.g. clicking the mouse, or passing a cursor over a hot-spot).

Algorithms in computer-assisted music composition

Composing music is an intricate task, which must be analysed within a culture specific and historical context. Generating musical ideas is a complex cerebral process, with a great many context-specific or individual approaches possible. The work of the composer often involves trying many different combinations of initial ideas against each other, and choosing one over others, using highly subjective or intuitive decision-making processes. Creating a musical sketch is a process of integrating different musical ideas into a coherent structure. Those compositional decisions are informed by the cultural and musical background and education of the composer. Computers can offer assistance in the traditional processes of music making by, for instance, the creation of printed musical scores, but beyond this, in the field of computer-assisted music and electronic music and multimedia art, a variety of approaches to music composition have been developed.

Musical styles have tended historically to evolve towards a crystallized form, and eventually to generate definable sets of rules. Computers can be programmed to create structures based on definable sets of rules, and thus can become an aid in the composition process by offering access to algorithmically generated musical material, based on either traditional (e.g. diatonic), serial, or other structures, or some set of newly conceived rules. Computers can also generate random numbers, and by using this randomly generated input, chance elements can be introduced into computer-aided compositional processes.

Algorithmic composition, sometimes also referred to as automated composition, basically refers to

> …the process of using some formal process to make music with minimal human intervention [Alpern, 1995].

i.e., the application of rigid, well-defined rules to the process of composing music.

One important strand of this computer-aided music composition is based on the internationally accepted MIDI protocol (the industrial, cross-platform standard for storing sequences of electronically-produced musical performance data)[1]. Algorithms can be constructed with the intention of generating MIDI performance information, which can then be used to trigger patterns of sound.

Apart from using computers for their labour-saving advantages, from the point of view of traditional approaches to music composition, there is inherent interest in the possibilities that large-scale computational power may offer in serial composition, chance outcomes in composition, and other algorithmic approaches to composition which generate new or unexpected musical material.

Once again, this is not a new field, and as far back as the Renaissance, composers frequently followed stringent compositional rules, which could be considered algorithmic, specifically in the *cantus firmus* composition process, whereby the succession of pitches from the original theme must be preserved during the whole work. An example is:

> …Guilluam Dufay´s Communio, from the Missa Sancta Jacobi. The original theme contains seventy-six notes. The first note and word of this theme is copied to the first note and word of the mass and the subsequent notes and text are copied into the piece until the original theme ends. This copying process can be viewed as an algorithm and could be implemented with a computer. [Beckert, 1995].

Canonic composition in the late 15th century could also be considered as algorithmic [Maurer, 1999].

> The prevailing method was to write out a single voice part and to give instructions to the singers to derive the additional voices from it. The instruction or rule by which these further

[1] Using a MIDI keyboard as a trigger device, the MIDI protocol allows the composer to record which keys were hit by the performer, the velocity of attack, duration, and when the keys were hit in relation to a digital clock; actual sounds are not recorded. This performance data is then used to re-trigger stored sounds in real time during playback of the music.

> parts were derived was called a *canon*, which means "rule" or "law." For example, the second voice might be instructed to sing the same melody starting a certain number of beats or measures after the original; the second voice might be an inversion of the first or it might be a retrograde (etc.). [Grout, 1996, 166].

Johann Kirnberger is considered to be the first author of a consciously algorithmic composition system, publishing in 1757 *Der allezeit fertige Polonoisen und Menuettencomponist*. [Barnig, 2000, 1]. In this work he constructed a system that contains a fixed number of small pieces or modules of pre-composed music, which are available for each section of the composition, and are identified with a number from 1 to 6. Chance is introduced through the use of dice, rolled to determine which piece in each section is to be used in the compositional outcome.

W.A. Mozart's *Musikalisches Würfelspiel* is another musical dice game, first published in 1793, two years after his death. Neither the original manuscript nor a direct reference to Mozart was ever found, however his authorship is no longer questioned by musicologists [Barnig, 2000, 1] [1].

Experimentation with similar approaches to composition has been attributed by Beckert [1995] to: A. Kircher (1650), Mizler (1793), J. Haydn (1793), F.G. Hayn (1798), J.C. Graf (1801), C.H. Fiedler (1801), L. Fischer (1801), A. Calegaris (1801) and G. Catrufo (1811).

Influenced by Zen Buddhism, John Cage utilized random elements in many of his compositions, in an effort to preclude ego influences from his compositional activity[2].

> The radical changes in his methods for harnessing chance operations in the 1950s reflect his abiding search for strategies which would a) remove opportunities for his own tasteful shaping, b) nurture thought-stilling perceptual acuity in performers and listeners, and c) ensure that his work truly tapped the universe's natural processes and did not impose artificial, thought-full contours upon them (Nelson, 1995).

In *Reunion*, for example, Cage has the piece performed by playing chess on a photo-receptor equipped chessboard:

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[1] The shareware application *CyberMozart 3.0.1* - recreates Mozart's Musical Dice Game. In Mozart 's game one threw two dice a total of sixteen times to choose items from his list of 176 measures. *CyberMozart* uses random number generation (press the "Compose" button) to create a new English Country Dance (the promotional web-site claims 'trillions' of possibilities). *Cybermozart* is available for download at: http://www.hitsquad.com/smm/programs/CyberMozart ; accessed 13-11-02.

[2] Nelson, M.; 1995. *Quieting the Mind, Manifesting Mind: The Zen Buddhist Roots of John Cage's Early Chance-Determined and Indeterminate Compositions* http://www.sun.rhbnc.ac.uk/Music/Archive/Disserts/mailto%20mdnelson@alumni.princeton.edu ; accessed 24-02-04.

> The players' moves trigger sounds, and thus the piece is different each time it is performed [Alpern, quoted in Maurer, 1999].

Cage also constructed compositions using other non-traditional, rule-based strategies, for instance by referring note organization to extraneous natural phenomena. *Atlas Eclipticalis* [1961] was composed by laying score paper on top of astronomical charts and placing notes simply where the stars occurred, in this way purposely delegating the compositional process to indeterminacy [Schwartz, 1993, 92].

The post-World War II era saw composers involved in twelve-tone method and serialism, attempting to control all parameters of music and to objectify and abstract the compositional process as much as possible.

> Decisions over everything from notes to rhythms to dynamic markings were often subject to pre-composed "series" and "matrices" of values, which, in effect, "automated" many of these parameters by determining the order in which each must occur in a piece. These series and matrices were, then, the "algorithms" which superceded the human creative process. Serialism can thus be labeled "algorithmic" or "automated" composition in a rather pure sense, especially when it strives to integrate as many musical parameters as possible. [Maurer, 1999].

Messiaen's 1949 piano etude, *Mode de valeurs et d'intensites*, is a notable example of this approach. It was built from a thirty-six pitch series, each pitch of which was given specific rhythmic, dynamic, registral, and attack characteristics with which it was to be used in the composition [Kostka, 1995, 526-527]. Another notable example of non-computer oriented "stochastic" composition can be found in Stockhausen's *Klavierstücke XI,* in which the sequence of various fragments of music are performed by a pianist in random sequence [Maurer, 1999].

Compositions based on strict algorithmic processes have continued to be devised well into the age of computer technology, which nevertheless do not rely on computers. For example, much of the early work by composer/multimedia artist Ron Pellegrino involved some algorithmic processes.

> *Metabiosis* is an installation from 1972. The installation was set-up with lenses that sensed a change in air flow. A light source was projected onto the lenses, so that as they moved, light was cast to varying parts of the room. On the walls were photoresistors that measured differences in light intensity; the energy coming from the photocells was then sent as control voltages to several analog synthesizers. The voltages were then used to trigger changes in frequency, timbre, and amplitude. The algorithm here involved the testing and measurement

> of air flow patterns as each new audience member entered or left the performance space. (Burns, 1996).

Harnessing computers for algorithmic compositional activity, with the benefit of hindsight, was inevitable in the digital age. The earliest experiment in computer-generated music composition was that by Lejaren Hiller and Leonard Isaacson at the University of Illinois, U.S.A., in 1955-56. They used the *Illiac* 'high-speed' digital computer, to program basic material and stylistic parameters, producing the *Illiac Suite* (1957). The 'score' for this piece was output by the computer, and then transposed into traditional musical notation for performance by a string quartet. [Maurer, 1999]. This piece is discussed in further detail below.

Iannis Xenakis is one of the best-known composers to have worked in the area of computer-generated music composition. One of the most important techniques developed by Xenakis is what he calls "stochastic music". The term "stochastic" is derived from the Greek "stochos" meaning "guess", and is used in physics to describe a process which uses an element of chance. Xenakis first used this term when he devised a method of composition based on probability theory. [Stephen, 1998, 1]. His work *"Morsima-Amorsima"* (1962) was completely composed by algorithmic composition, using a stochastic music program that compiled a score from a list of note densities and probabilistic weights, making decisions or choices with the help of a random number generator. [Beckert, 1995]. Xenakis' approach was different from that taken by the composers of the *Illiac Suite* however, in that, rather than have the computer completely create the musical score, he was relying on the high-speed computations to introduce indeterminacy and to develop musical material, which he then crafted into a finished musical structure. [Maurer, 1999].

Consideration of the concept of computer-aided music composition leads us to a central question:

Can an analysis of the creative process of composing music generate, or be mapped upon, mathematical formulas that when applied, in turn, may produce aesthetically pleasing results?

Beckert [1995] breaks this problem down into two parts, which can be summarized as:

1) *Practical Issues:* How to understand the compositional process well enough to express it as an algorithm; and

2) *Aesthetic Issues:* How to program a computer to choose between good and bad music.

The success of computer-assisted algorithmic composition will initially depend upon there being an accurate analogy between a composer's creative process and the implemented algorithm. To put it another way, is it possible to reduce the act of creativity to a set of mathematical parameters? From a programming point of view it begins with a process of deconstructing large problems into smaller, more manageable components.

Of course, the question must be asked: why even attempt to get machines to carry out what has traditionally been viewed as one of the most romantic, creative, indeed human of tasks – the creation of beautiful music? In her introduction Beckert [1995] naïvely stresses labor- and time-saving advantages in defense of algorithmic composition, and even seriously suggests it as a way of generating new melody "when uninspired or untalented". Such musical motivations could only be considered dubious, if the resulting "music" is uninteresting. What *is* of primary interest to musical artists in general, and to this inquiry into the processes and aesthetic possibilities of digital multimedia art, is the question:

Does algorithmic composition lead to the creation of engaging musical material that would otherwise not have been possible or likely? i.e. Does it generate new ideas?

Historically, the broad category of algorithmic music composition with computer assistance can be sub-divided into three main areas (Burns, 1993):

1. algorithms for compositional structure;
2. algorithms for sound synthesis;

and

3. algorithms for the correlation of sound synthesis with structure.

Beckert [1995] states that mathematical models are, in most cases, judged to be algorithmic, when or because they fulfill Knuth´s five criteria, (derived from "legendary" computer scientist Donald Knuth 's classic work, *The Art of Computer Programming: Fundamental Algorithms*). Knuth's criteria are:

1. Mathematical models are finite and have a calculable number of steps to complete the process.

2. The single steps follow a rigid mathematical formula.

3. The input may consist of initial values (parameters) or tables of values.

4. The model generates output, (in the contemporary world an example would be in the form of MIDI-files).

5. Most models are sufficiently basic, because they only use simple computations to generate the output.

To develop an algorithmic compositional system based on Knuth's criteria requires a highly intellectual approach, entirely free of culturally-based aesthetics or pre-conceptions. It requires a rigorous belief in the value of experimentation in music compostion, free from the imperative to entertain. It is a science of music. As such it is music-making with an eye, or at least an ear, to the future.

The forms of computer-assisted algorithmic composition [1]

a) Stochastic algorithms

The simplest and relatively primitive form of algorithmic composition is to program a computer to generate notes on a completely random basis, in order to piece together melodic fragments, i.e., many or all of the creative decisions in the stochastic method are simply left to chance. This method of randomly generating musical events may, in theory, produce some interesting outcomes, but ultimately has, in this simple form, proved to produce results which are not musically satisfying enough for it to have become a serious tool of contemporary music composition. Random sounds have the potential to become simply boring, even irritating, if extended over time.

However, a great amount of conceptual complexity can be introduced to stochastic computations with the introduction of statistical theory and Markov Chains to direct the output of the algorithm towards a more aesthetically satisfying or coherent result. Markov Chains are used to make compositional decisions, based on probability. Information is linked together in a series of states, based on the probability that state A will be followed by state B. The process is continually in transition, because state A is then replaced by state B, which is followed by another

[1] The tables used in this section are based on those offered by Beckert [1995], with some modification and expansion.

state, and so on. Each defined state is related to a table of possible following states, showing the corresponding probability that the transition to those states will occur. The states themselves can contain information about just one note, if the composer works on the note-level, or they can contain predefined melodies if the composer works on a higher level. This allows the composer to increase the likelihood that states are combined that will result in a pre-defined “pleasing” musical outcome.

The first composers to use a computer to generate music, Lejaren Hiller and Leonard Isaacson, were also the first to address the aesthetic challenge introduced by randomness, and begin to experiment with techniques involving Markov Chains, statistical analysis and other stochastic procedures. In the *Illiac Suite*, they attempted to simulate the compositional process itself on the computer by using a generator / modifier / selector approach [Maurer, 1999]. Hiller and Isaacson’s approach

1) uses Markov Chains and Markov Analysis to generate raw material as a compositional base; then

2) applies different techniques (such as permutation and geometric transformation) to manipulate or modify the generated material; and finally

3) uses selection rules to choose suitable material for structuring the compositional outcome (“critic” function).

Type of algorithm[1]	pure **stochastic** (based on one item in the probability distribution of an ordered set of observations)
Description	musical values such as notes are randomly generated
Complexity	low
Composer defined parameters	• number of generated notes • seed value or number of steps
Aesthetic quality of output	low
Advantages	• easy to program • easy to understand
Disadvantages	• low quality of output
Subtypes	• aleatoric algorithms • rule-based stochastic algorithms

[1] After Beckert, 1995

A sub-type of the stochastic model is the *aleatoric model*, in which indeterminacy or coincidence is applied to compositional processes that were originally deterministic. Under this model, building blocks or musical fragments of the composition are predefined by the composer, but the decision as to which elements are combined is based completely on chance or coincidence.

One example for this specific type of the stochastic-aleatoric way of generating music is the composition *Music of Changes* (1951) by John Cage. In this piece, a random number generator was used to make decisions as to which predefined elements were combined to make a composition – not in principle any great leap forward from Kirnberger of 1757. This so-called "Monte Carlo method" is a simple example of a rule-based stochastic method: music is generated by applying rule-based style selection to randomly generated input.

As was mentioned earlier (p.314), Xenakis was a pioneer in applying musically intelligent ideas to the use of stochastic music composition. Stephens [1998, 1] describes another interesting example of Xenakis' approach, which he called "The Theory of Sieves":

> This involves taking the full range of an instrument and creating a scale by dividing the range into steps. This creates a sequence of pitch, which works similarly to the even-temperament tuning scale, but instead of advancing in semitones the scale may advance by steps of a different size. Notes are then chosen by overlaying the resulting scale with a grid also divided into steps but of a different size. The notes that occur in both are considered to have "fallen through the sieve" and form a pitch class.
>
> Stochastic processes are also used to distribute events once they have been chosen using the theory of sieves or some other method. A typical example of the application of this procedure might read as follows: The composer sets a number of unchanging parameters, such as the length of time to be covered, the pitch range to be used, and the nature of events that can occur (such as loud pizzicato or quiet piano clusters) and the stochastic procedure is used to determine where within these parameters the events occur. Xenakis first used this method to control large masses of sound in his orchestral work Metastasis (1953-54).

b) Iterative algorithms

Another relatively simple, but effective way of using computers for calculatin musical events is known as *the iteration method*. In order to produce data that is useful in a musical sense, the systems of mathematical equations are put through a process of iteration, whereby the solution is calculated, and then repeatedly presented to or fed back into the equations. Each new result then in turn is offered as the input values for the next iteration, and so on. The complete set of these solutions over time is referred to as an “orbit” of the system.

Beckert [1995] lists three categories of behaviour into which these systems can fall upon iteration, usually referred to as *attractors* (since they represent the set of points towards which the values of the system tend).These categories are:

i) *constant*, cases where all points in the orbit are the same. These will be unlikely to produce interesting musical results.

ii) *oscillatory,* cases where all points belong to a repeating set of points. These have a much greater possibility of producing interesting repetitions, providing the period (the number of distinct points in the repeating set) is large enough. In practice the set is usually small, resulting in cyclic melodies that oscillate between only a few pitches, quickly invoking tedium.

iii) *chaotic*, or strange attractors, where the orbit behaves seemingly in a random manner, never visiting exactly the same point twice. “Chaotic” behaviour produces the most interesting results. The attractor of a chaotic orbit defines a certain range of values, amongst which the system will wander, often returning to similar, but not quite the same series of points. Interpreted musically, this can often be perceived as being variations on a theme. In this sense, chaotic orbits can be considered to have potential value as generators of interesting or recognisably musical material. The material thus produced has a high degree of correlation with its immediate past, but is, at the same time, continuously generating something new.

Di Scipio [1999] describes the use of chaotic dynamics in iterative algorithms to create complex auditory “images” or simulated textural and environmental soundscapes. Examples include sound textures which, it is claimed, are reminiscent of “rains, thunderstorms and more articulated phenomena of acoustic turbulence”.

> With rain-like sounds, it is important to introduce some perceptual clue of the different surfaces impacted upon by the raindrops. This can be done by choosing the appropriate iterate, *n*. Upon listening, larger values of *n* result into more resonant materials, while smaller values are more reminiscent of "deaf", very compact surfaces. Furthermore, by visiting the extreme regions in the system phase space, we can control the perceived size of droplets,

> resulting either into a drizzling or a driving rain. As the droplet size gets larger, eventually a hailstorm is approached.

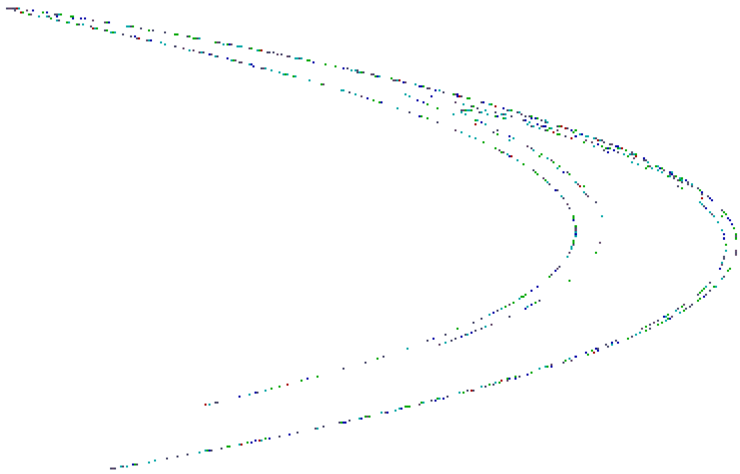
This research suggests the possibilities of using algorithmic techniques for new experiments in electroacoustic music and the creation of synthetic, but credible, auditory scenes in multimedia applications and virtual reality.

Type of algorithm[1]	**iterative** (the repetition of a sequence of steps under the control of a conditional)
Description	mathematical equations are put through a process of iteration, and the calculated solution is fed back into the equation
Complexity	Low
Composer defined parameters	• number of steps • constants
Aesthetic quality of output	high, because the material that produced has a high degree of correlation with its past, but is at the same time always producing something new.
Advantages	• easy to program • easy to understand • good possibilities for influence on the behaviour of the algorithm through the constants • high quality of output
Disadvantages	for some constants the algorithm will produce boring results
Subtypes	• *Henon Map*: a simple two dimensional system, based on a simplified model of a star's orbit around a gravitational center, discovered by the French astronomer Henon. • *Logistic Map*: equation is a simple, one-dimensional equation that is used to model population growth over time.

The Henon map is an example of a very simple dynamic system that exhibits "strange" behavior. The orbit traces out its characteristic banana shape, but on close inspection, the shape is made up of thicker and thinner parts which occur in a fractal configuration. Upon magnification, the thicker bands resolve to still other thick and

[1] After Beckert, 1995

thin components, and so it goes forever. The equations that generate this strange pattern perform the mathematical equivalent of repeated stretching and folding, over and over again.



Above: Henon Map

c) Serial algorithms

Polyphonic composition techniques of the 14-15th centuries (for example the canon) were precursors of the serial model, where a series of notes is first defined (melodic theme is composed). This series is then used as a basis for variations.

By the time of Wagner, diatonic music had matured to a peak of harmonic sophistication approaching chromaticism [Dorak, 2000]. It was as though the diatonic system was being pushed to its limits, and was reaching a point of over-complexity. Thus, with hindsight, the time was right when Schoenberg's 12-tone method opened the door to a new system of non-diatonic music, which became known as *serial music*. A “series” within the serial model is defined as having the following properties:

- The series comprises all twelve notes of the chromatic scale

- No note appears more than once within the series

- Each series has 48 possible versions, as the series may start on any semitone, and can be played:

i)	forwards,	
ii)	backwards,	(retrograde)
iii)	“upside down”,	(inversion)
iv)	upside down and backwards	(retrograde inversion)

(4 x 12 = 48)

Repetition of a note is permitted before the next note is played, providing it is in the same octave.

As the basic building block of serial music, the series has a range of functions during composition:

- directing the compositional process

All musical material that appears during the compositional process is derived from the original series, and therefore the original series directs the compositional process and determines compositional outcomes.

- developing new material

The series develops new material during the compositional process, according to the structure- and form-rules that are applied to it.

- giving the basis for deriving the permutation rules

The original series is permutated during the composition. The permutation rule can be derived from the original series by considering the mathematical relationships between the notes within the original series. This gives a complex hierarchical and\ self-referential system that is directed by the functional aspects of the original series.

To produce variations of the original series, one must substitute the series with ordinal numbers. Then manipulation is a simple matter of using the numbers as input into functions. The results are interpreted as the new series. Variations of the initial series are obtained by applying the following combinatorial permutations:

- inversion,

- retrograde or
- retrograde inversion

In a computer-assisted context, the original series may be generated with a random number generator, that follows the four rules listed above; or it may be chosen completely by the composer, if it is desired for example to establish a series of proportional values which are then applied repeatedly, starting with the initial value to obtain a series.

For each note-parameter in the series (pitch, duration, tempo, etc.) a material list is defined, containing the values of the original series. This list is then used for manipulation.

First, some of the elements from the list are repeated, using a multiplication function, then the elements are manipulated using a permutation rule. The output is a new series that is derived from the original series.

Serial Music composition requires precision and a rigidly-ordered, cerebral approach. Rules largely define the outcome which tends to be atonal.This very controlled and systematic approach to composition offered the perfect material for algorithmic experimentation.

Karlheinz Stockhausen was the first composer to make computer-assisted use of this principle of series, in *Studie I* (1953). Another composer who has used the serial model is Gottfried Michael Koenig. Since the 1960s, Koenig has explored possibilities of formalizing composition-theory in the form of computer programs.

Music based purely on the serial model is not judged by traditional aesthetic criteria, but instead by the functional context of the series within the composition. In other words, the artistic aim is not to produce a pleasing melody or any particular mood or emotional connotation, but rather it is of greater importance that the composition has integrity, based on the defined mathematical operations and functions. Thus, the computer becomes a valuable and impartial tool for calculating the variations of the series and even the series itself, if the composer chooses to base the music on a mathematical concept, or randomness, rather than some culturally-based aesthetic. Serialism is not purely a contrived exercise valuable only for it's intellectual astringency.

As serialism matured, composers such as Webern refined 12-tone composition into a means of producing extraordinarily beautiful and elegant works of art (Linton 1998). It is certainly possible to evoke emotional response with serial music. Wedin (1972) showed that variations in musical intensity, mode, pitch, rhythm, tempo, and texture have at least some influence on emotional responses. Composer George Rochberg constructed music out of both tonal and atonal languages, combining serialism with "emotional" music, dramatically reinterpreting the notion of stylistic uniformity that had been a hallmark of the Western aesthetic since antiquity (Linton 1998).

Type of algorithm[1]	**serial** (pitches, durations, timbres etc. must be ordered in a strictly defined manner before repeating)
Description	a series is defined, that is then used for creating variations. The series has the functions of: • directing the compositional process • developing new material • providing the basis for deriving permutation rules
Complexity	High
Composer defined parameters	• values of series may be defined by the composer • multiplication function • number of steps
Aesthetic quality of output	variable, unpredictable
Advantages	• easy to program • easy to state rules
Disadvantages	• strong tendency to produce atonal music
Subtypes	none

e) Genetic algorithms

A more recent approach to algorithmic music composition has become known as *genetic programming*. In general terms, genetic algorithms are said to evolve a “population” by examining the genetic code of each “individual” and assigning a fitness score that is based on the suitability of the individual for fulfilling certain pre-defined criteria. Successful individuals survive to become the parents of the next generation by a process of ‘breeding and mutation’.

[1] After Beckert, 1995

In a musical context, genetic algorithms generate their own musical materials, as well as forming their own grammar, by using a process of artificially-created "natural selection" to evolve other simple computer programs. In order to perform this process, a small set of functions and terminals, or constants, are used to describe the "domain" the composer wishes an evolved program to operate in. The subsequent process is theoretically similar to natural evolution. Such programs contain a "critic" function, which auditions the numerous automatically-produced outputs at various stages of the processing, in order to decide which are suitable (meet pre-defined criteria) for final output (i.e. rule-based selection is used to find a set of "good" phrases, the composer ultimately deciding, of course, which of these to discard and which to save)； [Maurer, 1999].

In genetic composition algorithms, the individuals are the different musical phrases, and the notes within the phrases are similar to the parameters on the chromosomes in the natural genetic encoding.

Alpern [1995] provides further insight:

> …if the human programmer wishes to evolve a program which can generate or modify music, one would give it functions which manipulate music, doing things such as transposition, note generations, stretching or shrinking of time values, etc. Once the functions have been decided on, the genetic programming system will create a population of programs which have been randomly generated from the provided function set. Then, a fitness measure is determined for each program. This is a number describing how well the program performs in the given problem domain. Since the initial programs are randomly generated, their performance will be very poor--however, a few programs are likely to do slightly better than the rest. These will be selected in pairs, proportionate to their fitness measure, and then a new population of programs will be created from these individuals, and the whole process will be repeated, until a solution is reached (in the form of a program which satisfies the critic), or a set of number of iterations has passed. Operations which may be performed in generating this new population include reproduction (passing an individual program on into the next generation unchanged), crossover (swapping pieces of code between two 'parent' programs in order to create two unique 'children'), mutation, permutation, and others. [Alpern, 1995].

Type of algorithm[1]	**genetic**
Description	a set of initial phrases is 'evolved' by applying a 'fitness' criteria, to create the final composition; uses a process that is analogous to natural selection/evolution
Complexity	high
Composer defined parameters	• initial set of phrases • fitness criteria • number of iterations
Aesthetic quality of output	high
Advantages	• makes allowance for aesthetics through critic function • only the building blocks and critic need to be provided • will find an optimal solution, if the critic is well-defined
Disadvantages	• difficult to define/program the critic • the solution space becomes big quickly and therefore searching through it will take a long time • searching through the solution space is difficult, because of its unstructured form
Subtypes	none

f) Fractal algorithms

The word "fractal" means "broken" and originally referred to fractional dimensions and irregular (hence "broken") surfaces. Fractal geometry is a branch of experiential mathematics that recognizes that objects in the real, or natural, world are not purely square, triangular, circular, spherical, cylindrical, etc., but, rather that the latter are only idealized geometric models. For example, the Earth is usually described as a sphere, when in fact it has a complex uneven shape.

Solomon [1998, 1] states that two hallmarks of what are now considered to be fractal systems are:

1) inherent nested, hierarchical organization,

and

2) self-similarity, i.e., copies within copies within copies...ad infinitum.

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[1] After Beckert, 1995

Fractals are not limited to the realms of mathematics and computer graphics, but rather abound in nature. Tree branching, cloud structures, galaxy clustering, fern shapes, the veins of human and leaf, music, coastlines, fluid turbulence, crystal growth patterns, snowflakes and very much more, involve self-similarity.

There are in general two methods to generate music with the use of fractals:

1) using fractal pictures

2) using mathematical equations only

If fractal pictures are used to calculate music, mathematical formulas are applied to the fractal images to cut samples out of it.

Beckert [1996] outlines the procedure as follows:

- the graph of the function is set on top of the fractal picture
- all points that lay within the graph are joined together in a set
- this set contains the colour-values of the covered points
- the points(that is, their colour-value) are read horizontal from left to right and they are used for the x-coordinates in a coordinate system.
- each colour-value represents a number. This number stands for the y- coordinate in a coordinate system.

By this way, a melody is generated that must later be modified (made “smoother”), presumably in a subjective or intuitive manner, in order to create a more “pleasing” result. Linear functions and polynomials are most useful for this method, but other functions are also said to give interesting results.

The generation of polyphonic music from fractal pictures involves a slightly different technique. A fractal picture is divided into rows, according to the predetermined number of voices designated for the composition by the composer. Those rows are subsequently used to generate the different voices, in the manner described above.

Type of algorithm [1]	**fractal equations**
Description	linear transformation is used to generate notes
Complexity	middle range
Composer defined parameters	• constants • number of steps or number of generated notes • constraints • seed values
Aesthetic quality of output	middle range
Advantages	• the composer has great influence upon the behaviour of the equations by setting the parameters • easy to program
Disadvantages	• difficult to understand • difficult to guess how the algorithm will behave
Subtypes	generating music with fractal pictures

Commercial and shareware algorithmic music applications

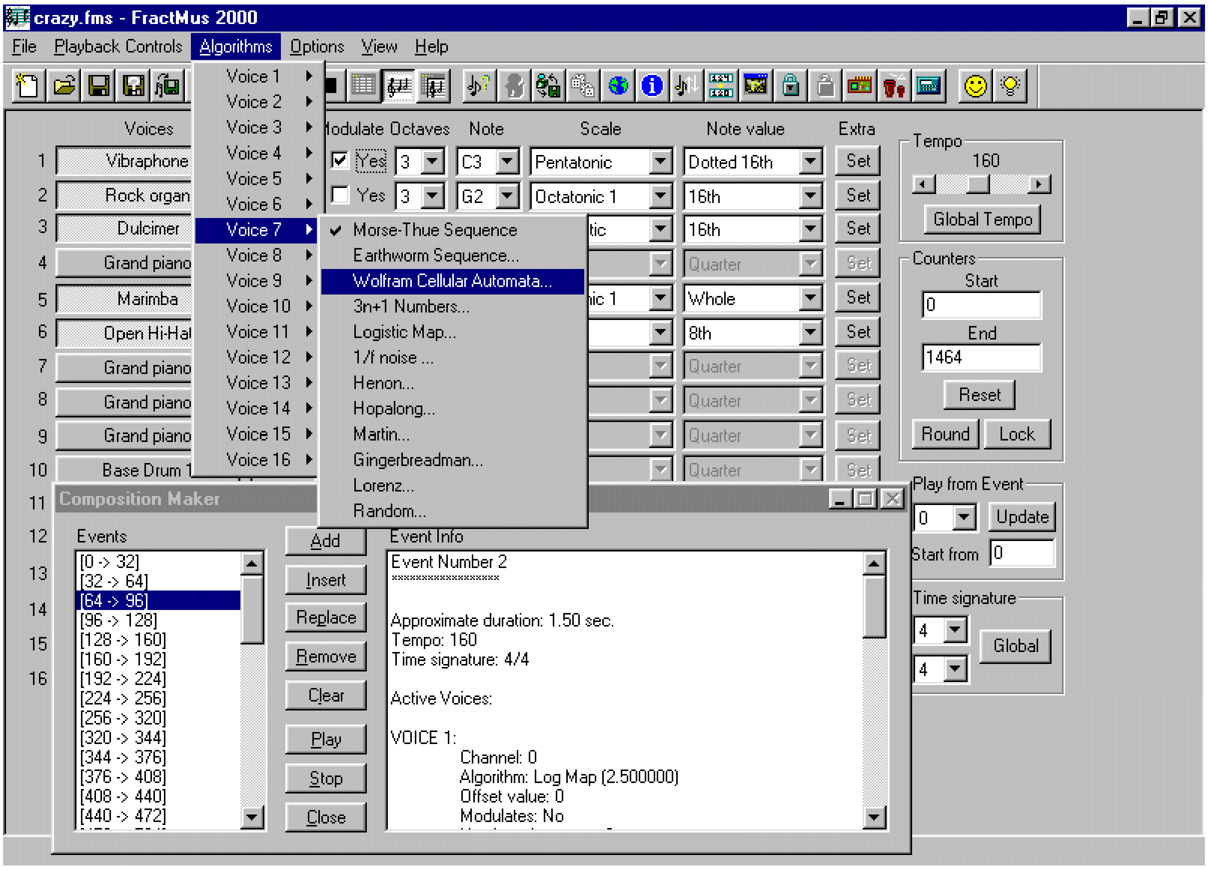
Put simply, 'algorithms' are formal procedures.

> These procedures permit computers to have a dialogue with us through numerical mediators which are both representatives of events in the world and the subjects of calculations. The mechanised ratiocination that takes place among hidden circuitry is linked to our quotidian world through responsive interfaces which accomplish rapid conversions between numbers and physical events [Blinkley, 1993].

Many computer programmers have found a fascination or seen potential profit in creating formal algorithmic music generating software applications, with graphical user interfaces which allow non-computer programmers to enter the world of algorithmic music composition.

FractMus 2000 is a freeware algorithmic-music generator, which creates melodies using mathematical formulas. This application generates notes using twelve algorithms from number theory, chaotic dynamics, fractals and cellular automata. This yields the composer an almost inexhaustible source of melodic material. Each voice can use any algorithm independently from the others.

[1] After Beckert, 1995



One can view a graphical representation of the algorithm by pressing the ***Visualize*** button. The programmer makes the following observation which (fortunately) applies to programs of this nature in general

> A word of caution: **YOU** are the composer, FractMus will create no masterpiece for you, nor it was designed for that. Think of it as a tool which gives you raw material that you can later use in your compositions. Writing a midi file and later importing it with music-editing programs such as *Finale* or *Encore* gives you the invaluable option of seeing your creation as a musical score, with all the correct pitches and durations, for later editing. FractMus uses only a few of the infinity of possible algorithms for note creation. Some of them exhibit fractal behavior, like the Morse-Thue sequence, Henon attractor, gingerbread man fractal, etc, while others use well-known formulas from chaotic dynamics, like the Logistic Map.
>
> In the end, it is always your inventiveness what makes a composition better or worse, FractMus just gives you the "inspiration". [1]

Koan Pro is one of the earliest and best-known examples of commercially available algorithmic applications, or "generative software". The latter term was coined by influential composer/producer Brian Eno. After a prototype of *Koan* was sent to him in early 1995, he developed a number of pieces which he released as *Generative Music 1*.

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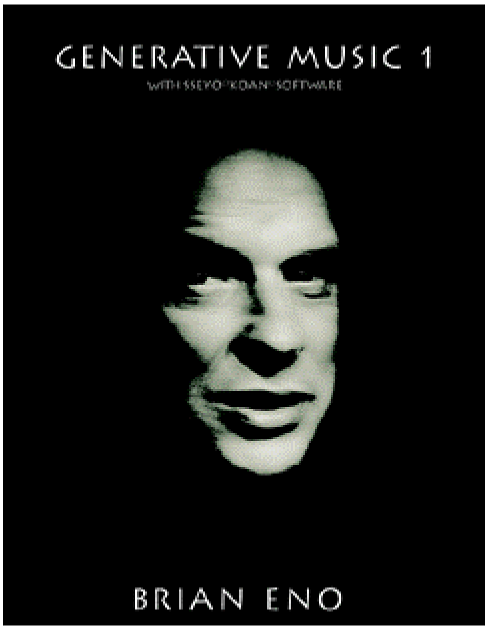
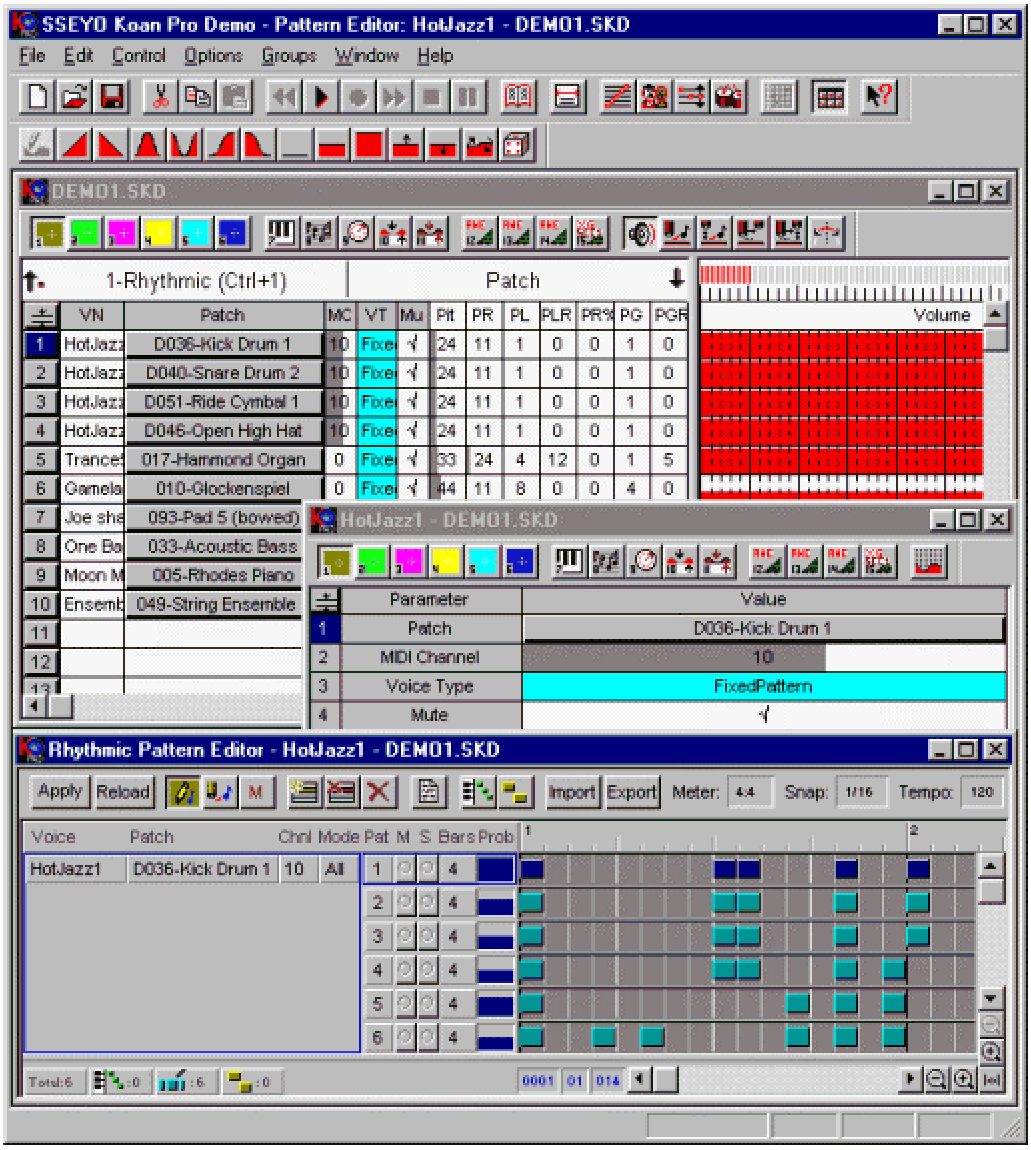
[1] URL: http://www.geocities.com/SiliconValley/Haven/4386/index.html ; accessed 5-12-02.



Brian Eno writes of *Koan*:

> "Some very basic forms of generative music have existed for a long time, but as marginal curiosities. Wind chimes are an example, but the only compositional control you have over the music they produce is in the original choice of notes that the chimes will sound. Recently, however, out of the union of synthesisers and computers, some much finer tools have evolved. *Koan Software* is probably the best of these systems, allowing a composer to control not one, but one hundred and fifty musical and sonic parameters within which the computer then improvises (as wind improvises the wind chimes).
>
> The works I have made with this system symbolize to me the beginning of a new era of music. Until 100 years ago, every musical event was unique: music was ephemeral and unrepeatable and even classical scoring couldn't guarantee precise duplication. Then came the gramophone record, which captured particular performances and made it possible to hear them identically over and over again.
>
> But now there are three alternatives: live music, recorded music and generative music. Generative music enjoys some of the benefits of both its ancestors. Like live music it is always different. Like recorded music it is free of time-and-place limitations - you can hear it when and where you want.
>
> I really think it is possible that our grandchildren will look at us in wonder and say: "you mean you used to listen to exactly the same thing over and over again?"

Koan Pro is a music authoring software package which combines MIDI sequencing with what the creators call “dynamic music creation tools”. With over 200 user-defined parameters, all of which affect the musical output, *Koan* has a complex user-interface.



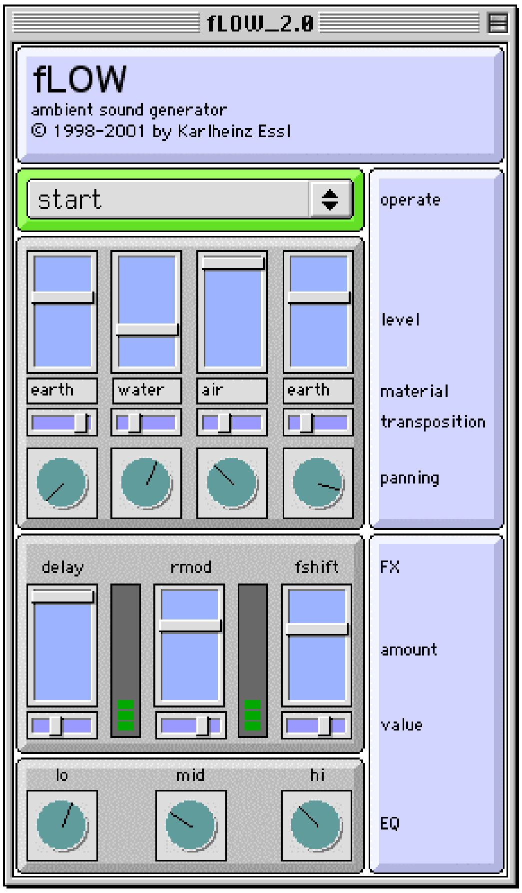
It is driven by the proprietary SSEYO *Koan Music Engine* (SKME). While this, in some ways, resembles a software synthesizer, it is capable of producing much more complex 'performances', with user-definable melodic, harmonic and rhythmic components. The composer determines the operating parameters (or algorithmic rules), which the software then applies, in order to create music in real-time. *Koan* operates with six voice types:

1. Ambient,
2. Rhythmic,
3. Fixed-Pattern,
4. Follows,
5. Repeat-Bar, and
6. Listening

and includes a pattern editor for creating 'dance grooves' with Fixed-Pattern riffs and sequences. The user can import riffs and patterns from externally produced MIDI files or record new (type 0) MIDI files. An 'auto-chording' feature is used to create 'thick' or 'thin' chords and arpeggios, or for adding effects, and the software includes several powerful randomisation and mutation functions, plus micro-level control over volume, pitch, start time and modulation.

Koan Vector Audio: This is a proprietary approach to producing ultra-compact music for minimum bandwidth utilization combined with the generative software. In other words, this is an algorithmic music composition application aimed at producing small file sizes suitable for use on the internet. Like MIDI, *Koan* vector audio contains note information, but also information on how to create the sounds, which drives Koan's inbuilt software synthesiser. No audio samples or audio sample set is required. As "open" text, it is easily added to a webpage, Flash movie or email and starts playing immediately. Koan audio vectors can be as small as 10 bytes.

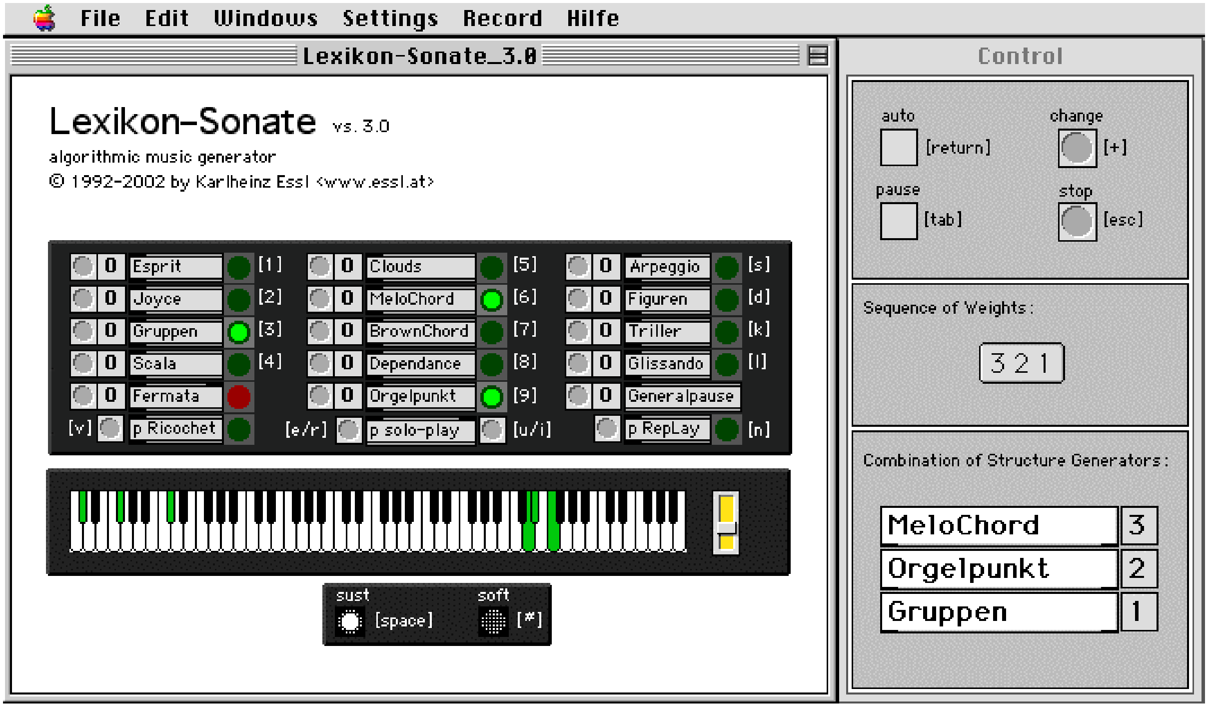
fLOW is a DSP computer program created by Dr. Karlheinz Essl (1998), running only on Apple *Macintosh* machines. It generates an ever-changing and never repeating soundscape, in real time, that fills the space with flooding sounds that resemble "metaphorically" (Essl's term) the timbres of water, fire, earth, and air. This so-called "ambient soundscape generator" adjusts itself through various parameters and controllers that are represented in real time on screen. It also functions as a multi-channel sound installation, which can be adapted to different spaces and locations.



right: screenshot showing the *flow* interface

Lexikon-Sonate, also created by Essl, is an interactive, real-time composition environment for musical composition and live performances. It is described by its author as “a work-in-progress that was started in 1992”.

> Instead of being a composition in which the structure is fixed by notation, it composes musical structures in real time, taking advantage of composition algorithms developed by Karlheinz Essl since 1985. With this algorithmic music generator one can easily create complex musical structures ‘on the fly’. Registered users can store them as MIDI files and process them afterwards using MIDI sequencers or notation programs. Lexikon-Sonate is an infinite music installation that can run on a computer for years without repeating itself. Lexikon-Sonate can also be used as an instrument for live performance of electronic music.
>
> 10 years after the Lexikon-Sonate project first started, it has been re-released as a software environment which for the first time offers the possibility to record the output to disk as a MIDI file. This allows registered user of the program to create customized musical structures which can be used for compositional purposes. [1]



Above: GUI for *Lexicon-Sonate,* a free-ware algorithmic music generating application

Random Music Machine is a *Macintosh*-only, user-configured, generative music program for defining algorithmic systems for generative music composition. In that respect it is similar to programs like *Koan*, but it uses its own unique compositional engine. It uses randomness in many of its generative functions - hence its name - but this is combined with non-random methods and user control settings, allowing for

[1] URL: http://www.hitsquad.com/smm/programs/Lexikon-Sonate_mac/ ; accessed 27-12-02

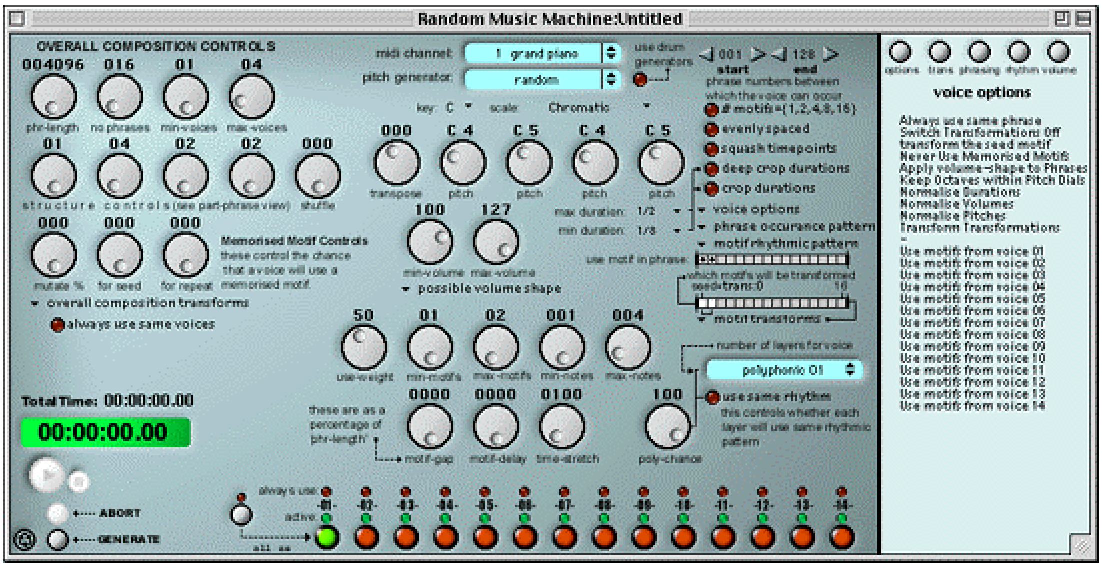
the adjustment of parameters that define how the program will create a music composition. Examples of user controls are:

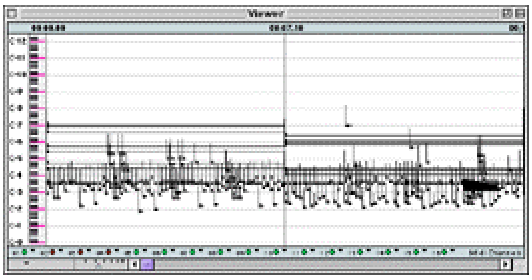
- multiple pitch generator types, including tonal melodies, random melodies, pink noise melodies, brown noise melodies, chaotic functions, unstable melodies, chord progressions, tonal chord progressions, random chords, discords, concords, rapid-grain-notes.
- multiple generative layers per voice, with either different of same rhythmic patterns - user can generate multiple melodic textures per voice, of chordal patterns (or stacks of chords)
- generators for beats and drums patterns which can be also used to generate melodic and note patterns.
- motif rhythmic pattern generators
- motif and phrase transformations, such a transpositions, inversions, retrograde, intervalic expansions, contractions, “triadize”, “chordize”, embellishments, add ornamentations - and controls for determining which motifs get transformed.
- multiple generators for dynamic and volume
- controls for phrase-length, motif-lengths, motif-stretching, overall compositional form and structure controls, controls for number of voices and probabilities for selecting a particular voice, voice occurrence, motif occurrence pattern, controls for voices using motifs from certain other voices - which get transposed to that voice's parameters,
- overall composition mutation options for tempo, dynamics, fade in/out.

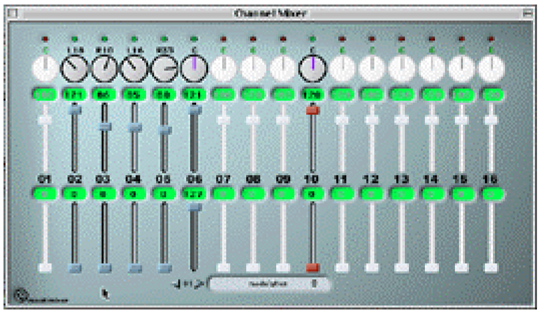
The user then clicks on the *play* button and the program will automatically generate a new composition. It can be configured to create music in many different styles and it will generate a new and original composition each time play is clicked. As the promotional web-site cynically states:

> …then you can have an endless supply of these new-originals - more throw-away music in the age of post-re-production - cyberdust-to-cyberdust - an age of auto-generation which upsets the out-moded concept of 'repeat-listening' - where everbody can be a composer and nobody is an audience. [1]

[1] URL: http://www.robotsoftware.co.uk/robot.htm ; accessed 27-12-02.

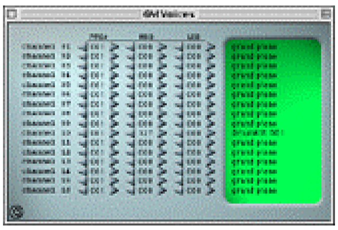


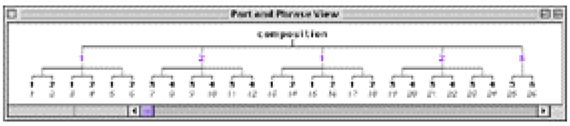




Viewer window : this shows a piano roll style view of the generated music. One can select which voices are displayed and can alter the characteristics of the displayed information

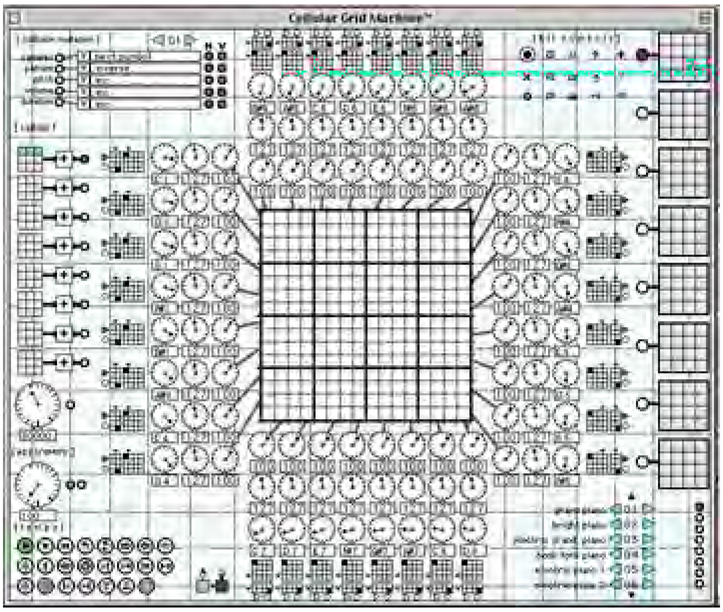
Mixer window: allows one to change volume levels, pan levels and alter continous controller levels that include modulation, reverb, chorus, variation, brightness... It also has user definable controllers.





Voice selection window: select the voice program numbers and bank numbers

Part/phrase viewer window: shows the higher level compositional structure



Cellular Grid Machine is an experiment in “a mutated method of music making”. It is a (software) machine that spits out music from its auto mutating 16x16 sound tracker mechanism. With cellular grid machine the user can work with a 16x16 tracker grid (screenshot above) in which midi trigger symbols can be placed - each type having its own response and midi output.

(e.g. arpeggios, trills, clouds, random, add-pitch, rapid-lines, soft, loud, reverse, restart, spark) - each tracker line initially controlling the symbols pitch, volume, duration, channel response and the tempo of movement.

When CGM is started, the tracker grid is continuously traversed in both the horizontal and vertical direction (can be a mixture of left / right / up / down) causing midi output whenever a trigger-symbol is crossed. Although the movement through the tracks can be in multiple directions, they share the same arrangement patterns of trigger symbols. - shared space, alternate times - while each track has its own individual control of how to play these trigger symbols. Further to this, multiple, time based, auto-mutate, reassemble, disassemble, re/de-construct processes can be activated to alter the grid pattern layout and the various parameter and control settings.

Also possible are collision-detection pattern / parameter mutations processes - and ‘flow and movement’ mutations; multiple tempos; games of life activations; the user can add things, remove things, rotate patterns, shift patterns, swap patterns, exchange patterns. As the promotional web-site again somewhat cynically describes:

> … all sorts of stupid things can act on the patterns and the parameters creating a strange and choreographic performance.
>
> *Cellular Grid Machine* has brought together different ideas and techniques that sit uncomfortably between being unusable and misplaced and unexpectedly inspired and ingenious.

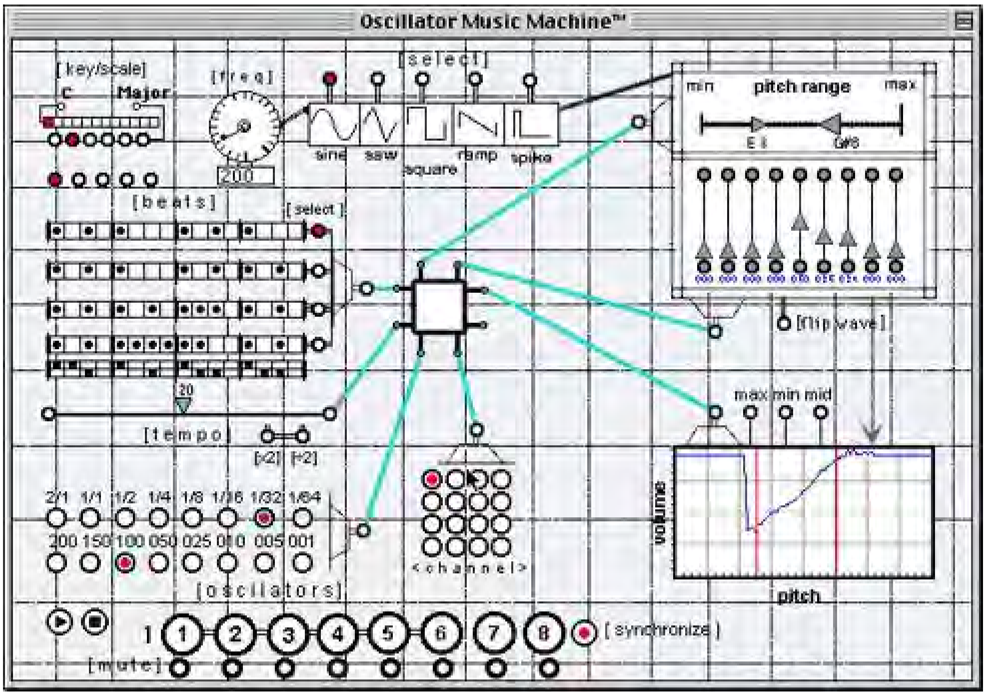
Cellular Grid Machine asserts its personality and pre-scripted method onto the music. To use it do not try to control totally the music being made, rather surrender in part to the machine. The music is the excrement from its mechanism. - [garbage in garbage out]. One can manipulate and control the machine, but the music may not always be in your hands.

Cellular Grid Machine celebrates the machine aesthetic, and our gradual fusion with machines. There is beauty in its overtly complex and formal arrangements of dials and controls and gadgets. Its movement, operations, things flashing and interactions are its theater.

CGM is auto-fasinating, allowing the generation of hours of continual mutating and self evolving forms that are created by processes which are stupid, illogical, pointless and non musical/

CGM uses the notion of Throw Away Music. Any music generated is for the moment. It's then lost or thrown away when the CGM is quit. This ephemeral nature is used to emphasis the semi-automatic-generation of music produced using CGM and to place significance on the continual synthesis of new material over the reuse/reproduction of the old. [1]

Oscillator Music Machine is another application created by RobotSoftware. It too is described as “a real-time, algorithmic music-generating machine”.



The creators claim that *Oscillator Music Machine* can been used as a tool for generating melodic ideas, compositions, in installations as continual process music, as a performance instrument and as a controller for effects modules and effects parameter, via routing its output through other programs for experimental electronica.

[1] URL: http://www.robotsoftware.co.uk/cgm.htm ; accessed 27-10-02.

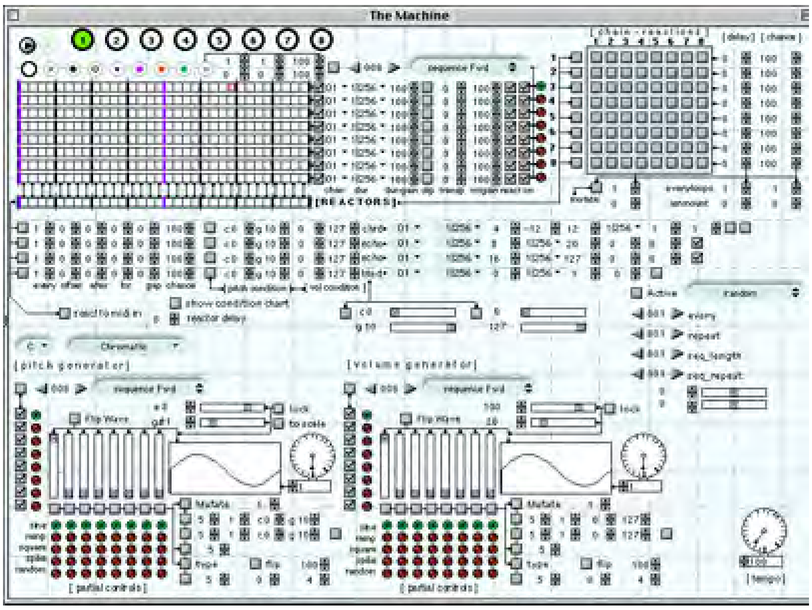
It has multiple generative tracks - with independent controls and independent tempos (or auto-synchronisation) to create evolving organic structures and asynchronous organisations of sound - with ever shifting element relations.

It allows the definition of ‘Oscillating Pitch Contours’, by choosing between 5 waveform types, inversions and 9 partials - allowing the creation of simple to complex-shaped pitch contours. This simple technique can create musical and “amusical” melodies very quickly.

For each voice the user defines 4 different tracker grids and articulation patterns - with a possible random selection feature and an auto-pattern mutation mode. This allows for indeterminate musical forms and chance layering of musical elements. This means that continuously changing music structures can be created very simply.

The user is also able to draw pitch>volume relationship curves, so that the volumes of the notes is determined by the pitch. This can create unusual dynamics for the pitch patterns. [1]

A third quirky piece of algorithmic music-generating software has been created by Robot Software***,*** entitled ***The Reactionary Music Machine.*** It is designed for XG based software synthesizers, running only on the MAC PPC platform. It also includes scaled-down version of RobotSoftware's MU128/XG editor. The promotional web-site states:



“It currently includes no documentation and development on it has been put on hold to concentrate on other things...but it works and is great for composers working with experimental aesthetics.

Reactionary Music Machine is the first in the series of Incredibly Strange Music Machines - esoteric and eccentric midi software.

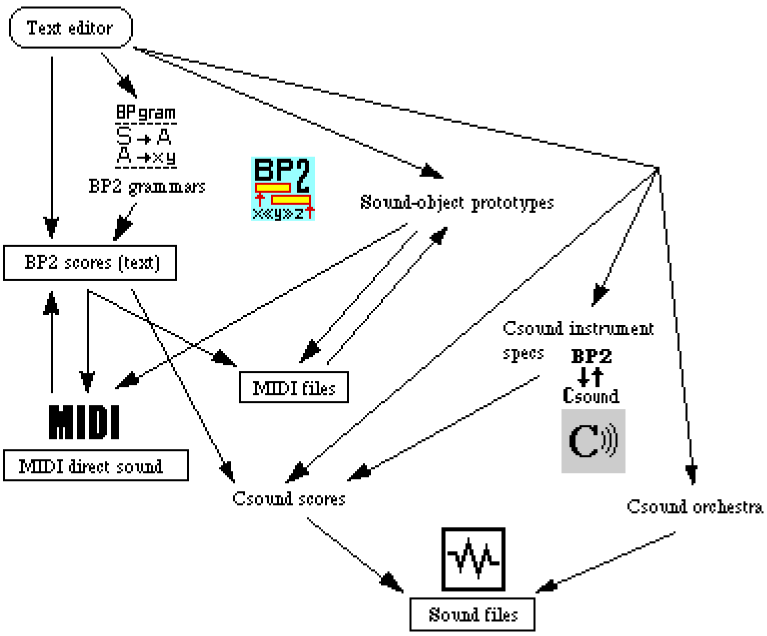
It is a real-time, self-interacting, algorithmic musical automaton, that can be user-configured and let loose to self-mutate and self mutilate its own settings creating evolving and catastrophic musical forms. Its name comes from its Reactor systems that can respond

[1] URL: http://www.robotsoftware.co.uk/omm.htm ; accessed 27-10-02.

to its generated music, and reactors can be "chain-reacted" to create unstable generative-feedback mayhem and frenzied glitch-moments." [1]

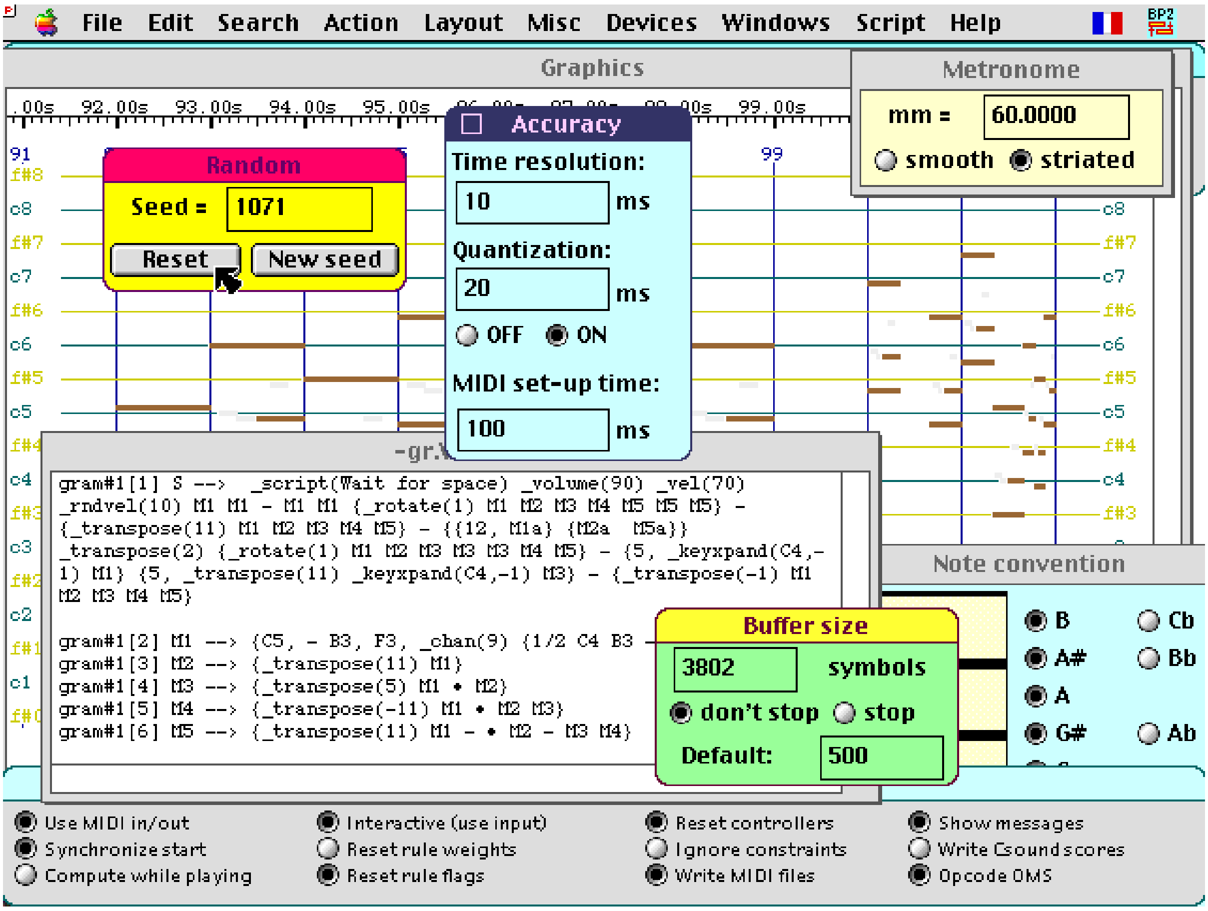
Bol Processor 2.9.3 created by Bernard Bel, won the Bourges 1997 international award (*ex aequo* with "Cecilia") in the category of computer-aided composition and realization software. *Bol Processor* produces music with a set or rules (a compositional grammar) or from text scores typed or captured from a MIDI instrument. The author states:

> Thanks to interactions with composers tired of Euro-centric music software, it gradually became a flexible environment for composing music that makes sequencers and other MIDI software feel dumb. For instance, it is very smart at solving complex rhythmic problems, as it takes care of integer-ratio arithmetic with the minimum supplied information. BP2's ability to solve systems of constraints, both at the symbolic and numeric levels, is a significant advantage on other programs (e.g. Max/Msp) that handle basic maths only. The authors' contention is that music composition is to a greater extent a constraint-driven design process than the application of stipulatory rules and methods.



The interaction between *Bol Processor* (BP2), MIDI and Csound. BP2 scores in text format may be typed in, produced by a grammar or entered from a MIDI device. These scores contain simple notes, sound-objects and performance controls. A BP2 score combined with sound-object prototypes works out a direct MIDI output. The same output can be saved to MIDI files which may in turn be exported to sound-object prototypes. The BP2 score combined with a formal description of Csound instruments yields a Csound score. Csound is then invoked to produce a sound file using the orchestra file and the Csound score. [Bel, 1998].

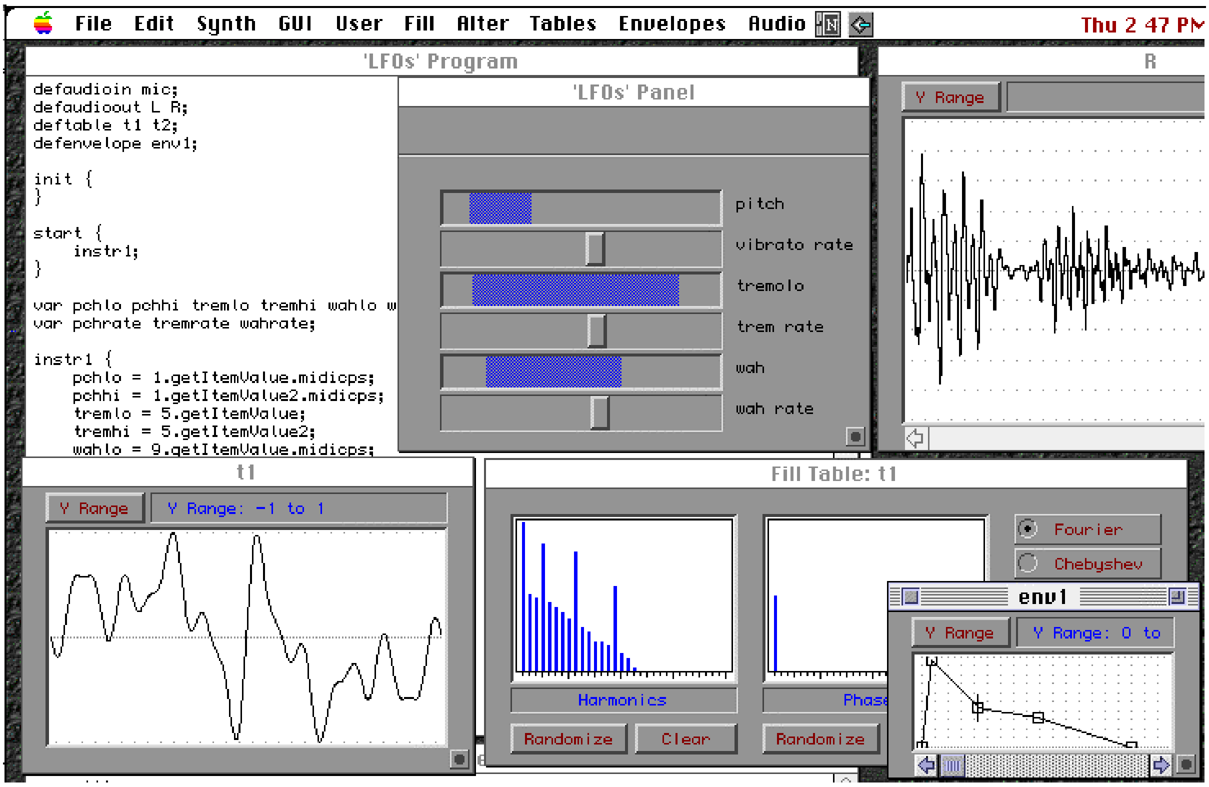
1 (URL: http://www.robotsoftware.co.uk/rxmm.html ; accessed 27-10-02).



Bol Processor provides desktop tools for the composition of serial music and a straightforward, relatively intuitive approach to the design of sophisticated time structures.

SuperCollider is a Mac-only freeware environment and programming language for real time audio synthesis. The user can write programs to generate or process sound in real time or non real time. *SuperCollider* can be controlled by MIDI, the mouse, Wacom graphics tablet, and over a network via Open Sound Control. SuperCollider can read and write sound files in AIFF, WAV, Sound Designer 2, and NeXT/Sun formats. *SuperCollider* supports sound cards using Steinberg's ASIO driver api.[1] *SuperCollider* was used to compose the soundtrack to the short experimental animation entitled *Cells* found on the accompanying interactive CD-ROM the Machine (accessed on the pay-TV section of the Recreation Centre).

[1] URL: http://www.audiosynth.com/ ; accessed 23-11-02.



Above: *SuperCollider* interface

Fractal music and the future of multimedia art

Reference is made elsewhere (see p. 140) to the grand concept elucidated in Herman Hesse's novel *"Magister Ludi",* a vision appropriated herein to serve as a unifying theme in this dissertation. On a musical plane, fractal music offers, for some thinkers at least, an insight into a glorious future for computer based multimedia art, in which microcosms of infinitely beautiful detail are nested within magnificently structured macrocosmic architecture. Murcherino [1], for example, has found his own visionary inspiration along these perhaps grandiose, but certainly optimistic lines, in the 'world within world' of a fractal consciousness. He sees this expressed in music when he writes:

> India's musical / cultural tradition mandates the construction of musical developments in a recursive, non-linear way. In Indian music, rhythms and melodies are spontaneously built out of a small pattern of set notes called the "raga". The raga is then syncopated, translated, rhythmically squeezed and manipulated in all manners, in order to synchronize beside benchmarks in the musical piece. This kind of juggling demands unconscious recursive "calculations", which are generated by the performer's intuitive filter of years of experience. Like our Western rock-and-roll, this Eastern form is a mixture of simple harmonies,

[1] Murcherino, Nicholas (undated). Recursion: A Paradigm for Future Music? http://alpha01.dm.unito.it/personalpages/cerruti/musicafutura.html ; accessed 29-01-03.

> improvisation and composition, but raga music eclipses Rock with its exotic metric and rhythmic demands on both performer and listener.
>
> Paradigm denotes a framework, one on which to build current and future research and experimentation. The conjecture (offered here) is the construction of a foundation based on: recursion combined with iteration.
>
> I submit an idea for a potential symbiosis that reflects the archetype of Indian music, in which relatively small parts of a composition could, in a self-referential manner, "fractalize" into a larger composition with computer assistance, analogous to the extemporaneous variations woven by raga artists.
>
> In taking advantage of the variety and self-similarity inherent in a fractal-related approach, future musical sequences could pattern a more naturalistic, improvisatory character than is currently inherent in much of the electronic repertoire. Fueled by commonplace access to multimedia computing, and with the signal processing power to easily manipulate snippets of melody, meter and rhythm, experimenters now have the means to take a step towards this horizon.
>
> …the mathematics of fractals can provide a gateway to experimentation for composers anticipating the next century's music…

Murcherino is describing musical sub-routines within the meta-art machine of the future.

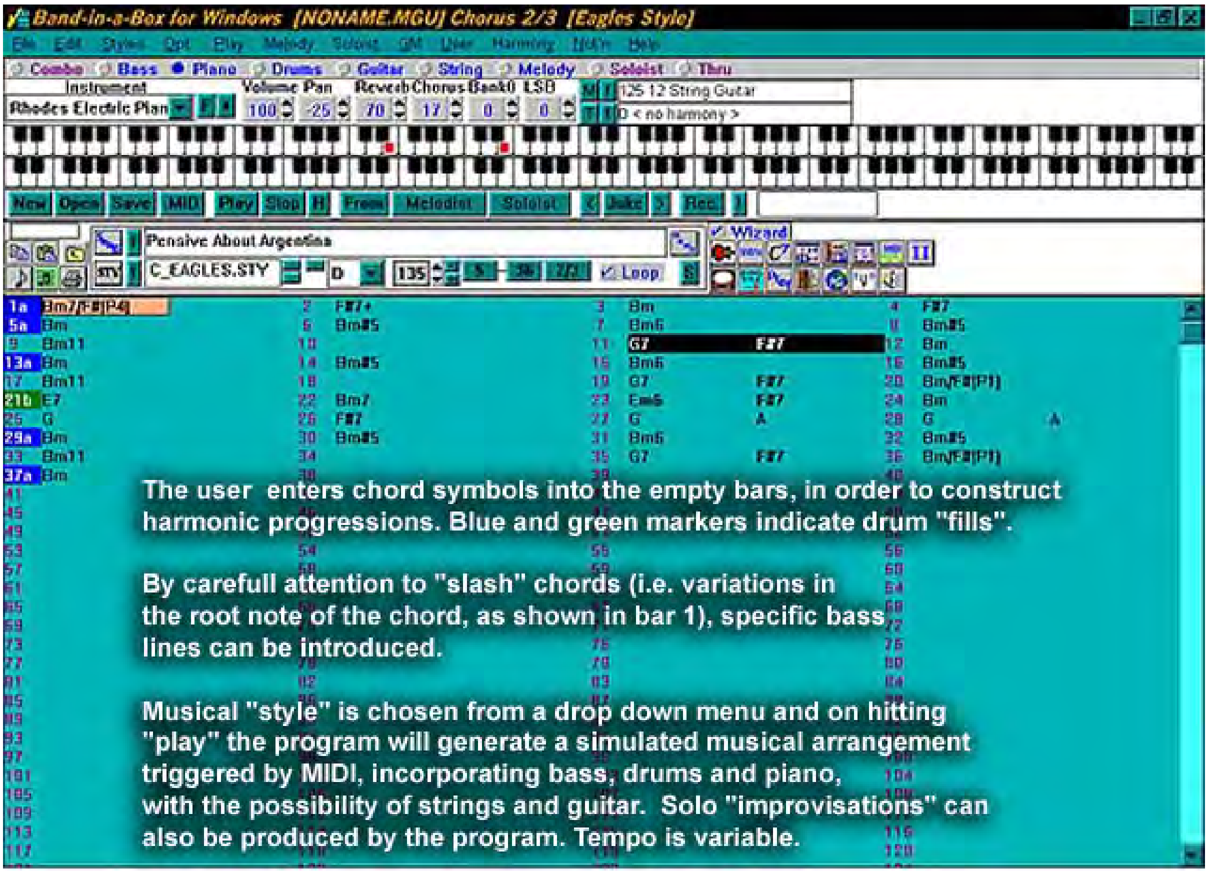
A Concluding note on computer-assisted Algorithmic Composition systems.

Each composer has a unique inspiration, which drew him/her into musical life, and each follows an equally unique compositional process. At this stage in its history it cannot be the aim of research into computer-assisted algorithmic composition to develop a universal composing system that matches the creative needs of all composers.

Algorithmic systems must be constructed that meet the specific creative needs of the individual composer, and the parameters tweaked according to that individuals capacity for creativity.

> Algorithmic composition is perhaps a little bit like religion or politics: one must find one's own path. [Beckert, 1995]

Once an algorithmic system of music composition has been packaged into a stand-alone software application, it is immediately creatively interesting to subvert the intentions of the programmer and discover accidental outcomes. For example, the “intelligent” accompaniment and MIDI song arranging application *Band-in-a-Box* (PG Music) is an example of a labour-saving computer program which can be a useful practical tool in certain contexts of deadline-driven commercial music, but which can equally be said to homogenize music composition and arrangement by reducing song-writing to formularized, cliché-ridden musical styles. (Used in *The Machine,* see p. 555.)



While designed originally as a way of using computers and MIDI to generate “minus-one” arrangements to assist in honing instrumental and improvisational skills, it becomes more interesting if the intentions of the programmer can be subverted, by constructing a harmonic progression and then trawling through the pre-programmed genres for interesting accidental musical effects which are occasionally generated by its cyclic variations on cliché, and then using cut-and-paste techniques to build a piece from these interesting accidents.

Music and Sound in multimedia from the perspective of the cinema tradition

Commercial imperatives drive filmmakers to push the available technology to the limits in order to gain the most sensory impact. As Branigan observes:

> If an art form is characterised by its ability to exploit best the available technical means of an epoch, then film qualifies as the most typical art of the twentieth century. [Branigan, 1995, 11].

Music has played an integral part in the motion picture experience almost since its inception. Early film projection technology was cumbersome and noisy and live "programme" music was introduced in early silent film screenings as much to mask the distracting sound of the projector, as it was to add to the aesthetic experience. [Deveau, 2000, quoting Kurt London].

Nevertheless the traditions of popular and high art entertainments - of stage musicals, light opera, and Wagner - were models on which this early incorporation of music were based, and rapid developments took place as the potential to enhance the cinematic experience with appropriate mood music was recognized. Small bands soon augmented pianists and organists, and at the peak of this period orchestras were employed and special compositions began to be produced. The art and craft of film with music - a true 'multi-medium' art - was born.

The potential of cinema began to unfold quite rapidly and the silent film era was a time of great creativity and technical experimentation with the psychological impact of juxtaposed film imagery. The first of the great directors, such as Griffiths and Eisenstein, recognised that cinema was more than the photography of motion, and began to explore the possibilities of narrative made available by film editing and montage.

The introduction of sound to film was hastened by a crisis in the fledgling industry, caused by the sudden popular appeal of a new technology – radio, which kept people at home, listening [Barry *et al.* 1964, 290].

The early sound films are best described as "filmed plays", as early directors began their explorations of the new medium conditioned by the theatre tradition, and also because of technical limitations[1]. The problematic nature of the recording system which required the camera to be isolated in a sound-proof booth; the lack of directionality and relatively poor quality of early microphones; and especially synchronization issues, were among the technical difficulties which governed and essentially limited the artistic possibilities of early sound films.

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[1] Sennett, Robert S. (undated) *Setting the Scene : the Great Hollywood Art Directors.* http://www.amctv.com/amc/behind/0,3121,CAT0-56-CAT1-180-CAT2-278-,00.html ; accessed 29-01-03.

Consequently some of the creative progress in editing made in the silent film era was temporarily sacrificed for the novelty of including sound. Early film theorists (e.g. Eisenstein) were critical of the development, expressing concerns that the combination of sound and image would, according to Dancyer [1993, 42]

> …counter the use of the shot as a building block that gains meaning when edited with other shots. They therefore argued that sound should not be used to enhance naturalism, but rather it be used in a nonsynchronised or asynchronous fashion. This contrapuntal use of sound would allow montage to continue to be used creatively.

Such concerns did not hinder the rapid evolution of the creative use of sound in films, however. This was the dawn of the age of streaming synchronised sound and image. This new era witnessed doors opening to experimentation involving the manipulation of data designed to activate sensory pathways to the brain - in order to stimulate thoughts and emotions, raise consciousness or raise the pulse rate, and to entertain the growing audiences for this dynamic new mixed-media form. This was the beginning of the technological integration of visual and auditory media, and the pioneering work of the time continues to be of interest to practicing multimedia artists.

One of the earliest directors of historical stature to see the creative and dramatic possibilities of sound editing was German director Fritz Lang, as typified by his classic film of 1931 *"M"*. Lang was especially significant for the way he edited sound in an innovative and, for the time, startlingly unrestricted way. He experimented with the use of sound to overcome or collapse both space and time. He was an important pioneer of the technique of using a single sound grab to create continuity between disparate visual shots, by creating a master shot with unifying reference audio and then insert-editing meaningfully related visual sequences. He thus achieved a narrative structure that takes place over time, or in many different locations, unifying such visual sequences with a continuous piece of underlying (or sometimes superimposed) sound and/or dialogue, which usually offered another level of meaning to the visuals. In "*M"*, he juxtaposes sound and images to create scenes of compelling emotional resonance. A device Lang used to create suspense in "*M*" was the murderer's recurrent whistling before each homicide. Unlike many film-makers of the early "talkie" period, he realized that sound was much more than dialogue. Artfully employed, sound evokes powerful emotions.[1]

The evolution of a practical understanding of the melding of sound and image was significantly influenced by Alfred Hitchcock - another giant in the evolution of the language of sound in film towards the artistic economies of semantic compression. During Hitchcock's long and illustrious career he constantly experimented with new techniques of building narrative and drama, and frequently turned to the manipulation of sound as a creative element, being responsible for many influential innovations in the use of non-naturalistic sound, and music in film, and especially the concept of using a sound transition to introduce an idea into a scene.

As Dancyer describes:

> The classic example is the scream in Hitchcock's *The 39 Steps* (1935). As we hear the scream, we see a train. Not only is a transition accomplished, but the simulation of human and mechanical elements makes the human response seem louder and more terrifying.
>
> [Dancyer, 1993, p.244.]

While stressing the importance of this evolutionary process in cinema a brief overview is all that can be attempted here. It is unfortunately beyond the scope of this study to delve exhaustively into the colourful detail of the history and evolution of the manipulation of sound and music in film.

In the current situation in commercial cinema production, at the highest technical and budgetary level of the industry, sound and image can be totally disassembled, so

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[1] *Fritz Lang: Genius of Film Noir*. Jones International and Jones Digital Century [1994-99]. http://www.digitalcentury.com/encyclo/update/fritzlan.htm ; accessed 8-12-02.

that the creative editing possibilities in both sound and picture are extremely broad and free.

It remains however important to the study of the potential of multimedia art to attempt to understand the peculiar experience of immersion in an artificial reality which results from the successful synthesis of film editing, sound editing and music composition in contemporary cinema. Expectations resulting from the universal popularity of cinema inevitably condition expectations of the use of sound in multimedia art.

The effect of audio design and music on the perception of image and narrative

Unfortunately this field cannot be covered in great depth within the context of attempting such a wide overview of multimedia art. But it is an issue of great importance, which has yet to be fully explored and understood, and must remain a central and fascinating consideration with regard to the aesthetics of multimedia. Spande [1] introduces the paradox of effective film music:

> In both its historical and immediate form, film music exhibits the properties of a *vanishing mediator*, a phenomenon which disappears necessarily from the field of its own effects. ...
>
> We watch the film and music blares from every speaker, yet it remains, in large, unnoticed in the way that a "normal" musical experience, such as might be had in a concert, is noticed and enjoyed. In a very necessary way, the film music becomes submerged, producing a field of effects, not the least of which is our enjoyable immersion in the movie. To remember back to the most enjoyable parts of the film rarely includes a similar remembrance of the music that accompanied them. Thus the film music often "disappears" from the effect (the memory) of the film/ film music complex.
>
> As such, the phenomenon of film music represents the quintessential postmodern object.

The function of music composition in film is a field of study at once fascinating, and elusive. Spande points out in another article [1996] that there are few films which do well without music, mentioning *The Asphalt Jungle* as the only notable exception. In Spandes view:

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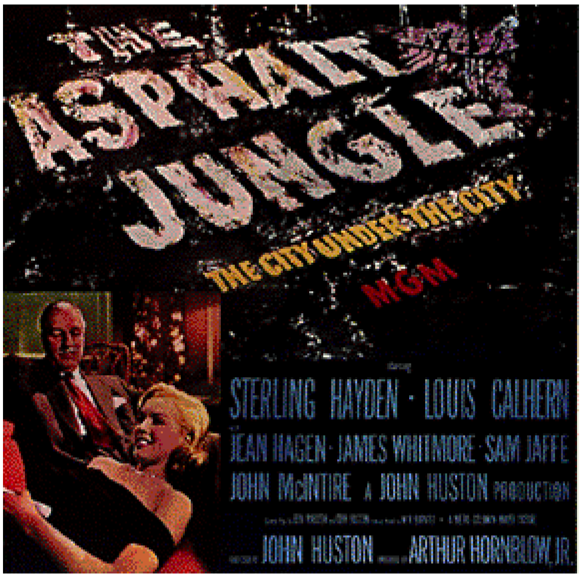
1 Robert Spande, (undated); *The Three Regimes: A Theory of Film Music.* http://www.uselessindustries.net/robobo/filmmusic.html ; accessed 27-02-04.

> The deeper reality of film music,...involves the way the film tries to evoke a sort of temporary and illusive "film-subjectivity," which closely mimics our everyday subjectivity. To do this, the film must rely on tactics far beyond simply spinning a good yarn. The film experience must imitate in some way all three overlapping dimensions of subjective reality: the symbolic, the real and the imaginary. The method by which this is done involves very heavily the use of film music.



Spande [1996] outlines his view of how cinematic narrative functions, referring in turn to the “symbolic, imaginary and real” in the audiences perception of the presented "reality". He goes on to make important points with regard to the evolution of musical codes in film music. He argues that music is not a universal language, but rather culture specific, and therefore must be said to be arbitrary. Through exposure to the films and television generated by our culture we learn to recognize stylistic cliches as signifiers of stock cinematic situations, and these become triggers which telegraph the intention of the director. For instance:

> In the realm of the horror movie..., two musical codes have co-evolved most forcefully. The first of these is the stinger, a high pitched musical "cattle prod" which usually accompanies a harsh, sudden change in the narrative. ..use of the stinger lets us know to be afraid. The startle of the stinger is not only "startling," but also symbolizes the fear and threat... The stinger is a direct representation of the action on the screen by the music. The action is scary = the music is scary.

One of the most famous early examples of the ‘stinger’ is the discordant violin sound used for shock effect by composer Bernhard Hermann in Alfred Hitchcock’s classic *“Psycho”* [Spande, 1996].

The second horror movie code referred to here by Spande is designated “the sublime”. It is a haunting melody, usually in a minor key, often tender and touching or evocative of innocence, childhood. This can suggest a reference to some deeply embedded childhood trauma which has set in motion the aberrant behaviour depicted in the film. For Spande, this music does not directly represent the horror but instead seems to say: "What this music represents is so horrible that representation is not even possible." By introducing these ideas, Spande implies that there are genre-specific codes in film music, i.e. that identifiable, stylised musical signifiers have evolved and are in use in commercial film music that can be codified. It follows from this observation that computer programs may be able to compose, or aid in the composition, of film music. To conclude, a suggestive example of such a software compositional aid, 'Scorebot', decribed by Pierce and Ng, must suffice:

> Given a musical theme and information which describes a scene (emotional content and timing and importance of events), the software can create a variation on the theme which conforms to the scene…at its core, is an original database application which stores and allows manipulation of data on two levels: the 'higher' functions which manipulate or generate the musical elements that will evoke certain moods, and the 'lower' functions which control data related to specific projects (the timing, mood, and thematic information that forms the basis for the actual score).

Sound and Music in Interactive Media

Sound is a vital element in interactive media. From the earliest days of coin-operated arcade games, sound design has been an integrated element, in order to enhance the immersive experience [Granner, 1999, 1]. According to Granner, arcade games have their own peculiar design imperatives, predicated by the acoustic environment in which they are played.

> Combined with the need to support "instant" gratification is the fact that coin-op games are played in public places, and in many (most!) of those places the kind and amount of noise tolerated from a coin-op device is extensively restricted. In the case where "anything goes," in the arcade or game room, the problem is that all the machines in the room are making a lot of noise, creating a din which dramatically reduces the effective signal-to-noise ratio of a soundtrack.
>
> In this rather hostile environment, the sound designer must "choose wisely" which sound elements to emphasize. Elements that would provide a delicate or "deep" background or ambience behind foreground sound elements will be lost either to having the game turned almost all the way down, or to a public version of stereo wars.
>
> These problems confront the sound designer of any type of coin-op game.

Granner lists a number of other problems that must be faced by the practical arcade game sound designer. Firstly, the outcome of introducing an interactive element is that the audio experience will be of indeterminate shape, necessitating the avoidance of long passages, forcing the composer to work in small modular units of sound. Secondly, due to the collaborative, teamwork nature of games development, other members of the team are exposed to long periods of repetitive exposure to the soundtrack, often resulting in frequent (perhaps premature) requests for rewrites. This brief reference to an important and interesting new field of research, must conclude with a quote from Ganner, outlining a third problem faced by game sound designers:

> More insidious is the creep in intensity; the new death-blow must be bigger, louder, more intense than the others. It is the job of the sound designer to balance these requests in such a way as to keep the developers happy. In a strange sense, the soundtrack represents the mood of the project, and this places the sound designer in the unenviable role of Master of Attitude Adjustment. The best way to serve this purpose is to bring in new stuff regularly. This minimizes the chance of the whole soundtrack becoming stale. [Ganner, 1999].

2.2.2 The Sound Channel

Digital field recording

It has been noted earlier in this dissertation (see p.193) that high-quality digital video cameras have impacted on multimedia art, bringing many of the possibilities of video art to the multimedia mix. An equally important aspect of this technological development being offered to the consumer is the simultaneous availability of high quality digitally recorded stereo sound - such cameras now capture, and store digitized audio, synchronously with the video image. There are also a range of commercially available Walkman-style portable Digital Audio Tape recorders, which equal the previous industry-standard Nagra field recorder in audio fidelity, but offer the advantages of lossless duplication.

The chance recording of ambient audio environments, random source music, and found sounds can offer a powerful contribution to the final creation of a successful virtual experience, especially with the capabilities offered by binaural recording to enhance the perceived fidelity of the simulated sensory experience in multimedia art.

Spatiality

We hear in three dimensions, and the localization, directionality and spatiality of human sound perception has had important survival value, whether it be in circumstances where the detection of the approach of predators outside our field of vision meant survival of the gene pool, or, in more recent times, the evolution of the aural component of city traffic survival skills. According to Vernonnen [1996, 1]

> The ear-body-brain combination correctly decodes a handful of simultaneous and possibly conflicting spatial cues, often to within a few degrees of precision and in a fraction of a second.

If art can be viewed as a celebration of human faculties, then it is not surprising that spatiality has long been considered a valuable aesthetic and dramatic value in the making of music, and more recently of huge importance in cinema. The sense of enhanced realism produced by strongly localized sound events has driven the evolution of cinema sound from monaural, to stereo, to multi-track surround sound.

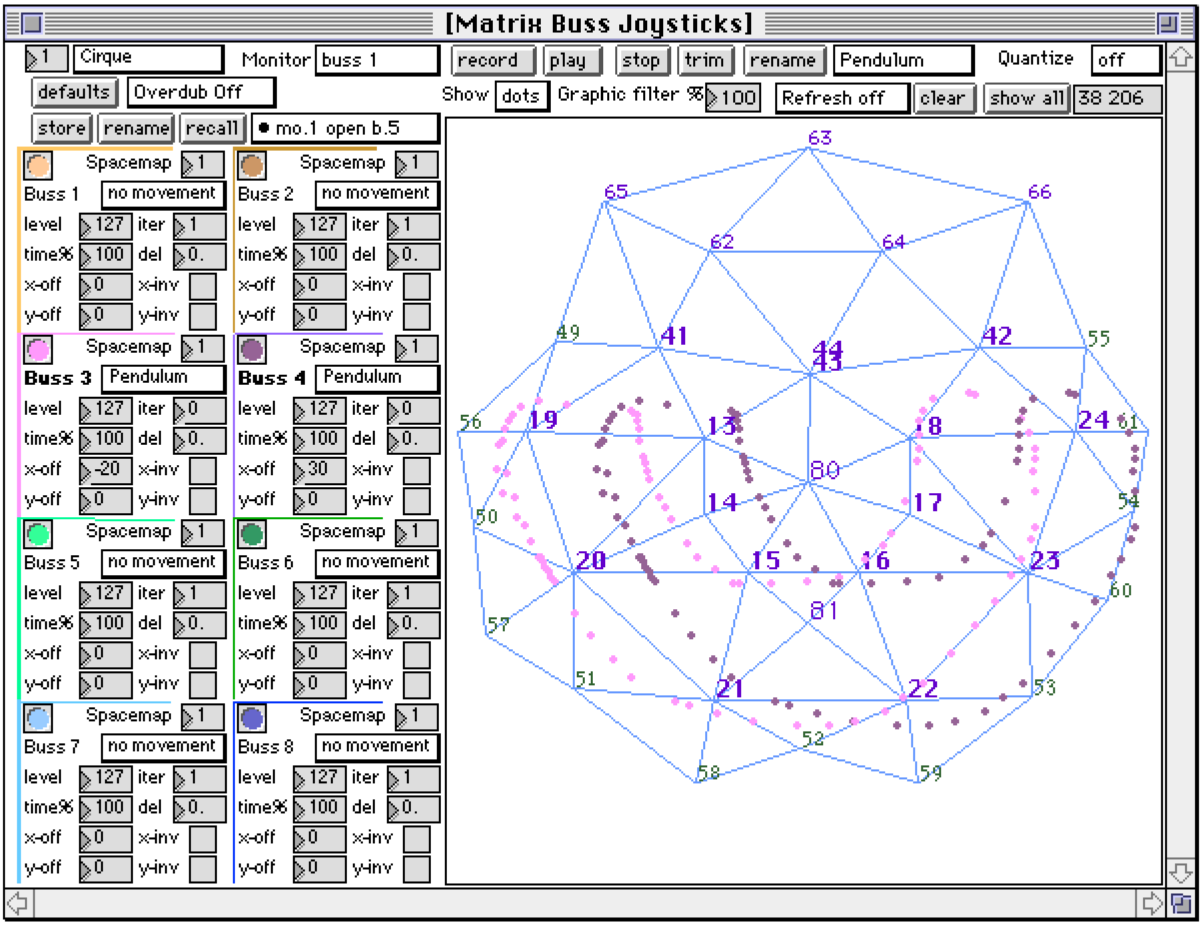
There is a long history of experimentation with spatiality in music. Harley, [1994, pp. 100-224 documents the history of space in musical thought in the 20th century, stressing the perceptual and temporal aspects of musical spatiality, a reflection of the close connection of space and time in human experience. Many leading twentieth century composers have used spatiality as a major ingredient in their composition and theoretical writing (for example Stockhausen, Cage, Xenakis and Boulez). For Vernonnen [p.1]:

> The invention of sound recording in the previous century was a cultural milestone, bringing certain music to places, times and people it had never touched. The most primitive apparatus was an instant curiosity piece, yet the superiority of the live musical performance was rarely questioned. What was lacking in recorded sound was a sensation of ambience, a separation of the instruments and a perception of the context of the performance. By the early twentieth century a simple theory of human sound localization had been developed, that enabled others to propose ways to convey this missing spatial detail.
>
> Thus stereo was born, but it has not always been applied conforming with any stereo theory.

Most practical spatial systems (including simple stereo) rely on psychoacoustics, endeavouring to create a natural impression, or an illusion of enveloping sound.

The advent of digital recording and signal processing technology has seen an accompanying strong emphasis on acoustic space simulation (digital reverberation, digital delay technology and more recently spatial systems (discussed below). The acoustic space in which music is performed is a significant factor in the success of the musical experience. The evolution of musical styles has been significantly influenced by the textural effect of reverberation, a by-product of architectural styles. Venonnen (ibid.) points out the importance of the acoustic space to the appreciation of musical genre, emphasizing this point by asking the reader to consider: listening to Gregorian chant in an open field and, listening to Balinese Gamelan music in a large cathedral.

The digital revolution has resulted in dynamic and prolific experimentation in the field of audio and music production technology. Multidimensional spatiality in the aural modality can make a powerful contribution to the ultimate goal of creating sensually immersive multimedia art experiences. It is not surprising therefore that spatiality in audio is a focus of contemporary research and development in digital sound processing. Such research goes beyond the recording and processing of sound to include total integrated delivery systems, which maximize the effectiveness of the encoded signal. Potyen [1995, 62] points out that such spatial audio systems can be seen as belonging to one of two major types, a) systems that attempt to enhance the sense of spaciousness (e.g. Multi-channel/multi-speaker systems such as Dolby Stereo Digital, DTS, IMAX); and b) systems designed to place individual sound sources at predictable locations in three-dimensional space. The latter extends to systems which attempt to achieve "auralization" - the process of rendering audible (by physical or mathematical modeling) a soundfield source in a space, in such a way as to simulate the binaural listening experience at a given position in a modeled space. Examples of type b) systems have so far has been limited to large scale entertainments such as the LCS Cue Control system installed at *Flintstones* theme park at Universal Studios in Los Angeles [Potyen, 1995, 63]. Below is a screenshot showing the interface from a LCS Cue Control system, which is offered on both *Macintosh* and an Intel processor based 'IBM compatible' running Microsoft *Windows* OS .



Vernonnen's overview of spatial sound (written in 1996) makes the important point that :

> There is no commercially available system that successfully conveys the natural spatial hearing experience... a close approach is made by binaural stereo and transaural stereo methods, both capable of delivering a quite natural sounding three dimensional effect when used with well recorded source material. The limitation is that one must wear headphones with binaural stereo, or in the case of transaural stereo, there is only one good listening position for hearing the full spatial effect through the loudspeakers.

It is logical to optimize audio response, and possible to maximize spatial and directional cues inherent in binaural recordings of sound environments, even to the point of creating the trans-aural effect (most vividly perceived using headphones), produced by the cancellation or minimization of cross-feeding of left ear and right ear information. The typical multimedia receptor experience takes place in a relatively standardized hardware environment. The typical multimedia workstation consists of computer, monitor and built in stereo speakers at approx. 1 to 1.5 metres from the receiver. For this reason multimedia delivery systems offer a good environment for utilizing binaural and transaural audio compared, say, with the typical home music system which is far less spatially standardized. Typically, the experience is a solo one. However it seems likely that this present standard will not be maintained, with many predictions visualizing a future in which computers and home entertainment systems merge. Vernonnen [*ibid*., p3] concludes his discussion of spatial sound with the observation that:

> The biggest current research effort in spatial sound is being directed toward head related functions, based on the shadowing effects of the head and on pinna cues. Using these functions it is now possible to generate binaural signals that convey three-dimensional sound over stereo headphones, as a synthetic parallel to conventional binaural recordings. The main problem is that to achieve acceptable accuracy, the equipment must be calibrated to the individual's spatial location versus spectral response, caused by the pinna. If a generic head related transfer function could be discovered that suited all individual variations in ear shape and size, it could be the equivalent of the Rosetta Stone for virtual binaural audio.... spatial sound is at the point now of a having sufficient technological tools to solve most of the problems, but lacking a consensus on how to employ these tools for optimum results. This has a resulted in a highly fragmented field, not aided at all by a general ignorance about spatial psychoacoustics and the competing marketing departments of key corporate players. What is required is the cultivation of an informed and general outlook on spatial sound, combined with an appreciation of past mistakes and achievements.

The above summary of technological approaches to the making of music in the digital age reveals a rich and complex field of creativity, stimulated by ongoing research and development on a number of technical fronts. Vernonnen’s comments on future directions for research into spatial simulation are suggestive of the possibilities that lay ahead for many new and wondrous technical developments in the exciting creative field of digital audio.

Interactivity as a Modality in Multimedia Art

Immersion and psychological space.

An analytical approach to multimedia art must embrace the function, meaning and artistic potential of the interactive element. Interactivity sets multimedia apart from other mixed media forms. The participant or user is offered the freedom to navigate through a structured perceptual environment or space, at his/her own pace, and mentally map out the territory delineated by the artist. “Cyber” comes from the Greek for “helmsman”. Immersion is achieved through manipulation of the main sensory channels, with the additional catalyst that, as in life, in the process of navigating through such a virtual reality, one is offered choices. To progress requires active decision-making. Decision-making is both engaging and empowering.

Ascott [1993,] claims that

> “Art is no longer a window onto the world but a doorway through which the observer is invited to enter into a world of interaction and transformation.”

The introduction of branching structures into narrative opens up new options that allow the artist to diverge from traditional linear mind-sets. This is one area where the new digital artform will clearly increasingly diverge from traditional art and media, and artists addressing the issues involved with interactivity can be considered true pioneers. Cameron, [2000, 1] distinguishes between *interpretation* and *interaction,* i.e. the interactivity required to intervene in a multimedia application and the interactivity required to read or interpret a text*:*

> Interactivity refers to the possibility of an audience actively participating in the control of an artwork or representation. Until now, what we call culture has not allowed for a great deal of interaction from the audience. The audience is given a space for interpretation and a space for reaction, but not for interaction. There are those who argue that interpretation is interaction, and so of course it is, but not in the sense intended here. Interactivity means the ability to intervene in a meaningful way within the representation itself, not to read it differently. Thus interactivity in music would mean the ability to change the sound, interactivity in painting to change colours, or make marks, interactivity in film the ability to change the way the movie comes out and so on. In its most fully realised form, that of the simulation, interactivity allows narrative situations to be described in potential and then set into motion - a process whereby model building supersedes storytelling, and the what-if engine replaces narrative sequence.

The benchmark of good interactivity design is successfully producing a state of *immersion,* (unselfconscious engagement) on the behalf of the participant/audience.

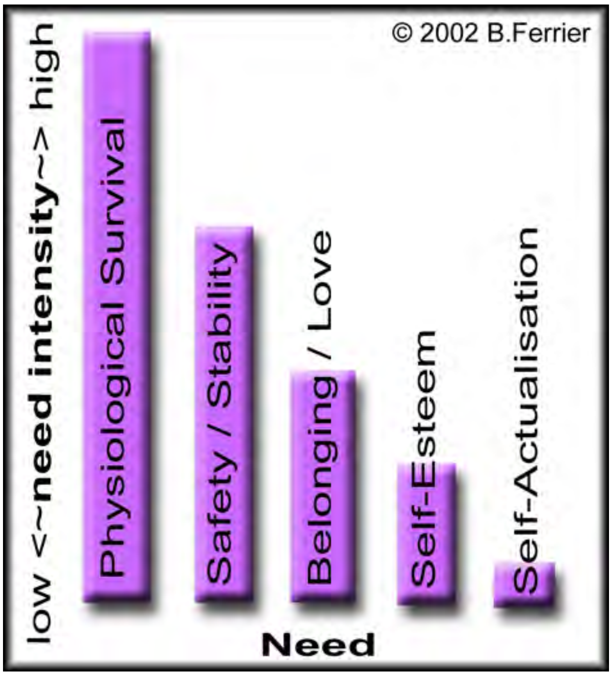
Maslow's Hierarchy of Need

Multimedia art can be seen as partly a process of psychological manipulation of the willing consumer. If one accepts the introduction of interactivity (or the designing of structures which allow user choice in outcomes) to be the defining bench-mark of true multimedia, then it is relevant to consider psychological theory in relation to decision-making and choice, in order to shed light on how the interactive process works. For instance, Mueller, [1996, 2] refers to classical psychological theories of human motivation as a possible approach to understanding the way users react to branching structures in interactive media, citing Maslow's *hierarchy of need*.

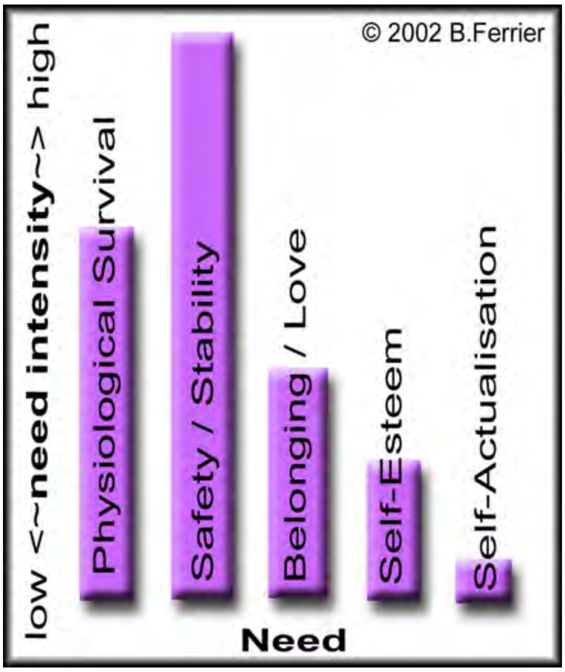
Abraham Maslow published his first conceptualization of this theory almost 60 years ago [Maslow, 1943] and it has since become one of the most popular and often cited theories of human motivation [Huett, 1998, 1].

Right: a graphical depiction of Maslow's hierarchy (Maslow, 1970, (colour added by author)

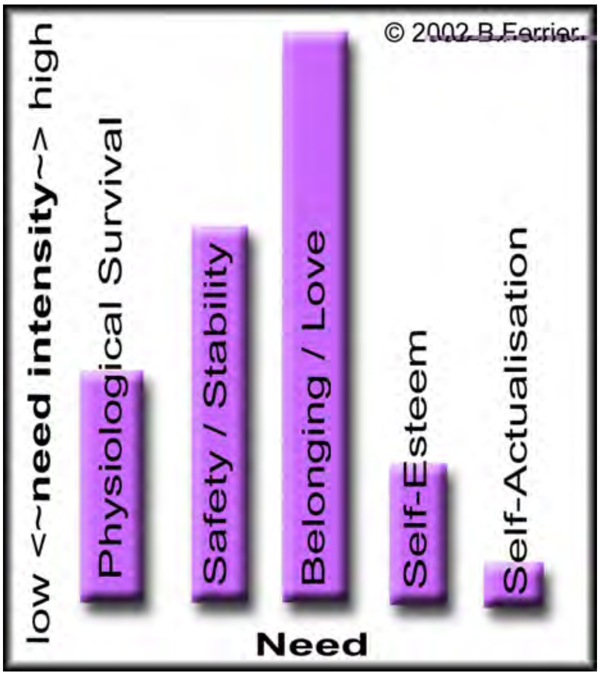
One level of the hierarchy must be partially satisfied in order for the next to take priority. Physiological needs (physical survival) tend to have the highest strength and thus form the base of the pyramid. The following series of graphs attempts to extrapolate on the implications of what happens once a need is to some degree satiated. The graphs follow the implication of Maslow's theory that all needs require some maintenance at any stage in the cycle. However one can imagine that, for example, the more stress experienced by not satisfying the primary physiological survival needs, the less energy is available to allocate to the other higher-order needs.



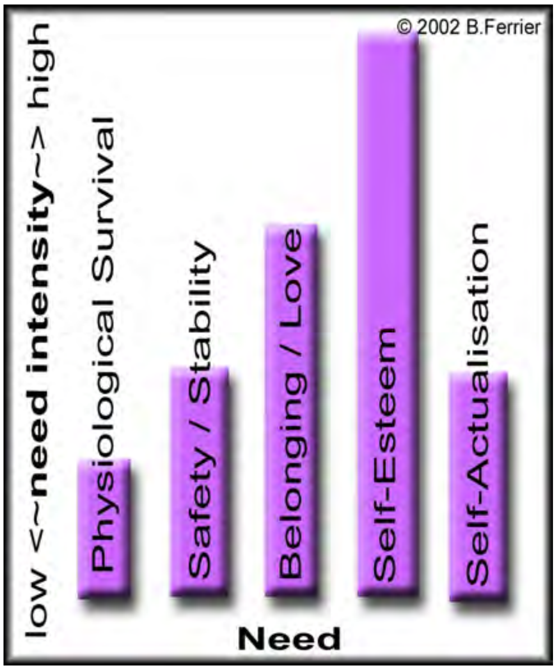
Physiological needs dominate



Move to the level of seeking safety, security freedom from fear

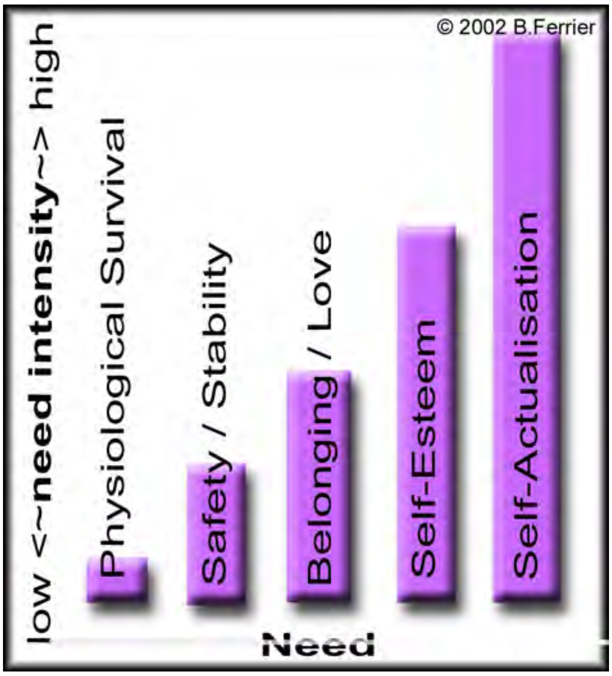


Having found safety, energy is concentrated on seeking family, relationship, belonging, love



With a secure family life and strong relationships in place, energy becomes focused on material achievement, career and self-projection into power relationships in society

Mueller [1996, 2] relates interactive choices made by multimedia users to Maslow's *hierarchy of human needs*.



Finally a stage is reached where the freedom to pursue personal fulfillment / spiritual needs exists.

Interactive 'performances that engage' can be viewed as responding to a similar or analogous hierarchy of necessity, or need intensity. Fundamental is survival: mental, physical, emotional. "Twitch" games utilize this imperative. Role-playing games such as *Ultima* [1] utilize the next level of needs, for belonging, community. Mueller suggest that, ideally, interactive rewards could take into account the higher level needs, for recognition and achievement, for knowledge and understanding, and finally self-actualization.

In relating these ideas to game design Mueller [1996, 6] explains:

> …when I read that 75% of people were still dealing with survival issues, I thought, oh God that's why we're in with all these shoot 'em up games. But it's not true, some people split from survival by doing other things.
>
> Then you've got bonding, recognition, achievement, which is what game play can be about. Knowledge and understanding, things you need to know and understand to self-actualise. These are important things to consider when you're structuring the branching out from the places that your game or interactive could take you. Because these are the things that resonate with an audience.

Problem posed by modular structures in interactive art

Cox, a museum curator and historian, points out that the imperative to produce modular structures, as part of the process of presenting information in a hyperlinked form, has its drawbacks.

> Nicholas Negroponte has written that "Interactive multimedia leaves very little to the imagination... When you read a novel, much of the colour, sound and motion come from you". Does using the CD-ROM require imagination? Do computers demand a new type of literacy or are they simply catering for users with a short attention span? Look at the length of the text pieces - they are all in 500-700 word digestible bites. You follow the themes that interest you by jumping from link to link in the hypertext, like we do on the Internet. I don't know what conclusion to draw here, only to highlight the problematic nature of serving up information in

[1] See page 400 for a discussion of this interactive game genre.

> short, bite-sized, undemanding grabs which leaves little room for conceptual thinking and thoeretical analysis of the issues. [1]

This echoes a similar difficulty with regard to sound design for interactive media, mentioned by Ganner [1991, 1] who notes the need to work in small discrete "chunks" of sound, precluding the development of long passages (see p. 459).

The accompanying interactive CD-ROM, *The Machine,* celebrates this modularity. The soundtrack for this work is made up of many short looping passages, chosen so that that there is a smooth flow of music, as the user navigates through the labrynth. A second layer of sounds synchronise with user-activated events. These sounds were chosen using the criterion that they should merge seamlessly with the back-ground sound loops, the end result being the creation of an interactively driven sound -scape that provides mood and glues the whole experience together. This form of audio design/music composition has its own aesthetic possibilities that differ from traditional linear music forms. It is difficult to develop climaxes as most sections are designed to loop until the user decides to progress. The looping of climactic music can become very irritating.

Technical, artistic and structural possibilities of interactive art

Despite the pressure in interactive media design to segment information into small self-contained modules, the new media forms clearly offer the potential for new kinds of artistic paradigms. Some of the possible technical, artistic and structural advantages offered by the utilization of branching structures are about working the little picture (communication) and big picture (meta-communication) simultaneously [Mueller, 1996]. These include designing the unexpected (creating novelty and surprise for entertainment or immersive value - 'little picture' supports 'big picture'), allowing for the continuous reframing of perspectives on the narrative structure, using structural options to influencing the mental process of the user, manipulating emotions, distortion or compression of time, all with the ultimate goal of revealing the meta-communication.

Rollings and Morris [2000, 33], comparing the use of linear and non-linear problem-solving in games design, point out that linear design requires a participant or game-player to tackle problem A, the problem B, etc. without offering choice. While the individual problems may be engaging, the linear structure precludes the more immersive possibility of strategic thinking, whereby the order in which the problems are tackled is an interesting game-play element on its own right. They state that

[1] Notes from a lecture on *Real Wild Child* : the CD-ROM, by Peter Cox, loaned to the author.

successful game design will allow for tactics (short-term decisions, e.g. which gun to use) and strategy (long-term decisions, e.g. which path to take for victory).

Interactivity can be both physical and conceptual. The design of these points of decision can be structured to illuminate concepts. There is a range of functionalities attributable to, or discernable during the process of navigating through branching structures [Mueller, 1996, 2]. Beginning with opening options, gamers proceed into a virtual unknown of changing circumstances (possibly chaos) through a stage of path finding, trail-blazing, mental mapping. Interactive navigation demands the ability to diverge from the linear and expected, and requires engaging in decision-making, in connecting ideas, and the ability to shift activities and focus as circumstances require.

Time and interactivity

This discussion of issues raised by the introduction of interactivity into electronic art can only briefly mention the use of time-based stylistic or structural elements, familiar from music analysis, functioning in interactive art. Dynamics, such as pacing, rhythm, tension and release, building to a climax (crescendo-decrescendo) are all important dynamic variables in the interactive script, adding dramatic tension and variety, and thus increasing or maintaining immersion, adding meaning, emphasis and accents to the cognitive experience. The structure of multi-layered meaning is in itself a means of communication, as well as a field of aesthetic activity.

St. Hippolyte [1995] points out an inherent problem facing designers of interactive narratives - the underlying linear chronology implicit in storytelling:

> If the telling of a story is linear, then that story has a default chronology, which is the chronology of the telling. That is, if nothing else is said, the first events described are the ones that happened earliest, the lasts events described are the ones that happened last, and everything that happens in between is described in order. This is simply the default; stories with flashbacks and foreshadows, i.e., events described out of sequence, can have a much more complex chronology. Many literary devices are available to bend time; but no device is necessary to describe a well-ordered sequence of events linearly: it is what happens automatically unless an effort is made to the contrary.
>
> If the telling of a story is nonlinear, however, there is no default chronology. The telling may go along many paths, each with its own implicit chronology. But if there are chronological rules to be enforced, the author must assure that the chronology that emerges from each telling fulfills those rules. Naturally, this translates into limits on the viewer's choices.

This discussion has attempted to flag some of the unique issues raised by an analysis the process of producing interactive, non-linear, digital multimedia art.

The structure of interactivity and interactive design paradigms

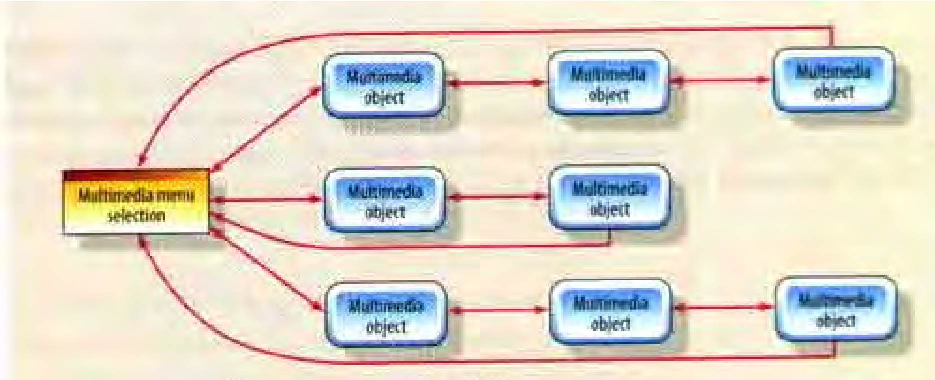
According to Mueller [*ibid*.], by incorporating interactivity into a work of multimedia art, the artist must develop a hierarchical structure for the piece, one which simultaneously addresses or covers: content, cross references and links, and user actions, reactions, expectations. The interactive design process will, of necessity, require visual aids such as concept maps, flow charts and storyboards, in order to systematize the complexity of larger-scale projects involving non-linear branching structures. Technically the process must then lead the artist to develop engagement devices and navigational design elements, define boundaries, decide on sub-divisions and propose multi-routed vectors.

Hofstetter [2000, 291-303], states there are five ways to design the flow of a multimedia application, in ascending order of complexity, and proposes some generic visualizations to represent each of the main types:

1. **Linear List** : The linear list design simply lets the user move forward to see new materials or backward to review.

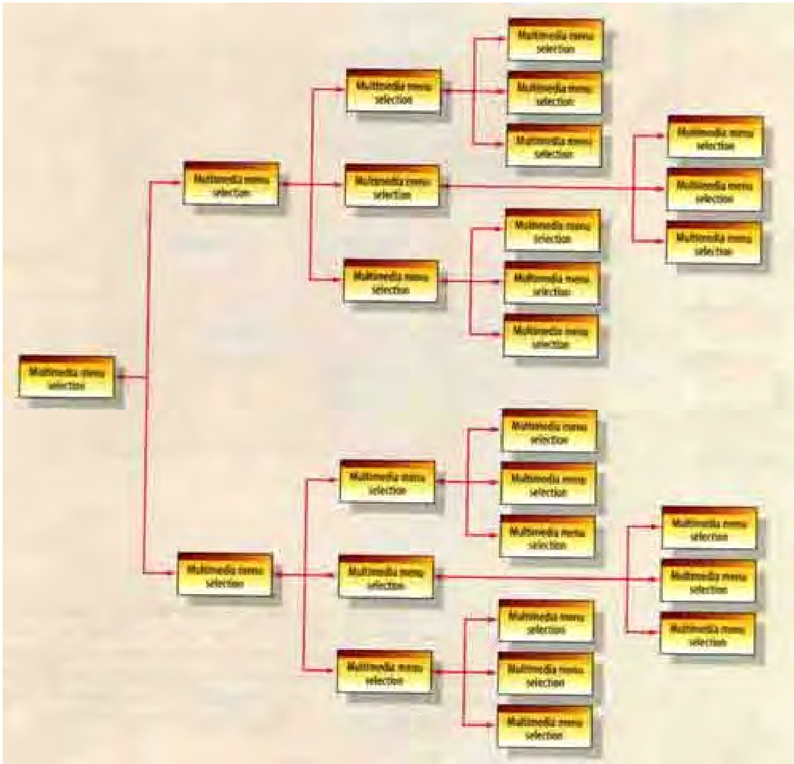
As objects in a linear multimedia piece will frequently contain animations and sound, one of the biggest challenges in this form of multimedia design is making looping sections which can stand alone, (i.e. which work despite the repetition of this looping media form set up to keep interest while the user makes choices) and yet which contribute solidly to the story-line [1].

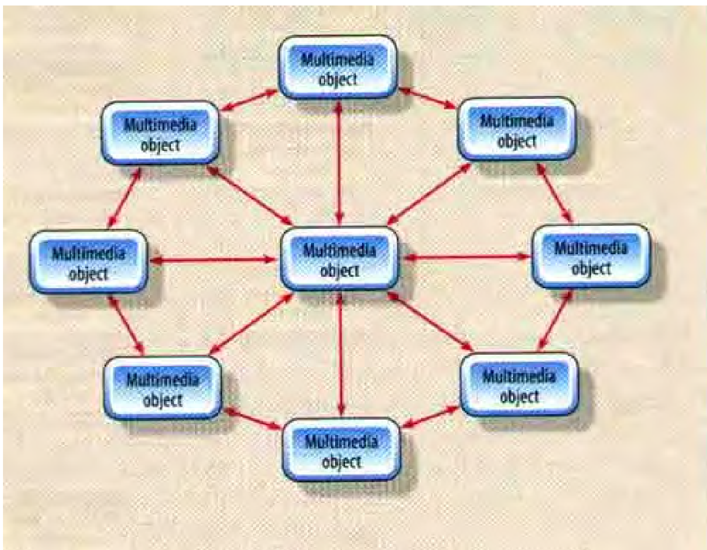
[1] In the accompanying CD-ROM *the Machine*, a section accessed through *the Recreation Centre*, entitled *Child Minding Centre,* contains a sample of linear progression through a series of small looping cartoons created in Macromedia *Flash*, designed as a children's cartoon, called *Emutown*. This section has not been completed, but offers enough material to give the user a sense of a story shape. It is different from a traditional story in that each story "module" must loop, (including the soundtrack)and be individually coherent.



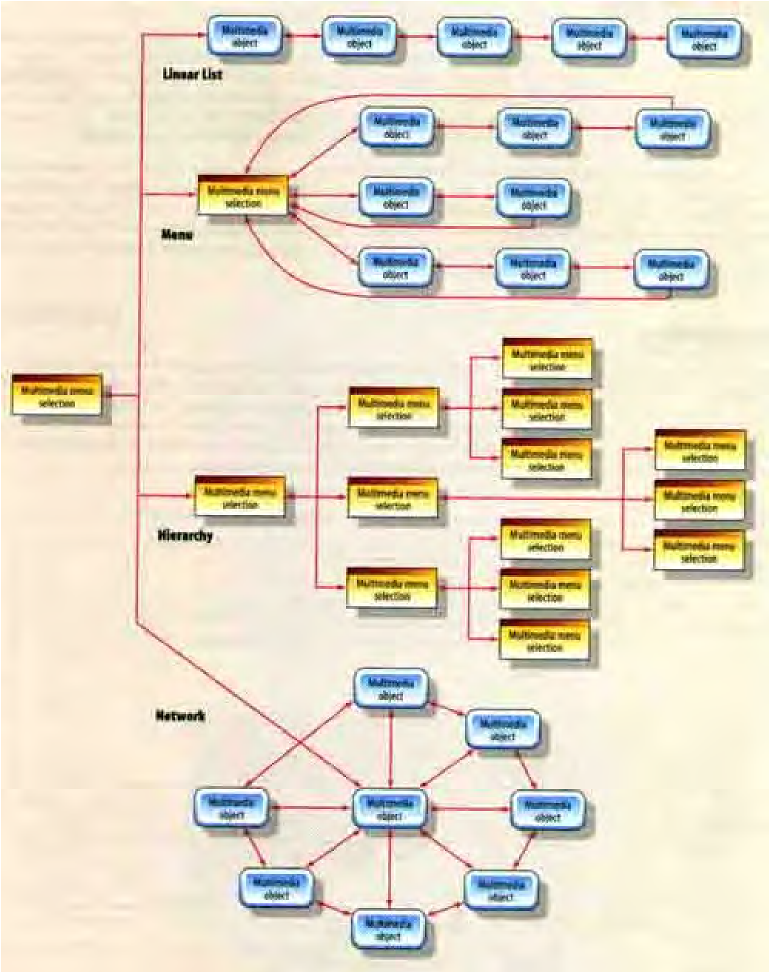
2. Menu (left) : The menu design offers the user a set of choices in a simple branching arrangement. This is a simplest, most common, multi media structure.

3. Hierarchy (right): The hierarchy presents the user with menus of submenus in a more complex branching structure.





4.Network (left): The network diagram contains multiple linked items that, according to Hofstetter, provide the richest kind of navigation. There is a section in *The Machine*, accessed through the *Crop Circle Mandala* Portal (a sub-branch of the Central Control Panel) that is called Control Panel 3, that is visualized in this form.



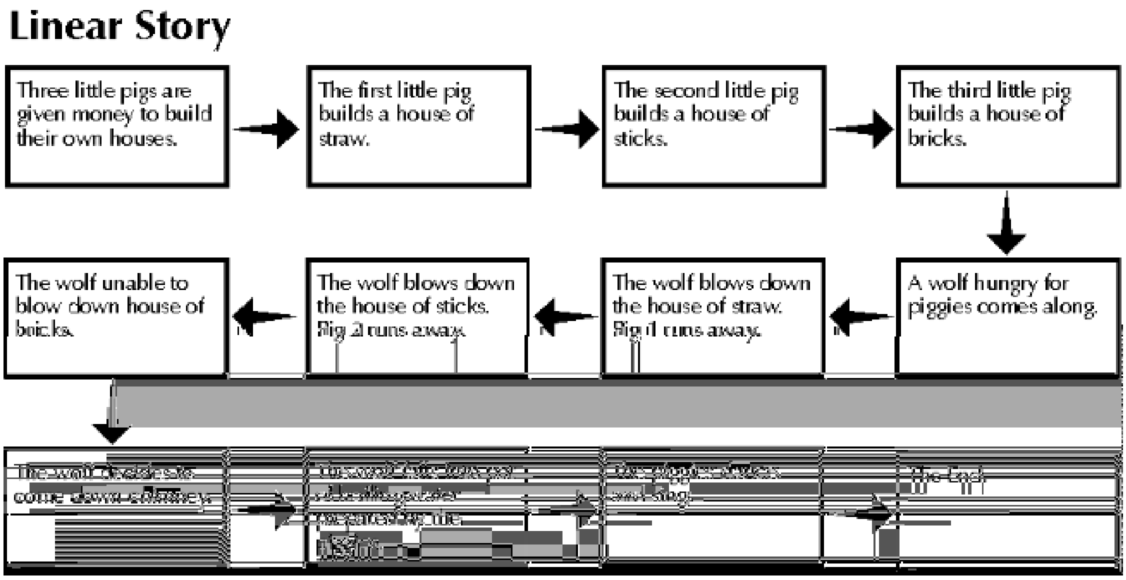
5.Hybrid (left): Hybrid designs facilitate more complex clusters of information and can employ linear lists, menus, hierarchies and networks where appropriate, according to the primary directive of the meta-communication. *The Machine* is an example of this kind of complex interactive structure.

Interactive Story Shapes

A brief discussion of analytical concepts of narrative was presented on page 210, which focused on narrative in film. *Interactive* narrative is a relatively new and difficult field which is beginning to be explored by theorists. St. Hippolyte, (1995) presents an interesting overview and states:

> The interactive dilemma may be described as the unavoidable tension between the author's goal to express something specific and the viewer's freedom to choose among different narrative possibilities…The solution to this problem almost invariably involves placing further limits on the viewer. There may exist messages that rely so little on the narrative itself that the viewer can be given the lattitude to change the story at will, but in all other cases the viewer's choices must be carefully constrained so that he does not obstruct the author's message… Hypertext narratives have their strongest impact when the links themselves communicate meaning, which occurs when the viewer has traversed enough paths and begins to visualize the narrative structure as a whole. The links can then form a kind of sketch, distilling the story into a single timeless image. On the other hand, if the links fail to carry any meaning of their own, if they are just links, the interactivity is reduced to a procedural convenience for reading the text electronically, and the ability of the viewer to affect the story vanishes.

Hofstetter's concept maps of linking systems, discussed above, have an interesting resonance when compared to the ideas of Phelps, with regard to interactive story shapes. Phelps [1998, 2-8] differentiates seven different story shapes that can occur in interactive digital media, again in ascending order of complexity, providing clear examples and visualizations of each.



Like Hofstetter, Phelps begins with a simple linear structure – seemingly no different to a book or film. However, Phelps [*ibid*., p.2] cites two specific and persuasive reasons why traditional linear structures have a place in a list of interactive media paradigms:

> **1.** Linearity does not preclude a work from being multi-media. A linear experience, allowing the user to define the duration of engagement with each segment, can be enhanced with sound, music and animated sequences, thus justifying its existence on a CD-ROM or Web site (illustrated in *Emutown*, accessed through *the Child Minding Centre*, on the accompanying CD-ROM, *The Machine*).
>
> **2.** Due to the overwhelming amounts of written work that has and is being produced, physical archival space for linear works is at a premium, requiring increased need for digitization, e.g. encyclopedias, dictionaries and collected works. Related to this are environmental considerations. Increasing volume of printed work puts a strain on forest resources, necessitating that ephemeral media e.g. newspapers and magazines, appear exclusively online.

Digital media are more durable. Apart from these archiving imperatives, the advantages of the linear structure are that it is a simple, reliable and resource-cheap way of approaching the challenge of producing an interactive story. In interactive games with a low budget, the linear structure is often the only affordable approach.[1]

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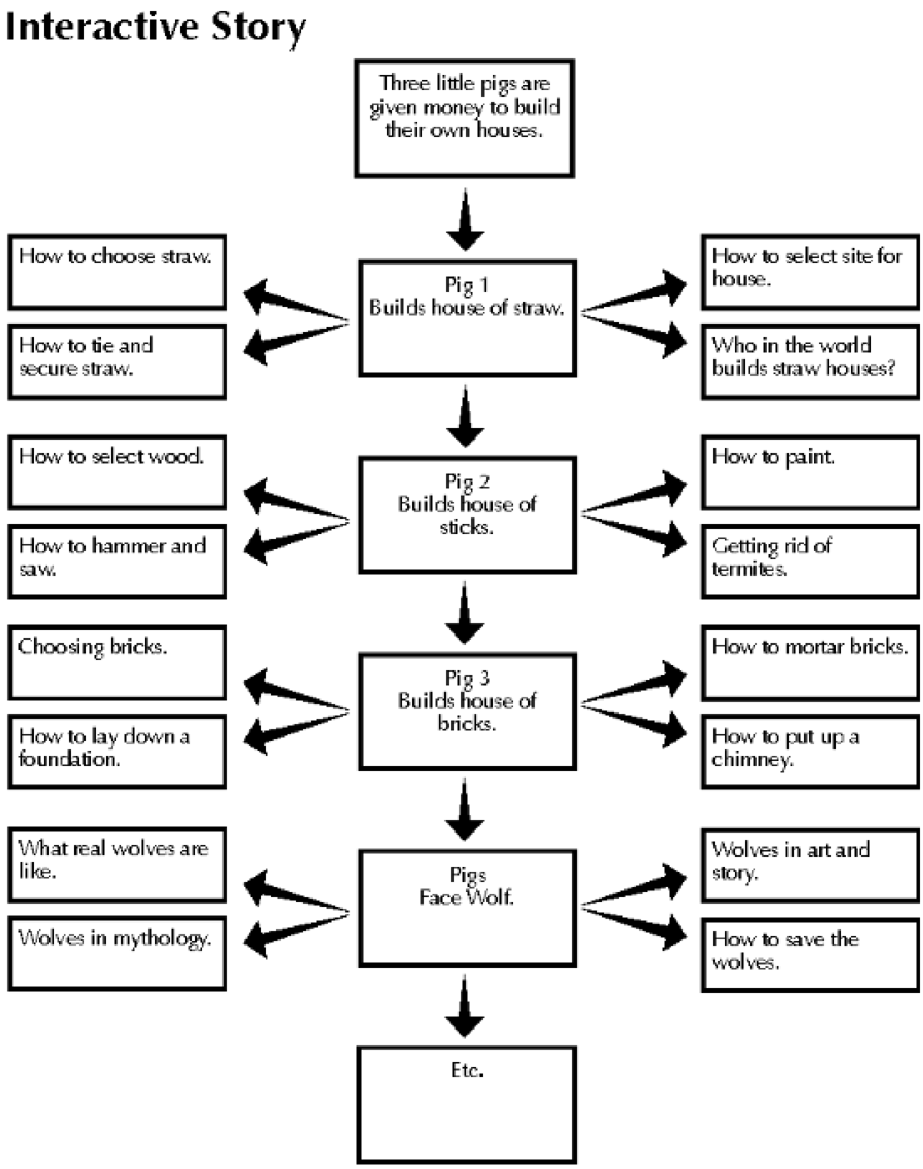
[1] http://www.igda.org/ ; accessed 24-02-04. *'Interactive Storytelling in Games: What it is, Why we need it, and How you do it'* ; presented by International Hobo at Digital Media World 2001, November 13th, 2001 (uncredited).

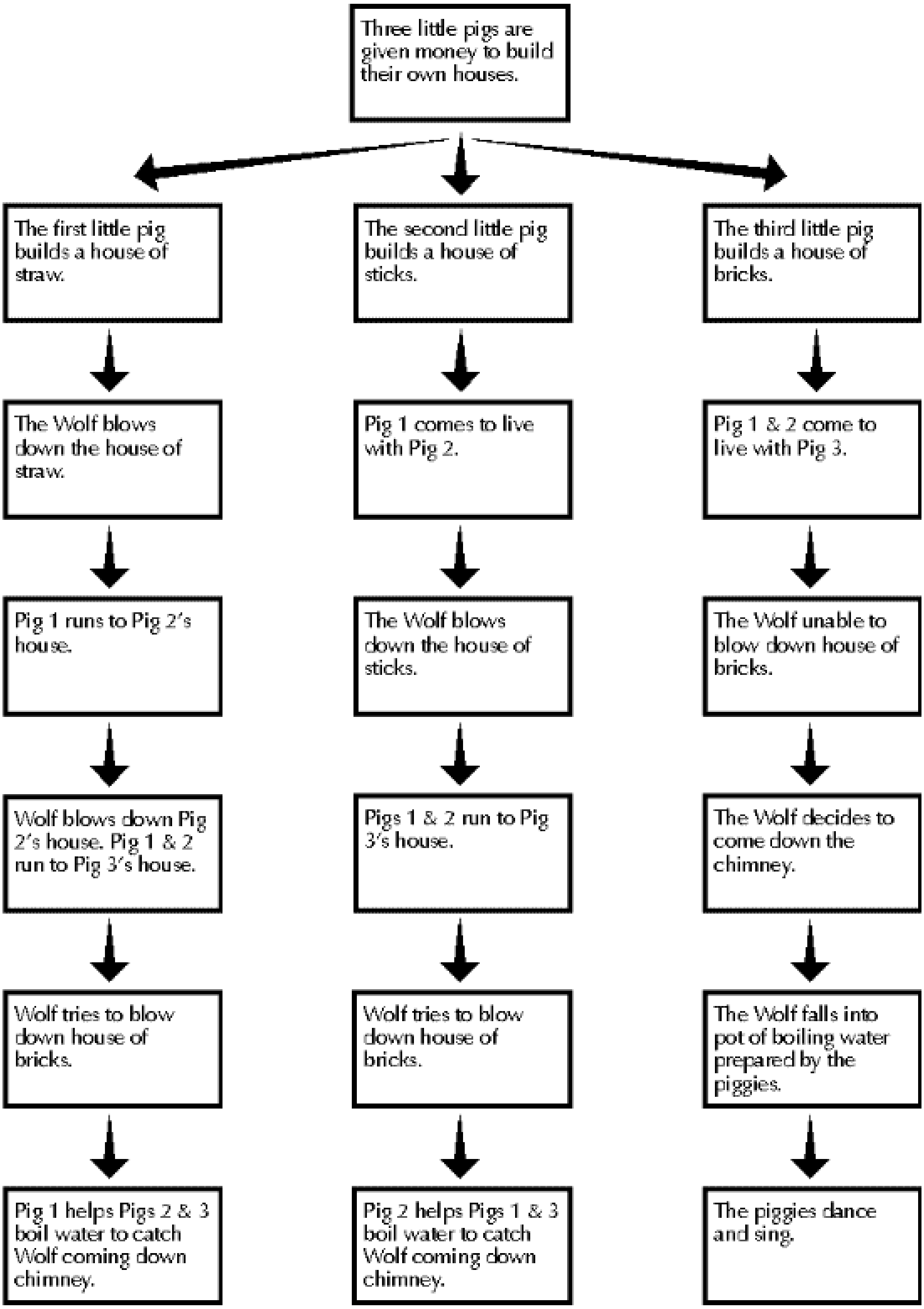


An *interactive narrative shape* (above) is one in which the story is still basically linear, but each screen of material offers an opportunity for the user an enriched experience. It is sometimes also referred to as an annotative structure, or advanced footnoting. The user can venture away from the main point to related sub-points, but it is easy to get back to the main theme. St. Hippolyte (1995) calls these *linear narratives with embedded nonlinear elements,* and claims that many examples can be found among children's multimedia titles, in which the the user can often click on a character or object and get a "sideshow" of some kind.

Multi-Linear Story

According to Phelps, [*ibid*., p.23], a *multi-linear narrative shape* is one in which several parallel stories will be presented without those stories directly interconnecting, (which must be differentiated from a *braided* multi-linear story, covered below).

Multi-linear Story



The reasons for using this structure may be that:

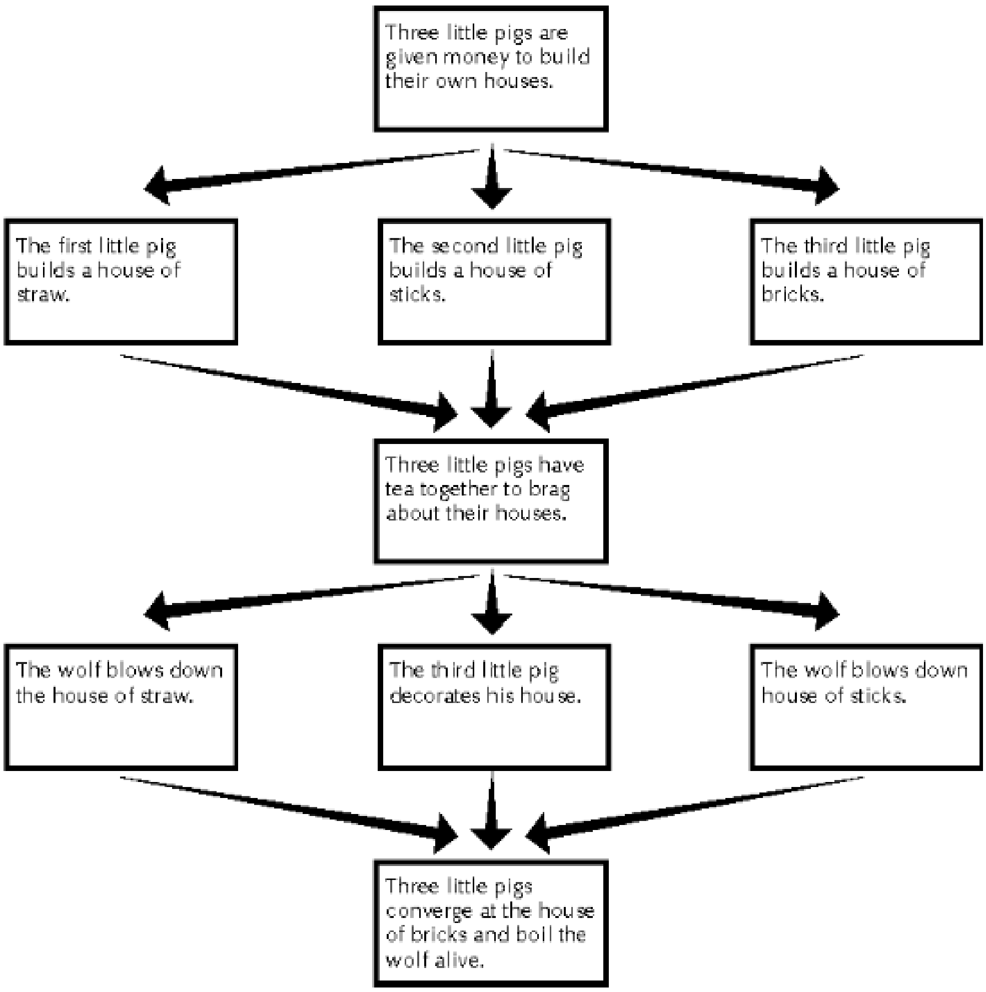
- the story takes the user through the same physical spaces, or the same series of events, but through the eyes of different characters, or
- the story might follow the same character through different events,

or perhaps even

- the audience is meant to compare and contrast these different lives and/or events.

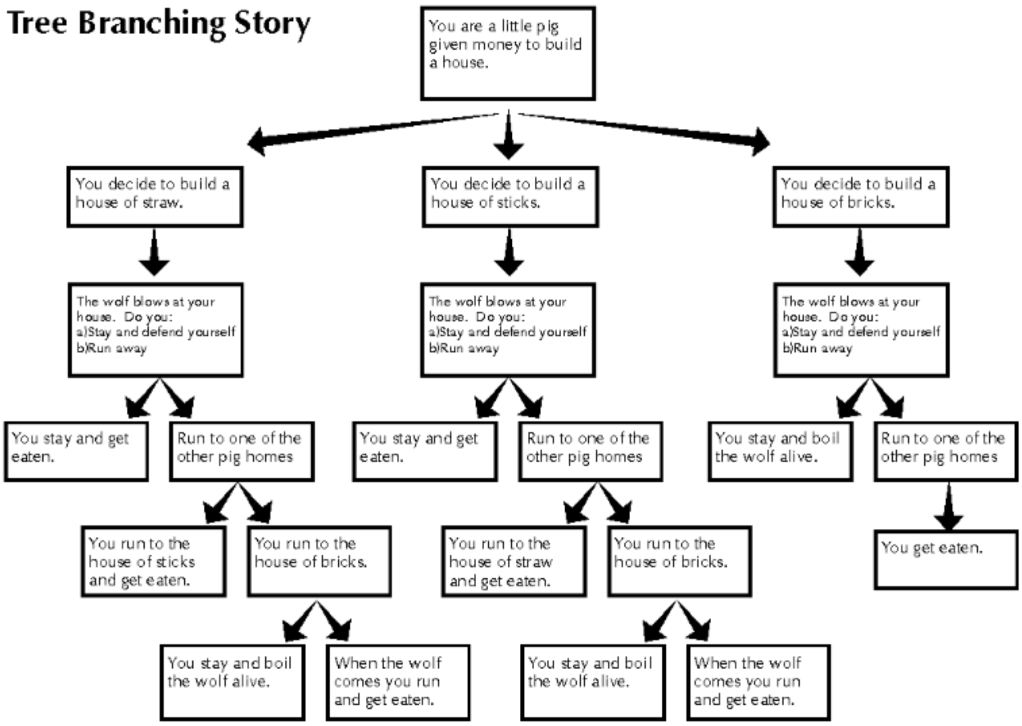
The user gets to choose the story direction, but once commited, must follow this through to its conclusion, without being able to jump directly between the various alternate branches. This facilitates a comparative learning, “what if” approach.

Braided Multi-linear Story



With a *braided multi-linear structure*, the author is less concerned with parallel development, and more with how lives and events can repeatedly interconnect.

With the braided multi-linear shape an initial situation is given, which then branches out in a number of thematically related plot directions, which subsequently converge upon another situation, that then again offers a number of directions to choose from. In this manner it is still possible to build dramatic tension, give the user a feeling of free movement, and yet even with that free movement provide a seamless story experience. Branches recombine at the key story points, creating a structure which is a balance between linear and branching structure. More significantly, it makes the work compellingly repeatable. (Phelps argues that most CD-ROM experiences and much hyper-fiction tend to encourage a one-time experience. Once the puzzle has been solved, there is no reason to go back and do it again at a later date.) This structure is used to some extent in interactive games[1] such as Konami's 1998 cinematic spy-thriller/action game *Metal Gear Solid* (criticized by some mainstream game critics as having too much story and too little gameplay)[2], and Ion Storm's intriguing, dark and challenging *Deus Ex* (2000), which set new standards for freedom of plot development in interactive computer games[3].



Phelps, [p.4] calls Tree-branching “the classic structure”. This form echoes Hofstetter’s simple hierarchical navigation system. At each scene or branch in the decision tree the audience is given a choice or several choices. Each of the chosen scenes in turn leads to further choices. Thus plot-based tree-branching fiction suffers from the unforgiving mathematics of exponential growth. As few as six decision points with only two choices at each implies sixty-four different story-lines [Farmer, 1996]. Traditionally, to avoid infinite choices being set up, one "correct" story path exists and all others must either converge or lead to a dead-end, thus pruning their potential early into the decision making process. Phelps explains:

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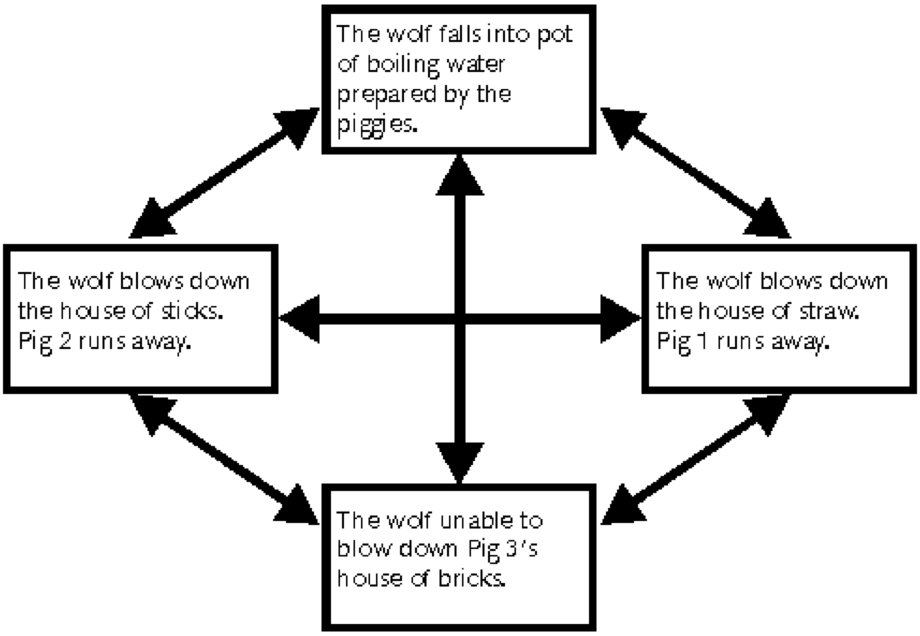
[1] http://www.igda.org/ ; accessed 24-02-04. *'Interactive Storytelling in Games: What it is, Why we need it, and How you do it'* ;presented by International Hobo at Digital Media World 2001, November 13th, 2001 (uncredited).
[2] e.g. http://www.gamespot.com/ps/adventure/metalgearsolid/review.html ; accessed 26-02-04.
[3] http://www.cdmag.com/articles/028/142/deusex_review.html ; accessed 26-02-04.

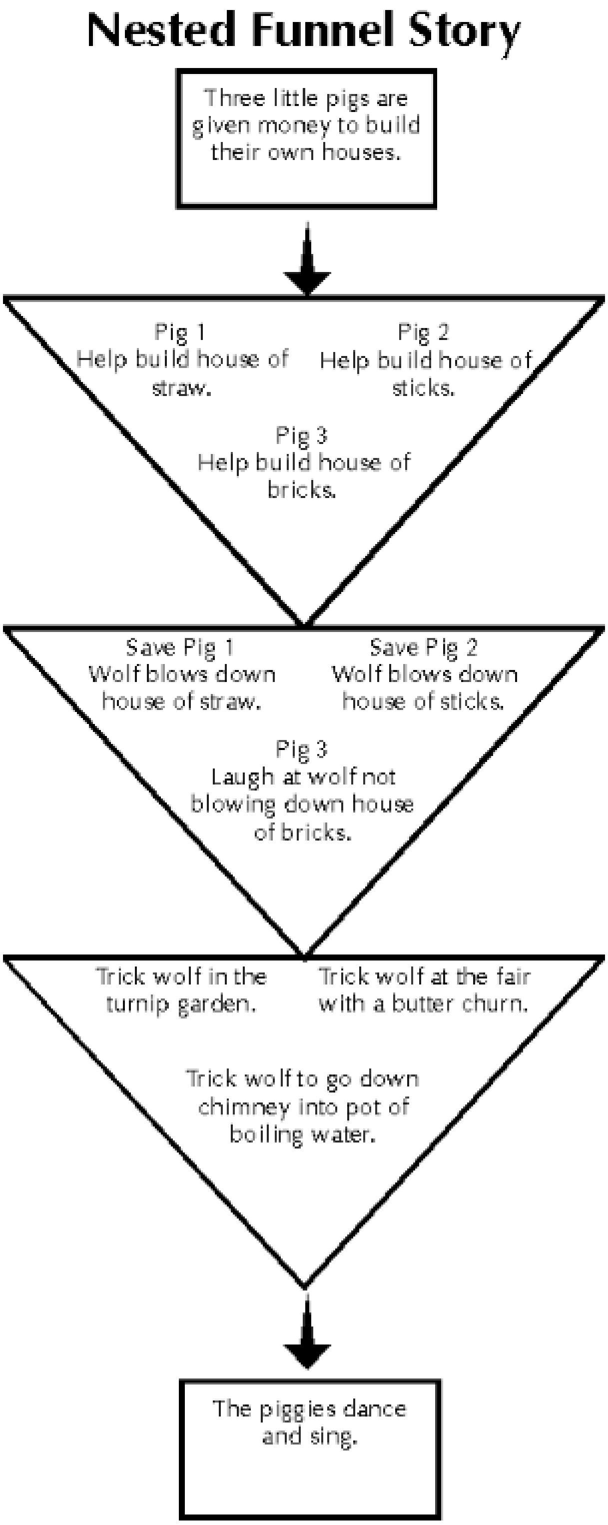
…the majority of linear narratives have important decision points where the story could have gone one way or another depending upon the choices made. Nevertheless, storytellers carefully craft the series of events in their stories deliberately making their characters' choices at each turn such that the entire narrative expresses a certain theme of concern to the teller. To enter into certain storytellers' works and change their choices is to subvert the theme and to damage the artistic integrity of a work... A better strategy is to create a story whose theme encompasses a few select endings and the story is carefully told such that the audience is satisfied with making a constrained set of choices that will lead to one of these endings. This is where the challenge and artistry of using the tree-branching narrative shape enters in.

Non-Linear Story



This interactive story shape is consistent with Hofstetter's depiction of a network navigation system (see p.361), which he claims is the "richest form of navigation". Yet according to Phelps this story-form has problems for narrative because non-linear, in its strictest sense, must ultimately mean non-chronological. Phelps' comments here reveal the difficulty in creating a satisfying narrative with this structure:

> I have found it unusual to discover anything like a complete story coming out of this shape, because lack of narrative progression provides too little support for developing character and plot. For a short work the audience may be willing to wade through all segments of a narrative in order to piece together a coherent story. For a longer work the tendency is for people to lose interest after only a few screens.



Hofstetter does not provide us with a navigation system that addresses the concept presented by Phelps' *nested funnel* shape, which is conditional on user reactions in terms of timing and duration. Within the nested funnel the user must either go through all scenes, do all set activities, or at least a significant set of scenes and/or activities within an act, before being allowed into the next act. Farmer [1996] calls this "conditional branching". *Myst* (discussed in detail below, p. 392) could be viewed as a nested funnel narrative. A popular puzzle for such story-based games has been to require players to find different objects, which will later be vital to their success in entering different levels of a game, hence the "scavenger hunt" character of many nested funnel stories.

St. Hippolyte (1995) prefers to call this approach to interactive storytelling a *maze* and claims that it is the most widely used model for handling interactivity within a narrative structure, only emerging since personal computers could handle higher quality graphics with reasonable speed.

> In contrast to hypertext, the links in a maze story are metaphorical rather than literal. They depend on, and are limited by, an allusion to a physical model of interconnection. This physical model translates the decision tree into a space of some sort (cave, island, haunted

> house) through which the viewer moves. The paths naturally converge, because they are physical paths in a closed space.
>
> The metaphorical nature of the links is also the maze model's answer to the interactive dilemma. The storyteller depicts the entire decision structure metaphorically (interconnected rooms in a house, for example), works the metaphor into the story as an omnipresent theme (you are always in the house), and uses other elements of the story to limit the viewer's freedom of choice without diminishing the metaphor (you can't get into a particular room because you don't have the key). The viewer ends up perceiving much more freedom of choice than he actually has; this in turn gives the author more control, and more opportunity for expression…. The player drives the narrative by moving through the space it occupies; thus, whereas traditional stories move through time, maze stories move through space. This substitution of space for time is an obstacle to establishing a chronology. It can leave the viewer with the sense that time has been drained out of the story, that much of the story took place long ago. It can also make the characters in the story seem outside of time, which means less real. This is the price the storyteller must pay to use this model.

Maze stories such as *Myst* have tended to solve the chronology problem of non-linearity by presenting two intertwined stories: the story from long ago, that the viewer gradually uncovers, and the “present” story of that uncovering. The former is unchanging, while the latter is created anew (interactively) with every telling. In this way the background or unchanging story can deliver the bulk of the narrative detail and creative expression, while the ever-fresh “real-time” story of interactive discovery can give life to the experience for the user.

Collaborative or multi-author tree fictions

Another way to view "interaction" is as a chance for collaborative authorship, in the sense that readers are invited to add the next fragment or branch. Farmer, (1996) gives the following example:

> The City First Bank president is "pantsed" [sic] on a downtown street at noon by an old college buddy. His first spoken response is:
>
> 'Guess you wonder why I called you here…' He gets his pants back on and stands in front of Mason, who "pantsed" him.
>
> He slugs him, which is a big mistake. The crowd attacks him, killing him.
>
> He dies and goes to heaven.
>
> He dies and goes to hell.
>
> He dies and ends up in limbo.
>
> He speaks…
>
> 'Someone call the police'
>
> 'Let me tell you about our new Savings Plan'"

Farmer sums up the major criticism of this form of interactive writing, interesting though it might be in an experimental sense:

> …my feeling is that the advantage of having a wider passion and experience in the authorial position is unlikely to offset the disadvantage of not having a single tight and even ruthless ego tailoring the material presented to the reader. Is it surprising that very little literature has been collaborative?

An interesting use of this form of writing occurs on the snarg.net web-site discussed on page 92.

Transgressive metalepsis and interactivity

It is an assumption underlying the taxonomy presented by Phelps that much interactive multimedia is still fundamentally a form of storytelling. If one accepts this idea, then it is interesting to apply concepts related to narrative, in order to reveal what if anything is different about an *interactive* form of storytelling Events which occur within the world of the narrative, or diegesis, are termed *diegetic.* Events occurring at the level of the act of narrating are considered *extradiegetic.* Genette [1980, 228] defines any shift between two levels of narrative, such as between the diegetic and non- or extradiegetic, as a *metalepsis,* and further defines shifts between the two levels of narrative that confront the suspension of disbelief or draw attention to technique as *transgressive metalepsis,* the latter usually have a comic effect. According to Murtaugh (1996), a quintessential example of such an effect in the context of film involves background music. If, for instance, as lush romantic music swells, one character says to the other *"Sorry? I couldn't hear you over the music"*, we are breaking the narrative tradition and drawing attention to the artificially introduced mood music, which in turn breaks our involvement in and identification with the narrative.

> "Branch Point" interactions, where the viewer acts to change the course of a narrative, are thus disruptive not only when they stop the story to wait for user input, but fundamentally by placing the consequence of an action performed by the viewer within the diegesis.
>
> By presuming to break the diegetic barrier, such approaches invariably undermine a narrative system, calling into question the boundaries between the narrative, story, narrating act, and ultimately the reader. While such an action may well be a potent tool for the author interested in calling these relationships into question, the technique shifts the viewer's attention from the narrative to its disruption. Furthermore, if such "effects" are perceived as novelties, they cannot form the backbone of an experience; the novelty wears thin quickly, the comic or fantastic quickly become tedious.
>
> Focusing only on this exceptional potential of interactivity needlessly limits its role to one of disrupting narrative. Instead, by accepting its inherently extradiegetic position, interactivity may be seen, like the background music or editing in film, to serve the narrative by focusing the viewer's attention on the product rather than the process of the narrating.

Multimedia artists must therefore be beware of a possible tension that can be set up by the introduction of interactivity into narrative between the simple objective of user immersion in ‘a good story’ and another kind of immersion encouraged by giving the user the power of interactive control.

A brief history of the commercial evolution of the interactive computer game

There are cultural traditions upon which multimedia art draws. It is not difficult to accept that there will be strong influences from historical precedents in visual/graphic art, music and creative writing. The cinema tradition likewise will influence the incorporation of narrative, animation and aesthetic approaches to video elements.

Multimedia art requires expensive hi-tech equipment and skilled creative teams for its creation and delivery, as well as packaging and marketing budgets. As a commercially driven phenomenon it will inevitably be subject to the pressures of market forces and must therefore succeed as popular/mass entertainment. Practitioners will strive to develop those aspects of the form which maximize its entertainment value. Live performance entertainers in popular music and cabaret fields have long known the power of audience participation (interactivity) as a device to engross or immerse the audience in the entertainment. User *interaction* is a necessary and fundamental property that differentiates the computer game from other forms of passive entertainment or passive reception of information, and from its various artistic antecedents. *Interaction* is the ability of the player to actively make one or more choices, via the user interface, at each move. Carefully planned interaction is critical to the successful reception of a game by players. [1]

Computer gaming is a branch of the entertainment industry now as big, if not bigger in commercial terms than the music industry, and it has historically been at the forefront of commercial applications of interactive multimedia design. Total computer (or “video”) game hardware and software sales for 2001 was in the vicinity of $8.5 billion in the United States [Gaudiosi, 2001, 33]. The *Businessworld Magazine’s Internet Edition* web-site[2] report on worldwide games consoles sales 2001, states:

> Microsoft said it had shipped 1.5 million units of the $299 **Xbox** by the end of December after it was introduced in North America on November 15. The US software giant plans to roll out the console in Japan on February 22 and in Europe on March 14. Microsoft said it aimed to ship 4.5 million to six million **Xbox** consoles worldwide by the end of its fiscal year in June.
>
> Nintendo has not yet released sales figures for **GameCube**, which hit shelves in Japan in September with a price tag of ¥25,000 and came out in North America in November at $199.

[1] http://www.thescratchpost.com/resources/games/games_dict_i.shtml ; accessed 2-12-02.
[2] January 11-12, 2002): http://bworld.net ; accessed 22-01-02.

> Sony makes around 1.8 million **PlayStation 2** consoles a month, the spokesman said. Despite the upbeat result for the holiday season, he said that Sony would keep its annual target of shipping 20 million consoles worldwide by the end of March, bringing total shipment to 30.61 million units. Sony also said sales of game software titles tripled in December from a year earlier, with the number of titles for PlayStation 2 and PlayStation 1 combined having reached more than 4,200 in Japan and 1,400 in North America. -- *Reuters* [1]

Interactive computer games come in a variety of genres. They are played in virtual adventure worlds and can be analyzed in terms of "the game plan within a continuous space" [Mueller, 1996, 6]. The most prevalent form is the "twitch" or reaction-based shooting game, which commonly draws on sci-fi traditions, and super-hero comic books/cartoons conventions as well as other forms of adventure cinema. It is most frequently centred around "the champion" or user-controlled surrogate combatant. It contains equipment (virtual objects) consistent with the physical interactive interface e.g. guns, and other directly controlled virtual weapons, user-controlled virtual vehicles, and other derivations of the astro-physics metaphor e.g. energy, fuel and or ammunition. The game must have rules of play and victory conditions, and a range of possible interactive movements. Participation in interactive games requires the willing (and often unconscious) decision not to question too closely assumptions underlying the gameworld in which the user is playing, e.g. accepting the existence of magic. As virtual worlds can defy newtonian physics, they can be enriched imaginatively and conceptually by offering "para-normal" or "super-natural" phenomena, or unrealistic fighting abilities (the ability of a action figure to single-handedly shoot his way through a small army of enemy agents etc.).

A definitive history of interactive computer games cannot be offered here, but the major milestones in computer game evolution are presented, as a background to further discussion of the computer game's place in the development of interactivity in multimedia art.

The interactive video game originated in two forms: as the mainstay of electronic recreation arcades, and games designed to be played on home computers [Haddon, 1988, 27]. Computer games have become pervasive cultural icons among the young male population of developed societies, known by the brand names of dedicated hardware devices such as *nintendo, sega and playstation*. [2]

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[1] http://itmatters.com.ph/news/news_01112002j.html ; accessed 17-05-02.
[2] http://www.videotopia.com/games2.htm ; accessed 12-11-02.

Nintendo *Game Boy* - 1989. Designed by Gumpei Yokoi and Nintendo R&D Team #1. The *Game Boy* was a portable system utilising a black and green LCD screen and was programmable via interchangeable cartridges. It contained a 1.1 Mhz 8-bit Microprocessor and was released at a price of $US100. *Game Boy* was popular with adults as well as children.

Sega *Genesis* - 1989. Recognizing that immersive games sold systems, Sega took elements from its 16-bit arcade machines and produced the Mega-Drive in 1989. It reached American stores as the *Genesis*, retailing for $US199. Featuring a version of the Motorola 68000 16-bit microprocessor that had powered the original Apple *Macintosh*, the new console was capable of running effective translations of Sega arcade hits.

Sony *Playstation* - 1995. The Sony P*laystation* used a 33 Mhz 32-bit microprocessor specifically designed to produce polygon graphics.

Dedicated hardware computer games consoles were originally marketed as extensions to the home TV set, with advertising frequently featuring a family group sitting with a games console in the television room, to counter parental fears that computer games would lead to isolation and lack of social engagement among children); [Haddon, *ibid*., 28]. Equally important in the exponential growth in popularity of this new entertainment form was the introduction and evolving sophistication of animated 3D graphics.

During the 1970s the first games consisted of combinations of chips on which there were fixed programs. A machine might for instance come with three games on it. The next major development, first in arcade games and subsequently in dedicated domestic games machines was a movement towards microprocessor-based games, which could be re-programmed. [Haddon, 1988, 28]. Lamothe [1999,10] credits the release of the arcade game *Pong,* (designed by Nolan Busnell and Alan Alcorn and released by *Atari* in 1972) with establishing the video arcade game as a commercial mass-market business “overnight”. The game play was extremely simple, permitting two players, both of whom controlled an on-screen vertical bar, which could “bounce back” a moving dot (which moved between the vertical bars – reminiscent of table-tennis, hence the name). According to Kuittinen [1997, 3], Busnell placed the first game machine in a local petrol station as a trial, but was called back when the machine ceased to operate – exceeding expectations, it was full of money. A home version was designed by Al Alcorn, Bob Brown, and Harold Lee and introduced in 1975.

Arcade version of Pong


Home console

McCauley, [1997, 27] cites the first financially successful venture in the history of commercially available interactive computer games on the personal computer platform as *Battlezone,* which was released in 1980 and only available on Atari PC computer operating systems. According to McCauley it stood out because:

> Its first person perspective was conveyed through vector graphics with colour courtesy of acetate overlays on the screen. The game play was basic, the controls unwieldy, but it was genuine 3D...

While depicting 2D visual elements, *Battlezone* had a sense of 3D depth. According to Kuittinen [1997, 2], the United States Armed Forces were so impressed by this game that they commissioned Atari to build specially modified and upgraded versions for use in tank training, an example of how a combination of government/military objectives and commercial forces have combined to fuel the evolution of multimedia.

The interface for *Battlezone*

The next significant step in the early history of interactive games came with *Elite*, written by David Braben and Ian Bell and released on the 8-bit BBC Micro computer in 1984. This release introduced 3D wireframe renderings of planets, spaceships and spacestations. [1]

[1] http://www.frontier.co.uk/games/elite/ ; accessed 12-9-02.

The BBC1 web-site describes the gameplay:

> Both the background and the gameplay are deceptively simple. The player takes the role of the owner and sole crew-member of a small trading spacecraft. The first challenge is simply to be able to dock your craft. In-flight controls consist of nose up, nose down, roll clockwise and roll anti-clockwise (note that there is no 'turn left' or 'turn right' control; these manoeuvres must be performed as a combination of rolling and climbing/diving). The space-station orbits a planet, and rotates around its central axis. Therefore docking requires the player to line up his ship correctly and match the speeds of rotation. Given the time of the game's release, this alone would have made an adequate game.
>
> Added to this is a combat simulation. Another innovative feature, a 3D radar display, comes into its own here. It is possible to jump to other systems, and once out of range of the station, the player is liable to be attacked by pirates or alien *Thargoids*. The manual details about 20 other spacecraft, mostly named after snakes, including your own Cobra Mk III, that can be met in the game, and there are several that are not mentioned there. [1]

For the time, the game was also considered compelling because it set a new benchmark for fast action. It went on to sell around 1,000,000 units.

After *Elite* the next significant landmark was *The Sentinel,* programmed by Geoff Crammond, and, while speed was not a major strength (its "set" was a multi-layered chessboard - the player moved from square to square and it took the computer seconds to redraw the view) - its innovation was the generation of filled, colour polygon graphics (instead of wireframes).

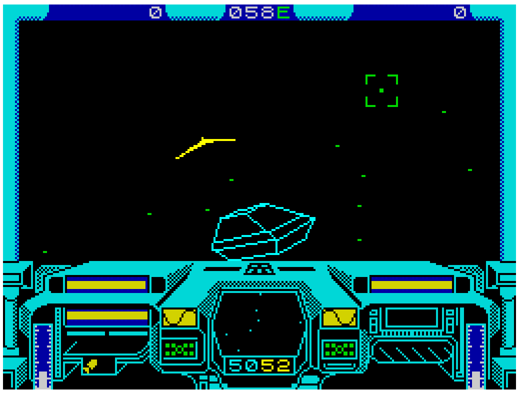
In the 1980s the first 16-bit computers became available, including the IBM PC, Macintosh, Atari ST and AMIGA 500. Once commercial personal computer technology went through the barrier of industry adoption of the 16-bit standard, new possibilities opened up for graphics, and the idea of filled colour polygons reached a new level of maturity. Games became more visually engaging [LaMothe, p1999, 10].

Defender of the Crown (1986/87) was the world's first 'interactive movie' [McCauley, 1997, 31]. (This interactive games genre is discussed in a later section, p. 407.)


[1] http://www.bbc.co.uk/dna/h2g2/alabaster/A711776 ; accessed 3-10-02.

Defender of the Crown featured attractive visuals created by Jim Sachs (see illustrations above), but it proved less compelling in terms of interactivity, when compared with its commercially more successful and durable competitor *Starglider,* which offered:

> ...an enormous space combat/flight sim with colour filled polygons oozing loveliness. You could fly through space, fly down onto planets, fly through underground tunnels, all the while having your eyeballs bathed in wondrous 16-bit 3D. [McCauley, 1997, 31]



Left: *Starglider* interface, featuring colour-filled polygons. Comparisons to the graphic quality of imagery in the less popular *Defender of the Crown* interactive story game shown above, suggest that while graphic quality is important, immersive, engaging interactivity is primary in a games popular appeal.

By the early 1990s, the IBM PC-compatible had become the market leader driven by the release of Windows 3.0, which popularized the graphical user interface originally pioneered by Apple *Macintosh*, but capitalized on the IBM PC's extensive user-base in the business sector. However PCs were still lagging behind Macintosh and Amiga in terms of the audio quality and the display of graphics. [LaMothe, 1999, 10].

McCauley [*ibid*., p. 32] claims that *Wolfenstein 3D* (one of the first shareware 3D games, released by *id Software* in 1991) was another watershed in terms of interface design, introducing the concept of "sprites". Sprites are proxies of a master template media file. An example of such a media file might be a short animation of feet performing a walk-cycle. Sprites are given co-ordinates in the game/drama on-screen space, and are "filled in" with specific information in near-to-real-time by referring back to the original media file, according to interactive input. In the case of our example walk-cycle, the co-ordinates are all

Above: *Wolfenstein 3D* Title Screen

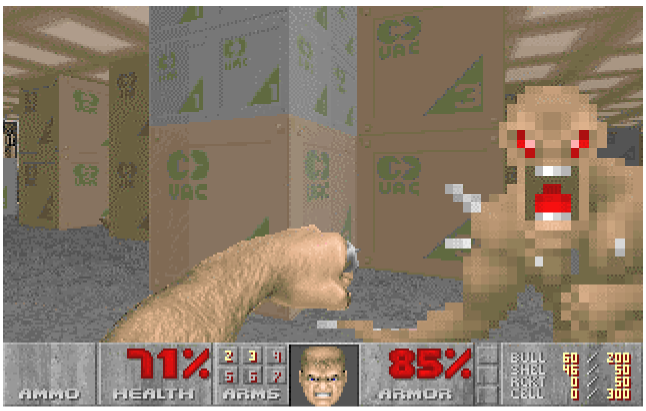
Above: Intermission Screen

that must change when the user interacts, allowing faster screen redrawing, i.e., the impact as a software programming device was to facilitate faster interactive feedback. *Wolfenstein* further raised the level of expectation and immersion with a fast-moving colourful 3D world.

The first person perspective was made vivid by the now virtually universal device in action games of showing the view down the barrel of the players revolver, with, in this case, hordes of virtual Nazi soldiers as the opponents. The shareware release (see below discussion of *Doom)* contained 1 episode, consisting of 10 missions (levels).

Above: *Wolfenstein 3D* interface in action. Note lack of surface textures.

The commercial release consisted of 6 episodes, including the shareware episode. Each episode had a different "boss" which had to be killed in the final mission in order to complete the episode. In order to complete an episode, only 9 of the 10 missions needed to be completed [1]. Hidden in one of the first eight missions was an entrance to a secret level [2].



According to LaMothe [1999, 11], a landmark event in the history of desktop computer games, which heralded the next era of interactive game playing sophistication, was the release in late1993 by id *Software* of the hugely successful *Doom.*

[1] : http://www.wikipedia.com/wiki/Wolfenstein+3D ; accessed 14-06-02.
[2] **Cheats:** Descriptions of undocumented game capabilities for achieving useful (but uncommon) results by the player. http://www.thescratchpost.com/resources/games/games_dict_c.shtml

Below is a quote from the id Software web-site's history section, which discusses how the internet-based shareware model, introduced with *Wolfenstein*, has fueled the popularity of gaming:

> Selling millions of copies and chalking up tens of millions of downloads as shareware, DOOM remains one of the most popular PC games ever. And the title's impact on the gaming world is still felt today.
>
> With DOOM, *id Software* put the shareware distribution model on the map, with the game's runaway success owing a debt to the growth of Internet distribution. The company has continued to support shareware and other non-traditional means of distribution, influencing the way companies market and sell video games.
>
> DOOM also introduced multiplayer gaming to the masses, allowing players to compete in intense 4-player LAN or head-to-head modem competitions.
>
> *id Software* didn't stop there, the team of innovators also made DOOM's source code available to their fan base, encouraging would-be game designers to modify the game and create their own levels, or "mods." Fans were free to distribute their mods of the game, as long as the updates were offered free of charge to other enthusiasts. The mod community took off, giving the game seemingly eternal life on the Internet. [1]

Above: Screenshots showing *Doom* in action. Note texture-mapped surfaces.

Lamothe credits *Doom* with establishing the IBM compatible PC as the worldwide game-playing (and game design/programming) platform of choice for the home computer market, and states that this has remained the case through to the present day. McCauley [1997, 32] describes the reasons for *Doom's* popularity in terms of visual appeal, comparing it to its predecessor:

> Wolfenstein had three major draw-backs in the realism stakes - the actual rooms had flat walls with only the occasional bit of detail in sprite from, everything was lit with exactly the same amount of light and the playing area was completely flat. *Doom* changed all that (with) texture mapping...resulting in detailed rooms with all manner of textures and designs on them. Extra tension was added with variable lighting levels, meaning that you might venture into a normal looking room, only to have the lights go off and all hell break loose, while an extra dimension was added in the form of multi-layered levels.

[1] http://www.idsoftware.com/business/home/history/ ; accessed 5-06-02.

The innovation of "levels", which McCauley refers to here, are levels of complexity and action that only become accessible by achieving certain skill levels (for example, achieving what could be called "kill ratios" within certain time spans). Multiple gaming levels have become powerful marketing techniques which allow the evolving development of gaming skills and offer immersion, empowerment and achievement or skill-recognition as rewards. They have fostered the development of a gaming culture based around each of the major popularly successful games. *Doom* also introduced the ability of the user's surrogate to walk up stairs (in the first person point of view) creating differing heights [1]

LaMothe [1999, 11] credits *Doom* with causing Microsoft to revaluate the importance of computer games to the computer industry, leading to the development of the DirectX – "a new set of graphics, sound, input, networking and 3D systems" for the PC platform/Windows environment.

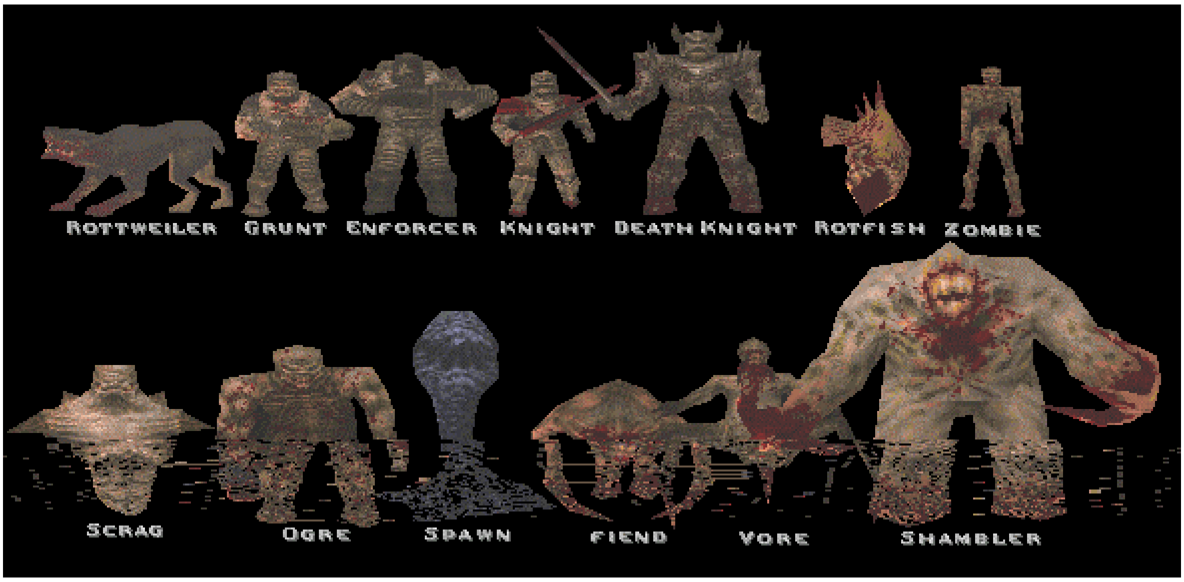
DirectX took some time to infiltrate the games design market and went through several upgrades. Previously, games software programming accessed DOS32 and LaMothe describes how, as late as 1996-1997, most games companies had not made the transition to DirectX (which utilizes the more powerful Win32), but states that since version 5.0 of DirectX (and to the present day) DirectX has been ubiquitous in the games design industry. DirectX has, at the time of writing, reached version 8.0. According to LaMothe [*ibid*., p.12] programming games in DirectX allows designers to leverage every advance in graphic and sound technology, immediately it is released.

As has been described in this short history of the evolution of interactive games so far, it is progress in multimedia technologies which have essentially driven advances in games design and popularity, especially with regard to 3D graphics.

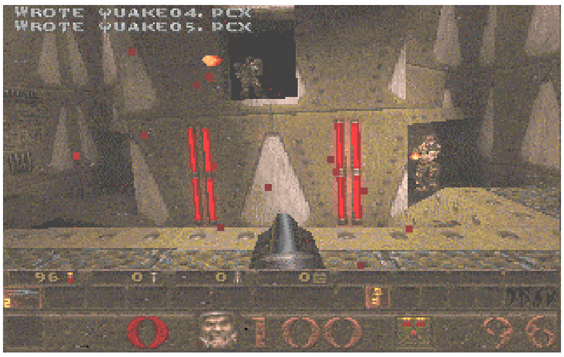
[1] http://www.mac-archive.com/wolfenstein/history.html ; accessed 7-06-02.

LaMothe describes yet another quantum leap in game design, which again came from id Software, with the release in 1996 of an epic, single-player game*, QUAKE*. Based on the company's newly developed proprietary graphics engine, *QUAKE* was promoted as the first "truly" 3D video game, and allowed players to interact with the virtual worlds created by the programmers with a freedom and visual depth previously unattainable. Gamers were free to look and move in any direction, giving the game a deeply immersive environment. The opposing monsters (adversaries) were complex 3D polygon models with detailed textures mapped to their surfaces (rather than sprites) and simulations of real-world physics programmed into the interactive actions, facilitating "the art of rocket jumping".

Below: *Quake* virtual opponents



Below: screenshots showing *QUAKE* in action.



Duke Nukem 3D, a first-person shooting game released in 1994 by Apogee, was strongly influenced by the technical innovations established by QUAKE, but is notable for the satire and (often crude) humour it introduced into what had previously been a fairly humourless genre, including a stream of one-liners from the title character. For instance, when the player comes upon a corpse that closely resembles the player character in the rival game *Doom*, Duke wryly comments: *"That's one doomed space marine"* – the authors assume an informed culture of gamers who understand the reference. This game also freely plunders from many sci-fi movie themes, such as the *Alien* film series (begun in 1979), and the film *Army of Darkness* (1991)[1]. Remediation of existing media into a games format is discussed in a later section (see p. 399).

The release of *Quake* was a major technical milestone and a convenient point at which to end this brief history of the evolution of interactive computer game design. The field has proliferated immensely in recent years and a comprehensive study of recent evolution of interactive computer games is a vast, and quite fascinating study in its own right. Many thousands of games have been designed and marketed with an ever-increasing level of graphic and interactive sophistication.

Above: Screen shot showing *Duke Nukem* in action.

Refer to page 412 for an in-depth analysis of a successful contemporary interactive computer game *Grand Theft Auto: Vice City*, which at the time of writing is a benchmark for current technical and aesthetic standards in this form.

Games Design Paradigms and Techniques

Present state-of-the-art gaming products offer the user an interactive interface which employs the opposing thumb-forefinger ability once thought to separate humans from apes, by utilizing the joy-stick. The associated software universally employ powerful graphics to facilitate immersion, with the intention of creating a virtual world which empowers the user, and underscored by looped music tracks which accent the action and create a mood of urgency, tension and impending drama.

[1] http://www.wikipedia.com/wiki/Duke+Nukem+3D ; accessed 14-06-02.

Rollings and Morris [2000, 23] claim that all existing games require the user to achieve one or more of the following goals:

- Collect something (point scoring games)
- Gain territory (war-games)
- Get somewhere first (a race, either literally or figuratively – e.g. an arms race)
- Discovery (exploration or deduction – very rarely an end in itself)
- Eliminate other players

They make the further generalizations that racing and conquest-type games involve visible objectives (the user can at any point see how well they are progressing or scoring), whereas collection games often don't involve visible objectives and therefore usually offer in-game rewards for points scored. Examples of the latter include most role-playing games (the user earns "experience points" to spend improving their character's skills, attributes, magic potency etc.); strategy games allow the user to gather resources to spend on fuel or ammunition units and skill up-grades; adventure games require the user to collect items to use in later puzzles. Rolling and Morris [*ibid.,* p24] also assert that few games involve just one of the preceding list of objectives.

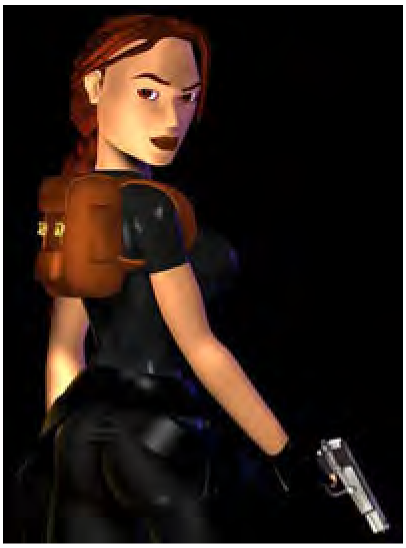
The standard interface designs of electronic interactive games have obvious antecedents in film, adventure cartoons and comics. However, the typical games narrative is traditionally static, involving the barest, stereotypical metaphor of "Good vs. Evil" / "Us vs. Them". The weapon metaphor is a powerful one, offering an intensity of engagement driven by the need to make rapid and continuous responses. This is in essence fantasy play, and while the representation of 'reality' offered is of vivid, remorseless battles to the death, it must be assumed that these gaming experiences are perceived for what they are - a game. Repetitive Strain Injury may be a downside (*Nintendo thumb* has already entered health practitioner's lexicon[1]) because the interface is - to use the jargon - transparent, i.e. the interactive process is simple and direct, allowing a spontaneous interaction with the game-play, generating a level of immersion in the game play which allows the user to become 'unaware' of the user interface or their own bodies).

Change in interface design is largely driven by technological advances and is constantly evolving. For instance, the controller interface for a Sony Playstation 1

[1] source BBC news URL: http://news.bbc.co.uk/hi/english/health/newsid_1079000/1079603.stm accessed 07-09-02.

has 14 separate controls which are assigned by the games developer to be triggered by the user for different tasks, such as causing the game action “proxy” (e.g. the girl warrior in the Sony *Playstation*-based *Tomb Raiders*) to jump, punch, shoot, duck etc.

Watching an experienced user control this interface is to witness a high level of specialized manual dexterity, hand-eye co-ordination, reflexes and conditioned responses.



Above: Virtual Character *Lara Croft* from *Tomb Raider III.* *Tomb Raider* Action

A deduction one can make from researching the history of the evolution of computer games is that, as the power and speed of desktop computer systems increase, their potential for delivering proportionally more vivid and engaging graphics, sound, motion and interactivity has created a looping or cyclical effect in the marketing of the software and the hardware - the market is motivated to buy the faster more powerful computers in order to access the pleasures of the more complex, visually more dynamic (and therefore more exciting and immersive) games packages. Presumably, exposure to a higher level of stimulus in succeeding generations of games packages creates further marketing leverage, by establishing a need in the consumer to seek ever more stimulating packages, reducing resistance to paying for upgrades in computer speed and power. This technology-driven commercial environment produces competitive cycles between the leading software and hardware manufacturers. All of this commercial activity has relevance to multimedia art in that it influences the power of and price of available hardware and software, and influences the expectations and level of aesthetic sophistication or taste and stimulus expectations of a generation of multimedia users.

Computer Games and Storytelling

Interactive electronic games can be viewed within the context of the ancient tradition of storytelling.

> It sounds like a simple question: "How do you tell an interactive story?" But in reality it contains a deep philosophical contradiction. If being interactive has any meaning, then it must be that the person who is experiencing (viewing, listening to, playing, reading) the interactive story affects the way the story goes, and perhaps even the way it comes out. That's what makes a story interactive. But to tell a story implies that you have a story to begin with. How do you tell a story when you do not and cannot know exactly what the story is in advance? …Interactive stories being written today range from hypertext novels to text adventures to interactive multimedia titles. Any and all media are up for grabs; for the computer every medium is just another bit stream anyway. But for all these stories in all these media, the range of underlying narrative structures is rather condensed. A small number of interactive models, i.e., mechanisms for limiting the viewer's choices and the story's outcomes, account for most of the interactive stories published to date. Many more such models are possible, however, meaning that a vast world of creative opportunities has yet to be exploited. In fact, it has yet to be even charted. (St. Hippolyte (1995))

The analysis offerred by Phelps (see p.362) makes some progress toward addressing the problem of understanding narrative within an interactive environment. On can take the view that, to varying degrees depending on genre, computer games are a new technological manifestation of the art of storytelling. As such they draw heavily upon cinema, and specifically the action hero movie (itself a 20th century rewrite of age-old mythology) for the model which has largely defined their form: many computer games are about action, adrenalin, ‘subject as hero - object as enemy’ role-play. This emphasis on physical interactivity, via mouse-gun or joy-stick, works against the development of a complex narrative. Through the manipulation of a certain indefinable subjective sense of power and excitement, designers of typical electronic twitch games offer the participant what one must call Pavlovian rewards.

This stimulus-response reinforcement has a potential to create habits and expectations in the minds of users regarding ways of engaging in entertainment, for instance to lower the threshold at which boredom and impatience sets in. The exploration of “real-time” 3D animation and highly complex interactivity have become the two most important legacies of the computer games industry to the evolution of multimedia art. With regards the latter, the interactive metaphor can usefully be viewed as an *engagement strategy* [Mueller, 1996, 6].

Mueller argues that the common elements of engagement strategy can be mental, physical, and/or emotional. One of the important variables which can be manipulated is *point of view* of the gamer. Laurel [1989, 6] defines three levels of point of view, or “personness”:

First-person experience is a role-playing one in which the user

> ...participates directly in the world of the movie...(i.e. the user) is a character in the action.

Second-person experience is where the user

> …stands outside the imaginary world but communicates to characters or entities inside it, or vice versa.

Third-person experiences is where the user

> …stands outside the action altogether.

(a situation Laurel notes "is hopefully never achieved in interactive media!")

This classification, is based in a notion of being either inside (good) or outside (bad) the action or diegesis of the movie, and seems to preclude an immersive narrative experience. Murtaugh [1996] points out that in Laurel's second-person experience, every instance of a "communication" between viewer and story represents a transgressive metalepsis -- a disruption of the narrative and that Laurel's notion of a first-person interactive experience denies the presence of a diegetic barrier altogether.

> In this case, there is no narrative; the viewer is put into an environment and acts -- his or her experience is the story. Removing the viewer from the "action" altogether (third-person) is presented as entirely undesirable as the experience reduces to nothing more than an ordinary cinematic experience, presumably something the interactive moviegoer would not tolerate. The fundamental problem with Laurel's taxonomy is the implication that one cannot provide an "immersive experience" without literally making the viewer a character. Undeniably, having viewers experience a story *as if they were present* in the diegesis is powerful. At the

same time, giving viewers a sense of presence need not bind their actions to consequences in the story.

Other common elements of engagement strategies which can be manipulated by the game designer to enhance immersion are:

- role play scenarios – enhanced by the skillful weaving of a developed narrative conceit. Participatory involvement of a player in a game, especially as related to decision-making from some game perspective, is enhanced by the richness, cohesive unity and uniqueness of the narrative being offered.

- (virtual) worlds, and the metaphorical environment: mental maps, navigational devices or metaphors. Engagement is increased by employing a richness of virtual environment e.g. 3D depth, aesthetic or sensual rewards of richly textured, immersive graphics, an appropriate audio experience and clearly stated metaphors. Ease of understanding of locality and the development of familiarity with environment enhance engagement.

- characters and character intention – traditional narrative devices which games, whether role-playing or not, must to some degree deploy. Character development engages, as the character becomes more distinctive, with prescribed patterns of behavior.

- puzzles, mazes, problem solving, reward objects – interactive engagement is enhanced by intellectual challenges

- action tasks, goals, rewards and the manipulation of feelings. *Reward / Penalty Structure* is the aspect of a game that deals with the relationship of rewards to penalties, for actions within the game. A reward / penalty structure is positive if there is no penalty for losing, but there is a reward for winning (most games are of this type). An important aspect of the reward / penalty structure is the personal attitude taken by the player when they lose, e.g., are they made to feel humiliated? [1]

- clear rules, cause and effect, other game-play conditions - actions or behavior incumbent on players, consistent operator and accounting system which cannot be altered, and as such offer a coherent structure and confident playing environment.

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[1] http://www.thescratchpost.com/resources/games/games_dict_r.shtml ; accessed 2-12-02.

- Simulation – building a complete immersive experience based on a real-life situation, wherein there is a wide range of possibilities for dynamic execution or manipulation of a model of some object or system. e.g. apart from games that put the player in the driver's seat, such as flight and race car simulators, there are games that recreate battles, hospital drama, city planning and space travel. *Capitalism* is an example - a financial strategy game that puts the player at the head of a corporation and simulates different business environments, as everyone competes to make money. Builds on pre-learnt cultural expectations and understanding of sytems.

- openings to new levels of difficulty; progressively confronting the user with more difficult challenges as skill levels improve - provides for on-going engagement, counters the tapering off of interest.

- Flow – a difficult to quantify but important quality, fundamental to a successful games experience; an overall seamlessness, internal logic and sense of coherence and unity.

Computer Game Genres

The following is an adaption of the list which Lamothe [1999, 13] offers of ten games genres, which is a useful aid for categorizing and distinguishing currently available interactive computer games:

1. *DOOM*-like, First-person "shoot 'em" games
(i.e. based on the conventions set introduced by *DOOM*):
These are shooting games, which offer a navigatable world, rendered in "full (real-time) 3D", which is viewed from the players (first person) perspective.
Examples: *DOOM, Hexen, Quake, Unreal, Duke Nukem 3D, Dark Forces*;
"Technologically the most difficult to develop".

2. Sports Games
Can be either 2D or 3D; one player or team play; graphically not as advanced as category 1 games but "the artificial intelligence in sports games is some of the most advanced of all the games genres".

3. Fighting Games
Typically played by one or two players; action is presented as a side-view of two 3D characters controlled by the player(s) or by a floating 3D camera; The game imagery can be 2D, 2.5D (multiple bitmap images of 3D objects gives the appearance of 3D;

see p. 125 for a discussion of pseudo-3D), or full, *x,y,z* co-ordinate-based 3D. *Tekken* (for the *Sony Playstation*) established this genre for the home console market; relatively less popular on the PC, “due to interface problems with controllers and the two-player fun factor”.

4. Arcade/shoot’em up/platform

Primarily ‘old-school’ 2D games, though some are being remade into 3D worlds, though gameplay remains 2D. Examples: *Asteroids, Pac Man, Jazz JackRabbit.*

5. Mechanical Simulations

Any kind of driving, flying boating, racing, skiing, and tank-battle simulation. Mostly 3d but with a lower standard of graphics than offered by some of the more sophisticated genres

6. Ecosystem Simulation

These games allow the player to control an artificial system of some kind, such as a city, a colony of ants or financial system. Users become immersed in virtual worlds over which they have power, requiring organizational skills, forward planning, strategy and analysis. Examples: *Populous, SimCity, SimAnt, Gazzillionaire*

***7*. Strategy or War Games**

Splintered into a number of sub-genres; usually turn-based (users take turns); require strategy / thinking /f orward planning to proceed.
Examples: *Warcraft, Diablo,* (real time, but requires a great deal of strategy), *Civilization, Final Fantasy VII.*

8. Interactive Stories

Based on the tradition of *Myst;* “Basically, these games are prerendered or on ‘tracks’”. The user moves through the game by solving puzzles. These games usually don’t allow free roaming game-play and could be described as interactive books. Usually created using *Macromedia Director*, or similar interactive authoring software, rather than a computer coding language (and are thus looked down on by professional computer games programmers); strong emphasis on atmosphere and aesthetics; slower pacing. (Discussed in more detail below, p.392).

9. Retro Games

Technologically up-graded remakes of previously successful, old-school or early generation games with cult-appeal and an established user base. (There is an overlap here with category 2).

10. Puzzle and Board Games
Adaptions to a computer environment of traditional "real world" board and puzzle games such as Monopoly, Chess, Mahjong, card games, Tetris etc. can be 2D or 3D.

Ecosystem Simulation Games: *The Sims*

LaMothe's categorization of computer games genres struggles to describe what he calls the *Ecosystem Simulation Game* genre. In a popular example, *The Sims*, the user 'micro-manages' the lives and domestic environment of simulated people, (or two, or four, or eight) attempting to ensure they have a successful career, satisfying life and happy relationships. There is no other game quite like *The Sims* in that, rather than glamour or adventure or magical hyper-realism, the paradigm is to immerse the user in a very low-key, realistic domestic environment. There is no specific goal, no real end-point to this game, nor a time-limit. A simulated "day" goes by in compressed time, and in a half an hour the user must make sure their *Sim* gets enough to eat and sleep, as well as appropriate social interaction, and other social functions including showering and simulated visits to the bathroom. In addition, the user defines the virtual environment – one must build your character's house, choose the wallpaper, windows and doors, appliances and furniture. Every decision the user makes affects the *Sims* "comfort zone", so choosing more expensive furniture increases the level of satisfaction a *Sim* gets when using it. The user defines the personal characteristics of the *Sim* through interactive choice and there is scope for simulated individuality:

> Well, you can let your Sim be a slacker, and a mean one to boot. Neglect your kids, skip work until you're fired, sleep all day. Or, meet and marry your next-door neighbor, have children, and work your way up the ladder of success in seven different job categories ranging from a military career to a job in the Entertainment industry…
>
> There's some cool features that work in mysterious ways. For example, until you teach your Sim to be a good cook, there's a very real possibility that while making dinner, he or she will set the stove on fire. If you haven't purchased a smoke alarm, the Sim will panic and might even catch on fire and die. If this happens, an urn is all that's left, and if another member of the household puts the urn outside, it will turn into a headstone. On some random dark night at the stroke of midnight, the ghost of the dearly departed will haunt the house.
>
> There are a number of upgrades for the Sims, and each one allows you to increase the scope of the game. So, in a very creepy way you can recreate your own life in a virtual neighborhood…[1]

[1] http://www.thescratchpost.com/reviews/games/nov01/reviews_game_sims.shtml ; accessed 30-11-02.

Multimedia applications are frequently sub-divided into different areas of activity or regions which are interactively accessed through menus or other navigational systems. Audio can be used as a navigational aid. *The Sims* relies significantly on the use of sound to clarify its structure, with an unusual emphasis on audio as interactive icon.

Left: Typical Sims environment, featuring its standard aerial view, brought to life with the sound of cooking.

Below: An example of the use of sound for interactive mapping – activities and icons become familiar through the triggering of associated sound files.

Right: At the users discretion, simulated character *Vanadia Quadd* has just pulled a packet of chips from the refrigerator, incurring the expenditure of $5. A cash register sound accompanies any expenditure of money in the Sims world. This short, distinct sound is instantly recognisable, and attention grabbing, serving as an audio icon to clearly signify a significant games event.

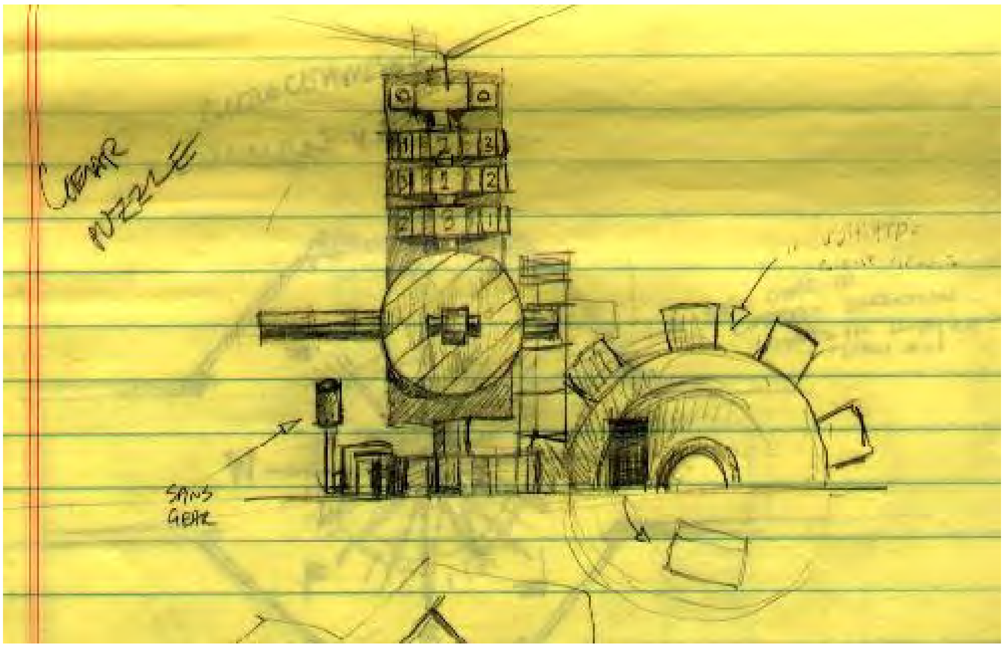
Right: Radios, TVs and computer games all feature in the Sims, and considerable original audio design work has gone into these 'entertainments'. The program allows the user to set the musical ambience. Each of the "change station" options designated above trigger an appropriate background audio file.

Interactive Stories or Puzzle Solving Games

A significant, though perhaps to date less virile strand to the computer games industry listed in LaMothe's categorization of games genres shown above, is the interactive story. This genre was proven to be a commercially viable new option by the success of the land-mark *Myst (*Broderbund,1993)*,* one of the all-time best selling computer games, and its glossy sequel *Riven* (2000). Such games characteristically offer an enigmatic, navigatable environment, and a story/plot which the user is required to discover by observation, deduction, collecting clues, solving puzzles, as the media-rich environment is explored. The image below is a screenshot from *Myst.*

Above: "View out of the lighthouse in Stoneship Age" [1]

[1] from the official Riven/Myst web-site http://www.riven.com/Online/Myst/SightsSounds/Lighthouse accessed 3-4-02.



Above: “Early design sketch of the gear puzzle on Myst Island” [1]

The profit-led march of technological advance is inevitably a driver of advances in multimedia design. Faster CPUs, larger and faster hard-drives, faster data transfer from CD-ROM players, improvements in video and audio cards etc., as they become available on consumer level computer systems, allow new levels of interactive sensory experience to be explored. Minimum System requirements for running the so-called “enigmatic, problem-solving environment” computer game *Riven* were hi-tech at the time of its release (2000), but vastly surpassed by today’s standard off-the-shelf computer:

For Windows CD-ROM	**For Macintosh CD-ROM**
Windows 95 required	MacOS required System 7.5 or higher
100 MHz Pentium or faster	90 MHz PowerPC or faster
16 MB RAM	9 MB RAM free
Minimum 75 MB hard disk space	Minimum 65 MB hard disk space
4X CD-ROM drive or faster	4X CD-ROM drive or faster
640x480 display, High Colour	640x480 display, thousands of colours
Windows compatible sound device	
Video and sound cards compatible with DirectX	

The game play itself is a device to motivate exploration of the multimedia sensory experience, whose ultimate purpose is to allow the end-user to luxuriate in the immersive sensory possibilities of multimedia.

[1] http://www.riven.com/Online/Myst/SightsSounds/Lighthouse ; accessed 3-4-02.

The visual and sound experiences offered in strategy games are strongly influenced by fantasy cinema (e.g. *Riven* opens with a cinematic video sequence which introduces the narrative conceits of the gameplay, delivering a sophisticated level of Hollywood-style production values in terms of animation and audio). Both *Myst* and *Riven* are permeated with an intriguing attention to detail, inventiveness and brooding sense of impending drama.

The device of an interactively (mouse-click) triggered cross-fade/dissolve between pre-rendered 3D images, is employed to give the user a sense of controlling the journey through a haunting virtual world. An eerie atmosphere is reinforced by a sound track of carefully chosen supporting sound events, and a strongly atmospheric ambient-style music score, creating a unique sense of suspended time. *Myst* is a serene but enigmatic and lonely place, through which the participant wanders alone, searching for clues to unlock the mystery of this surreal, deserted landscape. It demands of the user significant intellectual, intuitive, memory, observation and problem-solving skills to navigate successfully through its cryptic cyber-spaces (not to mention patience, persistence and significant amounts of leisure time).

Part of *Myst's* success is the strong sense of cohesion and unity in the design. The aesthetic values upon which its various artistic inputs are based are of a high order, within a commercial/popular art context, with the discipline of megabyte limitations honing the designs and stylizing the music. In the case of *Riven*, the designers have transenced the data delivery barrier, by spreading the game over five CDs. The necessity of continuously changing CDs breaks the sense of immersion, however.

The interactive computer games industry is a highly competitive marketplace and it is a challenging process to build and market a successful title. The significantly larger-budgeted and better-resourced *Riven* was less popularly successful than its innovative predecessor *Myst*[1]. Now that there is a larger scale storage delivery media available in DVD, which expands the technical boundaries by allowing seemless access to a much larger media asset register, it would seem likely there could be new impetus to the genre of personal computer-based interactive game defined by the success of *Myst.* Ultimately it is a compelling concept, packed with

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[1] This genre contributes a very small section of the overall games market. Broderbund sold over 4 million copies of *Myst* and total unit sales of all three Myst titles are in excess of 10 million, but Nintendo has sold more than 40 million copies of *Tetris* for the Game Boy module, and that does not include the Nintendo Entertainment System (NES), Super NES, PC, Macintosh, Commodore, PlayStation, Saturn, and arcade versions of the game.
http://www.gamerstoday.com/techies/features/game_millenium/ ; accessed 15-02-04.

creativeideas which create market appeal (combined of course with technical and artistic excellence in the execution) and thus an interesting future can be anticipated in this field.

The *Myst* web-site[1] credits a seven member team as the creators of the game, and lists the following software tools used in the creation of the game (descriptive entries in square brackets have been added).

Graphics and Construction tools:
HyperCard (Apple) [hyperlinked navigation authoring]
Think Pascal (Symantec)
Photoshop (Adobe) [digital image manipulation and processing]
Premier (Adobe) [video editing and processing]
Illustrator (Adobe) [vector graphics design]
Painter (Fractal Design) [digital simulation of natural media]
Morph (Gryphon Software) [key-frame image morphing application]

Music and Sound tools:
MasterTracks Pro 5 (Passport) [MIDI sequencing/composing application]
Proteus MPS Plus (E-Mu) [MIDI-triggerable sound module with musical sounds]
Pro Tools Audio Interface (Digidesign) [multi-track audio recording/editing software]
Sound Designer II (Digidesign) [Audio recording hardware card and driver software]
SoundEdit Pro (MacroMedia) Post-production sound editing application

Additional Tools:
Infini-D (Specular International); [3D modeling, animation and rendering]
De-Babelizer (Equilibrium Technologies); [File format conversion]
MovieShop (Apple) [Digital Video Editing]
ComboWalker (Apple)
Picture Compressor (Apple)
ConvertToMovie (Apple)
MoviePlayer (Apple) [Digital playback of media]
SoundToMovie (Apple)
QuickTime XCMDs (Apple)
Fontographer (Altsys) [Font design application; also converts fonts across platforms]
Tilod PICS Import/Export filters (John Knoll)

Images and animations were modeled and rendered on six Macintosh Quadras using *StrataVision 3d* by Strata, Inc. HyperCard was colorized using a proprietary version of Symplex System's HyperTint, written by John Miller.

This relatively small and closely integrated team, collaborating to produce an experimental interactive art piece, in the form of an interactive game, succeeded in striking a chord with the popular imagination. The designers employed a wide range

[1] http://www.riven.com/Online/Myst/MakingMyst#anchor80764 ; accessed 3-4-02.

of integrated software, addressing the various channels of communication and sensory input. This is illustrative of the conceptual framework that this dissertation has established: that we are witnessing the evolution of a meta-art machine. The work itself can be viewed as offering an historical landmark in interactive design.

The convention of using first person narrative which has arisen in both interactive electronic ‘twitch’ and puzzle game environments is new - film almost universally uses third person narrative. Twitch games are predominantly spatially oriented, with an emphasis on dynamic action and motion. Their more - in a narrative sense - sophisticated relative, the interactive story, is suggestive of possible future directions for the development of a multidimensional digital artform, offering a vision of moving beyond a mere spatial orientation towards the creation of multi-layered, crypto-space, rich in language, music, drama and conceptual texture. This phenomenon is currently restricted to the home-computer or games-console/TV playback experience, because interactive stories require a relatively high fidelity listening environment (as atmospheric sound and music play such an important part in creating an immersive experience), which is not available in the arcade environment. Also, the user must make a substantial investment in a software package (CD-ROM set) which tends to encourage a full exploration of the virtual experience (including watching introductory movies, experimenting with controls, researching on-line documentation etc.) in order to make the investment pay off. In the arcade game situation, the investment is made in small incremental amounts, (purchasing small amounts of game-time). [Ganner, 1999]. Ganner explains:

> The coin-op player makes that investment 2 quarters at a time. In most circumstances, whether he wins or loses, he'll be asked to invest another 2 quarters in about 3 minutes. So at the end of 3 minutes he has to evaluate whether this game is worth an additional investment. This means that the game has 3 minutes to seduce him. Put another way, the game must allow the player to feel sufficiently successful, to "obtain the amusement dividend," within the span of 3 minutes.

There is evidence that the interactive story genre of game design is evolving. In the first *Myst* and *Riven* games, players move along predetermined (pre-rendered) pathways by simply pointing and clicking the mouse. *Myst III "Exile"* offers the user the ability to look in every direction (rendering a 3D world in real time), giving a much more realistic ‘feel’ than the earlier versions. This step towards real-time 3d in interactive stories has a counterpoint – contemporary twitch games (such as *Grand Theft Auto: Vice City*; see p. 412 for an in-depth analysis) are increasingly being layered with strategy and problem solving engagement strategies – i.e the two forms are converging.

It must be said that, while the designers of *Myst III "Exile"* have achieved a sumptuous, imaginative visual experience with extraordinary attention to detail, the resulting rich-textured environments are in fact so cluttered with high-resolution objects and backgrounds that there is a possible negative outcome: players at times may have to resort to randomly clicking all over the screen, just to find the right path or clue. Getting the right balance in navigational design between enticing mystery and technical flow is one of the big challenges. When does cryptic become confusing?

In an age of ever-faster, ever-noisier games requiring players to eviscerate, decapitate and otherwise inflict mortal harm on their virtual opponents, *"Exile"* offers a relaxing, nonviolent atmosphere and pace, in which reflection and deliberation are rewarded with the unfolding of a classic tale of suffering and redemption. Aficionados (or addicts) of other more adrenalin-oriented games genres may be likely to find this less than rewarding however.

The narrative in *Myst III:"Exile"* builds on that established in the previous sequels of the game. It begins 10 years after the events in the first two *Myst* adventures, and tells the story of one man, Saavedro, who has watched his civilisation brought to ruin by the sons of Atrus (a character in previous versions). Saavedro seeks revenge on Atrus. As the game player, one must stop him and in order to do that, one will have to explore several Ages, navigate the puzzles in these Ages, and uncover the truth about the villain, these magical worlds, and their connections to Atrus. Against this background, players also embark on a quest to return an important book that was stolen. The interactions with the characters in *Exile* advance the plot, provide subtle clues and the back-story information, and draw the user deeper into the intrigue.

The game employs the “levels of difficulty" engagement strategy and in this case players must navigate through five entirely new “ages”, each with its own theme, and formidable challenges. These thematic connections offer a cultural depth not previously found in games genres, and inform the way the worlds look and how they function (their imagined physics and dramatic focus). For example, *Voltaic*, the age of electrical energy, is set on a spartan rock, and the visuals are resplendent with wires and turbines, tucked among crags and crevices. Each of the embedded puzzles in this “age” (level) has to do with the generation of power/energy, and a comparable theme applies within each of the other four ages. It is interesting to consider that in order to build on the success of earlier games, designers have attempted to evolve the conventions previously set up, and have made the puzzles very challenging:

> The story is well integrated into these puzzles. The villain, Saavedro, has come before you and has already solved these puzzles, in accordance with how the Age was designed by Atrus. As he moved through the Ages however, he tampered with the designs, destroying parts of the devices, so that solving the puzzles is made even more difficult for you. In addition to figuring out how to solve the puzzles, you have to also figure out what Saavedro did to screw it up for you, and then solve the puzzle working around his changes. At this point, you too may either throw up your hands in frustration, or better yet, get a copy of the strategy guide to help you get through the complex puzzle solving steps in order to move along in the story.
>
> All this puzzle complexity means you can easily get lost in the storyline and the rich, complex plot, as you set yourself to solving the puzzles. It is very easy to forget what all the puzzle solving is all about, as these complex designs begin to dominate your mental concentration.
>
> This leads to my strong recommendation for most gamers to not even try to get through this game WITHOUT the official strategy guide.[1]

Myst and its sequels have helped open up interactive gaming to other demographic groups not attracted by the violence of twitch games:

> The *Myst* game also strongly appeals to women, with 48% of all *Myst* players estimated to be women. It is estimated that women make up 15% of all PC game players. There are zero characters killed in all the *Myst* games, and there are zero times any nudity is depicted in any *Myst* game. The player can "die" only in one place in the *Exile* game, at the very end, and after selecting one of the various end-game scenarios.[2]

There are 21 minutes of fully orchestrated music in *Myst III: Exile*, and 59 total minutes of music on the *Myst III: Exile* soundtrack. While the game comes on 4 CDs, designers have taken advantage of larger hard-drive storage capacity available on contemporary computers and offer the choice of loading all the files on a hard drive to eliminate disc swapping, as one progresses through the game.

Remediation of other media forms within the computer games context.

With the globalization of media industries and the increasing consolidation of media resources into fewer and larger corporations, there comes a strong commercial drive to generate additional profits from media assets and copyright properties by remediating popular titles into other media forms. A clear example is the way in which film score composition has been significantly changed by this process, with trend towards consciously building a commercially viable soundtrack CD through the incorporation into the movie of popular songs by celebrity artists. As a branch of this economic strategy, CD-ROM games have been generated by repurposing successful movies.

[1] http://www.gtpcc.org/gtpcc/mystexile.htm ; accessed 15-02-04.
[2] ibid.

Poster for *Blade Runner: The Movie.*

Cover of *Blade Runner: the Game*.

An example is the game version of *Blade Runner*, published by Virgin Interactive Entertainment (1997), which is based on the existing story remediated from the science fiction film of the same name.

The CD-ROM game reproduces important narrative devices of the film, including the setting, characters and context. Set in Los Angeles, in the year 2019, both the film and the game depict a future in which off-world colonization has become a reality. Intelligent humanoids, known in the film as *replicants,* are employed as slave-labour on these extra-terrestrial worlds, but they are outlawed on Earth. *Blade Runners* are a special police squad whose job it is to detect and eliminate any infiltrating *replicants.* The film is structured along a linear narrative path where the central character, Deccard (played in the film by Harrison Ford) searches for clues to uncover the identities of four *replicants* who have infiltrated the Earth's security.

Narrative events unfold according to the normal filmic conventions of a pre-determined, fixed and unalterable sequence. The audience is taken (passively) on a journey that unfolds over linear time. The CD-ROM game version sets up these same conventions, but the player is positioned in the first person to control the order in which the action unfolds.

There are 6 different possibilities of how the gameplay may end, depending on how the user handles interaction (e.g. shoot first and ask questions later, or vice versa etc.). All "actors" (i.e. on-screen action figures) consist of voxel graphics, while the background locations are rendered videos.

While in *Blade-Runner* the objective is to kill the *replicant*s, it is actually a good example of the blurring or mixing of genres. As a computer game based on the detective genre, *Blade Runner* emphasizes investigation, problem solving and consideration of options rather than speed, reflex and hand-eye co-ordination.

On-line interactive multi-user role-play gaming

The impact of gaming on developed societies is significant when the phenomenon of on-line gaming is considered. CD-ROM games are shaped by the nature of the off-line medium itself, both because they become a single purchase commodity, and because of the finite (700 meg.) data storage space that they comprise. In order to consider the boundaries of interactivity presently structuring the CD-ROM it is useful to compare the off-line computer game with an interactive on-line game. The on-line game has its own “reality”, which is changing day by day, due to the interaction between players.

> In games such as EVERQUEST, ASHERON'S CALL and ULTIMA ONLINE, thousands of people participate simultaneously in a virtual world where the action goes on 24/7 - whether you're there or not. [Morgenstern, 2002, 55].

One of the most popular examples, *Ultima Online*, is interesting for the way that it demonstrates that interactivity and narrative can be constrained and structured by a text's medium. *Ultima* is a successful series of role-playing adventure games. Game play is set in the fictitious medieval lands of *Britannia*. The *Ultima Online* website describes it like this:

> From rambling, secluded villages to the polished stone towers of Lord British's Castle, the cities and villages of Britannia provide much for the citizens. In the north lie trade centers and the rugged towns of the workingman. To the east, scholars toil in pursuit of the magic of the realm. Within cities throughout the land, however, one finds the adventurous citizens of Britannia, plying wares, searching for adventure or merely sipping ale with friends.[1]

The introductory movie for *Ultima Online* sets up a "sword-and-sorcery" tale of a wizard who held the *Ultima* world captive in a crystal, in order to weave magic upon the inhabitants. A hero vanquished the wizard, but in the process the crystal was smashed into a thousand pieces (*shards*), each representing a parallel world, one of which is the trajectory one follows within the game space. A starting point for joining the game is the *Ultima Online* website. The user creates a character and guides it through a fantasy world where interactions are encountered with virtual merchants and monsters, as well as other player's "characters".

Above: the *Ultima Online* 'Atlas of *Britannia'*

The user must create an account and is guided through selecting a *shard* and creating a character. Users can start from character templates:

[1] source: http://guide.uo.com/atlas.html ; accessed 14-06-02 .

Shards are located around the world, and the user is advised to choose a *shard* that is located geographically close to ensure a “robust connection to the game”. The *shards* are currently located as follows:

North American East	**North American West**	**Western Europe**	**Japan**
Chesapeake Atlantic Catskills Lake Superior Great Lakes Siege Perilous AOL Legends	Baja Pacific Napa Valley Sonoma	Europa Drachenfels	Hokuto Yamato Asuka Wakoku Izumo Mizuho Mugen
Korea Arirang Balhae Baekdu	**Taiwan** Formosa	**Australia** Oceania	

Until the development of the internet-based *Ultima Online*, this game was set in one fixed world, and the player interacted with computer generated characters. With advances in broadband connections, gamers are taking advantages of online multiplayer games[1]. As it has evolved, a user may be online with tens of thousands of other players at any one time, each capable of making interactive choices and

[1] http://faculty.erau.edu/pratta/Archive/c/compgames.html ; accessed 02-01-03.

thus influencing the dynamics of game-play. Clearly this represents a different order of interactivity. The freedom of such a structure reduces the scope for an embedded narrative, at least in terms of setting parameters for an unfolding story with a definite end. This does not mean that *Ultima Online* is an example of fully or freely interactive multimedia, as the rules that govern interactions between characters mean that, although the path one follows is unrestricted, the number of approaches to the game are limited. The community of *Ultima Online* has its own standards and codes of conduct and etiquette, and how one chooses to react to those rules[1] will determine your role in *Britannian* society. As in real life, one must still earn a living, and the user is offered for example, a set of profession templates.

Ironically the whole process of online development can be short-circuited if you have the financial resources. The *Ultima Online* web-site advertises auctions. A well-developed castle in the *Ultima* world is on offer for around US$1000. In a news item about the game culture which has arisen around *Ultima Online*, *Windows Users News*, [March 1999, Volume 9] reports that:

Ultima Online passes 100,000 users Origin Systems' Ultima Online, now claims to be the largest fee based online-only game system in the world. They have passed 100,000 members each paying $9.95 a month to play the medieval fantasy adventure role-playing game Ultima. The game must first be purchased; it has a MSRP of $49.95. Origin Systems, www.origin.ea.com, is a subsidiary of Electronic Arts, www.ea.com, one of Origin Systems, www.origin.ea.com, is a subsidiary of Electronic Arts, www.ea.com, one of the largest publishers of interactive gaming software. During peak periods there are as many as 20,000 players using the systems at the same time. The average member uses the online system an average of 20 hours a week. More than 50% of the 100,000 members play everyday. [2]

[1] *Emanata* - originally coined by Mort Walker (Lexicon of Comicana) to describe the graphic devices used by cartoonists to describe "what is going on inside" their characters. In role-playing games, emanata may include: suspending play so that a player may describe what their character is thinking/feeling; having players keep journals or diaries in-character; having an NPC confidant draw the character out in private (eg. a priest/confessor); or simply good acting (for those players who have the appropriate skills). http://www.thescratchpost.com/resources/games/games_dict_e.shtml ; accessed 2-12-02.

[2] http://wun.tns.net/newsletters_txt/199903_nat_e.txt; accessed 17-05-02.

Interactivity must be conducted within certain boundaries which allow for more or less choice, control and agency. *Ultima Online* helps to illustrate how—at the moment—the different parameters of on-line and off-line multimedia can be used by designers to emphasize different kinds of interactivity. It is important, then, to specify what these are in any discussion of the particular kinds of interactivity offered in a particular multimedia text.

Turn-based Strategy Games

In the context of real time “twitch” gaming, no input (player does nothing) is interpreted by the game as a valid input – i.e. the on-screen game action will continue. This is contrasted with turn-based games in which the competition is between rival players - the game stalls until a player takes their turn, in the tradition of Chess. Strategy games emphasize cogitation, competitive planning and management, rather than manual dexterity and reflex, and game-play occurs in a complex virtual environment, which allows a great deal of interactive choice.

For example, in *Civilization III* (Sid Meier, 1990) - the user “stars” as leader of one of the Earth's past, famous civilizations. France, Japan, America, India are just a few of the game's fifteen civilizations to choose from. The user gains control of the civilization from its first conception in 4000 BC. Then, through expansion of cities, trade, diplomacy and war-mongering, the aim is to attempt to bring your civilization to be the most powerful in the virtual world, through strategic planning, and intelligent, competitive decision-making. The user must oversee a highly complex management system and for example must keep an eye on workers, expand with new cities, manage old cities, maximize production, minimize corruption, balance tax dollars between scientific discovery and “civilization happiness”, build armies, command armies, wage war, conduct diplomacy.

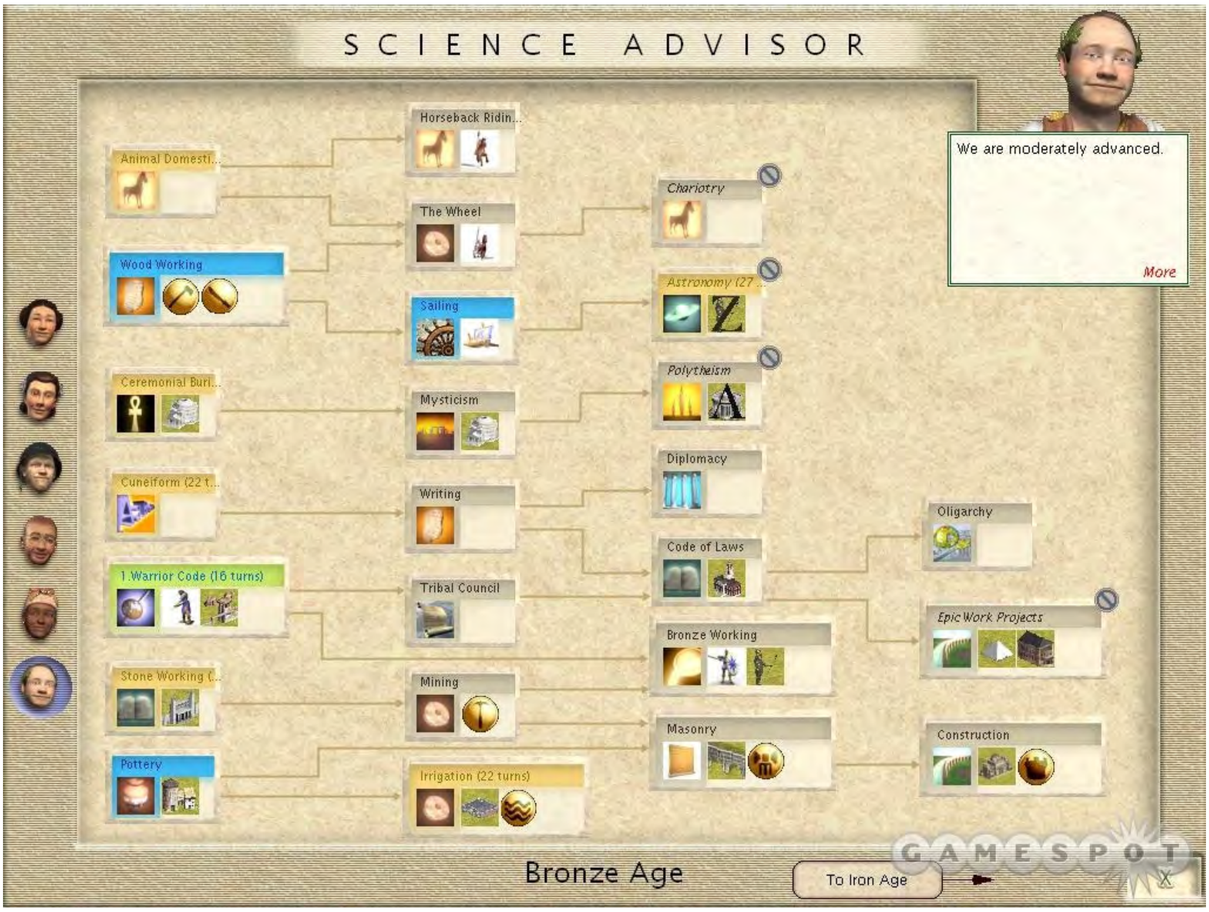
Civilization III introduces the concept of “Culture”. The more religious and educational structures a user defines for a city, the larger its cultural “influence radius” is. Culture as an interactive variable is, in this version of the game, basically, how strong an influence the culture of a city has on other competing civilizations. If one culture overwhelms another city's culture, then that city will defect to the dominant civilization – a way of defeating an enemy without warfare. User choice before the game is started regarding what type of “path” is taken is a major part of winning. Each civilization has certain strengths (within the *Civilization* paradigm) that will help the user on a chosen path. Indians, for example, are religious and commercial and therefore much more suited for a cultural victory than, say, a victory through “war-mongering”.

Above: Screenshot from the turn-based strategy game *Civilization III*, which uses an isometric grid, offering a diamond-shaped aerial ("tile") view of the virtual terrain. The user views the empire from a god-like, high altitude position that is also like looking down at a strategy plan model in a military command headquarters.[1]

Germans, on the other hand, are militaristic and scientific and a user choosing this Civilization will find advantage in dominating with armies rather than attempting cultural domination.

Most of the civilization management takes place in the *Advisor* screens (see Science Advisor screen below). If the user needs to make choices regarding how to spend tax money, for example, one must proceed to the *Domestic Advisor* screen. If a user needs to contact another civilization, the *Foreign Advisor* screen may be utilized.

[1] http://www.civfanatics.com ; accessed 14-02-04.



The music in *Civilization III* slowly cycles between eastern and western flavors and simply provides an unobtrusive but important ambient audio background. The sound effects are equally subtle but surprisingly realistic. Sounds of waves and birds give atmosphere and help make the world come alive during the inevitably long sessions of gaming.

The Civilization promotional web-site states [1]:

> *Civilization* instantly set the standard and defined a new genre of empire-building strategy games and is still recognized as one of the greatest games of all time. The game is an addictive blend of building, exploration, discovery and conquest. Players match wits against some of history's greatest leaders as they strive to build the ultimate civilization to stand the test of time.

While it is difficult to find any research-based statistics on the demographics of users of this genre of interactive gaming, they obviously must be considered as a quite different gaming community with their own set of motivations, values and skills. There are also quite a different set of gaming conventions set up by this genre of games to, say, DOOM-like shoot'em games.

[1] http://www.civ3.com/legacy.cfm ; accessed 16-04-02.

2.3.8 Interactive cinema : *the use of live video drama elements in interactive games*

The inevitable influence of cinema on the computer games industry has begun to be more evident with recent trends towards incorporating live action video into the structure of the game to enhance the narrative and to build characters [Gosh, 2002].

Above: promotional image from *the Pandora Directive*. Comparing it with screenshots from other games genres (e.g. *Ultima* or *the Sims*) the imagery draws on cinematic production values.

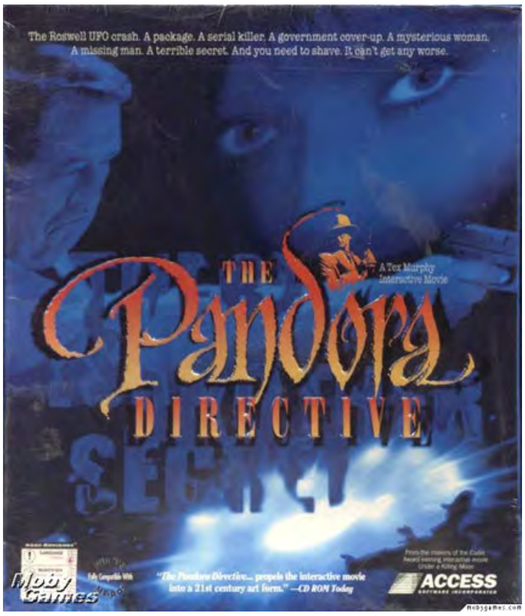
Games manufacturers have turned to experienced film directors to create convincing live action sequences, an example being the work of Australian film editor/director Adrian Carr for U.S.-based Access Software's *Tex Murphy* series of puzzle-solving games, including: *Mean Streets* (1989), *Martian Memorandum* (1991), *Under a Killing Moon* (1994), *The Pandora Directive* (1996), the latest of which is *Tex Murphy: Overseer* (1998).

The promotional web-site for Access Software's products describes some of the evolving characteristics of this kind of game:

> One of the improvements *The Pandora Directive* has over other Interactive Movies, is that it can be replayed several times. The preceding storyline is just one way in which the plot can develop. There are 3 different narrative paths through *The Pandora Directive*, each with its own unique scenes, plot twists and character developments. And, depending on how you choose to move through the story, there are 7 different endings. This allows you to replay the Interactive Movie several times, each time coming across new conversations, locations and puzzle solutions.

> *The Pandora Directive* offers two levels of game play. Level One is for those users who play games to see the story, not tackle the challenging puzzles. It has a complete on-line hint system that assists them if they get 'stuck'. Level Two is the Expert Gamers' Level. This level provides some extra puzzles, slightly more gameplay, more challenging solutions, and rewards the users with more points. In the beginning of the game, users can select which level they would like to play. All three paths and seven endings are accessible on both levels of play. [1]

This product uses conventions set up by the detective fiction genre to build a user interface which requires discovering and making deductions from clues to progress through the puzzle, as well as making choices from a menu of responses, which either access one of the three different channels of the branching structure or precipitate the live drama sequences. Makers have retained a "lowest common denominator" approach to technical standards and the game can be played on a very minimal computer system, Pentium 1 with sound card and CD-ROM drive.



These examples given above are of computer games extending in a cinema direction, so to speak. There are, by contrast, art works being produced that take narrative video forms and attempt to move in an interactive direction. For example, an interesting approach to interactive movies is embodied in the work of British interactive artist Chris Hales. Hales carries out practical experimentation in interactive moviemaking (with a strong visual emphasis) and is a research student in this subject at the Royal College of Art Film/TV department and a senior lecturer at UWE Bristol [2]. He has produced over ten "interactive movies".

> These are experimental works in which the movie IS the interface. They investigate different strategies, structures and styles for interacting with movies. The installation on which they are shown is a generic one-person interactive touchscreen cinema, allowing enjoyment of all the movies without being specific to any one. It is the content and purpose of the movies

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[1] source: http://www.microsoft.com/games/texmurphy/pandoradirective/ ; accessed 14-06-02 .
[2] http://www.duckspool.com/Tutors/Chris%20Hales/hales.htm ; accessed 15-01-03 .

> themselves that is important in my work, not the ingeniousness (or otherwise) of the installation on which they are show [1] .

They are contained on the CD-ROM format and made interactive using Macromedia *Director.*

Interactive film installations have been shown in numerous galleries worldwide and *The Twelve Loveliest Things I Know* won the Artistic Excellence prize at the Artec95 Biennale exhibition in Nagoya. In *The Twelve Loveliest Things I Know,* children were interviewed and asked to describe the twelve loveliest things they know. Examples include "the feel of petals," "a cloud that looked like a crocodile," "when you're hot and you have a nice cool drink." [2] The responses were gathered into themes. This exercise, shot on Hi-8 video, formed the basis of a personal interactive documentary which attempts to provide emotion and thoughtful reflection, a mosaic of memories, childhood experience, the contrasting of values between children and adults, their hopes and experiences [3].

When viewed through adult eyes, the responses take on a poignancy at the loss of such innocence. It could be called interactive electronic poetry, utilising multiple video images [Hales, 1996, 2]. In terms of interaction,

> …things that stand out intuitively as colourful or moving or that catch the eye can be 'touched'. Coloured images appear to an old man which represent themes collated from children: achievement, speed, success, caring, play etc. Themes are developed by intuitive

[1] http://csw.art.pl/new/99/7e_heldl.html ; accessed 15-01-03.
[2] http://www.wired.com/news/culture/0,1284,6611,00.html ; accessed 15-01-03.

[3] http://www.emaf.de/1995/vidcompE.html ; accessed 15-01-03.

> clicking. It is possible to experience fairground rides, flying (and crashing) a kite, walking a dog, sledging, bicycle riding and similar events. Choosing symbols of adulthood can bring harsh reality. If no touches are made, the film eventually reverts on its own to the central motif of the elderly man and his thoughts [1].

It is a relatively plot-less piece, that uses time limits on its interaction to keep things moving, as there are no instructions or obvious iconic devices used - though one learns that patches of colour, particularly red, are likely to yield to selection. It is not plot-driven, so if the user misses a choice there is no real cost. The piece cycles back through to the beginning for a second pass [1].

Another of Hales' works, entiltled *Jinxed!,* is an interactive comedy based firmly in the slapstick genre. It is shot in the Hi-8 format and converted to black and white (reducing the file sizes, hence allowing more video time on the CD), and treated with a film simulation effect to give a scratched and grainy appearance. The central character is 'a loser', who is woken abruptly in a dingy apartment by an telephone answering machine message, telling him his job interview has been brought forward by an hour. He desperately needs to attend the interview, and without the user's interaction he will simply wash, dress and exit. Objects in his apartment are however 'jinxed' and by clicking on them it is possible to activate alternate video sequences, and painfully and comically hinder the Loser in his attempts to get ready and leave. *Jinxed* utilises a digital effect during editing of the video images, with a animated 'bulge' filter applied in a relevant part of the on-screen video image indicating a time-limited interactive zone, i.e. "Jinxed" objects (e.g. a bar of soap and the iron) appear to bulge when they become interactive, and there is a simultaneous audio cue. If the user clicks on the soap for example, it falls to the floor and the Loser slips on it, in true slapstick fashion, hitting the bathroom floor awkwardly, but in each case with a humorous expression of pained resignation. Although this could be described as being plot-driven, each of the possible viewer selected incidents is "stand-alone". Once a "jinx" has occurred, the action character is back at the same point as if it hadn't occurred. His limp from the previous fall does

[1] ibid.

not seem to last. Nevertheless, Hales' work represents a tentative but stimulating starting point for the exploration of a potential new art form.

Games and Society

As a popular culture phenomenon, electronic games almost exclusively target the (predominantly male) youth-market and the emergent industry could be criticized for trading in gratuitous violence, which, it is feared, leads to de-sensitization [Janushewski and Truong, 1999, 4]. and also possibly addiction. Miller [2002] writes:

> Shawn Woolley loved an online computer game so much that he played it just minutes before his suicide.
>
> The 21-year-old Hudson man was addicted to *EverQuest*, says his mother, Elizabeth Woolley of Osceola. He sacrificed everything so he could play for hours, ignoring his family, quitting his job and losing himself in a 3-D virtual world where more than 400,000 people worldwide adventure in a never-ending fantasy.
>
> On Thanksgiving morning last year, Shawn Woolley shot himself to death at his apartment in Hudson. His mother blames the game for her son's suicide. She is angry that Sony Online Entertainment, which owns *EverQuest,* won't give her the answers she desires. She has hired an attorney who plans to sue the company in an effort to get warning labels put on the games. [2]

While, to such observers, the violent/addictive attributes of computer games are negative and emotive, the influence of the game metaphor does encourage curiosity and involvement in cultural activity that is in itself non-destructive, though it may deprive children of time for other more valuable or educational and self-fulfilling experiences. There is a complex debate regarding social issues and value judgements involved here, and, for instance it can be argued that simulated violence is a whole lot better than real violence and crime. There are no definitive scientific studies establishing a causal connection between simulated and real violence. According to Janushewski and Truong, [1999, 4].

> Although there have been studies that have found video game violence to have little negative effects on their players (Griffiths, 1999), there are also many studies that have found a positive correlation between negative behavior, such as aggression, and video and computer game violence (Anderson, 1986; Malouff et al, 1988). Numerous laboratory studies dating back to the original Bandura, Ross, and Ross (1961) study of modeling of aggression in children, have shown that exposure to aggressive models can lead to an increase in subsequent aggression. There is no real consensus on the affects of violence in video games on players… [Griffiths, 1999].

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[1] http://www.austega.com/interactive/hfdefn.htm#introduction ; accessed 15-01-03.
[2] Source: The Milwaukee Journal Sentinal, March 30, 2002 . URL: http://www.jsonline.com/news/state/mar02/31536.asp ; accessed 12-6-02 .

Clearly, the social / psychological / ethical issues are complex and difficult, but interactive computer games have now become a pervasive cultural reality. Such games have taken a firm hold on the collective imagination of popular youth culture, a fact verified by their success as a new form of entertainment. They are now a standard offering alongside movies at the neighbourhood video outlet and indeed, as an uncredited item in the Sydney Morning Herald [19-2-98, 12] records, 1998 was the first year that computer games rentals outstripped video copies of movies as a proportion of Australian distributors income.

Current Standards in Interactive Computer Games

This discussion of interactivity concludes with an analysis of a contemporary and highly popular computer game, in an attempt to establish a sense of the aesthetic and technical levels this branch of multimedia has reached, and how it might be considered as a contributing stream of development important to the future of interactive digital art.

Grand Theft Auto: *Vice City*

For PlayStation 2 (PS2). Developed By Rockstar North.

Published By Rockstar Games.

Technical specifications:
1 Player ONLY
Memory Card required (for PS2) - 500kb
Analog Control
Vibration Function
Pressure Sensitive

Rated MATURE: Violence, Strong Language, Strong Sexual Content.

In the US, *Grand Theft Auto: Vice City* initially received highly favourable reviews, with Time Magazine calling it a "national obsession" and claiming that the game "borders on art". Niche market gaming magazines have described it as "peerless". [1]

The (little picture) detail and (big picture) scale of this interactive game is an indicator of the current level of maturity and sophistication interactive gaming has reached in

[1] Dave McCarthy http://news.bbc.co.uk/1/hi/technology/2419007.stm ; accessed 17-12-02 .

terms of richness and complexity, technical expertise and plot development. The action is fast and exciting, the visuals vivid and strongly stylized.

A flowing, multi-layered, real-time 3D experience is produced by the user's rapid interactive responses, via the multi-functioned buttons and joysticks of the Sony Playstation 2 interface, which includes a vibration function for additional sensory feedback. Sound, visuals and interactivity blend into a highly charged and immersive multimedia experience.

(**Below:** Text in red indicates car controls, turquoise indicates control of virtual character "on foot").

On-screen the user is in the driver's seat, barely controlling wild, edgy cinematic car chase scenes. Vivid action sequences string together in a flow of unrealistically explosive sound and visuals, designed to produce an almost endless variety of optional story interludes. This interactive blitz is structured by nodal interludes which occur when the user completes a mission, or is prematurely "wasted" or "busted" by the law, each of which outcomes is a trigger to the key scripted plot nodes. These plot points, or pre-rendered "cut-scenes", are branching points in the interactive structure, and can be also be accessed (when indicated within the gaming geography) by pink light, sometimes appearing in a circle, like an apparition, outside a significant mission- or briefing-related destination.

The game is populated by a large variety of incidental pedestrian models.

The action is set against the backdrop of a teeming western metropolis, populated by a faceless caste of street life - people on roller skates, people lounging on benches, driving leisure boats around, people enjoying the sun on the beach, police chasing criminals down the street, etc. Life is 'on the edge' in this city, and users will not infrequently observe background action in which rival gangs get into random fights with each other, even if a user does not instigate a direct conflict interactively.

This game is emphatically violent, and certainly (one would hope) not pitched at children. In fact it is *about* violence, and to the uninitiated it may appear on first impressions to be a sick and twisted activity. However the embedded, culturally-sophisticated humour, biting social comment and satire, and rich media experience pull it back from that unsavoury brink.

It attempts to balance a dark and charmless plot, involving murder, extortion, revenge and organized crime, with an equal amount of tongue-in-cheek humour and style, making use of a number of the kitsch pop-culture stereotypes found in American film and television of the 1980s. The illicit drug focus owes a debt to films of the celebrated period such as *Scarface* and television shows like *Miami Vice* [1].

[1] Jeff Gerstman http://gamespot.com/gamespot/stories/reviews/0,10867,2895954-4,00.html ; accessed 16-12-02.

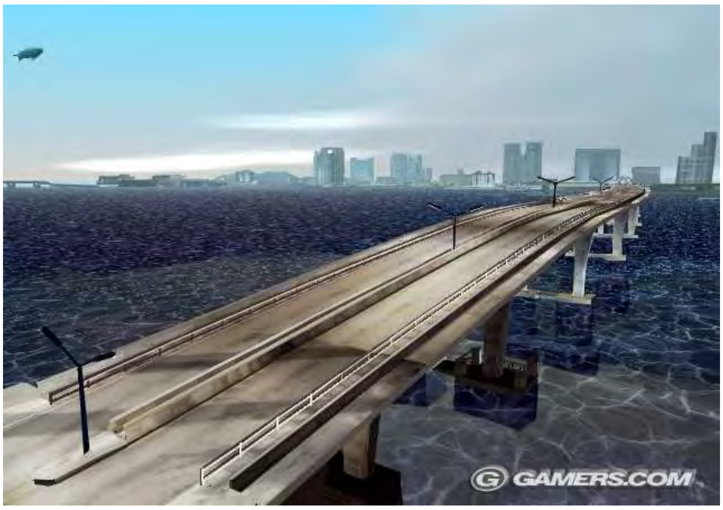
A strongly satirical/humorous element is developed in the audio track – the user soon discovers that (once stolen) each car is tuned to a different send-up radio station, adding a layer of elaborately scripted voice-over dialogues, which cleverly lampoon various media stereotypes. The soundtrack to the user's new 'life of crime' becomes a hit parade of kitsch but iconic 80s pop songs, alternating with satirical station breaks and "radio personality" voice-overs, to produce an energetic and evocative audio channel, which is ultimately a dominant element of this multimedia *tour-de-force*. The user can switch between stations, adding to the uniqueness of each driving sequence.

Above: The radio stations have actual geographical locations in *Vice City* e.g. heavy metal station *V-Rock*.

The Geography of Vice City

The 3D city is extensive and highly detailed. Action is spread over the two large and a few smaller islands which comprise *Vice City*, allowing for a finite geography (see next image). Spread across this 3D city there are a huge variety of interior areas for the user to explore. One can enter strip clubs, malls, restaurants and diners, drug stores and other retail outlets, dance clubs, hotels, police stations, and much more. Some interior locations are stark and eerily vacant or run-down, while others are up-market and full of activity. For instance, if the user navigates into the dance club, you'll see a large group of incidental male and female figures gyrating to the music. There are also a variety of atmosheric outdoor locations, such as golf courses and lonely dirt bike tracks that owe a debt to other action sport-oriented games genres, but also offer a tangibly cinematic atmosphere.

The island location allows for a variety of boating activities, and of course high-speed boat chases.



The weather is a variable factor and, for example, the bridges between the islands are initially closed due to an impending typhoon. It rains at random intervals, complete with rain-drops on the virtual camera lens.

The GTA *Vice City* Plot:

In the year 1986, in a fictional Vice City (a take on Miami, Florida, USA), the central character, and the user's vicarious persona or action figure, Tommy Vercetti, has just been released after serving 15-years in prison. The Forelli family, an over-the-top caricature of a mafia organization, appreciates Tommy's refusal to implicate them in exchange for a lesser sentence, so on his release they send him south to Vice City to head up some new operations, beginning with a large cocaine transaction. This is set up by his mob contact, a shady and corrupt lawyer named Ken Rosenberg.

Right: Screen-shot from a 2D nodal "cut-scene". The Ken Rosenberg character is the stereotypical neurotic and cowardly lawyer, with a nervous, high-pitched voice. His morals aren't high however, as he is suspected of cheating on his law school exams. It was 'street smarts' over intelligence or legal skill that got him where he is, as he hints at having employed 'heavies' on numerous occasions to intimidate and corrupt juries. Despite these intimidations he has had a poor success rate in trials. His "Blue-chip" clients include Georgio Forelli, cousin of Liberty City crime boss, Sonny Forelli. Rosenberg is always desperate for money and willing to try any scheme possible to get it, so long as it isn't his neck out on the line. [1]

[1] http://ps2.ign.com/articles/375/375249p1.html ; accessed 16-12-02.

However their first drug deal ends in an ambush, leaving Tommy with no money, no cocaine, and no clue as to who set him up. The "mob" leader, Sonny Forelli (screenshot left) is enraged by this development. Tommy has to make up for the loss under the threat that the gangsters will come down from their headquarters in Liberty City for revenge. From a literary standpoint the plot is not particularly innovative, relying heavily on lampooning hackneyed cinema clichés and crime character stereotypes.

Gameplay

Vice City is an interactive story in as much as, within the structure laid out by the story nodes, the user's actions, reactions and decisions decide how the plot unfolds for the virtual character Tommy. The user can gradually acquire gameplay skills that can improve the outcome, i.e. within the dark scenario which is established, Tommy's story can become a success story. The user controls the surrogate Tommy, with direction buttons, an accelerator, a jump button, and punch/kick/shoot controller. The 3D Tommy figure can walk and run in any direction, but will be stopped by hard surfaces. He shows a rare glimpse of humanity when he gets tired after long running stints, bending forward, hands on knees, to pant and catch his breath. His repertoire of preprogrammed actions allow him to punch and kick.

He can jump, allowing him to negotiate fences and ledges, and leap from balconies, efficiently, though with no great panache and the occasional grunt. In terms of methods of engagement, this design team has\ a good grasp of action.

Tommy can crouch and hide, and in the crouching position he can shoot with more control. And of course he can steal and drive any vehicle and operate any available weapon. The on-screen interface overlays (below) offer a pink circle in the lower left-hand corner, containing a constantly up-dating location map, which shows present location of the central character (white dot), a compass, and in most cases a coloured indicator showing the geographic location of

the mission objective (usually a darker pink dot on the outer circle, but sometimes an icon related to the mission - such as a blue tee-shirt when Tommy is required to get appropriate golf clothing in order to carry out an intimidation on the golf course). The user is thus given constant real-time feedback and orientation about progress within the game environment, and about resources and key mission objectives. In the upper right hand corner there is an enclosed icon showing current "twitch" weapon choice (this defaults to a fist if no weapons have been acquired). There is a read out left of the weapon icon offering real-time updates on :

- time expended on current mission,
- money acquired (green numbers),
- a health rating – signified by a pink heart (which diminishes from 100% with the wear and tear of a dangerous criminal life in *Vice City*); and

- a blue star rating, which increases according to the amount of police interest the user has drawn during interactive criminal activities and escapades. Earn two stars and the police will not cease their pursuit. Achieve three stars and undercover police enter the pursuit. Four stars introduces SWAT teams and roadblocks with tyre spikes. Five stars induces the FBI to enter the chase, who have high-powered guns and are more aggressive and skilled drivers. Six stars provokes the authorities to call out the army, which nearly always ends in Tommy being “wasted”. (Police interactivity is discussed in more detail below).

It is only possible to speculate as to the range of psychological processes that the user might experience as a result of partaking in this fantasy experience. It is implied in this design that increasing dramatic intensity can be induced in a user by relating interactive action to this multiple scoring system.

The script determines that the user, using the Tommy character as protagonist, must start an investigation into who staged the cocaine “rip off”, and extract his own revenge, and the hijacked goods. In the process this involves performing various adventure sub-missions designed to establish a power base and connections in the underworld, earn illegal money and generally “set up shop” as a mobster in *Vice City.*

During the course of performing these missions, the user, as Tommy, must interact with the plot, which involves performing an interactive role in the many sub-plots, by attempting to achieve prescribed goals, mostly of a criminal nature. The complexity and variety of possible interactive paths mean that any sense of involvement in an interactive story is mostly lost in the heat of reacting to the demands of these action sub-games. There is little real feeling set up during gameplay that the story will come to a climax or conclusion, and in this regard no sense of narrative sweep or intriguing character development. The user cannot re-define or challenge the moral climate, e.g. cannot change Tommy’s basically amoral character and choose to do good etc.

In the course of working through the sub-plots Tommy, complete with loud 80s Hawaiian shirt, must inevitably steal, kill and intimidate, and defy the consequences of law-enforcement. There are many criminal themes explored in the game solely for their action possibilities, and an essentially rebellious, anti-establishment celebration of breaking the ”normal” rules of the social compact. Thus Tommy can be a power figure for the disempowered, but only by choosing a life of crime. In this feast of anti-social behaviour, Tommy can get involved in a turf war between Cuban and Haitian gangs, befriend a Scottish heavy metal rock group named *Love Fist*,

become a pizza delivery boy, vandalise the local shopping mall, demolish a building to lower real estate prices, join a motorcycle outlaw gang, run an adult film studio,

rob a bank, and much, much more.

From time to time the user must learn new life-skills e.g. interactively change the action figures clothes to further mission objectives. The following is an excerpt from an on-line “walkthrough”[1]:

“**Shakedown”**

Difficulty: 4/10

Reward: $2000 and all other asset properties are now available for purchase

Walkthrough:

In this mission, the user must convince businesses in the North Point Mall to pay

[1] Computer games have generated an on-line culture, and the *walkthrough* is jargon for a gameplay guide. Source: http://db.gamefaqs.com/console/ps2/file/grand_theft_auto_3_property.txt','2898981','700','600 accessed 17-12-02.

> protection money. You have 5 minutes to go to the mall and break all the windows of the businesses inside. Head toward the mall and make sure you have full armour, health, and a lot of ammunition. Once you get inside, begin to shoot out all of the windows of the businesses in the mall. When you start shooting the windows, the fuzz will be after you, so try to avoid them. Once you have shot out all the windows, the mission is complete.

This sort of action demands efficient interactive navigation to different parts of the virtual city. This requires parallel use of high-level reaction based motor skills and abstract conceptualization of the virtual geography. Tommy has great facility (and no compunction) at stealing any vehicle that he comes across (via a special button programmed for interactive car stealing) – hence the series title. Tommy can steal taxis and buses, even police cars if he's quick and ruthless. He can hail passing cars and callously toss the protesting drivers out when they slow down, usually with a gruffly muttered one-liner, such as "My needs are greater than yours…" The user decides on the action, and then is rewarded/entertained by the resulting humorous or action-packed animation.

The user, with Tommy's help, quickly becomes a serial car-thief, with an eye for fast automobiles. The user soon recognises slower family-type sedans and avoids them with disdain.

Game Controls

The cars respond impressively to the hard-wired playstation controller in a hypnotic, adrenalin-charged (though somewhat cartoon-like) simulation of high-speed driving through a busy city, accented by powerful sound effects, and bathed in subtly changing atmospheric light. Both in terms of the driving metaphor and the general principle of seeking every possible coherent engagement strategy, the controller offers both ‘direction’ buttons and a joy-stick for steering, a pressure-sensitive accelerator button (which also triggers a change in the cars engine sound as the “revs” increase), a separate reversing accelerator button, a brake button and even a horn. On-screen the virtual cars themselves have simulated glowing head- and tail-lights, and the brake lights react to the controller brake button. These are all elements specifically designed to build the narrative image, and were likely to have been developed during team brainstorming sessions in pre-production. Being central to the core concept of the game, they are complex 3D models with many discrete parts. Windscreens crack and shatter, and buckled doors and bent fenders swing in the slipstream and eventually fly off. Cars break apart in spectacular fashion during collisions.

It’s probably not uncommon for users exploring the game for the first time to go through a stage of abandoning caution and relishing the wrecking of cars for the sheer fun of it, in a virtual, consequence-free world. There is a light-hearted, surreal “dodge ‘em car” feel about this whole recklessly destructive process, even though this frequently involves the accidental, (or even intentional), hit-and-run slaughter of innocent by-standers, and lethal head-on collisions, complete with screams, and the sounds of impact and smashing glass. Stolen vehicles rebound from outrageously violent collisions to continue on their hair-raising, high-speed journey, with just a few more buckled or dangling parts, and a little more smoke emission, giving the user a growing confidence as a death defying getaway driver. The engine hood flies off soon after the second or third collision. Street lighting poles defy physics and crash noisily to the ground on vehicular impact, without noticeably impeding the progress of the stolen vehicle, adding to the mayhem.

There is a collision detection/counter routine in the programming which requires that, at some point of accumulated damage, the stolen vehicle will explode, usually signalled by palls of heavier black smoke and flames coming from the engine. Tommy must be evacuated quickly, or be “wasted”, obliging the user to begin the mission again, an anti-reward. This emergency situation can become a regular trigger to an ‘upward shift’ in immersive intensity in the gameplay, and in fact, this narrative device implies that the level of user immersion is proportional to the user’s ability to self-induce a sense of internalized pressure to execute a rapid controller-triggered escape from the interactive scenario, under a timer routine. Of course, developing better virtual driving skills is one of the central concerns of the game –

expert driving draws less police attention and allows for more rapid execution of crime missions,

Virtual Cameras

Virtual camera angles can be instantly varied by the user (using the "Select" button), changing the participant's view into the action or 3D geometry, offering another dimension of control. First-person geometry shows the scene as it would be observed by the character as participant in that situation, i.e. a "through Tommy's eyes" point-of-view.

Third person geometry allows the player to see the action (including their surrogate action character's form) from a distance, such as in profile (right).

Vice City defaults to a view which could be called 'displaced first person' geometry, in which the player is looking at the action from slightly behind the action figure, as though following him (see next image).

Left : the default "displaced" first position camera angle, viewing the action figure from slightly behind.

The user can also toggle to a frontal view of the action figure (below).

If the Tommy character is not activated interactively with the controller buttons, he idles, and gets comically impatient and fidgety after a time, scratching his head, shifting his weight, stretching his legs, looking at his watch, and complaining, (below).

In between fidgeting he appears to keep the street under survellance, moving his head from side to side intently. A cast of hoboes, streetwalkers, joggers and tourists wander by, and the randomly cycling ambient sound samples, traffic noise, and one-liners build a satirical simulation of street life of the period.

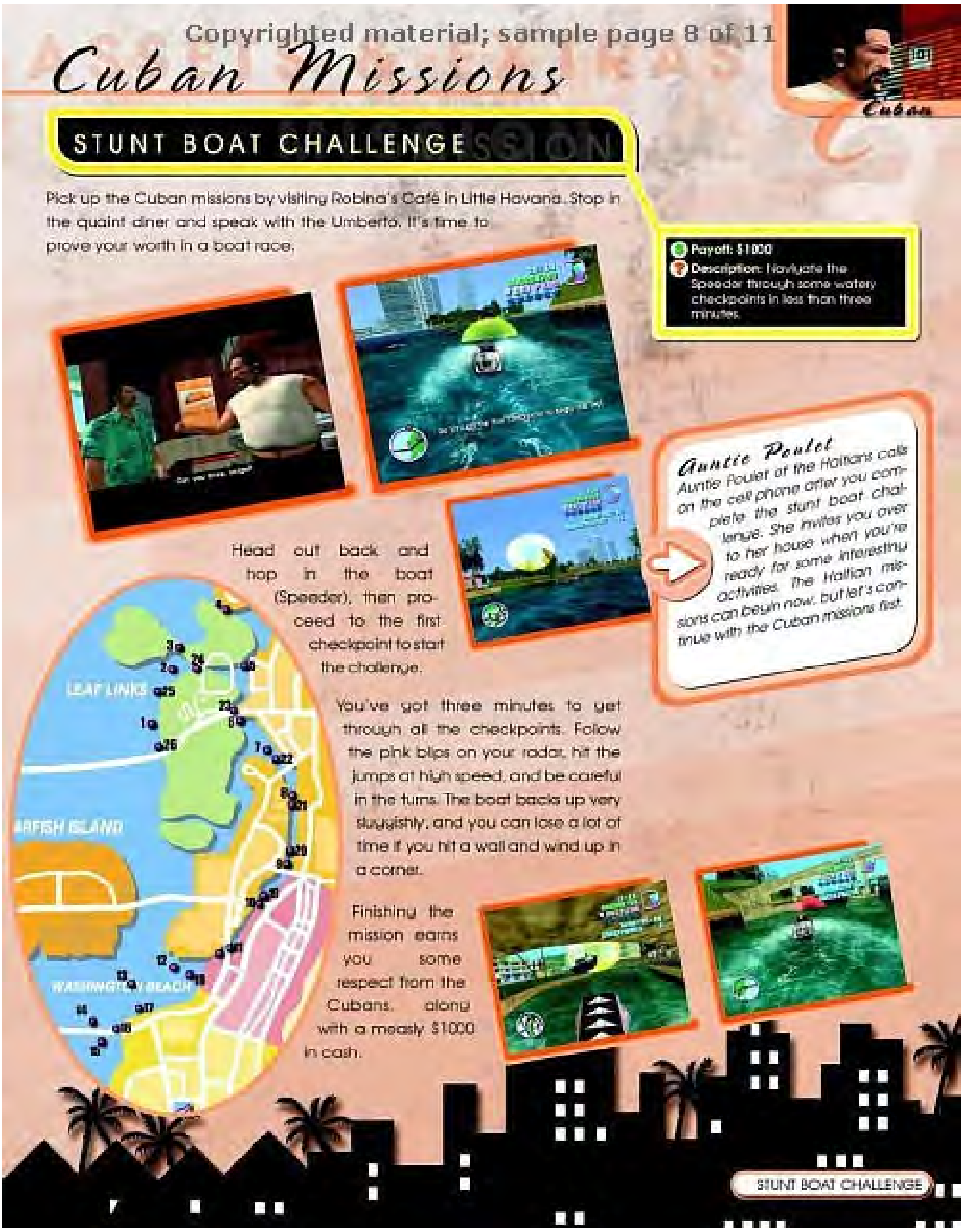
The various non-default interactive viewing options almost seem to be there “just because they can be”, and offering token variety or advantage for immersion in the game, i.e. this user soon reverted to the first person camera, because of it’s role-playing excitement.

Missions

To progress through the game hierarchy, the user must attempt to complete the over 80 crime missions. Missions are sub-plot challenges, and in fact games-within-the-game, i.e. mini-games, with an internal set of rules, strategies and interactive rewards. Most of the GTA missions are multiple-part affairs that involve much more than moving the action character from point A to point B and then back to point A. Some of these play activities are scripted as extensions of primary missions, such as having to take a getaway car to a *Pay ‘n’ Spray* car respray shop after “pulling a job" – like a coda or post script. However, other missions require the use of more advanced tactics. For example, one mission requires the user to conceal and detonate a bomb inside a shopping mall that is swarming with police adversaries. To do so, the user first must draw police attention and provoke them into to a chase. Leading the police into a garage, the user must then ambush them and steal one of their uniforms, so Tommy can then return to the heavily guarded mall in disguise. Once business is taken care of at the mall, users then have to escape, and get all the way back across town to the hideout, using an appropriate stolen vehicle, each of which responds differently to the interactive controller. An experienced user sizes up potential getaway cars for their game-play advantage, depending upon the interactive game-play scenario which is presented. High-powered sports cars are extremely fast and maneuverable, trucks are slower and more cumbersome but can effortlessly crash through roadblocks and bump aside on-coming police cars with impunity. Lighter delivery vans roll very easily,

All these variations on a theme – the auto metaphor, which is the core interactive experience "driving" this slickly-produced game . Missions include such remunative illegal fantasy play as: collecting drugs for the virtual rock group *Love Fist;* "taking out" a psychotic killer who is stalking the band; eliminating a gang of six thugs planning to heist a bank convoy, within a ten-minute time limit, using a machine gun; etc. The following is an excerpt from *Grand Theft Auto: Vice City Official Strategy Guide*[1].

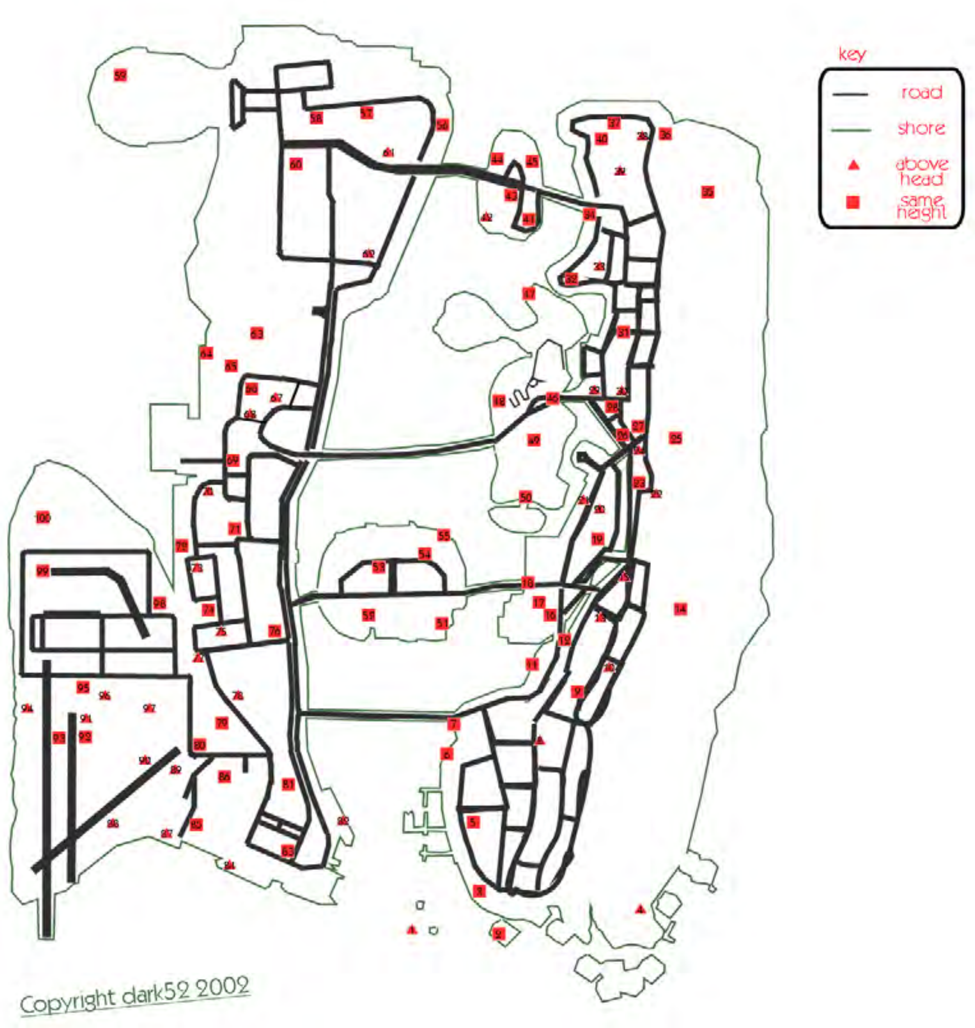
[1] found on the www.amazon.com site.



Around the city there are special zones where icons appear which will assist in the completion of missions. A glowing pink 3D heart appears which can replenish Tommy's diminishing "health" rating. Adrenaline pills appear which slows down on-screen action to a visually attractive slow motion, and temporarily gives the action figure much more strength in combat. Police "bribe" stars appear occasionally, which on contact lower the "wanted" level by one star, cynically reducing the police menace. Body armour can also be acquired or purchased, doubling the action hero's capacity to sustain injury.

Hidden Packages

There's over a hundred hours of gameplay if one tries to track down all the “hidden packages” and stunt “launchpads”. There are 100 hidden packages scattered around Vice City to challenge the users skills. Some can be collected by having the action figure simply walk up to them, others are on rooftops or in secret, inaccessible spots that require vehicle skill to reach. Each is worth a $1000 bonus, and for every ten secret packages the user collects there is a reward of special bonus weapons, such as the chainsaw, flamethrower and rocket launcher, as well as body armour and a military tank. However collecting them will increase challenge levels, taking the user into rival gang territory and also ramp up the danger of being “busted” by law enforcement officers.



Above: map of “hidden packages” in *Grand Theft Auto: Vice City.*

Assuming the user has the leisure time, several more hours can be spent just "cruising around" in virtual vehicles exploring the city, and discovering the interiors of buildings (one can even find beach balls floating in pools which can be kicked around by Tommy for fun), or engaging in "scaring up" the cash to buy assets such

as strip clubs and condominiums. The average mission duration can be considerable, though this can become a source of frustration to the novice, since failure in a mission (being "wasted" or "busted") means having to replay every part until you get it right. This can develop into a repetitively nightmarish scenario (reminiscent of the movie *Ground Hog Day*), for the unskilled user. The gameplay requires the user to spend the first portion of the game undertaking amoral missions for other criminal characters, in order to earn money for expenses, such as the cost of weapons and real estate assets. These include drive-by murders and an extortion /intimidation mission on the golf-course on behalf of Avery, the shady real estate mogul (featuring golf-cart / car chases – see next image). There is also a drug-related pizza delivery-boy assassination and remote-control helicopter bombings.

Above : Tommy on a mission, with the character *Mercedes* his passenger

But, assuming the user is comfortable with the amoral and essentially negative nature of the concept, and game-skills are well honed, the city is for the taking. You'll be empowered by your almost invincible character, hard at work, keeping your protection racket in line and gradually establishing yourself as the new underworld boss, with all the rewards of a life of crime unfettered by conscience.

"Rampages"

As a further skill-based engagement satategy, 35 skull-like “Rampage” icons are scattered around *Vice City*. Once one of these appears, walking the action figure “into” it triggers a bonus mission in which the user is challenged to kill or destroy a set number of adversaries or vehicles within a time limit. This is an example of playing the small picture and the big picture simultaneously, i.e. they are like mini-games but successful completion adds to the user's virtual assets and helps one achieve larger game goals. For example, Rampage 23 requires you to act the part of a cold-blooded serial sniper:

> **Location:** On the grass next to the Ocean View Medical building.
>
> **Weapon:** M4
>
> **Reward:** $1150
>
> Kill 25 gang members in 2 minutes to complete the mission. Simply stay on the grassy knoll, using the first person perspective to shoot everyone in sight. When you run out of targets hop down onto the street and run around killing any people you are able to find [1].

The reward for completion of all the rampages is $1 million.

Stunts

“Stunts” are programmed in to reward the highest level of gameplay skill. Each of the islands contain ramps, which, when approached at speed, cause the stolen vehicle to become airborne. “Insane Jumps” offer cash prizes for successful negotiation, while other more difficult manouevres known as “Unique Jumps” also trigger a slow-

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[1] From *The Complete Guide to GFA: Vice City*, Derwent Howard Publishing, Sydney, p.68

motion replay. The latter require high-level virtual driving skills, and are only successfully completed if the vehicles wheels land in an exact pre-determined “correct” position. The user must steal a high-powered sports car to complete these special objectives. To achieve these quite esoteric rewards requires the user to invest a very considerable time in order to acquire extremely high-level motor co-ordination-based game-skills, specific to this game, requiring very fast reaction times.

Acquiring Assets

Eventually, completion of missions allows the acquisition of enough virtual currency to be able to purchase various properties. For example:

Vercetti Estate
Location: Starfish Island
How to Acquire: Beat 'Rub Out'
Asset Completion Reward: Vercetti Estate will generate $5000 daily

Sunshine Autos is another Vice City business that the user can purchase for $50,000 (and which one will need in order to finish the game). *Sunshine Autos* is a car dealership, which can return a total of up to $9,000 per day. Also the user can gain possession of four rare cars, which can’t be found anywhere else in the game. This is the best real estate purchase available, as one pays a relatively low amount to buy it at $50,000, yet it returns the second highest daily revenue for any ownership. *Sunshine Autos* is located on the second island, at the lower end.

The addition of real estate and other acquired virtual assets creatively blurs the line between genres that the game falls into, giving it extra layers of engagement more in keeping with environmental simulation and strategy genres, above and beyond the “twitch” action elements.

Purchasing properties opens up new more complex, and more rewarding missions. For instance, when one buys the taxi company, a new interactive branch in the hierarchical design structure is opened up, featuring a series of taxi-related missions that are separate from the side missions that one can take on by simply entering any taxi as a passenger.

Once the user has completed a property's missions, that property will start earning money. This means that money eventually becomes a non-issue--as it should be for any self-respecting crime lord--since your various properties will always generate some cash. In this way the game retains and builds interest, as new freedoms to operate within the game structure are unlocked by successful execution of the sub-plots. The user must drive around and visit them to collect, from time to time. There are several other income generating properties to purchase, including a film studio, and the Malibu Club, which generates the highest income at $10,000 per day, but is the most expensive purchase at $120,000. Other specific secondary, non asset-based mission types are also available, such as vigilante missions, fire truck missions, and ambulance missions. It is vital therefore that the user purchase an optional additional memory card for the play-station so that levels of achievement within the gameplay, once reached, can be retained in memory and recalled at the next gaming session. Without this hardware capability it is virtually impossible to progress through the many levels of the game.

The Police

The police are your major enemy, and while slow to react to Tommy early in the game play (see left below), gradually become a formidable threat in the interactive action as Tommy draws more “heat” (see right below).

Armed with an MP5, Tommy prepares to take on Vice City SWAT.

They have the ability to shoot out the tires on your getaway vehicle, a skill which Tommy also needs to acquire to complete certain missions (see left screenshot below). Attempting to jump into a vehicle and leave the scene of a crime usually gives police adversaries enough time to shoot out one of the tires. As mentioned, at higher levels of response (signified on-screen by a row of blue stars which gradually light up as your criminal activities become more serious), the police will set up spike strips to take out all the getaway cars tires. (Below right, police interception).

They ignore most standard traffic violations however, but once the user has seriously transgressed the law, they begin pursuit of the action character and will set up standard roadblocks. As the users actions become more violently illegal more police resources are marshaled, and eventually the FBI join the chase. It is possible however to steal police cars and Tommy easily gets the upper hand in a physical confrontation with individual virtual officers, which are portrayed in a subtle way as less-than-human automata.

At higher levels of official notoriety, Police Helicopters will also give chase, especially if the user kills a policeman.SWAT teams will actually rappel out of the helicopters, making them even more dangerous (see next image). A direct hit in a helicopter’s cockpit with certain weapons, though, can bring one down. At the highest level of law enforcement response, the army rolls tanks onto the street,

making your chance of survival slim. Just brushing against a tank causes the stolen vehicle to explode.

All this means that the on-going encounters with the authorities in *Vice City* can be exciting and immersive, requiring lightning-quick reactions and decisions, and building a sense of ever-increasing challenge and drama, which triggers adrenalin and encourages total concentration and involvement in the gameplay.

However, one technical flaw is that the police adversaries don’t deal with elevation changes particularly well. If one runs into a parking garage and leaves via the ground level or a higher floor, the pursuing police aren't “smart enough” to find their way up and engage. They'll even keep firing in your general direction, even though there are several walls and ceilings between 'you' and the officers. But one encounters this sort of thing very rarely amidst many wild and intense chases and shoot-outs. Dan Houser, production manager and script-writer for the production company Rockstar, speaking of the evolution of the script, comments that [1]:

> The script is a joint effort involving mainly James Worrall, one of the level designers at Rockstar North, and myself for the cut-scenes and mission briefings; Lazlow and me for the radio stuff; and a whole group of us in New York and Edinburgh who work on the pedestrian comments. The most important bit, obviously, is the mission briefings, which we wanted this time to play like short scenes from movies, with certain bits feeling like TV shows.
>
> The plot and characters come first, along with a long list of missions from all of the level designers at Rockstar North, and then this is slowly turned into a script so the cut-scenes can be built. In order to keep it as authentic as possible we did a lot of research into the '80s. It was tough, but somehow I managed to survive watching all those movies and reading all those books.

Virtual Vehicles

Living up to it's name*, Grand Theft Auto: Vice City* offers the virtual car-thief a remarkable variety of virtual vehicles including a large range of sedans, sports cars, limousines and hot-rods, light trucks to heavy-loaders, buses, motorcycles and scooters, a variety of motor boats, dingys and yachts, golf buggies, helicopters and aeroplanes, including a seaplane, and even military tanks.

[1] The Economics of GTA: Vice City, Pt. 3 http://ps2.ign.com/articles/375/375488p1.html ; accessed 14-12-02.

The vehicles have been programmed to have different driving/flying capabilities (speed, manoueverability, resistance to damage etc.). For example the tanks move slowly but are virtually indestructible and any collision with a stolen tank results in the on-coming vehicle exploding in flames.

Character Development

Vice City script-writer Dan Houser, discussing character development in the script, says:

> The script sets up the interactivity. As the characters are now also talking during the missions, you can experience the characters development while playing the game. To a great extent movies and games are different disciplines. The one thing we have really learnt from movies is the need for great dialogue and we've tried to insure the characters are strong, with some great lines. [1]

Pivotal characters are stereo-typical caricatures, extending the overall satirical approach taken in the *Vice City* plot. Each is a cliché underworld figure, drawn from the rich cinema tradition of the gangster movie. Characters have no moral ambiguity or redeeming features, apart from their humorous value as exaggerated send-ups.

Examples of the extensive cast of featured characters that drive the interactive plot include [2]:

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[1] The Economics of GTA: Vice City, Pt. 3 http://ps2.ign.com/articles/375/375488p1.html ; accessed 14-12-02.

[2] This and the following short character descriptions are quoted from: *The Economics of GTA: Vice City, Pt. 2. What drives the characters in Rockstar's game?* (Part 2 of 3). http://ps2.ign.com/articles/375/375386p1.html ; accessed 14-12-02 .

Lance Vance

Lance Vance is a smooth talking, sharp suit wearing man with many questions. He is new to *Vice City* and little is known about him. What is known is that he has previously been trained as a helicopter pilot. His behaviour is anything but appropriate as he has been heard mouthing off in a lot of nightclubs in the short time he has been in *Vice City*. It is possible that some emotional insecurity makes him aggressive and needy when he drinks.

Steve Scott

Steve Scott is a film director who has been spotted at various parties where organized crime deals are done. Scott is a seedy man thought to be seeking any money he can to finance his films. He shows an unnatural obsession with sharks and mountains of mashed potatoes.

Ricardo Diaz

Alleged crimelord Diaz is as shady as they come in Vice City. He bribed the INS for his Green card back in 1978. He entered the country from Colombia and is considered extremely dangerous. Contacts in Colombia believe he is a major player in the narcotics industry and anti-government activity. Diaz saves face from the public by making large donations to charity and is an extremely popular philanthropist. He gives to various Ricardo Diaz foundations across *Vice City* and Central and South America, all of which are believed to be a front. Diaz is extremely short and is believed to suffer from a Napoleon complex. Medical records show he has over-active glands and sweats more than is socially acceptable. After a failed hair transplant in 1983, the doctor was later found dead in a swamp area east of city. The murder has since been unsolved.

Diaz is a gun collector and is always armed. He has a private army/militia and is heavily guarded at all times. A long-running battle for control of narcotics business in Vice City has suggested that he has bribed police and officials within the town. Diaz keeps the city under control as he is feared due to his reputation for unpredictable behaviour. He is currently thought to be responsible for 18 murders.

Interactive Weapons

This a violent, no-holds-barred world and, in order to be upwardly mobile, the career criminal needs weapons. Weapons materialize magically, hovering in a yellow halo, at certain trigger points in the plot, often when Tommy happens to be within the vicinity as police over-power a passing villain – an opportunistic Tommy gets the chance to acquire the felon’s armoury. The user must have Tommy pass through the halo to acquire the weapon. Weapons can also be purchased from a store called *Ammu-nation,* if the user has accumulated currency from successful misdemeanors.

Weapons are categorised into different classes of destructive power, and Tommy can only access one weapon from each class at any one time. For example, the default assault rifle is a Ruger, but later on in the gameplay you'll be able to acquire the speedier M16. Picking up an M16 will replace the Ruger, getting a golf club will replace your baseball bat, and so on. The selection of weapons includes pistols, submachine guns, rifles, shotguns, sniper rifles, and thrown explosives.

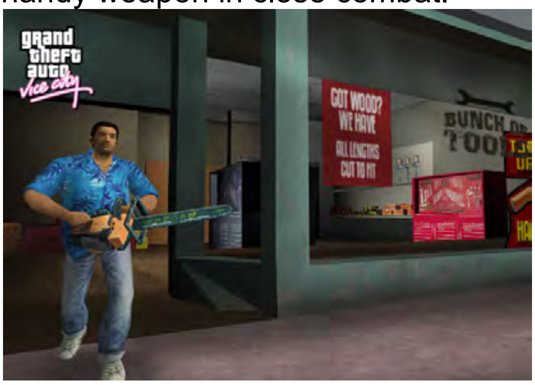
The user must progress from handling basic weapons to aquiring lethal skill with deadlier versions. For example, the Tec-9 is the default submachine gun, but later in the gameplay you'll be able to buy an MP5K, which has a much faster rate of fire necessary to complete certain rampages and high level missions.

Later still, you'll be able to wield rocket launchers, a flamethrower, an M60 machine gun, or even a Gatling gun, taking the whole game concept to a level of self-parody.

The physics of *Vice City* determine that different weapons have different weights, and Tommy's movement speed may be compromised by the weapon selected. Wielding a pistol or a submachine gun lets Tommy run at full speed. Carrying the shotgun or rifle prevents him from sprinting, but he can walk normally. The heavy weapons cause him to lumber around sluggishly.


Tommy can shatter car windows with a 'billy club'. Occupants leave hastily.

It is also possible to enter a hardware store and purchase a "melee weapon": screwdriver, a hatchet, a baseball bat, or a machete. You'll also find additional melee weapons in different parts of the city. Entering the golf course, for example, makes it easy to find a golf club, which proves a handy weapon in close combat.



The virtual weapons reach a level of absurd satire. For example, one can also acquire a chainsaw, which is a potentially dangerous weapon of intimidation in theory, but surprisingly unsatisfying in action. The appearance of the chainsaw in Tommy's hands inspires incidental pedestrians to flee in hysterical panic.

Getting “busted” (losing a violent confrontation with police and being captured but not terminally “wasted”) means that, once released from custody (and a new scene begins - returning to a node in the story branching structure), all weapons will have been confiscated. Being “wasted” means a trip to hospital for repair. Fines and medical fees eat automatically into your accumulated cash reserves.

Graphics

The game's textures are bright and colourful, stylised to convey in a cartoonish manner how '80s-era Miami might have looked, rendered “on-the-fly” (in real time) as the user interacts. Nearby buildings are laced with neon that glows attractively at night. There is a great deal of attention to detail, such as the glint of sunshine off the windows of nearby cars as vehicles turn corners, distant vapour trails from planes in the sky, camera lens flares and changing weather conditions. The only technical problem with the graphics is the frame rate's tendency to bog down when a large contingent of police forces are committed to pursuit, making escape that much more challenging. The entire look of the game is quite different from its predecessors, and technically, the game has a clean appearance, character models are cleanly executed, and the animation is smooth and seamless.

Some of the quirky animation highlights which are often unlocked quite unexpectedly include for instance hijacking a motorcycle from the front, which causes Tommy to execute a flying jump kick that knocks the rider completely off the bike. (Attempt to hijack a bike from the side, and Tommy will inevitably deliver an elbow to the face of the unsuspecting rider).

The “draw distance” allows the user to see the changing background detail of the city over a great distance. This is even more noticeable when you're finally flying high above the city (a skill-level not easily discovered by beginners) and one can see almost all the way across to the far perimeter of the city limits.

Sound and Music

Sound plays a very important part in setting the frenetic, humour-laced tone for the entire game and is a strong media element driving the sense of action. The immersion value of *Vice City* is greatly enhanced by this stylized but detailed audio design, keeping up a constant sense that the game world is “alive”: quips from incidental pedestrians (a seemingly endless bank of random humorous comments and reaction lines), distant honking horns, sirens and natural sounds, the rumble of motorcycles -- all of it executed intelligently to evoke and maximise ambience. Car and footstep sound effects change with ground surface texture.

The voice acting used throughout the story segments effectively conveys the crime-really-*does*-pay story, and the radio stations provide the kitsch but energized soundtrack to go with the action. The game's cast is drawn from the ranks of experienced movie talent: Tommy Vercetti's voice is provided by Ray Liotta (who featured in the major commercial movies *Blow* and *Muppets From Space*), bringing the surly, understated but super-macho character to life. The rest of the voice talent--which includes Gary Busey, Dennis Hopper, David Paymer, Danny Trejo, Luis Guzman, Philip Michael Thomas, and retired adult film actress Jenna Jameson--also have put in highly professional and convincingly stylised performances [1].

The game's action sound effects are powerful and effective. The violent sounds of explosions and gunfire add the necessary impact and build a sense of hyper-realism and urgency.

[1] Jeff Gerstmann http://gamespot.com/gamespot/stories/reviews/0,10867,2895954-4,00.html accessed 13-12-02.

As mentioned, the radio stations in Vice City are very important to the satirical aspect of the game and are well executed, with large production teams devoted to this task alone. The '80s music broadcasting from the send-up stations sets the frantic, rollicking urban tone for the entire game. The development team have acquired copyright clearance on tracks from bands and artists which include Quiet Riot, Hall and Oates, Bryan Adams, Yes, Lionel Richie, Kate Bush, Michael Jackson, David Lee Roth, Roxy Music, Toto, Mr. Mister, The Buggles, Jan Hammer, Herbie Hancock, Judas Priest, and Frankie Goes to Hollywood, plus many more. There are literally hours of music in the game, and once your interest drops in one genre, changing to another is a matter of pushing the *L1* button. Options include a heavy rock station, a pop station, an underground rap/electro station, a quiet love song station, a new wave station, a mariachi-style Latin American station and more. The rock station is one of the most humorous and “in your face”, as the voice-over artist does an effective and facile job of claiming to be a “hard-rockin' rebel” while actually coming across like a “total wimp”.

Vice City also has talk-back radio stations. *K Chat* is a shallow celebrity interview show hosted by a satirised “airhead” female presenter. Here is a transcript of a small section of an interview, which parodies “The Crocodile Hunter” television character of brash Australian Steve Erwin. The male voice-over of *Mr. Zoo* is thus executed in a Monty-Pythonesque, pseudo-Australian accent.

Amy: So, hello everyone and welcome back to K-Chat. Vice City's main place for things. I mean well, it's a place in Vice City where things go on like interviews or things or other things like that. But at the moment it's interviews! And I'm Amy Sheckenhausen, the best interviewer in Vice City and exclusive to K-Chat. Remember you only hear Amy on K-Chat. Our next guest is a man on a mission. And that's why he's got such a silly name. His mission is simple, zoos. His name is Mr Zoo.

Mr Zoo: G'day Aim.

Amy: Hi Mr Zoo.

Mr Zoo: Hi, the name's Pat, Pat Flanerdy. But I love zoos, I really do. That's why they also call me Mr Zoo.

Amy: OK, and which do you prefer?

Mr Zoo: Ah… what darlin’?

Amy: Which name, Mr Zoo, or Pat Flingerthingy?

Mr Zoo: Ah, I don't mind babe. Whatever you fancy. Fine by me as long as we talk about animals. I don't give a damn what you call me. As long as it ain't Sheila or something.

Amy: Ha, ha ha, you're silly Mr Zoo. Why would I call you Sheila?

Mr Zoo: I don't know love you tell me.

Vice City Public Radio features a single, endless public debate show called *Pressing Issues*, hosted by a character named Maurice Chavez, which is broken up by frequent satirical "pledge" breaks. The aim of the show is supposed to be the bringing together of representatives of radical points of view to debate one topic. However, each show's debate inevitably strays from the topic and deteriorates into a very personal slanging-match. Each character portrays a not-too-subtle caricature of a different social stereotype: politicians, preachers, over-bearing mothers etc., who frequently contradict themselves. The satire remains light-weight and obvious, never being allowed to distract from the action.

The commercial radio station's programming is broken up by regular send-up commercials, which are often heavy-handed, though also a little off-beat. What follows is a transcript from one such ad, a superficial and heavy-handed satire promoting the heavy metal band *Love Fist's* concert in Vice City:

Announcer: Lock up your daughters! Shoot your sons!

Boy: Dad! *(Gun shot)*

Announcer: Because Love Fist is coming to town!

(Fist, fist, fist till morning)

The world tour that has been banned through out the world comes to Vice City.

The monsters of rock and roll excess, Love Fist!

(The girls are going crazy cause we are getting lazy shooting love fist fuel all around)

Get ready for a night of death metal love ballads that'll have you shoving your fist in the air! It's the steel hearts, stone cold prostate tour!

Brought to you by *Giggle Cream*, because dessert should be funny! And the *Maibazu Thunder*, because after you get struck by lighting *(Lightning noise)* there's thunder!

> Come see the pounding rock from the band that brought you hits like "*Shin Stainer*", "*Liver Buster*" and "*Dangerous Man Dead Family*".
>
> Come get love fisted! Love Fist at the Vice City arena!
>
> Brought to you by V Rock!

Overall, the talk radio and the various radio station commercials are quite cleverly written, in a light-weight Americanised satirical style, though at times they seem too self-aware.

Of further note, the game defaults to stereo sound, but it also features DTS encoding. The game doesn't place any particular emphasis on the spatiality of the surround sound effects, but the DTS mode sounds cleaner and richer than the default stereo sound.

Cheats

Like most well-developed computer games, *Vice City* has a range of embedded "cheats". These are

> Descriptions of undocumented game capabilities for achieving useful (but uncommon) results by the player[1].

These presumably encourage less skilled or less patient players to remain immersed in the gameplay by allowing them to jump to higher levels of interactivity without committing the time required to acquire advanced skills and complete hierarchical missions, and also add to the sense of the "in the know" gamers belonging to a secret culture. For example, in *Vice City*, a certain combination of buttons will allow cars to become amphibian, and they can be driven out into the bay as though they are motor boats, a surreal sort of diversion in its own right, but allowing for fast navigation to other parts of the islands during timed missions. 'Cheats' allow the user

[1] http://www.thescratchpost.com/resources/games/games_dict_c.shtml ; accessed 3-01-03.

to by-pass or over-turn the given rules of the game universe, and for instance access weapons without having to earn currency, restore full "health" to the action figure, and to change the weather conditions at will. But beyond this, cheats also are a technique for introducing (as a further engagement strategy) absurd fun elements – the action figure can suddenly change into amusingly inappropriate clothing, the traffic flow can be made to go ridiculously faster, and 'cheats' can introduce low gravity conditions or hallucinogenic slow motion, or make the action figure magnetically attractive to female pedestrians (in which case he is soon surrounded by a large group of sexy young female figures whose urgent comments are a source of bizarre amusement, made even more ridiculous when Tommy 's over-eager right fist is triggered by the user). It is also possible to make cars randomly explode and even cause the whole teeming population of *Vice City* to "go crazy" like frenzied ants stirred up in a nest. This all delivered with a tone of irreverent self-mockery and a healthy sense of the absurd, which is one of the more charming and redeeming factors embedded in interactive games culture.

Moral Framework

Grand Theft Auto: Vice City is placed in a corrupt and dangerous fantasy world in which many of the normal consequences of violent street crimes, such as negligent hit-and-run driving, road-rage murder and drive-by shootings, go unpunished. Stealing cars is presented as the basic, unquestioned modus operandi. But in *Vice City,* beyond the level of spontaneous or accidental crimes, *planned* evil is rewarded, and all the tasks to be accomplished in the game play are illegal, immoral, and often theoretically cruel and sadistic. For an outrageous example of the cool and cynical disregard for the value of human life promoted by this game, the *Complete Game Guide to Grand Theft Auto: Vice City*[1], reports in a section entitled "Secret Stash" under the heading *Homeless Roadkill*:

> There is a tunnel near *8-Ball's* hideout. If you drive down it you will find four homeless bums. Drive over them and you will be able to use the bottles of wine they were swigging from as Molotov Cocktails (p.96)

Rather than a moralistic Robin Hood ethos, robbing from the poor is cynically encouraged, even when the user is well on the way to acquiring major assets. To rob one of fifteen small shops, the action figure must enter the store and point a gun at the terror-stricken owner without shooting. The longer the gun is held on the abject victim, the more money is received.

[1] published by the makers of "the Official Playstation 2 Magazine", Derwent Howard Publishing

Serious doubts are raised about the influence the slick and glossy presentation of this sort of moral climate might have on younger or unstable users. However, specialist commentator David McCarthy is reported as arguing that the story does in fact contain a basic moral framework:

> Some British parents are concerned the video game, which should not be sold to under-18s, may be accessed by children. Dave McCarthy, who works for a specialist video games magazine, told the BBC the game was "clearly" not suitable for children. "Players have to undertake various criminal activities, and acts of a morally dubious nature such as stealing cars, engaging the services of prostitutes, beating up people, hijacking police cars," he said.
>
> "It's fairly clear that this sort of thing isn't really suitable for under-18s." However, he feared some parents could nonetheless succumb to pressure and buy the game for their offspring, despite the warnings. "Parents buying games for kids tend to have a relatively lax attitude towards the certification of video games," he said.
>
> "Certainly they don't seem to view video game age certificates with the same degree of respect they would cinema, for example." However, he conceded the game placed any violence and law-breaking within a "basic moral framework".
>
> "Players are penalised for conducting larcenous acts," he said. “If you steal a car, the police will come after you and try and bring you to book." [1]

However without the police adversaries the game would lack a sense of drama, challenge and achievement, and so one could cynically argue that the embedded moral consequences are only a convenience to the gameplay conceit, rather than a serious attempt at setting up some kind of believable moral framework for young players. There are no difficult moral dilemmas to be resolved in the *Vice City* action or interactive plot, there are no “good guys”, and the only rewards for “good” behaviour come from having Tommy violently intervening in gang wars or muggings. If art is indeed a mirror to society, then *Vice City* can perhaps be defended for its biting satirical take on the corrupt underbelly of American urban culture and its lampooning of the clichés and superficial stereotypes of American mass media, without preaching or taking a high moral stance.

Does the introduction of interactivity necessarily imply greater moral jeopardy than the passive immersion in gratuitously violent and graphically pornographic cinema? Studies have yet to establish beyond doubt whether this interactive participation / immersion in a violent, decadent and amoral world encourages or triggers relatively higher levels of “real” amoral or anti-social (or even illegal, and destructive) behaviour in some users. Perhaps, on the contrary, it actually serves as a way of “letting off steam”, that may even channel or counter such tendencies.

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[1] Dave McCarthy http://news.bbc.co.uk/1/hi/technology/2419007.stm ; accessed 17-12-02.

Nevertheless the game encourages a callous and cynical detachment to the meaning of the violent, murderous action embedded within, and abandons any attempt at balancing the evil in the world depicted. There is a sense that the heavy-handed satire is simply an excuse for partaking in this celebration of negativity and amorality. There is a very strong "us vs them" feeling, and minority groups of all kinds are treated in crude stereotypical fashion. The game has been criticized for strong embedded racism, which has caused considerable public outrage:

> …the controversy over the portrayal of the ethnic minority group in *Vice City* has reached a new level, with protests at New York's City Hall, and a statement released yesterday by the Haitian Centers Council and Haitian Americans for Human Rights which condemned "a cultural attack on the millions of Haitians living in the United States."
>
> The Haitian organisations are expected to announce actions to be taken against Rockstar Games in the near future.
>
> Of course, the fact that the group referred to as "the Haitians" in the game is in fact a rival drug trafficking group, rather than being a generic term for an entire ethnic group, doesn't seem to have been considered. Never let it be said that the American conservative media would let the facts get in the way of a good dose of outrage.
>
> After all, that area of the media has been getting good mileage out of *Grand Theft Auto* recently. Only weeks ago, the game was widely condemned after two teenagers who shot and killed a driver on a local highway stated that they had been influenced by scenes in the game. The family of the murdered man is now suing publisher Take 2 Interactive in a $100 million lawsuit. [1]

For many users, *Grand Theft Auto: Vice City* undoubtedly remains in the realms of harmless fantasy play that is comfortably understood for what it is – a sophisticated, hard-hitting, commercially-honed leisure activity. *Vice City* is simple escapism - a challenging, involving and, for many, ultimately an entertaining and sustained distraction, with no pretensions of being morally or spiritually up-lifting. Other users may be less than satisfied by its lack of intellectual engagement, finding that it barely goes beyond a challenge to hand-eye co-ordination. Still others will find it quite simply offensive.

Grand Theft Auto: Vice City is a unique cultural product of a paradoxical and multi-layered society, and is a frightening reflection of this culture which spawned it. It depicts an urbanised world lacking in beauty, moral strength, a world with few redeeming qualities. It is more than a little tragic that such talent, resources and technical ingenuity have been devoted to such a negative concept – and yet this is a company operating in an unforgiving commercial climate, which has developed a highly successful formula and exploited the available technology brilliantly in terms of market penetration.

Vice City is created with input from media producers who are artists, yet is probably not viewed by its creators or consumers as "art". Yet, it is a work of the imagination, makes

[1] http://www.gamesindustry.biz/content_page.php?section_name=pub&aid=2626 : accessed 26/11/2003 .

social comment, and its immersive experience depends on the successful creation of a cohesive world with a strongly delineated narrative. It is interesting to speculate as to why *Vice City* might be judged to be less worthy of the status of a work of art than, say, the popular film Tarrantino's *Pulp Fiction*, which is also a cynical, humorous, action-packed satire and which also simultaneously celebrates and condemns decadent and amoral elements of urban society. Petersen's framework for analyzing art becomes less informative in the face of such ambiguous, post-modern complexity. However *Vice City* is certainly representational art (Type 1) and does "unmask, debunk and critique" in its satirical approach (Type III).

Despite its extremely dubious moral stance, it is a highly complex, technically brilliant, and carefully crafted example of interactive multimedia, which points the way to the possible levels of richness, mult-layered meaning and sophisticated engagement this fledgling art-form may yet reach in the hands of future artistic masters.

Virtual Reality

Virtual reality allows a user to explore a computer-generated world from within that simulated world. Visual, auditory and possibly kinaesthetic stimuli are arranged in such a way as to create in the user the perception of being within the virtual world. Virtual Reality must have three components: it is inclusive, it is interactive, and it happens in real time. [Sherman and Judkins, 1992, 17].

Historical precedents begin with the arcade game *Sensorama,* patented in 1962 by Hollywood based cinematographer, Morton Heilig. In describing Heilig 's ideas Packer and Jordan [2000] touch on the connection between virtual reality technology and the concept of a technological meta-artform:

> In the 1950's it occurred to cinematographer Morton Heilig that all the sensory splendor of life could be simulated with "reality machines." He proposed that an artist's expressive powers would be enhanced by a scientific understanding of the senses and perception. His premise was simple but striking for its time: if an artist controlled the multi-sensory stimulation of the audience, he could provide them with the illusion and sensation of first-person experience, of actually "being there."
>
> Inspired by short-lived curiosities such as Cinerama and 3D movies, it occurred to Heilig that a logical extension of cinema would be to immerse the audience in a fabricated world that engaged all the senses. He believed that by expanding cinema to involve not only sight and sound, but also taste, touch, and smell, the traditional fourth wall of film and theater would dissolve, transporting the audience into a habitable, virtual world. He called this cinema of the future "experience theater," constructing a quirky, nickelodeon-style arcade machine in 1962

> he aptly dubbed Sensorama, that catapulted viewers into multi-sensory excursions through the streets of Brooklyn, as well as other adventures in surrogate travel.

Sensorama was a multi-sensory experience that included mechanized motion, sound, tactile-feedback handlebars, 3-D stereoscopic views, and artificial breezes and wafting aromas such as jasmine and hibiscus.

Rheingold [1991, 49] quotes a portion of Heilig's patent application, which demonstrates that one strong motivator driving Heilig's ideas was his vision that virtual reality had an important future as a teaching device to simulate experience that was either too dangerous or too expensive for a novice to acquire in reality. Virtual Reality has the potential therefore to play two significant roles in society – that of a teaching device simulating experiences for aspiring pilots, astronauts, weapons operators, surgeons etc. clearly with important ramifications in modern warfare and space exploration, and a second direction focusing on its possibilities for providing entertainment.

In Ascott's [1993] opinion:

> The question of consciousness, the technology of consciousness, the transcendence of consciousness will be the themes of 21st century life. Fundamental to this evolution is the development of a telematic art in the cyberspace, and fundamental to that art are the experiments, concepts, dreams and audacity of artists working today with telecommunications systems and services.

Such visions of the future of art focus on experimentations with electronically created four-dimensional simulations of reality, or: virtual reality (time being the fourth dimension). The term 'virtual reality' (VR) was first used to designate specific audiovisual and tactile simulation technology, which enabled the user to experience simulated or artificial (computer-generated), three-dimensional 'realities', with the aid of data gloves, suits and helmets. Why the fascination with a 'virtual reality', when reality itself is so mystifying, complex, immediate and still so unexplored? Fricke, [1994, 227] contends that:

> The potential of virtual reality technology to free us from the constraints of time and space appeals to a human longing for transcendence. We want to experience other circumstances

> without any real threat of danger. We want to be gods, to be able to change shape and form at will. Virtual reality assures us---that we can reach the sun without melting our wings.

A technology-based art-form which plays with virtual realities is another way of describing the meta-art machine, and its potential must surely be exciting.

> The ability to shape temporal experience through the manipulation of a set of simultaneous and successive events is a power which sound producing instruments have afforded the aural composer/performer since pre-history.
>
> The development during the last decade of realtime videographic devices capable of instantaneous generation and manipulation of absolute (or abstract) images has given the visual artist the same power. In this decade, the rapid advancements being made in realtime computer graphics technology promise even more powerful visual instruments, particularly in the totally enveloping new medium of immersive Virtual Reality. [Scroggins, 1996].

It is only possible to touch on this important branch of multimedia experimentation here, but developments in the field of VR are likely to lead the way in the overall evolution of multimedia as technological meta-art.

Conclusion to Part One, *Chapter Two*

The second chapter of Part One presents an analysis of multimedia art-making activities from the point of view of the primary sensory modalities, which are sub-divided into the respective channels of communication. Each of these mini-studies begins with a condensed account of historical developments leading up to contemporary practice, and concludes with an overview of contemporary software approaches to these contributing artforms. Desktop multimedia production systems can be used to make artistically oriented media components, which stimulate the senses and inform through the available channels of communication. In doing so, the multimedia artist inevitably draws on and contributes to the evolving traditions outlined in these concise histories presented in Chapter Two. The premise that these technology systems could be the forerunner of future comprehensive art machines is retained as a unifying organizational concept.

An analysis of the visual modality begins with a discussion of the “image” channel. Hesses’ mystical vision of a future wholistic meta-art machine, which has served as an inspiration for this study, promoted a concept of universality, a coming together of all valuable knowledge to do with the human condition, with the idea of a merging of the spiritual, scientific, cultural and artistic. As an example of the kind of universal principles that future technologically-based meta-art practices might hopefully address or embrace, the concept of “Divine Proportion”, or *phi* is then introduced. Considered by many to be a precise way of establishing the most pleasing proportional relationship in art, using mathematics, it has long been used in art and architecture, across cultures. Included is a brief historical summary of thought and practice related to this concept, and reference to the observation of many occurrences in nature of such specific proportional relationships, which gave rise to the appellation *divine*.

Colour Theory is another area which offers systematized knowledge which is fundamental to artistic practice within the visual modality. Colour theory is presented as an interactive experience within the accompanying interactive CD-ROM, *the Machine.*

The analysis of image as semiotic text is made more conceptually complex by the introduction of coded interactive linking techniques. The use of image as navigational metaphor and nodal icon has produced a new media phenomenon known as *hyperimage* (a subset of hypermedia), blurring the boundaries of what an image can be. 2D, 2.5D (pseudo-3D) and 3D image formats are discussed in this context. Holmes' [2002] attempt at a graphical representation of this broadening of the concept of what can constitute an image, and his inclusive definition of hyperimage is introduced, which takes into account the *texton* condition (representing embedded programming or coded aspects of hyperimage) and the *scripton* condition (depicted as a ring of spheres qualified by graphic, musical, spatial and kinaesthetic attributes, representing: *2D imagery*, *3D* and *pseudo-3D imagery* and *movie* and *sound* possibilities available in contemporary hyperimage).

Within the visual modality, the motion picture 'channel' is then addressed. A dichotomy is established between narrative forms, found in cinema and much television (in which sound supports image in the process of building a coherent story), and a relative late-comer to motion picture arts, the music video, a style-conscious and frequently fragmented form in which image frequently supports the sound modality.

Because of traditional bandwidth limitations, multimedia art commonly uses short, segmented video elements as dynamic highlights, often consisting of non-narrative montages of images and video footage, supporting a voice-over, either as an introductory movie, or as a contribution to the "look and feel" or style of the product. As a short-form, semantically compressed video artefact, music video has been influential on the way multimedia video segments are constructed. An overview of contemporary thought regarding the influence and complex semiotics of music video is subsequently introduced. An analysis of music video is made more difficult by the tension between the genre's dual roles, for music video must work firstly as a promotional vehicle for corporate music industry "products", and yet music videos have a status as works of art in their own right. The music video phenomenon has emerged as a field of vibrant creative experimentation. While narrative does occur in music video and many music videos are at least partially influenced by traditional cinematic techniques, many music videos belong more to a category which could be called animated, or kinetic, graphic arts, in which linear narrative is abandoned for a rapid montage of impressionistic, surreal or even random, and often shocking imagery. Influential technical innovations introduced to contemporary visual language by innovative music video producers include: a shift away from dialogue-based continuity and linearity, virtuoso editing techniques with rapid, rhythmic, stream-of-consciousness cuts; hand-held camera techniques and the introduction of grunge (blur and graininess), as well as other anti-slick, non-mainstream technical approaches, which challenge or reject the sheen of commercialized television, usually as a political or anti-establishment statement. These experimental and anti-establishment techniques, and innovations in the pacing and rhythm of editing have fed back into cinema and television, especially in commercial advertising for youth markets, and have had a significant impact on the conventions and technical vocabulary of motion picture arts in general.

The analysis of music video alludes to the difficulty of neatly categorizing music videos into any one clearly definable category. The analysis covers discernible approaches to constructing music videos, including: nostalgia, romanticism, nihilism, music video as celebrity vehicle. Other motivating themes or issues recognized include: fashion, modes of sexuality and gender identity, the nature of "the gaze", celebration of celebrity, relationship with authority, and other youth culture issues.

The suggestion is then made that the accelerating take-up of DVD technology in consumer electronics markets of affluent cultures may influence or increase the importance of music video in relationship to the music industry, and even evolve towards new hybrid interactive genres of artistic experience. The new technology may prove to offer a vehicle for the realization of music video's early promise as a discrete field of artistic activity. Music video has the potential via DVD to become a consumable art product in its own right, beyond its present status as a unique form of television advertising. The fact that DVD technology is a suitable platform for the introduction of (so-far relatively simple) non-linear interactivity suggests the possibility that music video may even eventually emerge as a dynamic sub-genre of multimedia art, a claim supported in this dissertation by an analysis of a recent interactive, multimedia music video product, *Meet MediaBand*. A view is presented that the merger of music video with DVD technology may herald an interesting new interactive direction for music video, as a fledgling branch of an interactive cinema art.

Narrative development is a common structural element or communicative technique of much media production, and theories relating to narrative in cinema are next discussed, from the point of view of what light they might shed on multimedia art practice.

Another historical art stream that has potential to influence the video component of of the synthesis of arts known as multimedia, is a discrete field of experimentation which emerged after the release of the Sony *Porta-pak* hand-held domestic video recorder in 1965 – *Video Art*. Video art has an unusual history in that it has developed along a distinct non-commercial path, oriented towards highly self-conscious and conceptualised, frequently self-referencing public installations. Experimental Video Art stands outside the mainstream motion-picture tradition in that it does not necessarily depend on the use of actors, frequently contains no dialogue, may have no discernible narrative or plot, or adhere to any of the other familiar cinematic conventions. Cinema's ultimate goal is to entertain, whereas the brief historical survey of the evolution of this form that is undertaken herein reveals that video artist’s intentions are more varied. Some explore the technical and philosophical boundaries of the medium itself, while others rigorously attack or challenge the viewer's expectations of video, as they are shaped by conventional cinema and commercial television. This radically avant-garde field of experimentation has been stamped by an uncompromising anti-commerciality, seriousness of intent and rigorously intellectual, experimental stance, quite at odds with the “instant gratification”, lowest-common-denominator, pop-culture approach broadly adopted by mainstream commercial television interests.

With its confronting techniques of montage and unexpected juxtaposition of communicative imagery, Video Art can inform multi-media by virtue of its sophisticated, abstracted handling of political, humanistic and artistic concepts, and by standing back from and commenting on media experience in a strongly visual form.

In the context of pursuing the construct of multimedia as meta-art, another interesting and significant sub-stream of the motion picture channel designated herein as “experimental/abstract film and kinetic art”, is given an historical analysis and theoretical consideration. Since the advent of frame-based film-making technology, some artists have been drawn to the experimental interest offered by viewing motion picture for its possibilities of bringing temporality to non-representational, formalist visual art. This includes the production of abstract and kinetic films using various techniques of painting and drawing directly on the film cel. Many of these experiments have a strong connection with the previously discussed tradition of technology-based simulations of the synaesthesia experience.

Within the 'motion picture channel' of the visual modality another important contributing media form is known simply as ‘animation’, sub-divided into 2D and 3D forms. The history of 2D animation is outlined, which shows, once again, that technological advances (such as for instance Rotoscoping and Fleisher’s Set-back machine, and Disney’s multiplane camera) combined with commercial pressures to find streamlining economies in this very painstaking and hands-on production process, had a great deal of influence on the way the art form developed. Various stages in the evolution of 2D cartoon cel animation are cdescribed, including the economical “limited animation” style prominent in television cartoon series of the 70-80s. Limited animation was, however, used for consciously minimalist artistic ends to produce a visual style by the UPA cartoonists. Later experiments with compositing live actors with paint-and-ink cartoon characters, led the way forwards (the most notable example being *Who Framed Roger Rabbit?*), and in this context, blue-screening techniques are discussed.

With the dawning of the computer era, new techniques became available that streamlined traditional hand-drawn animation, but more importantly the merging of time-line based interpolation with computer-aided architectural drafting (CAD) software allowed a new form of animation based on 3-dimensional modeling, which is represented by the sophisticated contemporary 3D desk-top animation applications such as *3DS Max*, *Lightwave* and *Maya*. Various technical innovations drove forward the evolution of 3D animation, including inverse kinematics, NURBS and ray-tracing, all enhancing the realism of the new 3D animation form. A very brief history of the incursion of 3D into commercial cinema and television is presented, and then a fairly comprehensive range of contemporary commercial 3D applications ar given brief reviews.

The final section under the heading of *the Visual Modality* is devoted to 'the Text Channel'. Software that assists in narrative and character development for creative scriptwriting is discussed. The latter primarily works by facilitating the organisation of ideas, checklisting the “through-lines” implied by script and character decisions, also aiding in the cross-referencing and linking of ideas. This section concludes with a discussion of “text in motion” or kinetic typography, wherein text is considered as a design element in multimedia productions, beginning with the pioneering work of Saul Bass.

Having undertaken an overview of the contributing media forms within the visual modality, it was appropriate to attempt a similarly broad analysis of contributing digital arts within the aural modality. This discussion is opened by addressing the “music” channel, beginning with an account of the 80 year history of electronic music, a field strongly influenced by technological change which has flowered during the digital revolution. For the purposes of this discussion, electronic musical instruments are defined as:

Instruments that synthesize sounds from an electronic source.

Many of the concepts fundamental to an understanding of contemporary digital audio practice within multimedia art are historically indebted to earlier analog electronic music technologies and are best understood within this context.

Beginning with Helmholtz's resonator, the condensed historical overview presented in this section outlines early electronic instruments such as *the Theremin*, which were based on the *heterodyning* principle (created by two high frequency radio sound waves of similar but varying frequency combining to create a lower, audible frequency, equal to the difference between the two radio frequencies, and approximating the range of human hearing, 20 Hz to 20,000 Hz).

An account is presented of how the musical possibilities inherent in magnetic tape-recording technology were explored by exponents of the *musique concrete* style of electronic music, another field of sonic experimentation that made a strong contribution to the evolution of contemporary digital music practice. *Musique Concrète* was recorded directly to tape with real (concrete) sounds, retaining the living characteristics of the original sound. Sounds could be recorded, then played backwards or at fast or slow speeds (i.e. have their pitch and tempo raised or lowered), processed through a variety of methods, spliced, looped, and ultimately an experimental musical piece could be structured which could not be produced in the 'real' world. It concentrated on 'found sounds' or natural recordings rather than electronically produced sounds.

The two streams of electronic music techniques – Musique Concrète and sound synthesis, eventually gained mainstream acceptance through their introduction to popular music during the 60s. The Beatles, driven by the creative mind of John Lennon, were especially influential, and their music albums, from *Revolver* (1966) on, led the way in the exploration of Musique Concrète techniques within popular song formats, ensuring that the new tape-based experiments in sound were introduced to mass audiences. Musique Concrète was given an interesting commercialized technological expression through the development of the tape-based keyboard known as *the Mellotron,* which had a brief period of ascendence in popular music during this period.

Meanwhile the sound synthesis branch of electronic music went through a significant parallel technological revolution, fueled by popular music artists' interest in introducing innovative sound to commercial music and to film soundtracks.

Technologies previously confined to large physics laboratories in academic institutions were made available in relatively portable and affordable mass-produced forms. Robert Moog and Donald Buchla pioneered these markets, firstly with large modular systems, but later with cheaper, less cumbersome pre-wired anologue synthesizers (most notably the *Minimoog).* Larger companies such as *Arp* and *Roland* developed Moog's protoypes into more sophisticated and cost effective instruments and soon dominated the market.

The advent of the digital revolution saw commercial analogue synthesis technology being gradually displaced by a new wave of digital synthesis modules and keyboards. This brief history of the commercial evolution of creative sound synthesis technology is pertinant to the discussion of 'the music channel' of interactive multimedia meta-art, in as much as the software-based *virtual modular synthesizer*, a direct descendant of commercial sound synthesis harware, has emerged in recent years as an important creative audio tool among the arsenal of creative processes available to the computer-based multimedia artist.

Other areas where computers can facilitate the artistic manipulation of sound, include:

1) The capture or sampling of digital representations of real world sound (digital sound recording);

2) The processing and manipulation of the resulting digital waveforms, in order to in some way enhance or filter the aural experience.

Musical Instrument Digital Interface or MIDI technology is another important creative field within the digital audio production area of multimedia design and development. Physical modeling is a related technology, by which accurate digital models of the acoustic qualities of an instrumental sound during performance are developed, in an attempt to get around the problems of relative inflexibility inherent in MIDI keyboard - triggered digital audio sampling technology, when this is used as a means of producing music compositions. MIDI composition using Physical Modelling begins, not with an acoustic waveform, but with an analysis of the physical dynamics of the sound-producing mechanism itself. Jaffe and Smith’s (1995) exploration of physical modeling is discussed in the light of how such technologies embody the potential to extend the multimedia art machine into new realms of expression.

A new cultural manifestation – the body of music which has directly evolved out of the new digital music technologies, known as techno-music, is then considered, beginning as usual with a condensed historical account of its emergence through the creative subversion of mass-produced music-specific technologies. Beginning with the *disco* music of the early 1970s, and taking in the influence of the industrial synthesizer music of German avant-garde electro-pop bands, most notably Kraftwerk, this form of contemporary popular music is discussed as a form of experimentation in which repetition, tonal graduation, ambiguous structure and the sound of the machine were all crucial, if initially accidental, elements that helped form the basis of a new techno aesthetic.

Following the format that has heretofore been established, the discussion of the techno-music phenomenon led on to a consideration of representative, commercially available desk-top software that currently facilitates the creation of this form of music.

An area of music composition in which digital technology plays a significant and interesting creative role is that of algorithmic music composition, or automated composition. An extended discussion of issues surrounding this form of creative aural activity is presented. Algorithmic music composition is a process of applying rigid, well-defined rules to the process of composing music. An historical overview reveals that as far back as the Renaissance, composers frequently followed stringent compositional rules, which could be considered algorithmic. It is a conceptual approach which has fascinated composers down through the centuries, from Mozart's musical dice game, to 20th century masters such as Messiaen, Stockhausen and John Cage, who developed non-traditional, rule-based strategies. As a strictly rule-based technique of composition, serial composition especially lends itself to the digitization / automation of its processes.

It is argued that consideration of the concept of computer-aided music composition turns on a central question:

Can an analysis of the creative process of composing music generate, or be mapped upon, mathematical formulas that when applied, in turn, may produce aesthetically pleasing results?

This problem, it is argued, should be broken down into two parts:

Practical Issues: How to understand the compositional process well enough to express it as an algorithm; and

Aesthetic Issues: How to program a computer to choose between some concept of "good" and "bad" music.

Of primary interest to musical artists in general, and to this inquiry into the processes and aesthetic possibilities of digital multimedia art, is the question:

> Does algorithmic composition lead to the creation of engaging musical material that would otherwise not have been possible or likely? i.e. Does it generate new ideas?

Five forms of computer-assisted algorithmic music composition are subsequently identified and defined:

- **a)** Stochastic algorithms
- **b)** Iterative algorithms
- **c)** Serial algorithms
- **d)** Genetic algorithms
- **e)** Fractal algorithms

The discussion of algorithmic composition techniques concludes with a review of a representative sample of commercially available and shareware algorithmic music applications.

Consideration of creative activities within 'the music channel' of the auditory modality inmultimedia development is then moved foward with a discussion of the role of music in cinema as a major contributing historical tributary to multimedia audio art practice. To remain consistent, a brief historical overview of the evolution of audio art and sound editing in cinema is next developed, though it is acknowledged justice could not be done to this rich and complex field within such a broad study.

The evolution of genre-specific musical codes or meaningful conventions in film music is then considered. Examples are offered, following Spande [1996] of the evolution of genre-specific audio signifiers, which have become established as a kind of short-hand language, and form an important part of the tacit agreement between film makers and their audiences with regard to established expectations or ground-rules within specific film genres.

The discussion then proceeds to a consideration of the specific problems inherent in preparing audio and music within the kinds of non-linear interactive structure prevalent in multimedia art. From the earliest days of coin-operated arcade games, sound design has been an integrated element, in order to enhance the immersive experience. The audio in early arcade games needed to address a difficult acoustic environment. Ganner's [1999] argues that the outcome of introducing an interactive element is that the audio experience will be of indeterminate shape, making it difficult to compose long passages, forcing the composer to work in small modular units of sound.

The dissertation then focusses on 'the Sound Channel' of the Auditory Modality, in which recordings of natural sounds are used as recognizable signifiers within the audio design, with the aim of building an immersive virtual experience and aiding in meaningful communication. This channel offers the opportunity of introducing spatiality into the artificial perceptual experience. There is a long history of experimentation with spatiality in music (many leading twentieth century composers have used spatiality as a major ingredient in their composition and theoretical writing, for example Stockhausen, Cage, Xenakis and Boulez) and the advent of digital recording and signal processing technology has also seen a strong emphasis being placed in audio design on acoustic space simulation (digital reverberation, digital delay technology and other simulated spatial systems).

An analytical approach to multimedia art must embrace the function, meaning and artistic potential of the interactive element, for interactivity sets multimedia apart from other mixed media forms. The introduction of branching structures into narrative opens up new options and artistic paradigms by allowing the artist to diverge from traditional linear mind-sets. A distinction is made between *interpretation* and *interaction.* Interactivity requires the ability to intervene in a meaningful way within the representation itself, rather than to simply "read" it differently.

Reference is then made to classical psychological theories of human motivation as a useful starting point for understanding the way users react to branching structures in interactive media. The most significant of these is Abraham Maslow's *hierarchy of need.* This theory postulates that human motivation and decision-making is driven initially by the fundamental and strongest physiological level of need for food, water and shelter. Once this level of need is at least partially satisfied, a hierarchy of other higher-level needs come into play, beginning with a need for safety, stability and freedom from fear, and progressing to a need for belonging to a group and for love, followed by a need for self-esteem and recognition/respect, until finally the highest level of need, the need for self-actualization, creativity and fullfilment, begins to become a motivating factor. Each level of the hierarchy must be partially satisfied in order for the next to take priority.

Interactive ‘performances that engage’ can be viewed as responding to a similar or analogous hierarchy of necessity, or need intensity. Mueller [1996] relates Maslow’s hierarchy of need to the various forms of interactive computer gameplay, and, for example, argues that so-called “Twitch” or reaction-based shooting games utilize the fundamental imperative for mental, physical, emotional survival, while role-playing games such as *Ultima* are offered as an example of interactive gameplay utilizing the next level of need, for belonging and community.

Some of the possible technical, artistic and structural advantages offered by the utilization of branching structures are about working the little picture (communication) and big picture (meta-communication) simultaneously. Successful game design will utilize the immersive possibilities made available by requiring the user to employ strategic thinking, including both short-term decisions or tactics (e.g. which gun to use) and long-term strategy or game-plans (e.g. which path to take for victory).

By incorporating interactivity into a work of multimedia art, the artist must develop a non-linear, hierarchical structure for the piece. Hofstetter [2000] identifies five interactive design structures, in increasing order of complexity. These are :

- Linear List,
- Menu (simple branching arrangement),
- Hierarchy (more complex branching structures with menus of sub-menus),
- Network (multiple linked items – the richest form of navigational design), and
- Hybrid (which may contain elements of the preceding four generic design models).

Hofstetter’s analysis of interactive structures are shown to have some congruence with the ideas of Phelps [1998], who identifies seven interactive story “shapes”, again in an ascending order of complexity.

1. Simple linear structure – none of the advantages of non-linearity, but can be enhanced by sound and animation

2. Interactive narrative – again linear, but each screen offers options for an enriched experience
3. Multi-linear narrative – several parallel stories are presented without those stories directly interconnecting
4. Braided multi-linear structure - an initial situation is given which then branches out in a number of thematically related plot directions, which subsequently converge upon another situation, that then again offers a number of directions to choose from; this form builds dramatic tension, gives the user a feeling of free movement, and yet provides a seamless story experience.
5. Tree-branching – the classic interactive structure: each scene offers several choices, which in turn lead to further choices – leading to exponentially accumulating outcomes. Traditionally, to avoid infinite choices being set up, one "correct" story path exists and all others lead to death, thus pruning their potential early into the decision making process.
6. Non-Linear Story (consistent with Hofstetter’s network model) is the richest form of interactive narrative but is problematic because of its non-chronological nature, providing limited options for character and plot development.
7. Nested Funnel Story – a linear progression but dependent on user input for timing and duration (e.g. *Myst*).

Another way to view "interaction" is as a chance for collaborative authorship, in the sense that readers are invited to add the next fragment or branch of an unfolding narrative. Experiments in this form of interactive authorship have yet to produce literary output that can rival highly skilled literary craft however.

A tension is set up by the introduction of interactivity into narrative because of a possible conflict between the simple objective of user immersion in ‘a good story’ and another kind of immersion encouraged by giving the user the power of interactive control. Genette [1980, p 228] defines any shift between the diagesis of the narrative and the extradiagetic as a *metalepsis,* and further defines as *transgressive metalepsis* any shift between these two levels of narrative (i.e. simple immersion in the story and the act of technical interaction with its narrative development) when it confronts the suspension of disbelief, or draws attention to structure or technique.

Having established some general principles regarding interactivity in art, a brief history of computer games between the early ‘70s (beginning with the release of

Busnell and Alcorn's *Pong)*, through to the maturation period of the mid-90s is then offered, which reveals that it is progress in multimedia technologies which has essentially driven advances in games design allowing ever increasing levels of media richness, virtual realism and immersive user interactivity. Most important in the exponential growth in popularity of this new entertainment form was the introduction and evolving sophistication of animated 3D graphics, allowing players to interact with the virtual worlds created by the programmers with a freedom combined with a visual richness and depth previously unattainable. A major milestone in this development was the adoption of the 16-bit standard for personal computer technology in the 1980s. Commercial pressure to compete in this lucrative market drove the development of Microsoft Corporation's *DirectX* technology which now dominates the software-based sector of the industry.

Engagement strategies which can be manipulated by the game designer to enhance immersion in the game were then considered, concluding that engagement is increased by employing a richness of virtual environment as a background to the construction of role playing scenarios. Interactive engagement is enhanced by intellectual challenges, action tasks, goals, rewards and the manipulation of feelings. Clear rules, cause and effect relationships, and a reward/penalty structure are important game-play conditions. Openings to new levels of difficulty; progressively challenging the user in more difficult challenges as skill levels improve, provides for on-going engagement, countering the tapering-off of interest.

Genres of interactive electronic games are then considered. Nine distinct categories are identified:

1. First-person reaction-based shooting games (in the tradition brought to maturity with id Software's *DOOM*)
2. Sports Games
3. Fighting Games (two-player, side view, martial-arts style games - most successful on dedicated consoles such as the Sony *Playstation*)
4. Remakes of 'old school' arcade games
5. Mechanical Simulations e.g flight simulators
6. Ecosystem simulations
7. Strategy/War Games
8. Interactive Stories
9. Puzzle & Board Games

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Volume 2

A Folio of Original Multimedia Art:

(includes supporting documentation/exegesis).

<u>Part 1 :</u>

"Take Your Space Now"

– a healing journey into the healing energies of colour and sound.

(a four-part video animation, based on the paintings of Australian artist Duane Radford, presented on DVD).

<u>Part 2 :</u>

"The Machine"

- an interactive art experience, presented on CD-ROM.

Barry Ferrier, B.A. (Syd.), B.Litt. (Hons.), Dip. Music (Comp.)

This dissertation and folio of original multimedia art (in two volumes)
presented to the School of Contemporary Arts,
Southern Cross University, Lismore, N.S.W.,
in fulfillment of the requirements
for the degree of Doctor of Philosophy.

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The DVD and CD-ROM comprising the creative folio are included in Vol. 2.

2.1 Take Your Space Now – *A Journey through the Healing Energies of Colour and Sound*

Background to the creation of this 4-part experimental video animation:

In late 1996, I was commissioned by the Australian painter known as “Duane” (Duane Radford) to create an animated video, based on a series of approximately 60 original oil paintings. A self-taught artist, Duane describes producing these works in a trance-like state of deep meditation – they are “channeled” in the manner of automatic writing. Duane is a spiritual and mystically-oriented person, and a believer in what has become known as “New Age” beliefs. She has her own particular slant to these beliefs however, and follows a mission to be a positive healing force, in a society she views as psychologically troubled and out of balance. As a healer, she has produced a range of organic herbal healing products (on a farm near Maryborough, Queensland) which, with her husband David, she successfully markets nation-wide. She also runs a gallery centred around her visual art, and markets large, high quality prints of these works.

The paintings themselves are challenging works, with a strong, individualistic sense of colour, using a brush technique quite often reminiscent of the old masters, for instance in the handling of such elements as clouds.

Mercurial Landscape (Part 1)

In most cases they are representational, in that they do depict recognizable, though visionary, environments or spaces. This collection of paintings is a unique body of work at once futuristic, yet with a classical, painterly colouring that is often quite somber. Duane’s paintings have a healing agenda – she believes they can trigger emotional responses in the viewer and produce a kind of catharsis. Duane believes that each of the paintings will cause different reactions among viewers of the paintings depending upon their individual emotional needs and personal journey.

Duane explained by phone during initial contact that, during meditation, she had received what she described as a vision that she should pursue the idea of making a new product out of these paintings. The piece, to be created through remediation of her art into a cinematic form, was conceived by the artist as offering the viewer a healing experience, based on Duane's ideas about the healing properties of colour vibrations and sound harmonics. The commission required using the paintings as the source material, but encompassed a transformation of these images by new media technology, in order to bring the dimensions of time, movement and poetic sound to the works. The resulting work of video art would become *a journey into the healing energies of colour and sound* (Duane's sub-title to the final piece – below is a screenshot from the DVD introduction).

By a series of chance encounters (brought about by her quest to find the right collaborators for this innovative task), she eventually came into contact with the present writer. Duane is a determinedly intuitive person, with a strong belief in the concept of synchronicity, and it was on such a basis that she offered this commission. Her patience and support in allowing this project to reach a resolution through long periods of experimentation, delays brought on by ill-health, technical breakdowns and the contingencies of a busy working life are here acknowledged gratefully. It was a thoroughly respectful and trusting, though active collaboration.

The project interested me for a number of reasons. Since 1985, when I was commissioned by the Queensland Perfroming Arts Trust to create the *Dreams and Machines* production at the Q.P.A.C. Concert Hall, I have had an interest in ideas surrounding the man-machine art process - when we attempt to express the art impulse through the mediation of technology. *Dreams and Machines* attempted to highlight and celebrate the possibilities opened up by a relationship between the performing arts and technology – in this case Fairlight Computer Music Instrument Fairlight Computer Video Instrument and prototype Fairlight Voice-tracker, state-of-the-art music and video effects technology. In this project were the seeds of my on-going research interest in ideas arising from a collaboration between a natural-media artist and a digital artist. I was also very interested in exploring techniques and concepts available through contemporary digital animation techniques, both 2D and 3D which were only just emerging. The collection of images being offered as source material were striking.

Duane's paintings are mystical, have a dreamy, non-rational quality, and are painted with a confident use of intense but earthy colours and colour relationships. Their subject matter transcends social issues, politics and the mundanity of contemporary life and exists in a visionary, otherworldly space. They are representational, but what they represent is a physicalisation of imaginative, transcendental visions of aspirational spiritual states and yearnings.

Relating Duane's work to the concepts of art presented in *Part One* of this dissertation, there are strong connections with the Jungian ideas (see p. 42) embodied in the quoted passage from Aniela Jaffe's writing :

> …when conscious or rational knowledge has reached its limits and mystery sets in, …(some artists) tends to fill the inexplicable and mysterious with the contents of (the) unconscious.

Clearly, they are conceived within a "New Age" metanarrative[1]. The New Age is a free-flowing spiritual movement; a network of believers and practitioners who share somewhat similar beliefs and practices. Unlike most formal religions, it has no holy text, central organization, membership, formal clergy, geographic centre, dogma, creed, etc., but its teachings are a loosely knit spectrum of spiritual and occult ideas derived from a variety of esoteric mystical traditions[2]. The New Age movement emerged as a recognizable popular movement during the late 1960s - early 70s,

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[1] See p. 48 for a discussion of meta-narratives.

2 *The roots of the NEW AGE* (uncredited, 1996); http://www.xs4all.nl/~wichm/newage3.html ; accessed 8-03-04 partly as a reaction against what some perceived as the failure of both Christianity (or other mainstream religions) and Secular Humanism to provide spiritual and ethical guidance for the future, in a world racked by personal alienation, violent conflict and social injustice. The New Age movement owes a significant debt to the Theosophical tradition, which was forged in the late 19th century by Madame H.P. Blavatsky out of the European esoteric tradition, incorporating Spiritualism and the Oriental Renaissance into one coherent system that itself took on aspects of a religion, promoting ethical notions of responsibility for personal action through *reincarnation* and *karma*. The wholistic New Age world-view and philosophy are traceable to many other esoteric sources and mystical traditions: Astrology, Channeling, Gnostic traditions, Wicca and other Neo-pagan traditions, as well as eastern mysticism including Hinduism, Taosim, etc.[1] It has remained a discernible force in western society, especially the U.S.:

> The New Age has not traveled to the end of the road yet. It is reaching a stage of maturity in which wheat is being separated from the corn. Yet, it still comprises a broad spectrum of activities, from the commercial rip-off, to unselfish dedication to serve mankind spiritually.

The New Age/healing agenda at the core Duane's work is summed up in her title, and in the one inflexible demand she placed upon the work, that I include a "statement from the artist" at the outset, which she had received in a state of meditation, regarding the carthartic nature of the experience of viewing the paintings (and therefore, logically, the video piece). This "mission" statement begins with:

> These paintings hold the key to unlock the doors held deep within you, and create an emotional response to release the experience and move on with your journey, at a pace that is both comfortable and right for you.
>
> This video should be viewed again and again for your ever-changing personal needs.

I had a great deal of difficulty satisfying this request in a manner with which I was comfortable, and must admit to never feeling entirely at one with the idea, believing that it may in fact prejudice non-"new age" viewers against the experience. I made several attempts at building a sequence which included these words against a backdrop of the paintings, believing it was a good opportunity to show the paintings "whole". (Up until the change of format to DVD, which allowed the addition of the slideshows, the opportunity to see the each painting "complete" was not available, due to Duane's requirement to stay "inside" the paintings, discussed in more depth below).

[1] *New Age Spirituality*. http://www.religioustolerance.org/newage.htm accessed 1-02-03

There is a resonance with Duane's work to Peat's concept [2002] of the *alchemy of creativity*. A quote from Peat has been offered (see p.47), and it is useful to re-iterate a point made in this quote, which offers a valuable way of viewing Duane's paintings:

> Art, in its widest sense, is a form of play that lies at the origin of all making, of language, and of the mind's awareness of its place within the world. Art, in all its forms, makes manifest the spiritual dimension of the cosmos and expresses our relationship to the natural world.

To analyse Duane's body of paintings in terms of Pietersen's four meta-paradigms of art [1] reveals the difficulty of neatly pidgeon-holing artworks within rigid theoretical structures. Nevertheless Pietersen's terminology is appropriate. Duane's paintings:

- have qualities consistent with the Type 1 visionary/metaphysical mode (mystical and incorporating encompassing concepts),
- but also have a strong presence within the Type III experiential/interpretive mode (they are personal, expressive, revelatory and poetic).
- Duane's healing mission also allows for a connection to be made with Type IV art – the missionary/developmental mode (concerned with ideological truths, humanistic and reformist values; they are transcendant and the artist seeks to influence life/world/society according to valued ideals and principles).

Duane's works are a celebration of painterly colour and harmony, and constitute a highly individualistic artist's unique vision and statement. Thus a work of animated video art created entirely out of this resource will always be a unique work also. The digitized images that became the raw material of the work are atypical of what might be called "the neon look" of computer-generated techno-music animations, which commonly feature the output of CG animation software applications. Duane's paintings have nothing in common with the flat hard-edged style of typical cel-frame animation in the Disney tradition. They don't have the cold, dark, metallic sheen of sci-fi films, and they are closer to painterly art of past centuries than visually linking to the look and feel of music video. I took the view that using these works as source material for a work of video art or kinetic animation offered a means of "grounding" or authenticating this work, by offering a basis in natural, painterly colour, and by virtue of the single-mindedness and integrity of the vision which inspired and produced them.

[1] See p. 49.

What gives rigour to practical experimentation or research in the field of computer animation? I sought an idea-driven approach and a framework for some meaningful research. In terms of Neitzsche's concept of a Appollonian/Dionysian polarity in art (see p. 66), there is a "path of least resistance" due to the logical, hierarchical nature of digital media processes, which can easily lead the artist towards predominantly Appollonian classical, ordered structures and approaches to digital art creation. Does the software control the artist or the artist the software? There is a possible danger lurking for the unwary, that an undisciplined attitude could lead to the production of homogenized art products, heavily dependent on software engineering techniques. A digital artist must be conscious of the possible criticism that the true producers of computer art are the software engineers. To illustrate: experienced 3D artists, for example, can frequently discern from the look of desk-top computer animations, exactly which software application has produced a product – the strengths and weaknesses of the software itself produces a discernible visual style. As a counterbalance to this problem, the idea of using Duane's painterly, Dionysian images as source material suggested the possibility of producing some interesting "non-computerish" results. In other words, the series of paintings are unique both as individual works and as a body of work, and animating them inevitably would lead to the production of a unique work of experimental digital art.

While I could immediately recognise ways in which I could animate some of the individual paintings which appealed to me, what I did not immediately understand was the challenge I would face in developing a structure to what became a very large-scale work. There was a gradual and ongoing process of developing a relationship with each of the paintings. The relationship began with the production of a short test animation as an attempt to indicate to Duane the range of possibilities. At this stage I kept most of the animations simple and conservative, with standard "pans and zooms". I did include a section of more unorthodox experimental nature, which (happily) greatly appealed to my prospective employer as being more imaginative and in keeping with her futuristic vision for the piece. (Use the scene selection menu to access Part Two: *Gamma Rays*. This section is the very first sequence I produced for Duane, at the outset of the project. It uses a stock transition available in Adobe *Premiere*, to emphasise the diagonals which dominate the painting).

The animations were all produced using digital techniques available through relatively cheap commercial animation and video production software. Short rendered segments based on manipulating a digital scan of an individual painting were woven together into sequences, and music was later composed, to integrate and glue the sections together, and add ambience and drama. The digital animations were output to mini-DV tape format for storage. Most of the animated segments were also archived on CD. The final product has evolved into a DVD, containing four sections of animation and music. These range in duration from 17 minutes to approximately 23 minutes. It was also planned to include an inserted booklet explaining the project and showing small prints of the paintings. The sheer body of media elements, experiments, versions, re-edits etc. made the project very demanding in terms of physical storage /archiving, organization, and especially continuity.

2. Development Phase

The images were supplied in the form of high quality photographic prints, no larger and mostly smaller than A4 size. The photographs were originally scanned with a drum scanner at 300 dpi and the digitized images stored on a CD. Later scans were made with a domestic scanner at 600dpi resolution.

One significant challenge lay in the fact that the paintings were conceived and executed as discrete works of art. While unified by the individualistic approach and technique of the artist, they were not consciously planned as a series. An essential imperative was therefore to develop a coherent artistic agenda to underpin or give structure to this weaving together of a unified linear, time-based work.

My analysis led me to conclude that Duane’s paintings fall into five broad groupings of style or content:

1. Images of space, galaxies, nebulae
2. Images of underground caverns of volcanic activity
3. Mystical landscapes
4. Underwater worlds
5. Abstract expressionist/non-representational images exploring colour interplay

Category 1: Images of space, galaxies, nebulae. Some examples:

Rose Nebulae (Part 1) *Triffid Nebulae (Part 1)*

Category 2: Images of underground caverns of volcanic activity:

Vulcan's Workshop (First Image of Part 2). *Hall of the Mountain King (Part 2)*

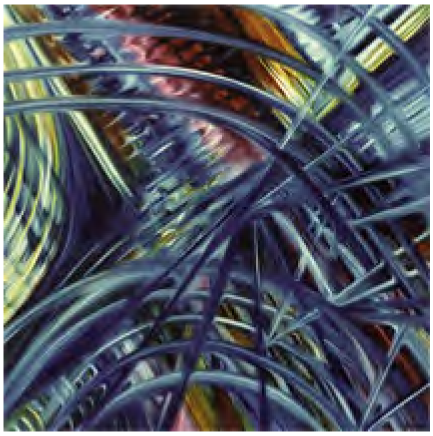
Category 3: Mystical Landscapes:

Forgotten City (First Image of Part 1) *Dominion of Venus (Part 1)*

Category 4: Underwater worlds:

Neptune's Retreat (First Image of Part 4) *Above Atlantis (Part 4)*

Category 5: Abstract or expressionist/non-representational images exploring colour interplay:



Rainbow Fantasy (Part 1) *Nerve Centre (Part 3)*

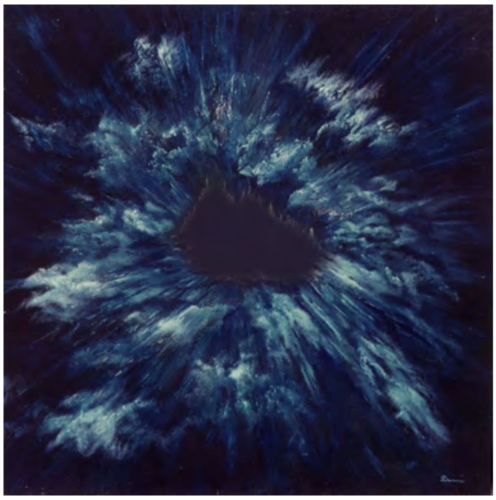
3. Image Manipulation Software Used in the Project

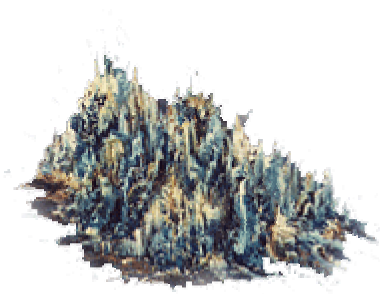
Most of the image preparation was carried out using *Adobe Photoshop 5.0,* though version 5.5 , 6.0 and 7.0 were employed towards the finalisation stage, as they became available at the various teaching institutions where I have worked over the long production period.

Basic processes included:

- Varying resolution (resampling)
- Cropping to change aspect ratio
- Some digital signal processing – mainly altering histogram "levels" to increase contrast, to better convey the richness of colour on a television screen.

In some cases areas of the image were separated from the original image for animation purposes.



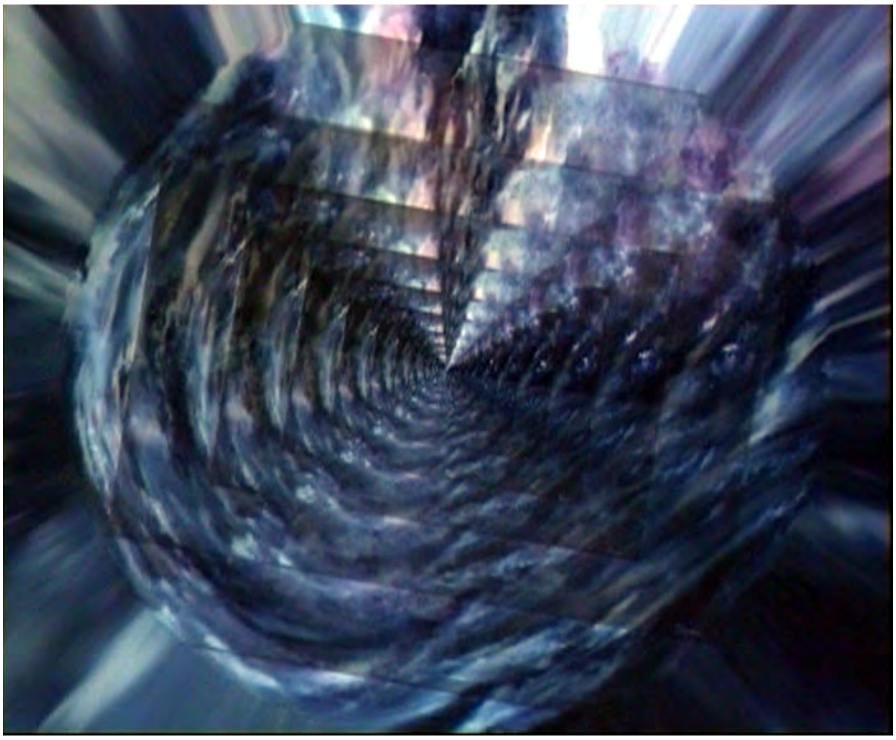


In the case of the painting *Halleys Comet* (Part 3), Duane had already depicted a moving comet in the painting – it was necessary to remove the comet in order to animate it. It took many attempts to produce a comet that was believable, yet retained the mystic quality of the original image. Duane rejected several of my early attempts.

Halley's comet (Part 3).

The original painted comet, which was removed.

Some effects were also created using digital image manipulation processes available in *Photoshop.* Sequences of images were created using the "actions" process. This stores a series of parameters, keystrokes and mouse actions for reuse on a single mouseclick, which allows the user to "batch render" image sequences, with gradual progressive changes, for later use in animations:



Forgotten City sequence; Part 1.

Video:

- Adobe *After Effects 4.1 (*through to *5.5*): spatial transformations & special effects processes. Much use was made of the MetaCreations *Final Effects* plug-in package which allowed the simple creation of starfields, image-mapped 3D spheres and rudimentary partical systems

- MetaCreations *Goo* : time-based bitmap distortion. This cheap but ingenious software was aimed at a non-professional market, but allowed for very creative experimentation in keyframed distortion and morphing of bitmaps using multiple variable parameters to create unique time-based effects.

- *Various 3-D animation applications:* Creating 3D terrain models from the greyscale values of the paintings, remapping the paintings onto the terrains and animating camera "fly-throughs".

- Adobe *Premiere 4.2 (*through to much later *7.0)*: editing animations together into sequences, experimental transitions especially using gradient maps; other sfx

4. Early Problems

The paintings did not conform to television aspect ratio. The following image shows television aspect ratio measured in pixels, at a resolution of 72 dpi:

The majority of the paintings were not even close to this aspect ratio:

Hall of the Mountain King (Part 2)

Rainbow Fantasy (Part 1)

In creating the first attempts at basic animation, the approach taken was to begin or end on a view of the whole painting (which seemed desirable). This approach produced a documentary-style presentation, which reduced immersion in the experience of viewing the images, by offering both an imaginative and an analytical "space" simultaneously, producing a state related to that which Genette [1980, 228] calls *transgressive metalepsis* (see p. 371). The client instinctively withdrew from this approach, consistently affirming that she did not want a documentary approach, but sought a more immersive, less analytical experience.

After viewing test animations which showed full views of the painted images, with black borders filling out the screen, the client's instructions were that animations should never show beyond the "edges" of the images, but rather explore the paintings from within the picture plane. Duane wanted the video to be a journey through the imaginary pictorial space, and she chose to keep the viewer "within" this imaginary world of her painter's imagination. To paraphrase her rationale for making a video of her paintings which rarely showed the whole painting:

When you explore a garden you cannot see the whole design, but you gradually build a mental picture as you wander through and discover each part.

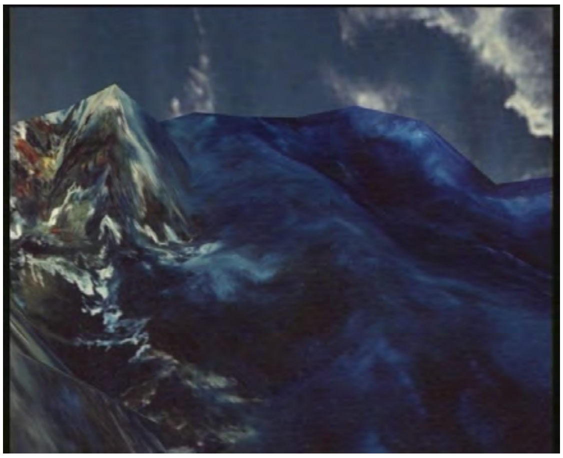
5. Formal Structure

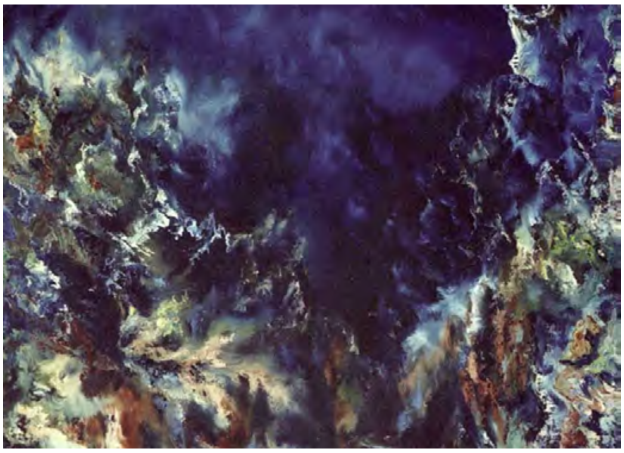
The most difficult challenge was finding a way to develop meaningful sequences of images. Introducing the time element implies narrative. As time passes, we tend to expect a story, a beginning, middle and end, a climax etc, especially when using representational imagery. However, there was no innate sequential structure in the collection of paintings as a whole. The paintings had in fact been presented as a series of four sequences in a photographic collection (which was eventually used as the basis of the slideshows of the paintings found on the DVDs) however these had been prepared in collaboration with a third party and Duane instructed me to avoid these later categorizations.

I have long had an interest in the phenomenon of synaesthesia, having been exposed to academic studies related to this mysterious human quirk, while pursuing my undergraduate degree in psychology. I claim no synaesthetic ability, and so I undertook extensive research into synaesthesia (which appears in *Part One* of this dissertation) in the hope that it might reveal or establish commonly held musical key-colour relationships among leading exponents of this concept or "skill". I had planned to use this as a structural element in the composition of the score, as the paintings each have a dominant colour and could be classified into groupings based on this colour dominance, allowing a relationship to be established between the resulting clusters of colour-dominant paintings and specific musical keys. However my research revealed no clearly established or universal correspondences being reported between colours and musical keys, and I eventually abandoned this idea altogether as being an arbitrary and extraneous imposition (see the section on synaesthesia, beginning on page 107, and specifically a comparison between the musical sound-colour correspondences used by J.B. Castel, Rimsky-Korsakoff and Scriabin on which appears on page 119).

Finally, a simple narrative device helped to get the project underway. Taking the cue from the number of paintings which depicted deep space, nebulae, etc. the video gradually began to be conceived as a journey through deep space, visiting "worlds of colour". These "worlds" were created by building animated planets, using the technique of wrapping a representative painting around a 3D wire frame sphere. By zooming and panning against an animated starfield background, the illusion of flying towards these "painted" planets could quite simply be created.

The virtual camera eventually zooms down towards the planet and cross-fades with the 2D image-scan of the painting (or, as in the next image, a 3D terrain model derived from a painting and with the painting remapped onto a 3D terrain model as a surface texture [1]), suggesting that the viewer has descended onto the landscape of the imaginary planet.





From the *Mystery of the Deep* sequence; Part 3

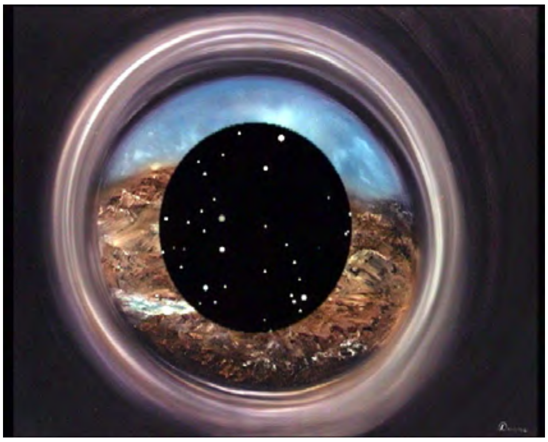
The camera then explores the painting as though it is a landscape.

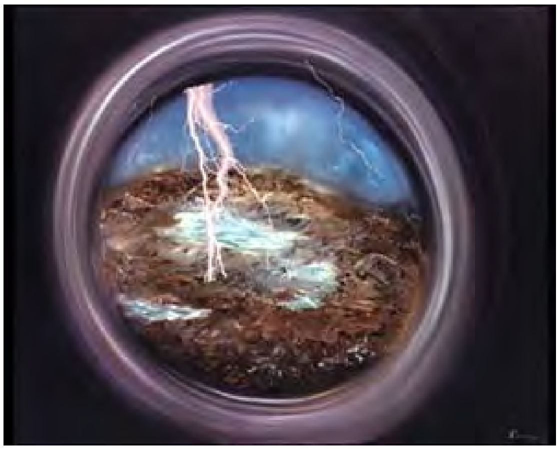
In the example shown above, the painting, entitled *Mystery of the Deep* (Part Three), suggests an underwater world populated by coral, and so the narrative device can be further extended by exploring animation techniques and approaches associated with water, or which created the illusion of the movement of water, bubbles etc.

[1] See page 512 for an explanation of this process.

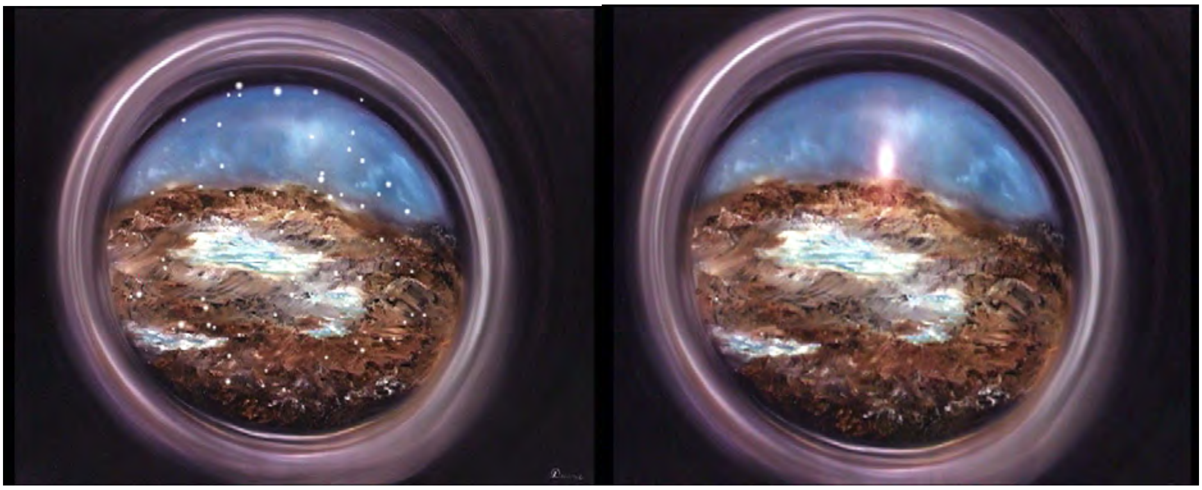
In many cases, if two or more paintings were reasonably similar in colouring I extended this narrative/structural device by imagining that these similar paintings depicted different parts of the original “painted” 3D planet – i.e. the camera would zoom down to, say, a blue world and then explore all the blue-themed paintings, as though they depicted different geographical locations in this one blue world.

Whenever the painting contained some strong visual structure or emphasized shape or line, I endeavoured to respond when possible to this, using transitions or effects which echoed the dominant shape or visual structure. Below we have an example of using the standard *iris* transition, available in most digital video editing applications, to move into the sequence of animations based on *Porthole to Earth* (Part 1).
At the same time it briefly creates the illusion of an eye.



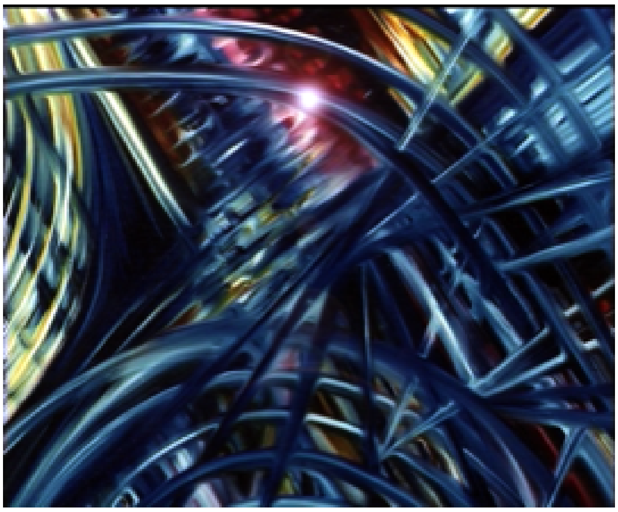


The sequence which emerged out of this transition is an early and somewhat naïve example of how a single painting could suggest a narrative exploration of the image, to produce an animation sequence. Somehow I had to find a reason for movement and animation events.



From the *Porthole to Earth* sequence; Part 1

Some of the paintings are closer to abstract than representational, though mostly they retain a sense of 3D depth. This sometimes meant that, what would begin as a representational (fantasy) journey into the landscapes of an imaginary, colour-themed world, might of necessity dissolve into an exploration of a non-representational, abstract space, as depicted in some of the paintings, or which resulted from some of the animation techniques.



Left, the image entitled *Nerve Centre*, while essentially abstract, was brought to life through the incorporation of imaginary animated nerve "pulses". (Part 3).

In the case of such abstract sequences, I would ultimately attempt to bring the viewer back to the "representational world" of the colour theme, as though they had slipped into a dream, and had then returned to (the imaginary) representational reality.

I chose the "*Through the Black Hole*" sequence as the opener, as I imagined it as a place where matter might form, and so seemed an apporopriate place to begin the imaginary journey.

The "worlds of colour" narrative device proved impossible to sustain, in that it became repetitive and obvious, but it was important as a means of getting the project underway. It occurs from time to time, but ultimately there is no definable formal structure to the finished work – handling of linear time was often a matter of intuitive decision-making, or chance, and there is undeniably some blatantly random juxtaposition of animation sequences. Most of the individual clips or sequences were, by necessity, created independently, i.e. by experimentally pursuing an animation idea related to an individual painting and without reference to other animations sequences. They do have an individual sense of cohesion, unity and integrity however. So in that sense I worked in a similar fashion to Duane, in that she painted each painting as a stand-alone piece, rather than conceiving the body of work as a structured sequence of inter-related images. Yet the total body of work *is* unified, by an individual aesthetic and a highly personal approach to art and life, and this too can be said of the resulting sequences of animated video art I ultimately created. I did have the advantage of perceiving the inter-relatedness of the various sequences as a challenging problem, and, as I gained confidence and experience, was able to better foresee ways of making connections between animated clips, and to gradually develop a more “big-picture” approach that facilitated the editing process.

While many of the source images are representational, their intention is clearly expressionist, within a New Age belief system. Hence a second guiding principle became: focus the animation process on developing the expressionist intentions of the paintings.

Ultimately the project became ‘technique driven’, however: discovering a new animation technique and then recognising its potential in relation to a particular image, was often way the creative process unfolded. In this sense, it was truly experimental art.

The client’s sub-title for the project, "...a journey into the healing energies of colour and sound", became the strongest indicator of the required “look and feel” and to some degree timing (in the sense of pace) of the product. I tried to find some authoritative references regarding current knowledge of the healing properties of colour, but sources I discovered proved unscientific and of little real value to my rationalist approach. I do however have sympathy with the essentially healing and up-lifting spirit in which these sorts of ideas are presented, and I attempted to absorb this as a kind of a pre-verbal and emotive response to the timeless flow of universal energy that (to various degress of success) Duane depicts.

When researching the work of Jackson Pollock I perceived links between action painting and Zen calligraphy[1] and it began to occur to me that there could well be a strong connection here also to Duane's painting style, in the sense that, while her paintings are rarely totally abstract, they present a performance, inspired by a transcendental state. In the course of our collaboration she revealed that she never "corrected" her work or overpainted, and that as soon as that seemed necessary she abandoned a work in progress.

Having said that the process became technique driven, it was also of paramount importance to allow the paintings to speak for themselves, and a simple exploration of the painted space using spatial (co-ordinate) transformations (panning), and scaling (zooming) was ultimately the most common technique employed. As I became more familiar with the paintings I began to notice the astonishing detail concealed within the brushwork – faces, forms, animal and human shapes are cleverly hidden within the swirling brushtrokes.

It is surprisingly difficult to find reference on the internet to similar sorts of work to *Take Your Space Now.* The work does have connections with other historical experiments in combining image and sound however. A link could be made with the ideas of Thomas Wilfred (see p.124). He conceptualised an artform of colour-music projections, which he called *Lumia*. He described it in terms of polymorphous, fluid streams of slowly metamorphosing colour. *Take Your Space Now* could certainly be called a form of *Lumia,* and, having read of Wilfred's ideas, a new freedom was found in developing the animations, unburdened by my initial need for formal structure and rational meaning.

It is also possible to make a connection between the final form of *Take Your Space Now* and the ideas of Leopold Survage, who created a series of sequential natural media paintings which he planned to photograph for an unrealised abstract film of "symphonies in movement" he called *Rythme Coloré.* Sauvage was quoted in *Part One* (p.220) as saying:

> Coloured music is in no way an illustration or an interpretation of a musical work. It is an autonomous art, although based on the same psychological principles as music.

What could be called an inversion of *Coloured Music* is Disney's *Fantasia*, mentioned in *Part One* in the context of synaesthetic art (p. 105). It is a fantasy

[1] See pp. 51-52.

animation "responding" to previously created music. Though it was only recognized in retrospect, there are indeed some parallels or resonances with the approaches taken in the production of *Take Your Space Now. Fantasia* has a fragmented, episodic narrative, and long sequences of stylized, dream-like, and somewhat abstract action, that attempts to "express" the music in a visual form. *Take Your Space Now* can be considered to have much in common, though it reverses the relationship between the aural and the visual, as Suavage had imagined.

It occurred to me after carrying out my research into synaesthesia, and the history of technological approaches to simulating the synaesthetic experience, that Duane's belief in the link between colour and sound represents a kind of intuitive connection with the ideas and activities of a great many artists and thinkers down through the ages.[1]

Originally Duane imagined the video piece to be one single program. In attempting to give each painting equal "weight" or animation time in the piece, we ended up with experimental animations totaling well over an hour in duration. The piece therefore became conceived of, for a considerable period of time during development, as a two-part programme (at this stage envisioned as a double VHS tape, and accordingly Duane had a cover designed for a double video library case, with enclosed booklet). Near the end of the production process the two parts had become approximately 45 minutes each in duration. This seemed to be just too long a period to remain immersed in such an unstructured, non-narrative media experience. When Duane finally came to understand the nature and possibilities of DVD, we reached an agreement to further sub-divide the piece into 4 sections, available interactively on the DVD. With the benefit of hindsight it would have been better to have had this "architectural" goal in mind all along, in terms of making the four sections strongly thematic and aiding in the development of a tighter structure. However the natural evolution of the project led us to the ultimate finished structure – thus is the nature of experimental work.

6. 2D Animation Techniques

Normal television resolution is 72 dots (or pixels) per inch (dpi). Scanning images at a higher resolution (300 to 600 dpi) allows the obvious possibility of simulated camera movements – simulated zooms could be achieved without pixellation, and

[1] See p.117 for a history of the links between synaesthesia and art.

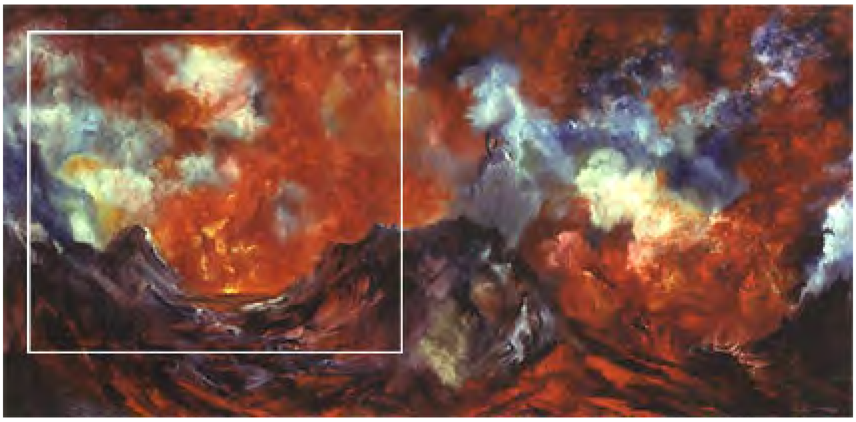
the camera could pan across the image using co-ordinate transformation (discussed in more detail below).

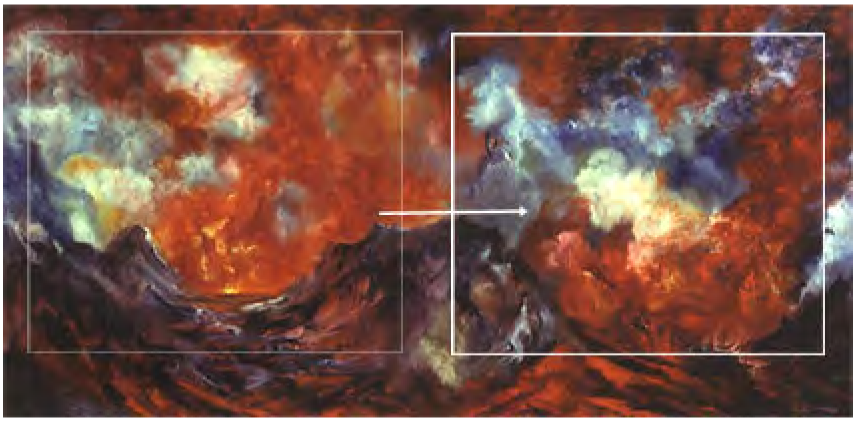
Resulting techniques require the ability of video software applications such as Adobe After Effects to “in-between”, or interpolate parameters such as x-y co-ordinates, on a timeline, based on a frame-rate and duration.

Scanning at higher than television pixel resolution allows for:

i). Panning ii). Zooming
iii). Combined panning & zooming iv).Rotation
v). Combined panning, zooming and rotation

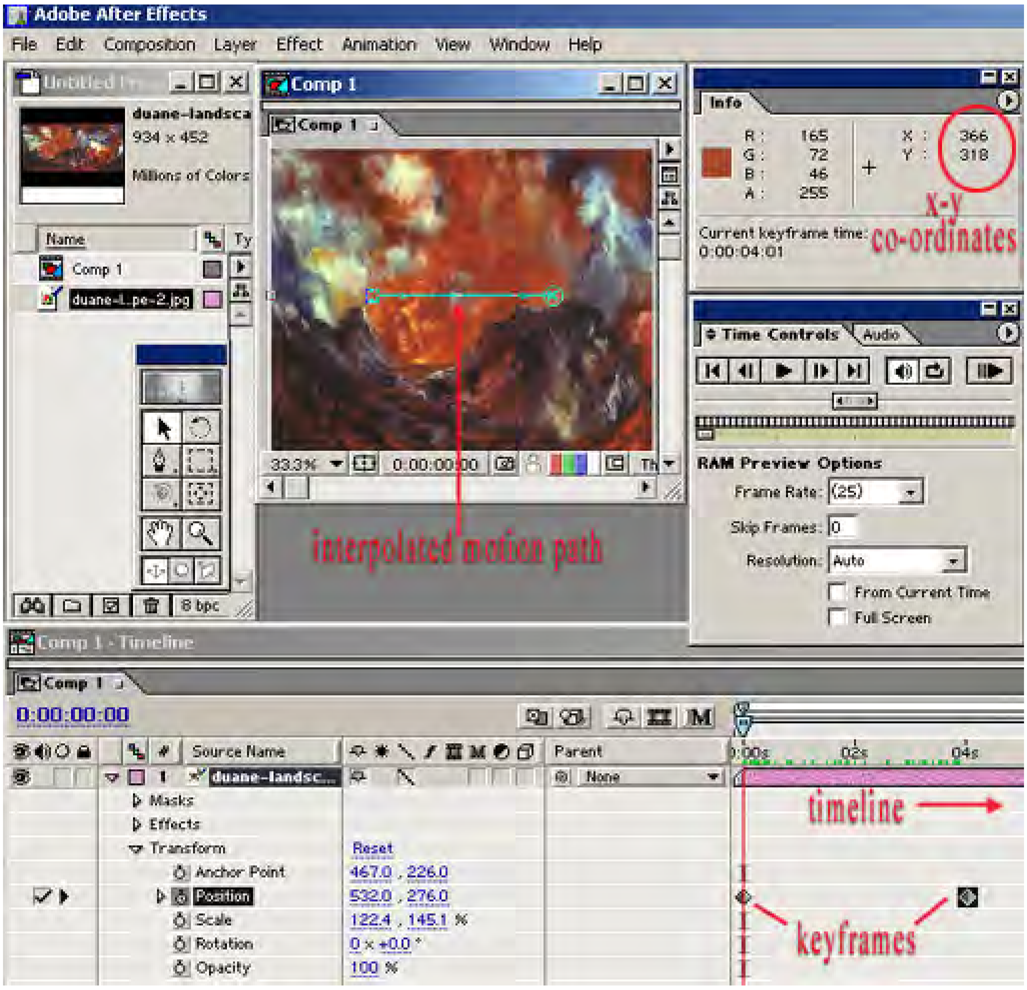
Panning





Mercurial Landscape (Part 1)

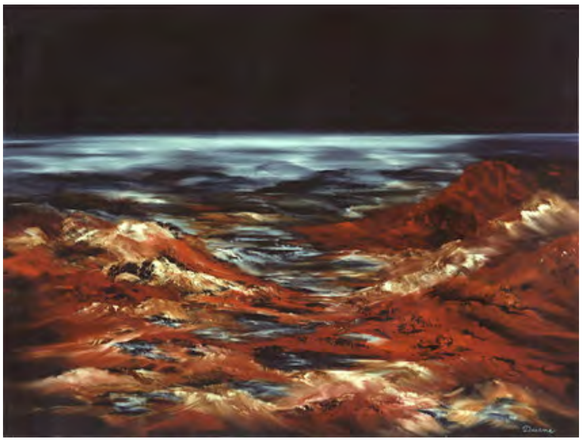
The beginning and end co-ordinates are specified in panels available in the software. At a frame rate of 25 fps, and based on the duration also designated by the user, the software calculates from the pixel data available from the high-resolution scan which pixels are visible at each frame of the animation, producing an illusion of movement, once the in-between frames are generated.



Above: A screenshot of the Adobe *After Effects* interface, set up to render a 4 second animation (note the keyframes against the timeline, bottom right hand corner; 04s above the timeline representing 4 seconds of elapsed time). The animation is based on an interpolated motion path, built on two keyframes which specify differing x-y co-ordinates, referenced to duration and frame-rate. The software generates the in-between frames, producing an illusion of panning on playback. Early work was carried out on a then state-of-the-art 132 megahertz Apple PowerPC, which could take hours to render a short piece. Some animations took up to a week to render, which was challenging if the experiment was slightly flawed.

Through familiarity with the visual language of film and television the viewer interprets this movement described above as a virtual camera movement, in this case a left to right pan.

The creative challenge is in the discovering of opportunities to animate elements (create visual movement over time) within the picture plane. A simple example: many of the images of space allowed for animated starfields.

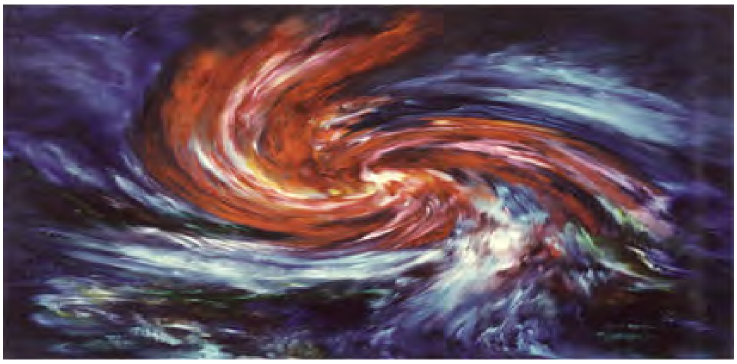


In the Beginning (Part 1)

By applying a mask to an area as shown above, it can be made transparent, revealing an animated starfield on a layer “behind” the scanned painting. This animation effect is created using an algorithmic process, available as a plug-in software application, within Adobe *After Effects*, using the MetaCreations’ *Final Effects* plug-in interface. The starfield animation option takes the pixel data of a bit-mapped image and distributes it into animated points of colour or 3d balls of colour, which appear to move toward or away from the camera over time. All the parameters (size of stars, density of stars, speed and direction of movement etc.) are controllable by the user. I took care to always use one of the artist’s images as source material, even when producing animated starfields, so that subtle variations in the colours of background stars are generated by the colouring of the original painted images.

Both *Adobe After Effects* and MetaCreations *Goo* software allow the application of various distortion effects on the pixel grid of bitmaps, which can then be key-framed to produce simulated movement over the designated frame series, creating the illusion of animation over time. This is possible using algorithmic “filters” which process images by applying pre-programmed mathematical manipulation of pixel data according to variable user-definable parameters. The visual effects which can be produced can seem quite appropriately hallucinogenic and surreal within the context of this fantasy style project..

Above: A reduction of the original scan of the painting "*Birth of an Island*" (Part 1)

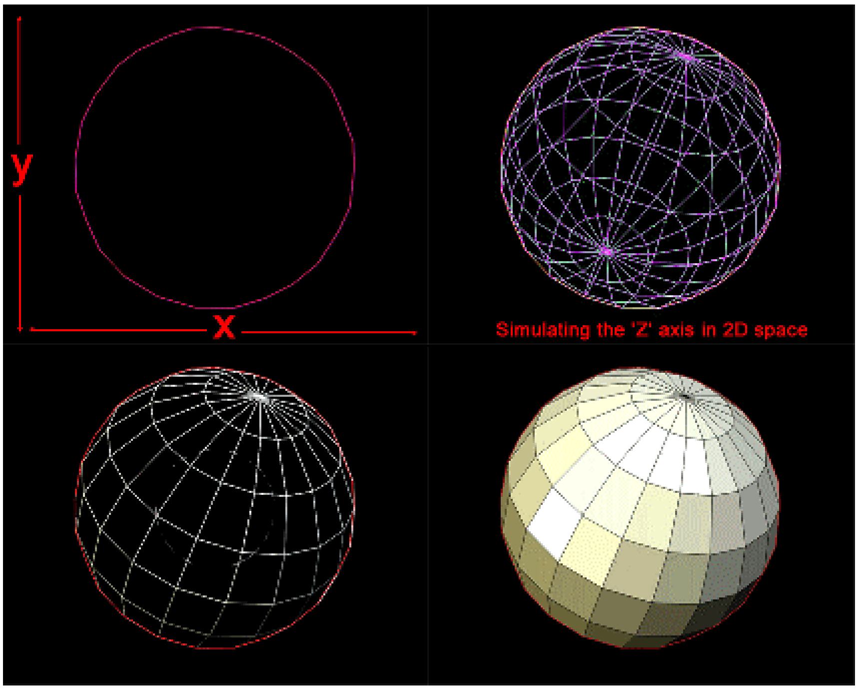


Above: a frame of an animation, created using the "twirl" filter, which distorts an image progressively into a vortex or spiral, by shifting pixels data to new co-orinates based on the algorithmic matrix. The software will interpolate the in-between states based on user-defined variable parameters set at beginning and end keyframes on the timeline. On playback the viewer sees a gradual morphing of the image between the beginning and end states

Above: The screenshot of a frame created by the same process, in this case using the "pinch" filter, which appears to drag the pixels progressively towards a central point, discarding pixels approaching the epicenter, to create an illusion of compression, and interpolating new pixels towards the perimeters, to create an illusion of stretching the canvas.

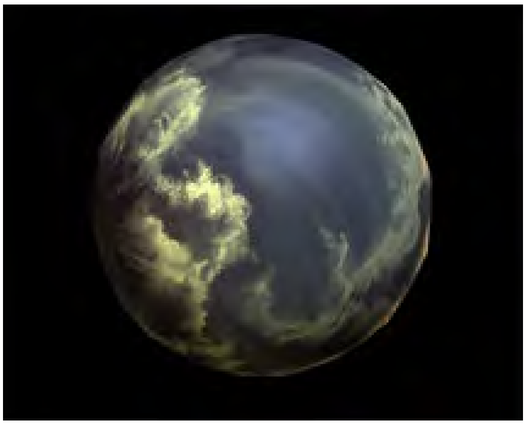
7. 3D sphere mapping

A process used frequently in the project is 3d sphere “mapping”, produced in this project using the MetaCreations’ *“Final Effects*” plug-in, within the Adobe *After Effects* video compositing environment, which allowed me to produce a 3d result without the need to purchase an expensive mainstream 3D modeling / animation application.



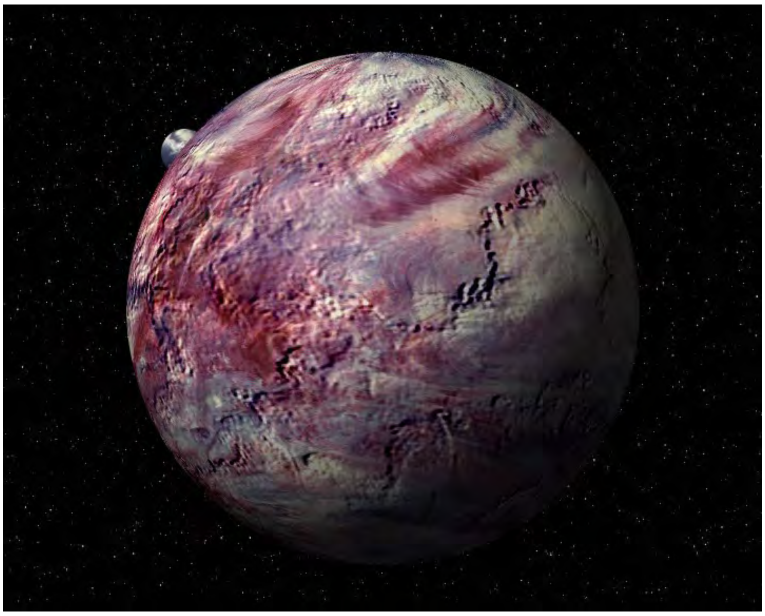
Typically, 3D software can depict 3D imagery based on stored 3-dimensional co-ordinate data. Simple or primitive 3D geometric shapes are most easily representative because of their regularity. Thus it is relatively simple to create the illusion of a sphere, using this system of mapped 3-dimensional co-ordinates and simple geometrical formulae. Two-dimensional images can then be mapped onto this simulated 3D model. The software has been programmed to extrapolate a new image based on this co-ordinate system, by depicting how the 2D image would look if its co-ordinates were mapped onto the co-ordinates of the imaginary sphere in 3D space. The resulting textured 3-d model can then be animated, by varying co-ordinates, scale, rotation, lighting and other measurable parameters over time, and rendering the resulting output frames into a video sequence.

Above: a reduced-size version of the bitmap image derived by scanning from the photo of the original painting *Heavenly Embrace*. (Part 1).

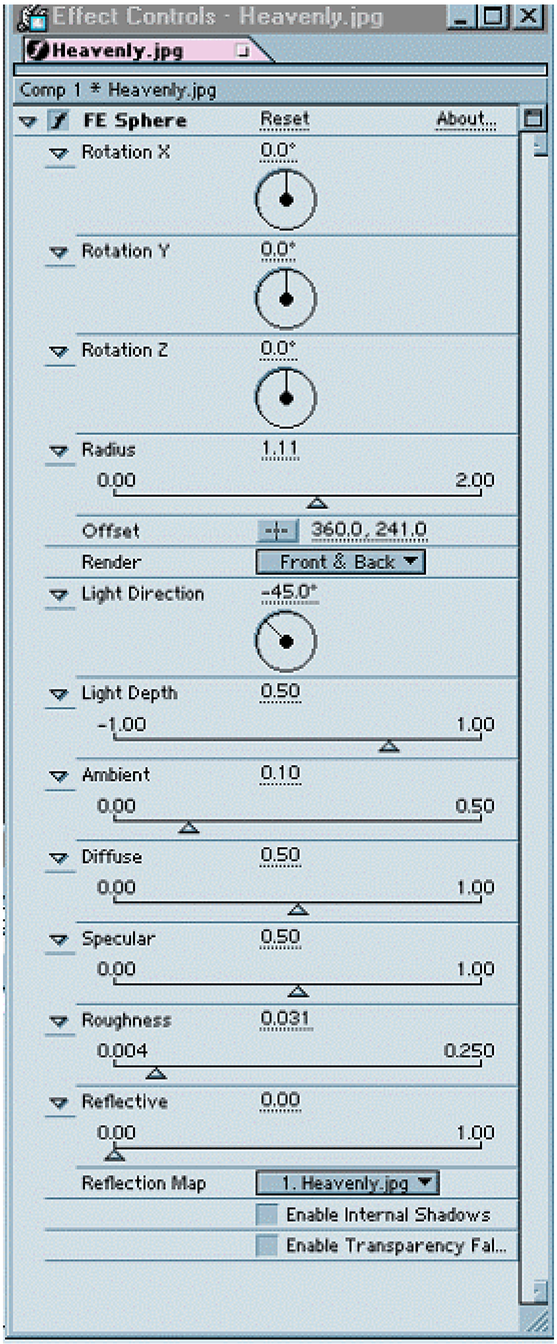


Above: a screenshot of a frame from the resulting animation, in which the bitmap is remapped to the 3d "wireframe-model" co-ordinates. which can then be rotated, can move in simulated space, and vary in radius to produce the illusion of advancing or retreating within the picture plane (zooming).

An improvement on this technique is to use the original 2D image as a "bump map", which gives a more textured 3D surface to the planet (below). The premise of *bump mapping* is to shade or texture a model's surface as though it had tiny bumps. What is important is that the surface geometry itself does not change, however the illumination algorithm uses vectors to simulate the effect of light, as though reflecting from tiny perturbations across the surface, adding complexity and realism at relatively low computational cost.



Duane's painting *Felinity* mapped to a sphere in 3D space *(Part 3).*



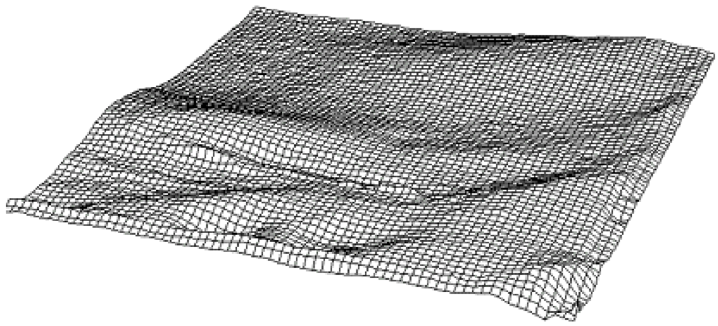
Left: Control panel for *Final Effects* "plug-in" showing animatable variables for the sphere mapping plug-in. **Note**: variable rotation co-ordinates, variable radius for simulating zooms, and detailed control of simulated lighting conditions, including specular highlight and lighting direction.

Bump mapping differs from *displacement mapping* in that the latter technique is a direct approach to rendering surface detail, whereas the former technique simply modifies a parameter to an illumination equation. The use of *displacement mapping* can further augment the visual features of a 3D object, without having to complicate its underlying mathematical model.[1]

3D terrain models

The MetaCreations *Infini-D* package was used to generate 3D "landform" models from the grayscale values of the image scans. The software looks at the grayscale values of a de-saturated version of the image, and constructs a 3-d terrain model by assigning grayscale variables a relative co-ordinate value, such that the lighter grayscale values become peaks, the darker values become valleys in the theoretical terrain model. These 3d models, converted to the universal .dxf 3-dimensional

[1] http://xarch.tu-graz.ac.at/autocad/wiki/DisplacementMapping credited to Hin Jang, 2001. accessed 8-3-04.

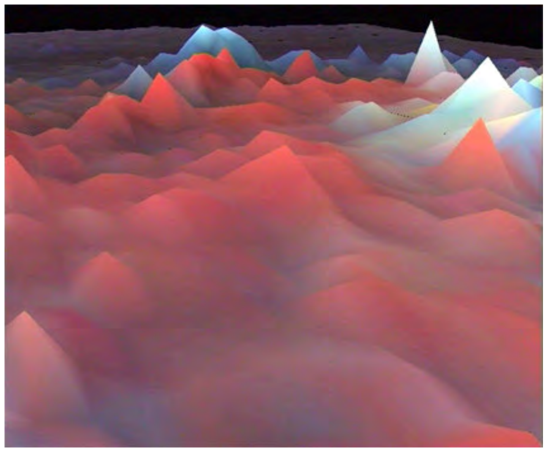


wireframe (x-y-z co-ordinate) format, were then textured by applying the image which generated the gray-scale values back onto the resulting 3D model. A virtual camera was then be animated in *Infini-D* or a more sophisticated 3-D environment, producing a simulated fly-through. Later experiments were carried out with *Kinetix 3DstudioMax.*

Right: 3-D terrain model generated from the grayscale values of the scanned painting, with the painting remapped onto the model, creating a unique 3D simulated landscape, that is entirely dependent on the colour values of the painting .

A fly-through can then be created by animating a virtual camera in relation co-ordinates in the new world-space. The computer generates video frames depicting what a viewer would see "in theory", based on this input data.

(from the *Journey's End* sequence, Part 1.)



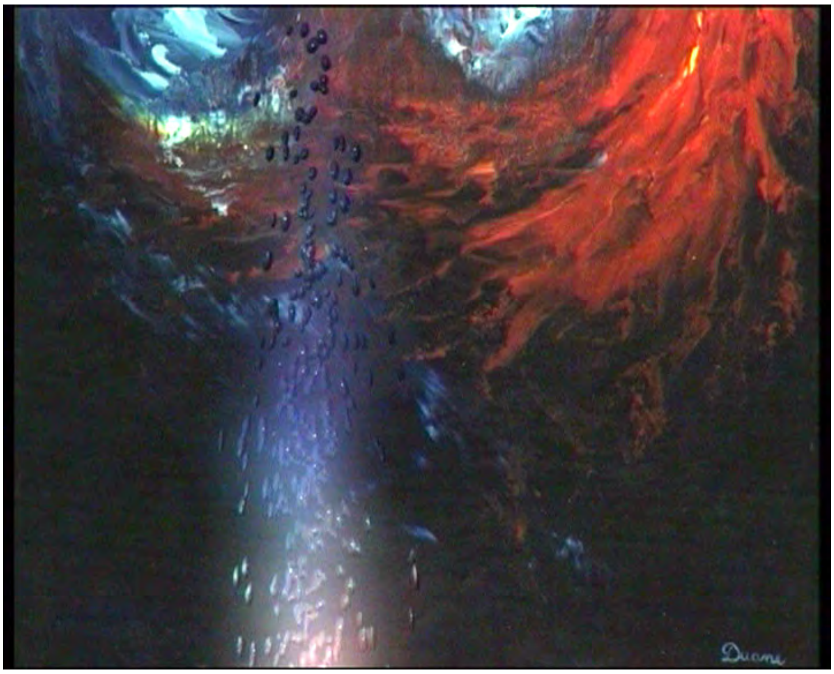
This technique also was also used to produce a stalagmite type effect as pictured below. (*Hall of the Mountain King* sequence*;* Part 2):

Below is a still from yet another example of a grey-scale 3D terrain model fly-through, with the 2D image mapped back onto the terrain models surface. Many attempts were made to execute this effect successfully, but an aestheticically pleasing result proved only occasionally possible.

Lord of the Rings sequence *(*Part 3)

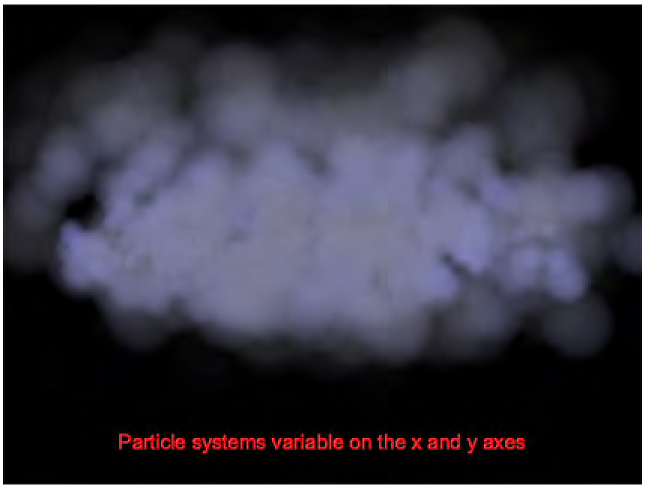
8. Particle Systems

Particle systems simulate growth processes and dynamic behavior using simple shapes, such as spheres. Simple particle effects can be composited onto 2D images using the MetaCreations' *Final Effects* plug-in system, within the Adobe *After Effects* environment. (*Journey's End* sequence; Part 1):

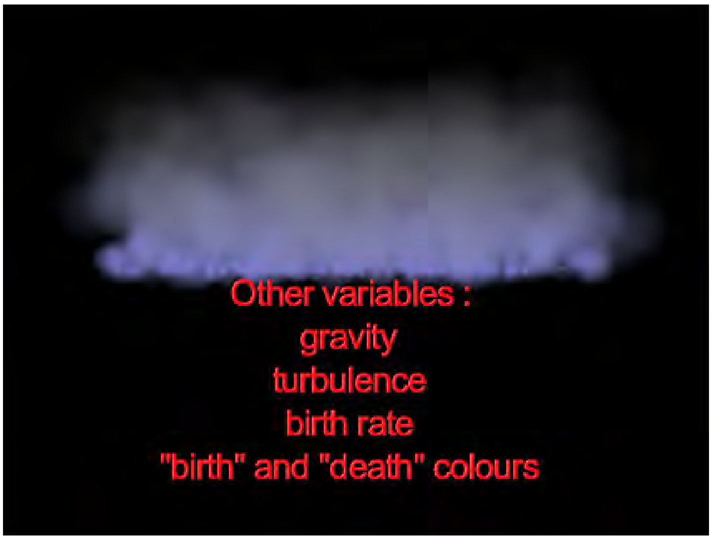


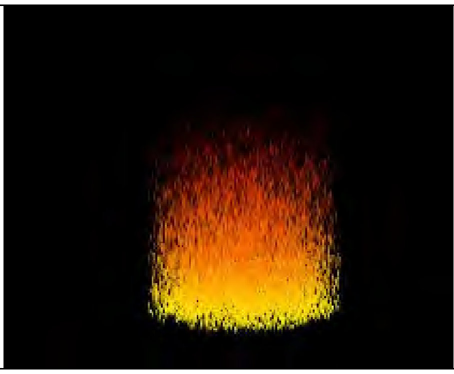
This allows the user to simulate the movement of 3D clouds, fire, smoke, bubbles and other visual effects, and to some degree simulate 'atmosphere' (haze etc.), all variable over time, on top of a 2D image.

Above: Screenshot of 'particle effects' simulating turbulent smoke.

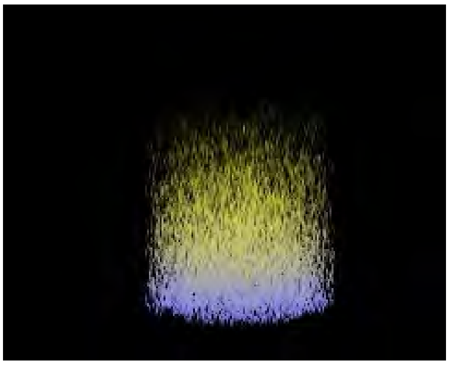


Above: When a larger X-axis variable is input, it widens the scale over which the particles emerge.



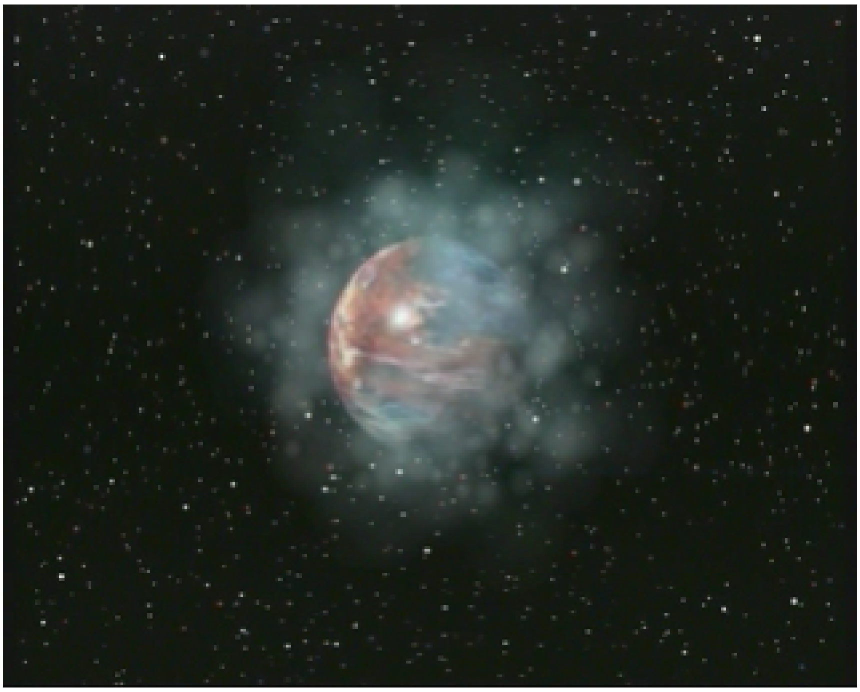


Above: Simulated fire produced by *Final Effects* Particle Systems.

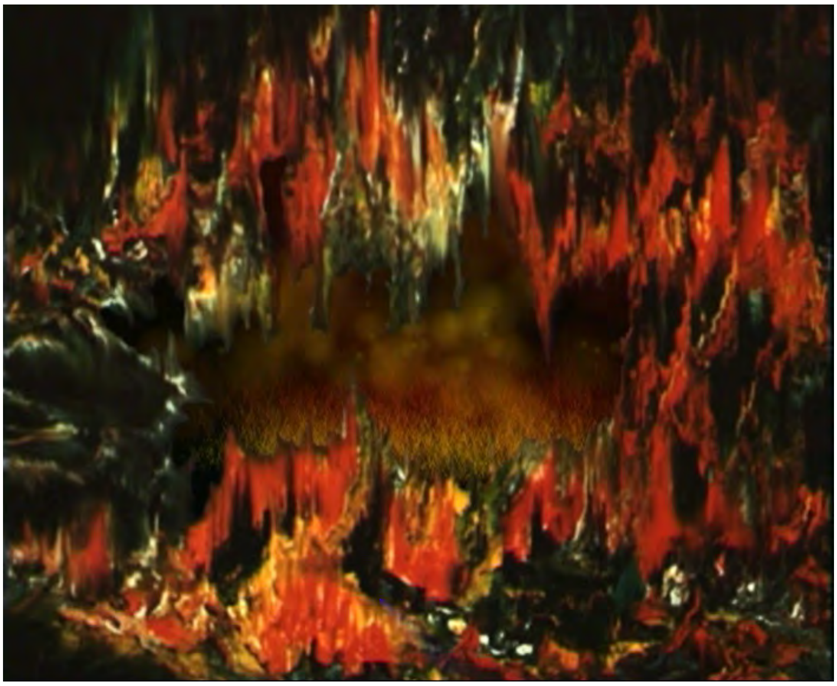


Above: Varying the "Birth" (purple) and "Death" (yellow) colours of the dynamic particles.

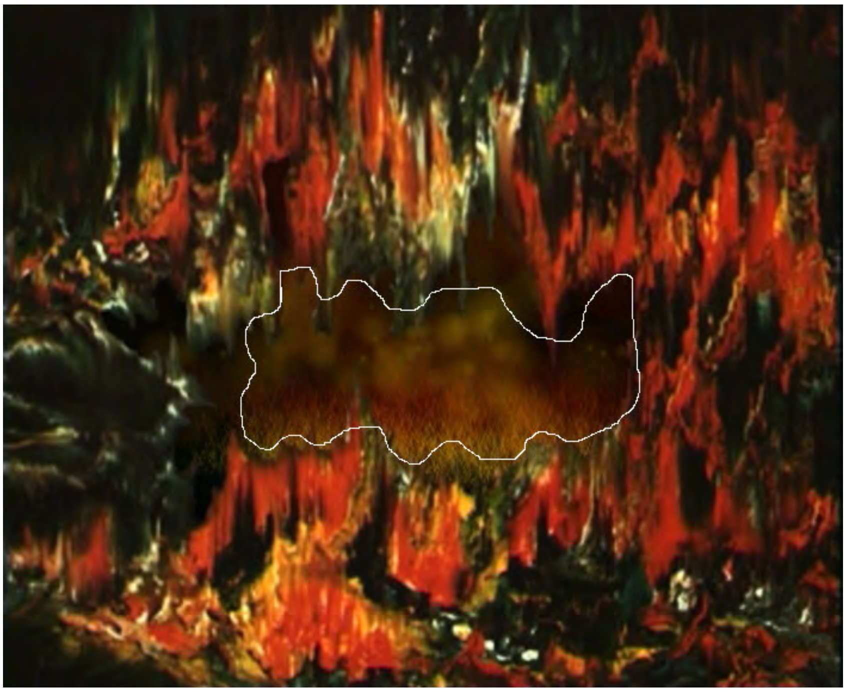
Below: An example of particle effects used to simulate atmosphere:



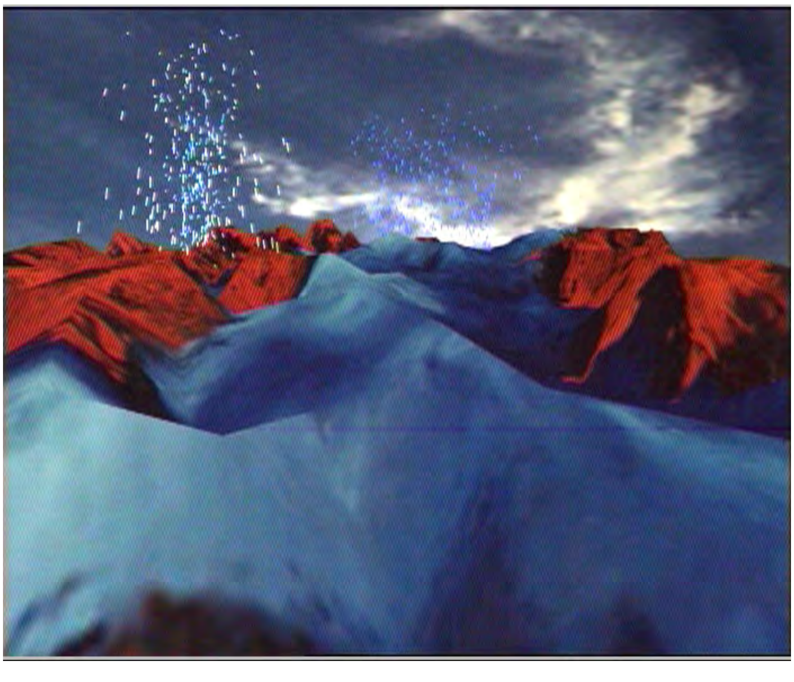
Birth of an Island sequence *(Part 1)*



Above: An example of using particle systems to simulate fire. In this case an area of the image is masked out with an alpha channel to contain the effect within a suitable part of the image (see below), to further enhance the sense of fantasy. (*The Gathering ;* Part 4*)*



Infini-D also has some basic particle effects which were a little different to those available in *Final Effects*, which I experimented with in the following sequence.



Above: Combination of 3D terrain model remapped with scanned painting as a surface texture, with particle effects producing simulated volcanic activity. (*Thor, God of Thunder*; Part 3) .

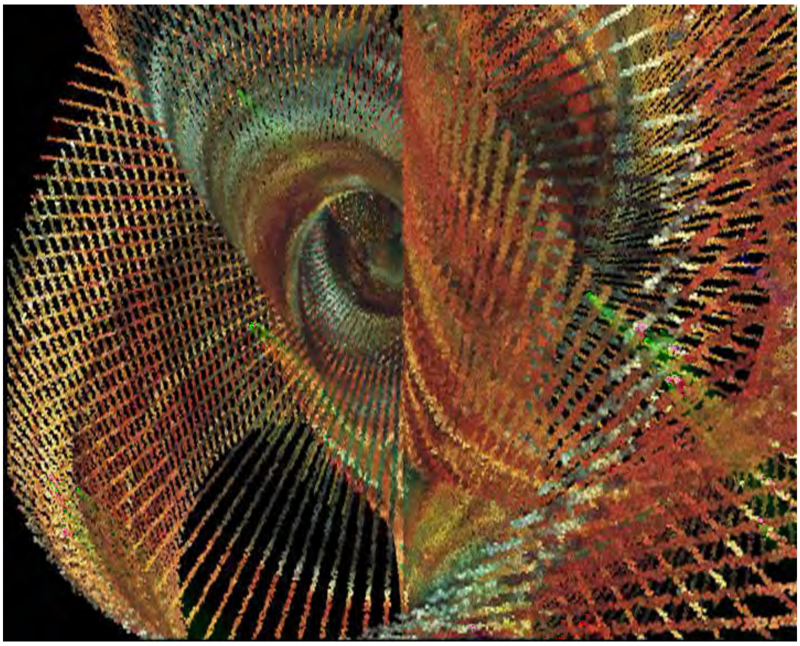
The backround sky used in this image above is another of the scanned paintings *Heavenly Embrace* – at all times only the paintings were used to produce imagery to retain internal integrity within the project.

Right: A reduced sized scan of the painting *Heavenly Embrace,* used as a sky background in the previous screenshot. (Part1)

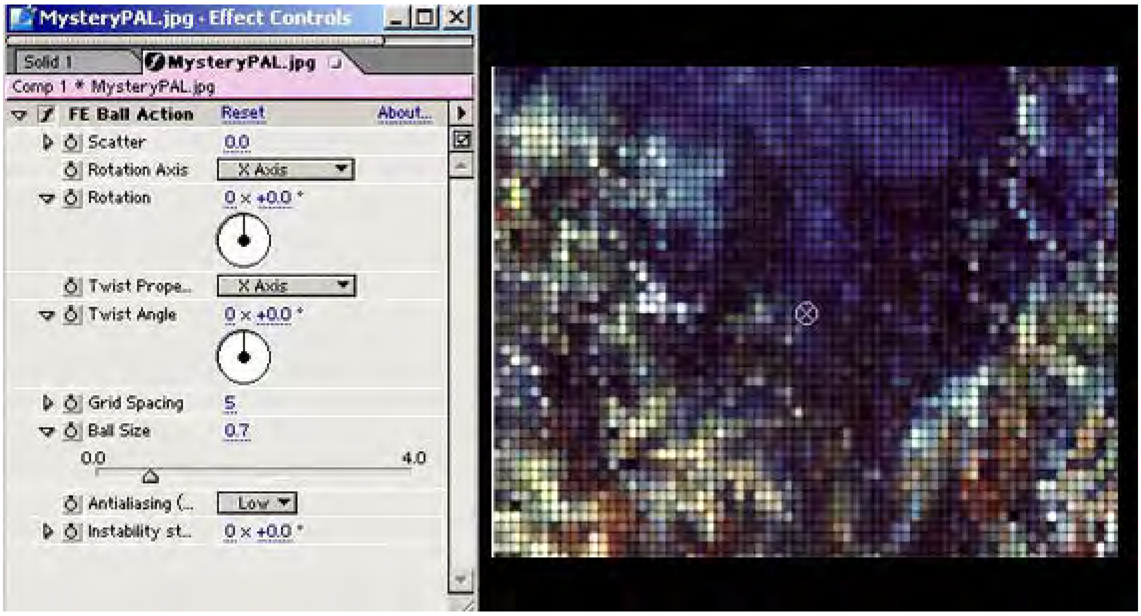
Left: A reduced sized scan of the painting *Aquarius.* (Part 3)

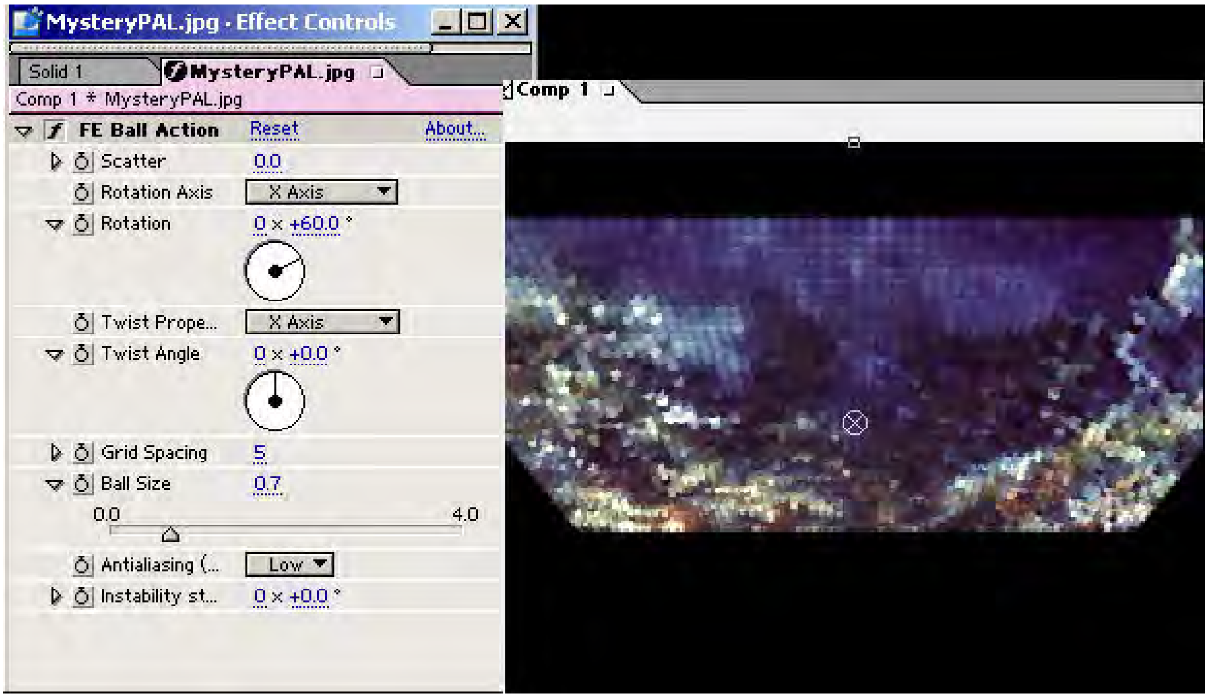
Final Effects includes another plug-in algorithm with variable parameters which maps the image onto a matrix of 3D balls. These can then be animated over Time, by varying rotations in the x, y and z axes, producing convoluted spiral planes.

It is difficult to get a satisfying result from this filter, much experimentation is required, but it can produce dramatic three dimensional results. The next series of images gives an idea of how this plug-in operates. The first image in this series shows the "ball action" applied with a small sphere size, but with no axis rotation, giving a mosaic -like effect.

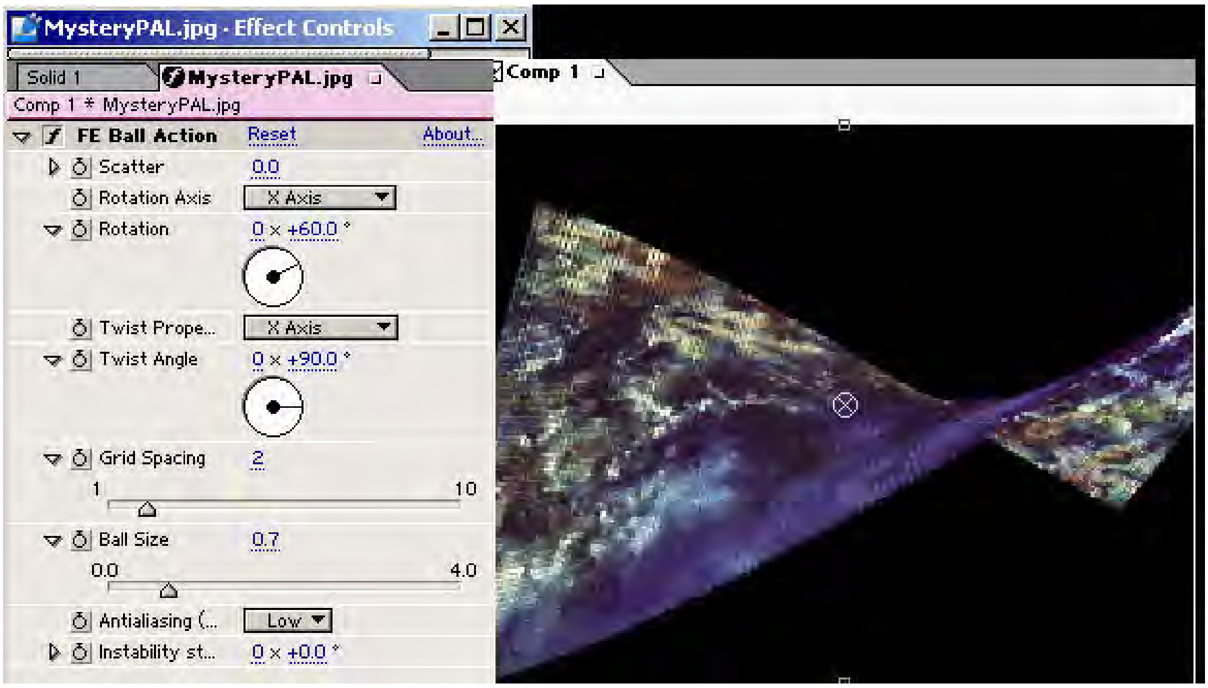


Above: A still from an animation sequence based on the *Aquarius* painting, using the *Final Effects* 'Ball Action' particle system plug-in. (Part 3)



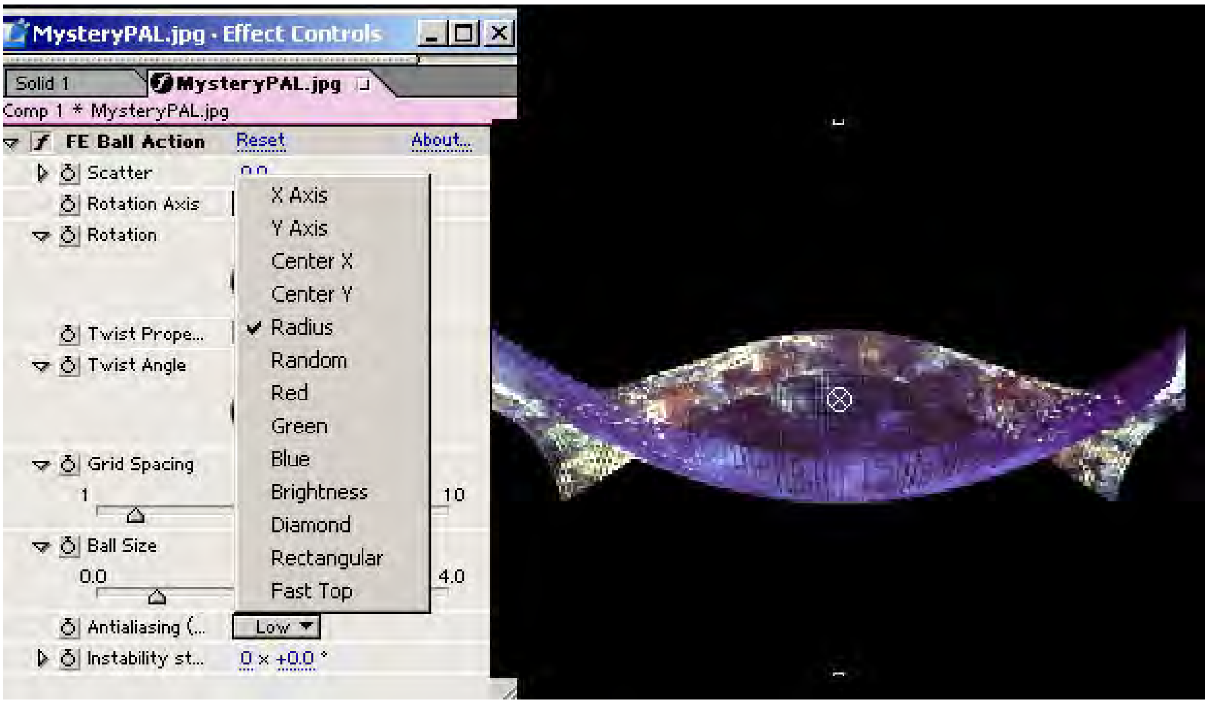


Above: *Ball Action* rotated 60 degrees on the x-axis only. A one-axis rotation gives the illusion of a tilt to the image plane.



Above: *Ball Action* rotated 60 degrees on the x-axis, with twist properties applied to the X-axis and a twist angle of 90 degrees.

In the next screen-shot, notice the pop-up down menu, showing the many variations and multi-combinations of axes that can be manipulated using the *twist properties* animation parameter, each producing unique structures and visual effects which morph over time.



Above: *Ball Action* rotated 60 degrees on the x-axis, with twist properties this time set to *radius*, and a twist angle of 90 degrees.

If the “Ball Action” animation filter is applied to a previously rendered animation which contains colour changes, the colour changes will radiate through the morphing rings and ripples of ball shapes, like a pulsing bank of light. In the above image the ball size is reduced, producing a net-like effect.

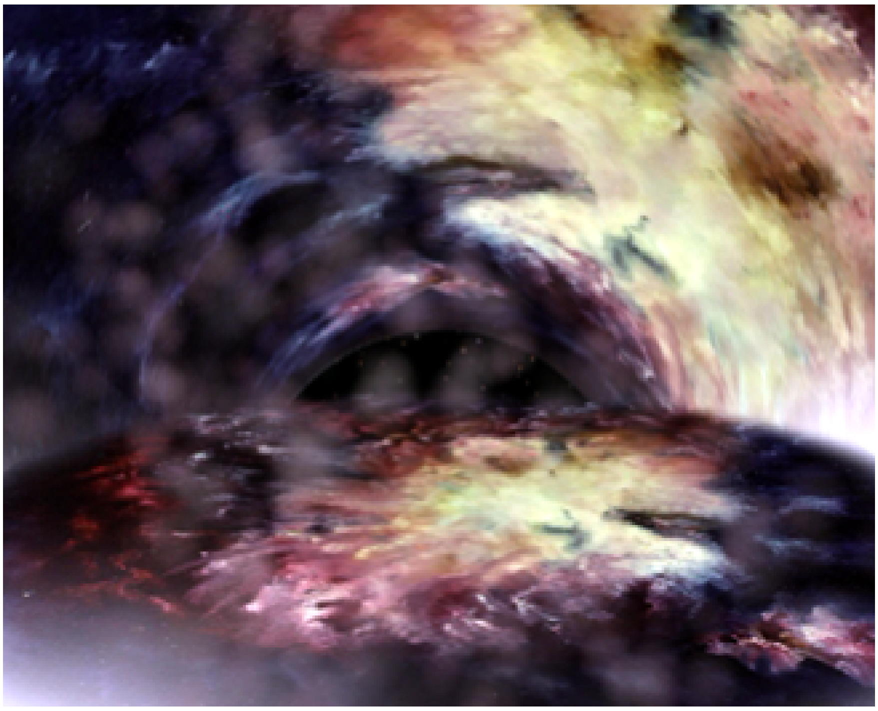
9. More complex compositing, multiple layering and the use of multiple effects

While the possibilities presented by simple distortions and transformations were continually exlored, I also experimented with more complex layering and combinations of effects.

An example is the animated sequence that is based on the following image, produced by a combination of several techniques. In the second of the next two images we see the painting *Supernova* mapped onto the inside of a 3D cylinder, producing a sort of cave effect.

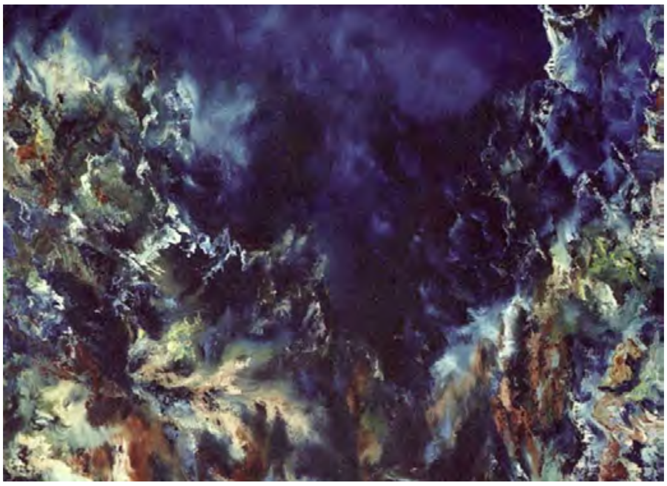
Left: *Supernova* (Part3)

(Referring to the image below) a second instance of the *Supernova* image is projected onto the “floor”, using an image distortion effect which creates the illusion of perspective (available in *After Effects*), while in the background we have an animated starfield (generated from the same image), with the “stars” moving at high speed. This is all overlaid by simulated haze using one of the *Final Effects* particle system plug-ins. The accompanying synthesized orchestral music score combines with the visual image to suggest a sort of dramatic, otherworldly genesis. A simulated, highly activated starfield is placed on a layer behind these two composited images, so that when they are animated to give the illusion of an opening tunnel, the viewer can see through to this third dynamic layer.



From the *Supernova* sequence which opens Part 3

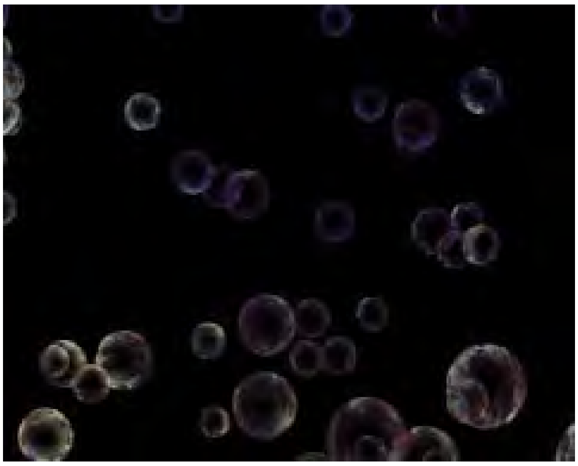
Another example of creating an animated sequence using a combination of effects and techniques is that created from the scan of the painting entitled *Mystery of the Deep* (the original painting is 90cm x 60 cm, the following scanned version is cropped to PAL aspect ratio).



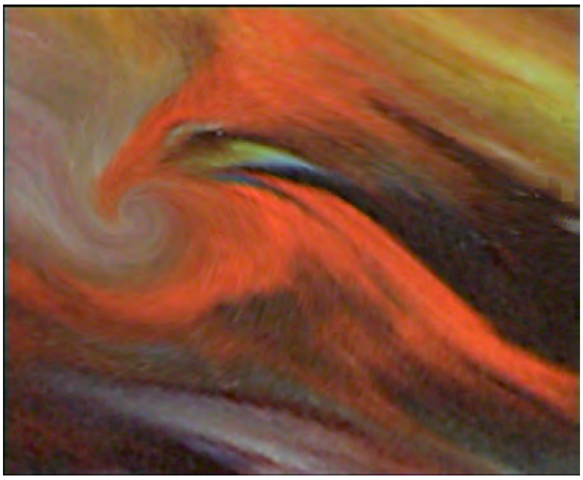
Left: *Mystery of the Deep*, Part 3

The next image (below left) was created by using the same perspective distortion effect mentioned in relation to the previous image, but taken to an extreme, and with a wave filter applied, to simulate the surface of a lake. In the lower half of the image using a mask I have applied another variation on particle systems which produces the illusion of bubbles (complete with semi-opaque reflections of the original image mapped onto the bubbles). The upper portion of the animation is overlaid by a particle system simulation of steam rising. This is an example of one effect suggesting another – it was produced after I had made a reasonably successful experiment with using the Java “Lake applet” as a video animation effect (see next section for more detail on Java techniques used in the project). As the original painting suggested underwater coral to me, I began to explore ideas associated with water – thus these two animations emerged and contributed to the final sequence. I was satisfied to have produced a quite lengthy narrative sequence based on this single image, by sequencing these and several other animation ideas together. This is possibly the clearest version of the narrative device I was trying to establish of firstly showing the planet, then zooming down to the planet to view a 3D terrain

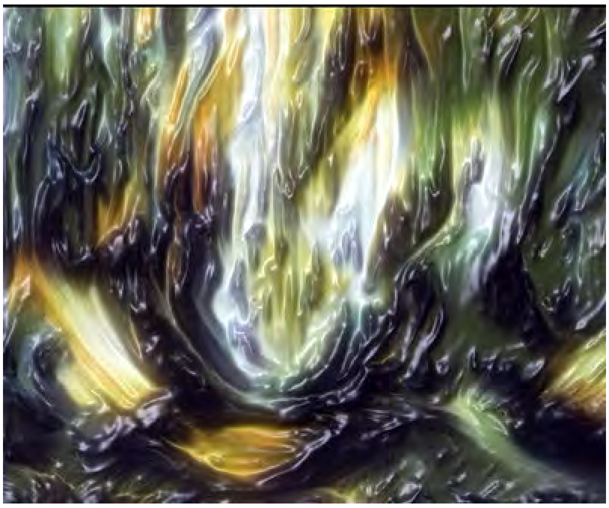
model of the painted image, then exploring the image itself, with some morphing to keep a sense of movement, then (in this case) exploring an underwater world, returning to the original image, then dissolving into a whole sequence of abstract and transcendental manipulations of the painting. Manipulating images using multiple parameters in MetaCreations *Goo* produced many interesting accidental effects, such as the illusion of the painting morphing into a shape reminiscent of a bird's head (below right).

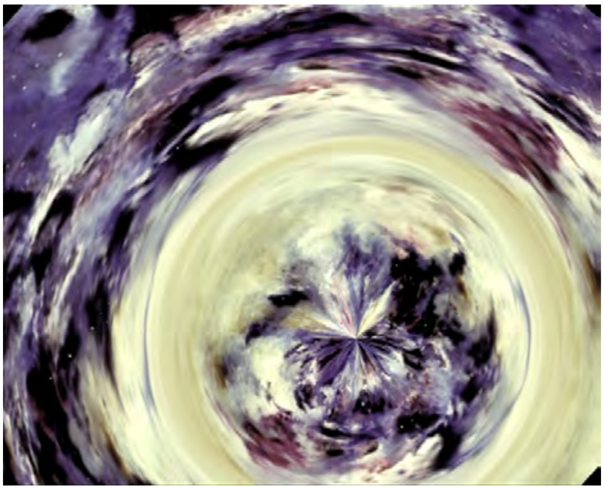


Above: an example of the 'bubbles' simulation algorithm (see above image).



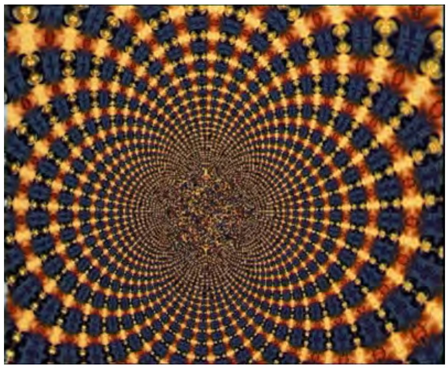
Above: *Goo* morphing, producing an interesting accidental result. *Rainbow Fantasy*, Part 1.





Above: "Glass" filter; *Spirit of the Forest*, Part 4. **Above:** "Vortex" filter; *Doradus*; Part 1.

Above: Metacreation's "*Goo*" morphing. *Spirit of the Forest;* Part 4.

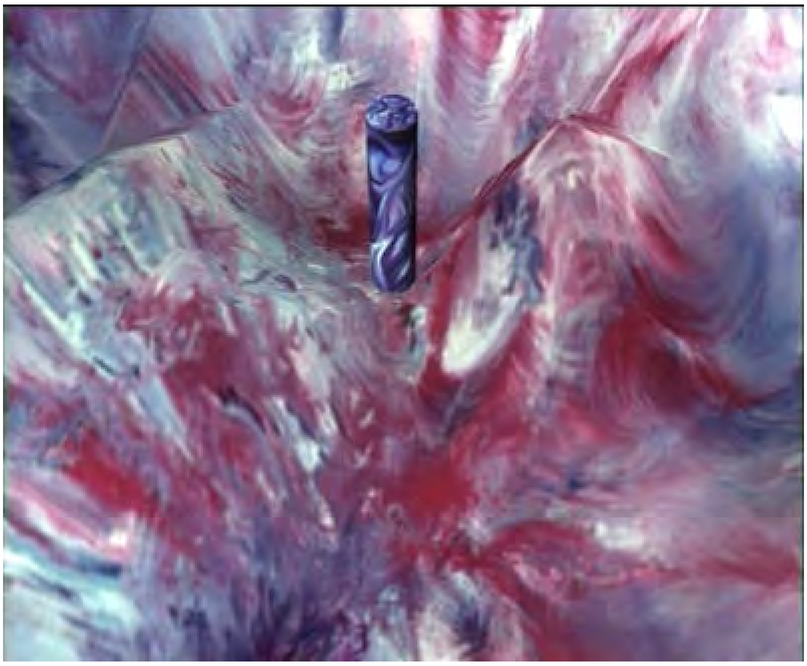


Above: *Goo* also offered a complex variable fractal parameter which was used sparingly.

Duane's striking painting entitled *Tears* has a brooding, totemic feeling to it, and in my attempts to bring a more three dimensional impact to *Take Your Space Now*, I experimented with making a 3D "totem pole" out of this image. My inspiration was the monolith in the opening sequence to Stanley Kubrick's classic film *2001: a Space Oddyssey.*

I later placed the cylindrical "pole" in a 3D terrain created from, and image-mapped with the painting *Felinity* and constructed an elaborate camera sequence, featuring the *Tears* totem in a long simulated "dolly" shot (see next image).

Left: *Tears* , Part 3.



Above : from the *Felinity* - *Tears* sequence (Part 3)

10. Animations using Java Applets

Another source of animation effects became possible with the purchase of a Matrox dual head video display card, with a port which carries a video signal for projecting the screen image on a television. This allowed me to output animations made using Java applets to digital video tape. There is a range of visual effects possible through Java programming, which target the internet. These Java applets produce some visual effects I have not seen available in commercial compositing and video editing programs or plug-ins. A particularly good source of these was found at www.anfy.com, the web-site of an Italian group of programmers with some innovative ideas for web animation. This site offers a freeware version of a standalone application which produces some very striking visual effects, with embedded advertising which is removed on paying the registration fee and receiving a serial number. I was able to use the freeware version to output some interesting experimental animations, rendering them at 640x480 pixels and outputting them to mini-DV through the TV output connectors on the Matrox card. These had to be re-captured and re-sized in *After Effects* for PAL display, and in some cases rendering them at such a large size (much bigger than would be practical for internet delivery) was so processor intensive that they only played in a very stuttery, slow-to-render fashion. I used time compression in *After Effects* to increase the frame rate in order to get a smoother flow.

Above: Java applets such as the "Lake" applet used to produce animated effects unavailable in desktop video compositing /editing applications. The animation was output to DV tape, re-digitized, resized and the particle system “steam “ effect overlaid. Below: The "Tunnel" java applet by Anfy. Care must be taken to edit the 2D image so that a seamless blend is made at the folding point in the simulated tunnel.

11. Final Edit

For the first two years of project development I worked part-time on a Macintosh 7600 with a 132 megahertz PowerPC processor, connected to a 10 gigabyte external hardrive, partitioned into a RAID array, with 196 megabytes of RAM. I was using a TARGA 1000 Pro video card, which was restricted to analogue S-Video output (no firewire). By present standards this is an archaic system, and it was indeed painfully slow to render video animations, but it was the best desk-top video capture/output solution that was available when I first began this work. Progress was painfully slow at first, as I went through a lengthy and frustrating system optimization period, working towards a workable production method, and simultaneously trying to gain a workable understanding of the many digital processes and technical concepts. I was working solo, and technical support from the software and hardware vendors was very limited as, at the time, few people had experience with such systems. I had no previous knowledge of digital animation techniques and had to explore my new copy of Adobe *Premiere* with the aid of the manual (something digital artists abhor), and struggle with the Apple extensions manager, trying to find the minimal of combination of operating system extensions for this kind of work. Extension conflicts, crashes, loss of data were frequent. Because of the tight data storage capacity (impressive though it was for the time), I developed a process of outputting animations to a Sony DHR-1000 mini-DV video tape editing deck, via analogue S-video connectors. This lack of adequate hard-drive storage capacity meant that my attempts to compile animations into useable sequences were thus confined to linear editing the chosen animation segments on mini-DV tape, using the DH-1000 linear digital deck, connected by firewire and LANC control, to a Sony VX-1000 digital camera. Using this technology I was able to string sequences together and output them to VHS tape to mail to Duane for comment.

At a number of stages I attempted to compile the various experimental animations into a final running order. During this process I attempted to create large-scale dynamics through pacing and contrast. The whole project generated a massive amount of output, with a total of over 60 one-hour mini-dv tapes used, though many of these variations of the same material, as I attempted to define an optimum sequence or edit, as well as back-up/safety copies.

The first time I attempted create a long integrated sequence I produced a 30-minute edit/compilation, which attempted to make a coherent linear experience out of the many discrete short animations. This helped me grasp the difficulty of making a viable linear structure. It also contained countless jarring "jump-cuts" and revealed the need for soft-transitions between segments. I then recorded the whole piece, as it stood at the time, into a large hard-disk drive system at Bond University's School of Film and TV, where I was working at the time. (At this time my home system didn't have the hard-drive capacity to store the large amount of data comprising both the contributing animated clips and the final render). I replaced jump cuts with dissolves, experiencing one of many steep learning curves with another video editing / compositing software package called *Speed Razor,* mainly using a gradient map transition algorithm, which allowed me to design transitions based on the dominant shapes I perceived within the painted imagery or animations.

Right: Custom designed gradient map produced in *Photoshop* and used as the reference for a transition algorithm. The lightest areas appear to dissolve first, darkest areas last, during the transition between two images or animations. This allows the user to prepare gradient maps, which echo the shapes inherent in the video source material, adding to a sense of consistency or unity of design.

I managed to get a 30-minute sequence completed this way, which gave me a feel for the difficult task ahead. The rendering time on this system was still a major limiting factor, and made it difficult to make much headway. As it was the central storage device for the non-linear video editing teaching facility, it was only possible to have the hard-drive empty for a few days between semesters.
I eventually purchased a large enough hard-drive for my home studio, to allow me to work on this process independent of the time pressure involved with working in a busy teaching laboratory. I purchased a Canopus *Raptor* digital firewire video processing hardware card and a Pentium III computer, and continued the process begun in the Bond University teaching lab. The whole process was entirely subjective, intuitive and experimental, working in fitful bursts over the months as my

heavy teaching commitments allowed. This was very limiting in terms of getting a feel for the overall sweep of the project.

Some animation experiments were inevitably more successful than others, and I ended up in a situation where I had about a forty-minute sequence that was interesting, and of what I considered to be an acceptable standard, and another forty minute segment which consisted essentially of “out-takes”, or what I considered to be ‘B-grade’ animations. On recognition of this, I began another round of animation, this time on the Microsoft *Windows NT* / PC platform (which I reluctantly moved to once I began work at Bond), consciously attempting to bring the new animations up to the standard of the first tape. Happily, I found the new sequences frequently outmatched the original sequence as my confidence and experience increased.

This history is discernible in the final four sequences of animation. *Part One* was essentially produced on the *Macintosh* platform. *Part Two* is essentially produced on the PC, and Parts *Three* and *Four* contain animations produced on both platforms. Nearly all of the “3D terrain-model” experiments were produced on a *Macintosh*, because the *Infini-D* software, though not aimed at a professional market, made it very simple to undertake this process, and is a *Macintosh*-only product available for free on a computer magazine CD. Likewise, most of the morphing bit-map effects were created on the *Mac*, as I only had access, until very recently, to a *Macintosh* version of MetaCreation’s *Goo.*

Towards the end of the development process (early 2002) I purchased a Canopus *Storm* ‘real-time’ video capture/render card and the necessary Pentium III dual processor system, and further hard-drive capacity, which sped up the editing/collating process considerably by minimising the lengthy rendering component, though more challenging learning curves were traversed. It also made available a new range of transitions, and once again I made heavy use of the custom gradient map transition algorithm when collating sequences. I became very dependent on the Canopus real-time dsp processes as the pressure to finish this extremely drawn-out project mounted. I produced the titles for the credits sequence using *Premiere* 6.0’s simple titling facility, and the titles transition facility available with the Canopus software.

12. 1 The Soundtrack

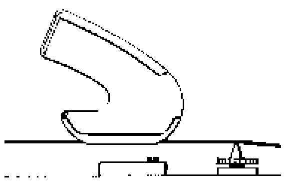
The most difficult challenge faced in composing the score for this work was the sheer scale, or that is, the long duration of the piece. It was high on my agenda to avoid a typical “doof” –style, techno-music approach i.e. strict metronomic beat and looping rock or techno drum-beats etc. Firstly, this seemed too obvious and clichéd, and for me placed the animations in the wrong genre. The few times I did experiment with drum beats and loops it seemed to somehow cheapen the effect of the animations, detract from the timeless, transcendental quality, by introducing associations with contemporary culture and images assoicated with dance clubs and urban society etc. which detracted from the overall look and feel of the project. I wanted to find a way of reflecting the individualistic approach of the painter in the process of creating the recorded score. The paintings are semi-representational – they depict, in most cases, recognizable but abstracted fantasy spaces with strong expressionist and transcendental qualities. In many cases if one was to not consider the names and poetic references given to the paintings, they can be viewed as abstract textures of colour. I hoped to reflect this individualistic blend between abstract and representational with an analogous approach in the music score – I imagined a mystical, layered, shifting soundscape with strong emphasis on texture and a minimum of traditional musical reference points (in the sense of melody and harmonic structure). I (somewhat ambitiously) visualized an hallucinogenic effect, that was musically coherent, yet seemed to “come from another world”, i.e. avoided genre references.

I also consciously attempted to avoid a programmed, MIDI-sequencing approach, which tends to emphasise the tyranny of the bar division. I so set out to introduce an improvisational quality, which reflected what I understood to be Duane’s approach to developing the paintings – spontaneous and intuitive, expressionist, “in the moment”, aiming to avoid getting caught up in reworking and corrections.

For my initial attempts at developing a sound-track for the piece, I improvised long sections of music on electronic keyboards, using a Roland JV-1010 connected to a weighted MIDI trigger keyboard, while watching a fairly complete rough cut of edited sequences of animation. I consciously avoided having any preconceived idea of how to approach these improvisations. I began by simply choosing sounds from the large banks of pre-programmed sounds based on the suggestions implicit in the patch names, auditioning any patch that suggested transcendental, ambient or sci-fi. If I found a sound patch that seemed to offer creative possibilities for producing a transcendental, or otherworldly ambient audio-scape, I would roll the video-tape and press record (using a Sony portable DAT recorder), and followed a personal brief to avoid resorting to over-dubbing and drop-ins, corrections etc. I built up a library of quite lengthy improvisations based on responding to the technical characteristics of the chosen JV-1010 sound-patch that had appealed to me subjectively, and I queued my responses to the timing of events as they unfolded on the video rough cut. While making these audio recordings I simultaneously also made MIDI recordings of my keyboard performances, using E-magic's *Logic Audio*. By making MIDI recordings it allowed me to later re-voice some of the improvisations and develop variations in texture and voicing, which then enabled me to re-use these improvisations as thematic material, re-introducing them in an attempt to develop a sense of continuity and structure.

This approach was substantially the way I worked in sections 1 and 2.

In Part One (near the end of the *Skyway* sequence, Part1) I overdubbed an acoustic steel-string guitar part, using an E-Bow in section 1, over the improvised keyboard part. The E-bow is normally associated with electric guitar, but I have discovered it works quite magically on the 2nd or B-string, and to a lesser degree on the upper (higher pitched) E-string of a steel-string acoustic guitar. The technique does not work on the wound strings of an acoustic guitar.



> "The E-Bow does its work on the string itself, by producing an energy field that vibrates and sustains the guitar string... Powered by a 9-volt battery, the E-Bow is held in place of the guitar pick, giving instant access to sounds (reminiscent of) violin, cello, flute and horn as well as unique sounds of its own". http://www.ebow.com/brochure.htm accessed 14-01-03. (The E-bow is featured in the soundtrack to *Take Your Space Now* beginning just before the *Journeys End* sequence, Part 1)

There are two horizontal depressions or slots either side of a small central guide light. The guide light shines on the target string providing a visual reference of positioning (but this is really a cosmetic addition, giving the illusion that this simple technology is more hi-tech). However, the adjacent strings either side of the target string rest in the slots, and the e-bow slides along these two strings, which act like rails, keeping the e-bow positioned exactly over the target string. There is an alternative switchable setting which emphasizes a higher harmonic.

The e-bow produces a sustained and eerie sound, presumably by employing an oscillating magnetic field, causing the guitar string to vibrate, and because of the lack of attack in the sound envelope it is reminiscent of backwards guitar (traditionally achieved by reversing the playback of a tape recording).

A second “live” recorded melodic element which was introduced is the ethnic-sounding flute melodies which appear from time to time, which were played by the author on a Irish whistle or flageolet. (featured in several places in the soundtrack for example the *Thor, God of Thunder* sequence; Part 3).

At a later stage I developed the arrangement further by introducing other sampled sounds, building a more complex texture and more synchronization points between the soundtrack and significant or dramatic movement in the animations. The goal was to enhance the sense of synchronization between the soundtrack and the animations, by introducing synchronized accented sound effects at key dramatic or movement points in the animation. So, in effect, there is a usually a “bed-track” created by improvising with keyboard pads, which creates an overall sense of atmosphere and mood, and over-layed episodic, “event” sounds which attempt to place animation events in a representational “space” by attaching an analogous or fantasy sound effect.

In sections 3 and 4, the improvised keyboard elements are shorter episodes cut from longer improvised pieces, and these sections have a somewhat greater emphasis on collage, utilizing a very large library of specially chosen sound samples which I have collected over the number of years that this project took to complete, both by purchasing sample CDs and by downloading from free sources on the internet. Composition using layered collages of improvised keyboard episodes and sound samples were achieved within Adobe *Premiere*. I selected the sampled sounds intuitively, trying to avoid repetition and to build a sombre orchestral texture which was sublime and transcendant, and as far as possible not genre-specific. I strove for a balance which enhanced the animation by connecting sound with movement, conscious however at all times of the danger of “mickey-mousing”. Because of the “deep space” nature of large sections of the animations, an “other wordly" sci-fi type of atmosphere was most suitable, if not inevitable. In response to the healing aspect and feminine qualities of the paintings I was prompted to to make a feature of female voice samples (especially Part One), to introduce a 'humanizing' element among the grandiose sc-fi textures.

By way of contrast. I purposely set out to make the musical sequence for the credits sequence cheerful, up-beat and in the nature of a "wake-up call" from the long sections of hypnotic, ambient or sci-fi texture in the body of the work.

By the time I got down to composing the score I was deeply enmeshed in the challenges of three part-time university teaching positions. Also, since adolescence I have worked weekends as a professional musician, and it was necessary to sustain this in the lifestyle mix also, because of the income breaks that part-time workers must address. (Though I would continue to play music in any case). Hence it was frustratingly difficult to get the sustained periods of concentrated work required to fulfil my own ambitions with regard to the soundtrack. Ultimately I think the soundtrack is pragmatic, with some very successful sections. So, as for many such projects, lack of development time and budget were the major limitations constraining the work. My arsenal of electronic sound generating devices was somewhat "retro" (i.e. dated), so the production values are sometimes obviously low-budget. I choose to enjoy the cheesy sci-fi effect. In an ideal world I would have composed and recorded an orchestral score and perhaps employed an orchestrator, or at least had more input from live musicians.

Working as a one man media production operation (including manufacture) has been challenging, but inevitably I have gained much insight into digital art processes/practice from the experience and it has informed my tertiary teaching in an on-going relationship.

Software used for Composition and Audio Processing:

- Emagic *Logic Audio*: MIDI sequencing. The main use of this programme was to make MIDI recordings of improvisations created while viewing edited sequences in order to be able to produce re-orchestrated variations on some of the thematic material that was produced.

- Sonic Foundry *Soundforge 4.0* and *Peak 2.0* : sound editing/dsp.

- Sonic Foundry *"Acid"* : Loop-based composition. A small amount of music was produced using this software, most notably the popular style techno track used during the credits.

- Propellerhead's *Reason 1.0 - 2.5* : Some additional compositional processes were undertaken using this application at the end of the composition process, to fill in gaps with the idea of introducing a different sort of sound.

- Adobe *Premiere 4.2 - 6.0* : final layering of sound – quite a lot of the music composition was created directly within this application by collaging together sounds from the large library of especially collected sound samples.

13. Transferring to the DVD format

The closing stage of the project – authoring a DVD version of the project - proved to be another frustrating and drawn-out technical challenge. The competing formats and confusing and contradictory claims in the literature made it very difficult to develop a confident strategy. When finally I naïvely purchased a Pioneer DVD-R format DVD burner, which came with a very inadequate authoring software application, it initially seemed to produce a rather poor quality DVD-viewing experience. I came to the incorrect conclusion that the format I had chosen was the problem and later purchased a Sony DVD burner, the "super-drive", which can burn both DVD-R and DVD+R media, and also handles re-writable media. I was dancing to marketer's drums, but with one advance, at least it made the testing process considerably cheaper (as burning two "test" non-rewritable DVDs consumed $20 worth of media and would have ultimately run into hundreds of dollars.)

The mpeg compression process produced many unexpected outcomes in processing the animated digital video files. For example, mpeg compression frequently seemed to lighten the black sky backgrounds in the "deep space" sequences, giving them a milky, dark-greenish quality and revealing a tiling effect as the compression processed pixel data into a lesser range of colour values. This necessitated painstakingly reviewing the DVD playback, and returning to the original video files and cutting them into sections, so that contrast could be increased via dsp, necessitating another round of time-consuming test-rendering/review. This resulted in an uneasy compromise in every case, and though visually an improvement, giving a deeper black and reducing the aliasing effect. This compromise didn't work for the client, who then decided to abandon the DVD version and asked me to return to a VHS tape format. This decision was later reversed, and further research into mpeg cpmression ratios did finally aled to the production of a single DVD disc containing the four animated sections and the other interactive options (Intrdouction sequence, Duane's message, slideshows, and credits sequence).

However, just when I was so close to completing this final stage in the production of the DVD version for this PhD submission, I experienced the ultimate, dreaded computer disaster – total break-down of my computer system without adequate back-up of the data. Some of the final versions of the music had been saved to CD fortunately. I had purchased an Apple E-Mac 1.25 gigahertz computer running OSX, in order to complete the Macintosh version of the interactive CD-ROM *The Machine*, and now, while waiting for a resolution of the warranty issues and to set up a new Windows-based machine, I was forced to return to the Macintosh platform and painstakingly piece together a new DVD from various mini-DV tapes. I built a new version of the DVD using Apple's *DVD Studio Pro* authoring package. Though this was a painful experience to have to endure at this late stage in the production process, it was not without its rewards, as this software gave me greater control of the quality parameters producing a far superior result to the output I had previously achieved using Sonic Foundary *DV Architect* on a Windows platform. The latter software only allowed the option of rendering the slideshows to video tracks, which consumed a considerable portion of the available DVD data capacity. Using *DVD Studio Pro*, the user has the option of not rendering the slide shows to video, freeing up considerable space on the DVD which allowed for a higher compression rate and thus better quality images in the mpeg version for DVD. The audio was compressedusing DVD Studio Pro's "A-Pack" application. The DVD disc was created using a variable bit rate, two-pass encoding, with a maximum data rate of 7 megabits per second and a minimum of 6 megabits per second.

To conclude this section: the long and winding journey which led to the creation of the final DVD was at once fascinating, challenging, exhausting, frustrating, rewarding – as no doubt many major artistic undertakings are. There are many small details within the work, both visually and musically, that I would like to change or improve. Many of the small flaws that I perceive in the work are a result of technical limitations, which in turn were a result of budgetary limitations, as this has been a self-funded project (apart from the fee paid to me by the client, before the project blew out into its final epic scale). The most difficult challenge was to bring this huge work to a conclusion.   I must acknowledge that the opportunity to freely work with these images was a great privilege. Despite its flaws, there are a great many sections that I enjoy and feel achieve what Duane set out to achieve.

The breadth and depth of the learning experience always justified the project, as it kept me at the cutting edge of a wide variety of software applications as these evolved, and which in turn enriched both my creative and teaching skills in this field.

2.2 *The Machine* – an interactive art experience presented on CD-ROM.

Background to the experimental work

The final component to this PhD submission is an interactive CD-ROM entitled *the Machine.* This is a symbolic/satirical multimedia piece which reflects or embodies many of the ideas considered in this dissertation .

The user encounters a virtual mechanized factory operated by our imaginary hosts, Acme Corp. The purpose of this factory is enigmatic. This work is labrynthine and of curiously diverse nature. There is meaning and there is playfulness. It is multi-layered multimedia art, containing a synthesis of media, presented in the visual, auditory and textual channels. It is interactive – each individual user's engagement with the ideas expressed through this non-linear series of multimedia animations will be unique. The virtual world depicted exhibits decay and a patina of corrosion. It is infested with virtual vermin. *The Machine* is conceived as having an artificial intelligence, and various efforts were made to create the illusion that this intelligence is "alive", has a personality, and interacts with the user. The work is influenced by a post-modern sensibility, and is not meant to be taken too seriously.

The Machine began as a tool for instructing university students in a variety of multimedia design and programming techniques using Macromedia *Director* (and the proprietory object-oriented programming language *Lingo).* I use it to demonstrate multimedia concepts and, because of this, I instinctively went for a satirical, sometimes shocking approach, to make it (hopefully) more engaging in a teaching context, deconstructing my work techniques for students in class. As it developed however, it began to take a shape of its own as a practical creative experiment in interactive ideas to accompany the theory outlined in *Part One* of this PhD submission, regarding interactive art. It explores a range of possible interactive genres and is rich in image, sound design, animation ideas and interactivity. I have always admired the work of the Surrealist artist Max Ernst. I have been influenced by the surrealist collages which Ernst pioneered, in building the abstract and conceptual pieces on this CD, which in many cases are really animated collages, with a surreal flavour. These collages use a combination of animated gifs, traditional *Director* bit-map animation (using co-ordinate transformation, scaling, rotation and simple colour animation) plus animated *Flash* cartoons. The latter include their own separately constructed soundtracks that allow extra layers of sound to be mixed in with the two available tracks in the *Director* sound design, as it evolved during production. This was always an experimental process of exploring the logical conclusions of interactive and animations ideas, as they came to me.

2.2.1 Speech Synthesis

The opening interface reveals (once the on-slider is activated) that *the Machine* has a controlling intelligence that can speak to the user. Its ever-present eye tracks the user's mouse movements. *The Machine* project focuses significantly on an exploration of the creative potential of the speech synthesis capabilities of recent versions of Apple *Macintosh* and Microsoft *Windows* operating systems (now controllable within *Director,* using *Lingo)* allowing an interesting text/speech channel to be incorporated in the multimedia mix. In *the Machine,* rollovers and mouse clicks are linked to a series of pre-written scripts "voiced" by the speech synthesis capability, that are chosen randomly via *Lingo*. Every time the user returns to a particular menu or rollover, there is a good chance that a different script will be activated, giving reasonable variation in what the computer says to the user, and hopefully allowing the user to return to *the Machine* without repetition of the voice synthesis scripts causing a rapid reduction in interest[1]. This approach also created the opportunity to play with developing a character for *the Machine.* The speech synthesis facility for both platforms are, of course, North American in origin. It was an on-going challenge to subvert the American accent by experimenting with phonetic spelling of words, in

[1] **Note:** To fully explore the voiceover scripts it is necessary to allow the voice synthesis elements to "play out" once activated by the rollover, before exploring the interface further, i.e. if a voice-over is activated, one must keep the mouse relatively motionless until the voice over has played, if the full text is to be received. It is not intended that a regular user *must* take this approach however. The random chunks of embedded script that occur by chance, as mouse movements are made while exploring the interface, produce an equally interesting (though obviously more fragmented) result.

order to avoid some of the (to an Australian ear sensitive to American cultural dominance via media channels) more irritating “Americanisms”. It was also interesting to make *the Machine* character a little cynical and to experiment with vernacular and ways of introducing variations in the pre-recorded voice patterns, in order to distance *the Machine* character as much as possible from its bland, stereotypical sci-fi monotone, and to lighten the whole multimedia experiment with a little humour. Hesitations, stutters, clearing of the virtual throat, and care with the placing of commas and question marks (which affect the timing and sometimes the pitch of the synthesized playback) were utilized to vary speech patterns and to help build a more entertaining character[1]. Here is an example of the Lingo script for the Machine’s voice, chosen randomly from the following three possibilities.

```
on mouseup
  puppetsound 2, "E-TIMER"
  n=random(3)
  if sprite(33).loch>305 and n=1 then

    voicespeak ("Oh, um, well, arrm, Gweetings and, um, like, gosh,  hello.
Ahm, if one may introduce oneself, urm, you may as well call me Machine, I
don't care, something like that will do, anyway. Your User status is,
v.i.p., recreational day tour visitor. Right you are then. I've got to do a
sort of, um, formal introduction now, so If you get a bit impatient try
splatting some of those, bloody cockroaches, but I've got to sort of do my
arm, introduction thing, you know? I take a few minutes, um, to arm, warm
up. Yo. I am, like, sort of self-programmed to arm, guide you on this
journey through my, well my mindscape, or that is, my, ahm, brainsphere?
Cool? I mean, it's meant to be fun in here, got that? You've read the big
fat book, now enjoy the movie! OK? Sorry if I, um, repeat myself sometimes,
I'm, like, a machine, you know, and the computer to um, human interface
bizzo is always a bit, you know, dodgy, don't you find? Hey, lighten up
dude, it's got to be like, fun, is that a deal? I guess You may as well
proceed now by, ahm, clicking on, you know, the main, ahm, conveyer belt
power switch thing?  and we'll have a look around the acme intelligent
machine, ok? oh, and by the way, Cockroaches have souls, too, you know, so
while I'd appreciate if you keep them out of my eye, please don't be
unnecessarily cruel with your mouse, alright?")

  end if

  if sprite(33).loch>305 and n=2 then

    voicespeak(" Oh, err, Hi, ahm or that is, hello. Like, ahm, I am
Machine, or that is, you can call me Machine, if you like, I mean what's
the difference what you call me?  Really? Your user status is, v.i.p.,
recreational day tour visitor. Cool. Well, um, I'm supposed to give an
introduction here, so you could probably chase some cockroaches, while I
get it over with, if you get bored, or what ever, ok? I'm supposed to, ahm,
sort of be your tour guide, through, you know, my brainspace, yeah. It will
be an experience, so keep your wits about you, and remember to breathe.
There is a manual, but it's about as thick as a telephone book and, while
it does in fact have a lot of pictures, let's face it, it's very long-
winded and pompous so I suggest you just go with the interactive diversion,
ok? So, hey, let's do it! Right then, ahm, you might as well, ahm click on
the main conveyer belt power switch thingo and I'll take you on the
```

[1] If the user holds the mouse down with the cursor positioned over *the Machine’s* eye, exaggerated variations of vowels sounds are triggered. Those that rise in pitch are created simply by placing a question mark at the end of a string of vowels e.g. aaaaaaaaaaaaaaaaaaaaaaaaaah? (*Windows* only)

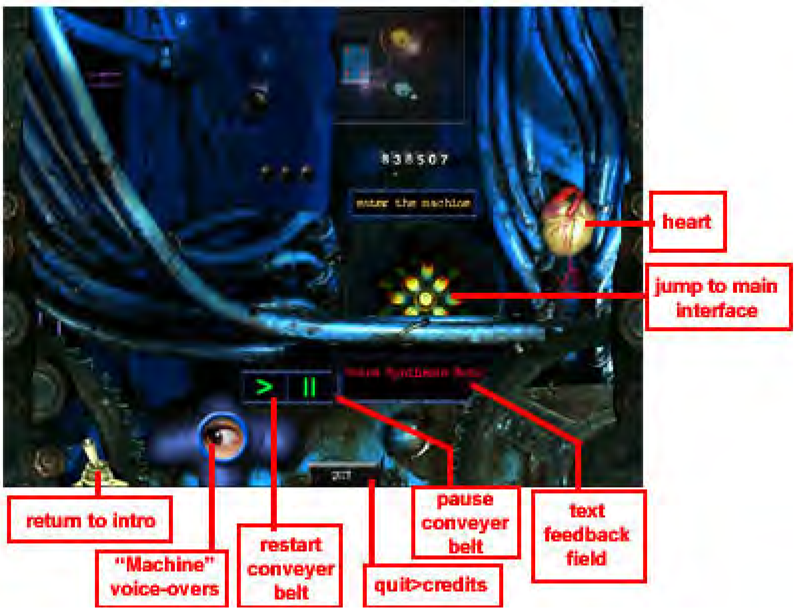
```
official guided tour of the acme corporation's contemporary cutting edge machine, ok?. Are you with me friend? Let's go then. Hay, Click away.")

  end if

  if sprite(33).loch>305 and n=3 then

    voicespeak(" Oh, arm, gosh, like, Hello Mouse ka tear , welcome, arm, cool, I guess, like, good of you to drop by. Your User  status is, v.i.p., recreational day tour visitor. Excellent. So here is the formal sort of intro thing, which I really don't enjoy much. I, um, suggest you splat some of those frigging cockroaches, while I get the intro blurb bit, out of the way, OK? Oh, and, um, like I'm sorry if I repeat myself sometimes, my maker was a bit of a, you know, sloppy, um, programmer, you know how it is? Well, arm, I'm sort of the master of ceremonies around here, if you know what I mean, so, arm, get that ol mouse finger ready and, hey. wanna look around some of my brain? I mean, you've waded through all that passive linnyarity of words, let's have some fun. Interactive Multimedia is meant to be fun, isn't it?. Or did I already say that? Super Cool. Let's, like, go then? I'll show you around the acme corporation's latest marvel of artificial intelligence and bionic programming, ok? Well, click on the main conveyer belt power switch, yes, yes, that one, in the middle, ho hum, oh, hurry up. and shit, stop harrassing those cockroaches will you, man, like, stay focussed. ok?")

  end if

end
```

As the Microsoft *Windows* OS has but four available voices (compared with the *Macintosh's* 22), and the two competing software solutions are by no means identical in the results generated, much testing was carried out in order to develop a reasonably consistent cross platform result. Voices can be selected using the *Lingo* command **voiceset ( ),** with a voice number being placed within the brackets. Unfortunately there is no consistency between voice numbering systems on the two platforms.

The addition of the voice synthesis component into the multimedia "mix" of *the Machine* opened up the possibility that the spoken word channel could become a central aspect of the experience, rather than being limited to a minor contribution (if restricted to recorded sound activated through the *Director* sound channels and, for example, *Lingo* **puppetsound** commands). The user will be very unlikely to discover all of the hidden mouse-activated comments *the Machine* can make, even if consciously seeking to discover them. The voice-over layer evolved into a dominant element of the production, adding a rich and complex new element of satire and cynicism, which is fascinating in the way random snippets of the voice-over script options overlap and merge, as the user explores the interface. But of course, it is a novelty that can rapidly lose its appeal, and I would hesitate to take this approach again, without the available speech synthesis software going through substantial expansion and development.

Themes, interface design and structure

On opening the *The Machine* the user confronts a cockroach-infested switch interface. Mouse movements bring the sleeping *Machine* intelligence to life. Activating the start-up slider-switch induces *the Machine* to make an introductory welcome speech. *The Machine* is very sensitive to its inferior status, and yet also very proud of its logic and consistency. It finds humans irritating and somewhat sadistic, but tries to be polite. It reacts to poking the cursor in its eye with pain and indignity. It has a lot of trouble with cockroaches also. The cockroaches in turn flee from the cursor, knowing the evil human as an enemy. The cockroaches can be eliminated, one by one, if the user is fast enough in the execution of co-ordinated hand/eye motor-skills via the mouse-button.

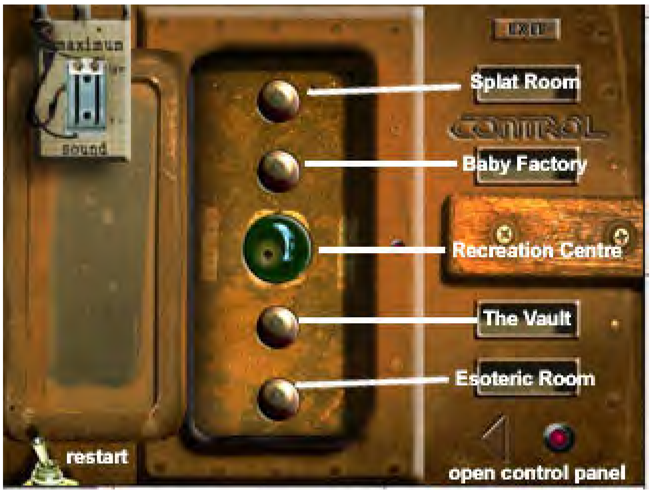


On activating the conveyer belt switch in the centre of the introductory screen, the user progresses to a dark, virtual factory inhabited by a staff of invisible workers, and controlled and monitored by *the Machine* intelligence. It is a mechanized world, but contains organic components including, for instance, a beating heart. The user confronts a conveyor belt, producing unknown production units. I visualize these as being some kind of digital meta-art entities, referred to in this thesis, in the discussion of Herman Hesse's *Magister Ludi* (p. 140), though this idea is not necessary to the experience. The use of the portable network graphic (.png) file format, with its excellent alpha channel control of transparency, allowed the introduction of separate animatable shadows, which gave a stronger feeling of 3D depth to the whole project. The new 3D capabilities of *Director* were not accessed for this project, as the learning curve is steep and long, the *Lingo* verbose, and, attractive as it is, to introduce it would have added greatly to the production time. (There is however minimal use of 3D text elements within the sub-movie *About Colour* accessed from the *Recreation Centre.)*

The next interactive step requires the user to discover that the rotating hub reveals itself as an interactive hotspot, which takes the user through one of three linking scenes to the main control panel (or menu). Again the navigation pathway is chosen randomly using *Lingo*. The simple *Lingo* code for this is:

```
On mouseup
Go marker (word random(3) of  “eye animachine transistors”)
End
```

(with “eye animachine transistors” being the names of 3 separate marker points along the *Director* timeline, containing three different animation sequences, and different voiceover scripts). Again, introducing variety and avoiding predictable repetition was the objective. This randomization technique is used when navigating between each of the three main menus or *Control Centres.*



The user now arrives at the main interface (*Control Panel 1*), which among its options allows the user to interactively control the volume level of the music/sound effects channel. The decaying, rusty metal is used as the main motif in the interface design, consciously avoiding a slick sci-fi, computerish look. From this menu the user can access:

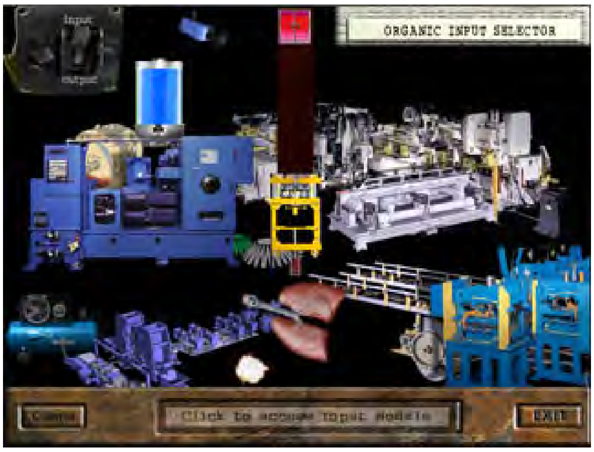
1. *The Splat Room*
2. *The Baby Factory*
3. *The Reacreation Centre (Control Panel 2)*
4. *The Vault*
5. *The Esoteric Room (Control Panel3)*

and can

6. *Quit* the Program

Each of 1-5 interactive options available from *the Control Centre* is accessed by an interactive controlled, animated journey through a labrynthine network of virtual tunnels. There are randomly chosen "voice-over" speech synthesis scripts triggered by rollovers on each of the interactive buttons and the drop-down sound control slider.

The Splat Room has a window (left), with a view of the alley at the rear of the Acme Factory. It functions as a very basic send-up ‘twitch’ game called *Splat*. Working against the clock, the user user must shoot down a swarm of genetically modified locusts. It is fairly difficult *not* to win. The intention is to add other games here, one of which is nearly complete but was not ready to be included in this prototype.



The Baby Factory (above) is sub-section of the Acme *Machine*, a rather dark destination, obliquely satirizing the human obsession with controlling life through such scientific techniques as genetic modification, cloning, artificial intelligence and robotics. This section thus also draws on the *Frankenstein’s Monster* tradition in literature and cinema - the paradoxical fascination with, and horror at, the ramifications of tinkering with creation. The randomized speech synthesis creates an especially complex, ever-changing satirical voice-over, as the user peruses the equally complex animated collage which constitutes the input module (above left). Here the user can use the drop-down *organic input selector* to switch between :

- soft tissue input,
- skeletal input, and
- vital organ input, or
- suspend input

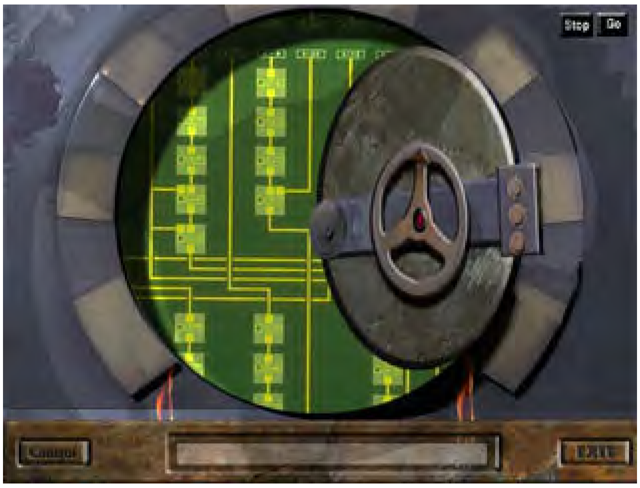
modifying input to the *Acme Baby Factory* production line. In the output module (above right) the user can select (using a row of toggle switches) a baby type from:

- the standard model,
- the realist model,
- the optimist model,

and also choose

- intelligence boost, and
- “coolness” boost,

allowing for a customizable baby. Perhaps the meta-art entities/objects produced within our Acme meta-art machine are actually a superior post-human synthesis of culture, technology and organism? While the ideas explored in this section are promising, the execution has not come up to expectations, and will be developed further with time.



The Vault (left) is deep in the basement of *the Machine*, a storage facility for a linear collection of surreal and experimental looping animations, (some created with Macromedia *Flash*) with a satirical and political slant. Every factory, even a meta-art factory must have a storage and archival division.

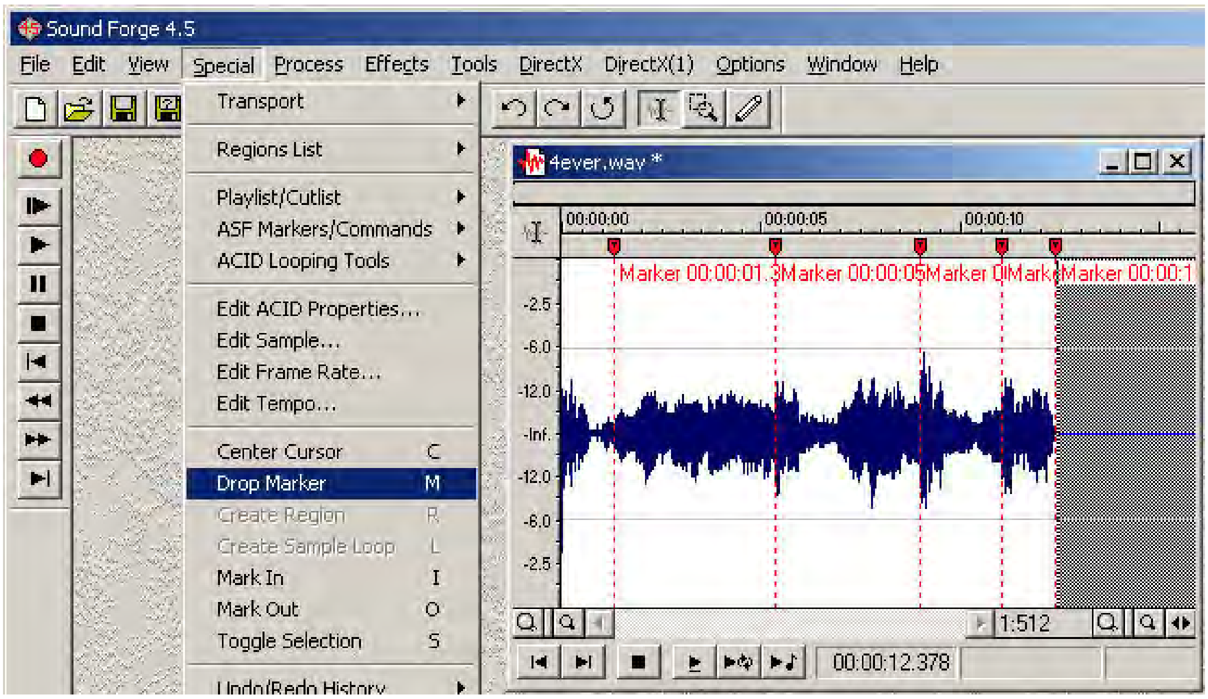
The Esoteric Room (Control Panel 3, right*)* is high in the upper levels of *the Machine*, and allows access to a place of light and inspiration, arranged as a network surrounding a visual portal. It was created as a counterpoint to the rather dark vision presented throughout the rest of *the Machine,* and makes a marginally more positive philosophical/poetic comment on the human condition.

The major secondary sub-node of *the Machine* is *The Recreation Centre*, which develops the "virtual factory" metaphor further, by offering virtual factory staff within *the Machine* (and visitors) some diversion during production breaks, in order to raise morale and increase production. From *The Recreation Centre* the user can access:

1. The TV Room
2. The Staff Room
3. The Child-Minding Centre
4. The Random Insult Generator
5. The Pointless Exercise Room (the Anagram Machine)
6. The Staff Education centre

The Staff Room gives us a glimpse of office romance. Here I began to explore a simple 'comic book' style of animation, which I also use for teaching students about storyboarding cartoons and creating animatics. The intention for this section is too create randomly chosen variations of characters on each loop, (as the Acme Factory is a rather promiscuous workplace). Only two variations have been produced so far. The stills for the first scene were created using Curious Labs *Poser 4.0*. (see p.266), (left, top of next page) and for the second, *Poser 5.0*. (see next page, right). They offer a clear example of how software development can empower the artist to producea more sophisticated result.

Technically this section also demonstrates to students an example of using non-audio markers embedded in a digital audio file as synchronization points between audio and visual animation events in *Director*. *Director* handles audio through the operating system audio capabilities, while controlling graphic animation within the stand-alone projector. There traditionally has been a danger that (depending on available RAM and the speed of the central processor) audio and visual animations created on the *Director* timeline can get out of "sync" during playback. By embedding marker points in the audio files and programming *Director* to wait for these to be read before moving the playback forward at crucial points in the animation, the multimedia author can ensure tight audio synchronization when it matters. The markers can be inserted into the audio file using standard audio editing applications such as *Peak* on a *Macintosh* and *Soundforge* on a *PC,* as illustrated below.



The Child Minding Centre allows access to a children's cartoon (created in Macromedia *Flash*) entitled *Emutown*. This is presented as a linear sequence of looping animations, allowing the user to control the duration of the experience of each linear "module" in the story. (See p 363 for a discussion of linear structure within an interactive story). The *Lingo* **voiceset()** command was used to change the voice synthesis element to a female story-teller. The following images are example stills from this piece:

Emutown is a work in progress, but enough of the story is sketched in to suggest / demonstrate the idea. The web-address www.emutown.com has been registered, with the intention of making an on-line version, and this pilot uploaded for testing.

The Anagram Machine is simply a diversion for factory workers within *the Machine*, celebrating this intriguing phenomenon hidden within word structures, which this author finds fascinating and amusing.

The Random Insult Generator allows factory workers (or frustrated users) to let off steam and express their anger at dealing with authority. It uses random access to lists of words based on the sentence "Go … yourself, you …, …, …, ...". Each of the spaces is filled by a random selection from a list of 50 appropriately insulting / descriptive / scatological words. It is possible in this manner to generate 312,500,000 (50^5) unique insults on the click of a button (explicit language warning), a *Lingo* exercise students find amusing. The following is the lingo script for this:

```
on mouseUp
  puppetsound 2, "fart2"
```

```
  puppettransition 51
  put "Go" into field "insult"
  put " " & word random(55) of "on,-make-an-idiot-of on,-a-round-of-
applause-for screw disappear on,-delude on-inflicting throw-up-on on,-suck-
up-to on,-big-note smell abort defile pander-to fumigate on,-torture-us-
with-a-dose-of defenestrate take-a-good-look-at swallow defecate throttle
choke applaud pre-digest masturbate play-with fornicate-with excrete on,-
congratulate regurgitate self-medicate bury mollycoddle wet fall-in-love-
with fossilize noodle interview disconbobulate deodourize delouse flatter
disinfect preen ogle fuck fondle root wank strangle stuff piss-on crap
dismember shit worship decontaminate" after field "insult"
  put " " & "yourself," & " " & "you" & " " & word random(52) of "peurile
nauseatingly lukewarm malignant poisonous slippery preposterous noxious
fetid turgid dissipated smug rotting puny monstrous ghastly mangle-headed
diseased loathsome ulcerous dirty schmoozing  ruthless shifty  blatant
cretinous obsequious puke-dribbling charmless irksomely repulsively maudlin
bloated  defective  stolid stupid lazy dreary  useless disgustingly
gormless revolting ugly greasy foul-smelling gutless gruesome pathetically
jaded evil morbid hopeless"  after field "insult"
  put "," & " " & word random(50) of "putrifying  primitive frowsy brutish
cloying demented noisome septic posturing clueless insignificant festering
virus-infected insincere self-pitying  two-faced shabby no-good deceitful
snakey relentlessly-self-absorbed heartless rorting shameless fart-faced
pimple-nosed unfeeling unmoving insidious negative acne-cratered
unscrupulous mutated one-sided ill-informed short-sighted self-obsessed
fucking pox-infested sleazy conceited suppurating  self-opinionated
narcissistic self-indulgent groveling tasteless crawling insidious
spineless"  & "," after field "insult"
  put " "  & word random(50) of "parasitical burbling slagging vain
cancerous life-leeching mediocre pooh-reeking dumb-ass fat-assed gangrenous
unendurable child-molesting vomit-featured excuse-for-a sycophantic lard-
brained rancid brain-dead goop-oozing social-climbing  malodorous slut-
licking bottom-feeding obsessive-compulsive low-life craven  alcoholic
depressing dysfunctional shit-for-brains pussy-footing sewer-breath spew-
breath sexist racist yuppie dribbling pig-humping  egomaniac constipated
jerk-off lying no-class pus-infested squirming  semi-retarded mother-
fucking hairy-backed festering ass-licking" after field "insult"
  put " " & word random(53) of "bucket-of-spit philistine fiasco
offspring-of-necrophilia piece-of-offal joke cockroach-dropping reject-of-
failure-school plague-of-boils bucket-of-shit load-of-rubbish case-of-
worthless-crap curmudgeon dose-of-rat-chunder nobody vermin colostomy-bag
pustule germ insect piece-of-shit goats-penis gutter-trash  scum-bag sheep
goose-turd  toad biomass couch-potato disaster flop over-compensator
bastard moron pervert air-head loser weasel dickhead dork maggot slurry-of-
turds mongrel twit nerd slob creep blood-sucker worm slug prick fuckwit
deviant" & "!" after field "insult"
  put word random(19) of " a s d f g h j k l p o i u y t r e w q" into
field "squarehead"
  put word random(19) of " a s d f g h j k l p o i u y t r e w q" into
field "seal"
  sound(1).volume=60
  voicespeak(field "insult")
end
```

The lists of letters at the end are randomly changing images produced by using image fonts.

The Education Centre offers virtual factory workers a chance to broaden their minds with an interactive sub-movie entitled *About Colour*, teaching the user about the use of colour in design, one of the universal set of principles underlying any work of visual art, and therefore any meta-art product (see p.180 and p. 188). Technically it is an example of opening another *Director* movie from within a *Director* projector. It is also an example of instructional multimedia and is therefore the only "serious" section of *the Machine*. This was an interesting and challenging project in its own right, especially because I planned to make a cross-platform version. It comprises of many experiments in interactive animation, and this piece also employs voice synthesis.

The TV Room (left) allows access to a virtual TV for relaxation, with a free-to-air section and a pay TV section. You must turn on the TV using the virtual remote control. Each TV channel presents a comment on a media or television genre. Some of these are *Quicktime* videos, some collage animations based on animated gifs and *Lingo* generated text, and still others Macromedia *Flash* cartoons. Most are short looping pieces. Many contain random variations. These animations were carefully composed so that the random elements flowed seamlessly to make a textured, everchanging sound composition. There are 9 channels on free-to-air *GenreTV* (simply because in creating the *universal remote* I arbitrarily built in 9 buttons, and I then had to populate them with animation ideas! The resulting short pieces have been developed to various levels, but are really a set of simple of animated collages, set up as "gags".)

Channel 1 is the adults-only channel, "off air" at the moment, but with some advertising. It incorporates an oblique comment on internet pornography and censorship. From a musical point of view, this is an interesting example of looping composition and should be allowed to run for several minutes to experience the seemless flow of musical layers and loops, accompanying the randomized visual animation loops.

Channel 2 is the ambient/relaxation channel and features a virtual fish-tank, for meditation purposes (the idea of meditation tapes and meditating to an electronic device always seemed a dubious one to this writer).

Channel 3 features a light-hearted take on self-improvement programmes, entitled *Yoga for Beginners (*see image above*)*.

Channel 4 is the Open University channel broadcasting a feature entitled *the Psychology of Meta-Art.* Here we have an example of a randomly varying selection of key text lines being voiced by the speech synthesizer.

Channel 5 is the cartoon channel.

Channel 6 is the Discovery Channel, this week featuring *The Last Frog*, a brief comment on impending environmental catastrophe and unsustainable development.

Channel 7 is the classics channel with a feature on *the Great Bard*.

Channel 8 features the station I.D. for the Gothic Channel, specializing in the horror movie genre.

Channel 9 is the news channel, with a somewhat macabre twist, referencing terrorism, and once again featuring randomly triggered elements.

The pay-TV section, *EyeTV*, is accessed by placing a 1951 Australian penny in the virtual slot on the television set, allowing access to a further 9 channels. Of these:

Channel 1 of *EyeTV* features an example of the blue screen compositing technique (see p. 242). Video footage of a sea eagle, shot against a blue summer sky, was superimposed over footage shot from an aeroplane as it passed through a cloudbank. The original blue-sky background to the flight of the bird was “keyed-out” using Adobe *After Effects*.

Channel 2 is the Science Show, featuring an experimental abstract animation, entitled *Cells,* again created using *After Effects.* The soundtrack was composed in the algorithmic music composition program *SuperCollider* (see p. 340)

Channel 3 features my first experiment with the 3D character animation program *Poser* , and is entitled *Elfy’s Ballet.*

Channel 4 and 5 feature respectively two untitled pieces which are also experiments in abstract animation using *After Effects,* and fall within the genre of kinetic animation explored in *Part One* (see p. 220).

Channel 6 is a comment on common media marketing approaches in print and television which exploit gender stereotypes of glamour and idealized female beauty. The animated distortions of female “supermodels” were created using MetaCreations *SuperGoo*.

Channel 7 is the sports channel, continuing the theme taken up on Channel 6 regarding gender stereotypes and sexuality. It uses randomization of key-words, which appear over the animation.

Channel 8 offers advertising for the *Acme Corp. Baby Factory,* featuring a graduate from the factory speaking words from the Great Bard. It was produced in Curious Labs *Poser,* composited in *After Effects.*

Special Feature: In Ya Face

Channel 9 of *EyeTV* features a Special Feature – a 17 minute mixed media animation entitled *In Ya Face : Episode One – Two Headz are Better Than One,* starring the fabulous Head Brothers, Boof and Dick. This piece is a satirical cartoon for mature audiences, created with some input from a colleague (musician, vocalist and jingle writer) Ian Mason, who wrote some additional dialogue and was the voice artist for most of the main characters, and also contributed the satirical theme song *Fucked Up,* performed in the Flamingo Bar by the cartoon cabaret artist *Mr. Mayhem, (*a caricature of Mr. Mason himself).

This cartoon, while a major experimental piece in its own right, must be regarded as very much as a work in progress, a rough draft or pilot version. It was developed after Tony Skinner, a seasoned producer at Channel 9 Melbourne[1] was given a copy of a showreel of my experiments in animation and had viewed my inya-face.com web-site, both of which I had produced while I was working at Bond University. Mr. Skinner encouraged me to develop an idea for a television pilot out of some of the sections he found commercially appealing, with the idea of producing a short trailer and using it to sell “the idea” or raise funding for a series production. He saw it being marketed for pay TV in Europe. He liked a sequence I had created using a very primitive cartoon seagull flying through a variety of strange locations, and some other experiments with two male characters created in MetaCreations *Poser, (*as well as some of the wackier ideas on the iny-face.com site). While these animations were

[1] Tony Skinner had made contact with me at a time when he had responsibility for developing Web TV for the Packer network, before the Howard Liberal Government effectively legislated data-casting out of existence.

not originally conceived as a unified story, in fact had nothing to do with each other, Mr. Skinner is a highly regarded and very experienced television professional, and for him the rough-and-ready, low-tech look, and irreverent zaniness with no regard for genre or style, had something a little different to offer. He was convincing in his argument that creating something out of this material was a worthwhile effort. This is *very* limited animation! (see p.238 for a discussion of limited animation techniques) and for instance, the two central characters, Boof and Dick, soon discover they can fly (because walking is much harder to animate convincingly!) I made several incremental attempts at cutting the original animations into a semblance of continuity and then, with the assistance of Ian Mason, improvised some dialogue which sowed the seeds of the characters which eventually developed out of this experimentation. Mason’s input brought a crude and blatant, but boisterously irreverent influence to my more quirky and surrealist style, (though I was never comfortable with the sexist connotations).

For my own interest I then decided to try to put together a complete 17-minute “episode” (for a 30-minute time-slot, minus commercial breaks), in order to brainstorm ideas and get a better feel for what was implied by the sketchy story which emerged from this preliminary exercise. The final pilot cartoon was made by editing the various sections together and trying to build on and elaborate a story, gradually adding new sections when ideas occurred, or when it was deemed necessary to move the story forward – a sort of reverse-engineered script-writing, which proved to be an interesting and lateral way of working.

Boof and Dick Head are somewhat down-and-out brothers, in the “rough diamond” mould. Boof Head is a confident, optimistic motivator (with a touch of ‘punk’ about him) and Dick Head a laconic, naively idealistic loser (who wears a *drize-a-bone* coat and *akhubra* hat). Of course it is easy to see them as a sort of Ocker (colloquial Australian) *Beavis and Butthead* characters of *MTV* fame*,* but with development, that outcome could be avoided, as seeds of their own unique personalities have been sown here*.* They pass through the ups and downs of each episode as rudely cynical, yet ultimately somewhat gullible observers of the seedy side of urban life, who enjoy the colourful quirks of reality and altered states of reality. A song was added at Mr. Skinner's request, and my collaborator Ian Mason at that time had just recorded a demonstration tape of a rather appropriately stupid song. This I reluctantly used because of time constraints, and eventually by default it became the theme song in the final sound mix.

Story Summary: Boof and Dick wake in their dingy apartment after a rather heavy night. Whether it's some kind of synchronized hallucination brought on by lack of sleep and strong substance abuse, or a strange supernatural psychic experience, they find themselves hypnotically drawn by the Voice of the Demi-God of Commercial Television down a terrifying psychedelic vortex into another reality, where a weird new "trip" awaits them. They soon discover another recent arrival in this enigmatic parallel world, a talking bird named Steven Seagull (see image below), with a rather obsessive personality problem and "chip on his shoulder".

Steven is a victim of a magician's trick gone wrong. He feels his youth was stolen from him. He has already become "street-wise" and becomes the Head Brothers' guide through this dark parody world, where everyone is trying to escape reality and anything can happen. Steven is a flighty bird and disappears skyward after the trio witness a suicide from a tall building. Badly needing a drink, the Head Brothers enter the Flamingo Lounge, a haven of underworld crime figures and bizarre behaviour, where the dancing is 'off the planet' and the in-house cabaret band punch out an R 'n' B beat behind the portly, decrepit vocalist Mr. Mayhem, in a funky tale of tragedy and brain damage. They also meet a mysterious temptress named Magdelena, who later, at her exotic apartment upstairs, flaunts herself in a wildly sensuous dance routine. The beautiful but sinister Magdelena slips a suspicious substance into their cocktails, sending the Head Brothers into a state of even further suspended reality – an altered state of consciousness within an altered state of consciousness. We must wait for the next episode to find out what happens to the (now so vulnerable) fabulous Head Brothers at the hands of Magdelena, and how Steven Seagull's identity crisis progresses.

Much of the animation featured in this cartoon was achieved using MetaCreations' *Poser*[1], a non-professional level 3D character animation programme which allows the user to extensively customize a library of human character and animal models and keyframe their movements. It suffers from limitations in the lighting and effects department. Hand drawn cell-frame animations are also featured, and a number of other fairly crude compositing techniques and special effects are introduced from time to time. Some of the backgrounds were created using *Bryce 3D* (see p.254).

One of the more interesting elements in the cartoon is the dialogue which occurs in the *Flamingo Bar* scene. A courting couple meet at a table in the bar and have an interminable, surreal conversation, which is interspersed with glimpses of other intimate moments happening among the cast of other misfits in the bar. I had found a stock audio CD among the 'bargain CDs' at a Harvey Norman Department Store, which contained a collection of samples of an actor satirically mimicking famous quotes from ex-U.S. President, George Bush Snr.

These comprise the majority of the lines spoken by the male character of the courting couple (left). I envisioned a conversation between two people who weren't actually listening to each other. The female's lines are a combination of lines from stock audio sample CDs of 'hip-hop' vocals, and selected lines found by trawling Internet sites offering free collections of comic vocal sounds.
After assembling all this vocal material I experimented with intercutting until I had developed a flowing conversation, again a sort of reverse engineering of the dialogue, which I then used to animate the 'lip-synced' bar sequences. The old 'drunk ' who crawls from the dance-floor to the bar, lip-syncs to the voice of another unkown voice-over artist, impersonating a later U.S. President, Bill Clinton which amused me by adding a nice consistency to the experiment.

The *Inya Face* cartoon is really an elaborate animatic, rather than a work that anywhere near approaches a professional level 3D animation, but nonetheless it has its moments. Reducing the frame size to 320x240 and frame rate to 10 fps (to allow

[1] It is now marketed by Curious Arts.

for playback from CD) caused a further dramatic drop in the viewing quality of the animation. The soundtrack works quite well, but the lip-sync is primitive, and the characters move like the puppets from the 70s cult TV show Thunderbirds. It was an experiment which gave me considerable amusement as well as a valuable insight into animation processes, and experience with new approaches to script development. It has now found its niche within the Machine's TV Room.

Sound Design

The sound design is an important element of *the Machine* and aims at creating a seamless, flowing audio experience that unfolds like a composed piece of music, as the user makes interactive choices and follows the multiple possible interactive pathways through the piece. The inherent problem posed by modular structures in interactive art is consciously addressed in this piece. Much work has been put into creating an evocative, textured, constantly changing audio atmosphere or soundscape. In this environment where looping sections of sound must be built up in order to give the user the opportunity and freedom to make interactive choices in their own time, repetition of audio loops can soon cause the onset of boredom, if not irritation. *Lingo* is used in *the Machine* to choose randomly from lists of carefully-chosen non-pitched sounds, “over-dubbed” on each looping point, to give the illusion that the soundscape is constantly, though subtly changing. Rollovers introduce the voice synthesis, and so *Lingo* is used to constantly adjust the background sound level to accomodate for this extra interactively-triggered audio event, and make sure the words are clearly audible. An audio volume slider for the soundtrack is included.

The voice synthesis element was a late addition to the audio design, and there is a rich written text element that pre-dates the voice synthesis which prompts the user from the navigation bar, and in fact, it is possible to comprehend and navigate through *the Machine* without the voice synthesis layer. The voice synthesis element allows for a much richer audio environment at relatively minimal file size cost, as the large quantity of complex dialogue options made possible through this approach, are stored as small text files within the *Director* projector, rather than as pre-recorded sound files. A final element of the voice synthesis and soundtrack worthy of discussion is the “song” which acts as the soundtrack to the ‘credits’ section of *the Machine,* which is accessed by clicking the ‘exit’ or ‘quit’ buttons. It occurred to me to get the machine character to “sing”. I experimented with creating an instrumental backing track for a song, (imported as a wave file, with markers embedded at the bar lines). I then used *Lingo* scripts to create a voice-over, that works as a vocal track for the song, so that *the Machine* is in effect singing “live”, each vocal line being triggerred by the embedded markers. A counter routine checks how many times the song has played and *Lingo* chooses the appropriate verse lines on each repeat, depending on the integer in the counter field. This song is called *Android* and I initially composed it for a musical I wrote in the early 80s called *Goodnight World* (mentioned in the introduction). I also included a version in the *Dreams and Machines* show for the Queensland Performing Arts Centre, and so it has a long history, reflecting my on-going interest in technology and art, and the lyrics remain relevant, especially in this present context. The simple and purposely "mechanical" backing track was developed using P.G. Music's algorithmic M.I.D.I. musical arrangement application *Band-in-a-Box,* using the inbuilt and very kitsch Bee Gees "style"[1]. The resulting M.I.D.I. file was then imported into Propellerheads' *Reason* and appropriate instrumental sounds were selected, and a wave file generated. Because of data limitations, the track is imported at 22.5 khz mono. The idea of the lyrics being synthesized "in real time" (rather than pre-recorded) appeals to me, although there are synchronization problems (which will inevitably be worse on slower computers) – and in fact it is interesting that the vocals seem to vary slightly every time it is played. Extra "machine" sounds are triggered or "over-dubbed" at each vocal marker point.

Technical Issues

The Machine was produced using the Microsoft *Windows* version of Macromedia *Director MX* on the *Windows 2000* platform, using a Pentium III, 933 megahertz computer and 1 gigabyte of RAM. My supervisor for this PhD project is a *Macintosh* only man, at an Apple-only university, and the inconvenience posed for him in reviewing *the Machine* as a *Windows-* only product suggested strongly to me that it may be possible that other readers of this dissertation might be in the same position, so I determined to make a cross-platform solution. Despite all the claims to the contrary made by Macromedia, this is an extremely challenging and frustrating (if not ultimately almost impossible) task, with very limited technical support literature to aid the developer. The first major problem encountered in developing the *Macintosh* version was that, while the Director project file opened as promised on a *Macintosh* version of the software, none of the native *Director* transitions worked using versions of the Mac *OS X* operating system (and transitions feature heavily in this production). This proved to be a known bug and the only workaround offered by Macromedia was to insert frames at the points of transition, which proved to be

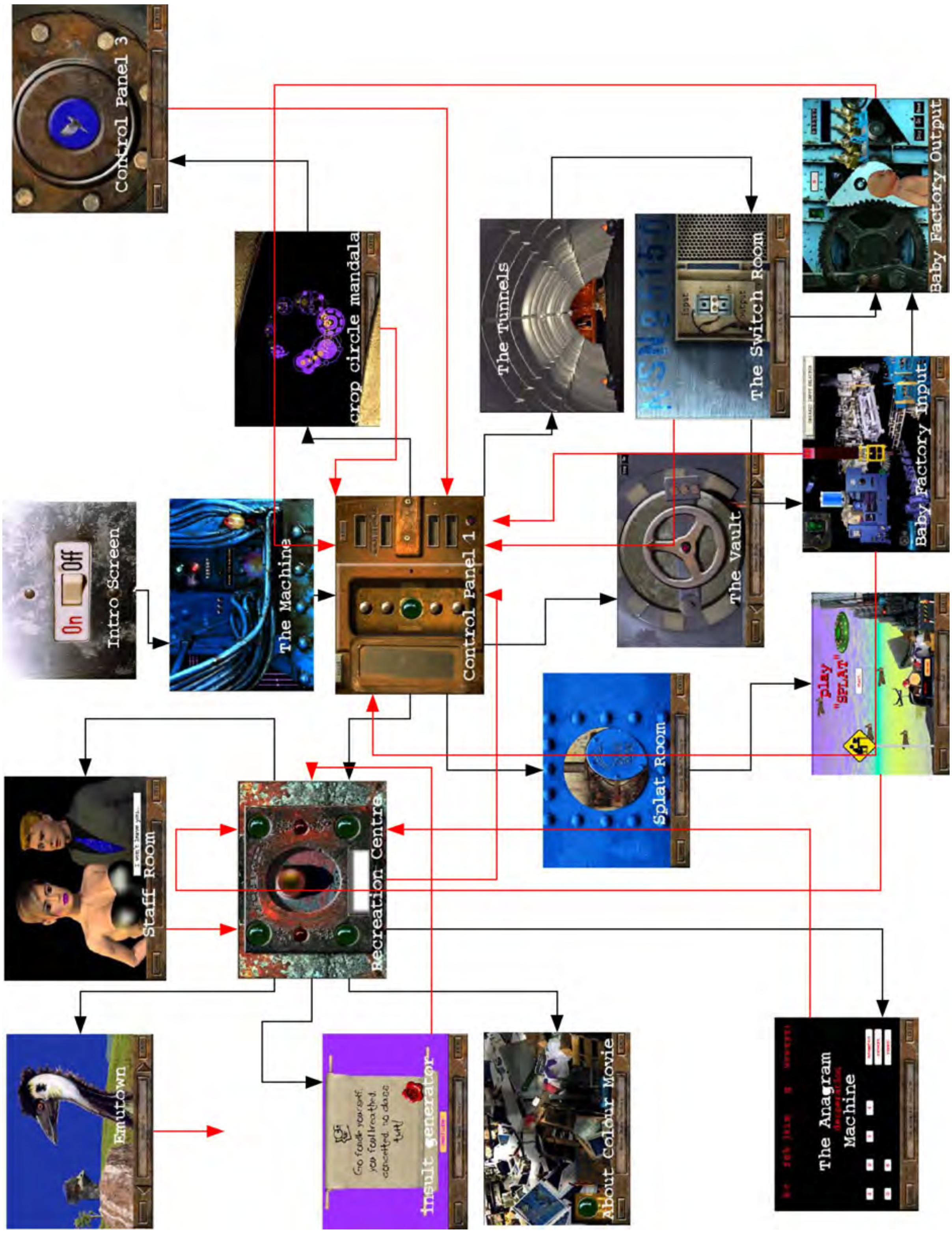
[1] Pre-programmed musical clichés are algorithmically generated and triggerred by inserting chord symbols into a musical chord chart. Parts for drums, bass piano and strings were generated and saved as MIDI files. See p.343.

structurally impossible, in the context of this particular production. Outputting as a Macintosh *OS 9* (classic mode) version projector seemed to be the only acceptable solution. After I spent a lot of frustrating time working on an OS 9 version, Macromedia released *Director MX 2004,* which can output both a *Windows* and *Macintosh* version of a completed project. This allowed me to solve the transitions problem, but many other issues arose. I spent considerable time re-developing the voice synthesis scripts for the *Macintosh* OS, as the phrasing, and timing of pauses when a comma is inserted in the script are very different to the output of the *Windows* voice synthesis engine, and also, unlike the Microsoft solution, the *Macintosh* version does not respond to a question mark with upward increments in pitch, which meant that many of the more amusing accidental effects created within the *Windows* version simply did not work, and seemed incomprehensible, requiring other solutions. Then I discovered that there seemed to be the possibility of Apple not keeping a uniform regime with regard to the voice numbering system between different versions of their OSX operating sytem, as testing a CD burnt using on the ”Jaguar” version of OSX triggered a female robot-type voice when played on the “Panther” version of the OS, which destroyed the intended effects and made the voice-over monotonous and uninteresting. Colour issues were also a major cross-platform problem, as graphics created in a *Windows* environment tend to appear lighter and seem slightly washed-out when viewed on a *Macintosh*, which detracted from the dark, mysterious look I had intended. I compensated for this where I could by darkening the overlayed transparent .png shadows, of which I had made a feature in the original version, and inserting more of these where it seemed necessary.

The Machine exemplifies some of the challenges confronting producers of interactive art. Some of the criticisms that I can anticipate are: it is esoteric, complex and labrynthine, rather than being a genre-specific and accessible, commercially-focused construct. It was developed with no clearly defined target demographic, perhaps the biggest mistake a commercially-oriented multimedia producer can make if popular / commercial success is the goal. It is possible that a user will not attempt to explore it more than a few times at best. It defies many of the normal interactive conventions and testing showed that some people failed to discover interactive nodes. There is a great quantity of diverse content produced over a long period and therefore it is unrealistic to expect that an even production standard could be maintained throughout. As there are a great many possible interactive choices and paths, the prospective user may quite easily miss crucial speech-synthesized dialogue or important highlights etc. and remain mystified. Notwithstanding these possible criticisms, *the Machine* has been a very useful experimental project in terms of teaching multimedia. *The Machine* is not meant to be a commercially released product, and so enjoys the luxury of remaining experimental, having been developed purely as a means of gaining practical experience in the issues and technical challenges that interactive art presents (as outlined in the theoretical component of this study). *The Machine* files contained on the CD-ROM should ideally be copied to a hard drive for playback, as the work pushes the limits of multimedia delivery (as intended). Playing the CD-ROM version obviously depends on the speed of the CD drive, and the user may experience pauses where data is loading, and slower transitions, with the occasional unpredictable glitch. The work requires a relatively fast processor (800+ megahertz), with a high-end video card, and ample RAM to play back as intended – for example testing on a 400 megahertz G4 Macintosh with 256 meg of RAM revealed serious problems with the playback speed of the voice synthesis, glitches in the audio, unacceptably slow transitions, and a far from satisfactory overall interactive experience – it frequently simply wouldn't play at all. A 450 megahertz G4 with more RAM produced a noticeably better, but still less than optimal interactive experience. Attempting to play *the Machine* on a PC lap-top set up for business applications was a total failure. However a much better and quite acceptable result was achieved on other more recent, though quite standard "off-the-shelf" PCs with *Windows XP* and a fast processor. The ReadMe file on the CD-ROM gives appropriate web addresses for downloading the *QuickTime* installer, which will allow uninstalled PC users to access to the *QuickTime* movies, which are primarily a feature of the *TV Room* section. The PC version of the included ReadMe file also gives the web address for downloading the Microsoft speech synthesis installer (for pre-XP versions of *Windows,* requiring 60 megabytes of hardrive space) if it is absent from the users computer. An alert message prompts the user if speech synthesis is not initialised on the host machine.

The majority of animated imagery and sound on the CD is original, but all non-original media used in the creation of the piece is available for free download from Internet sources. It is believed all non-original media is copyright free.*The Machine* is ambitious in terms of the complexity and multi-layered nature of the graphics processing and I was never totally happy with the result. There are a few points where this may cause the application to crash occasionally, though this result is inconsistent and uncommon. Nevertheless, it serves its purpose well as an experiment in interactive techniques using current levels of computing technology and this industry-standard software, and, as I have stated, it has proved to be a very valuable teaching aid.

Below: Flow Chart of the interactive structure of *The Machine.*



Conclusion to Part Two

In this section the processes involved in the creation of two works of digital art have been discussed. This dissertation concludes with a summary of these processes and the issues which arose during there production:

The first is *"Take Your Space Now – A Journey through the Healing Energies of Colour and Sound",* created in collaboration with Australian oil painter Duane Radford and comprising an animated journey through her body of work. An introduction to this artist, her work, and her ideas for the project was presented. This commission required using the paintings that comprise Duane's life's work as the source material, but encompassed a transformation of these images by new media technology, in order to bring the dimensions of time, movement and poetic sound to the works. The resulting work of video art was envisaged as *a journey into the healing energies of colour and sound,* within a "New Age" metanarrative.

An analysis of Duane's body of paintings in terms of Pietersen's four metaparadigms of art reveals that they have qualities consistent with the Type 1 visionary / metaphysical mode (mystical and incorporating encompassing concepts), but also have a strong presence within the Type III experiential/interpretive mode in that they are personal, expressive, revelatory and poetic. Duane's healing mission allows for a further connection to be made with Type IV art (the missionary / developmental mode, concerned with ideological truths, humanistic and reformist values) in that they are transcendant and the artist seeks to influence life/world/society according to valued ideals and principles.

The most significant challenge in producing this work was the fact that the paintings were conceived and executed as discrete works of art and not originally planned as a series. It was necessary to develop a coherent artistic approach to give structure to this weaving together of a unified linear, time-based work of animated digital art.

The paintings were seen to fall into five broad groupings of style or content:

1. Images of space, galaxies, nebulae
2. Images of underground caverns of volcanic activity
3. Mystical Landscapes
4. Underwater worlds
5. Abstract expressionist/non-representational images exploring colour interplay

An account was given of the processes that went into producing the finished animations, beginning with the preparation of the scanned photographic images for video animation using Adobe PhotoShop (resampling and scaling the images and modifying the histogram to better convey the richness of colour onscreen). The images did not conform to 4:3 aspect ratio of television, and it was decided in collaboration with Duane to remain at all times “within” the picture plane, and in this way keep the viewer within the fantasy world of the painter’s imagination. It was Duane’s stated aim to avoid a documentary style approach – the artist wished to create a work of animated video art that was self-contained.

Extensive research into synaesthesia revealed no scientifically established correspondences between colours and musical keys or sounds, and the idea of using synaesthetic correspondences as a structural device was abandoned as an arbitrary and extraneous imposition. However the concept of *simulating* the synaesthetic experience links the work to the ideas and activities of a great many artists and thinkers down through the ages who have dreamed of a synaesthetic art experience. These links are established in the account of this tradition presented in Part One of the dissertation.

The client’s sub-title for the project, *“…a journey into the healing energies of colour and sound”* was a reference for the required “look and feel” and timing of the product. However research into the healing properties of colour also proved to be of little practical value in the search for a structure.

Finally, a simple narrative device was established which helped to get the project underway. Taking the cue from the number of Duane’s paintings which depicted space, galaxies and nebulae, etc., the video was eventually conceived as a journey through deep space, visiting “worlds of colour”. These worlds were created by building animated planets, using the technique of mapping digitized images of the paintings to a primitive 3D wire frame sphere model. By rotating the sphere, and zooming and panning the virtual camera against an animated starfield background, the illusion of flying towards these “painted” planets was established. After a digital transition “morphs” this image into the 2D image-scan of the painting, the “virtual camera” then explores the painting as though it is a landscape. In many cases, if two or more paintings were reasonably similar in colouring, this narrative/structural device was extended by imagining that these similar paintings depicted different parts of the original “painted” 3D planet – i.e. the camera would zoom down to, say, a blue world and then explore all the blue-themed paintings, as though they depicted different geographical locations in this one blue world.

Strong visual structure in shape or line was echoed when possible by the choice of transitions and visual effects. This representational (fantasy) journey into the landscapes of an imaginary, colour-themed world sometimes required transitions into a non-representational, abstract space, in order to incorporate the more abstract paintings or animation techniques into the piece.

A guiding principle was to focus the animation process on developing the expressionist intentions of the paintings. The production process became driven by experimentation with animation techniques, though it remained of paramount importance to allow the paintings to speak for themselves, and a simple exploration of the painted space using spatial transformations was ultimately the most common technique employed.

Thomas Wilfred’s conceptualization of an artform of colour-music projections, that he called *Lumia*, described in terms of “polymorphous, fluid streams of slowly metamorphosing colour” was an inspiration for the evolution of *Take Your Space Now*. The ideas of Leopold Sauvage (who created a series of sequential natural media paintings which he planned to photograph for an unrealised abstract film of "symphonies in movement" he called *Rythme Coloré*) also have a resonance with the final realization of this art piece.

A detailed (though simplified) explanation of the range of digital animation techniques that were employed in this production was then undertaken, including:

- standard key-framed transformations,
- effects created using “plug-in” digital signal processing algorithms,
- other simple 3d techniques such as terrain mapping
- particle effects
- complex compositing, multiple layering and the use of multiple effects
- Java-based animations

The difficult and lengthy task of assembling the many resulting experimental animation clips into a coherent flowing edit was then described. The gradual evolution in available desktop technology was a factor in the final resolution of this challenge, which in the early period required long and repetitive rendering tests, but became somewhat easier with the later availability of real-time video editing software and hardware. Gradually the piece took shape through a dialogue with Duane conducted via the mail.

A similar challenge was faced in composing the musical score for this work, because of the escalating scale, and final long duration of the piece. A techno-music approach with strict metronomic beat and looping rock or techno drum-beats was largely avoided. Extemporized electronic keyboard elements were combined with elaborate collages of sci-fi style audio/music samples, building up a multi-layered, atmospheric texture, over which synchronized sound effects were introduced to tie the soundtrack to prominent animation “events”. The unique sound of the e-bowed guitar was introduced as were some live ethnic flute lines. Operatic-style female voice samples are featured as a reference “humanizing” element, among the sweeping sci-fi style ambient textures.

The final stage of transferring the work to DVD was then outlined, again proving to be a significant technical challenge due to evolving but immature technology available in desk-top computing.

The second work in the creative component that is described in this doctoral submission is an interactive CD-ROM entitled *the Machine*. This work had its genesis as a simple tool for instructing students in multimedia design and development, but evolved into a practical exposition of ideas presented in the exegesis, which emerged from a study of interactivity in electronic art. This post-modern work appeals to the users visual and auditory senses with strong emphasis on cross-channel languages of interpretation between the image, text, speech, music, natural sound and motion graphic channels. It is an experiment in interactive art. It employs text-to-speech voice synthesis to simulate an artificial intelligence, and presents a dark but essentially light-hearted satire on media and technology themes. By experimenting with phonetic spelling and the introduction of intentional mis-pronunciation, vernacular and simulated voice impediments in the speech synthesis scripts, an attempt was made to build a character for the virtual computer intelligence and interactive “tour-guide”, while also to subverting the inevitable americanisation built into the text-to-speech voice synthesis software. Frame-driven scripts (activated by the playback head) and mouse-driven events (clicks and rollovers) were linked to a series of optional voice synthesis scripts. These were chosen (each time a mouse or frame event was activated) by using *Lingo* randomization techniques, producing a complex and constantly changing textual / speech element, which became a major experimental focus of the work. An interesting experiment in creative use of the voice synthesis is the inclusion of a song *Android* which features the voice synthesis element being triggered in synchronization with a pre-recorded musical backing track.

Many of the individual elements within the work can be described as surrealist collages of looping digital media. Another major focus of the work was the development of a seamless and evocative audio experience within a user-controlled interactive context. *Lingo* randomization techniques were once again employed to to produce variations in the looping sections of sound.

The Machine is divided into two main areas. The first is the virtual factory leading to the main control panel, from where the user may access five sub-nodes:

1. **The Splat Room** (a send-up computer game)

2. **The Baby Factory** (a dark comment on bio-engineering technology)

3. **The Vault** (a collection of political and surreal cartoons)

4. **Esoteric Room** – some poetic comment;

and finally the a second area and major sub-node

5. **the Recreation Centre** *(Control Panel 2)*

The user can quit *the Machine* from any access point. Each of the five sub- sections is accessed though a sequence of labrythine virtual tunnels. The second main sub-section of *The Machine* is accessed via *the Recreation Centre* node, from where the user has access to six further sub-sections:

1. **The TV Room** (a series of satirical comments on media issues, but also containing some experimental abstract/kinetic animations and a feature animation, *In Ya Face: Two Headz are Better than One)*

2. **The Staff Room** (sound files with non-audio markers; “comic strip” animation)

3. **The Child-Minding Centre** (an example of linear story structure in an interactive context, featuring the interactive *Flash* animation entitled *Emutown)*

4. **The Random Insult Generator** (featuring randomised access to text strings)

5. **The Pointless Exercise Room** (the Anagram Machine)

6. **The Staff Education and Training Centre,** from where a separate interactive program or sub-movie is presented, which is an interactive introduction to the use of colour in Graphic Design

Addressing the inherent problems of interactive, non-linear sound design was a major focus of this experimentation, with the overall goal of producing a seamless, dynamically changing, non-linear audio experience to enhance the mood, drama and humour of the piece. In general, wherever it could be managed during the production of this interactive CD-ROM, randomization techniques are used to offer variety and reduce repetition, and the sound-track regularly incorporates randomized atonal sound effects to camouflage or vary the sound loops at interactive nodes.